

1

Number System

FUNDAMENTAL CONCEPTS

I. Numbers

In Hindu-Arabic system, we have ten digits, namely 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9.

A number is denoted by a group of digits, called **numeral**.

For denoting a numeral, we use the place-value chart, given below.

	Ten-Crores	Crores	Ten-Lakhs	Lakhs	Ten-Thousands	Thousands	Hundreds	Tens	Ones
(i)				5	2	8	6	7	9
(ii)			4	3	8	0	9	6	7
(iii)		3	5	2	1	8	0	0	9
(iv)	5	6	1	3	0	7	0	9	0

The four numerals shown above may be written in words as:

- (i) Five lakh twenty-eight thousand six hundred seventy-nine
- (ii) Forty-three lakh eighty thousand nine hundred sixty-seven
- (iii) Three crore fifty-two lakh eighteen thousand nine
- (iv) Fifty-six crore thirteen lakh seven thousand ninety

Now, suppose we are given the following four numerals in words:

- (i) Nine crore four lakh six thousand two
- (ii) Twelve crore seven lakh nine thousand two hundred seven
- (iii) Four lakh four thousand forty
- (iv) Twenty-one crore sixty lakh five thousand fourteen

Then, using the place-value chart, these may be written in figures as under:

	Ten-Crores	Crores	Ten-Lakhs	Lakhs	Ten-Thousands	Thousands	Hundreds	Tens	Ones
(i)		9	0	4	0	6	0	0	2
(ii)	1	2	0	7	0	9	2	0	7
(iii)				4	0	4	0	4	0
(iv)	2	1	6	0	0	5	0	1	4

II. Face value and Place value (or Local Value) of a Digit in a Numeral

- (i) The face value of a digit in a numeral is its own value, at whatever place it may be.

Ex. In the numeral 6872, the face value of 2 is 2, the face value of 7 is 7, the face value of 8 is 8 and the face value of 6 is 6.

- (ii) In a given numeral:

Place value of ones digit = (ones digit) $\times$ 1,

Place value of tens digit = (tens digit) $\times$ 10,

Place value of hundreds digit = (hundreds digit) $\times$ 100 and so on.

Ex. In the numeral 70984, we have

Place value of 4 = $(4 \times 1) = 4$,

Place value of 8 = $(8 \times 10) = 80$,

Place value of 9 = $(9 \times 100) = 900$,

Place value of 7 = $(7 \times 10000) = 70000$.

Note: Place value of 0 in a given numeral is 0, at whatever place it may be.

III. Various Types of Numbers

1. **Natural Numbers:** Counting numbers are called natural numbers.

Thus, 1, 2, 3, 4, are all natural numbers.

2. **Whole Numbers:** All counting numbers, together with 0, form the set of whole numbers.

Thus, 0, 1, 2, 3, 4, are all whole numbers.

- 3. Integers:** All counting numbers, zero and negatives of counting numbers, form the set of integers.
Thus,, $-3, -2, -1, 0, 1, 2, 3, \dots$ are all integers.
Set of positive integers = $\{1, 2, 3, 4, 5, 6, \dots\}$
Set of negative integers = $\{-1, -2, -3, -4, -5, -6, \dots\}$
Set of all non-negative integers = $\{0, 1, 2, 3, 4, 5, \dots\}$
- 4. Even Numbers:** A counting number divisible by 2 is called an even number.
Thus, $0, 2, 4, 6, 8, 10, 12, \dots$ etc. are all even numbers.
- 5. Odd Numbers:** A counting number not divisible by 2 is called an odd number.
Thus, $1, 3, 5, 7, 9, 11, 13, \dots$ etc. are all odd numbers.
- 6. Prime Numbers:** A counting number is called a prime number if it has exactly two factors, namely itself and 1.
Ex. All prime numbers less than 100 are:
 $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97$
- 7. Composite Numbers:** All counting numbers, which are not prime, are called composite numbers.
A composite number has more than 2 factors.
- 8. Perfect Numbers:** A number, the sum of whose factors (except the number itself), is equal to the number, is called a perfect number, e.g. 6, 28, 496
The factors of 6 are 1, 2, 3 and 6. And, $1 + 2 + 3 = 6$.
The factors of 28 are 1, 2, 4, 7, 14 and 28. And, $1 + 2 + 4 + 7 + 14 = 28$.
- 9. Co-primes (or Relative Primes):** Two numbers whose H.C.F. is 1 are called co-prime numbers,
Ex. (2, 3), (8, 9) are pairs of co-primes.
- 10. Twin Primes:** Two prime numbers whose difference is 2 are called twin-primes,
Ex. (3, 5), (5, 7), (11, 13) are pairs of twin-primes.
- 11. Rational Numbers:** Numbers which can be expressed in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$, are called rational numbers.
Ex. $\frac{1}{8}, \frac{-8}{11}, 0, 6, 5\frac{2}{3}$ etc.
- 12. Irrational Numbers:** Numbers which when expressed in decimal would be in non-terminating and non-repeating form, are called irrational numbers.
Ex. $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \pi, e, 0.231764735\dots$

IV. Important Facts:

- All natural numbers are whole numbers.
- All whole numbers are not natural numbers.
0 is a whole number which is not a natural number.
- Even number + Even number = Even number
Odd number + Odd number = Even number
Even number + Odd number = Odd number
Even number - Even number = Even number
Odd number - Odd number = Even number
Even number - Odd number = Odd number
Odd number - Even number = Odd number
Even number $\times$ Even number = Even number
Odd number $\times$ Odd number = Odd number
Even number $\times$ Odd number = Even number
- The smallest prime number is 2.
- The only even prime number is 2.
- The first odd prime number is 3.
- 1 is a unique number - neither prime nor composite.
- The least composite number is 4.
- The least odd composite number is 9.
- Test for a Number to be Prime:**
Let p be a given number and let n be the smallest counting number such that $n^2 \geq p$.
Now, test whether p is divisible by any of the prime numbers less than or equal to n . If yes, then p is not prime otherwise, p is prime.

Ex. Test, which of the following are prime numbers?

(i) 137

(ii) 173

(iii) 319

(iv) 437

(v) 811

- Sol.** (i) We know that $(12)^2 > 137$.
Prime numbers less than 12 are 2, 3, 5, 7, 11.
Clearly, none of them divides 137.
 $\therefore$ 137 is a prime number.
- (ii) We know that $(14)^2 > 173$.
Prime numbers less than 14 are 2, 3, 5, 7, 11, 13.
Clearly, none of them divides 173.
 $\therefore$ 173 is a prime number.
- (iii) We know that $(18)^2 > 319$.
Prime numbers less than 18 are 2, 3, 5, 7, 11, 13, 17.
Out of these prime numbers, 11 divides 319 completely.
 $\therefore$ 319 is not a prime number.
- (iv) We know that $(21)^2 > 437$.
Prime numbers less than 21 are 2, 3, 5, 7, 11, 13, 17, 19.
Clearly, 437 is divisible by 19.
 $\therefore$ 437 is not a prime number.
- (v) We know that $(30)^2 > 811$.
Prime numbers less than 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
Clearly, none of these numbers divides 811.
 $\therefore$ 811 is a prime number.

V. Important Formulae:

- | | |
|---|--|
| (i) $(a + b)^2 = a^2 + b^2 + 2ab$ | (ii) $(a - b)^2 = a^2 + b^2 - 2ab$ |
| (iii) $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$ | (iv) $(a + b)^2 - (a - b)^2 = 4ab$ |
| (v) $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ | (vi) $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$ |
| (vii) $a^2 - b^2 = (a + b)(a - b)$ | (viii) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ |
| (ix) $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$ | (x) $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$ |
| (xi) $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ | |
| (xii) If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$ | |

TESTS OF DIVISIBILITY

1. Divisibility By 2:

A number is divisible by 2 if its unit digit is any of 0, 2, 4, 6, 8.

Ex. 58694 is divisible by 2, while 86945 is not divisible by 2.

2. Divisibility By 3:

A number is divisible by 3 only when the sum of its digits is divisible by 3.

Ex. (i) In the number 695421, the sum of digits = 27, which is divisible by 3.

$\therefore$ 695421 is divisible by 3.

(ii) In the number 948653, the sum of digits = 35, which is not divisible by 3.

$\therefore$ 948653 is not divisible by 3.

3. Divisibility By 9:

A number is divisible by 9 only when the sum of its digits is divisible by 9.

Ex. (i) In the number 246591, the sum of digits = 27, which is divisible by 9.

$\therefore$ 246591 is divisible by 9.

(ii) In the number 734519, the sum of digits = 29, which is not divisible by 9.

$\therefore$ 734519 is not divisible by 9.

4. Divisibility By 4:

A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

Ex. (i) 6879376 is divisible by 4, since 76 is divisible by 4.

(ii) 496138 is not divisible by 4, since 38 is not divisible by 4.

5. Divisibility By 8:

A number is divisible by 8 if the number formed by its last three digits is divisible by 8.

Ex. (i) In the number 16789352, the number formed by last 3 digits, namely 352 is divisible by 8.
 $\therefore$ 16789352 is divisible by 8.

(ii) In the number 576484, the number formed by last 3 digits, namely 484 is not divisible by 8.
 $\therefore$ 576484 is not divisible by 8.

6. Divisibility By 10:

A number is divisible by 10 only when its unit digit is 0.

Ex. (i) 7849320 is divisible by 10, since its unit digit is 0.

(ii) 678405 is not divisible by 10, since its unit digit is not 0.

7. Divisibility By 5:

A number is divisible by 5 only when its unit digit is 0 or 5.

Ex. (i) Each of the numbers 76895 and 68790 is divisible by 5.

8. Divisibility By 11:

A number is divisible by 11 if the difference between the sum of its digits at odd places and the sum of its digits at even places is either 0 or a number divisible by 11.

Ex. (i) Consider the number 29435417.

(Sum of its digits at odd places) – (Sum of its digits at even places)
 $= (7 + 4 + 3 + 9) - (1 + 5 + 4 + 2) = (23 - 12) = 11$, which is divisible by 11.
 $\therefore$ 29435417 is divisible by 11.

(ii) Consider the number 57463822.

(Sum of its digits at odd places) – (Sum of its digits at even places)
 $= (2 + 8 + 6 + 7) - (2 + 3 + 4 + 5) = (23 - 14) = 9$, which is not divisible by 11.
 $\therefore$ 57463822 is not divisible by 11.

9. Divisibility By 25:

A number is divisible by 25 if the number formed by its last two digits is either 00 or divisible by 25.

Ex. (i) In the number 63875, the number formed by last 2 digits, namely 75 is divisible by 25.
 $\therefore$ 63875 is divisible by 25.

(ii) In the number 96445, the number formed by last 2 digits, namely 45 is not divisible by 25.
 $\therefore$ 96445 is not divisible by 25.

10. Divisibility By 7 or 13:

Divide the number into groups of 3 digits (starting from right) and find the difference between the sum of the numbers in odd and even places. If the difference is 0 or divisible by 7 or 13 (as the case may be), it is divisible by 7 or 13.

Ex. (i) $4537792 \rightarrow 4 / 537 / 792$

$(792 + 4) - 537 = 259$, which is divisible by 7 but not by 13.
 $\therefore$ 4537792 is divisible by 7 and not by 13.

(ii) $579488 \rightarrow 579 / 488$

$579 - 488 = 91$, which is divisible by both 7 and 13.
 $\therefore$ 579488 is divisible by both 7 and 13.

11. Divisibility By 16:

A number is divisible by 16, if the number formed by its last 4 digits is divisible by 16.

Ex. (i) In the number 463776, the number formed by last 4 digits, namely 3776, is divisible by 16.
 $\therefore$ 463776 is divisible by 16.

(ii) In the number 895684, the number formed by last 4 digits, namely 5684, is not divisible by 16.
 $\therefore$ 895684 is not divisible by 16.

12. Divisibility By 6: A number is divisible by 6, if it is divisible by both 2 and 3.

13. Divisibility By 12: A number is divisible by 12, if it is divisible by both 3 and 4.

14. Divisibility By 15: A number is divisible by 15, if it is divisible by both 3 and 5.

15. Divisibility By 18: A number is divisible by 18, if it is divisible by both 2 and 9.

16. Divisibility By 14: A number is divisible by 14, if it is divisible by both 2 and 7.

17. Divisibility By 24: A given number is divisible by 24, if it is divisible by both 3 and 8.

18. Divisibility By 40: A given number is divisible by 40, if it is divisible by both 5 and 8.

19. Divisibility By 80: A given number is divisible by 80, if it is divisible by both 5 and 16.

Note: If a number is divisible by p as well as q , where p and q are co-primes, then the given number is divisible by pq .

If p and q are not co-primes, then the given number need not be divisible by pq , even when it is divisible by both p and q .

Ex. 36 is divisible by both 4 and 6, but it is not divisible by $(4 \times 6) = 24$, since 4 and 6 are not co-primes.

VI. Factorial of a Number

Let n be a positive integer.

Then, the continued product of first n natural numbers is called factorial n , denoted by $n!$ or $\underline{n}$.

Thus, $n! = n(n-1)(n-2) \dots \dots \dots 3.2.1$

Ex. $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Note: $0! = 1$

VII. Modulus of a Number

$$|x| = \begin{cases} x, & \text{when } x \geq 0 \\ -x, & \text{when } x < 0 \end{cases}$$

Ex. $|-5| = 5$, $|4| = 4$, $|-1| = 1$, etc.

VIII. Greatest Integral Value

The greatest integral value of an integer x , denoted by $[x]$, is defined as the greatest integer not exceeding x .

Ex. $[1.35] = 1$, $\left[\frac{11}{4}\right] = \left[2\frac{3}{4}\right] = 2$, etc.

IX. Multiplication BY Short cut Methods

1. Multiplication By Distributive Law:

(i) $a \times (b + c) = a \times b + a \times c$ (ii) $a \times (b - c) = a \times b - a \times c$

Ex. (i) $567958 \times 99999 = 567958 \times (100000 - 1) = 567958 \times 100000 - 567958 \times 1$
 $= (56795800000) - 567958 = 56795232042$.

(ii) $978 \times 184 + 978 \times 816 = 978 \times (184 + 816) = 978 \times 1000 = 978000$.

2. Multiplication of a Number By 5^n :

Put n zeros to the right of the multiplicand and divide the number so formed by 2^n .

Ex. $975436 \times 625 = 975436 \times 5^4 = \frac{9754360000}{16} = 609647500$.

X. Division Algorithm or Euclidean Algorithm

If we divide a given number by another number, then:

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

Important Facts:

1. (i) $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n .
- (ii) $(x^n - a^n)$ is divisible by $(x + a)$ for all even values of n .
- (iii) $(x^n + a^n)$ is divisible by $(x + a)$ for all odd values of n .

2. To find the highest power of a prime number p in $n!$

Highest power of p in $n! = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \dots + \left[\frac{n}{p^r}\right]$, where $p^r \leq n < p^{r+1}$

SOLVED EXAMPLES

Ex. 1. Simplify: (i) $8888 + 888 + 88 + 8$

(ii) $715632 - 631104 - 9874 - 999$

(LIC, ADO, 2007)

Sol. (i) 8888 (ii) Given $\text{exp} = 715632 - (631104 + 9874 + 99)$
 888 $= 715632 - 641077 = 74555$.

$$\begin{array}{r} 8888 \\ 888 \\ 88 \\ + 8 \\ \hline 9872 \end{array}$$

$$\begin{array}{r} 631104 \\ 9874 \\ + 99 \\ \hline 641077 \end{array}$$

$$\begin{array}{r} 715632 \\ - 641077 \\ \hline 74555 \end{array}$$

Ex. 2. What value will replace the question mark in each of the following questions?

(i)? - 1936248 = 1635773 (ii) 9587 - ? = 7429 - 4358

Sol. (i) Let $x - 1936248 = 1635773$. Then, $x = 1635773 + 1936248 = 3572021$.

(ii) Let $9587 - x = 7429 - 4358$.

Then, $9587 - x = 3071 \Rightarrow x = 9587 - 3071 = 6516$.

Ex. 3. What could be the maximum value of Q in the following equation?

$$5P9 + 3R7 + 2Q8 = 1114$$

Sol. We may analyse the given equation as shown:

Clearly, $2 + P + R + Q = 11$.

So, the maximum value of Q can be $(11 - 2)$, i.e. 9 (when $P = 0$, $R = 0$).

Ex. 4. Simplify: (i) 5793405×9999 (ii) 839478×625

Sol. (i) $5793405 \times 9999 = 5793405 \times (10000 - 1) = 57934050000 - 5793405 = 57928256595$.

(ii) $839478 \times 625 = 839478 \times 5^4 = 839478 \times \left(\frac{10}{2}\right)^4 = \frac{839478 \times 10^4}{2^4} = \frac{8394780000}{16} = 524673750$.

Ex. 5. Evaluate: (i) $986 \times 137 + 986 \times 863$ (ii) $983 \times 207 - 983 \times 107$

Sol. (i) $986 \times 137 + 986 \times 863 = 986 \times (137 + 863) = 986 \times 1000 = 986000$.

(ii) $983 \times 207 - 983 \times 107 = 983 \times (207 - 107) = 983 \times 100 = 98300$.

Ex. 6. Simplify: (i) 1605×1605 (ii) 1398×1398

Sol. (i) $1605 \times 1605 = (1605)^2 = (1600 + 5)^2 = (1600)^2 + 5^2 + 2 \times 1600 \times 5$
 $= 2560000 + 25 + 16000 = 2576025$.

(ii) $1398 \times 1398 = (1398)^2 = (1400 - 2)^2 = (1400)^2 + 2^2 - 2 \times 1400 \times 2$
 $= 1960000 + 4 - 5600 = 1954404$.

Ex. 7. Evaluate: (i) $475 \times 475 + 125 \times 125$ (ii) $796 \times 796 - 204 \times 204$

Sol. (i) We have $(a^2 + b^2) = \frac{1}{2}[(a+b)^2 + (a-b)^2]$

$$\therefore (475)^2 + (125)^2 = \frac{1}{2}[(475+125)^2 + (475-125)^2] = \frac{1}{2}[(600)^2 + (350)^2]$$

$$= \frac{1}{2}[360000 + 122500] = \frac{1}{2} \times 482500 = 241250$$

(ii) $796 \times 796 - 204 \times 204 = (796)^2 - (204)^2 = (796 + 204)(796 - 204)$
 $= (1000 \times 592) = 592000$.

$$[\because (a-b)^2 = (a+b)(a-b)]$$

Ex. 8. Simplify: (i) $(387 \times 387 + 113 \times 113 + 2 \times 387 \times 113)$

(ii) $(87 \times 87 + 61 \times 61 - 2 \times 87 \times 61)$

Sol. (i) Given Exp. $= (387)^2 + (113)^2 + 2 \times 387 \times 113 = (a^2 + b^2 + 2ab)$, where $a = 387$ and $b = 113$
 $= (a + b)^2 = (387 + 113)^2 = (500)^2 = 250000$.

(ii) Given Exp. $= (87)^2 + (61)^2 - 2 \times 87 \times 61 = (a^2 + b^2 - 2ab)$, where $a = 87$ and $b = 61$
 $= (a - b)^2 = (87 - 61)^2 = (26)^2 = (20 + 6)^2 = (20)^2 + 6^2 + 2 \times 20 \times 6 = (400 + 36 + 240)$
 $= (436 + 240) = 676$.

Ex. 9. Find the square root of $4a^2 + b^2 + c^2 + 4ab - 2bc - 4ac$.

(Campus Recruitment, 2010)

Sol. $\sqrt{4a^2 + b^2 + c^2 + 4ab - 2bc - 4ac} = \sqrt{(2a)^2 + b^2 + (-c)^2 + 2 \times 2a \times b + 2 \times b \times (-c) + 2 \times (2a) \times (-c)}$
 $= \sqrt{(2a + b - c)^2} = (2a + b - c)$.

Ex. 10. A is counting the numbers from 1 to 31 and B from 31 to 1. A is counting the odd numbers only. The speed of both is the same. What will be the number which will be pronounced by A and B together?

(Campus Recruitment, 2010)

Sol. The numbers pronounced by A and B in order are:

A	1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31
B	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Clearly both A and B pronounce the number 21 together.

$$\begin{array}{r} \textcircled{1} \textcircled{2} \\ 5 \ P \ 9 \\ 3 \ R \ 7 \\ \hline 2 \ Q \ 8 \\ \hline 1 \ 1 \ 1 \ 4 \end{array}$$

Ex. 11. Simplify: (i) $\frac{789 \times 789 \times 789 + 211 \times 211 \times 211}{789 \times 789 - 789 \times 211 + 211 \times 211}$ (ii) $\frac{658 \times 658 \times 658 - 328 \times 328 \times 328}{658 \times 658 + 658 \times 328 + 328 \times 328}$

Sol. (i) Given exp. = $\frac{(789)^3 + (211)^3}{(789)^2 - (789 \times 211) + (211)^2} = \frac{a^3 + b^3}{a^2 - ab + b^2}$, (where $a = 789$ and $b = 211$)
 $= (a + b) = (789 + 211) = 1000.$

(ii) Given exp. = $\frac{(658)^3 - (328)^3}{(658)^2 + (658 \times 328) + (328)^2} = \frac{a^3 - b^3}{a^2 + ab + b^2}$, (where $a = 658$ and $b = 328$)
 $= (a - b) = (658 - 328) = 330.$

Ex. 12. Simplify: $\frac{(893 + 786)^2 - (893 - 786)^2}{(893 \times 786)}.$

Sol. Given exp. = $\frac{(a + b)^2 - (a - b)^2}{ab}$ (where $a = 893, b = 786$) = $\frac{4ab}{ab} = 4.$

Ex. 13. Which of the following are prime numbers?

(i) 241

(ii) 337

(iii) 391

(iv) 571

Sol. (i) Clearly, $16 > \sqrt{241}.$

Prime numbers less than 16 are 2, 3, 5, 7, 11, 13.

241 is not divisible by any of them.

$\therefore$ 241 is a prime number.

(ii) Clearly, $19 > \sqrt{337}.$ Prime numbers less than 19 are 2, 3, 5, 7, 11, 13, 17.

337 is not divisible by any one of them.

$\therefore$ 337 is a prime number.

(iii) Clearly, $20 > \sqrt{391}.$ Prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19.

We find that 391 is divisible by 17.

$\therefore$ 391 is not a prime number.

(iv) Clearly, $24 > \sqrt{571}.$ Prime numbers less than 24 are 2, 3, 5, 7, 11, 13, 17, 19, 23.

571 is not divisible by any one of them.

$\therefore$ 571 is a prime number.

Ex. 14. If Δ stands for the operation 'adding first number to twice the second number', then find the value of $(1 \Delta 2) \Delta 3.$

Sol. $(1 \Delta 2) \Delta 3 = (1 + 2 \times 2) \Delta 3 = 5 \Delta 3 = 5 + 2 \times 3 = 5 + 6 = 11.$

Ex. 15. Given that $1^2 + 2^2 + 3^2 + \dots + 10^2 = 385$, then find the value of $2^2 + 4^2 + 6^2 + \dots + 20^2.$

Sol. $2^2 + 4^2 + 6^2 + \dots + 20^2 = 2^2 (1^2 + 2^2 + 3^2 + \dots + 10^2) = 2^2 \times 385 = 4 \times 385 = 1540.$

Ex. 16. Which of the following numbers is divisible by 3?

(i) 541326

(ii) 5967013

Sol. (i) Sum of digits in 541326 = $(5 + 4 + 1 + 3 + 2 + 6) = 21$, which is divisible by 3.

Hence, 541326 is divisible by 3.

(ii) Sum of digits in 5967013 = $(5 + 9 + 6 + 7 + 0 + 1 + 3) = 31$, which is not divisible by 3.

Hence, 5967013 is not divisible by 3.

Ex. 17. What least value must be assigned to * so that the number 197*5462 is divisible by 9?

Sol. Let the missing digit be $x.$

Sum of digits = $(1 + 9 + 7 + x + 5 + 4 + 6 + 2) = (34 + x).$

For $(34 + x)$ to be divisible by 9, x must be replaced by 2.

Hence, the digit in place of * must be 2.

Ex. 18. Which of the following numbers is divisible by 4?

(i) 67920594

(ii) 618703572

Sol. (i) The number formed by the last two digits in the given number is 94, which is not divisible by 4.

Hence, 67920594 is not divisible by 4.

(ii) The number formed by the last two digits in the given number is 72, which is divisible by 4.

Hence, 618703572 is divisible by 4.

Ex. 19. Which digits should come in place of * and \$ if the number 62684*\$ is divisible by both 8 and 5?

Sol. Since the given number is divisible by 5, so 0 or 5 must come in place of \$. But, a number ending with 5 is never divisible by 8. So, 0 will replace \$.
Now, the number formed by the last three digits is 4*0, which becomes divisible by 8, if * is replaced by 4.
Hence, digits in place of * and \$ are 4 and 0 respectively.

Ex. 20. Show that 4832718 is divisible by 11.

Sol. (Sum of digits at odd places) – (Sum of digits at even places) = $(8 + 7 + 3 + 4) - (1 + 2 + 8) = 11$, which is divisible by 11.
Hence, 4832718 is divisible by 11.

Ex. 21. Is 52563744 divisible by 24?

Sol. $24 = 3 \times 8$, where 3 and 8 are co-primes.
The sum of the digits in the given number is 36, which is divisible by 3. So, the given number is divisible by 3.
The number formed by the last 3 digits of the given number is 744, which is divisible by 8. So, the given number is divisible by 8. Thus, the given number is divisible by both 3 and 8, where 3 and 8 are co-primes. So, it is divisible by 3×8 , i.e. 24.

Ex. 22. What are the values of M and N respectively if M39048458N is divisible by both 8 and 11, where M and N are single-digit integers?

Sol. Since the given number is divisible by 8, it is obvious that the number formed by the last three digits, i.e. 58N is divisible by 8, which is possible only when $N = 4$.
Now, (sum of digits at even places) – (sum of digits at odd places)

$$= (8 + 4 + 4 + 9 + M) - (4 + 5 + 8 + 0 + 3)$$

$$= (25 + M) - 20 = M + 5, \text{ which must be divisible by 11.}$$

So, $M = 6$.

Hence, $M = 6, N = 4$.

Ex. 23. Find the number of digits in the smallest number which is made up of digits 1 and 0 only and is divisible by 225.

Sol. $225 = 9 \times 25$, where 9 and 25 are co-primes.
Clearly, a number is divisible by 225 if it is divisible by both 9 and 25.
Now, a number is divisible by 9 if the sum of its digits is divisible by 9 and a number is divisible by 25 if the number formed by the last two digits is divisible by 25.
 $\therefore$ The smallest number which is made up of digits 1 and 0 and divisible by 225 = 1111111100.
Hence, number of digits = 11.

Ex. 24. If the number 3422213pq is divisible by 99, find the missing digits p and q.

Sol. $99 = 9 \times 11$, where 9 and 11 are co-primes.
Clearly, a number is divisible by 99 if it is divisible by both 9 and 11.
Since the number is divisible by 9, we have:

$$(3 + 4 + 2 + 2 + 2 + 1 + 3 + p + q) = \text{a multiple of 9}$$

$$\Rightarrow 17 + (p + q) = 18 \text{ or } 27$$

$$\Rightarrow p + q = 1 \quad \dots(i) \quad \text{or} \quad p + q = 10 \quad \dots(ii)$$
 Since the number is divisible by 11, we have:

$$(q + 3 + 2 + 2 + 3) - (p + 1 + 2 + 4) = 0 \text{ or a multiple of 11}$$

$$\Rightarrow (10 + q) - (7 + p) = 0 \text{ or } 11$$

$$\Rightarrow 3 + (q - p) = 0 \text{ or } 11$$

$$\Rightarrow q - p = -3 \quad \text{or} \quad q - p = 8$$

$$\Rightarrow p - q = 3 \quad \dots(iii) \quad \text{or} \quad q - p = 8 \quad \dots(iv)$$
 Clearly, if (i) holds, then neither (iii) nor (iv) holds. So, (i) does not hold.
Also, solving (ii) and (iii) together, we get: $p = 6.5$, which is not possible.
Solving (ii) and (iv) together, we get: $p = 1, q = 9$.

Ex. 25. x is a positive integer such that $x^2 + 12$ is exactly divisible by x. Find all the possible values of x.

Sol. $\frac{x^2 + 12}{x} = \frac{x^2}{x} + \frac{12}{x} = x + \frac{12}{x}$.

Clearly, 12 must be completely divisible by x.

So, the possible values of x are 1, 2, 3, 4, 6 and 12.

Ex. 26. Find the smallest number to be added to 1000 so that 45 divides the sum exactly.

Sol. On dividing 1000 by 45, we get 10 as remainder.
 $\therefore$ Number to be added = $(45 - 10) = 35$.

Ex. 27. What least number must be subtracted from 2000 to get a number exactly divisible by 17?

Sol. On dividing 2000 by 17, we get 11 as remainder.
 $\therefore$ Required number to be subtracted = 11.

Ex. 28. Find the number which is nearest to 3105 and is exactly divisible by 21.

Sol. On dividing 3105 by 21, we get 18 as remainder.
 $\therefore$ Number to be added to 3105 = $(21 - 18) = 3$.
 Hence, required number = $3105 + 3 = 3108$.

Ex. 29. Find the smallest number of five digits which is exactly divisible by 476.

Sol. Smallest number of 5 digits = 10000.
 On dividing 10000 by 476, we get 4 as remainder.
 $\therefore$ Number to be added = $(476 - 4) = 472$.
 Hence, required number = 10472.

Ex. 30. Find the greatest number of five digits which is exactly divisible by 47.

Sol. Greatest number of 5 digits is 99999.
 On dividing 99999 by 47, we get 30 as remainder.
 $\therefore$ Required number = $(99999 - 30) = 99969$.

Ex. 31. When a certain number is multiplied by 13, the product consists entirely of fives. Find the smallest such number.

Sol. Clearly, we keep on dividing 55555..... by 13 till we get 0 as remainder.
 $\therefore$ Required number = 42735.

Ex. 32. When a certain number is multiplied by 18, the product consists entirely of 2's. What is the minimum number of 2's in the product?

Sol. We keep on dividing 22222..... by 18 till we get 0 as remainder.
 Clearly, number of 2's in the product = 9.

Ex. 33. Find the smallest number which when multiplied by 9 gives the product as 1 followed by a certain number of 7s only.

Sol. The least number having 1 followed by 7s, which is divisible by 9, is 177777, as $1 + 7 + 7 + 7 + 7 + 7 = 36$ (which is divisible by 9).
 $\therefore$ Required number = $177777 \div 9 = 19753$.

Ex. 34. What is the unit's digit in the product?

$$81 \times 82 \times 83 \times \dots \times 89?$$

Sol. Required unit's digit = Unit's digit in the product $1 \times 2 \times 3 \times \dots \times 9 = 0$
 $[\because 2 \times 5 = 10]$

Ex. 35. Find the unit's digit in the product $(2467)^{153} \times (341)^{72}$.

Sol. Clearly, unit's digit in the given product = unit's digit in $7^{153} \times 1^{72}$.
 Now, 7^4 gives unit digit 1.
 $\therefore 7^{152}$ gives unit digit 1.
 $\therefore 7^{153}$ gives unit digit $(1 \times 7) = 7$. Also, 1^{72} gives unit digit 1.
 Hence, unit digit in the product = $(7 \times 1) = 7$.

Ex. 36. Find the unit's digit in $(264)^{102} + (264)^{103}$.

Sol. Required unit's digit = unit's digit in $(4)^{102} + (4)^{103}$.
 Now, 4^2 gives unit digit 6.
 $\therefore (4)^{102}$ gives unit digit 6.
 $(4)^{103}$ gives unit digit of the product (6×4) i.e., 4.
 Hence, unit's digit in $(264)^{102} + (264)^{103}$ = unit's digit in $(6 + 4) = 0$.

Ex. 37. Find the total number of prime factors in the expression $(4)^{11} \times (7)^5 \times (11)^2$.

Sol. $(4)^{11} \times (7)^5 \times (11)^2 = (2 \times 2)^{11} \times (7)^5 \times (11)^2 = 2^{11} \times 2^{11} \times 7^5 \times 11^2 = 2^{22} \times 7^5 \times 11^2$.
 $\therefore$ Total number of prime factors = $(22 + 5 + 2) = 29$.

$$\begin{array}{r} 42735 \\ 13 \overline{)555555} \\ \underline{52} \\ 35 \\ \underline{26} \\ 95 \\ \underline{91} \\ 45 \\ \underline{39} \\ 65 \\ \underline{65} \\ 0 \end{array}$$

$$\begin{array}{r} 12345679 \\ 18 \overline{)22222222} \\ \underline{18} \\ 42 \\ \underline{36} \\ 62 \\ \underline{54} \\ 82 \\ \underline{72} \\ 102 \\ \underline{90} \\ 122 \\ \underline{108} \\ 142 \\ \underline{126} \\ 162 \\ \underline{162} \\ 0 \end{array}$$

Ex. 38. What is the number of zeros at the end of the product of the numbers from 1 to 100?

Sol. Let $N = 1 \times 2 \times 3 \times \dots \times 100$.

Clearly, only the multiples of 2 and 5 yield zeros on multiplication.

In the given product, the highest power of 5 is much less than that compared to 2. So, we shall find the highest power of 5 in N .

$$\text{Highest power of 5 in } N = \left[\frac{100}{5} \right] + \left[\frac{100}{5^2} \right] = 20 + 4 = 24.$$

Hence, required number of zeros = 24.

Ex. 39. What is the number of zeros at the end of the product $5^5 \times 10^{10} \times 15^{15} \times \dots \times 125^{125}$?

Sol. Clearly, the highest power of 2 is less than that of 5 in N .

So, the highest power of 2 in N shall give us the number of zeros at the end of N .

$$\begin{aligned} \text{Highest power of 2} &= \text{Number of multiples of 2} + \text{Number of multiples of 4 (i.e. } 2^2) + \\ &\quad \text{Number of multiples of 8 (i.e. } 2^3) + \text{Number of multiples of 16 (i.e. } 2^4) \\ &= [(10 + 20 + 30 + \dots + 120) + (20 + 40 + 60 + \dots + 120) + (40 + 80 + 120) + 80] \\ &= (780 + 420 + 240 + 80) = 1520. \end{aligned}$$

Hence, required number of zeros = 1520.

Ex. 40. On dividing 15968 by a certain number, the quotient is 89 and the remainder is 37. Find the divisor.

$$\text{Sol. Divisor} = \frac{\text{Dividend} - \text{Remainder}}{\text{Quotient}} = \frac{15968 - 37}{89} = 179.$$

Ex. 41. A number when divided by 114, leaves remainder 21. If the same number is divided by 19, find the remainder. (S.S.C., 2010)

Sol. On dividing the given number by 114, let k be the quotient and 21 the remainder.

$$\text{Then, number} = 114k + 21 = 19 \times 6k + 19 + 2 = 19(6k + 1) + 2.$$

$\therefore$ The given number when divided by 19 gives remainder = 2.

Ex. 42. A number being successively divided by 3, 5 and 8 leaves remainders 1, 4 and 7 respectively. Find the respective remainders if the order of divisors be reversed.

$$\begin{array}{l} \text{Sol.} \quad \begin{array}{r|l} 3 & x \\ \hline 5 & y - 1 \\ \hline 8 & z - 4 \\ \hline & 1 - 7 \end{array} \quad \begin{array}{l} \therefore z = (8 \times 1 + 7) = 15; \\ y = (5z + 4) = (5 \times 15 + 4) = 79; \\ x = (3y + 1) = (3 \times 79 + 1) = 238. \end{array} \end{array}$$

$$\text{Now,} \quad \begin{array}{r|l} 8 & 238 \\ \hline 5 & 29 - 6 \\ \hline 3 & 5 - 4 \\ \hline & 1 - 2 \end{array}$$

$\therefore$ Respective remainders are 6, 4, 2.

Ex. 43. Three boys A, B, C were asked to divide a certain number by 1001 by the method of factors. They took the factors in the orders 13, 11, 7; 7, 11, 13 and 11, 7, 13 respectively. If the first boy obtained 3, 2, 1 as successive remainders, then find the successive remainders obtained by the other two boys B and C.

$$\begin{array}{l} \text{Sol.} \quad \begin{array}{r|l} 13 & x \\ \hline 11 & y - 3 \\ \hline 7 & z - 2 \\ \hline & 1 - 1 \end{array} \quad \begin{array}{l} \therefore z = 7 \times 1 + 1 = 8, \\ y = 11z + 2 = 11 \times 8 + 2 = 90; \\ x = 13y + 3 = 13 \times 90 + 3 = 1173. \end{array} \end{array}$$

$$\text{Now,} \quad \begin{array}{r|l} 7 & 1173 \\ \hline 11 & 167 - 4 \\ \hline 13 & 15 - 2 \\ \hline & 1 - 2 \end{array} \quad \text{So, B obtained 4, 2 and 2 as successive remainders.}$$

$$\text{And,} \quad \begin{array}{r|l} 11 & 1173 \\ \hline 7 & 106 - 7 \\ \hline 13 & 15 - 1 \\ \hline & 1 - 2 \end{array} \quad \text{C obtained 7, 1 and 2 as successive remainders.}$$

Ex. 44. In a division sum, the divisor is ten times the quotient and five times the remainder. If the remainder is 46, determine the dividend.

Sol. Remainder = 46 ; Divisor = $5 \times 46 = 230$; Quotient = $\frac{230}{10} = 23$.

$\therefore$ Dividend = Divisor $\times$ Quotient + Remainder = $230 \times 23 + 46 = 5336$.

Ex. 45. If three times the larger of the two numbers is divided by the smaller one, we get 4 as quotient and 3 as remainder. Also, if seven times the smaller number is divided by the larger one, we get 5 as quotient and 1 as remainder. Find the numbers.

Sol. Let the larger number be x and the smaller number be y .

Then, $3x = 4y + 3 \Rightarrow 3x - 4y = 3$... (i)

And, $7y = 5x + 1 \Rightarrow -5x + 7y = 1$... (ii)

Multiplying (i) by 5 and (ii) by 3, we get:

$15x - 20y = 15$... (iii) and $-15x + 21y = 3$... (iv)

Adding (iii) and (iv), we get: $y = 18$.

Putting $y = 18$ in (i), we get: $x = 25$.

Hence, the numbers are 25 and 18.

Ex. 46. A number when divided by 6 leaves remainder 3. When the square of the same number is divided by 6, find the remainder.

Sol. On dividing the given number by 6, let k be the quotient and 3 the remainder.

Then, number = $6k + 3$.

Square of the number = $(6k + 3)^2 = 36k^2 + 9 + 36k = 36k^2 + 36k + 6 + 3$
 $= 6(6k^2 + 6k + 1) + 3$, which gives a remainder 3 when divided by 6.

Ex. 47. Find the remainder when $9^6 + 7$ is divided by 8.

Sol. $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n .

So, $(9^6 - 1)$ is divisible by $(9 - 1)$, i.e. $8 \Rightarrow (9^6 - 1) + 8$ is divisible by 8 $\Rightarrow (9^6 + 7)$ is divisible by 8.

Hence, required remainder = 0.

Ex. 48. Find the remainder when $(397)^{3589} + 5$ is divided by 398.

Sol. $(x^n + a^n)$ is divisible by $(x + a)$ for all odd values of n .

So, $[(397)^{3589} + 1]$ is divisible by $(397 + 1)$, i.e. 398

$\Rightarrow [(397)^{3589} + 1] + 4$ gives remainder 4 when divided by 398

$\Rightarrow [(397)^{3589} + 5]$ gives remainder 4 when divided by 398.

Ex. 49. If 7^{126} is divided by 48, find the remainder.

Sol. $7^{126} = (7^2)^{63} = (49)^{63}$.

Now, since $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n ,

so $[(49)^{63} - 1]$ or $(7^{126} - 1)$ is divisible by $(49 - 1)$ i.e. 48.

$\therefore$ Remainder obtained when $(7)^{126}$ is divided by 48 = 1.

Ex. 50. Find the remainder when $(257^{166} - 243^{166})$ is divided by 500.

Sol. $(x^n - a^n)$ is divisible by $(x + a)$ for all even values of n .

$\therefore (257^{166} - 243^{166})$ is divisible by $(257 + 243)$, i.e. 500.

Hence, required remainder = 0.

Ex. 51. Find a common factor of $(127^{127} + 97^{127})$ and $(127^{97} + 97^{97})$.

Sol. $(x^n + a^n)$ is divisible by $(x + a)$ for all odd values of n .

$\therefore (127^{127} + 97^{127})$ as well as $(127^{97} + 97^{97})$ is divisible by $(127 + 97)$, i.e. 224.

Hence, required common factor = 224.

Ex. 52. A 99-digit number is formed by writing the first 59 natural numbers one after the other as:

1234567891011121314.....5859

Find the remainder obtained when the above number is divided by 16.

Sol. The required remainder is the same as that obtained on dividing the number formed by the last four digits i.e. 5859 by 16, which is 3.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (□) against the correct answer in each of the following:

1. What is the place value of 5 in 3254710? (CLAT, 2010)
 - (a) 5
 - (b) 10000
 - (c) 50000
 - (d) 54710
2. The face value of 8 in the number 458926 is (R.R.B., 2006)
 - (a) 8
 - (b) 1000
 - (c) 8000
 - (d) 8926
3. The sum of the place values of 3 in the number 503535 is (M.B.A., 2005)
 - (a) 6
 - (b) 60
 - (c) 3030
 - (d) 3300
4. The difference between the place values of 7 and 3 in the number 527435 is
 - (a) 4
 - (b) 5
 - (c) 45
 - (d) 6970
5. The difference between the local value and the face value of 7 in the numeral 32675149 is
 - (a) 5149
 - (b) 64851
 - (c) 69993
 - (d) 75142
 - (e) None of these
6. The sum of the greatest and smallest number of five digits is (M.C.A., 2005)
 - (a) 11,110
 - (b) 10,999
 - (c) 109,999
 - (d) 111,110
7. If the largest three-digit number is subtracted from the smallest five-digit number, then the remainder is
 - (a) 1
 - (b) 9000
 - (c) 9001
 - (d) 90001
8. The smallest number of 5 digits beginning with 3 and ending with 5 will be (R.R.B., 2006)
 - (a) 31005
 - (b) 30015
 - (c) 30005
 - (d) 30025
9. What is the minimum number of four digits formed by using the digits 2, 4, 0, 7? (P.C.S., 2007)
 - (a) 2047
 - (b) 2247
 - (c) 2407
 - (d) 2470
10. All natural numbers and 0 are called the numbers. (R.R.B., 2006)
 - (a) rational
 - (b) integer
 - (c) whole
 - (d) prime
11. Consider the following statements about natural numbers:
 - (1) There exists a smallest natural number.
 - (2) There exists a largest natural number.
 - (3) Between two natural numbers, there is always a natural number.
 Which of the above statements is/are correct?
 - (a) None
 - (b) Only 1
 - (c) 1 and 2
 - (d) 2 and 3
12. Every rational number is also
 - (a) an integer
 - (b) a real number
 - (c) a natural number
 - (d) a whole number
13. The number π is (R.R.B., 2005)
 - (a) a fraction
 - (b) a recurring decimal
 - (c) a rational number
 - (d) an irrational number
14. $\sqrt{2}$ is a/an
 - (a) rational number
 - (b) natural number
 - (c) irrational number
 - (d) integer
15. The number $\sqrt{3}$ is
 - (a) a finite decimal
 - (b) an infinite recurring decimal
 - (c) equal to 1.732
 - (d) an infinite non-recurring decimal
16. There are just two ways in which 5 may be expressed as the sum of two different positive (non-zero) integers, namely $5 = 4 + 1 = 3 + 2$. In how many ways, 9 can be expressed as the sum of two different positive (non-zero) integers?
 - (a) 3
 - (b) 4
 - (c) 5
 - (d) 6
17. P and Q are two positive integers such that $PQ = 64$. Which of the following cannot be the value of $P + Q$?
 - (a) 16
 - (b) 20
 - (c) 35
 - (d) 65
18. If $x + y + z = 9$ and both y and z are positive integers greater than zero, then the maximum value x can take is (Campus Recruitment, 2006)
 - (a) 3
 - (b) 7
 - (c) 8
 - (d) Data insufficient
19. What is the sum of the squares of the digits from 1 to 9?
 - (a) 105
 - (b) 260
 - (c) 285
 - (d) 385
20. If n is an integer between 20 and 80, then any of the following could be $n + 7$ except
 - (a) 47
 - (b) 58
 - (c) 84
 - (d) 88
21. Which one of the following is the correct sequence in respect of the Roman numerals: C, D, L and M ? (Civil Services, 2008)
 - (a) $C > D > L > M$
 - (b) $M > L > D > C$
 - (c) $M > D > C > L$
 - (d) $L > C > D > M$
22. If the numbers from 1 to 24, which are divisible by 2 are arranged in descending order, which number will be at the 8th place from the bottom? (CLAT, 2010)

- (a) 10 (b) 12
(c) 16 (d) 18
23. $2 - 2 + 2 - 2 + \dots$ 101 terms =? (P.C.S., 2008)
(a) -2 (b) 0
(c) 2 (d) None of these
24. 98th term of the infinite series 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, is (M.C.A., 2005)
(a) 1 (b) 2
(c) 3 (d) 4
25. If x, y, z be the digits of a number beginning from the left, the number is
(a) xyz (b) $x + 10y + 100z$
(c) $10x + y + 100z$ (d) $100x + 10y + z$
26. If x, y, z and w be the digits of a number beginning from the left, the number is
(a) $xyzw$
(b) $wzyx$
(c) $x + 10y + 100z + 1000w$
(d) $10^3x + 10^2y + 10z + w$
27. If n and p are both odd numbers, which of the following is an even number?
(a) $n + p$ (b) $n + p + 1$
(c) $np + 2$ (d) np
28. For the integer n , if n^3 is odd, then which of the following statements are true?
I. n is odd. II. n^2 is odd.
III. n^2 is even.
(a) I only (b) II only
(c) I and II only (d) I and III only
29. If $(n - 1)$ is an odd number, what are the two other odd numbers nearest to it?
(a) $n, n - 1$ (b) $n, n - 2$
(c) $n - 3, n + 1$ (d) $n - 3, n + 5$
30. Which of the following is always odd?
(a) Sum of two odd numbers
(b) Difference of two odd numbers
(c) Product of two odd numbers
(d) None of these
31. If x is an odd integer, then which of the following is true?
(a) $5x - 2$ is even (b) $5x^2 + 2$ is odd
(c) $5x^2 + 3$ is odd (d) None of these
32. If a and b are two numbers such that $ab = 0$, then (R.R.B., 2006)
(a) $a = 0$ and $b = 0$ (b) $a = 0$ or $b = 0$ or both
(c) $a = 0$ and $b \neq 0$ (d) $b = 0$ and $a \neq 0$
33. If A, B, C, D are numbers in increasing order and D, B, E are numbers in decreasing order, then which one of the following sequences need neither be in a decreasing nor in an increasing order?
(a) E, C, D (b) E, B, C
(c) D, B, A (d) A, E, C
34. If m, n, o, p and q are integers, then $m(n + o)(p - q)$ must be even when which of the following is even?
(a) m (b) p
(c) $m + n$ (d) $n + p$
35. If n is a negative number, then which of the following is the least?
(a) 0 (b) $-n$
(c) $2n$ (d) n^2
36. If $x - y = 8$, then which of the following must be true?
I. Both x and y are positive.
II. If x is positive, y must be positive.
III. If x is negative, y must be negative.
(a) I only (b) II only
(c) I and II (d) III only
37. If x and y are negative, then which of the following statements is/are always true?
I. $x + y$ is positive.
II. xy is positive.
III. $x - y$ is positive.
(a) I only (b) II only
(c) III only (d) I and III only
38. If $n = 1 + x$, where x is the product of four consecutive positive integers, then which of the following is/are true?
I. n is odd. II. n is prime.
III. n is a perfect square.
(a) I only (b) I and II only
(c) I and III only (d) None of these
39. If $x = \frac{2}{5}y + 3$, how does y change when x increases from 1 to 2?
(a) y increases from -5 to $-\frac{5}{2}$
(b) y increases from $\frac{2}{5}$ to 5
(c) y increases from $\frac{5}{2}$ to 5
(d) y decreases from -5 to $-\frac{5}{2}$
40. If x is a rational number and y is an irrational number, then
(a) both $x + y$ and xy are necessarily rational
(b) both $x + y$ and xy are necessarily irrational
(c) xy is necessarily irrational, but $x + y$ can be either rational or irrational
(d) $x + y$ is necessarily irrational, but xy can be either rational or irrational
41. The difference between the square of any two consecutive integers is equal to
(a) sum of two numbers
(b) difference of two numbers
(c) an even number
(d) product of two numbers

42. Between two distinct rational numbers a and b , there exists another rational number which is (P.C.S., 2006)
- (a) $\frac{a}{2}$ (b) $\frac{b}{2}$
 (c) $\frac{ab}{2}$ (d) $\frac{a+b}{2}$
43. If $B > A$, then which expression will have the highest value (given that A and B are positive integers)? (Campus Recruitment, 2007)
- (a) $A - B$ (b) AB
 (c) $A + B$ (d) Can't say
44. If $0 < x < 1$, which of the following is greatest? (Campus Recruitment, 2007)
- (a) x (b) x^2
 (c) $\frac{1}{x}$ (d) $\frac{1}{x^2}$
45. If p is a positive fraction less than 1, then
- (a) $\frac{1}{p}$ is less than 1 (b) $\frac{1}{p}$ is a positive integer
 (c) p^2 is less than p
 (d) $\frac{2}{p} - p$ is a positive number
46. If x is a real number, then $x^2 + x + 1$ is
- (a) less than $\frac{3}{4}$
 (b) zero for at least one value of x
 (c) always negative
 (d) greater than or equal to $\frac{3}{4}$
47. Let n be a natural number such that $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{n}$ is also a natural number. Which of the following statements is not true? (A.A.O. Exam, 2009)
- (a) 2 divides n (b) 3 divides n
 (c) 7 divides n (d) $n > 84$
48. If n is an integer, how many values of n will give an integral value of $\left(\frac{16n^2 + 7n + 6}{n}\right)$?
- (a) 2 (b) 3
 (c) 4 (d) None of these
49. If $p > q$ and $r < 0$, then which is true?
- (a) $pr < qr$ (b) $p - r < q - r$
 (c) $p + r < q + r$ (d) None of these
50. If $X < Z$ and $X < Y$, which of the following is necessarily true?
- I. $Y < Z$ II. $X^2 < YZ$
 III. $ZX < Y + Z$
 (a) Only I (b) Only II
 (c) Only III (d) None of these
51. In the relation $x > y + z$, $x + y > p$ and $z < p$, which of the following is necessarily true? (Campus Recruitment, 2008)
- (a) $y > p$ (b) $x + y > z$
 (c) $y + p > x$ (d) Insufficient data
52. If a and b are positive integers and $\frac{(a-b)}{3.5} = \frac{4}{7}$, then (Campus Recruitment, 2010)
- (a) $b > a$ (b) $b < a$
 (c) $b = a$ (d) $b \geq a$
53. If $13 = \frac{13w}{(1-w)}$, then $(2w)^2 = ?$ (Campus Recruitment, 2009)
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) 1 (d) 2
- Directions (Questions 54–57):** For a 5-digit number, without repetition of digits, the following information is available. (B.B.A., 2006)
- (i) The first digit is more than 5 times the last digit.
 (ii) The two-digit number formed by the last two digits is the product of two prime numbers.
 (iii) The first three digits are all odd.
 (iv) The number does not contain the digits 3 or 0 and the first digit is also the largest.
54. The second digit of the number is
- (a) 5 (b) 7
 (c) 9 (d) Cannot be determined
55. The last digit of the number is
- (a) 0 (b) 1
 (c) 2 (d) 3
56. The largest digit in the number is
- (a) 5 (b) 7
 (c) 8 (d) 9
57. Which of the following is a factor of the given number?
- (a) 2 (b) 3
 (c) 4 (d) 9
58. The least prime number is
- (a) 0 (b) 1
 (c) 2 (d) 3
59. Consider the following statements:
1. If x and y are composite numbers, then $x + y$ is always composite.
 2. There does not exist a natural number which is neither prime nor composite.
 Which of the above statements is/are correct?
- (a) 1 only (b) 2 only
 (c) Both 1 and 2 (d) Neither 1 nor 2
60. The number of prime numbers between 0 and 50 is
- (a) 14 (b) 15
 (c) 16 (d) 17

61. The prime numbers dividing 143 and leaving a remainder of 3 in each case are
 (a) 2 and 11 (b) 11 and 13
 (c) 3 and 7 (d) 5 and 7
62. The sum of the first four primes is
 (a) 10 (b) 11
 (c) 16 (d) 17
63. The sum of all the prime numbers from 1 to 20 is
 (a) 75 (b) 76
 (c) 77 (d) 78
64. A prime number N , in the range 10 to 50, remains unchanged when its digits are reversed. The square of such a number is
 (a) 121 (b) 484
 (c) 1089 (d) 1936
65. The remainder obtained when any prime number greater than 6 is divided by 6 must be
 (a) either 1 or 2 (b) either 1 or 3
 (c) either 1 or 5 (d) either 3 or 5
 (Campus Recruitment, 2007)
66. Which of the following is not a prime number?
 (a) 21 (b) 23
 (c) 29 (d) 43
 (CLAT, 2010)
67. Which of the following is a prime number?
 (a) 19 (b) 20
 (c) 21 (d) 22
 (CLAT, 2010)
68. Which of the following is a prime number?
 (a) 115 (b) 119
 (c) 127 (d) None of these
 (Campus Recruitment, 2008)
69. Which of the following is a prime number?
 (a) 143 (b) 289
 (c) 117 (d) 359
 (R.R.B., 2006)
70. The smallest value of natural number n , for which $2n + 1$ is not a prime number, is
 (a) 3 (b) 4
 (c) 5 (d) None of these
71. The smallest three-digit prime number is
 (a) 101 (b) 103
 (c) 107 (d) None of these
72. How many of the integers between 110 and 120 are prime numbers?
 (a) 0 (b) 1
 (c) 2 (d) 3
 (M.B.A., 2006)
73. Four prime numbers are arranged in ascending order. The product of first three is 385 and that of last three is 1001. The largest prime number is
 (a) 9 (b) 11
 (c) 13 (d) 17
 (R.R.B., 2006)
74. The sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the numbers is
 (a) 7 (b) 29
 (c) 41 (d) 67
75. Which one of the following is a prime number?
 (a) 161 (b) 221
 (c) 373 (d) 437
76. The smallest prime number, that is the fifth term of an increasing arithmetic sequence in which all the four preceding terms are also prime, is
 (a) 17 (b) 29
 (c) 37 (d) 53
77. The number of prime numbers between 301 and 320 are
 (a) 3 (b) 4
 (c) 5 (d) 6
78. Consider the following statements:
 1. If $p > 2$ is a prime, then it can be written as $4n + 1$ or $4n + 3$ for a suitable natural number n .
 2. If $p > 2$ is a prime, then $(p - 1)(p + 1)$ is always divisible by 4.
 Of these statements,
 (a) (1) is true but (2) is false
 (b) (1) is false but (2) is true
 (c) (1) and (2) are false
 (d) (1) and (2) are true
79. What is the first value of n for which $n^2 + n + 41$ is not a prime?
 (a) 1 (b) 10
 (c) 20 (d) 40
80. Let $X_k = (p_1 p_2 \dots p_k) + 1$, where $p_1, p_2, \dots, p_k$ are the first k primes.
 Consider the following:
 1. X_k is a prime number.
 2. X_k is a composite number.
 3. $X_k + 1$ is always an even number.
 Which of the above is/are correct?
 (a) 1 only (b) 2 only
 (c) 3 only (d) 1 and 3
81. $6 \times 3 (3 - 1)$ is equal to
 (a) 19 (b) 20
 (c) 36 (d) 53
 (CLAT, 2010)
82. $1234 + 2345 - 3456 + 4567 = ?$ (Bank Recruitment, 2010)
 (a) 4590 (b) 4670
 (c) 4680 (d) 4690
 (e) None of these
83. $5566 - 7788 + 9988 = ? + 4444$ (Bank Recruitment, 2010)
 (a) 3223 (b) 3232
 (c) 3322 (d) 3333
 (e) None of these

84. $38649 - 1624 - 4483 = ?$ (Bank Recruitment, 2009)
 (a) 32425 (b) 32452
 (c) 34522 (d) 35422
 (e) None of these
85. $884697 - 773697 - 102479 = ?$ (Bank Recruitment, 2009)
 (a) 8251 (b) 8512
 (c) 8521 (d) 8531
 (e) None of these
86. $10531 + 4813 - 728 = ? \times 87$ (Bank Recruitment, 2008)
 (a) 168 (b) 172
 (c) 186 (d) 212
 (e) None of these
87. What is 394 times 113?
 (a) 44402 (b) 44522
 (c) 44632 (d) 44802
 (e) None of these
88. $1260 \div 14 \div 9 = ?$ (Bank P.O., 2009)
 (a) 9 (b) 10
 (c) 81 (d) 810
 (e) None of these
89. $136 \times 12 \times 8 = ?$ (Bank P.O., 2009)
 (a) 12066 (b) 13046
 (c) 13064 (d) 13066
 (e) None of these
90. $8888 + 848 + 88 - ? = 7337 + 737$ (Bank P.O., 2009)
 (a) 1450 (b) 1550
 (c) 1650 (d) 1750
 (e) None of these
91. $414 \times ? \times 7 = 127512$ (Bank P.O., 2009)
 (a) 36 (b) 40
 (c) 44 (d) 48
 (e) None of these
92. Product of 82540027 and 43253 is
 (a) 3570103787831 (b) 3570103787832
 (c) 3570103787833 (d) 3570103787834
93. $(46351 - 36418 - 4505) \div ? = 1357$ (Bank P.O., 2009)
 (a) 2 (b) 3
 (c) 4 (d) 6
 (e) None of these
94. $6 \times 66 \times 666 = ?$ (Bank Recruitment, 2007)
 (a) 263376 (b) 263763
 (c) 263736 (d) 267336
 (e) None of these
95. If you subtract -1 from $+1$, what will be the result?
 (a) -2 (b) 0
 (c) 1 (d) 2 (R.R.B., 2006)
96. $8 + 88 + 888 + 8888 + 88888 + 888888 = ?$
 (a) 897648 (b) 896748
 (c) 986748 (d) 987648
 (e) None of these
97. From the sum of 17 and -12 , subtract 48. (E.S.I.C., 2006)
 (a) -43 (b) -48
 (c) -17 (d) -20
98. $60840 \div 234 = ?$
 (a) 225 (b) 255
 (c) 260 (d) 310
 (e) None of these
99. $3578 + 5729 - ? \times 581 = 5821$
 (a) 3 (b) 4
 (c) 6 (d) None of these
100. $-95 \div 19 = ?$
 (a) -5 (b) -4
 (c) 0 (d) 5
101. 12345679×72 is equal to
 (a) 888888888 (b) 888888888
 (c) 898989898 (d) 999999998
102. $8899 - 6644 - 3322 = ? - 1122$
 (a) 55 (b) 65
 (c) 75 (d) 85
 (e) None of these
103. $74844 \div ? = 54 \times 63$ (Bank P.O., 2009)
 (a) 22 (b) 34
 (c) 42 (d) 54
 (e) None of these
104. $1256 \times 3892 = ?$
 (a) 4883852 (b) 4888532
 (c) 4888352 (d) 4883582
 (e) None of these
105. What is 786 times 964? (Bank P.O., 2008)
 (a) 757704 (b) 754164
 (c) 759276 (d) 749844
 (e) None of these
106. What is 348 times 265? (S.B.I.P.O., 2008)
 (a) 88740 (b) 89750
 (c) 92220 (d) 95700
 (e) None of these
107. $(71 \times 29 + 27 \times 15 + 8 \times 4)$ equals (S.S.C., 2007)
 (a) 2496 (b) 3450
 (c) 3458 (d) None of these
108. $? \times (|a| \times |b|) = -ab$
 (a) 0 (b) -1
 (c) 1 (d) None of these
109. $(46)^2 - (?)^2 = 4398 - 3066$
 (a) 16 (b) 28
 (c) 36 (d) 42
 (e) None of these
110. $(800 \div 64) \times (1296 \div 36) = ?$
 (a) 420 (b) 460
 (c) 500 (d) 540
 (e) None of these

- 111.** $5358 \times 51 = ?$
 (a) 273258 (b) 273268
 (c) 273348 (d) 273358
- 112.** $587 \times 999 = ?$
 (a) 586413 (b) 587523
 (c) 614823 (d) 615173
- 113.** $3897 \times 999 = ?$
 (a) 3883203 (b) 3893103
 (c) 3639403 (d) 3791203
 (e) None of these
- 114.** $72519 \times 9999 = ?$
 (a) 725117481 (b) 674217481
 (c) 685126481 (d) 696217481
 (e) None of these
- 115.** $2056 \times 987 = ?$
 (a) 1936372 (b) 2029272
 (c) 1896172 (d) 1923472
 (e) None of these
- 116.** $1904 \times 1904 = ?$
 (a) 3654316 (b) 3632646
 (c) 3625216 (d) 3623436
 (e) None of these
- 117.** $1397 \times 1397 = ?$
 (a) 1951609 (b) 1981709
 (c) 18362619 (d) 2031719
 (e) None of these
- 118.** $107 \times 107 + 93 \times 93 = ?$
 (a) 19578 (b) 19418
 (c) 20098 (d) 21908
 (e) None of these
- 119.** $217 \times 217 + 183 \times 183 = ?$ (R.R.B., 2007)
 (a) 79698 (b) 80578
 (c) 80698 (d) 81268
 (e) None of these
- 120.** $106 \times 106 - 94 \times 94 = ?$
 (a) 2400 (b) 2000
 (c) 1904 (d) 1906
 (e) None of these
- 121.** $8796 \times 223 + 8796 \times 77 = ?$
 (a) 2736900 (b) 2738800
 (c) 2658560 (d) 2716740
 (e) None of these
- 122.** $287 \times 287 + 269 \times 269 - 2 \times 287 \times 269 = ?$
 (a) 534 (b) 446
 (c) 354 (d) 324
 (e) None of these
- 123.** $\{(476 + 424)^2 - 4 \times 476 \times 424\} = ?$
 (a) 2906 (b) 3116
 (c) 2704 (d) 2904
 (e) None of these
- 124.** The value of 112×5^4 is
 (a) 6700 (b) 70000
 (c) 76500 (d) 77200
- 125.** Multiply 5746320819 by 125.
 (a) 718,290,102,375 (b) 728,490,301,375
 (c) 748,290,103,375 (d) 798,290,102,975
- 126.** $935421 \times 625 = ?$
 (a) 575648125 (b) 584638125
 (c) 584649125 (d) 585628125
- 127.** $(999)^2 - (998)^2 = ?$ (R.R.B., 2008)
 (a) 1992 (b) 1995
 (c) 1997 (d) 1998
- 128.** $(80)^2 - (65)^2 + 81 = ?$
 (a) 306 (b) 2094
 (c) 2175 (d) 2256
 (e) None of these
- 129.** $(24 + 25 + 26)^2 - (10 + 20 + 25)^2 = ?$
 (a) 352 (b) 400
 (c) 752 (d) 2600
 (e) None of these
- 130.** $(65)^2 - (55)^2 = ?$
 (a) 10 (b) 100
 (c) 120 (d) 1200
- 131.** If a and b be positive integers such that $a^2 - b^2 = 19$, then the value of a is (S.S.C., 2010)
 (a) 9 (b) 10
 (c) 19 (d) 20
- 132.** If a and b are positive integers, $a > b$ and $(a + b)^2 - (a - b)^2 > 29$, then the smallest value of a is
 (a) 3 (b) 4
 (c) 6 (d) 7
- 133.** $397 \times 397 + 104 \times 104 + 2 \times 397 \times 104 = ?$
 (a) 250001 (b) 251001
 (c) 260101 (d) 261001
- 134.** If $(64)^2 - (36)^2 = 20 \times x$, then $x = ?$
 (a) 70 (b) 120
 (c) 180 (d) 140
 (e) None of these
- 135.** $\frac{(489 + 375)^2 - (489 - 375)^2}{(489 \times 375)} = ?$
 (a) 144 (b) 864
 (c) 2 (d) 4
 (e) None of these
- 136.** $\frac{(963 + 476)^2 + (963 - 476)^2}{(963 \times 963 + 476 \times 476)} = ?$
 (a) 2 (b) 4

- (c) 497 (d) 1449
(e) None of these
137. $\frac{768 \times 768 \times 768 + 232 \times 232 \times 232}{786 \times 768 - 768 \times 232 + 232 \times 232} = ?$
(a) 1000 (b) 536
(c) 500 (d) 268
(e) None of these
138. $\frac{854 \times 854 \times 854 - 276 \times 276 \times 276}{854 \times 854 + 854 \times 276 + 276 \times 276} = ?$
(a) 1130 (b) 578
(c) 565 (d) 1156
(e) None of these
139. $\frac{753 \times 753 + 247 \times 247 - 753 \times 247}{753 \times 753 \times 753 + 247 \times 247 \times 247} = ?$
(a) $\frac{1}{1000}$ (b) $\frac{1}{506}$
(c) $\frac{253}{500}$ (d) None of these
140. $\frac{256 \times 256 - 144 \times 144}{112}$ is equal to (S.S.C., 2010)
(a) 420 (b) 400
(c) 360 (d) 320
141. If $a = 11$ and $b = 9$, then the value of $\left(\frac{a^2 + b^2 + ab}{a^3 - b^3} \right)$ is (S.S.C., 2010)
(a) $\frac{1}{2}$ (b) $\frac{1}{20}$
(c) 2 (d) 20
142. If $a + b + c = 0$, $(a + b)(b + c)(c + a)$ equals (M.C.A., 2005)
(a) $ab(a + b)$ (b) $(a + b + c)^2$
(c) $-abc$ (d) $a^2 + b^2 + c^2$
143. If $a = 7$, $b = 5$, $c = 3$, then the value of $a^2 + b^2 + c^2 - ab - bc - ca$ is
(a) -12 (b) 0
(c) 8 (d) 12
144. Both addition and multiplication of numbers are operations which are
(a) neither commutative nor associative
(b) associative but not commutative
(c) commutative but not associative
(d) commutative and associative
145. Which of the following digits will replace the H marks in the following equation?
 $9H + H8 + H6 = 230$
(a) 3 (b) 4
(c) 5 (d) 9
(e) None of these

146. Find the missing number in the following addition problem:

$$\begin{array}{r} 8 \quad 3 \quad 5 \\ 4 \quad * \quad 8 \\ + \quad 9 \quad * \quad 4 \\ \hline 2 \quad 2 \quad * \quad 7 \end{array}$$

- (a) 0 (b) 4
(c) 6 (d) 9
147. What number should replace M in this multiplication problem?

$$\begin{array}{r} 3 \quad M \quad 4 \\ \times \quad 4 \\ \hline 1 \quad 2 \quad 1 \quad 6 \end{array}$$

- (a) 0 (b) 2
(c) 4 (d) 8
148. If p and q represent digits, what is the maximum possible value of q in the statement (S.S.C., 2010)
 $5p9 + 327 + 2q8 = 1114$?
(a) 6 (b) 7
(c) 8 (d) 9
149. What would be the maximum value of Q in the following equation?
 $5P7 + 8Q9 + R32 = 1928$
(a) 6 (b) 8
(c) 9 (d) Data inadequate
(e) None of these
150. What should come in place of * mark in the following equation?
 $1*5\$4 \div 148 = 78$
(a) 1 (b) 4
(c) 6 (d) 8
(e) None of these
151. If $6*43 - 46@9 = 1904$, which of the following should come in place of *?
(a) 4 (b) 6
(c) 9 (d) Cannot be determined
(e) None of these
152. What should be the maximum value of Q in the following equation?
 $5P9 - 7Q2 + 9R6 = 823$
(a) 5 (b) 6
(c) 7 (d) 9
(e) None of these
153. In the following sum, '?' stands for which digit?
 $? + 1? + 2? + ?3 + ?1 = 21?$
(a) 4 (b) 6
(c) 8 (d) 9
(e) None of these

Directions (Questions 154–155): These questions are based on the following information:

$CBA + CCA = ACD$, where A , B , C and D stand for distinct digits and $D = 0$.

154. B takes the value
 (a) 0 (b) 5
 (c) 9 (d) 0 or 9
155. C takes the value
 (a) 0 (b) 2
 (c) 2 or 3 (d) 5
156. A 3-digit number $4a3$ is added to another 3-digit number 984 to give the four-digit number $13b7$, which is divisible by 11. Then, $(a + b)$ is (M.B.A., 2006)
 (a) 10 (b) 11
 (c) 12 (d) 15
157. If $ab \overline{)252} \overline{)ba}$, the values of a and b are (I.A.M., 2007)
- $$\begin{array}{r} 24 \\ 12 \\ \underline{12} \\ \times \end{array}$$
- (a) 1, 2 (b) 2, 3
 (c) 1, 3 (d) None of these
- 158.
- $$\begin{array}{r} * * * \\ \times * \\ \hline 8 * * 1 \end{array}$$
- In the above multiplication problem, $*$ is equal to
 (a) 1 (b) 3
 (c) 7 (d) 9
159. If $*$ means adding 6 times the second number to the first number, then $(1 * 2) * 3$ equals
 (a) 21 (b) 31
 (c) 91 (d) 93
160. If $1 \times 2 \times 3 \times \dots \times n$ is denoted by $\underline{n}$, then $\underline{8} - \underline{7} - \underline{6}$ is equal to
 (a) $6 \times 7 \times \underline{8}$ (b) $7 \times 8 \times \underline{7}$
 (c) $6 \times 8 \times \underline{6}$ (d) $7 \times 8 \times \underline{6}$
161. The highest power of 9 dividing $99!$ completely is
 (a) 11 (b) 20
 (c) 22 (d) 24
162. For an integer n , $n! = n(n-1)(n-2)\dots\dots\dots 3.2.1$.
 (P.C.S., 2008)
 Then, $1! + 2! + 3! + \dots\dots\dots + 100!$ when divided by 5 leaves remainder
 (a) 0 (b) 1
 (c) 2 (d) 3
163. The number of prime factors in the expression $6^{10} \cdot 7^{17} \cdot 11^{27}$ is equal to
 (a) 54 (b) 64
 (c) 71 (d) 81
164. What is the number of prime factors contained in the product $30^7 \times 22^5 \times 34^{11}$?
 (a) 49 (b) 51
 (c) 52 (d) 53
165. What number multiplied by 48 will give the same product as 173 multiplied by 240?

- (a) 495 (b) 545
 (c) 685 (d) 865
166. A positive number, which when added to 1000, gives a sum which is greater than when it is multiplied by 1000. This positive integer is
 (a) 1 (b) 3
 (c) 5 (d) 7
167. 7 is added to a certain number ; the sum is multiplied by 5 ; the product is divided by 9 and 3 is subtracted from the quotient. Thus, if the remainder left is 12, what was the original number? (S.S.C., 2005)
 (a) 20 (b) 30
 (c) 40 (d) 60
168. Symbiosis runs a Corporate Training Programme. At the end of running the first programme, its total takings were ₹ 38950. There were more than 45 but less than 100 participants. What was the participant fee for the programme? (SNAP, 2005)
 (a) ₹ 410 (b) ₹ 450
 (c) ₹ 500 (d) ₹ 510
169. The sum of four consecutive even numbers A, B, C and D is 180. What is the sum of the set of next four consecutive even numbers? (Bank Recruitment, 2008)
 (a) 196 (b) 204
 (c) 212 (d) 214
 (e) None of these
170. A young girl counted in the following way on the fingers of her left hand. She started calling the thumb 1, the index finger 2, middle finger 3, ring finger 4, little finger 5, then reversed direction, calling the ring finger 6, middle finger 7, index finger 8, thumb 9 and then back to the index figure for 10, middle finger for 11, and so on. She counted upto 1994. She ended on her
 (a) thumb (b) index finger
 (c) middle finger (d) ring finger
171. Given $n = 1 + x$ and x is the product of four consecutive integers. Then which of the following is true?
 I. n is an odd integer. II. n is prime.
 III. n is a perfect square.
 (a) Only I is correct
 (b) Only III is correct
 (c) Both I and II are correct
 (d) Both I and III are correct
172. If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$? (L.I.C.A.D.O., 2007)
 (a) 110 (b) 113
 (c) 121 (d) Cannot be determined
 (e) None of these
173. Given that $(1^2 + 2^2 + 3^2 + \dots + 20^2) = 2870$, the value of $(2^2 + 4^2 + 6^2 + \dots + 40^2)$ is
 (a) 2870 (b) 5740
 (c) 11480 (d) 28700

- 174.** The value of $5^2 + 6^2 + \dots + 10^2 + 20^2$ is
 (a) 755 (b) 760
 (c) 765 (d) 770
- 175.** Given that $1 + 2 + 3 + 4 + \dots + 10 = 55$, then the sum $6 + 12 + 18 + 24 + \dots + 60$ is equal to
 (a) 300 (b) 330
 (c) 455 (d) 655
- 176.** If m and n are natural numbers such that $2^m - 2^n = 960$, what is the value of m ? (M.A.T., 2007)
 (a) 10 (b) 12
 (c) 15 (d) Cannot be determined
- 177.** On multiplying a number by 7, all the digits in the product appear as 3's. The smallest such number is (C.P.O., 2006)
 (a) 47619 (b) 46719
 (c) 48619 (d) 47649
- 178.** The number of digits in the smallest number, which when multiplied by 7 yields all nines, is
 (a) 3 (b) 4
 (c) 5 (d) 6
- 179.** A boy multiplies 987 by a certain number and obtains 559981 as his answer. If in the answer both 9's are wrong but the other digits are correct, then the correct answer will be
 (a) 553681 (b) 555181
 (c) 555681 (d) 556581
- 180.** The numbers 1, 3, 5,, 25 are multiplied together. The number of zeros at the right end of the product is (R.R.B., 2006)
 (a) 0 (b) 1
 (c) 2 (d) 3
- 181.** The numbers 1, 2, 3, 4,, 1000 are multiplied together. The number of zeros at the end (on the right) of the product must be
 (a) 30 (b) 200
 (c) 211 (d) 249
- 182.** First 100 multiples of 10 i.e. 10, 20, 30,, 1000 are multiplied together. The number of zeros at the end of the product will be
 (a) 100 (b) 111
 (c) 124 (d) 125
- 183.** The number of zeros at the end of the product $5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$ is
 (a) 5 (b) 7
 (c) 8 (d) 10
- 184.** The number of zeros at the end of $60!$ is
 (a) 12 (b) 14
 (c) 16 (d) 18
- 185.** The numbers 1, 3, 5, 7,, 99 and 128 are multiplied together. The number of zeros at the end of the product must be
 (a) Nil (b) 7
 (c) 19 (d) 22
- 186.** The numbers 2, 4, 6, 8,, 98, 100 are multiplied together. The number of zeros at the end of the product must be
 (a) 10 (b) 11
 (c) 12 (d) 13
- 187.** Let S be the set of prime numbers greater than or equal to 2 and less than 100. Multiply all the elements of S . With how many consecutive zeros will the product end?
 (a) 1 (b) 4
 (c) 5 (d) 10
- 188.** Find the number of zeros at the end of the result $3 \times 6 \times 9 \times 12 \times 15 \times \dots \times 99 \times 102$.
 (a) 4 (b) 6
 (c) 7 (d) 10
- 189.** The unit's digit of 13^{2003} is (A.A.O. Exam, 2010)
 (a) 1 (b) 3
 (c) 7 (d) 9
- 190.** The digit in the unit's place of the number 123^{99} is (I.A.M., 2007)
 (a) 1 (b) 4
 (c) 7 (d) 8
- 191.** Match List I with List II and select the correct answer:
- | List I
(Product) | List II
(Digit in the unit's place) |
|---------------------|--|
| A. $(1827)^{16}$ | (1) 1 |
| B. $(2153)^{19}$ | (2) 3 |
| C. $(5129)^{21}$ | (3) 5 |
| | (4) 7 |
| | (5) 9 |
| A B C | A B C |
| (a) 1 4 3 | (b) 4 2 3 |
| A B C | A B C |
| (c) 1 4 5 | (d) 4 2 5 |
- 192.** The digit in the unit's place of the number $(67)^{25} - 1$ must be
 (a) 0 (b) 6
 (c) 8 (d) None of these
- 193.** The unit's digit in the product $274 \times 318 \times 577 \times 313$ is
 (a) 2 (b) 3
 (c) 4 (d) 5
- 194.** In the product $459 \times 46 \times 28^* \times 484$, the digit in the unit place is 8. The digit to come in place of * is
 (a) 3 (b) 5
 (c) 7 (d) None of these
- 195.** The digit in the unit place of the number represented by $(7^{95} - 3^{58})$ is
 (a) 0 (b) 4
 (c) 6 (d) 7
- 196.** Unit's digit in $(784)^{126} + (784)^{127}$ is
 (a) 0 (b) 4
 (c) 6 (d) 8

- 197.** The digit in the unit's place of $[(251)^{98} + (21)^{29} - (106)^{100} + (705)^{35} - 16^4 + 259]$ is
 (a) 1 (b) 4
 (c) 5 (d) 6
- 198.** The digit in the unit's place of the product $(2464)^{1793} \times (615)^{317} \times (131)^{491}$ is
 (a) 0 (b) 2
 (c) 3 (d) 5
- 199.** If x is an even number, then x^{4n} , where n is a positive integer, will always have
 (a) zero in the unit's place
 (b) 6 in the unit's place
 (c) either 0 or 6 in the unit's place
 (d) None of these
- 200.** If m and n are positive integers, then the digit in the unit's place of $5^n + 6^m$ is always
 (a) 1 (b) 5
 (c) 6 (d) $n + m$
- 201.** The number formed from the last two digits (ones and tens) of the expression $2^{12n} - 6^{4n}$, where n is any positive integer is (S.S.C., 2005)
 (a) 10 (b) 00
 (c) 30 (d) 02
- 202.** The last digit in the decimal representation of $\left(\frac{1}{5}\right)^{2000}$ is (Hotel Management, 2009)
 (a) 2 (b) 4
 (c) 5 (d) 6
- 203.** Let x be the product of two numbers 3,659,893,456,789,325,678 and 342,973,489,379,256. The number of digits in x is (A.A.O., 2010)
 (a) 32 (b) 34
 (c) 35 (d) 36
- 204.** Let a number of three digits have for its middle digit the sum of the other two digits. Then it is a multiple of (C.P.F., 2008)
 (a) 10 (b) 11
 (c) 18 (d) 50
- 205.** What least value must be given to n so that the number 6135n2 becomes divisible by 9? (L.I.C.A.D.O., 2008)
 (a) 1 (b) 2
 (c) 3 (d) 4
- 206.** Find the multiple of 11 in the following numbers. (R.R.B., 2006)
 (a) 112144 (b) 447355
 (c) 869756 (d) 978626
- 207.** 111,111,111,111 is divisible by
 (a) 3 and 37 only
 (b) 3, 11 and 37 only
 (c) 3, 11, 37 and 111 only
 (d) 3, 11, 37, 111 and 1001
- 208.** Which of the following numbers is not divisible by 18?
 (a) 34056 (b) 50436
 (c) 54036 (d) 65043
- 209.** The number 89715938* is divisible by 4. The unknown non-zero digit marked as * will be
 (a) 2 (b) 3
 (c) 4 (d) 6
- 210.** Which one of the following numbers is divisible by 3?
 (a) 4006020 (b) 2345678
 (c) 2876423 (d) 9566003
- 211.** A number is divisible by 11 if the difference between the sums of the digits in odd and even places respectively is
 (a) a multiple of 3
 (b) a multiple of 5
 (c) zero or a multiple of 7
 (d) zero or a multiple of 11
- 212.** Which one of the following numbers is divisible by 11?
 (a) 4823718 (b) 4832718
 (c) 8423718 (d) 8432718
- 213.** Which one of the following numbers is divisible by 15?
 (a) 17325 (b) 23755
 (c) 29515 (d) 30560
- 214.** 7386038 is divisible by
 (a) 3 (b) 4
 (c) 9 (d) 11
- 215.** Consider the following statements:
 The numbers 24984, 26784 and 28584 are
 (1) divisible by 3 (2) divisible by 4
 (3) divisible by 9
 Which of these are correct?
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) 1, 2 and 3
- 216.** Which of the following numbers is a multiple of 8?
 (a) 923872 (b) 923972
 (c) 923862 (d) 923962
- 217.** If 78*3945 is divisible by 11, where * is a digit, then * is equal to
 (a) 0 (b) 1
 (c) 3 (d) 5
- 218.** If m and n are integers divisible by 5, which of the following is not necessarily true?
 (a) $m + n$ is divisible by 10
 (b) $m - n$ is divisible by 5
 (c) $m^2 - n^2$ is divisible by 25
 (d) None of these
- 219.** An integer is divisible by 16 if and only if its last X digits are divisible by 16. The value of X would be

- (a) 3 (b) 4
(c) 5 (d) 6
- 220.** Which of the following numbers is divisible by 3, 7, 9 and 11?
(a) 639 (b) 2079
(c) 3791 (d) 37911
- 221.** A number $476^{**}0$ is divisible by both 3 and 11. The non-zero digits in the hundred's and ten's place respectively are
(a) 7, 4 (b) 5, 3
(c) 5, 2 (d) None of these
- 222.** How many of the following numbers are divisible by 3 but not by 9?
2133, 2343, 3474, 4131, 5286, 5340, 6336, 7347, 8115, 9276
(a) 5 (b) 6
(c) 7 (d) None of these
- 223.** If the number $357^{*}25^{*}$ is divisible by both 3 and 5, then the missing digits in the unit's place and the thousandth's place respectively are
(a) 0, 6 (b) 5, 1
(c) 5, 4 (d) None of these
- 224.** 6897 is divisible by
(a) 11 only (b) 19 only
(c) both 11 and 19 (d) neither 11 nor 19
- 225.** Which of the following numbers is exactly divisible by 24?
(a) 35718 (b) 63810
(c) 537804 (d) 3125736
- 226.** The number 311311311311311311 is
(a) neither divisible by 3 nor by 11
(b) divisible by 11 but not by 3
(c) divisible by 3 but not by 11
(d) divisible by both 3 and 11
- 227.** 325325 is a six-digit number. It is divisible by
(a) 7 only (b) 11 only
(c) 13 only (d) all 7, 11 and 13
- 228.** If the seven-figure number $30X0103$ is a multiple of 13, then X is
(a) 1 (b) 6
(c) 7 (d) 8
- 229.** If a number is divisible by both 11 and 13, then it must be necessarily
(a) 429 (b) divisible by (11×13)
(c) divisible by $(11 + 13)$ (d) divisible by $(13 - 11)$
- 230.** Which of the following numbers are completely divisible by 7?
I. 195195 II. 181181
III. 120120 IV. 891891
(a) Only I and II (b) Only II and III
(c) Only I and IV (d) Only II and IV
(e) All are divisible
- 231.** If x and y are two digits of the number $653xy$ such that the number is divisible by 80, then $x + y$ is equal to
(a) 3 (b) 4
(c) 5 (d) 6
- 232.** The six-digit number $5ABB7A$ is a multiple of 33 for non-zero digits A and B . Which of the following could be possible value of $A + B$? (A.A.O. Exam, 2010)
(a) 8 (b) 9
(c) 10 (d) 14
- 233.** Which of the following numbers is divisible by 99?
(a) 114345 (b) 913464
(c) 135792 (d) 3572404
- 234.** The digits indicated by $*$ in 3422213^{**} so that this number is divisible by 99 are (R.R.B., 2010)
(a) 1, 9 (b) 3, 7
(c) 4, 6 (d) 5, 5
- 235.** If $37X3$ is a four-digit natural number divisible by 7, then the place marked as X must have the value
(a) 0 (b) 3
(c) 5 (d) 9
- 236.** If the seven-digit number $876p37q$ is divisible by 225, then the values of p and q respectively are
(a) 0 and 0 (b) 9 and 0
(c) 0 and 5 (d) 9 and 5
- 237.** If a number $774958A96B$ is divisible by 8 and 9, the respective values of A and B will be
(a) 5 and 8 (b) 7 and 8
(c) 8 and 0 (d) None of these
- 238.** How many of the following numbers are divisible by 132?
264, 396, 462, 792, 968, 2178, 5184, 6336
(a) 4 (b) 5
(c) 6 (d) 7
- 239.** If x and y are positive integers such that $(3x + 7y)$ is a multiple of 11, then which of the following is also a multiple of 11?
(a) $5x - 3y$ (b) $9x + 4y$
(c) $4x + 6y$ (d) $x + y + 6$
- 240.** If n be any natural number then by which largest number $(n^3 - n)$ is always divisible? (S.S.C., 2010)
(a) 3 (b) 6
(c) 12 (d) 18
- 241.** If a and b are two odd positive integers, by which of the following integers is $(a^4 - b^4)$ always divisible? (S.S.C., 2010)
(a) 3 (b) 6
(c) 8 (d) 12
- 242.** The difference between the squares of any two consecutive integers is equal to
(a) an even number
(b) difference of given numbers
(c) sum of given numbers
(d) product of given numbers

- 243.** The number $6n^2 + 6n$ for natural number n is always divisible by (M.A.T., 2007)
 (a) 6 only (b) 6 and 12
 (c) 12 only (d) 18 only
- 244.** The difference of a number consisting of two digits and the number formed by interchanging the digits is always divisible by
 (a) 5 (b) 7
 (c) 9 (d) 11
- 245.** The sum of a number consisting of two digits and the number formed by interchanging the digits is always divisible by
 (a) 7 (b) 9
 (c) 10 (d) 11
- 246.** The largest natural number, which exactly divides the product of any four consecutive natural numbers, is (S.S.C., 2007)
 (a) 6 (b) 12
 (c) 24 (d) 120
- 247.** If n is a whole number greater than 1, then $n^2 (n^2 - 1)$ is always divisible by
 (a) 8 (b) 10
 (c) 12 (d) 16
- 248.** If n is any odd number greater than 1, then $n (n^2 - 1)$ is
 (a) divisible by 24 always (b) divisible by 48 always
 (c) divisible by 96 always (d) None of these
- 249.** The sum of the digits of a 3-digit number is subtracted from the number. The resulting number is always
 (a) not divisible by 9 (b) divisible by 9
 (c) not divisible by 6 (d) divisible by 6
- 250.** A number is multiplied by 11 and 11 is added to the product. If the resulting number is divisible by 13, the smallest original number is
 (a) 12 (b) 22
 (c) 26 (d) 53
- 251.** The product of any three consecutive natural numbers is always divisible by
 (a) 3 (b) 6
 (c) 9 (d) 15
- 252.** The sum of three consecutive odd numbers is always divisible by
 I. 2 II. 3
 III. 5 IV. 6
 (a) Only I (b) Only II
 (c) Only I and II (d) Only I and III
- 253.** The greatest number by which the product of three consecutive multiples of 3 is always divisible is
 (a) 54 (b) 81
 (c) 162 (d) 243
- 254.** If p is a prime number greater than 3, then $(p^2 - 1)$ is always divisible by
 (a) 6 but not 12 (b) 12 but not 24
 (c) 24 (d) None of these
- 255.** The difference between the squares of two consecutive odd integers is always divisible by
 (a) 3 (b) 6
 (c) 7 (d) 8
- 256.** A 4-digit number is formed by repeating a 2-digit number such as 2525, 3232 etc. Any number of this form is exactly divisible by (S.S.C., 2005, 2010)
 (a) 7 (b) 11
 (c) 13 (d) Smallest 3-digit prime number
- 257.** A 6-digit number is formed by repeating a 3-digit number; for example, 256256 or 678678 etc. Any number of this form is always exactly divisible by
 (a) 7 only (b) 11 only
 (c) 13 only (d) 1001
- 258.** The sum of the digits of a natural number $(10^n - 1)$ is 4707, where n is a natural number. The value of n is (Hotel Management, 2010)
 (a) 477 (b) 523
 (c) 532 (d) 704
- 259.** $(x^n - a^n)$ is divisible by $(x - a)$
 (a) for all values of n
 (b) only for even values of n
 (c) only for odd values of n
 (d) only for prime values of n
- 260.** Which one of the following is the number by which the product of 8 consecutive integers is divisible?
 (a) 4 ! (b) 6 !
 (c) 7 ! (d) 8 !
 (e) All of these
- 261.** Consider the following statements:
 For any positive integer n , the number $10^n - 1$ is divisible by
 (1) 9 for $n = \text{odd only}$ (2) 9 for $n = \text{even only}$
 (3) 11 for $n = \text{odd only}$ (4) 11 for $n = \text{even only}$
 Which of the above statements are correct?
 (a) 1 and 3 (b) 2 and 3
 (c) 1 and 4 (d) 2 and 4
- 262.** If n is any positive integer, $3^{4n} - 4^{3n}$ is always divisible by
 (a) 7 (b) 12
 (c) 17 (d) 145
- 263.** If the square of an odd natural number is divided by 8, then the remainder will be
 (a) 1 (b) 2
 (c) 3 (d) 4
- 264.** The largest number that exactly divides each number of the sequence $1^5 - 1, 2^5 - 2, 3^5 - 3, \dots, n^5 - n, \dots$ is
 (a) 1 (b) 15
 (c) 30 (d) 120
- 265.** The difference of the squares of two consecutive even integers is divisible by
 (a) 3 (b) 4
 (c) 6 (d) 7

- 266.** The difference of the squares of two consecutive odd integers is divisible by
 (a) 3 (b) 6
 (c) 7 (d) 8
- 267.** The smallest 4-digit number exactly divisible by 7 is (P.C.S., 2009)
 (a) 1001 (b) 1007
 (c) 1101 (d) 1108
- 268.** What least number must be added to 1056 to get a number exactly divisible by 23? (P.C.S., 2009)
 (a) 2 (b) 3
 (c) 21 (d) 25
- 269.** Which of the following numbers should be added to 8567 to make it exactly divisible by 4? (Bank Recruitment, 2008)
 (a) 3 (b) 4
 (c) 5 (d) 6
 (e) None of these
- 270.** Find the least 6-digit number which is exactly divisible by 349. (R.R.B., 2006)
 (a) 100163 (b) 101063
 (c) 160063 (d) None of these
- 271.** Which is the greatest 5-digit number exactly divisible by 279? (R.R.B., 2006)
 (a) 99603 (b) 99550
 (c) 99882 (d) None of these
- 272.** The least number, which must be added to the greatest 6-digit number so that the sum may be exactly divisible by 327 is
 (a) 194 (b) 264
 (c) 292 (d) 294
- 273.** The least number more than 5000 which is divisible by 73 is
 (a) 5009 (b) 5037
 (c) 5073 (d) 5099
- 274.** The nearest integer to 58701 which is exactly divisible by 567 is
 (a) 55968 (b) 58068
 (c) 58968 (d) None of these
- 275.** The smallest number which must be subtracted from 8112 to make it exactly divisible by 99 is
 (a) 91 (b) 92
 (c) 93 (d) 95
- 276.** The smallest number that must be added to 803642 in order to obtain a multiple of 11 is
 (a) 1 (b) 4
 (c) 7 (d) 9
- 277.** The number of times 99 is subtracted from 1111 so that the remainder is less than 99 is
 (a) 10 (b) 11
 (c) 12 (d) 13
- 278.** The smallest number by which 66 must be multiplied to make the result divisible by 18 is
 (a) 3 (b) 6
 (c) 9 (d) 18
- 279.** The smallest 6-digit number exactly divisible by 111 is
 (a) 111111 (b) 110011
 (c) 100011 (d) 110101
 (e) None of these
- 280.** The sum of all 2-digit numbers divisible by 5 is
 (a) 945 (b) 1035
 (c) 1230 (d) 1245
 (e) None of these
- 281.** How many 3-digit numbers are completely divisible by 6?
 (a) 149 (b) 150
 (c) 151 (d) 166
- 282.** The number of terms between 11 and 200 which are divisible by 7 but not by 3 are (C.P.F., 2008)
 (a) 18 (b) 19
 (c) 27 (d) 28
- 283.** Out of the numbers divisible by 3 between 14 and 95 if the numbers with 3 at unit's place are removed, then how many numbers will remain? (R.R.B., 2006)
 (a) 22 (b) 23
 (c) 24 (d) 25
- 284.** How many numbers less than 1000 are multiples of both 10 and 13?
 (a) 6 (b) 7
 (c) 8 (d) 9
- 285.** How many integers between 100 and 150, both inclusive, can be evenly divided by neither 3 nor 5?
 (a) 26 (b) 27
 (c) 28 (d) 33
- 286.** If all the numbers from 501 to 700 are written, what is the total number of times the digit 6 appears? (Civil Services, 2007)
 (a) 138 (b) 139
 (c) 140 (d) 141
- 287.** How many 3-digit numbers are there in between 100 and 300, having first and the last digit as 2?
 (a) 9 (b) 10
 (c) 11 (d) 12
- 288.** The total number of integers between 200 and 400, each of which either begins with 3 or ends with 3 or both is (S.S.C., 2007)
 (a) 10 (b) 100
 (c) 110 (d) 120
- 289.** While writing all the numbers from 700 to 1000, how many numbers occur in which the first digit is greater than the second digit, and the second digit is greater than the third digit?
 (a) 61 (b) 64
 (c) 78 (d) 85

- 290.** A 9-digit number in which zero does not appear and no digits are repeated has the following properties: The number comprising the left most two digits is divisible by 2, that comprising the left most three digits is divisible by 3, and so on.
The number is
(a) 183654729 (b) 381654729
(c) 983654721 (d) 981654723
- 291.** If 11,109,999 is divided by 1111, then what is the remainder? (M.A.T., 2007)
(a) 1098 (b) 1010
(c) 1110 (d) 1188
- 292.** A number divided by 68 gives the quotient 260 and remainder zero. If the same number is divided by 65, the remainder is
(a) 0 (b) 1
(c) 2 (d) 3
- 293.** Which of the following prime numbers while dividing 2176 leaves 9 as remainder?
(a) 17 (b) 29
(c) 167 (d) 197
- 294.** Match List I with List II and select the correct answer:
- | List I | List II |
|---|--------------------------|
| (a, b as given in Euclidean algorithm
$a = bq + r$) | (Values of q and r) |
| A. $a = -112, b = -7$ | 1. $q = -13, r = 1$ |
| B. $a = 118, b = -9$ | 2. $q = 14, r = 3$ |
| C. $a = -109, b = 6$ | 3. $q = -19, r = 5$ |
| D. $a = 115, b = 8$ | 4. $q = 16, r = 0$ |
| A B C D | A B C D |
| (a) 3 1 4 2 | (b) 3 2 4 1 |
| (c) 4 1 3 2 | (d) 4 2 3 1 |
- 295.** The number 534677 is divided by 777. The difference of divisor and remainder is
(a) 577 (b) 676
(c) 687 (d) 789
- 296.** In a division sum, the quotient, dividend and remainder are 15, 940 and 25 respectively. The divisor is
(a) 31 (b) 50
(c) 60 (d) 61
- 297.** In a division sum, the divisor is 12 times the quotient and 5 times the remainder. If the remainder is 48, then the dividend is
(a) 2404 (b) 3648
(c) 4808 (d) 4848
- 298.** The divisor is 25 times the quotient and 5 times the remainder. If the quotient is 16, then the dividend is
(a) 400 (b) 480
(c) 6400 (d) 6480
- 299.** A number when divided by the sum of 555 and 445 gives two times their difference as quotient and 30 as the remainder. The number is
(a) 1220 (b) 1250
(c) 22030 (d) 220030
- 300.** In doing a question of division with zero remainder, a candidate took 12 as divisor instead of 21. The quotient obtained by him was 35. The correct quotient is
(a) 0 (b) 12
(c) 13 (d) 20
- 301.** In a division problem, the divisor is 7 times of quotient and 5 times of remainder. If the dividend is 6 times of remainder, then the quotient is equal to
(a) 0 (b) 1
(c) 7 (d) None of these
(Hotel Management, 2007)
- 302.** On dividing a number by 19, the difference between quotient and remainder is 9. The number is
(a) 352 (b) 361
(c) 370 (d) 371
- 303.** A number when divided by 136 leaves remainder 36. If the same number is divided by 17, the remainder will be (S.S.C., 2010)
(a) 2 (b) 3
(c) 7 (d) 9
- 304.** A number when divided by 195 leaves a remainder 47. If the same number is divided by 15, the remainder will be (Hotel Management, 2010)
(a) 1 (b) 2
(c) 3 (d) 4
- 305.** A certain number when divided by 899 gives a remainder 63. What is the remainder when the same number is divided by 29? (R.R.B., 2008)
(a) 5 (b) 25
(c) 27 (d) None of these
- 306.** A number when divided by 5 leaves the remainder 3. What is the remainder when the square of the same number is divided by 5?
(a) 0 (b) 3
(c) 4 (d) 9
- 307.** The difference between two numbers is 1365. When the larger number is divided by the smaller one, the quotient is 6 and the remainder is 15. What is the smaller number?
(a) 240 (b) 270
(c) 295 (d) 360
- 308.** When n is divided by 4, the remainder is 3. What is the remainder when $2n$ is divided by 4?
(a) 1 (b) 2
(c) 3 (d) 6
- 309.** When a number is divided by 13, the remainder is 11. When the same number is divided by 17, the remainder is 9. What is the number?

- (a) 339 (b) 349
(c) 369 (d) Data inadequate
(e) None of these
- 310.** In a division sum, the remainder was 71. With the same divisor but twice the dividend, the remainder is 43. Which one of the following is the divisor?
(a) 86 (b) 93
(c) 99 (d) 104
- 311.** When a certain positive integer P is divided by another positive integer, the remainder is r_1 . When a second positive integer Q is divided by the same integer, the remainder is r_2 and when $(P + Q)$ is divided by the same divisor, the remainder is r_3 . Then the divisor may be
(a) $r_1 r_2 r_3$ (b) $r_1 + r_2 + r_3$
(c) $r_1 - r_2 + r_3$ (d) $r_1 + r_2 - r_3$
(e) Cannot be determined
- 312.** Two numbers when divided by a certain divisor leave the remainders 4375 and 2986 respectively but when the sum of two numbers is divided by the same divisor, the remainder is 2361. The divisor in question is
(a) 4675 (b) 4900
(c) 5000 (d) None of these
- 313.** A number divided by 13 leaves a remainder 1 and if the quotient, thus obtained, is divided by 5, we get a remainder of 3. What will be the remainder if the number is divided by 65?
(a) 16 (b) 18
(c) 28 (d) 40
- 314.** The numbers 2272 and 875 are divided by a three-digit number N , giving the same remainder. The sum of the digits of N is
(a) 10 (b) 11
(c) 12 (d) 13
- 315.** A number when divided by three consecutive numbers 9, 11, 13 leaves the remainders 8, 9 and 8 respectively. If the order of divisors is reversed, the remainders will be (R.R.B., 2008)
(a) 10, 8, 9 (b) 10, 1, 6
(c) 8, 9, 8 (d) 9, 8, 8
- 316.** After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?
(a) 41 (b) 53
(c) 75 (d) 80
- 317.** A number is successively divided by 8, 7 and 3 giving residues 3, 4 and 2 respectively and quotient 31. The number is
(a) 3555 (b) 5355
(c) 5535 (d) 5553
- 318.** A number when divided by 3 leaves a remainder 1. When the quotient is divided by 2, it leaves a remainder 1. What will be the remainder when the number is divided by 6?
(a) 2 (b) 3
(c) 4 (d) 5
- 319.** When the square of any odd number, greater than 1, is divided by 8, it always leaves remainder (I.A.M., 2007)
(a) 1 (b) 6
(c) 8
(d) Cannot be determined
- 320.** The numbers from 1 to 29 are written side by side as follows:
1234567891011121314.....2829
If this number is divided by 9, then what is the remainder? (M.A.T., 2006)
(a) 0 (b) 1
(c) 3 (d) None of these
- 321.** If 17^{200} is divided by 18, the remainder is
(a) 1 (b) 2
(c) 16 (d) 17
- 322.** What is the remainder when 2^{31} is divided by 5?
(a) 1 (b) 2
(c) 3 (d) 4
- 323.** Consider the following statements:
(1) $a^n + b^n$ is divisible by $a + b$ if $n = 2k + 1$, where k is a positive integer.
(2) $a^n - b^n$ is divisible by $a - b$ if $n = 2k$, where k is a positive integer.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 324.** $(7^{19} + 2)$ is divided by 6. The remainder is
(a) 1 (b) 2
(c) 3 (d) 5
- 325.** If $(10^{12} + 25)^2 - (10^{12} - 25)^2 = 10^n$, then the value of n is
(a) 5 (b) 10
(c) 14 (d) 20
- 326.** $(3^{25} + 3^{26} + 3^{27} + 3^{28})$ is divisible by
(a) 11 (b) 16
(c) 25 (d) 30
- 327.** $(4^{61} + 4^{62} + 4^{63} + 4^{64})$ is divisible by
(a) 3 (b) 11
(c) 13 (d) 17
- 328.** $(9^6 + 1)$ when divided by 8, would leave a remainder of
(a) 0 (b) 1
(c) 2 (d) 3
- 329.** If n is even, $(6^n - 1)$ is divisible by
(a) 6 (b) 30
(c) 35 (d) 37
- 330.** If $(12^n + 1)$ is divisible by 13, then n is
(a) 1 only (b) 12 only
(c) any odd integer (d) any even integer

- 331.** 25^{25} is divided by 26, the remainder is
 (a) 1 (b) 2
 (c) 24 (d) 25
- 332.** If $(67^{67} + 67)$ is divided by 68, the remainder is
 (a) 1 (b) 63
 (c) 66 (d) 67
- 333.** One less than $(49)^{15}$ is exactly divisible by
 (a) 8 (b) 14
 (c) 50 (d) 51
- 334.** The remainder when 7^{84} is divided by 342 is
 (a) 0 (b) 1
 (c) 49 (d) 341
- 335.** The remainder when 2^{60} is divided by 5 equals
 (a) 0 (b) 1
 (c) 2 (d) 3
- 336.** By how many of the following numbers is $2^{12} - 1$ divisible?
 2, 3, 5, 7, 10, 11, 13, 14
 (a) 4 (b) 5
 (c) 6 (d) 7
- 337.** The remainder when $(15^{23} + 23^{23})$ is divided by 19, is
 (a) 0 (b) 4
 (c) 15 (d) 18
- 338.** When 2^{256} is divided by 17, the remainder would be
 (a) 1 (b) 14
 (c) 16 (d) None of these
- 339.** $7^{6n} - 6^{6n}$, where n is an integer > 0 , is divisible by
 (a) 13 (b) 127
 (c) 559 (d) All of these
- 340.** It is given that $(2^{32} + 1)$ is exactly divisible by a certain number. Which of the following is also definitely divisible by the same number?
 (S.S.C., 2007)
 (a) $2^{16} + 1$ (b) $2^{16} - 1$
 (c) 7×2^{33} (d) $2^{96} + 1$
- 341.** The number $(2^{48} - 1)$ is exactly divisible by two numbers between 60 and 70. The numbers are
 (A.A.O. Exam, 2010)
 (a) 63 and 65 (b) 63 and 67
 (c) 61 and 65 (d) 65 and 67
- 342.** n being any odd number greater than 1, $n^{65} - n$ is always divisible by
 (a) 5 (b) 13
 (c) 24 (d) None of these
- 343.** Let $N = 55^3 + 17^3 - 72^3$. Then, N is divisible by
 (a) both 7 and 13 (b) both 3 and 13
 (c) both 17 and 7 (d) both 3 and 17

Directions (Questions 344–345): These questions are based on the following information:

Given $N = |1| + |2| + |3| + \dots + |99| + |100|$.

- 344.** Find the last two digits of N .
 (a) 00 (b) 13
 (c) 19 (d) 23
- 345.** Find the remainder when N is divided by 168.
 (a) 33 (b) 67
 (c) 129 (d) 153
- 346.** What is the remainder when 4^{61} is divided by 51?
 (a) 20 (b) 41
 (c) 50 (d) None of these
- 347.** What is the remainder when 17^{36} is divided by 36?
 (a) 1 (b) 7
 (c) 19 (d) 29
- 348.** Which one of the following is the common factor of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$?
 (a) $(47 - 43)$ (b) $(47 + 43)$
 (c) $(47^{43} + 43^{43})$ (d) None of these
- 349.** Find the product of all odd natural numbers less than 5000.
 (a) $\frac{5000!}{2500 \times 2501}$ (b) $\frac{5000!}{2^{2500} \times 2500!}$
 (c) $\frac{5000!}{2^{5000}}$ (d) None of these
- 350.** How many zeros will be required to number the pages of a book containing 1000 pages?
 (a) 168 (b) 184
 (c) 192 (d) 216
- 351.** If $a^2 + b^2 + c^2 = 1$, what is the maximum value of abc ?
 (a) $\frac{1}{3}$ (b) $\frac{1}{3\sqrt{3}}$
 (c) $\frac{2}{\sqrt{3}}$ (d) 1
- 352.** Find the unit's digit in the sum of the fifth powers of the first 100 natural numbers.
 (a) 0 (b) 2
 (c) 5 (d) 8
- 353.** If the symbol $[x]$ denotes the greatest integer less than or equal to x , then the value of $\left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{50}\right] + \left[\frac{1}{4} + \frac{2}{50}\right] + \dots + \left[\frac{1}{4} + \frac{49}{50}\right]$ is
 (a) 0 (b) 9
 (c) 12 (d) 49
- 354.** When $100^{25} - 25$ is written in decimal notation, the sum of its digits is
 (a) 444 (b) 445
 (c) 446 (d) 448
- 355.** What is the number of digits in the number $(1024)^4 \times (125)^{11}$?
 (a) 35 (b) 36
 (c) 37 (d) 38

356. How many numbers will be there between 300 and 500, where 4 comes only one time?

- (a) 89 (b) 99
(c) 110 (d) 120

[UPSSSC Lower Subordinate (Pre.) Exam, 2016]

357. Which is not a prime number?

[Indian Railways Gr. 'D' Exam, 2014]

- (a) 13 (b) 19
(c) 21 (d) 17

358. If $x = a(b - c)$, $y = b(c - a)$, $z = c(a - b)$, then the value of $\left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 + \left(\frac{z}{c}\right)^3$ is

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) $\frac{2xyz}{abc}$ (b) $\frac{xyz}{abc}$
(c) 0 (d) $\frac{3xyz}{abc}$

359. Among the following statements, the statement which is not correct is:

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) Every natural number is an integer.
(b) Every natural number is a real number.
(c) Every real number is a rational number.
(d) Every integer is a rational number.

360. If $a + b + c = 6$ and $ab + bc + ca = 10$ then the value of $a^3 + b^3 + c^3 - 3abc$ is

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) 36 (b) 48
(c) 42 (d) 40

361. If $(1001 \times 111) = 110000 + (11 \times \underline{\hspace{1cm}})$, then the number in the blank space is

- (a) 121 (b) 211
(c) 101 (d) 1111

[CTET, 2016]

Direction (Question 362): The following question consists of a question and two statements I and II given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Given answer

- (A) The data in statement I alone is sufficient to answer the question while II alone is not sufficient to answer the questions.
(B) Data in statement II alone is sufficient to answer the question while data in statement I alone is not sufficient to answer the question.
(C) The data in statement I alone or statement II alone is sufficient to answer the question.
(D) The data in both Statement I and II is insufficient to answer the question.
(E) The data in both Statement I and II is sufficient to answer the question.

362. If (the place value of 5 in 15201) + (the place value of 6 in 2659) = $7 \times \underline{\hspace{1cm}}$, then the number of the blank space is:

- (a) 800 (b) 80
(c) 90 (d) 900

[CTET, 2016]

363. The sum of digits of a two – digit number is 12 and the difference between the two – digits of the two – digit number is 6. What is the two – digit number?

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) 39 (b) 84
(c) 93
(d) Other than the given options
(e) 75

364. The difference between the greatest and the least four digit numbers that begins with 3 and ends with 5 is

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) 900 (b) 909
(c) 999 (d) 990

365. The sum of the perfect squares between 120 and 300 is

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) 1204 (b) 1024
(c) 1296 (d) 1400

366. If $p^3 - q^3 = (p - q)(p - q)^2 - xpq$, then find the value of x

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) 1 (b) -3
(c) 3 (d) -1

367. What minimum value should be assigned to *, so that $2361*48$ is exactly divisible by 9?

[ESIC—UDC Exam, 2016]

- (a) 2 (b) 3
(c) 9 (d) 4

368. If p, q, r are all real numbers then $(p - q)^3 + (q - r)^3 + (r - p)^3$ is equal to

[SSC—CAPF/CPO Exam, 2016]

- (a) $(p - q)(q - r)(r - p)$ (b) $3(p - q)(q - r)(r - p)$
(c) 1 (d) 0

369. If $(a^2 - b^2) \div (a + b) = 25$, then $(a + b) = ?$

- (a) 30 (b) 25
(c) 125 (d) 150

[RRB—NTPC Exam, 2016]

370. How many prime numbers are there between 100 to 200?

[CMAT, 2017]

- (a) 21 (b) 20
(c) 16 (d) 11

371. The least number of five digit is exactly divisible by 88 is

- (a) 10032 (b) 10132
(c) 10088 (d) 10023

[SSC Multi-Tasking Staff Exam -2017]

- 372.** The number of three digit numbers which are multiples of 9 are

[CLAT, 2016]

- (a) 100 (b) 99
(c) 98 (d) 101

- 373.** Two consecutive even positive integers, sum of the squares of which is 1060, are

[CLAT, 2016]

- (a) 12 and 14 (b) 20 and 22
(c) 22 and 24 (d) 15 and 18

Directions (Question 374): The question below consists of a question and two statements numbered I and II given below it. You have to decide whether the data given in the statements are sufficient to answer the questions. Read both the statements and given answer.

- (A) If the data in statement I alone is sufficient to answer the question, while the data in statement II alone is not sufficient to answer the question.
(B) If the data in statement II alone is sufficient to answer the question, while the data in statement I alone is not sufficient to answer the question.
(C) If the data either in statement I alone or statement II alone is sufficient to answer the questions.
(D) If the data given in both Statement I and II together are not sufficient to answer the question.
(E) If the data in both Statement I and II together are necessary to answer the question.

- 374.** What is the number of trees planted in the field in row and column?

[IBPS—Bank Spl. Officer (Marketing) Exam, 2016]

- I. Number of columns is more than the number of rows by 4.
II. Number of columns is 20.

- 375.** If n is a natural number and $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$, where p_1, p_2, p_3 are distinct prime factors, then the number of prime factors for n is

[CDS, 2016]

- (a) $x_1 + x_2 + x_3$ (b) x_1, x_2, x_3
(c) $(x_1 + 1)(x_2 + 1)(x_3 + 1)$ (d) None of the above

- 376.** Consider the following statements for the sequence of numbers given below:

11, 111, 1111, 11111,

1. Each number can be expressed in the form $(4m + 3)$, where m is a natural number.
2. Some numbers are squares.

Which of the above statements is/are correct?

[CDS, 2016]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) neither 1 nor 2

- 377.** If the sum of two numbers is 14 and their difference is 10. Find the product of these two numbers.

[UPSSSC Lower Subordinate (Pre.) Exam, 2016]

- (a) 24 (b) 22
(c) 20 (d) 18

- 378.** If $m = -4$, $n = -2$, then the value of $m^3 - 3m^2 + 3m + 3n + 3n^2 + n^3$ is

- (a) -120 (b) -124
(c) -126 (d) -128

[SSC—CGL (Tier-I) Exam, 2015]

- 379.** If the sum of two numbers is 14 and their difference is 10, find the product of these two numbers.

[UPSSSC—Lower Subordinate (Pre.) Exam 2016]

- (a) 18 (b) 20
(c) 24 (d) 22

- 380.** What is the sum of all natural numbers from 1 to 100?

[CLAT-2016]

- (a) 5050 (b) 6000
(c) 5000 (d) 5052

ANSWERS

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (c) | 4. (d) | 5. (c) | 6. (c) | 7. (c) | 8. (c) | 9. (a) | 10. (c) |
| 11. (b) | 12. (b) | 13. (d) | 14. (c) | 15. (d) | 16. (b) | 17. (c) | 18. (b) | 19. (c) | 20. (d) |
| 21. (c) | 22. (c) | 23. (c) | 24. (b) | 25. (d) | 26. (d) | 27. (a) | 28. (c) | 29. (c) | 30. (c) |
| 31. (b) | 32. (b) | 33. (d) | 34. (a) | 35. (c) | 36. (d) | 37. (b) | 38. (c) | 39. (a) | 40. (d) |
| 41. (a) | 42. (d) | 43. (d) | 44. (d) | 45. (d) | 46. (d) | 47. (d) | 48. (c) | 49. (a) | 50. (b) |
| 51. (b) | 52. (b) | 53. (c) | 54. (d) | 55. (b) | 56. (d) | 57. (b) | 58. (c) | 59. (d) | 60. (b) |

61. (d)	62. (d)	63. (c)	64. (a)	65. (c)	66. (a)	67. (a)	68. (c)	69. (d)	70. (b)
71. (a)	72. (b)	73. (c)	74. (d)	75. (c)	76. (b)	77. (c)	78. (b)	79. (d)	80. (d)
81. (c)	82. (d)	83. (c)	84. (e)	85. (c)	86. (a)	87. (b)	88. (b)	89. (e)	90. (d)
91. (c)	92. (a)	93. (c)	94. (c)	95. (d)	96. (d)	97. (a)	98. (c)	99. (c)	100. (a)
101. (b)	102. (a)	103. (a)	104. (c)	105. (a)	106. (c)	107. (a)	108. (b)	109. (b)	110. (e)
111. (a)	112. (a)	113. (b)	114. (a)	115. (b)	116. (c)	117. (a)	118. (c)	119. (b)	120. (a)
121. (e)	122. (d)	123. (c)	124. (b)	125. (a)	126. (b)	127. (c)	128. (d)	129. (d)	130. (d)
131. (b)	132. (b)	133. (b)	134. (d)	135. (d)	136. (a)	137. (a)	138. (b)	139. (a)	140. (b)
141. (a)	142. (c)	143. (d)	144. (d)	145. (e)	146. (c)	147. (a)	148. (b)	149. (c)	150. (a)
151. (e)	152. (c)	153. (c)	154. (c)	155. (b)	156. (a)	157. (a)	158. (d)	159. (b)	160. (c)
161. (d)	162. (d)	163. (b)	164. (d)	165. (d)	166. (a)	167. (a)	168. (a)	169. (c)	170. (b)
171. (d)	172. (b)	173. (c)	174. (a)	175. (b)	176. (a)	177. (a)	178. (d)	179. (c)	180. (a)
181. (d)	182. (c)	183. (c)	184. (b)	185. (b)	186. (c)	187. (a)	188. (c)	189. (c)	190. (c)
191. (c)	192. (b)	193. (a)	194. (a)	195. (b)	196. (a)	197. (b)	198. (a)	199. (c)	200. (a)
201. (b)	202. (d)	203. (b)	204. (b)	205. (a)	206. (d)	207. (d)	208. (d)	209. (c)	210. (a)
211. (d)	212. (b)	213. (a)	214. (d)	215. (d)	216. (a)	217. (d)	218. (a)	219. (b)	220. (b)
221. (c)	222. (b)	223. (d)	224. (c)	225. (d)	226. (a)	227. (d)	228. (d)	229. (b)	230. (e)
231. (d)	232. (b)	233. (a)	234. (a)	235. (a)	236. (c)	237. (c)	238. (a)	239. (a)	240. (b)
241. (c)	242. (c)	243. (b)	244. (c)	245. (d)	246. (c)	247. (c)	248. (a)	249. (b)	250. (a)
251. (b)	252. (b)	253. (c)	254. (c)	255. (d)	256. (d)	257. (d)	258. (b)	259. (a)	260. (e)
261. (d)	262. (c)	263. (a)	264. (c)	265. (b)	266. (d)	267. (a)	268. (a)	269. (e)	270. (a)
271. (c)	272. (d)	273. (b)	274. (c)	275. (c)	276. (c)	277. (b)	278. (a)	279. (c)	280. (a)
281. (b)	282. (a)	283. (c)	284. (b)	285. (b)	286. (c)	287. (b)	288. (c)	289. (d)	290. (b)
291. (c)	292. (a)	293. (d)	294. (c)	295. (b)	296. (d)	297. (d)	298. (d)	299. (d)	300. (d)
301. (b)	302. (d)	303. (a)	304. (b)	305. (a)	306. (c)	307. (b)	308. (b)	309. (b)	310. (c)
311. (d)	312. (c)	313. (d)	314. (a)	315. (b)	316. (b)	317. (b)	318. (c)	319. (a)	320. (c)
321. (a)	322. (c)	323. (c)	324. (c)	325. (c)	326. (d)	327. (d)	328. (c)	329. (c)	330. (c)
331. (d)	332. (c)	333. (a)	334. (b)	335. (b)	336. (a)	337. (a)	338. (a)	339. (d)	340. (d)
341. (a)	342. (c)	343. (d)	344. (b)	345. (a)	346. (d)	347. (a)	348. (b)	349. (b)	350. (c)
351. (b)	352. (a)	353. (c)	354. (a)	355. (b)	356. (b)	357. (c)	358. (d)	359. (c)	360. (a)
361. (c)	362. (a)	363. (d)	364. (d)	365. (d)	366. (b)	367. (b)	368. (b)	369. (b)	370. (a)
371. (a)	372. (a)	373. (c)	374. (d)	375. (b)	376. (a)	377. (a)	378. (c)	379. (c)	380. (a)

SOLUTIONS

1.

TL	L	T-Th	Th	H	T	O
3	2	5	4	7	1	0

Place value of 5 = $(5 \times 10000) = 50000$.

2. The face value of 8 in the given numeral is 8.
3. Sum of the place values of 3 = $(3000 + 30) = 3030$.
4. Difference between the place values of 7 and 3 in given numeral = $(7000 - 30) = 6970$.
5. Difference between the local value and the face value of 7 in the given numeral = $(70000 - 7) = 69993$.
6. Required sum = $(99999 + 10000) = 109999$.
7. Required difference = $(10000 - 999) = 9001$.
8. Required number = 30005.
9. Required number = 2047.
10. All natural numbers and 0 are called the whole numbers.
11. Clearly, there exists a smallest natural number, namely 1. So statement (1) is true.
Natural numbers are counting numbers and the counting process never ends. So, the largest nature number is not known. Thus, statement (2) is false.
There is no natural number between two consecutive natural numbers. So, statement (3) is false.
12. Clearly, every rational number is also a real number.
13. Clearly, π is an irrational number.
14. Since $\sqrt{2}$ is a non-terminating and non-repeating decimal, so it is an irrational number.
15. $\sqrt{3}$ is an infinite non-recurring decimal.
16. We can write $9 = (1 + 8)$; $9 = (2 + 7)$;
 $9 = (3 + 6)$, $9 = (4 + 5)$.
Thus, it can be done in 4 ways.
17. We may have (64 and 1), (32 and 2), (16 and 4) and (8 and 8).
In any case, the sum is not 35.
18. We may take the least values of y and z as $y = 1$ and $z = 1$.
So, the maximum value of x is 7.
19. We know that: $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$.
 $\therefore (1^2 + 2^2 + 3^2 + \dots + 9^2) = \left(\frac{1}{6} \times 9 \times 10 \times 19 \right) = 285$.
20. $(n + 7) = 88 \Rightarrow n = (88 - 7) = 81$, which is false as $20 < n < 80$.
So, the required number is 88.
21. We have, $M = 1000$; $D = 500$; $C = 100$ and $L = 50$.
 $\therefore M > D > C > L$ is the correct sequence.
22. These numbers are 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 2. 8th number from the bottom is 16.
23. The given series is such that the sum of first hundred terms is zero, and 101st term is 2. So, the sum of 101 terms is 2.
24. Given series is 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4,
 $\therefore$ 96th term is 4. So, $T_{97} = 1$ and $T_{98} = 2$. Hence, the 98th term is 2.
25. Let the hundred's, ten's and one's digits be x , y and z respectively.
Then, the given number is $100x + 10y + z$.
26. Let the thousand's, hundred's, ten's and one's digits be x , y , z , w respectively.

Then, the number is $1000x + 100y + 10z + w = 10^3x + 10^2y + 10z + w$.

27. We know that the sum of two odd numbers is even.
 $\therefore (n \text{ is odd, } p \text{ is odd}) \Rightarrow (n + p) \text{ is even.}$
28. n^3 is odd $\Rightarrow n$ is odd and n^2 is odd.
 $\therefore$ I and II are true.
29. $(n - 1)$ is odd $\Rightarrow (n - 1) - 2$ and $(n - 1) + 2$ are odd $\Rightarrow (n - 3)$ and $(n + 1)$ are odd.
30. The product of two odd numbers is always odd.
31. x is odd $\Rightarrow x^2$ is odd $\Rightarrow 5x^2$ is odd $\Rightarrow (5x^2 + 2)$ is odd.
32. $ab = 0 \Rightarrow a = 0$ or $b = 0$ or both are zero.
33. $A < B < C < D \Rightarrow D > B > A$
Also, $A < B < C < D$ and $D > B > E \Rightarrow E < B < C < D \Rightarrow E < C < D$ and $E < B < C$.
Clearly, we cannot arrange A, E, C in an increasing or decreasing order.
34. When m is even, then $m(n + 'o')(p - q)$ is even.
35. $n < 0 \Rightarrow 2n < 0$, $-n > 0$ and $n^2 = (-n)^2 > 0$.
Thus, out of the numbers 0, $-n$, $2n$ and n^2 we find that $2n$ is the least.
36. It is given that $x - y = 8$.
I. We may have $x = 5$ and $y = -3$.
So, it is not necessary that both x and y are positive.
II. If x is positive, then it is not necessary that y is positive, as $x = 5 \Rightarrow y = -3$.
III. If $x < 0$, then $y = x - 8$ which is clearly less than 0.
So, if x is negative, then y must be negative.
37. If $x < 0$ and $y < 0$, then clearly $xy > 0$.
So, whenever x and y are negative, then xy is positive.
Note that $(x < 0, y < 0)$ does not imply that $(x + y)$ is positive.
e.g. If $x = -2$ and $y = -3$, then $(x + y) = -5 < 0$.
Note that $(x < 0, y < 0)$ does not imply that $(x - y) > 0$.
e.g. If $x = -5$ and $y = -2$, then $x - y = -5 - (-2) = -5 + 2 = -3 < 0$.
38. Let $n = 1 + x = 1 + m(m + 1)(m + 2)(m + 3)$, where m is a positive integer.
Then, clearly two of $m, (m + 1), (m + 2), (m + 3)$ are even and so their product is even. Thus, x is even and hence $n = 1 + x$ is odd.
Also, $n = 1 + m(m + 3)(m + 1)(m + 2) = 1 + (m^2 + 3m)(m^2 + 3m + 2)$
 $\Rightarrow n = 1 + y(y + 2)$, where $m^2 + 3m = y$
 $\Rightarrow n = 1 + y^2 + 2y = (1 + y)^2$, which is a perfect square.
Hence, I and III are true.
39. $1 \leq x \leq 2 \Rightarrow 1 \leq \frac{2}{5}y + 3 \leq 2$
 $\Rightarrow (1 - 3) \leq \frac{2}{5}y \leq (2 - 3) \Rightarrow \frac{5}{2}(1 - 3) \leq y \leq \frac{5}{2}(2 - 3)$
 $\Rightarrow -5 \leq y \leq \frac{-5}{2}$.
Hence, y increases from -5 to $\frac{-5}{2}$.
40. (a) Let $x = 0$ and $y = \sqrt{2}$. Then, x is rational and y is irrational.
 $\therefore x + y = 0 + \sqrt{2} = \sqrt{2}$ which is irrational.
Thus, $x + y$ is not rational.

(b) Let $x = 0$ and $y = \sqrt{2}$. Then, x is rational and y is irrational.

$\therefore xy = 0 \times \sqrt{2} = 0$, which is rational.

Hence, xy is not irrational.

(c) As shown in (b), xy is not necessarily irrational.

(d) $x + y$ is necessary irrational. But xy can be either rational or irrational.

Hence, (d) is true.

41. Let x and $(x + 1)$ be two consecutive integers. Then
 $(x + 1)^2 - x^2 = (x + 1 + x)(x + 1 - x) = (x + 1 + x) \times 1$
 $= (x + 1 + x) = \text{sum of given numbers.}$

42. If a and b are two rational numbers, then $\frac{a+b}{2}$ is a rational number lying between a and b .

43. $B > A \Rightarrow A < B \Rightarrow (A - B) < 0$.

Since A and B are both positive integers, we have
 $(A + B) > 0$ and $AB > 0$.

If $(A = 1 \text{ and } B = 2)$, we have $AB < (A + B)$.

If $(A = 2 \text{ and } B = 3)$, we have $(A + B) < AB$.

Thus, we cannot say which one of $A + B$ and AB has the highest value.

44. $0 < x < 1 \Rightarrow x^2 < x < 1 \quad \dots(i)$
 $\Rightarrow \frac{1}{x^2} > \frac{1}{x} > 1 > x > x^2 \quad [\text{using } (i)]$

Hence, $\frac{1}{x^2}$ is the greatest.

45. $p < 1 \Rightarrow \frac{1}{p} > 1 \Rightarrow \frac{2}{p} > 2 \Rightarrow \frac{2}{p} - p > 2 - p > 0 \quad [\because p < 1]$

Hence, $\left(\frac{2}{p} - p\right)$ is a positive number.

46. $(x^2 + x + 1) = \left(x^2 + x + \frac{1}{4}\right) + \frac{3}{4}$
 $= \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4} \quad \left[\because \left(x + \frac{1}{2}\right)^2 \geq 0\right]$

Hence, $(x^2 + x + 1)$ is greater than or equal to $\frac{3}{4}$.

47. $\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{7}\right) + \frac{1}{n} = \frac{(21 + 14 + 6)}{42} + \frac{1}{n} = \left(\frac{41}{42} + \frac{1}{n}\right)$.

This sum is a natural number when $n = 42$.

So, each one of the statements that 2 divides n ; 3 divides n and 7 divides n is true.

Hence, $n > 84$ is false.

48. $\frac{(16n^2 + 7n + 6)}{n} = \left(\frac{16n^2}{n} + \frac{7n}{n} + \frac{6}{n}\right) = \left(16n + 7 + \frac{6}{n}\right)$.

For $\left(16n + 7 + \frac{6}{n}\right)$ to be an integer, we may have $n = 1$

or $n = 2$ or $n = 3$ or $n = 6$.

Hence, 4 values of n will give the desired result.

49. $(p > q \text{ and } r < 0) \Rightarrow pr < qr$ is true.

50. $(X < Z \text{ and } X < Y) \Rightarrow X^2 < YZ$.

51. $x + y > p$ and $p > z \Rightarrow x + y > z$.

52. $\frac{(a-b)}{3.5} = \frac{4}{7} \Rightarrow (a-b) = \frac{4}{7} \times \frac{7}{2} = 2 \Rightarrow b < a$.

53. $\frac{13}{1} = \frac{13w}{(1-w)} \Rightarrow \frac{w}{(1-w)} = 1 \Rightarrow w = 1 - w \Rightarrow 2w = 1 \Rightarrow w = \frac{1}{2}$.
 $\therefore (2w)^2 = 4w^2 = 4 \times \frac{1}{4} = 1$.

Questions 54 to 57

Let the digits of the number in order be A, B, C, D, E .

Then, $A > 5 \Rightarrow E = 1$.

A, B, C are all odd and none of the digits is 3 $\Rightarrow A, B, C$ are the digits from 5, 7, 9.

Since A is the largest digit, so $A = 9$.

$\therefore B = 5$ or 7 and $C = 5$ or 7 .

Now, the number DE is the product of two prime numbers.
 $E = 1$ and $D = 2, 4, 6$ or 8 .

41 and 61 are prime numbers and 81 cannot be expressed as product of two primes.

Only 21 can be expressed as the product of two prime numbers ($21 = 3 \times 7$).

So, $D = 2$.

Hence, the number is 95721 or 97521.

54. The second digit of the number is either 5 or 7.

55. The last digit of the number is 1.

56. The largest digit in the number is 9.

57. The number is odd. So, it is not divisible by 2 or 4.
Sum of digits $= 9 + 7 + 5 + 2 + 1 = 24$, which is divisible by 3 but not by 9.

So, the given number is divisible by 3.

58. The least prime number is 2.

59. **Statement 1.** Let $x = 4$ and $y = 15$. Then, each one of x and y is a composite number.

But, $x + y = 19$, which is not composite.

$\therefore$ Statement 1 is not true.

Statement 2. We know that 1 is neither prime nor composite.

$\therefore$ Statement 2 is not true.

Thus, neither 1 nor 2 is correct.

60. Prime numbers between 0 and 50 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43 and 47.

Their number is 15.

61. Each of the required numbers must divide $(143 - 3) = 140$ exactly.

Now, $140 = 5 \times 7 \times 4$.

Hence, the required prime numbers are 5 and 7.

62. Sum of first four prime numbers $= (2 + 3 + 5 + 7) = 17$.

63. Sum of all prime numbers from 1 to 20 $= (2 + 3 + 5 + 7 + 11 + 13 + 17 + 19) = 77$.

64. Clearly, 11 is a prime number which remains unchanged when its digits are interchanged. And, $(11)^2 = 121$.

Hence, the square of such a number is 121.

65. Let the required prime number be p . Let p when divided by 6 give n as quotient and r as remainder. Then

$p = 6n + r$, where $0 \leq r < 6$

Now, $r = 0, r = 2, r = 3$ and $r = 4$ do not give p as prime.

$\therefore r \neq 0, r \neq 2, r \neq 3$ and $r \neq 4$.

Hence, $r = 1$ or $r = 5$.

66. Clearly, $21 = 3 \times 7$, so 21 is not prime.

67. Clearly, 19 is a prime number.

68. We know that 115 is divisible by 5. So, it is not prime.

119 is divisible by 7. So, it is not prime.

$127 < (12)^2$ and prime numbers less than 12 are 2, 3, 5, 7, 11.

Clearly, 127 is not exactly divisible by any of them. Hence, 127 is a prime number.

69. Clearly, 143 is divisible by 11. So, 143 is not prime.
289 is divisible by 17. So, 289 is not prime.
117 is divisible by 3. So, 117 is not prime.
 $359 < (20)^2$ and prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19.
And, 359 is not exactly divisible by any of them. Hence, 359 is a prime number.
70. Putting $n = 1, 2, 3, 4$ respectively in $(2n + 1)$ we get:
 $(2 \times 1 + 1) = 3$, $(2 \times 2 + 1) = 5$, $(2 \times 3 + 1) = 7$ and $(2 \times 4 + 1) = 9$,
where 3, 5, 7 are prime numbers.
 $\therefore$ The smallest value of n for which $(2n + 1)$ is not prime, is $n = 4$.
71. Clearly, 100 is divisible by 2. So, 100 is not prime.
 $(101) < (11)^2$ and prime numbers less than 11 are 2, 3, 5, 7. Clearly, 101 is not divisible by any of 2, 3, 5 and 7.
Hence, 101 is the smallest 3-digit prime number.
72. Each one of 112, 114, 116, 118 is divisible by 2. So, none is prime.
Each one of 111, 114, 117 is divisible by 3. So, none is prime.
Clearly, 115 is divisible by 5. So, it is not prime.
Each one of 112 and 119 is divisible by 7. So, none is prime.
Hence, there is only 1 prime number between 110 and 120, which is 113.
73. Let the given prime numbers be p, q, r and s . Then
 $p \times q \times r = 385$ and $q \times r \times s = 1001$
 $\Rightarrow \frac{p \times q \times r}{q \times r \times s} = \frac{385}{1001} = \frac{5}{13} \Rightarrow \frac{p}{s} = \frac{5}{13} \Rightarrow p = 5$ and $s = 13$.
Hence the largest of these prime numbers is 13.
74. Let the required prime numbers be $x, y, y + 36$. Then,
 $x + y + y + 36 = 100 \Rightarrow x + 2y = 64$
Let $x = 2$. Then, $2y = 62 \Rightarrow y = 31$.
So, these prime numbers are 2, 31 and 67.
In given choices 67 is the answer.
75. $\sqrt{437} > 20$
All prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19. 161 is divisible by 7, 221 is divisible by 13 and 437 is divisible by 19. 373 is not divisible by any of the above prime numbers.
 $\therefore$ 373 is prime.
76. The required arithmetic sequence of five prime numbers is 5, 11, 17, 23, 29 and therefore, the required 5th term is 29.
77. Each of the numbers 302, 303, 304, 305, 306, 308, 309, 310, 312, 314, 315, 316 and 318 is clearly a composite number.
Out of 307, 311, 313, 317 and 319 clearly every one is prime.
Hence, there are 5 prime numbers between 301 and 320.
78. Clearly, $3 \neq 4n + 1$ and $3 \neq 4n + 3$ for any natural number n .
 $\therefore$ Statement (1) is false.
Putting $p = 3, 5, 7, 11, 13, 17$ etc. we get:
 $(p - 1)(p + 1) = (2 \times 4), (4 \times 6), (6 \times 8), (10 \times 12), (12 \times 14), (16 \times 18)$ etc., each one of which is divisible by 4.
 $\therefore$ Statement (2) is true.
Hence, (1) is false and (2) is true.
79. $n = 1 \Rightarrow (n^2 + n + 41) = (1 + 1 + 41) = 43$, which is prime.
 $n = 10 \Rightarrow (n^2 + n + 41) = (100 + 10 + 41) = 151$, which is prime.
 $n = 20 \Rightarrow (n^2 + n + 41) = (400 + 20 + 41) = 461$, which is prime.
 $n = 40 \Rightarrow (n^2 + n + 41) = (1600 + 40 + 41) = 1681$, which is divisible by 41.
Thus, 1681 is not a prime number.
Hence $n = 40$ for which $(n^2 + n + 41)$ is not prime.

80. $X_2 = (2 \times 3) + 1 = 7$, which is prime and $X_2 + 1 = 8$, which is even.
 $X_3 = (2 \times 3 \times 5) + 1 = 31$, which is prime and $X_3 + 1 = 32$, which is even.
 $X_4 = (2 \times 3 \times 5 \times 7) + 1 = 211$, which is prime and $X_4 + 1 = 212$, which is even and so on.
Thus X_k is prime and $(X_k + 1)$ is even.
Hence, 1 and 3 are true statements.
81. $6 \times 3(3 - 1) = 6 \times 3(2) = 6 \times 6 = 36$.
82. Given Expression $= (1234 + 2345 + 4567) - 3456$
 $= (8146 - 3456) = 4690$.

$$\begin{array}{r} 1234 \\ 2345 \\ + 4567 \\ \hline 8146 \\ - 3456 \\ \hline 4690 \end{array}$$

83. Let $5566 - 7788 + 9988 = x + 4444$. Then
 $(5566 + 9988) - 7788 = x + 4444$
 $\Rightarrow 15554 - 7788 = x + 4444 \Rightarrow x + 4444 = 7766$
 $\Rightarrow x = (7766 - 4444) = 3322$.

$$\begin{array}{r} 7766 \\ - 4444 \\ \hline 3322 \end{array}$$

84. Given Expression $= 38649 - (1624 + 4483)$
 $= (38649 - 6107) = 32542$.
85. Given Expression $= 884697 - (773697 + 102479)$
 $= 884697 - 876176$
 $= 8521$.

$$\begin{array}{r} 884697 \\ - 876176 \\ \hline 8521 \end{array}$$

86. Let $10531 + 4813 - 728 = x \times 87$. Then
 $(15344 - 728) = 87 \times x \Rightarrow x = \frac{14616}{87} = 168$.

$$\begin{array}{r} 10531 \\ + 4813 \\ \hline 15344 \\ - 728 \\ \hline 14616 \end{array}$$

$$\begin{array}{r} 168 \\ 87 \overline{)14616} \\ \underline{87} \\ 591 \\ \underline{522} \\ 696 \\ \underline{696} \\ \hline \end{array}$$

87. $394 \times 113 = 394 \times (100 + 10 + 3)$
 $= (394 \times 100) + (394 \times 10) + (394 \times 3)$
 $= 39400 + 3940 + 1182$
 $= 44522$.

$$\begin{array}{r} 39400 \\ 3940 \\ + 1182 \\ \hline 44522 \end{array}$$

88. $1260 \div 14 \div 9 = \left(1260 \times \frac{1}{14} \times \frac{1}{9}\right) = 10$.

89. $136 \times 12 \times 8 = 136 \times 96 = 136 \times (100 - 4)$
 $= (136 \times 100) - (136 \times 4)$
 $= 13600 - 544 = 13056$.

$$\begin{array}{r} 13600 \\ - 544 \\ \hline 13056 \end{array}$$

90. Let $8888 + 848 + 88 - x = 7337 + 737$.
Then $9824 - x = 8074$

$$\Rightarrow x = 9824 - 8074 \quad \Rightarrow x = 1750.$$

$\begin{array}{r} 9824 \\ - 8074 \\ \hline 1750 \end{array}$	$\begin{array}{r} 8888 \\ 848 \\ + 88 \\ \hline 9824 \end{array}$	$\begin{array}{r} 7337 \\ + 737 \\ \hline 8074 \end{array}$
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91. Let $414 \times x \times 7 = 127512$. Then

$$x = \frac{127512}{414 \times 7} = \frac{18216}{414} = \frac{2024}{46} = \frac{1012}{23} = 44.$$

$$\begin{array}{r} 44 \\ 23 \overline{)1012} \\ \underline{92} \\ 92 \\ \underline{92} \\ 0 \end{array}$$

92. $82540027 \times 43253 = 82540027 \times (40000 + 3000 + 200 + 50 + 3)$
 $= (82540027 \times 40000) + (82540027 \times 3000) + (82540027 \times 200)$
 $+ (82540027 \times 50) + (82540027 \times 3)$

$$\begin{array}{r} 3301601080000 \\ 247620081000 \\ 16508005400 \\ 4127001350 \\ + 247620081 \\ \hline 3570103787831 \end{array}$$

Shortcut Method:

Product of unit's digits of the given numbers $= 7 \times 3 = 21$
 Clearly, the required product will have 1 as the unit's digit, which is 3570103787831.

93. Let $\frac{46351 - 36418 - 4505}{x} = 1357$. Then

$$x = \frac{46351 - (36418 + 4505)}{1357}$$

$$\Rightarrow x = \frac{(46351 - 40923)}{1357} = \frac{5428}{1357} = 4.$$

$\begin{array}{r} 46351 \\ - 40923 \\ \hline 5428 \end{array}$	$\begin{array}{r} 36418 \\ + 4505 \\ \hline 40923 \end{array}$
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94. $6 \times 66 \times 666 = 6 \times (6 \times 11) \times (6 \times 111)$
 $= (6 \times 6 \times 6) \times (11 \times 111) = (216 \times 1221)$
 $= (1221 \times 216) = 1221 \times (200 + 10 + 6)$
 $= (1221 \times 200) + (1221 \times 10) + (1221 \times 6)$
 $= 244200 + 12210 + 7326 = 263736.$

$$\begin{array}{r} 244200 \\ 12210 \\ + 7326 \\ \hline 263736 \end{array}$$

95. $(+1) - (-1) = (+1 + 1) = 2.$

96.

$$\begin{array}{r} 888888 \\ 88888 \\ 8888 \\ 888 \\ 88 \\ + 8 \\ \hline 987648 \end{array}$$

97. Given Expression $= 17 + (-12) - 48 = 17 - 60 = -43.$

98. $\frac{60840}{234} = \frac{30420}{117} = \frac{3380}{13} = 260.$

99. Let $3578 + 5729 - x \times 581 = 5821$. Then

$$x \times 581 = (3578 + 5729) - 5821 \Rightarrow x \times 581 = (9307 - 5821)$$

$$= 3486 \Rightarrow x = \frac{3486}{581} = 6.$$

100. $-95 \div 19 = \frac{-95}{19} = -5.$

101. $12345679 \times 72 = 12345679 \times (70 + 2)$
 $= (12345679 \times 70) + (12345679 \times 2)$
 $= (864197530 + 24691358)$
 $= 888888888.$

$$\begin{array}{r} 864197530 \\ + 24691358 \\ \hline 888888888 \end{array}$$

102. $8899 - 6644 - 3322 = x - 1122 \Rightarrow 2255 - 3322 + 1122 = x$
 $\Rightarrow x = 3377 - 3322 = 55.$

103. Let $74844 \div x = 54 \times 63$. Then, $\frac{74844}{x} = 54 \times 63$
 $\Rightarrow x = \frac{74844}{54 \times 63} = \frac{8316}{6 \times 63} = \frac{1386}{63} = \frac{198}{9} = 22.$

104. $1256 \times 3892 = (1000 + 200 + 50 + 6) \times 3892$
 $= (1000 \times 3892) + (200 \times 3892) +$
 $(50 \times 3892) + (6 \times 3892)$
 $= 3892000 + 778400 + 194600 + 23352$
 $= 4888352.$

$$\begin{array}{r} 3892000 \\ 778400 \\ 194600 \\ + 23352 \\ \hline 4888352 \end{array}$$

105. $786 \times 964 = (800 - 14) \times 964$
 $= (800 \times 964) - (14 \times 964)$
 $= (771200 - 13496) = 757704.$

$$\begin{array}{r} 771200 \\ - 13496 \\ \hline 757704 \end{array}$$

106. $348 \times 265 = (350 - 2) \times 265 = (350 \times 265) - (2 \times 265)$
 $= \{(300 + 50) \times 265\} - 530 = (300 \times 265)$
 $+ (50 \times 265) - 530$
 $= 79500 + 13250 - 530 = 92750 - 530$
 $= 92220.$

107. $(71 \times 29 + 27 \times 15 + 8 \times 4) = (80 - 9) \times 29 + 405 + 32 =$
 $(80 \times 29) - (9 \times 29) + 437 = 2320 - 261 + 437 = 2757 - 261$
 $= 2496.$

108. Let $x \times (|a| \times |b|) = -ab$. Then, $x = \frac{-ab}{|ab|} = -1.$

109. Let $(46)^2 - x^2 = 4398 - 3066$
 Then, $(46)^2 - x^2 = 1332 \Rightarrow x^2 = (46)^2 - 1332$
 $\therefore x^2 = (50 - 4)^2 - 1332 = (50)^2 + 4^2 - 2 \times 50 \times 4 - 1332$
 $\Rightarrow x^2 = 2500 + 16 - 400 - 1332 = 2516 - 1732 = 784$
 $\Rightarrow x = \sqrt{784} = 28.$

$\begin{array}{r} 2 \overline{)784} \quad (28 \\ 4 \\ \underline{4} \\ 384 \\ 384 \\ \hline 0 \end{array}$	$\begin{array}{r} 2516 \\ - 1732 \\ \hline 784 \end{array}$
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110. $(800 \div 64) \times (1296 \div 36) = \frac{800}{64} \times \frac{1296}{36} = 450.$

$$\begin{aligned} \text{111. } 5358 \times 51 &= 5358 \times (50 + 1) = 5358 \times 50 + 5358 \times 1 \\ &= \frac{5358 \times 100}{2} + 5358 \\ &= \frac{535800}{2} + 5358 = 267900 + 5358 = 273258. \end{aligned}$$

112. $587 \times 999 = 587 \times (1000 - 1) = (587 \times 1000) - (587 \times 1)$
 $= 587000 - 587 = 586413.$

113. $3897 \times 999 = 3897 \times (1000 - 1) = (3897 \times 1000) - (3897 \times 1)$
 $= 3897000 - 3897 = 3893103.$

114. $72519 \times 9999 = 72519 \times (10000 - 1)$
 $= (72519 \times 10000) - (72519 \times 1)$
 $= 725190000 - 72519 = 725117481.$

$$\begin{array}{r} 725190000 \\ - 72519 \\ \hline 725117481 \end{array}$$

115. $2056 \times 987 = 2056 \times (1000 - 13)$
 $= (2056 \times 1000) - (2056 \times 13)$
 $= 2056000 - 26728 = 2029272.$

116. $1904 \times 1904 = (1904)^2 = (1900 + 4)^2$
 $= (1900)^2 + 4^2 + 2 \times 1900 \times 4$
 $= 3610000 + 16 + 15200 = 3625216.$

117. $1397 \times 1397 = (1397)^2 = (1400 - 3)^2$
 $= (1400)^2 + 3^2 - 2 \times 1400 \times 3 = 1960000 + 9 - 8400$
 $= 1951609.$

118. $(107 \times 107) + (93 \times 93) = (107)^2 + (93)^2$
 $= (100 + 7)^2 + (100 - 7)^2$
 $= (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
 $= 2[(100)^2 + 7^2] = 2[10000 + 49]$
 $= 2 \times 10049 = 20098.$

119. $(217 \times 217) + (183 \times 183) = (217)^2 + (183)^2$
 $= (200 + 17)^2 + (200 - 17)^2 = (a + b)^2 + (a - b)^2$
 $= 2(a^2 + b^2)$, where $a = 200$, $b = 17$
 $= 2 [(200)^2 + (17)^2] = 2 [40000 + 289]$
 $= (2 \times 40289) = 80578.$

120. $(106 \times 106 - 94 \times 94) = (106)^2 - (94)^2$
 $= (106 + 94)(106 - 94) = (200 \times 12) = 2400.$

$$\begin{aligned} \text{121. } (8796 \times 223 + 8796 \times 77) &= 8796 \times (223 + 77) \\ &\quad \text{[by distributive law]} \\ &= (8796 \times 300) = 2638800. \end{aligned}$$

$$\begin{aligned} 122. \quad & (287 \times 287 + 269 \times 269 - 2 \times 287 \times 269) \\ &= (287)^2 + (269)^2 - (2 \times 287 \times 269) \\ &= (287 - 269)^2 = (18)^2 = 324. \\ & \quad [\because a^2 + b^2 - 2ab = (a - b)^2] \end{aligned}$$

123. $(476 + 424)^2 - 4 \times 476 \times 424 = (a + b)^2 - 4ab = (a - b)^2$
 where $a = 476$ & $b = 424 = (476 - 424)^2 = (52)^2 = (50 + 2)^2$
 $= (50)^2 + 2^2 + 2 \times 50 \times 2$
 $= (2500 + 4 + 200) = 2704.$

124. $(112 \times 5^4) = 112 \times \left(\frac{10}{2}\right)^4 = \frac{112 \times 10000}{16} = 70000.$

$$\begin{aligned} \text{125. } (5746320819 \times 125) &= \frac{5746320819 \times (125 \times 8)}{8} \\ &= \frac{5746320819 \times 1000}{8} = \frac{5746320819000}{8} = 718290102375. \end{aligned}$$

$$\begin{aligned} \text{126. } 935421 \times 625 &= \frac{935421 \times 625 \times 16}{16} = \frac{935421 \times 10000}{16} \\ &= \frac{9354210000}{16} = 584638125. \end{aligned}$$

127. $(999)^2 - (998)^2 = (999 + 998)(999 - 998) = (1997 \times 1) = 1997.$

128. $(80)^2 - (65)^2 + 81 = (80 + 65)(80 - 65) + 81 = (145 \times 15) + 81 = (2175 + 81) = 2256.$

129. $(24 + 25 + 26)^2 - (10 + 20 + 25)^2 = (75)^2 - (55)^2$
 $= (75 + 55)(75 - 55) = (130 \times 20) = 2600.$

130. $(65)^2 - (55)^2 = (65 + 55)(65 - 55) = (120 \times 10) = 1200.$

131. $(a^2 - b^2) = 19 \Rightarrow (a + b)(a - b) = 19$.
Clearly, $a = 10$ and $b = 9$.

132. Since $(a + b)^2 - (a - b)^2 = 4ab$, so the given expression should be a multiple of 4.

So, the least value of $4ab$ is 32 and so the least value of ab is 8.

Hence, the smallest value of a is 4 and that of b is 2.
Hence, $a = 4$.

133. Given Expression = $(397)^2 + (104)^2 + 2 \times 397 \times 104$
 $= (397 + 104)^2 = (501)^2 = (500 + 1)^2$
 $= (500)^2 + 1^2 + 2 \times 500 \times 1 = 250000 + 1 + 1000 = 251001.$

134. $(64)^2 - (36)^2 = 20 \times x \Rightarrow (64 + 36)(64 - 36) = 20 \times x$
 $\Rightarrow x = \frac{100 \times 28}{20} = 140.$

135. Given Expression

$$= \frac{(a+b)^2 - (a-b)^2}{ab} \text{ (where } a = 489, b = 375) = \frac{4ab}{ab} = 4$$

136. Given Expression = $\frac{(a+b)^2 + (a-b)^2}{(a^2 + b^2)}$,
 (where $a = 963$ and $b = 476$) = $\frac{2(a^2 + b^2)}{(a^2 + b^2)} = 2$.

137. Given Expression

$$= \frac{(a^3 + b^3)}{(a^2 - ab + b^2)}, (\text{where } a = 768 \text{ and } b = 232) = (a + b)$$
$$= (768 + 232) = 1000.$$

$$\begin{aligned} &= (768 + 232) = 1000. \\ \text{138. Given Expression} &= \frac{(854)^3 - (276)^3}{(854)^2 + (854 \times 276) + (276)^2} \\ &= \frac{(a^3 - b^3)}{(a^2 + ab + b^2)}, \text{ where } a = 854 \text{ and } b = 276 \\ &= (a - b) = (854 - 276) = 578. \end{aligned}$$

139. Given Expression = $\frac{(753)^2 + (247)^2 - (753 \times 247)}{(753)^3 + (247)^3}$

$$= \frac{(a^2 + b^2 - ab)}{(a^3 + b^3)}, \text{ where } a = 753 \text{ and } b = 247$$
$$= \frac{1}{(a + b)} = \frac{1}{(753 + 247)} = \frac{1}{1000}.$$

140. Given Expression = $\frac{(256)^2 - (144)^2}{112} = \frac{(256 + 144)(256 - 144)}{112}$
 $= \frac{(400 \times 112)}{112} = 400.$

$$141. \frac{(a^2 + b^2 + ab)}{(a^3 - b^3)} = \frac{a^2 + b^2 + ab}{(a - b)(a^2 + b^2 + ab)} = \frac{1}{(a - b)} = \frac{1}{(11 - 9)} = \frac{1}{2}.$$

142. $a + b + c = 0 \Rightarrow a + b = -c, (b + c) = -a$ and $(c + a) = -b$
 $\Rightarrow (a + b)(b + c)(c + a) = (-c) \times$
 $(-a) \times (-b) = -(abc).$

$$\begin{aligned} \text{143. } (a^2 + b^2 + c^2 - ab - bc - ca) &= (a + b + c)^2 - 3(ab + bc + ca) \\ &= (7 + 5 + 3)^2 - 3(35 + 15 + 21) \\ &= (15)^2 - 3 \times 71 \\ &= (225 - 213) = 12. \end{aligned}$$

144. Both addition and multiplication of numbers are commutative and associative.

145. $9H + H8 + H6 = 230 \Rightarrow \{(9 \times 10) + H\} + (10H + 8) + (10H + 6) = 230$
 $\Rightarrow 21H + 104 = 230 \Rightarrow 21H = 126$
 $\Rightarrow H = 6.$

146. Let the missing digit be x . Then, 1 (carried over) + 3 + x + $x = 10 + x \Rightarrow x = 6.$

147. Clearly, $M = 0$ since $304 \times 4 = 1216.$

148. $5p9 + 327 + 2q8 = 1114 \Rightarrow (500 + 10p + 9) + (327) + (200 + 10q + 8) = 1114$

$$\begin{aligned} \Rightarrow 10(p + q) + 1044 &= 1114 \\ \Rightarrow 10(p + q) &= 70 \Rightarrow (p + q) = 7 \\ \Rightarrow \text{Maximum value of } q &\text{ is } 7 \\ [\text{As minimum value of } p &= 0] \end{aligned}$$

149. $5P7 + 8Q9 + R32 = 1928 \Rightarrow (500 + 10P + 7) + (800 + 10Q + 9) + (100R + 30 + 2) = 1928$

$$\begin{aligned} \Rightarrow 10P + 10Q + 100R + 1348 &= 1928 \\ \Rightarrow 10(P + Q + 10R) &= 580 \\ \Rightarrow P + Q + 10R &= 58 \Rightarrow R = 5 \text{ and} \end{aligned}$$

$$\begin{aligned} P + Q &= 8 \text{ or } R = 4 \text{ and } P + Q = 18 \\ \Rightarrow \text{Maximum value of } Q &\text{ is } 9 \\ [\text{for } P = 9 \text{ in second case}] \end{aligned}$$

150. Let $\frac{1x5y4}{148} = 78.$

$$\begin{aligned} \text{Then, } 10000 + 1000x + 500 + 10y + 4 &= 148 \times 78 \\ \Rightarrow 10000 + 1000x + 500 + 10y + 4 &= 11544 \\ &= 10000 + 1000 + 500 + 40 + 4 \\ \Rightarrow 1000x &= 1000 \Rightarrow x = 1. \end{aligned}$$

151. Let $6x43 - 46y9 = 1904.$

Clearly, $y = 3$ and $x = 5.$

Hence * must be replaced by 5.

152. $5P9 - 7Q2 + 9R6 = 823$

$$\begin{aligned} \Rightarrow (500 + 10P + 9) - (700 + 10Q + 2) + (900 + 10R + 6) &= 823 \\ \Rightarrow (500 + 900 - 700) + 10(P + R - Q) + (9 + 6 - 2) &= 823 \\ \Rightarrow 700 + 10(P + R - Q) &= 810 = 700 + 110 \\ \Rightarrow 10(P + R - Q) &= 110 \\ \Rightarrow P + R - Q &= 11 \\ \Rightarrow Q &= (P + R - 11). \end{aligned}$$

To get maximum value of Q we take $P = 9$ and $R = 9.$

This gives $Q = (9 + 9 - 11) = 7.$

Hence, the maximum value of Q is 7.

153. Let the required digit be x . Then

$$\begin{aligned} x + 1x + 2x + x3 + x1 &= 21x \\ \Rightarrow x + 10 + x + 20 + x + 10x + 3 + 10x + 1 &= 200 + 10 + x \\ \Rightarrow 22x &= 210 - 34 = 176 \Rightarrow x = 8. \end{aligned}$$

Hence, the required digit is 8.

154. It is given that $D = 0$. So, we have $A = 5.$

So, 1 is carried over.

$$1 + B + C = 10 + C \Rightarrow 1 + B = 10 \Rightarrow B = 9.$$

Now, $1 + C + C = A \Rightarrow 1 + 2C = 5 \Rightarrow 2C = 4 \Rightarrow C = 2.$

Hence $B = 9.$

$$\begin{array}{r} \textcircled{1} \textcircled{1} \\ CBA = 5 \\ + CCA = 5 \\ \hline AC0 \end{array}$$

155. We have, $C = 2.$

156. Since 13b7 is divisible by 11, we have

$$(7 + 3) - (b + 1) = 0 \Rightarrow 9 - b = 0 \Rightarrow b = 9.$$

Putting $b = 9$, $a + 8 = 9$ we get $a = 1$. Hence, $(a + b) = (1 + 9) = 10.$

$$\begin{array}{r} 4a3 \\ 984 \\ \hline 13b7 \end{array}$$

157. Clearly, we have $ab \times b = 24$ and $ab \times a = 12.$

$$\therefore \frac{ab \times b}{ab \times a} = \frac{24}{12} \Rightarrow \frac{b}{a} = \frac{2}{1} \Rightarrow a = 1, b = 2.$$

158. Clearly, $111 \times 1 = 111 \neq 8111.$

But, $999 \times 9 = 8991$. Hence, we have 9 in place of *.

159. $(1 * 2) = 1 + 6 \times 2 = 1 + 12 = 13.$

$$(1 * 2) * 3 = 13 * 3 = 13 + 6 \times 3 = 13 + 18 = 31.$$

160. $|\underline{8} - \underline{7} - \underline{6}| = 8 \times 7 \times |\underline{6} - 7 \times \underline{6} - \underline{6}|$

$$= (56 - 7 - 1) \times |\underline{6} - 48 \times \underline{6} - 6 \times 8 \times \underline{6}|.$$

$$\begin{aligned} 161. \text{ Highest power of 3 in } 99! &= \left[\frac{99}{3} \right] + \left[\frac{99}{3^2} \right] + \left[\frac{99}{3^3} \right] + \left[\frac{99}{3^4} \right] \\ &= \left[\frac{99}{3} \right] + \left[\frac{99}{9} \right] + \left[\frac{99}{27} \right] + \left[\frac{99}{81} \right] \\ &= 33 + 11 + 3 + 1 = 48. \end{aligned}$$

Since $9 = 3^2$, so highest power of 9 dividing $99! = \frac{48}{2} = 24.$

162. Every number from $\underline{5}$ onwards is completely divisible by 5.

$\therefore (\underline{5} + \underline{6} + \underline{7} + \dots + \underline{100})$ is completely divisible by 5.

And,

$$(\underline{1} + \underline{2} + \underline{3} + \underline{4}) = (1 + 2 + 3 \times 2 \times 1 + 4 \times 3 \times 2 \times 1) = (1 + 2 + 6 + 24) = 33.$$

Clearly, 33 when divided by 5 leaves a remainder 3.

Hence, $(\underline{1} + \underline{2} + \underline{3} + \underline{4} + \underline{5} + \dots + \underline{100})$ when divided by 5 leaves a remainder 3.

163. $6^{10} \times 7^{17} \times 11^{27} = (2 \times 3)^{10} \times 7^{17} \times 11^{27} = 2^{10} \times 3^{10} \times 7^{17} \times 11^{27}.$
 Number of prime factors in the given expression = $(10 + 10 + 17 + 27) = 64.$

164. $(30)^7 \times (22)^5 \times (34)^{11} = (2 \times 3 \times 5)^7 \times (2 \times 11)^5 \times (2 \times 17)^{11}$
 $= 2^{(7+5+11)} \times 3^7 \times 5^7 \times 11^5 \times 17^{11}$
 $= (2^{23} \times 3^7 \times 5^7 \times 11^5 \times 17^{11}).$

Number of prime factors = $(23 + 7 + 7 + 5 + 11) = 53.$

165. Let $x \times 48 = 173 \times 240$. Then, $x = \frac{173 \times 240}{48} = 173 \times 5 = 865.$

166. $(1000 + x) > (1000 \times x)$. Clearly, $x = 1.$

167. Let the original number be x . Then,

$$\frac{(x+7) \times 5}{9} - 3 = 12 \Rightarrow \frac{5x+35}{9} = 15 \Rightarrow 5x+35 = 135$$

$$\Rightarrow 5x = 100 \Rightarrow x = 20.$$

168. 38950 is completely divisible by 410 only.

Hence ₹ 410 is the correct answer.

169. Let the four consecutive even numbers be $a, a + 2, a + 4$ and $a + 6.$

$$\text{Then, } a + a + 2 + a + 4 + a + 6 = 180 \Rightarrow 4a = 168$$

$$\Rightarrow a = 42.$$

So, these numbers are 42, 44, 46 and 48.

Sum of next four consecutive even numbers = $(50 + 52 + 54 + 56) = 212.$

170. Number on thumbs = 1, 9, 17, 25,

This is an AP in which $a = 1$ and $d = (9 - 1) = 8.$

$$\therefore T_n = a + (n-1)d = 1 + 8(n-1) = (8n-7).$$

$$8n-7 = 1994 \Rightarrow 8n = 2001 \Rightarrow n = 250.$$

$$T_{250} = 1 + (250-1) \times 8 = 1 + 249 \times 8 = 1993.$$

So, 1993 lies on thumb and 1994 on index finger.

- 171.** Out of four consecutive integers two are even and therefore, their product is even and on adding 1 to it, we get an odd integer. So, n is odd. Some possible values of n are as under:

$$n = 1 + (1 \times 2 \times 3 \times 4) = (1 + 24) = 25 = 5^2,$$

$$n = 1 + (2 \times 3 \times 4 \times 5) = (1 + 120) = 121 = (11)^2,$$

$$n = 1 + (3 \times 4 \times 5 \times 6) = (1 + 360) = 361 = (19)^2,$$

$$n = 1 + (4 \times 5 \times 6 \times 7) = 841 = (29)^2 \text{ and so on.}$$

Hence, n is odd and a perfect square.

- 172.** $(x^2 + y^2) = (x + y)^2 - 2xy = (15)^2 - 2 \times 56$
 $= (225 - 112) = 113.$
- 173.** $(2^2 + 4^2 + 6^2 + \dots + 40^2) = (1 \times 2)^2 + (2 \times 2)^2 + (3 \times 2)^2$
 $+ \dots + (20 \times 2)^2 = 2^2 \times (1^2 + 2^2 + 3^2 + \dots + 20^2)$
 $= (4 \times 2870) = 11480.$
- 174.** $5^2 + 6^2 + \dots + 10^2 + 20^2 = (1^2 + 2^2 + 3^2 + \dots + 10^2) - (1^2 + 2^2 + 3^2 + 4^2) + 400$
 $= \frac{1}{6} n(n+1)(2n+1) - (1+4+9+16) + 400, \text{ where } n = 10$
 $= \left(\frac{1}{6} \times 10 \times 11 \times 21 \right) - 30 + 400 = (385 - 30 + 400) = 755.$
- 175.** $(6 + 12 + 18 + 24 + \dots + 60) = 6 \times (1 + 2 + 3 + 4 + \dots + 10) = 6 \times 55 = 330.$
- 176.** $2^m > 960$ when the least value of m is 10.
 Then, $2^{10} = 1024$ and $1024 - 960 = 64 = 2^6$.
 $\therefore m = 10$ and $n = 6.$
- 177.** We keep on dividing 33333... by 7 till we get 0 as remainder.

$$\begin{array}{r} 47619 \\ 7 \overline{)333333} \\ \underline{28} \\ 53 \\ \underline{49} \\ 43 \\ \underline{42} \\ 13 \\ \underline{7} \\ 63 \\ \underline{63} \\ \times \end{array}$$

$\therefore$ Required number = 47619

- 178.** We keep on dividing 99999... by 7 till we get 0 as remainder.

$$\begin{array}{r} 142857 \\ 7 \overline{)999999} \\ \underline{7} \\ 29 \\ \underline{28} \\ 19 \\ \underline{14} \\ 59 \\ \underline{56} \\ 39 \\ \underline{35} \\ 49 \\ \underline{49} \\ \times \end{array}$$

$\therefore$ Required number = 142857.

Number of digits = 6.

$$\begin{array}{r} 56 \\ 987 \overline{)559981} \\ \underline{4935} \\ 6648 \\ \underline{5922} \\ 7261 \\ \underline{2961} \\ \times \end{array}$$

Clearly, 7261 must be replaced by 2961, which is possible if 6648 is replaced by 6218, which in turn is possible if 5599 is replaced by 5556.

Thus, the correct number is 555681.

- 180.** Clearly, multiples of 2 and 5 together yield 0.
 Since the product of odd numbers contains no power of 2, so the given product does not give 0 at the unit place.
- 181.** Let $N = 1 \times 2 \times 3 \times 4 \times \dots \times 1000 = 1000!$
 Clearly, the highest power of 2 in N is very high as compared to that of 5.
 So, the number of zeros in N will be equal to the highest power of 5 in N .

$\therefore$ Required number of zeros

$$= \left[\frac{1000}{5} \right] + \left[\frac{1000}{5^2} \right] + \left[\frac{1000}{5^3} \right] + \left[\frac{1000}{5^4} \right]$$

$$= 200 + 40 + 8 + 1 = 249.$$

- 182.** Let $N = 10 \times 20 \times 30 \times \dots \times 1000 = 10^{100} \times (1 \times 2 \times 3 \times 4 \times \dots \times 100) = 10^{100} \times 100!$
 Number of zeros in $100!$ = Highest power of 5 in $100!$
 $= \left[\frac{100}{5} \right] + \left[\frac{100}{5^2} \right] = 20 + 4 = 24.$
 $\therefore$ Number of zeros in $N = 100 + 24 = 124.$

- 183.** Let $N = 5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$
 $= 5^{10} \times (1 \times 2 \times 3 \times 4 \times \dots \times 10) = 5^{10} \times 10!$

$$\text{Highest power of 2 in } 10! = \left[\frac{10}{2} \right] + \left[\frac{10}{2^2} \right] + \left[\frac{10}{2^3} \right] = 5 + 2 + 1 = 8.$$

$$\text{Highest power of 5 in } 10! = \left[\frac{10}{5} \right] = 2.$$

$$\therefore N = 2^8 \times 5^{12} \times k.$$

Since highest power of 2 is less than that of 5, so required number of zeros = 8.

- 184.** Clearly, highest power of 2 is much higher as compared to that of 5 in $60!$, so

Required number of zeros = Highest power of

$$5 = \left[\frac{60}{5} \right] + \left[\frac{60}{5^2} \right] = 12 + 2 = 14.$$

- 185.** Let $N = (1 \times 3 \times 5 \times 7 \times \dots \times 99) \times 128.$

Clearly, N contains 10 multiples of 5 (5, 15, 25, 35, ..., 95) and only one multiple of 2 i.e. 128 or 2^7 .

Clearly, highest power of 5 in N is greater than that of 2.
 $\therefore$ Number of zeros in N = Highest power of 2 in $N = 7.$

- 186.** $N = 2 \times 4 \times 6 \times 8 \times \dots \times 98 \times 100$

$$= 2^{50} \times (1 \times 2 \times 3 \times \dots \times 49 \times 50) = 2^{50} \times 50!$$

Clearly, the highest power of 2 in N is much higher than that of 5.

$\therefore$ Number of zeros in N = Highest power of 5 in $N =$

$$\left[\frac{50}{5} \right] + \left[\frac{50}{5^2} \right] = 10 + 2 = 12.$$

- 187.** Clearly, the list of prime numbers from 2 to 99 has only 1 multiple of 2 and only 1 multiple of 5.
 So, number of zeros in the product = 1.

- 188.** Let $N = 3 \times 6 \times 9 \times 12 \times \dots \times 102 = 3^{34} \times (1 \times 2 \times 3 \times 4 \times \dots \times 34) = 3^{34} \times 34!$
Clearly, highest power of 2 in $34!$ is much greater than that of 5.
So, number of zeros in $N =$ Highest power of 5 in $34! = \left[\frac{34}{5} \right] + \left[\frac{34}{5^2} \right] = 6 + 1 = 7$.
- 189.** 3^4 gives unit digit 1. So, $(3^4)^{500}$ gives unit digit 1.
And, 3^3 gives unit digit 7.
 $\therefore (13)^{2003}$ gives unit digit $= (1 \times 7) = 7$.
- 190.** 3^4 gives unit digit 1. So, $(3^4)^{24}$ gives unit digit 1.
And, 3^3 gives unit digit 7.
 $\therefore 3^{99} = (3^4)^{24} \times 3^3$ gives unit digit (1×7) i.e. 7.
- 191.** (A) 7^4 gives unit digit 1. So, $7^{16} = (7^4)^4$ gives unit digit 1.
 $\therefore (1827)^{16}$ gives unit digit 1. So, $A \rightarrow (1)$.
(B) 3^4 gives unit digit 1. So, $(3^4)^4$ gives unit digit 1.
 $\therefore 3^{19} = 3^{16} \times 3^3$ gives unit digit $= (1 \times 7) = 7$.
 $\therefore (2153)^{19}$ gives unit digit 7. So, $B \rightarrow (4)$.
(C) 9^2 gives unit digit 1. So, $(9^2)^{10}$ gives unit digit 1.
 $\therefore 9^{21} = (9^2)^{10} \times 9$ gives unit digit $= (1 \times 9) = 9$.
 $\therefore (5129)^{21}$ gives unit digit $= 9$. So, $C \rightarrow (5)$.
Hence, $A \quad B \quad C$ is the correct result.
 $\quad \quad \quad 1 \quad 4 \quad 5$
- 192.** Unit digit of $(67)^{25} =$ Unit digit of 7^{25} .
Unit digit of 7^4 is 1 and so the unit digit of $(7^4)^6$ is 1.
 $\therefore$ Unit digit of $7^{25} = (1 \times 7) = 7$.
Hence, the unit digit of $(7^{25} - 1)$ is $(7 - 1) = 6$.
- 193.** Unit digit in the given product $=$ Unit digit of $(4 \times 8 \times 7 \times 3)$, which is 2.
- 194.** Unit digit in the given product $= 8$.
Unit digit of $(9 \times 6 \times x \times 4)$ is 8. So, $x = 3$.
- 195.** Unit digit of 7^4 is 1. So, the unit digit of $(7^4)^{23}$ is 1.
 $\therefore$ Unit digit of $7^{95} =$ Unit digit of $(7^{92} \times 7^3) = (1 \times 3) = 3$.
Unit digit of 3^4 is 1. So, the unit digit of $(3^4)^{14}$ is 1.
 $\therefore$ Unit digit of $3^{58} =$ Unit digit of $(3^{56} \times 3^2) = (1 \times 9) = 9$.
Hence, the unit digit of $(7^{95} - 3^{58}) = (3 - 9) = 4$.
- 196.** Unit digit of 4^2 is 6. So, unit digit in $(4^2)^{63}$ is 6.
 $\therefore$ Unit digit of $(784)^{126} =$ Unit digit in 4^{126} , which is 6.
Unit digit in $4^{127} =$ Unit digit in $(4^{126} \times 4) =$ Unit digit in (6×4) , which is 4.
 $\therefore$ Unit digit in $(784)^{127}$ is 4.
Hence, unit digit of $\{(784)^{126} + (784)^{127}\} =$ Unit digit of $(6 + 4) =$ Unit digit of 10, which is 0.
- 197.** Unit digit in $[(251)^{98} + (21)^{29} - (106)^{100} + (705)^{35} - (16)^4 + 259] =$ Unit digit in $(1 + 1 - 6 + 5 - 6 + 9) = 4$.
- 198.** Unit digit in the given product
 $=$ Unit digit in $(4^{1793} \times 5^{317} \times 1^{491})$
Unit digit in 4^2 is 6 and so the unit digit in $(4^2)^{896}$ is 6.
 $\therefore$ Unit digit of $4^{1793} =$ Unit digit in $(6 \times 4) =$ Unit digit in 24, which is 4.
Unit digit in 5^{317} is 5 and the unit digit in 1^{491} is 1.
 $\therefore$ Unit digit in the given product $=$ Unit digit in $(4 \times 5 \times 1)$, which is 0.
- 199.** Let $x = 2y$. Then, $x^{4n} = (2y)^{4n} = \{(2y)^4\}^n = (16y^4)^n$.
 $y = 1, 2, 3, 4, 5, 6, 7, 8, 9$ gives unit digit as 6 in $(16y^4)^n$.
But, $y = 5$, gives unit digit 0 in $(16y^4)^n$.
Hence, the unit digit is 0 or 6.
- 200.** In 5^n we have 5 as unit digit and in 6^m we have 6 as unit digit.
 $\therefore$ Unit digit in $(5^n + 6^m) =$ Unit digit in $(5 + 6) =$ Unit digit in 11 $= 1$.
- 201.** We have: $(2^{12n} - 6^{4n}) = (2^{12n} - 2^{4n} \times 3^{4n}) = 2^{4n} (2^{8n} - 3^{4n})$.
Putting $n = 1$, we get the number $2^4 (2^8 - 3^4)$
 $= 16 (256 - 81) = (16 \times 175) = 2800$
Hence, the number formed by last two digits is 00.
- 202.** $\left(\frac{1}{5}\right)^{2000} = (0.2)^{2000}$.
Last digit of $(0.2)^{2000} =$ Last digit of $(0.2)^4 = 6$.
- 203.** Sum of digits in the two numbers $= 19 + 15 = 34$.
So, the product will have 33 or 34 digits.
Since $36 \times 34 = 1224$ (i.e. product has $2 + 2 = 4$ digits), so the number of digits in x is 34.
- 204.** We know that a number having x, y, z as its digits, is a multiple of 11, if $z + x - y = 0$
Hence, $y = z + x$.
- 205.** $6135n2$ is divisible by 9 if $(6 + 1 + 3 + 5 + n + 2) = (17 + n)$ is divisible by 9.
This happens when the least value of n is 1.
- 206.** In 978626, we have $(6 + 6 + 7) - (2 + 8 + 9) = 0$.
Hence, 978626 is completely divisible by 11.
- 207.** Sum of all digits $= 12$, which is divisible by 3. So, the given number is divisible by 3.
(Sum of digits at odd places) $-$ (Sum of digits at even places) $= 6 - 6 = 0$.
So, the given number is divisible by 11.
The given number when divided by 37 gives 3003003003.
So, the given number is divisible by 37.
The given number when divided by 111 gives 1001001001.
Clearly, it is divisible by 111 as well as by 1001.
Hence, the given number is divisible by each one of 3, 11, 37, 111 and 1001.
- 208.** We have $18 = 2 \times 9$, where 2 and 9 are co-primes.
But, 65043 is not divisible by 2. So, it is not divisible by 18.
- 209.** Let the missing digit be x . Then, $(80 + x)$ must be divisible by 4. Hence, $x = 4$.
- 210.** Sum of the digits in 4006020 is 12, which is divisible by 3. Hence, 4006020 is divisible by 3.
- 211.** Clearly, (d) is true.
- 212.** For 4823718 we have
 $(8 + 7 + 2 + 4) - (1 + 3 + 8) = (21 - 12) = 9$, which is not a multiple of 11.
 $\therefore$ 4823718 is not divisible by 11.
Consider the number 4832718.
We have $(8 + 7 + 3 + 4) - (1 + 2 + 8) = (22 - 11) = 11$, which is a multiple of 11.
Hence, 4832718 is divisible by 11.
- 213.** Consider the number 17325.
Its unit digit is 5. So, it is divisible by 5.
Sum of its digits $= (5 + 2 + 3 + 7 + 1) = 18$, which is divisible by 3.
So, the given number is divisible by 3.
And, since 5 and 3 are co-primes, so the given number is divisible by (5×3) , i.e. 15.
- 214.** Given number is 7386038.
Sum of its digits $= 35$, which is not divisible by any of 3 and 9.
So, the given number is not divisible by any of 3 and 9.
Also, 38 is not divisible by 4. So, the given number is not divisible by 4.

Also, $(8 + 0 + 8 + 7) - (3 + 6 + 3) = (23 - 12) = 11$.

So, the given number is divisible by 11.

- 215.** $(2 + 4 + 9 + 8 + 4) = 27$, $(2 + 6 + 7 + 8 + 4) = 27$ and $(2 + 8 + 5 + 8 + 4) = 27$.

So, each one is divisible by 3 and 9 both.

Also, 84 is divisible by 4. So, each one is divisible by 4.

Hence, each given number is divisible by 3, 9 and 4.

So, the statements 1, 2 and 3 are all correct.

- 216.** We know that 872 is divisible by 8. Hence 923872 is divisible by 8.

- 217.** Here $(5 + 9 + x + 7) - (4 + 3 + 8) = 6 + x$. So, we must have $x = 5$.

- 218.** Take $m = 15$ and $n = 20$. Then, each one of m and n is divisible by 5. But, $(m + n)$ is not divisible by 10.

Hence, $(m + n)$ is divisible by 10 is not true.

- 219.** An integer is divisible by 16, if the number formed by last 4 digits is divisible by 16.

- 220.** Clearly, 639 is not divisible by 7.

Consider 2079. Sum of its digits $= (2 + 0 + 7 + 9) = 18$.

So, it is divisible by both 3 and 9.

Also, $(79 - 2) = 77$, which is divisible by 7.

So, 2079 is divisible by 7.

Also, $(9 + 0) - (7 + 2) = 0$. So, 2079 is divisible by 11.

Hence, 2079 is divisible by each one of 3, 7, 9 and 11.

- 221.** Let the given number be $476xy0$. Then

$$(0 + x + 7) - (y + 6 + 4) = 0 \Rightarrow x - y - 3 = 0 \Rightarrow x - y = 3.$$

Also, $(4 + 7 + 6 + x + y + 0) = (17 + x + y)$ must be divisible by 3.

Since $x \neq 0$, $y \neq 0$, so $x + y \neq 1$.

$\therefore x + y = 4$ or 7 or 10 etc.

$(x + y = 4 \text{ and } x - y = 3) \Rightarrow x = 7/2$, which is not admissible.

$(x + y = 7 \text{ and } x - y = 3) \Rightarrow x = 5$ and $y = 2$.

- 222.** Sum of the digits in respective numbers is:

9, 12, 18, 9, 21, 12, 18, 21, 15 and 24.

Out of these 12, 21, 12, 21, 15, 24 are divisible by 3 but not by 9.

So, the number of required numbers is 6.

- 223.** Let the unit's place be x and the thousand's place be y . Then, $357y25x$ is divisible by 5 only when $x = 0$ or $x = 5$. Also, this number is divisible by 3 only when sum of its digits is divisible by 3.

So, $(22 + x + y)$ must be divisible by 3.

$\therefore x + y = 2$

Taking $x = 0$, we get $y = 2$.

So, the unit place = 0 and thousand's place = 2.

- 224.** Clearly, $(7 + 8) - (9 + 6) = 0$. So, 6897 is divisible by 11.

Also, $\frac{6897}{19} = 363$. So, 6897 is divisible by 19.

Hence, 6897 is divisible by both 11 and 19.

- 225.** We have $24 = 3 \times 8$, where 3 and 8 are co-primes.

Clearly, 718 is not divisible by 8. So, 35718 is not divisible by 8.

810 is not divisible by 8. So, 63810 is not divisible by 8.

804 is not divisible by 8. So, 537804 is not divisible by 8.

736 is divisible by 8. So, 3125736 is divisible by 8.

Also, sum of its digits $= (3 + 1 + 2 + 5 + 7 + 3 + 6) = 27$, which is divisible by 3.

So, 3125736 is divisible by 3 also.

Hence, it is divisible by 24.

- 226.** Sum of the digits of the given number $= (7 \times 3) + (14 \times 1) = (21 + 14) = 35$, which is not divisible by 3.

So the given number is not divisible by 3.

Also, $(1 + 3 + 1 + 1 + 3 + 1 + 1 + 3 + 1 + 1 + 3) - (1 + 1 + 3 + 1 + 1 + 3 + 1 + 1 + 3 + 1) = (19 - 16) = 3$, which is neither 0 nor divisible by 11.

So, the given number is not divisible by 11 also.

Hence, it is divisible by neither 3 nor 11.

- 227.** $(325 - 325) = 0$, which is divisible by 7.

So, the given number is divisible by 7.

$(5 + 3 + 2) - (2 + 5 + 3) = 0$. So, the given number is divisible by 11.

And, $\frac{325325}{13} = 25025$. So, 325325 is divisible by 13.

Hence, it is divisible by all 7, 11 and 13.

- 228.** We first divide the number into groups of 3 digits from the right $\rightarrow 3 \text{ } 0X0 \text{ } 103$

Difference of sum of numbers at odd and even places $= (103 + 3) - 0X0 = 106 - 0X0$, which must be divisible by 13.

$106 - 0X0$ is divisible by 13 only for $X = 8$.

- 229.** We know that 11 and 13 are co-prime.

So, a number divisible by both 11 and 13 will be divisible by (11×13) .

- 230.** I. We have $(195 - 195) = 0$

$\therefore 195195$ is divisible by 7.

II. We have $(181 - 181) = 0$

$\therefore 181181$ is divisible by 7.

III. We have $(120 - 120) = 0$

$\therefore 120120$ is divisible by 7.

IV. We have $(891 - 891) = 0$

$\therefore 891891$ is divisible by 7.

Hence, all are divisible by 7.

- 231.** Since $653xy$ is divisible by 80, we must have $y = 0$.

Now, $653x0$ must be divisible by both 5 and 16.

Clearly, it is divisible by 5 for all values of x .

Now, the number $53x0$ must be divisible by 16.

The least value of x is clearly 6. So, $x + y = 6 + 0 = 6$.

- 232.** We know that $33 = 11 \times 3$, where 11 and 3 are co-primes.

So, the given number must be divisible by both 11 and 3.

Since $5ABB7A$ is divisible by 11, we have

$(A + B + A) - (7 + B + 5) = (2A - 12)$ is either 0 or 11.

$$\Rightarrow 2A - 12 = 0 \text{ or } 2A - 12 = 11 \Rightarrow A = 6 \left[\because A \neq \frac{23}{2} \right]$$

So, the number becomes $56BB76$, which is divisible by 3.

$\therefore (5 + 6 + B + B + 7 + 6) = (24 + 2B)$ must be divisible by 3.

$$\therefore 2B = 6 \Rightarrow B = 3 \quad \left[\because B \neq 0 \text{ and } B \neq \frac{3}{2} \right]$$

Hence, $(A + B) = (6 + 3) = 9$.

- 233.** We have, $99 = (11 \times 9)$, where 11 and 9 are co-primes.

Consider the number 114345.

Clearly, $(5 + 3 + 1) - (4 + 4 + 1) = 0$. So, 114345 is divisible by 11.

Also, sum of its digits $= (1 + 1 + 4 + 3 + 4 + 5) = 18$, which is divisible by 9.

$\therefore 114345$ is divisible by 9.

Hence, it is divisible by (11×9) , i.e. 99.

- 234.** Let the unit's digit be x and ten's digit be y . Then, the number is $3422213yx$.

Also, $99 = (11 \times 9)$, where 11 and 9 are co-primes.

Since the given number is divisible by 9, it follows that

$(3 + 4 + 2 + 2 + 2 + 1 + 3 + y + x) = (17 + y + x)$ must be divisible by 9.

So, $y + x = 1$ or $y + x = 10$.

Again, the given number is divisible by 11.

So, $(x + 3 + 2 + 2 + 3) - (y + 1 + 2 + 4) = x - y + 3$ is either 0 or 11.

$\therefore (x - y + 3 = 0 \text{ or } x - y + 3 = 11) \Rightarrow (y - x = 3 \text{ or } x - y = 8)$

Now, $(y + x = 1 \text{ and } y - x = 3) \Rightarrow y = 2 \text{ and } x = -1$.

$$(y + x = 1 \text{ and } x - y = 8) \Rightarrow x = \frac{9}{2}$$

$$(y + x = 10 \text{ and } y - x = 3) \Rightarrow y = \frac{13}{2}$$

$$(y + x = 10 \text{ and } x - y = 8) \Rightarrow x = 9 \text{ and } y = 1.$$

Thus, $x = 9, y = 1$. So, the required number is 342221319.

235. 37×3 is divisible by 7

$\Rightarrow (7 \times 3 - 3)$ is either 0 or divisible by 7

$\Rightarrow 7 \times 0$ is divisible by 7.

$\Rightarrow X = 0$ or $X = 7$.

236. $225 = 9 \times 25$, where 9 and 25 are co-primes.

So, a number is divisible by 225 if it is divisible by both 9 and 25.

Given number is divisible by 25, only if $7q$ is divisible by 25, i.e. if $q = 5$.

Sum of digits of given number $= (8 + 7 + 6 + p + 3 + 7 + 5) = 36 + p$, which must be divisible by 9.

This is possible if $p = 0$.

Hence, $p = 0, q = 5$.

237. Since 774958A96B is divisible by 8, so the number 96B must be divisible by 8. So, $B = 0$, as 960 is divisible by 8. Now, 774958A960 is divisible by 9. So, $(55 + A)$ must be divisible by 9.

This happens when $A = 8$. Hence, $A = 8$ and $B = 0$.

238. $132 = 11 \times 3 \times 4$. So, the required number must be divisible by 3, 4 and 11.

264 $\rightarrow$ divisible by 3, 4 and 11.

396 $\rightarrow$ divisible by 3, 4 and 11.

462 $\rightarrow$ not divisible by 4.

792 $\rightarrow$ divisible by 3, 4 and 11.

968 $\rightarrow$ not divisible by 3

2178 $\rightarrow$ not divisible by 4

5184 $\rightarrow$ not divisible by 11.

6336 $\rightarrow$ divisible by 3, 4 and 11.

Hence, out of the given numbers 4 are divisible by 3, 4 and 11.

239. Let $3x + 7y = 11k$. Then, $y = \frac{(11k - 3x)}{7}$.

$$\text{Then, } 5x - 3y = 5x - \frac{3(11k - 3x)}{7} = \frac{35x - 33k + 9x}{7}$$

$$= \frac{44x - 33k}{7} = \frac{11(4x - 3k)}{7}, \text{ which is divisible by 11.}$$

240. $(1^3 - 1) = 0, (2^3 - 2) = 6, (3^3 - 3) = 24, (4^3 - 4) = 60$ and so on, each one of which is divisible by 6.

241. Let $a = 2k + 1$ & $b = 2m + 1$. Then,

$$\begin{aligned} a^4 - b^4 &= (a^2 - b^2)(a^2 + b^2) = (a + b)(a - b)(a^2 + b^2) \\ &= (2k + 2m + 2)(2k - 2m)(4k^2 + 4m^2 + 2 + 4k + 4m) \\ &= 8(k + m + 1)(k - m)(2k^2 + 2m^2 + 1 + 2k + 2m), \end{aligned}$$

which is divisible by 8.

242. Let the two consecutive integers be a and $(a + 1)$. Then, $(a + 1)^2 - a^2 = a^2 + 1 + 2a - a^2 = (2a + 1) = (a + a + 1) =$ sum of given integers.

243. $6n^2 + 6n = 6n(n + 1)$

$$(n = 1 \Rightarrow 6n^2 + 6n = 12), (n = 2 \Rightarrow 6n^2 + 6n = 36), (n = 3 \Rightarrow 6n^2 + 6n = 72),$$

each one of which is divisible by 6 and 12 both.

244. Let the ten's digit be x and the unit's digit be y . Then, $(10x + y) - (10y + x) = 9x - 9y = 9(x - y)$, which is divisible by 9.

245. Let the ten's digit be x and the unit's digit be y .

Then, $(10x + y) + (10y + x) = 11(x + y)$, which is divisible by 11.

246. Let $P = n(n + 1)(n + 2)(n + 3)$. Then, $n = 1$ gives $P = (1 \times 2 \times 3 \times 4) = 24$.

Hence, the required number is 24.

247. Let $N = n^2(n^2 - 1) = n^2(n - 1)(n + 1)$.

$$\text{Then, } n = 2 \Rightarrow N = 2^2 \times (2 - 1) \times (2 + 1) = (4 \times 1 \times 3) = 12.$$

Hence, the required number is 12.

248. Let $n = (2m + 1)$. Then,

$$N = n(n^2 - 1) = n(n - 1)(n + 1) = (2m + 1)(2m)(2m + 2) = 4m(m + 1)(2m + 1).$$

$$\text{Now, } m = 1 \Rightarrow N = (4 \times 1 \times 2 \times 3) = 24.$$

So, N is always divisible by 24.

249. Let the hundreds, tens and unit digits be x, y and z respectively.

$$\text{Then, } (100x + 10y + z) - (x + y + z) = 99x + 9y = 9(11x + y).$$

So, the resulting number is divisible by 9.

250. Let the required number be x . Then $\frac{11x + 11}{13} = a$ a whole number.

So, $(11x + 11)$ must be divisible by 13. By hit and trial, we get $x = 12$.

Hence, the smallest original number is 12.

251. Let the required product be $n(n + 1)(n + 2)$. Then,

$$n = 1 \Rightarrow n(n + 1)(n + 2) = (1 \times 2 \times 3) = 6.$$

$$n = 2 \Rightarrow n(n + 1)(n + 2) = (2 \times 3 \times 4) = 24.$$

$$n = 3 \Rightarrow n(n + 1)(n + 2) = (3 \times 4 \times 5) = 60.$$

So, each such product is divisible by 6.

252. Let the required odd numbers be $n, (n + 2)$ and $(n + 4)$.

Then $n + (n + 2) + (n + 4) = 3n + 6 = 3(n + 2)$, which is always divisible by 3.

But $n = 1 \Rightarrow 3(n + 2) = 3 \times 3 = 9$ which is not divisible by any of 2, 5 and 6.

Hence only II is true.

253. Three consecutive multiples of 3 are $3m, 3(m + 1)$ and $3(m + 2)$.

$$\text{Their product} = 3m \times 3(m + 1) \times 3(m + 2) = 27 \times m \times (m + 1) \times (m + 2).$$

$$\text{Putting } m = 1, \text{ this product is } (27 \times 1 \times 2 \times 3) = 162.$$

So, this product is always divisible by 162.

254. $p = 5 \Rightarrow (p^2 - 1) = (25 - 1) = 24$, which is divisible by 24.

$$p = 7 \Rightarrow (p^2 - 1) = (49 - 1) = 48, \text{ which is divisible by 24.}$$

$$p = 11 \Rightarrow (p^2 - 1) = (121 - 1) = 120, \text{ which is divisible by 24.}$$

Hence, $(p^2 - 1)$ is always divisible by 24.

255. Let the two consecutive odd integers be $(2m + 1)$ and $(2m + 3)$.

$$\text{Then, } (2m + 3)^2 - (2m + 1)^2 = [(2m + 3) + (2m + 1)][(2m + 3) - (2m + 1)] = (4m + 4) \times 2 = 8m + 8 = 8(m + 1), \text{ which is always divisible by 8.}$$

- 256.** Clearly, 2525 is not divisible by any of the numbers 7, 11 and 13.

The smallest 3-digit prime number is 101.

$$\begin{array}{r} 25 \\ 101 \overline{)2525} \\ \underline{202} \\ 505 \\ \underline{505} \\ 0 \end{array} \quad \begin{array}{r} 32 \\ 101 \overline{)3232} \\ \underline{303} \\ 202 \\ \underline{202} \\ 0 \end{array}$$

Hence (d) is true.

- 257.** Numbers like 2525, 3636 etc. are divisible by 101.
Numbers like 256256, 678678 etc. are divisible by 1001.
Numbers like 32163216, 43754375 etc. are divisible by 10001 and so on.
- 258.** 10^n has $(n + 1)$ digits. Then, 9 will appear n times in $(10^n - 1)$.
So, sum of digits in $(10^n - 1) = 9n$.
 $\therefore 9n = 4707 \Rightarrow n = \frac{4707}{9} = 523$.
- 259.** We know that $(x^n - a^n)$ is always divisible by $(x - a)$ for all values of n .
- 260.** The product of 8 consecutive numbers is divisible by each one of $8!, 7!, 6!, 5!, 4!, 3!$ and $2!$
- 261.** We know that $(x^n - a^n)$ is divisible by $(x - a)$, when n is even and $(x^n - a^n)$ is divisible by $(x + a)$, when n is even.
 $\therefore (10^n - 1)$ is divisible by $(10 - 1) = 9$, when n is even.
And, $(10^n - 1)$ is divisible by $(10 + 1) = 11$, when n is even.
 $\therefore$ Statements 2 and 4 are correct.
- 262.** Putting $n = 1$, we get $(3^4n - 4^3n) = (3^4 - 4^3) = (81 - 64) = 17$, which is divisible by 17.
- 263.** Let the odd natural number be $(2n + 1)$.
 $n = 1$ gives $(2n + 1)^2 = (2 + 1)^2 = 9$.
This when divided by 8 gives 1 as remainder.
 $n = 2$ gives $(2n + 1)^2 = 25$.
This when divided by 8 gives 1 as remainder and so on.
- 264.** Required number $= (2^5 - 2) = (32 - 2) = 30$.
- 265.** Two consecutive even integers are $2n$ and $(2n + 2)$.
 $\therefore (2n + 2)^2 - (2n)^2 = (4n^2 + 4 + 8n - 4n^2) = 4 + 8n = 4(1 + 2n)$, which is always divisible by 4.
- 266.** Two consecutive odd integers are $(2m + 1)$ and $(2m + 3)$.
 $\therefore (2m + 3)^2 - (2m + 1)^2 = (2m + 3 + 2m + 1)(2m + 3 - 2m - 1) = (4m + 4) \times 2 = 8(m + 1)$, which is always divisible by 8.
- 267.** The smallest 4-digit number is 1000.
This when divided by 7 leaves 6 as remainder.
 $\therefore$ 1001 is the smallest 4-digit number exactly divisible by 7.
- 268.** On dividing 1056 by 23, we get 21 as remainder.
 $\therefore$ Required number to be added $= (23 - 21) = 2$.

$$\begin{array}{r} 23 \overline{)1056} (45 \\ \underline{92} \\ 136 \\ \underline{115} \\ 21 \end{array}$$

- 269.** On dividing 8567 by 4, the remainder is 3.
To make it divisible by 4, we must add 1 to it.
- 270.** The least 6-digit number $= 100000$.
Required number $= 100000 + (349 - 186) = 100000 + 163 = 100163$.

$$\begin{array}{r} 349 \overline{)100000} (286 \\ \underline{698} \\ 3020 \\ \underline{2792} \\ 2280 \\ \underline{2094} \\ 186 \end{array}$$

- 271.** The greatest 5-digit number $= 99999$.
On dividing 99999 by 279, we get 117 as remainder.
 $\therefore$ Required number $= (99999 - 117) = 99882$.

$$\begin{array}{r} 279 \overline{)99999} (358 \\ \underline{837} \\ 1629 \\ \underline{1395} \\ 2349 \\ \underline{2232} \\ 117 \end{array}$$

- 272.** The greatest 6-digit number is 999999.
Required number to be added $= (327 - 33) = 294$.

$$\begin{array}{r} 327 \overline{)999999} (3058 \\ \underline{981} \\ 1899 \\ \underline{1635} \\ 2649 \\ \underline{2616} \\ 33 \end{array}$$

- 273.** On dividing 5000 by 73, we get
Required number $= 5000 + (73 - 36) = 5037$.

$$\begin{array}{r} 73 \overline{)5000} (68 \\ \underline{438} \\ 620 \\ \underline{584} \\ 36 \end{array}$$

- 274.** On dividing 58701 by 567, we get 300 as remainder.
 $\therefore$ Required number $= 58701 + (567 - 300) = 58701 + 267 = 58968$.

$$\begin{array}{r} 567 \overline{)58701} (103 \\ \underline{567} \\ 2001 \\ \underline{1701} \\ 300 \end{array}$$

- 275.** On dividing 8112 by 99, we get 93 as remainder.
So, the required number to be subtracted is 93.

$$\begin{array}{r} 99 \overline{)8112} (81 \\ \underline{792} \\ 192 \\ \underline{99} \\ 93 \end{array}$$

- 276.** On dividing 803642 by 11, we get 4 as remainder.
Required number to be added $= (11 - 4) = 7$.
- 277.** On dividing 1111 by 99, the quotient is 11 and the remainder is 22.
Hence, the required number is 11.

$$\begin{array}{r} 99 \overline{)1111} (11 \\ \underline{99} \\ 121 \\ \underline{99} \\ 22 \end{array}$$

- 278.** $66 = 11 \times 6$
In order to get a number divisible by 18, the above product must be multiplied by 3.
Hence 66 must be multiplied by 3.
- 279.** The smallest 6-digit number is 100000. On dividing 100000 by 111, we get 100 as remainder.
So, the number to be added $= (111 - 100) = 11$.
Hence, the required number $= 100011$.

$$\begin{array}{r} 111 \overline{)100000} (900 \\ \underline{999} \\ 100 \end{array}$$

- 280.** All 2-digit numbers divisible by 5 are 10, 15, 20, 25, ..., 95.

This is an A.P. in which $a = 10$, $d = 5$ and $T_n = 95$.

$$T_n = 95 \Rightarrow a + (n - 1)d = 95 \Rightarrow 10 + (n - 1) \times 5 = 95$$

$$\Rightarrow (n - 1) = \frac{85}{5} = 17 \Rightarrow n = 18.$$

$$\therefore \text{Sum} = \frac{n}{2}(a + l) = \frac{18}{2}(10 + 95) = (9 \times 105) = 945.$$

- 281.** 3-digit numbers divisible by 6 are 102, 108, 114, ..., 996.

This is an A.P. in which $a = 102$, $d = 6$ and $T_n = 996$.

$$\therefore T_n = a + (n - 1)d \Rightarrow 102 + (n - 1) \times 6 = 996 \Rightarrow (n - 1) \times 6 = 894$$

$$\Rightarrow (n - 1) = 149 \Rightarrow n = 150.$$

Hence, there are 150 such numbers.

- 282.** Multiples of 7 between 11 and 200 are 14, 21, 28, 35, 42, ..., 189, 196.

$$T_m = 196 \Rightarrow 14 + (m - 1) \times 7 = 196 \Rightarrow (m - 1) \times 7 = 182 \Rightarrow (m - 1) = 26 \Rightarrow m = 27.$$

Multiples of 7 and 3 both, i.e. that of 21 are 21, 42, 63, ..., 189.

$$T_n = 189 \Rightarrow 21 + (n - 1) \times 21 = 189 \Rightarrow (n - 1) \times 21 = 168 \Rightarrow (n - 1) = 8 \Rightarrow n = 9.$$

$$\therefore \text{Required number of terms} = (27 - 9) = 18.$$

- 283.** Numbers between 14 and 95 and divisible by 3 are 15, 18, 21, 24, ..., 93.

$$T_m = 93 \Rightarrow 15 + (m - 1) \times 3 = 93 \Rightarrow (m - 1) \times 3 = 78$$

$$\Rightarrow (m - 1) = 26 \Rightarrow m = 27.$$

Numbers to be deleted are 33, 63, 93.

They are 3 in number.

$$\text{Required number of numbers} = (27 - 3) = 24.$$

- 284.** Required numbers are multiples of (10×13) , i.e. 130.

These numbers are 130, 260, 390, 520, 650, 780 and 910. They are 7 in number.

- 285.** Number of integers between 100 and 150 (including both) = 51.

Numbers divisible by 3 are 102, 105, 108, ..., 150.

Let $T_m = 150$. Then, $a + (m - 1)d = T_m$.

$$\therefore 102 + (m - 1) \times 3 = 150 \Rightarrow (m - 1) \times 3 = 48 \Rightarrow m - 1 = 16 \Rightarrow m = 17.$$

Numbers divisible by 5 are 100, 105, 110, 115, ..., 150.

Let $T_n = 150$. Then, $a + (n - 1)d = T_n$.

$$\therefore 100 + (n - 1) \times 5 = 150 \Rightarrow (n - 1) \times 5 = 50 \Rightarrow (n - 1) = 10 \Rightarrow n = 11.$$

Numbers divisible by both 3 and 5 are 105, 120, 135, 150. They are four in number.

$$\text{Number of numbers divisible by 3 or 5} = (17 + 11 - 4) = 24.$$

$$\text{Number of numbers neither divisible by 3 nor by 5} = (51 - 24) = 27.$$

- 286.** Numbers from 501 to 599 which have 6 as digit are 506, 516, 526, 536, 546, 556, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 576, 586 and 596, i.e. 6 occurs 20 times.

$$\text{Number of times 6 occurs from 600 to 699} = 100 + 20 = 120.$$

$$\therefore \text{Total number of times 6 occurs} = 20 + 120 = 140.$$

- 287.** Such numbers are 202, 212, 222, 232, 242, 252, 262, 272, 282, 292. There are 10 such numbers.

- 288.** Such numbers are 203, 213, 223, 233, 243, 253, 263, 273, 283, 293 and all numbers from 300 to 399. Clearly, number of such numbers = $10 + 100 = 110$.

- 289.** When the second digit is 1, third digit can be 0, i.e. there is one such number.

When the second digit is 2, third digit can be 0 or 1, i.e. there are 2 such numbers.

When the second digit is 3, third digit can be 0, 1 or 2 i.e. there are 3 such numbers, and so on.

When the first digit is 7, second digit can be 1, 2, 3, 4, 5 or 6. So, there are

$$(1 + 2 + 3 + 4 + 5 + 6) = 21 \text{ such numbers between 700 and 799.}$$

When the first digit is 8, second digit can be 1, 2, 3, 4, 5, 6 or 7. So, there are

$$(1 + 2 + 3 + 4 + 5 + 6 + 7) = 28 \text{ such numbers between 800 and 899.}$$

When the first digit is 9, second digit can be 1, 2, 3, 4, 5, 6, 7 or 8. So, there are

$$(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8) = 36 \text{ such numbers between 900 and 999.}$$

$$\text{Hence, the required number} = (21 + 28 + 36) = 85.$$

- 290.** (a) In 183654729, 1836547 is not divisible by 7.

(b) In 381654729, 38 is divisible by 2, 381 is divisible by 3, 3816 is divisible by 4, 38165 is divisible by 5, 381654 is divisible by 6, 3816547 is divisible by 7, 38165472 is divisible by 8 and 381654729 is divisible by 9.

(c) In 983654721, 983 is not divisible by 3.

(d) In 981654723, 9816547 is not divisible by 7.

- 291.** On dividing 11109999 by 1111, we get:

So, the required remainder is 1110.

$$\begin{array}{r} 1111 \overline{) 11109999} (9999 \\ \underline{9999} \\ 11109 \\ \underline{9999} \\ 11109 \\ \underline{9999} \\ 11109 \\ \underline{9999} \\ 1110 \end{array}$$

- 292.** Number = $(68 \times 260) = 17680$. On dividing this number by 65, we get zero as remainder.

$$\begin{array}{r} 65 \overline{) 17680} (272 \\ \underline{130} \\ 468 \\ \underline{455} \\ 130 \\ \underline{130} \\ 0 \end{array}$$

- 293.** $(2176 - 9) = 2167 = (11 \times 197)$.

So, the required number is 197.

294. (a) $\begin{array}{r} -7 \overline{) -112} (16 \\ \underline{-112} \\ 0 \end{array}$

(b) $\begin{array}{r} -9 \overline{) 118} (-13 \\ \underline{-9} \\ 28 \\ \underline{-27} \\ 1 \end{array}$

(c) $\begin{array}{r} 6 \overline{) -109} (-19 \\ \underline{-6} \\ -49 \\ \underline{-54} \\ 5 \end{array}$

(d) $\begin{array}{r} 8 \overline{) 115} (14 \\ \underline{8} \\ 35 \\ \underline{32} \\ 3 \end{array}$

$$q = 16, r = 0$$

$$\therefore A \rightarrow 4$$

$$q = -19, r = 5$$

$$\therefore C \rightarrow 3$$

$$q = -13, r = 1.$$

$$\therefore B \rightarrow 1$$

$$q = 14, r = 3$$

$$\therefore D \rightarrow 2$$

295. On dividing 534677 by 777 we get 101 as remainder.
 $\therefore (\text{Divisor}) - (\text{Remainder}) = (777 - 101) = 676.$

$$\begin{array}{r} 777 \overline{)534677} (688 \\ \underline{4662} \\ 6847 \\ \underline{6216} \\ 6317 \\ \underline{6216} \\ 101 \end{array}$$

296. $\frac{(\text{Dividend}) - (\text{Remainder})}{\text{Quotient}} = \text{Divisor}.$

$$\therefore \text{Divisor} = \frac{(940 - 25)}{15} = \frac{915}{15} = 61.$$

297. Remainder = 48, Divisor = $(5 \times 48) = 240.$

$$12 \times \text{Quotient} = \text{Divisor} \Rightarrow \text{Quotient} = \frac{240}{12} = 20.$$

$$\text{Dividend} = (240 \times 20) + 48 = 4848.$$

298. Quotient = 16, Divisor = $(25 \times 16) = 400.$

$$5 \times \text{Remainder} = \text{Divisor} \Rightarrow \text{Remainder} = \frac{400}{5} = 80.$$

$$\text{Dividend} = (400 \times 16) + 80 = 6480.$$

299. Divisor = $(555 + 445) = 1000$, Quotient = $2(555 - 445) = 2 \times 110 = 220$ and Remainder = 30.

$$\therefore \text{Required number} = (1000 \times 220) + 30 = 220000 + 30 = 220030.$$

300. Divisor taken = 12, Quotient obtained = 35, Remainder = 0.

$$\therefore \text{Dividend} = (12 \times 35) = 420.$$

$$\text{Now, dividend} = 420, \text{divisor} = 21, \text{remainder} = 0.$$

$$\therefore \text{Quotient} = \frac{420}{21} = 20.$$

301. Divisor = $7 \times \text{quotient} = 5 \times \text{remainder}$ and Dividend = $6 \times \text{remainder}.$

$$\begin{array}{r} 5x \overline{)6x(1} \\ \underline{5x} \\ x \end{array}$$

Let remainder be x . Then, divisor = $5x$ and dividend = $6x$.

On dividing $6x$ by $5x$, we get 1 as quotient and x as remainder.

$$\therefore \text{Quotient} = 1.$$

302. Since the required number is a 3-digit number, so on dividing by 19, it would yield a two-digit quotient which means that the quotient is greater than the remainder.

Let the remainder be x . Then, quotient = $x + 9$.

$$\text{So, number } N = 19(x + 9) + x = 20x + 171.$$

$$\therefore (N - 171) \text{ must be divisible by } 20.$$

$$\text{Clearly, } (371 - 171) = 200, \text{ which is divisible by } 20.$$

$$\text{Hence, the required number} = 371.$$

303. Let the number be x . Let x when divided by 136 give q as quotient and 36 as remainder. Then, $x = 136q + 36 = (17 \times 8q) + (17 \times 2) + 2 = 17 \times (8q + 2) + 2.$

So, the given number when divided by 17 gives 2 as remainder.

304. Let the number be x and the quotient be q .

$$\text{Then, } x = 195q + 47 = (15 \times 13q) + (15 \times 3) + 2 = 15(13q + 3) + 2.$$

So, the given number when divided by 15 gives 2 as remainder.

305. Let the number be x and the quotient be q .

$$\text{Then, } x = 899q + 63 = (29 \times 31q) + (29 \times 2) + 5 = 29(31q + 2) + 5.$$

So, the given number when divided by 29 gives 5 as remainder.

306. Let the number be x and on dividing by 5, we get q as quotient and 3 as remainder.

$$\begin{aligned} \text{Then, } x = 5q + 3 \Rightarrow x^2 &= (5q + 3)^2 = (25q^2 + 30q + 9) \\ &= 5(5q^2 + 6q + 1) + 4. \end{aligned}$$

Thus, on dividing x^2 by 5, we get 4 as remainder.

307. Let the smaller number be x . Then, larger number = $(x + 1365).$

$$\therefore x + 1365 = 6x + 15 \Rightarrow 5x = 1350 \Rightarrow x = 270.$$

Hence, the smaller number = 270.

308. Let $n = 4k + 3$.

$$\text{Then, } 2n = 2(4k + 3) = 8k + 6 = 4 \times 2k + 4 \times 1 + 2 = 4(2k + 1) + 2.$$

Thus, on dividing $2n$ by 4, we get 2 as remainder.

309. Let $x = 13p + 11$ and $x = 17q + 9$.

$$\text{Then, } 13p + 11 = 17q + 9 \Rightarrow 17q - 13p = 2 \Rightarrow q = \frac{2 + 13p}{17}$$

The least value of p for which $q = \frac{2 + 13p}{17}$ is a whole number, is $p = 26$.

$$\therefore x = (13 \times 26 + 11) = 338 + 11 = 349.$$

310. Let the dividend be $(x + 71)$ and the divisor be y .

$$\text{Then, } [2(x + 71) - 43] \text{ is divisible by } y \Rightarrow (2x + 142 - 43) \text{ is divisible by } y$$

$$\Rightarrow (2x + 99) \text{ is divisible by } y.$$

$$\therefore \text{Divisor} = 99$$

Shortcut Method:

$$\text{Divisor} = (2 \times 71 - 43) = (142 - 43) = 99.$$

311. Let $P = x + r_1$ and $Q = y + r_2$, where each of x and y are divisible by the common divisor.

$$\text{Then, } P + Q = (x + r_1) + (y + r_2) = (x + y) + (r_1 + r_2).$$

$(P + Q)$ leaves remainder r_3 when divided by the common divisor.

$$\Rightarrow [(x + y) + (r_1 + r_2) - r_3] \text{ is divisible by the common divisor.}$$

Since $(x + y)$ is divisible by the common divisor, so divisor = $r_1 + r_2 - r_3$.

312. As proved in the above question, divisor = $4375 + 2986 - 2361 = 5000.$

313. The number is of the form $(13k + 1)$, where k is of the form $(5m + 3).$

$$\therefore \text{Number} = 13k + 1 = 13(5m + 3) + 1 = 65m + 40.$$

Clearly when the number is divided by 65, we get 40 as remainder.

314. Clearly, $(2272 - 875) = 1397$, is exactly divisible by N .

$$\text{Now, } 1397 = 11 \times 127.$$

$\therefore$ The required 3-digit number is 127, the sum of whose digits is 10.

315. $z = 13 \times 1 + 8 = 21.$

$$\begin{array}{r} 9 \mid x \\ 11 \mid y - 8 \\ 13 \mid z - 9 \\ \hline 1 - 8 \end{array}$$

$$y = 11 \times z + 9 = 11 \times 21 + 9 = 240.$$

$$x = 9 \times y + 8 = 9 \times 240 + 8 = 2168.$$

Now, when order of divisors is reversed, we have:

$$\begin{array}{r|l} 13 & 2168 \\ \hline 11 & 166 - 10 \\ \hline 9 & 15 - 1 \\ \hline & 1 - 6 \end{array}$$

∴ Respective remainders are 10, 1 and 6.

$$\begin{array}{r|l} 3 & x \\ \hline 4 & y - 2 \\ \hline 7 & z - 1 \\ \hline & 1 - 4 \end{array} \quad \begin{array}{l} z = 7 \times 1 + 4 = 11. \\ y = 4 \times z + 1 = 4 \times 11 + 1 = 45. \\ x = 3 \times y + 2 = 3 \times 45 + 2 = 137. \end{array}$$

When 137 is divided by 84, the remainder obtained is 53.

$$\begin{array}{r|l} 8 & x \\ \hline 7 & y - 3 \\ \hline 3 & z - 4 \\ \hline & 31 - 2 \end{array} \quad \begin{array}{l} z = 3 \times 31 + 2 = 95. \\ y = 7 \times z + 4 = 7 \times 95 + 4 = 669. \\ x = 8 \times y + 3 = 8 \times 669 + 3 = 5355. \end{array}$$

∴ Required number = 5355.

$$\begin{array}{r|l} 3 & x \\ \hline 2 & y - 1 \\ \hline & 1 - 1 \end{array} \quad \begin{array}{l} y = 2 \times 1 + 1 = 3 \\ x = 3 \times y + 1 = 3 \times 3 + 1 = 10. \end{array}$$

Clearly, 10 when divided by 6, leaves a remainder 4.

319. Let the number be $N = 2x + 1$.

$$N^2 = (2x + 1)^2 = 4x^2 + 1 + 4x = 4x(x + 1) + 1.$$

Clearly, $4x(x + 1)$ is always divisible by 8 since one of x and $(x + 1)$ is even which when multiplied by 4 is always divisible by 8.

Hence, required remainder = 1.

320. Sum of digits of numbers from 1 to 10 = 46.

Sum of digits of numbers from 11 to 20 = 56.

Sum of digits of numbers from 21 to 29 = 63.

Sum of digits of the given number = $46 + 56 + 63 = 165$.

So, the required remainder is the remainder obtained on dividing 165 by 9, which is 3.

321. When n is even, $(x^n - a^n)$ is divisible by $(x + a)$.

$$\therefore (17^{200} - 1^{200}) \text{ is divisible by } (17 + 1)$$

$$\Rightarrow (17^{200} - 1) \text{ is divisible by } 18$$

$$\Rightarrow \text{On dividing } 17^{200} \text{ by } 18, \text{ we get } 1 \text{ as remainder.}$$

322. $2^{31} = 2 \times 2^{30} = 2 \times (2^2)^{15} = 2 \times 4^{15}$.

When n is odd, $(x^n + a^n)$ is divisible by $(x + a)$.

$$\therefore (4^{15} + 1^{15}) \text{ is divisible by } (4 + 1)$$

$$\Rightarrow (4^{15} + 1) \text{ is divisible by } 5 \Rightarrow (2^{30} + 1) \text{ is divisible by } 5$$

$$\Rightarrow \text{On dividing } 2^{30} \text{ by } 5, \text{ we get } (5 - 1) \text{ i.e. } 4 \text{ as remainder.}$$

$$\therefore \text{Remainder obtained on dividing } 2^{31} \text{ by } 5$$

$$= \text{Remainder obtained on dividing } (2 \times 4) \text{ i.e. } 8 \text{ by } 5 = 3.$$

323. Clearly,

(1) when n is odd, $(a^n + b^n)$ is divisible by $(a + b)$. So, (1) is true.

(2) $(a^n - b^n)$ is divisible by $(a - b)$ for all values of n . So, (2) is true.

324. $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n .

$$\therefore (7^{19} - 1^{19}) \text{ is divisible by } (7 - 1)$$

$$\Rightarrow (7^{19} - 1) \text{ is divisible by } 6$$

$$\Rightarrow \text{On dividing } (7^{19} + 2) \text{ by } 6, \text{ remainder obtained} = 3.$$

325. $(10^{12} + 25)^2 - (10^{12} - 25)^2 = 4 \times 10^{12} \times 25$

$$[\because (a + b)^2 - (a - b)^2 = 4ab]$$

$$= 10^{12} \times 100 = 10^{12} \times 10^2 = 10^{14}.$$

Hence, $n = 14$.

326. $(3^{25} + 3^{26} + 3^{27} + 3^{28}) = 3^{25} (1 + 3 + 3^2 + 3^3)$

$$= 3^{25} (1 + 3 + 9 + 27) = 3^{25} \times 40$$

$$= (3 \times 10) \times (3^{24} \times 4) = 30 \times (3^{24} \times 4),$$

which is divisible by 30.

327. $(4^{61} + 4^{62} + 4^{63} + 4^{64}) = 4^{61} (1 + 4 + 4^2 + 4^3) = 4^{61} \times 85$, which is divisible by 17.

328. $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n

$$\therefore (9^6 - 1^6) \text{ is divisible by } (9 - 1)$$

$$\Rightarrow (9^6 - 1) \text{ is divisible by } 8$$

$$\Rightarrow \text{On dividing } (9^6 + 1) \text{ by } 8, \text{ we get } 2 \text{ as remainder.}$$

329. When n is even, $(x^n - a^n)$ is divisible by both $(x - a)$ and $(x + a)$.

So, $(6^n - 1)$ is divisible by both $(6 - 1)$ and $(6 + 1)$

$$\Rightarrow (6^n - 1) \text{ is divisible by both } 5 \text{ and } 7$$

$$\Rightarrow (6^n - 1) \text{ is divisible by } (5 \times 7), \text{ i.e. } 35.$$

[∵ 5 and 7 are co-primes]

330. $(x^n + a^n)$ is divisible by $(x + a)$ when n is odd.

$$\therefore (12^n + 1) \text{ is divisible by } (12 + 1) \text{ i.e. } 13 \text{ when } n \text{ is odd.}$$

331. $(x^n + a^n)$ is divisible by $(x + a)$ when n is odd.

$$\therefore (25^{25} + 1^{25}) \text{ is divisible by } (25 + 1)$$

$$\Rightarrow (25^{25} + 1) \text{ is divisible by } 26$$

$$\Rightarrow \text{On dividing } 25^{25} \text{ by } 26, \text{ we get } (26 - 1) = 25 \text{ as remainder.}$$

332. $(x^n + a^n)$ is divisible by $(x + a)$ when n is odd.

$$\therefore (67^{67} + 1^{67}) \text{ is divisible by } (67 + 1)$$

$$\Rightarrow (67^{67} + 1) \text{ is divisible by } 68$$

$$\Rightarrow \text{On dividing } (67^{67} + 67) \text{ by } 68, \text{ we get } (67 - 1) = 66 \text{ as remainder.}$$

333. $(49^{15} - 1) = (7^2)^{15} - 1 = 7^{30} - 1$.

Now, when n is even, $(x^n - a^n)$ is divisible by both $(x - a)$ and $(x + a)$.

$$\therefore (7^{30} - 1) \text{ is divisible by both } (7 - 1) \text{ and } (7 + 1) \text{ i.e. by both } 6 \text{ and } 8. \text{ Thus, } [(49)^{15} - 1] \text{ is divisible by both } 6 \text{ and } 8.$$

334. $7^{84} = (7^3)^{28} = (343)^{28}$.

Now, $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n .

$$\therefore [(343)^{28} - 1] \text{ is divisible by } (343 - 1)$$

$$\Rightarrow [(343)^{28} - 1] \text{ is divisible by } 342$$

$$\Rightarrow (7^{84} - 1) \text{ is divisible by } 342$$

$$\Rightarrow \text{On dividing } 7^{84} \text{ by } 342, \text{ we get } 1 \text{ as remainder.}$$

335. $2^{60} = (2^2)^{30} = 4^{30}$.

When n is even, $(x^n - a^n)$ is divisible by $(x + a)$.

$$\therefore (4^{30} - 1^{30}) \text{ is divisible by } (4 + 1)$$

$$\Rightarrow (4^{30} - 1) \text{ is divisible by } 5 \Rightarrow (2^{60} - 1) \text{ is divisible by } 5.$$

$$\Rightarrow \text{On dividing } 2^{60} \text{ by } 5, \text{ we get } 1 \text{ as remainder.}$$

336. $(2^{12} - 1) = (4096 - 1) = 4095$, which is clearly divisible by 3, 5, 7 and 13 i.e. four numbers in all.

337. $(x^n + a^n)$ is divisible by $(x + a)$ when n is odd.

$$\therefore (15^{23} + 23^{23}) \text{ is divisible by } (15 + 23)$$

$\Rightarrow (15^{23} + 23^{23})$ is divisible by 38 and hence by 19
 $\Rightarrow$ On dividing $(15^{23} + 23^{23})$ by 19, we get 0 as remainder.

338. When n is even, $(x^n - a^n)$ is divisible by $(x + a)$.

Now, $2^{256} = (2^4)^{64} = (16)^{64}$.

$\therefore (16^{64} - 1^{64})$ is divisible by $(16 + 1)$

$\Rightarrow (16^{64} - 1)$ is divisible by 17

$\Rightarrow (2^{256} - 1)$ is divisible by 17

$\Rightarrow$ On dividing 2^{256} by 17, we get 1 as remainder.

339. When n is even, $(x^n - a^n)$ is divisible by both $(x - a)$ as well as $(x + a)$.

Now, $(7^{6n} - 6^{6n}) = [(7^3)^{2n} - (6^3)^{2n}] = [(343)^{2n} - (216)^{2n}]$.

$\therefore (7^{6n} - 6^{6n})$ is divisible by both $(7 - 6)$ and $(7 + 6)$

$\Rightarrow (7^{6n} - 6^{6n})$ is divisible by 13.

And, $[(343)^{2n} - (216)^{2n}]$ is divisible by both $(343 - 216)$ and $(343 + 216)$

$\Rightarrow (7^{6n} - 6^{6n})$ is divisible by both 127 and 559.

340. Let $2^{32} = x$. Then, $(2^{32} + 1) = (x + 1)$.

Let $(x + 1)$ be completely divisible by the natural number N .
 Then, $(2^{96} + 1) = [(2^{32})^3 + 1] = (x^3 + 1) = (x + 1)(x^2 - x + 1)$,
 which is completely divisible by N since $(x + 1)$ is divisible by N .

341. $(2^{48} - 1) = [(2^6)^8 - 1] = (64)^8 - 1$.

When n is even, $(x^n - a^n)$ is completely divisible by both $(x - a)$ and $(x + a)$.

$\therefore (64^8 - 1^8)$ is divisible by both $(64 - 1)$ and $(64 + 1)$

$\Rightarrow (2^{48} - 1)$ is divisible by both 63 and 65.

342. $n^{65} - n = n(n^{64} - 1) = n(n^{32} - 1)(n^{32} + 1)$
 $= n(n^{16} - 1)(n^{16} + 1)(n^{32} + 1)$
 $= n(n^8 - 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$
 $= n(n^4 - 1)(n^4 + 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$
 $= n(n^2 - 1)(n^2 + 1)(n^4 + 1)(n^8 + 1)(n^{16} + 1)(n^{32} + 1)$
 $= (n - 1)n(n + 1)(n^2 + 1)(n^4 + 1)(n^8 + 1)$
 $(n^{16} + 1)(n^{32} + 1)$.

Clearly, $(n - 1)$, n and $(n + 1)$ are three consecutive numbers and they have to be multiples of 2, 3 and 4 as n is odd.

Thus, the given number is definitely a multiple of 24.

343. Let $a = 55$, $b = 17$.

Then, $N = a^3 + b^3 - (a + b)^3 = a^3 + b^3 - [a^3 + b^3 + 3ab(a + b)]$
 $= -3ab(a + b) = -3 \times 55 \times 17 \times 72$.

Clearly, N is divisible by both 3 and 17.

344. Taking out 10 common from $[\underline{10} + \underline{11} + \dots + \underline{100}]$, we get this expression in the form of a multiple of 10 which has zeros as its last two digits. So, the last two digits of the expression $[\underline{10} + \underline{11} + \dots + \underline{100}]$ are zeros.
 Thus, the last two digits of N must be the last two digits of the sum $(\underline{1} + \underline{2} + \dots + \underline{9})$.

Now, $\underline{1} + \underline{2} + \dots + \underline{9} = 1 + 2 + 6 + 24 + 120 + 720 + 5040 + 40320 + 362880$
 It has clearly 13 as the last two digits.

So, the last two digits of N are 13.

345. $168 = 2^3 \times 3 \times 7$ and $\underline{7} = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 7 \times 5 \times 3^2 \times 2^4 = 168 \times 30$
 Hence, $\underline{7}$ and all the factorials greater than $\underline{7}$ are divisible by 168.

Now, $N = \underline{1} + \underline{2} + \underline{3} + \dots + \underline{99} + \underline{100}$

$= \underline{1} + \underline{2} + \underline{3} + \underline{4} + \underline{5} + \underline{6} + \dots + \text{a multiple of } 168$.

So, the remainder obtained on dividing N by 168 is the same as that obtained on dividing $(\underline{1} + \underline{2} + \underline{3} + \underline{4} + \underline{5} + \underline{6})$ by 168.

Now, $\underline{1} + \underline{2} + \underline{3} + \underline{4} + \underline{5} + \underline{6} = 1 + 2 + 6 + 24 + 120 + 720 = 873 = (168 \times 5) + 33$.

Hence, the required remainder is 33.

346. $4^{61} = 4 \times 4^{60} = 4 \times (4^4)^{15} = 4 \times (256)^{15}$.

Now, $(x^n - a^n)$ is divisible by $(x - a)$ for all values of n .

$\therefore (256^{15} - 1)$ is divisible by $(256 - 1)$ i.e. 255 and hence by 51.

$\Rightarrow$ On dividing $(256)^{15}$ by 51, we get 1 as remainder

$\Rightarrow$ On dividing 4^{60} by 51, we get 1 as remainder

$\Rightarrow$ On dividing 4^{61} by 51, remainder obtained $= (4 \times 1) = 4$.

347. $17^{36} = (17^2)^{18} = (289)^{18}$.

Now, $[(289)^{18} - 1]$ is divisible by $(289 - 1)$, i.e. 288

$\Rightarrow (17^{36} - 1)$ is divisible by 288 and hence by 36

$\Rightarrow$ On dividing 17^{36} by 36, we get 1 as remainder.

348. When n is odd, $(x^n + a^n)$ is always divisible by $(x + a)$.

$\therefore$ Each one of $(47^{43} + 43^{43})$ and $(47^{47} + 43^{47})$ is divisible by $(47 + 43)$.

349. Product of all odd natural numbers less than 5000

$= 1 \times 3 \times 5 \times 7 \times \dots \times 4999$

$= \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times \dots \times 5000}{2 \times 4 \times 6 \times 8 \times \dots \times 5000}$

$= \frac{5000!}{2^{2500} (1 \times 2 \times 3 \times 4 \times \dots \times 2500)} = \frac{5000!}{2^{2500} \cdot 2500!}$.

350. The pages of the book may be divided into 10 groups:

$(1 - 100)$, $(101 - 200)$, $(201 - 300)$, ..., $(901 - 1000)$.

Clearly, for the first group, one needs 11 zeros.

For second to ninth groups, one needs 20 zeros each.

For the tenth group, one needs 21 zeros.

So, total number of zeros required $= 11 + 8 \times 20 + 21 = 192$.

351. $a^2 + b^2 + c^2 = 1$.

So, the maximum value of $a^2 b^2 c^2 = \left(\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3}\right) = \frac{1}{27}$.

($\because$ when sum of three positive quantities is fixed, the product will be maximum when the quantities are equal)

Hence, maximum value of $abc = \frac{1}{\sqrt[3]{27}} = \frac{1}{3\sqrt{3}}$.

352. Let $N = 1^5 + 2^5 + 3^5 + \dots + (100)^5$.

Then, $N = (1^5 + 2^5 + 3^5 + \dots + 10^5) + (11^5 + 12^5 + \dots + 20^5) + (21^5 + 22^5 + \dots + 30^5) + \dots + (91^5 + 92^5 + \dots + 100^5) = N_1 + N_2 + N_3 + \dots + N_{10}$.

Since each one of $N_1, N_2, N_3, \dots, N_{10}$ has the same

unit's digit of its terms, so unit's digit of each one of $N_1, N_2, \dots, N_{10}$ is also the same.

$\therefore$ Unit's digit in $N = 10 \times \text{Unit's digit of } N_1 = 0$.

- 353.** Clearly, each of the 38 terms $\left(\frac{1}{4}\right), \left(\frac{1}{4} + \frac{1}{50}\right), \left(\frac{1}{4} + \frac{2}{50}\right), \dots, \left(\frac{1}{4} + \frac{37}{50}\right)$ has a value lying between 0 and 1, while each one of the 12 terms $\left(\frac{1}{4} + \frac{38}{50}\right), \left(\frac{1}{4} + \frac{39}{50}\right), \dots, \left(\frac{1}{4} + \frac{49}{50}\right)$ has a value lying between 1 and 2.

Hence, the given expression $= (0 \times 38) + (1 \times 12) = 12$.

- 354.** $100^{25} - 25 = (10^2)^{25} - 25 = 10^{50} - 25$
 $= \underbrace{1000 \dots 00}_{50 \text{ zeros}} - 25 = \underbrace{9999 \dots 9975}_{48 \text{ times}}$
 $\therefore$ Sum of digits $= (48 \times 9) + 7 + 5 = 432 + 7 + 5 = 444$.

- 355.** $(1024)^4 \times (125)^{11} = (2^{10})^4 \times (5^3)^{11} = 2^{40} \times 5^{33} = 2^7 \times (2^{33} \times 5^{33}) = 2^7 \times 10^{33} = 128 \times 10^{33}$.

Clearly, the number has 1, 2, 8 and thirty-three zeros, i.e. $(3 + 33) = 36$ digits in all.

- 356.** From 300 to 399, we note that when '4' comes only one time = 19 such instances.

From 400 to 499, we note that when '4' comes only one time = 80 such instances.

So, total $= (19 + 80) = 99$ such instances

- 357.** $21 = 3 \times 7$ is not a prime number because 21 is a composite number.

- 358.** Given $x = a(b - c)$, $y = b(c - a)$; $z = (a - b)$

$$x = a(b - c)$$

$$\Rightarrow \frac{x}{a} = b - c \dots (i)$$

$$\text{Similarly, } y = b(c - a)$$

$$\Rightarrow \frac{y}{b} = c - a \text{ (ii) and similarly } z = c(a - b) \frac{z}{c} = c - a \text{ (iii)}$$

Adding (i), (ii) and (iii) we get

$$\therefore \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = b - c + c - a + a - b = 0$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0$$

$$\therefore \left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 + \left(\frac{z}{c}\right)^3$$

$$= 3 \times \frac{x}{a} \times \frac{y}{b} \times \frac{z}{c} = \frac{3xyz}{abc}$$

$$[\text{If } a + b + c = 0, a^3 + b^3 + c^3 = 3abc]$$

- 359.** Every real number is a rational number is not a correct statement.

- 360.** Given

$$a + b + c = 6$$

$$ab + bc + ca = 10$$

$$\therefore (a + b + c)^2 = 36$$

$$\Rightarrow a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = 36$$

$$\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 36$$

$$\Rightarrow a^2 + b^2 + c^2 + 2 \times 10 = 36$$

$$\Rightarrow a^2 + b^2 + c^2 = 16$$

$$\text{As we know } \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} = (a + b + c)$$

$$\frac{a^3 + b^3 + c^3 - 3abc}{16 - (ab + bc + ca)} = 6$$

$$\Rightarrow \frac{a^3 + b^3 + c^3 - 3abc}{16 - 10} = 6$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 6 \times 6$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 36$$

- 361.** Let the blank space is x

$$\therefore 1001 \times 111 = 110000 + 11x$$

Now find the value of x

$$(1000 + 1) \times 111 = 110000 + 11x$$

$$111000 + 111 = 110000 + 11x$$

$$111111 = 110000 + 11x$$

$$11x = 111111 - 110000$$

$$11x = 1111$$

$$x = \frac{1111}{11} = 101$$

- 362.** The place value of 5 in 15201 = 5000

$$\text{Place value of 6 in 2659} = 600$$

$$\therefore \text{ we have } 5000 + 600 = 7x$$

$$7x = 5600$$

$$x = \frac{5600}{7} = 800$$

- 363.** Let the two - digit number be $10a + b$ where $a > b$

According to the question,

$$a + b = 12 \dots (i)$$

$$a - b = 6 \dots (ii)$$

On adding equation (i) and (ii).

$$2a = 19$$

$$\Rightarrow a = 9$$

From equation (i),

$$9 + b = 12$$

$$\Rightarrow b = 12 - 9 = 3$$

$$\therefore \text{ Number is } 10a + b$$

$$= 9 \times 10 + 3 = 93$$

$$\therefore \text{ When } a < b. \text{ Then required number is } = 39$$

- 364.** Greatest four digit number that begins with 3 and ends with 5 = 3995

Least four digit number that begins with 3 and ends with 5 = 3005

$$\Rightarrow (p-q)(p^2+q^2-pq) = (p-q)\{(p-q)^2 - xpq\}$$

$$\left\{ \because a^3 - b^3 = (a-b)(a^2 - ab + b^2) \right\}$$

by cancelling same terms of both sides

$$\Rightarrow p^2 + q^2 - pq = p^2 + q^2 - 2pq - xpq$$

$$\{(a-b^2) = a^2 + b^2 - 2ab\}$$

$$\Rightarrow pq = -xpq$$

$$\Rightarrow x = -1$$

- 367.** 2361*48 will be divisible by 9 if the sum of the digits of the given number is divisible by 9.

$2 + 3 + 6 + 1 + * + 4 + 8$ i.e. $(24 + *)$ is divisible by 9. Clearly, $* = 3$ because 27 is divisible by 9.

- 368.** Time taken by A to complete colouring a picture = $\frac{3}{4}$ hours = 0.75 hours

Time taken by B to complete colouring a picture $\frac{7}{12}$ = hours = 0.58 hours

Time taken by C to complete colouring a picture $\frac{5}{8}$ = hours = 0.625 hours

Time taken by D to complete colouring a picture $\frac{6}{7}$ = hours = 0.85 hours

Hence, B took least time in colouring the picture.

- 369.** Let number be a
According to question, $\frac{4}{5}$ of $a - 45\%$ of $a = 56$

$$\frac{4}{5}a - \frac{45}{100} \times a = 56$$

$$\frac{35a}{100} = 56 \Rightarrow a = \frac{100 \times 56}{35}$$

$$\Rightarrow a = 160$$

$$= 65\% \text{ of } a = \frac{65}{100} \times 160 = 104$$

- 370.** Given $x + y : y + z : z + x = 6 : 7 : 8$

$$\frac{x+y}{6} = \frac{y+z}{7} = \frac{z+x}{8} = a$$

(let)

$$\Rightarrow x + y = 6a \dots (i)$$

$$y + z = 7a \dots (ii)$$

$$z + x = 8a \dots (iii)$$

On adding all three equations

$$x + y + y + z + z + x = 6a + 7a + 8a$$

$$\Rightarrow 2(x + y + z) = 21a$$

$$\Rightarrow 2 \times 14 = 21a \quad [\because x + y + z = 14]$$

$$\therefore a = \frac{2 \times 14}{21} = \frac{4}{3}$$

$$\therefore x + y = 6a = 6 \times \frac{4}{3} = 8$$

$$\therefore z = (x + y + z) - (x + y)$$

$$= 14 - 8 = 6$$

- 371.** A megabyte is 1.048, 576 bytes or 1,024 kilobytes. It conveniently expression the binary multiples inherent in digital computer memory architectures. However, megabyte architectures. However, megabyte is also taken to mean 1000×1024 (1024000) bytes.

- 372.** The first 3-digit number which is divisible by 9 is 108 and last three digit number which is divisible by 9 is 999.

So, we have an AP with $a = 108$, $d = 9$ and $a_n = 999$

$$\therefore a_n = a + (n-1)d$$

$$\Rightarrow 999 = 108 + (n-1)9$$

$$\Rightarrow 999 - 108 = (n-1)9$$

$$\Rightarrow 891 = (n-1)9$$

$$\Rightarrow (n-1) = \frac{891}{9} = 99$$

$$\Rightarrow n + 99 + 1 = 100$$

- 373.** Let, the two consecutive even numbers are a and $(a+2)$, respectively

According to question,

$$a^2 + (a+2)^2 = 1060 \quad \left\{ \because (a+b)^2 = a^2 + b^2 + 2ab \right\}$$

$$\Rightarrow a^2 + a^2 + 4 + 4a = 1060$$

$$\Rightarrow 2a^2 + 4a - 1056 = 0$$

$$\Rightarrow a^2 + 2a - 528 = 0$$

$$\Rightarrow a^2 + 24a - 22a - 528 = 0$$

$$\Rightarrow a(a+24) - 22(a+24) = 0$$

$$\Rightarrow (a-22)(a+24) = 0$$

$$\Rightarrow a = -24, 22$$

- 374.** The data in both the statements I and II are not sufficient to answer the question.

- 375.** Given $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$ where p_1, p_2, p_3 are distinct prime factors

Number of prime factors form $= (x_1 \times x_2 \times x_3) = x_1 x_2 x_3$

Hence, option (b) is correct

- 376.** 11, 111, 1111, 11111,

$$\text{Let } m = 2 \Rightarrow 4 \times 2 + 3 = 11$$

$$m = 27 \Rightarrow 4 \times 27 + 3 = 111$$

Each number can be expressed in the form $(4m+3)$ where m is a natural number

Hence, statement 1 is only correct.

- 377.** Maximum daily wages of an officers

= H.C.F. of ₹ 4,956; and ₹ 3,894

= ₹ 354

Illustration:

$$\begin{array}{r} 1 \\ 3894 \overline{) 4956} \\ \underline{3894} \\ 1062 \\ 1062 \\ \underline{0} \\ 0 \end{array}$$

$$\begin{array}{r} 3 \\ 1062 \overline{) 3894} \\ \underline{3186} \\ 708 \\ 708 \\ \underline{0} \\ 0 \end{array}$$

$$\begin{array}{r} 1 \\ 708 \overline{) 1062} \\ \underline{708} \\ 354 \\ 354 \\ \underline{0} \\ 0 \end{array}$$

$$\begin{array}{r} 2 \\ 354 \overline{) 708} \\ \underline{708} \\ \underline{0} \\ 0 \end{array}$$

Maximum daily wages of an officer = ₹ 354

Number of Maximum days to attend the duty = $\frac{4956}{354} = 14$

Number of days present on duty = $\frac{3894}{354} = 11$

Number of absent days = $14 - 11 = 3$ days

378. Let, the two natural numbers be a and b

According to given information

$$\therefore a^2 - b^2 = 19$$

$$\Rightarrow (a + b)(a - b) = 19 \times 1$$

$$\Rightarrow a + b = 19 \dots (i)$$

$$a - b = 1 \dots (ii)$$

On adding,

$$2a = 20$$

$$\Rightarrow a = 10$$

From equation (i),

$$\therefore 10 + b = 19$$

$$\Rightarrow b = 19 - 10 = 9$$

$$\therefore a^2 + b^2 = (10)^2 + (9)^2$$

$$= 100 + 81 = 181$$

379. Let the two numbers be a and b

$$\therefore a + b = 14 \dots (i)$$

$$a - b = 10 \dots (ii)$$

By adding equation (i) and (ii) we get

$$2a = 24$$

$$\therefore a = 12 \text{ and } b = 2$$

$$\therefore \text{product of these two numbers} = 12 \times 2 = 24$$

380. Let A is to be added then $2x^2 + 3x - 5 + A = x^2 + x + 1$

$$\Rightarrow A = x^2 - x + 1 - (2x^2 + 3x - 5) \Rightarrow A = x^2 - x + 1 - 2x^2 - 3x + 5$$

$$= -x^2 - 4x + 6$$

2

H.C.F. and L.C.M. of Numbers

IMPORTANT FACTS AND FORMULAE

- I. Factors and Multiples:** If a number a divides another number b exactly, we say that a is a *factor* of b . In this case, b is called a *multiple* of a .
- II. Highest Common Factor (H.C.F.) or Greatest Common Measure (G.C.M.) or Greatest Common Divisor (G.C.D.):** The H.C.F. of two or more than two numbers is the greatest number that divides each of them exactly. There are two methods of finding the H.C.F. of a given set of numbers:
- 1. Factorization Method:** Express each one of the given numbers as the product of prime factors. The product of least powers of common prime factors gives H.C.F.
 - 2. Division Method:** Suppose we have to find the H.C.F. of two given numbers. Divide the larger number by the smaller one. Now, divide the divisor by the remainder. Repeat the process of dividing the preceding number by the remainder last obtained till zero is obtained as remainder. The last divisor is the required H.C.F.
- Finding the H.C.F. of more than two numbers:* Suppose we have to find the H.C.F. of three numbers. Then, H.C.F. of [(H.C.F. of any two) and (the third number)] gives the H.C.F. of three given numbers. Similarly, the H.C.F. of more than three numbers may be obtained.
- III. Least Common Multiple (L.C.M.):** The least number which is exactly divisible by each one of the given numbers is called their L.C.M.
- 1. Factorization Method of Finding L.C.M.:** Resolve each one of the given numbers into a product of prime factors. Then, L.C.M. is the product of highest powers of all the factors.
 - 2. Common Division Method (Short-cut Method) of Finding L.C.M.:** Arrange the given numbers in a row in any order. Divide by a number which divides exactly at least two of the given numbers and carry forward the numbers which are not divisible. Repeat the above process till no two of the numbers are divisible by the same number except 1. The product of the divisors and the undivided numbers is the required L.C.M. of the given numbers.
- IV. Product of two numbers = Product of their H.C.F. and L.C.M.**
- V. Co-primes:** Two numbers are said to be co-primes if their H.C.F. is 1.
- VI. H.C.F. and L.C.M. of Fractions:**
- 1. H.C.F. =**
$$\frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}}$$
 - 2. L.C.M. =**
$$\frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$$
- VII. H.C.F. and L.C.M. of Decimal Fractions:** In given numbers, make the same number of decimal places by annexing zeros in some numbers, if necessary. Considering these numbers without decimal point, find H.C.F. or L.C.M. as the case may be. Now, in the result, mark off as many decimal places as are there in each of the given numbers.
- VIII. Comparison of Fractions:** Find the L.C.M. of the denominators of the given fractions. Convert each of the fractions into an equivalent fraction with L.C.M. as the denominator, by multiplying both the numerator and denominator by the same number. The resultant fraction with the greatest numerator is the greatest.

SOLVED EXAMPLES

Ex. 1. Find the H.C.F. of $2^3 \times 3^2 \times 5 \times 7^4$, $2^2 \times 3^5 \times 5^2 \times 7^3$, $2^3 \times 5^3 \times 7^2$.

Sol. The prime numbers common to given numbers are 2, 5 and 7.

$$\therefore \text{H.C.F.} = 2^2 \times 5 \times 7^2 = 980.$$

Ex. 2. Find the H.C.F. of:

(a) 42, 63 and 140

(b) 108, 288 and 360

(L.I.C.A.D.O., 2008)

Sol. (a) $42 = 2 \times 3 \times 7$, $63 = 3^2 \times 7$ and $140 = 2^2 \times 5 \times 7$.

$$\therefore \text{H.C.F.} = 7.$$

(b) $108 = 2^2 \times 3^3$, $288 = 2^5 \times 3^2$ and $360 = 2^3 \times 5 \times 3^2$.
 $\therefore$ H.C.F. = $2^2 \times 3^2 = 36$.

Ex. 3. Find the H.C.F. of 513, 1134 and 1215.

Sol.

$ \begin{array}{r} 1134 \overline{)1215} \left(1 \\ \underline{1134} \\ 81 \overline{)1134} \left(14 \\ \underline{81} \\ 324 \\ \underline{324} \\ \times \end{array} $	So, required H.C.F. = H.C.F. of 513 and 81. $ \begin{array}{r} 81 \overline{)513} \left(6 \\ \underline{486} \\ 27 \overline{)81} \left(3 \\ \underline{81} \\ \times \end{array} $
--	--

$\therefore$ H.C.F. of 1134 and 1215 is 81. $\therefore$ H.C.F. of given numbers = 27.

Ex. 4. Reduce $\frac{391}{667}$ to lowest terms.

Sol. H.C.F. of 391 and 667 is 23.

On dividing the numerator and denominator by 23, we get:

$$\frac{391}{667} = \frac{391 \div 23}{667 \div 23} = \frac{17}{29}.$$

Ex. 5. Find the L.C.M. of $2^2 \times 3^3 \times 5 \times 7^2$, $2^3 \times 3^2 \times 5^2 \times 7^4$, $2 \times 3 \times 5^3 \times 7 \times 11$.

Sol. L.C.M. = Product of highest powers of 2, 3, 5, 7 and 11 = $2^3 \times 3^3 \times 5^3 \times 7^4 \times 11$.

(L.I.C.A.D.O., 2008)

Ex. 6. Find the L.C.M. of

(a) 87 and 145 (b) 72, 108 and 2100

Sol. (a) $87 = 3 \times 29$ and $145 = 5 \times 29$.

$\therefore$ L.C.M. = $3 \times 5 \times 29 = 435$.

(b) $72 = 2^3 \times 3^2$, $108 = 3^3 \times 2^2$, $2100 = 2^2 \times 5^2 \times 3 \times 7$.

$\therefore$ L.C.M. = $2^3 \times 3^3 \times 5^2 \times 7 = 37800$.

Ex. 7. Find the L.C.M. of 16, 24, 36 and 54.

Sol.

2	16	24	36	54
2	8	12	18	27
2	4	6	9	27
3	2	3	9	27
3	2	1	3	9
	2	1	1	3

$\therefore$ L.C.M. = $2 \times 2 \times 2 \times 3 \times 3 \times 2 \times 3 = 432$.

Ex. 8. Find the H.C.F. and L.C.M. of $\frac{2}{3}$, $\frac{8}{9}$, $\frac{16}{81}$ and $\frac{10}{27}$.

Sol. H.C.F. of given fractions = $\frac{\text{H.C.F. of } 2, 8, 16, 10}{\text{L.C.M. of } 3, 9, 81, 27} = \frac{2}{81}$.

L.C.M. of given fractions = $\frac{\text{L.C.M. of } 2, 8, 16, 10}{\text{H.C.F. of } 3, 9, 81, 27} = \frac{80}{3}$.

Ex. 9. Find the H.C.F. and L.C.M. of 0.63, 1.05 and 2.1.

Sol. Making the same number of decimal places, the given numbers are 0.63, 1.05 and 2.10.

Without decimal places, these numbers are 63, 105 and 210.

Now, H.C.F. of 63, 105 and 210 is 21.

$\therefore$ H.C.F. of 0.63, 1.05 and 2.1 is 0.21.

L.C.M. of 63, 105 and 210 is 630.

$\therefore$ L.C.M. of 0.63, 1.05 and 2.1 is 6.30.

Ex. 10. Two numbers are in the ratio of 15: 11. If their H.C.F. is 13, find the numbers.

Sol. Let the required numbers be $15x$ and $11x$.

Then, their H.C.F. is x . So, $x = 13$.

$\therefore$ The numbers are $(15 \times 13$ and $11 \times 13)$, i.e., 195 and 143.

Ex. 11. Two numbers are in the ratio of 3: 4. Their L.C.M. is 84. Find the numbers.

(S.S.C., 2010)

Sol. Let the numbers be $3x$ and $4x$. Then, their L.C.M. = $12x$.

So, $12x = 84$ or $x = 7$.

$\therefore$ The numbers are 21 and 28.

Ex. 12. The H.C.F. of two numbers is 11 and their L.C.M. is 693. If one of the numbers is 77, find the other.

(P.C.S., 2009)

Sol. Other number = $\left(\frac{11 \times 693}{77} \right) = 99$.

Ex. 13. The sum of two numbers is 462 and their highest common factor is 22. What is the minimum number of pairs that satisfy these conditions?

(M.A.T., 2004)

Sol. Let the required numbers be $22a$ and $22b$.

Then, $22a + 22b = 462 \Rightarrow a + b = 21$.

Now, co-primes with sum 21 are (1, 20), (2, 19), (4, 17), (5, 16), (8, 13) and (10, 11).

$\therefore$ Required numbers are $(22 \times 1, 22 \times 20)$, $(22 \times 2, 22 \times 19)$,

$(22 \times 4, 22 \times 17)$, $(22 \times 5, 22 \times 16)$, $(22 \times 8, 22 \times 13)$ and $(22 \times 10, 22 \times 11)$.

Clearly, the number of such pairs is 6.

Ex. 14. The sum and difference of the L.C.M and H.C.F. of two numbers are 592 and 518 respectively. If the sum of the numbers be 296, find the numbers.

(Section Officer's, 2006)

Sol. Let L and H denote the L.C.M and H.C.F of the two numbers.

Then, $L + H = 592$

...(i)

And, $L - H = 518$

....(ii)

Adding (i) and (ii), we get: $2L = 1110$ or $L = 555$.

$\therefore H = 592 - 555 = 37$.

So, H.C.F. = 37 and L.C.M. = 555.

Let the numbers be x and $(296 - x)$.

Then, $x(296 - x) = 555 \times 37 \Rightarrow x^2 - 296x + 20535 = 0$

$\Rightarrow x^2 - 185x - 111x + 20535 = 0 \Rightarrow x(x - 185) - 111(x - 185) = 0$

$\Rightarrow (x - 185)(x - 111) = 0 \Rightarrow x = 185$ or $x = 111$.

Hence the numbers are 111 and 185.

Ex. 15. Find the greatest possible length which can be used to measure exactly the lengths 4 m 95 cm, 9 m and 16 m 65 cm.

Sol. Required length = H.C.F. of 495 cm, 900 cm and 1665 cm.

$495 = 3^2 \times 5 \times 11$, $900 = 2^2 \times 3^2 \times 5^2$, $1665 = 3^2 \times 5 \times 37$.

$\therefore$ H.C.F. = $3^2 \times 5 = 45$.

Hence, required length = 45 cm.

Ex. 16. Find the greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively.

(Section Officer's, 2006)

Sol. Required number = H.C.F. of $(1657 - 6)$ and $(2037 - 5)$ = H.C.F. of 1651 and 2032

$$\begin{array}{r} 1651 \overline{) 2032} \quad (1 \\ \underline{1651} \\ 381 \overline{) 1651} \quad (4 \\ \underline{1524} \\ 127 \overline{) 381} \quad (3 \\ \underline{381} \\ \times \end{array}$$

$\therefore$ Required number = 127.

Ex. 17. Find the largest number which divides 62, 132 and 237 to leave the same remainder in each case.

Sol. Required number = H.C.F. of $(132 - 62)$, $(237 - 132)$ and $(237 - 62)$

= H.C.F. of 70, 105 and 175 = 35.

Ex. 18. The H.C.F. of two numbers, each having three digits is 17 and their L.C.M. is 714. Find the sum of the numbers.

(C.P.O., 2007)

Sol. Let the numbers be $17a$ and $17b$.

Then, $17a \times 17b = 17 \times 714 \Rightarrow ab = 42$.

Now, co-primes with product 42 are (1, 42), (2, 21), (3, 14) and (6, 7).

So, the numbers are $(17 \times 1, 17 \times 42)$, $(17 \times 2, 17 \times 21)$, $(17 \times 3, 17 \times 14)$ and $(17 \times 6, 17 \times 7)$.

Since both the numbers are 3-digit numbers, the suitable pair is $(17 \times 6, 17 \times 7)$, i.e., (102, 119).

$\therefore$ Required sum = $102 + 119 = 221$.

Ex. 19. Find the least number which is exactly divisible by 32, 36, 45, 60 and 80.

(R.R.B., 2006)

Sol. Required number = L.C.M. of 32, 36, 45, 60, 80

2	32	–	36	–	45	–	60	–	80
2	16	–	18	–	45	–	30	–	40
2	8	–	9	–	45	–	15	–	20
3	4	–	9	–	45	–	15	–	10
5	4	–	3	–	15	–	5	–	10
3	4	–	3	–	3	–	1	–	2
2	4	–	1	–	1	–	1	–	2
	2	–	1	–	1	–	1	–	1

L.C.M. = $2 \times 2 \times 2 \times 3 \times 5 \times 3 \times 2 \times 2 = 1440$.

Hence, required number = 1440.

Ex. 20. Find the least number which when divided by 6, 7, 8, 9 and 12 leaves the same remainder 1 in each case.

Sol. Required number = (L.C.M. of 6, 7, 8, 9, 12) + 1.

3	6	–	7	–	8	–	9	–	12
2	2	–	7	–	8	–	3	–	4
2	1	–	7	–	4	–	3	–	2
	1	–	7	–	2	–	3	–	1

$\therefore$ L.C.M. = $3 \times 2 \times 2 \times 7 \times 2 \times 3 = 504$.

Hence, required number = $(504 + 1) = 505$.

Ex. 21. Find the smallest number which when increased by 10 is completely divisible by 12, 15, 18, 20 and 24.

(P.C.S., 2008)

Sol. Required number

= (L.C.M. of 12, 15, 18, 20, 24) – 10

= $(2 \times 2 \times 3 \times 5 \times 3 \times 2) - 10$

= $360 - 10 = 350$.

2	12	–	15	–	18	–	20	–	24
2	6	–	15	–	9	–	10	–	12
3	3	–	15	–	9	–	5	–	6
5	1	–	5	–	3	–	5	–	2
	1	–	1	–	3	–	1	–	2

Ex. 22. Find the greatest number of five digits which is divisible by 15, 21 and 36.

(P.C.S., 2010)

Sol. Greatest number of five digits = 99999.

Required number must be divisible by L.C.M. of 15, 21, 36, i.e., 1260.

On dividing 99999 by 1260, we get 459 as remainder.

$\therefore$ Required number = $(99999 - 459) = 99540$.

Ex. 23. Find the smallest number of five digits exactly divisible by 16, 24, 36 and 54.

Sol. Smallest number of five digits is 10000.

Required number must be divisible by L.C.M. of 16, 24, 36, 54, i.e., 432.

On dividing 10000 by 432, we get 64 as remainder.

$\therefore$ Required number = $10000 + (432 - 64) = 10368$.

Ex. 24. Find the largest number which when subtracted from 10000, the remainder is divisible by 32, 36, 48 and 50.

(R.R.B., 2006)

Sol. Required number

= $10000 - (\text{L.C.M. of } 32, 36, 48, 50)$

= $10000 - (2 \times 2 \times 3 \times 4 \times 3 \times 2 \times 5)$

= $10000 - 864 = 9136$.

2	32	–	36	–	48	–	54
2	16	–	18	–	24	–	27
3	8	–	9	–	12	–	27
4	8	–	3	–	4	–	9
3	2	–	3	–	1	–	9
	2	–	1	–	1	–	3

Ex. 25. Find the largest number of five digits which, when divided by 16, 24, 30 or 36, leaves the same remainder 10 in each case. (C.P.O., 2007)

Sol. Largest number of 5 digits = 99999. L.C.M. of 16, 24, 30 and 36 = 720.

On dividing 99999 by 720, remainder obtained is 639.

$\therefore$ Largest number of 5 digits divisible by 16, 24, 30 and 36 = $(99999 - 639) = 99360$.

Hence, required number = $(99360 + 10) = 99370$.

Ex. 26. Find the least number which when divided by 20, 25, 35 and 40 leaves remainders 14, 19, 29 and 34 respectively.

Sol. Here, $(20 - 14) = 6$, $(25 - 19) = 6$, $(35 - 29) = 6$ and $(40 - 34) = 6$.

$\therefore$ Required number = $(\text{L.C.M. of } 20, 25, 35, 40) - 6 = 1394$.

Ex. 27. What is the least number which when divided by the numbers 3, 5, 6, 8, 10 and 12 leaves in each case a remainder 2 but when divided by 13 leaves no remainder? (S.S.C., 2005)

Sol. L.C.M. of 3, 5, 6, 8, 10 and 12 = 120.

So, the required number is of the form $120k + 2$.

Least value of k for which $(120k + 2)$ is divisible by 13 is $k = 8$.

$\therefore$ Required number = $(120 \times 8 + 2) = 962$.

Ex. 28. The traffic lights at three different road crossings change after every 48 sec., 72 sec. and 108 sec. respectively. If they all change simultaneously at 8:20:00 hours, then at what time will they again change simultaneously? (R.R.B., 2006, M.A.T., 2005)

Sol. Interval of change = $(\text{L.C.M. of } 48, 72, 108) \text{ sec.} = 432 \text{ sec.}$

So, the lights will again change simultaneously after every 432 seconds, i.e., 7 min. 12 sec.

Hence, next simultaneous change will take place at 8:27:12 hrs.

Ex. 29. Seema, Meena and Reema begin to jog around a circular stadium and they complete their revolutions in 54 seconds, 42 seconds and 63 seconds respectively. After how much time will they come together at the starting point? (Bank. P.O., 2010)

Sol. L.C.M. of 54, 42 and 63 = 378.

So, the three girls will come together at the starting point in 378 seconds i.e., 6 min 18 sec.

Ex. 30. Arrange the fractions $\frac{17}{18}, \frac{31}{36}, \frac{43}{45}, \frac{59}{60}$ in the ascending order.

Sol. L.C.M. of 18, 36, 45 and 60 = 180. Now,

$$\frac{17}{18} = \frac{17 \times 10}{18 \times 10} = \frac{170}{180}; \quad \frac{31}{36} = \frac{31 \times 5}{36 \times 5} = \frac{155}{180}; \quad \frac{43}{45} = \frac{43 \times 4}{45 \times 4} = \frac{172}{180}; \quad \frac{59}{60} = \frac{59 \times 3}{60 \times 3} = \frac{177}{180}.$$

Since, $155 < 170 < 172 < 177$, so, $\frac{155}{180} < \frac{170}{180} < \frac{172}{180} < \frac{177}{180}$.

Hence, $\frac{31}{36} < \frac{17}{18} < \frac{43}{45} < \frac{59}{60}$.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. Find the factors of 330. (CLAT, 2010)

- (a) $2 \times 4 \times 5 \times 11$ (b) $2 \times 3 \times 7 \times 13$
(c) $2 \times 3 \times 5 \times 13$ (d) $2 \times 3 \times 5 \times 11$

2. Find the factors of 1122. (CLAT, 2010)

- (a) $3 \times 9 \times 17 \times 2$ (b) $3 \times 11 \times 17 \times 2$
(c) $9 \times 9 \times 17 \times 2$ (d) $3 \times 11 \times 17 \times 3$

3. 252 can be expressed as a product of primes as (IGNOU, 2002)

- (a) $2 \times 2 \times 3 \times 3 \times 7$ (b) $2 \times 2 \times 2 \times 3 \times 7$
(c) $3 \times 3 \times 3 \times 3 \times 7$ (d) $2 \times 3 \times 3 \times 3 \times 7$

4. Which of the following has most number of divisors? (M.B.A. 2002)

- (a) 99 (b) 101
(c) 176 (d) 182

5. A number n is said to be perfect if the sum of all its divisors (excluding n itself) is equal to n . An example of perfect number is

- (a) 6 (b) 9
(c) 15 (d) 21

6. $\frac{1095}{1168}$ when expressed in simplest form is

- (a) $\frac{13}{16}$ (b) $\frac{15}{16}$
 (c) $\frac{17}{26}$ (d) $\frac{25}{26}$
7. Reduce $\frac{128352}{238368}$ to its lowest terms. (IGNOU, 2003)
 (a) $\frac{3}{4}$ (b) $\frac{5}{13}$
 (c) $\frac{7}{13}$ (d) $\frac{9}{13}$
8. The simplest reduction to the lowest terms of $\frac{116,690,151}{427,863,887}$ is (SNAP, 2004)
 (a) $\frac{3}{11}$ (b) $\frac{7}{11}$
 (c) $\frac{11}{3}$ (d) None of these
9. The highest common factor of 0 and 6 is (P.C.S., 2008)
 (a) 0 (b) 3
 (c) 6 (d) Undefined
10. The H.C.F. of $2^2 \times 3^3 \times 5^5$, $2^3 \times 3^2 \times 5^2 \times 7$ and $2^4 \times 3^4 \times 5 \times 7^2 \times 11$ is
 (a) $2^2 \times 3^2 \times 5$ (b) $2^2 \times 3^2 \times 5 \times 7 \times 11$
 (c) $2^4 \times 3^4 \times 5^5$ (d) $2^4 \times 3^4 \times 5^5 \times 7 \times 11$
11. The H.C.F. of $2^4 \times 3^2 \times 5^3 \times 7$, $2^3 \times 3^3 \times 5^2 \times 7^2$ and $3 \times 5 \times 7 \times 11$ is
 (a) 105 (b) 1155
 (c) 2310 (d) 27720
12. H.C.F. of $4 \times 27 \times 3125$, $8 \times 9 \times 25 \times 7$ & $16 \times 81 \times 5 \times 11 \times 49$ is
 (a) 180 (b) 360
 (c) 540 (d) 1260
13. Find the highest common factor of 36 and 84. (R.R.B., 2003)
 (a) 4 (b) 6
 (c) 12 (d) 18
14. Even numbers are formed by taking at least two at a time from the numbers 0, 4, 8, 9. Their H.C.F. is (Hotel Management, 2007)
 (a) 2 (b) 4
 (c) 10 (d) None of these
15. The H.C.F. of 204, 1190 and 1445 is
 (a) 17 (b) 18
 (c) 19 (d) 21
16. Which of the following is a pair of co-primes?
 (a) (16, 62) (b) (18, 25)
 (c) (21, 35) (d) (23, 92)
17. The H.C.F. of 2923 and 3239 is
 (a) 37 (b) 47
 (c) 73 (d) 79
18. The H.C.F. of 3556 and 3444 is
 (a) 23 (b) 25
 (c) 26 (d) 28
19. The L.C.M. of $2^3 \times 3^2 \times 5 \times 11$, $2^4 \times 3^4 \times 5^2 \times 7$ and $2^5 \times 3^3 \times 5^3 \times 7^2 \times 11$ is
 (a) $2^3 \times 3^2 \times 5$ (b) $2^5 \times 3^4 \times 5^3$
 (c) $2^3 \times 3^2 \times 5 \times 7 \times 11$ (d) $2^5 \times 3^4 \times 5^3 \times 7^2 \times 11$
20. Find the lowest common multiple of 24, 36 and 40.
 (a) 120 (b) 240
 (c) 360 (d) 480
21. The L.C.M. of 22, 54, 108, 135 and 198 is
 (a) 330 (b) 1980
 (c) 5940 (d) 11880
22. The L.C.M. of 148 and 185 is
 (a) 680 (b) 740
 (c) 2960 (d) 3700
23. The H.C.F. of $\frac{a}{b}, \frac{c}{d}, \frac{e}{f}$ is equal to (I.A.M. 2007)
 (a) $\frac{\text{L.C.M. of } a, c, e}{\text{H.C.F. of } b, d, f}$ (b) $\frac{\text{H.C.F. of } a, c, e}{\text{L.C.M. of } b, d, f}$
 (c) $\frac{\text{H.C.F. of } a, c, e}{\text{H.C.F. of } b, d, f}$ (d) $\frac{ace}{bdf}$
24. The H.C.F. of $\frac{2}{3}, \frac{8}{9}, \frac{64}{81}$ and $\frac{10}{27}$ is :
 (a) $\frac{2}{3}$ (b) $\frac{2}{81}$
 (c) $\frac{160}{3}$ (d) $\frac{160}{81}$
25. The H.C.F. of $\frac{9}{10}, \frac{12}{25}, \frac{18}{35}$ and $\frac{21}{40}$ is
 (a) $\frac{3}{5}$ (b) $\frac{252}{5}$
 (c) $\frac{3}{1400}$ (d) $\frac{63}{700}$
26. The L.C.M. of $\frac{1}{3}, \frac{5}{6}, \frac{2}{9}, \frac{4}{27}$ is
 (a) $\frac{1}{54}$ (b) $\frac{10}{27}$
 (c) $\frac{20}{3}$ (d) None of these
27. The L.C.M. of $\frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{9}{13}$ is
 (a) 36 (b) $\frac{1}{36}$
 (c) $\frac{1}{1365}$ (d) $\frac{12}{455}$

28. The L.C.M. of $\frac{3}{4}, \frac{6}{7}, \frac{9}{8}$ is (L.I.C.A.D.O., 2008)
- (a) 3 (b) 6
(c) 9 (d) 18
29. The H.C.F. of 1.75, 5.6 and 7 is
- (a) 0.07 (b) 0.7
(c) 3.5 (d) 0.35
30. The G.C.D. of 1.08, 0.36 and 0.9 is (Hotel Management, 2002)
- (a) 0.03 (b) 0.9
(c) 0.18 (d) 0.108
31. The H.C.F. of 0.54, 1.8 and 7.2 is
- (a) 1.8 (b) 0.18
(c) 0.018 (d) 18
32. The L.C.M. of 3, 2.7 and 0.09 is
- (a) 2.7 (b) 0.27
(c) 0.027 (d) 27
33. If A, B and C are three numbers, such that the L.C.M. of A and B is B and the L.C.M. of B and C is C, then the L.C.M. of A, B and C is
- (a) A (b) B
(c) C (d) $\frac{A+B+C}{3}$
34. H.C.F. of 3240, 3600 and a third number is 36 and their L.C.M. is $2^4 \times 3^5 \times 5^2 \times 7^2$. The third number is (M.A.T., 2005)
- (a) $2^2 \times 3^5 \times 7^2$ (b) $2^2 \times 5^3 \times 7^2$
(c) $2^5 \times 5^2 \times 7^2$ (d) $2^3 \times 3^5 \times 7^2$
35. Three numbers are in the ratio 1: 2: 3 and their H.C.F. is 12. The numbers are
- (a) 4, 8, 12 (b) 5, 10, 15
(c) 10, 20, 30 (d) 12, 24, 36
36. The ratio of two numbers is 3: 4 and their H.C.F. is 4. Their L.C.M. is
- (a) 12 (b) 16
(c) 24 (d) 48
37. The sum of two numbers is 216 and their H.C.F. is 27. The numbers are
- (a) 27, 189 (b) 81, 189
(c) 108, 108 (d) 154, 162
38. The sum of two numbers is 528 and their H.C.F. is 33. The number of pairs of numbers satisfying the above conditions is
- (a) 4 (b) 6
(c) 8 (d) 12
39. The number of number-pairs lying between 40 and 100 with their H.C.F. as 15 is
- (a) 3 (b) 4
(c) 5 (d) 6
40. The H.C.F. of two numbers is 12 and their difference is 12. The numbers are
- (a) 66, 78 (b) 70, 82
(c) 94, 106 (d) 84, 96
41. The product of two numbers is 4107. If the H.C.F. of these numbers is 37, then the greater number is
- (a) 101 (b) 107
(c) 111 (d) 185
42. The product of two numbers is 2028 and their H.C.F. is 13. The number of such pairs is (A.A.O., 2010; P.C.S., 2009)
- (a) 1 (b) 2
(c) 3 (d) 4
43. Three numbers which are co-prime to each other are such that the product of the first two is 551 and that of the last two is 1073. The sum of the three numbers is
- (a) 75 (b) 81
(c) 85 (d) 89
44. The ratio of two numbers is 13: 15 and their L.C.M. is 39780. The numbers are (P.C.S., 2009)
- (a) 884, 1020 (b) 884, 1040
(c) 670, 1340 (d) 2652, 3060
45. Three numbers are in the ratio of 3: 4: 5 and their L.C.M. is 2400. Their H.C.F. is
- (a) 40 (b) 80
(c) 120 (d) 200
46. The L.C.M. and ratio of four numbers are 630 and 2: 3: 5: 7 respectively. The difference between the greatest and least numbers is (I.A.M., 2007)
- (a) 6 (b) 14
(c) 15 (d) 21
47. The H.C.F. and L.C.M. of two numbers are 12 and 336 respectively. If one of the numbers is 84, the other is (S.S.C., 2010)
- (a) 36 (b) 48
(c) 72 (d) 96
48. If the product of two numbers is 324 and their H.C.F. is 3, then their L.C.M. will be (P.C.S., 2008)
- (a) 972 (b) 327
(c) 321 (d) 108
49. If H.C.F. of p and q is x and $q = xy$, then the L.C.M. of p and q is (C.D.S., 2004)
- (a) pq (b) qy
(c) xy (d) py
50. The sum of two numbers is 2000 and their L.C.M. is 21879. The two numbers are
- (a) 1993, 7 (b) 1991, 9
(c) 1989, 11 (d) 1987, 13
51. The H.C.F. and L.C.M. of two numbers are 84 and 21 respectively. If the ratio of the two numbers is 1: 4, then the larger of the two numbers is
- (a) 12 (b) 48
(c) 84 (d) 108

52. The L.C.M. of two numbers is 495 and their H.C.F. is 5. If the sum of the numbers is 10, then their difference is
 (a) 10 (b) 46
 (c) 70 (d) 90
53. The product of the L.C.M. and H.C.F. of two numbers is 24. The difference of two numbers is 2. Find the numbers.
 (a) 2 and 4 (b) 6 and 4
 (c) 8 and 6 (d) 8 and 10
54. If the sum of two numbers is 36 and their H.C.F. and L.C.M. are 3 and 105 respectively, the sum of the reciprocals of the two numbers is (S.S.C., 2010)
 (a) $\frac{2}{35}$ (b) $\frac{3}{35}$
 (c) $\frac{4}{35}$ (d) None of these
55. The L.C.M. of two numbers is 12 times their H.C.F. The sum of H.C.F. and L.C.M. is 403. If one number is 93, find the other. (M.B.A., 2007)
 (a) 124 (b) 128
 (c) 134 (d) None of these
56. The H.C.F. and L.C.M. of two numbers are 50 and 250 respectively. If the first number is divided by 2, the quotient is 50. The second number is
 (a) 50 (b) 100
 (c) 125 (d) 250
57. The product of two numbers is 1320 and their H.C.F. is 6. The L.C.M. of the numbers is
 (a) 220 (b) 1314
 (c) 1326 (d) 7920
58. Product of two co-prime numbers is 117. Their L.C.M. should be
 (a) 1 (b) 117
 (c) equal to their H.C.F. (d) cannot be calculated
59. The L.C.M. of three different numbers is 120. Which of the following cannot be their H.C.F.? (Campus Recruitment, 2010)
 (a) 8 (b) 12
 (c) 24 (d) 35
60. The H.C.F. of two numbers is 8. Which one of the following can never be their L.C.M.?
 (a) 24 (b) 48
 (c) 56 (d) 60
61. If the L.C.M. of three numbers is 9570, then their H.C.F. can be
 (a) 11 (b) 12
 (c) 19 (d) 21
62. The H.C.F. of two numbers is 23 and the other two factors of their L.C.M. are 13 and 14. The larger of the two numbers is
 (a) 276 (b) 299
 (c) 322 (d) 345
63. About the number of pairs which have 16 as their H.C.F. and 136 as their L.C.M., we can definitely say that
 (a) no such pair exists
 (b) only one such pair exists
 (c) only two such pairs exist
 (d) many such pairs exist
64. The H.C.F. and L.C.M. of two numbers are 21 and 4641 respectively. If one of the numbers lies between 200 and 300, the two numbers are (M.A.T., 2006)
 (a) 273, 357 (b) 273, 359
 (c) 273, 361 (d) 273, 363
65. Two numbers, both greater than 29, have H.C.F. 29 and L.C.M. 4147. The sum of the numbers is
 (a) 666 (b) 669
 (c) 696 (d) 966
66. L.C.M. of two prime numbers x and y ($x > y$) is 161. The value of $3y - x$ is
 (a) -2 (b) -1
 (c) 1 (d) 2
67. The greatest number that exactly divides 105, 1001 and 2436 is
 (a) 3 (b) 7
 (c) 11 (d) 21
68. 21 mango trees, 42 apple trees and 56 orange trees have to be planted in rows such that each row contains the same number of trees of one variety only. Minimum number of rows in which the trees may be planted is (M.B.A., 2005)
 (a) 3 (b) 15
 (c) 17 (d) 20
69. The greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm, 12 m 95 cm is (R.R.B., 2008)
 (a) 15 cm (b) 25 cm
 (c) 35 cm (d) 42 cm
70. The capacity of two pots is 120 litres and 56 litres respectively. Find the capacity of a container which can exactly measure the contents of the two pots. (R.R.B., 2005)
 (a) 7500 cc (b) 7850 cc
 (c) 8000 cc (d) 9500 cc
71. A daily wage labourer was engaged for a certain number of days for ₹ 5750, but being absent on some of those days he was paid only ₹ 5000. What was his maximum possible daily wage? (C.P.O., 2006)
 (a) ₹ 125 (b) ₹ 250
 (c) ₹ 375 (d) ₹ 500
72. A person has to completely put each of three liquids: 403 litres of petrol, 465 litres of diesel and 496 litres of Mobil Oil in bottles of equal size without mixing any of the above three types of liquids such that

each bottle is completely filled. What is the least possible number of bottles required?

(Civil Services, 2007)

- (a) 34 (b) 44
(c) 46 (d) None of these

73. The maximum number of students among whom 1001 pens and 910 pencils can be distributed in such a way that each student gets the same number of pens and same number of pencils is

- (a) 91 (b) 910
(c) 1001 (d) 1911

74. A rectangular courtyard 3.78 metres long and 5.25 metres wide is to be paved exactly with square tiles, all of the same size. What is the largest size of the tile which could be used for the purpose?

- (a) 14 cm (b) 21 cm
(c) 42 cm (d) None of these

75. The least number of square tiles required to pave the ceiling of a room 15 m 17 cm long and 9 m 2 cm broad is

(M.B.A., 2006)

- (a) 656 (b) 738
(c) 814 (d) 902

76. Three sets of English, Mathematics and Science books containing 336, 240 and 96 books respectively have to be stacked in such a way that all the books are stored subjectwise and the height of each stack is the same. Total number of stacks will be

(S.S.C., 2007)

- (a) 14 (b) 21
(c) 22 (d) 48

77. Four metal rods of lengths 78 cm, 104 cm, 117 cm and 169 cm are to be cut into parts of equal length. Each part must be as long as possible. What is the maximum number of pieces that can be cut?

(Civil Services, 2009)

- (a) 27 (b) 36
(c) 43 (d) 480

78. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

- (a) 4 (b) 7
(c) 9 (d) 13

79. If r is the remainder when each of 7654, 8506 and 9997 is divided by the greatest number d ($d > 1$), then $d - r$ is equal to

(A.A.O., 2010)

- (a) 14 (b) 18
(c) 24 (d) 28

80. Let N be the greatest number that will divide 1305, 4665 and 6905, leaving the same remainder in each case. Then sum of the digits in N is

(S.S.C., 2004)

- (a) 4 (b) 5
(c) 6 (d) 8

81. A milkman has 3 jars containing 57 litres, 129 litres and 177 litres of pure milk respectively. A measuring can, after a different number of exact measurements of milk in each jar, leaves the same amount of milk

unmeasured in each jar. What is the volume of the largest such can?

(J.M.E.T., 2004)

- (a) 12 litres (b) 16 litres
(c) 24 litres (d) None of these

82. The greatest number which can divide 1356, 1868 and 2764 leaving the same remainder 12 in each case is

- (a) 64 (b) 124
(c) 156 (d) 260

83. Which greatest number will divide 3026 and 5053 leaving remainders 11 and 13 respectively?

(C.P.O., 2006)

- (a) 15 (b) 30
(c) 45 (d) 60

84. Find the greatest number that will divide 964, 1238 and 1400 leaving remainders 41, 31 and 51 respectively.

(I.I.F.T., 2005)

- (a) 61 (b) 71
(c) 73 (d) 81

85. Which of the following fractions is the largest?

- (a) $\frac{7}{8}$ (b) $\frac{13}{16}$
(c) $\frac{31}{40}$ (d) $\frac{63}{80}$

86. What is the least natural number which leaves no remainder when divided by all the digits from 1 to 9?

(C.D.S., 2004)

- (a) 1800 (b) 1920
(c) 2520 (d) 5040

87. What will be the least number which when doubled will be exactly divisible by 12, 18, 21 and 30?

- (a) 196 (b) 630
(c) 1260 (d) 2520

88. The sum of two numbers is 45. Their difference is $\frac{1}{9}$ of their sum. Their L.C.M. is

(S.S.C., 2007)

- (a) 100 (b) 150
(c) 200 (d) 250

89. The smallest fraction, which each of $\frac{6}{7}$, $\frac{5}{14}$, $\frac{10}{21}$ will divide exactly is

- (a) $\frac{30}{7}$ (b) $\frac{30}{98}$
(c) $\frac{60}{147}$ (d) $\frac{50}{294}$

90. The least number of five digits which is exactly divisible by 12, 15 and 18 is

- (a) 10010 (b) 10015
(c) 10020 (d) 10080

91. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is

- (a) 9000 (b) 9400
(c) 9600 (d) 9800

92. The number between 4000 and 5000 which is divisible by 12, 18, 21 and 32 is (P.C.S., 2006)
 (a) 4023 (b) 4032
 (c) 4203 (d) 4302
93. The number nearest to 43582 divisible by each of 25, 50 and 75 is (C.P.O., 2007)
 (a) 43500 (b) 43550
 (c) 43600 (d) 43650
94. The least number which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4 and 3 is
 (a) 3 (b) 13
 (c) 23 (d) 33
95. The greatest number which when subtracted from 5834, gives a number exactly divisible by each of 20, 28, 32 and 35 is (S.S.C., 2010)
 (a) 1120 (b) 4714
 (c) 5200 (d) 5600
96. The least number which is a perfect square and is divisible by each of the numbers 16, 20 and 24, is
 (a) 1600 (b) 3600
 (c) 6400 (d) 14400
97. The smallest number which when diminished by 7, is divisible by 12, 16, 18, 21 and 28 is
 (a) 1008 (b) 1015
 (c) 1022 (d) 1032
98. The least number which when increased by 5 is divisible by each one of 24, 32, 36 and 54 is
 (a) 427 (b) 859
 (c) 869 (d) 4320
99. The least number, which when divided by 12, 15, 20 and 54 leaves in each case a remainder of 8 is
 (a) 504 (b) 536
 (c) 544 (d) 548
100. A number less than 500, when divided by 4, 5, 6, 7 leaves remainder 1 in each case. The number is (Hotel Management, 2007)
 (a) 211 (b) 420
 (c) 421 (d) 441
101. What is the greatest number of 3 digits which when divided by 6, 9 and 12 leaves a remainder of 3 in each case? (M.B.A., 2007)
 (a) 903 (b) 939
 (c) 975 (d) 996
102. The largest four-digit number which when divided by 4, 7 or 13 leaves a remainder of 3 in each case, is
 (a) 8739 (b) 9831
 (c) 9834 (d) 9893
103. Let the least number of six digits, which when divided by 4, 6, 10 and 15, leaves in each case the same remainder of 2, be N. The sum of the digits in N is (S.S.C., 2007)
 (a) 3 (b) 4
 (c) 5 (d) 6
104. The least multiple of 13, which on dividing by 4, 5, 6, 7 and 8 leaves remainder 2 in each case is
 (a) 840 (b) 842
 (c) 2520 (d) 2522
105. Find the least number which when divided by 12, leaves a remainder of 7; when divided by 15, leaves a remainder of 10 and when divided by 16, leaves a remainder of 11. (L.D.C., 2006)
 (a) 115 (b) 235
 (c) 247 (d) 475
106. The least number, which when divided by 48, 60, 72, 108 and 140 leaves 38, 50, 62, 98 and 130 as remainders respectively is (P.C.S., 2011)
 (a) 11115 (b) 15110
 (c) 15120 (d) 15210
107. Find the least multiple of 23, which when divided by 18, 21 and 24 leaves remainders 7, 10 and 13 respectively.
 (a) 3002 (b) 3013
 (c) 3024 (d) 3036
108. What is the third term in a sequence of numbers that leave remainders of 1, 2 and 3 when divided by 2, 3 and 4 respectively? (M.A.T., 2004)
 (a) 11 (b) 17
 (c) 19 (d) 35
109. Find the greatest number of 4 digits which when divided by 4, 5, 6, 7 and 8 leaves 1, 2, 3, 4 and 5 as remainders. (M.C.A., 2005)
 (a) 9237 (b) 9240
 (c) 9840 (d) 9999
110. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is (Section Officers', 2005; L.I.C., 2007)
 (a) 1677 (b) 1683
 (c) 2523 (d) 3363
111. Find the least number which when divided by 16, 18, 20 and 25 leaves 4 as remainder in each case, but when divided by 7 leaves no remainder.
 (a) 17004 (b) 18000
 (c) 18002 (d) 18004
112. A gardener has to plant trees in rows containing equal number of trees. If he plants in rows of 6, 8, 10 or 12, then five trees are left unplanted. But if he plants in rows of 13 trees each, then no tree is left. What is the number of trees that the gardener plants? (J.M.E.T., 2004)
 (a) 485 (b) 725
 (c) 845 (d) None of these
113. When Seeta made necklaces of either 16 beads, 20 beads or 36 beads, not a single bead was left over. What could be the least number of beads Seeta had? (Bank Recruitment, 2008)
 (a) 700 (b) 720
 (c) 750 (d) 780

- 114.** An electronic device makes a beep after every 60 sec. Another device makes a beep after every 62 sec. They beeped together at 10 a.m. The next time, when they would beep together at the earliest is
(M.B.A., 2007)
- (a) 10.30 a.m. (b) 10.31 a.m.
(c) 10.59 a.m. (d) 11 a.m.
- 115.** Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?
(M.B.A., 2006)
- (a) 4 (b) 10
(c) 15 (d) 16
- 116.** Four bells begin to toll together and toll respectively at intervals of 6, 7, 8 and 9 seconds. In 1.54 hours, how many times do they toll together and in what interval (seconds)?
(R.R.B., 2006)
- (a) 14, 504 (b) 14, 480
(c) 12, 504 (d) 16, 580
- 117.** Four different electronic devices make a beep after every 30 minutes, 1 hour, $1\frac{1}{2}$ hour and 1 hour 45 minutes respectively. All the devices beeped together at 12 noon. They will again beep together at
(a) 12 midnight (b) 3 a.m.
(c) 6 a.m. (d) 9 a.m.
- 118.** Three girls start jogging from the same point around a circular track and each one completes one round in 24 seconds, 36 seconds and 48 seconds respectively. After how much time will they meet at one point?
(Specialist Officers', 2009)
- (a) 2 minutes 20 seconds (b) 2 minutes 24 seconds
(c) 3 minutes 36 seconds (d) 4 minutes 12 seconds
- 119.** Three persons walking around a circular track complete their respective single revolutions in $15\frac{1}{6}$ seconds, $16\frac{1}{4}$ seconds and $18\frac{2}{3}$ seconds respectively. They will be again together at the common starting point after an hour and
(a) 10 seconds (b) 20 seconds
(c) 30 seconds (d) 40 seconds
- 120.** A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and C in 198 seconds, all starting at the same point. After what time will they meet again at the starting point?
(a) 26 minutes 18 seconds (b) 42 minutes 36 seconds
(c) 45 minutes (d) 46 minutes 12 seconds
- 121.** Three wheels can complete 40, 24 and 16 revolutions per minute respectively. There is a red spot on each wheel that touches the ground at time zero. After how much time, all these spots will simultaneously touch the ground again?
- (a) $7\frac{1}{2}$ sec (b) 18 sec
(c) $7\frac{1}{2}$ min (d) 18 min
- 122.** A pendulum strikes 5 times in 3 seconds and another pendulum strikes 7 times in 4 seconds. If both pendulums start striking at the same time, how many clear strikes can be listened in 1 minute?
(a) 195 (b) 199
(c) 200 (d) 205
- 123.** Find the HCF of 132, 204 and 228.
(Indian Railways—Gr. 'D' Exam, 2014)
- (a) 12 (b) 18
(c) 6 (d) 21
- 124.** If three numbers are $2a$, $5a$ and $7a$, what will be their LCM?
(Indian Railways—Gr. 'D' Exam, 2014)
- (a) $70a$ (b) $65a$
(c) $75a$ (d) $70a^3$
- 125.** The product of two whole numbers is 1500 and their HCF is 10. Find the LCM.
(Indian Railways—Gr. 'D' Exam, 2014)
- (a) 15000 (b) 150
(c) 150 (d) 15
- 126.** A number x is divided by 7. When this number is divided by 8, 12 and 16. It leaves a remainder 3 in each case. The least value of x is:
(SSC—CHSL (10 + 2) Exam, 2015)
- (a) 148 (b) 149
(c) 150 (d) 147
- 127.** The number of pair of positive integers whose sum is 99 and HCF is 9 is
(SSC—CHSL (10 + 2) Exam, 2015)
- (a) 5 (b) 4
(c) 3 (d) 2
- 128.** The ratio of two numbers is 3 : 4 and their LCM is 120. The sum of numbers is
(SSC—CHSL (10 + 2) Exam, 2015)
- (a) 70 (b) 140
(c) 35 (d) 105
- 129.** The greatest four digit number which is exactly divisible by each one of the numbers 12, 18, 21 and 28
(SSC—CHSL (10 + 2) Exam, 2015)
- (a) 9288 (b) 9882
(c) 9828 (d) 9928
- 130.** The traffic lights at three different signal points change after every 45 seconds, 75 seconds and 90 seconds respectively. If all change simultaneously at 7 : 20 : 15 hours, then they will change again simultaneously at
(CLAT, 2016)
- (a) 7 : 28 : 00 hours (b) 7 : 27 : 45 hours
(c) 7 : 27 : 30 hours (d) 7 : 27 : 50 hours

ANSWERS

1. (d) 2. (b) 3. (a) 4. (c) 5. (a) 6. (b) 7. (c) 8. (a) 9. (d) 10. (a)
 11. (a) 12. (a) 13. (c) 14. (a) 15. (a) 16. (b) 17. (d) 18. (d) 19. (d) 20. (c)
 21. (c) 22. (b) 23. (b) 24. (b) 25. (c) 26. (c) 27. (a) 28. (d) 29. (d) 30. (c)
 31. (b) 32. (d) 33. (c) 34. (a) 35. (d) 36. (d) 37. (a) 38. (a) 39. (b) 40. (d)
 41. (c) 42. (b) 43. (c) 44. (d) 45. (a) 46. (c) 47. (b) 48. (d) 49. (d) 50. (c)
 51. (c) 52. (a) 53. (b) 54. (c) 55. (a) 56. (c) 57. (a) 58. (b) 59. (d) 60. (d)
 61. (a) 62. (c) 63. (a) 64. (a) 65. (c) 66. (a) 67. (b) 68. (c) 69. (c) 70. (c)
 71. (b) 72. (b) 73. (a) 74. (b) 75. (d) 76. (a) 77. (b) 78. (a) 79. (a) 80. (a)
 81. (c) 82. (a) 83. (c) 84. (b) 85. (a) 86. (c) 87. (b) 88. (a) 89. (a) 90. (d)
 91. (c) 92. (b) 93. (d) 94. (c) 95. (b) 96. (b) 97. (b) 98. (b) 99. (d) 100. (c)
 101. (c) 102. (b) 103. (c) 104. (d) 105. (b) 106. (b) 107. (b) 108. (d) 109. (a) 110. (b)
 111. (d) 112. (c) 113. (b) 114. (b) 115. (d) 116. (c) 117. (d) 118. (b) 119. (d) 120. (d)
 121. (a) 122. (b) 123. (a) 124. (a) 125. (b) 126. (d) 127. (a) 128. (a) 129. (c) 130. (b)

SOLUTIONS

$$\begin{array}{r}
 2 \overline{) 330} \\
 3 \overline{) 165} \\
 5 \overline{) 65} \\
 11
 \end{array}$$

$$\therefore 330 = 2 \times 3 \times 5 \times 11$$

$$\begin{array}{r}
 2 \overline{) 1122} \\
 3 \overline{) 165} \\
 11 \overline{) 187} \\
 17
 \end{array}$$

$$\therefore 1122 = 2 \times 3 \times 11 \times 17$$

3. Clearly, $252 = 2 \times 2 \times 3 \times 3 \times 7$.

4. $99 = 1 \times 3 \times 3 \times 11$;

$$101 = 1 \times 101;$$

$$176 = 1 \times 2 \times 2 \times 2 \times 2 \times 11;$$

$$182 = 1 \times 2 \times 7 \times 13.$$

So, divisors of 99 are 1, 3, 9, 11, 33 and 99;

divisors of 101 are 1 and 101;

divisors of 176 are 1, 2, 4, 8, 11, 16, 22, 44, 88 and 176;

divisors of 182 are 1, 2, 7, 13, 14, 26, 91 and 182.

Hence, 176 has the most number of divisors.

5. n	Divisors excluding n	Sum of divisors
6	1, 2, 3	6
9	1, 3	4
15	1, 3, 5	9
21	1, 3, 7	11

Clearly, 6 is a perfect number.

$$\begin{array}{r}
 6. 1095 \overline{) 1168} (1 \\
 \underline{1095} \\
 73 \\
 73 \overline{) 1095} (15 \\
 \underline{73} \\
 365 \\
 \underline{365} \\
 \times
 \end{array}$$

So, H.C.F. of 1095 and 1168 = 73.

$$\therefore \frac{1095}{1168} = \frac{1095 \div 73}{1168 \div 73} = \frac{15}{16}.$$

$$\begin{array}{r}
 7. 128352 \overline{) 238368} (1 \\
 \underline{128352} \\
 110016 \overline{) 128352} (1 \\
 \underline{110016} \\
 18336 \overline{) 110016} (6 \\
 \underline{110016} \\
 \times
 \end{array}$$

So, H.C.F. of 128352 and 238368 = 18336.

$$\therefore \frac{128352}{238368} = \frac{128352 \div 18336}{238368 \div 18336} = \frac{7}{13}.$$

$$\begin{array}{r}
 8. 116690151 \overline{) 427863887} (3 \\
 \underline{350070453} \\
 77793434 \overline{) 116690151} (1 \\
 \underline{77793434} \\
 38896717 \overline{) 77793434} (2 \\
 \underline{77793434} \\
 \times
 \end{array}$$

So, H.C.F. of 116,690,151 and 427,863,887 = 38896717.

$$\frac{116,690,151}{427,863,887} = \frac{116690151 \div 38896717}{427863887 \div 38896717} = \frac{3}{11}.$$

9. Since division by 0 is undefined, so 0 cannot be a factor of any natural number.

Hence, H.C.F. of 0 and 6 is undefined.

10. H.C.F. = Product of lowest powers of common factors
 $= 2^2 \times 3^2 \times 5$.

11. H.C.F. = Product of lowest powers of common factors
 $= 3 \times 5 \times 7 = 105$.

12. $4 \times 27 \times 3125 = 2^2 \times 3^3 \times 5^5$;
 $8 \times 9 \times 25 \times 7 = 2^3 \times 3^2 \times 5^2 \times 7$;
 $16 \times 81 \times 5 \times 11 \times 49 = 2^4 \times 3^4 \times 5 \times 7^2 \times 11$.
 $\therefore$ H.C.F. = $2^2 \times 3^2 \times 5 = 180$.

13. $36 = 2^2 \times 3^2$; $84 = 2^2 \times 3 \times 7$.
 $\therefore$ H.C.F. = $2^2 \times 3 = 12$.

14. Since all the numbers formed are even, 2 is a common factor.

Also, H.C.F. of two of the numbers i.e., 48 and 490, is 2.

So, the H.C.F. of all the numbers formed is 2.

15. $204 = 2^2 \times 3 \times 17$;
 $1190 = 2 \times 5 \times 7 \times 17$; $1445 = 5 \times 17^2$.
 $\therefore$ H.C.F. = 17.

16. H.C.F. of 18 and 25 is 1. So, they are co-primes.

$$\begin{array}{r} 17. \quad 2923 \overline{) 3239} \left(1 \right. \\ \quad \underline{2923} \\ \quad \quad 316 \overline{) 2923} \left(9 \right. \\ \quad \quad \quad \underline{2844} \\ \quad \quad \quad \quad 79 \overline{) 316} \left(4 \right. \\ \quad \quad \quad \quad \quad \underline{316} \\ \quad \quad \quad \quad \quad \quad \times \end{array}$$

$\therefore$ H.C.F. = 79.

$$\begin{array}{r} 18. \quad 3444 \overline{) 3556} \left(1 \right. \\ \quad \underline{3444} \\ \quad \quad 112 \overline{) 3444} \left(30 \right. \\ \quad \quad \quad \underline{3360} \\ \quad \quad \quad \quad 84 \overline{) 112} \left(1 \right. \\ \quad \quad \quad \quad \quad \underline{84} \\ \quad \quad \quad \quad \quad \quad 28 \overline{) 84} \left(3 \right. \\ \quad \quad \quad \quad \quad \quad \quad \underline{84} \\ \quad \quad \quad \quad \quad \quad \quad \quad \times \end{array}$$

$\therefore$ H.C.F. = 28.

19. L.C.M. = Product of highest powers of prime factors = $2^5 \times 3^4 \times 5^3 \times 7^2 \times 11$.

$$\begin{array}{r|rrrr} 20. & 24 & - & 36 & - & 40 \\ & 12 & - & 18 & - & 20 \\ & 6 & - & 9 & - & 10 \\ & 3 & - & 9 & - & 5 \\ & 1 & - & 3 & - & 5 \end{array}$$

L.C.M. = $2 \times 2 \times 2 \times 3 \times 3 \times 5 = 360$.

$$\begin{array}{r|rrrr} 21. & 22 & - & 54 & - & 108 & - & 135 & - & 198 \\ & 11 & - & 27 & - & 54 & - & 135 & - & 99 \\ & 11 & - & 9 & - & 18 & - & 45 & - & 33 \\ & 11 & - & 3 & - & 6 & - & 15 & - & 11 \\ & 11 & - & 1 & - & 2 & - & 5 & - & 11 \\ & 1 & - & 1 & - & 2 & - & 5 & - & 1 \end{array}$$

L.C.M. = $2 \times 3 \times 3 \times 3 \times 11 \times 2 \times 5 = 5940$.

22. H.C.F. of 148 and 185 is 37. $\therefore$ L.C.M. = $\left(\frac{148 \times 185}{37} \right) = 740$.

23. H.C.F. of fractions = $\frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}}$.

$$\text{H.C.F. of } \frac{a}{b}, \frac{c}{d}, \frac{e}{f} = \frac{\text{H.C.F. of } a, c, e}{\text{L.C.M. of } b, d, f}.$$

24. Required H.C.F. = $\frac{\text{H.C.F. of } 2, 8, 64, 10}{\text{L.C.M. of } 3, 9, 81, 27} = \frac{2}{81}$.

25. Required H.C.F. = $\frac{\text{H.C.F. of } 9, 12, 18, 21}{\text{L.C.M. of } 10, 25, 35, 40} = \frac{3}{1400}$.

26. Required L.C.M. = $\frac{\text{L.C.M. of } 1, 5, 2, 4}{\text{H.C.F. of } 3, 6, 9, 27} = \frac{20}{3}$.

27. Required L.C.M. = $\frac{\text{L.C.M. of } 2, 3, 4, 9}{\text{H.C.F. of } 3, 5, 7, 13} = \frac{36}{1} = 36$.

28. Required L.C.M. = $\frac{\text{L.C.M. of } 3, 6, 9}{\text{H.C.F. of } 4, 7, 8} = \frac{18}{1} = 18$.

29. Given numbers with two decimal places are: 1.75, 5.60 and 7.00. Without decimal places, these numbers are: 175, 560 and 700, whose H.C.F. is 35.

$\therefore$ H.C.F. of given numbers = 0.35.

30. Given numbers are 1.08, 0.36 and 0.90. H.C.F. of 108, 36 and 90 is 18.

$\therefore$ H.C.F. of given numbers = 0.18.

31. Given numbers are 0.54, 1.80 and 7.20. H.C.F. of 54, 180 and 720 is 18.

$\therefore$ H.C.F. of given numbers = 0.18.

32. Given numbers are 3.00, 2.70 and 0.09. L.C.M. of 300, 270 and 9 is 2700.

$\therefore$ L.C.M. of given numbers = 27.00 = 27.

33. L.C.M. of A and B is B; L.C.M. of B and C is C $\Rightarrow$ L.C.M. of A, B and C is C.

34. $3240 = 2^3 \times 3^4 \times 5$; $3600 = 2^4 \times 3^2 \times 5^2$;
H.C.F. = $36 = 2^2 \times 3^2$.

Since H.C.F. is the product of lowest powers of common factors, so the third number must have $(2^2 \times 3^2)$ as its factor.

Since L.C.M. is the product of highest powers of common prime factors, so the third number must have 3^5 and 7^2 as its factors.

$\therefore$ Third number = $2^2 \times 3^5 \times 7^2$.

35. Let the required numbers be x , $2x$ and $3x$. Then, their H.C.F. = x . So, $x = 12$.

$\therefore$ The numbers are 12, 24 and 36.

36. Let the numbers be $3x$ and $4x$. Then, their H.C.F. = x . So, $x = 4$.
So, the numbers are 12 and 16.
L.C.M. of 12 and 16 = 48.
37. Let the required numbers be $27a$ and $27b$.
Then, $27a + 27b = 216 \Rightarrow a + b = 8$.
Now, co-primes with sum 8 are (1, 7) and (3, 5).
 $\therefore$ Required numbers are $(27 \times 1, 27 \times 7)$ and $(27 \times 3, 27 \times 5)$ i.e., (27, 189) and (81, 135).
Out of these, the one available in the given alternatives is the pair (27, 189).
38. Let the required numbers be $33a$ and $33b$.
Then, $33a + 33b = 528 \Rightarrow a + b = 16$.
Now, co-primes with sum 16 are (1, 15), (3, 13), (5, 11) and (7, 9).
 $\therefore$ Required numbers are $(33 \times 1, 33 \times 15)$, $(33 \times 3, 33 \times 13)$, $(33 \times 5, 33 \times 11)$, $(33 \times 7, 33 \times 9)$.
The number of such pairs is 4.
39. Numbers with H.C.F. 15 must contain 15 as a factor.
Now, multiples of 15 between 40 and 100 are 45, 60, 75 and 90.
 $\therefore$ Number-pairs with H.C.F. 15 are (45, 60), (45, 75), (60, 75) and (75, 90).
[$\because$ H.C.F. of (60, 90) is 30 and that of (45, 90) is 45]
Clearly, there are 4 such pairs.
40. Out of the given numbers, the two with H.C.F. 12 and difference 12 are 84 and 96.
41. Let the numbers be $37a$ and $37b$.
Then, $37a \times 37b = 4107 \Rightarrow ab = 3$.
Now, co-primes with product 3 are (1, 3).
So, the required numbers are $(37 \times 1, 37 \times 3)$ i.e., (1, 111).
 $\therefore$ Greater number = 111.
42. Let the numbers be $13a$ and $13b$.
Then, $13a \times 13b = 2028 \Rightarrow ab = 12$.
Now, co-primes with product 12 are (1, 12) and (3, 4).
So, the required numbers are $(13 \times 1, 13 \times 12)$ and $(13 \times 3, 13 \times 4)$.
Clearly, there are 2 such pairs.
43. Since the numbers are co-prime, they contain only 1 as the common factor.
Also, the given two products have the middle number in common.
So, middle number = H.C.F. of 551 and 1073 = 29;
First number = $\left(\frac{551}{29}\right) = 19$;
Third number = $\left(\frac{1073}{29}\right) = 37$.
 $\therefore$ Required sum = $(19 + 29 + 37) = 85$.
44. Let the numbers be $13x$ and $15x$.
Then, their L.C.M. = $195x$.
So, $195x = 39780$ or $x = 204$.
 $\therefore$ The numbers are 2652 and 3060.
45. Let the numbers be $3x$, $4x$ and $5x$.
Then, their L.C.M. = $60x$.

So, $60x = 2400$ or $x = 40$.

$\therefore$ The numbers are (3×40) , (4×40) and (5×40) .

Hence, required H.C.F. = 40.

46. Let the numbers be $2x$, $3x$, $5x$ and $7x$ respectively.

Then, their L.C.M. = $(2 \times 3 \times 5 \times 7)x = 210x$.

[$\because$ 2, 3, 5, 7 are prime numbers]

So, $210x = 630$ or $x = 3$.

$\therefore$ The numbers are 6, 9, 15 and 21.

Required difference = $21 - 6 = 15$.

47. Other number = $\left(\frac{12 \times 336}{84}\right) = 48$.

48. L.C.M. = $\frac{324}{3} = 108$.

49. Product of numbers = H.C.F. $\times$ L.C.M.

$$\Rightarrow pq = x \times \text{L.C.M.}$$

$$\Rightarrow \text{L.C.M.} = \frac{pq}{x} = \frac{p(xy)}{x} = py.$$

50. Let the numbers be x and $(2000 - x)$.

Then, their L.C.M. = $x(2000 - x)$.

$$\text{So, } x(2000 - x) = 21879$$

$$\Leftrightarrow x^2 - 2000x + 21879 = 0$$

$$\Leftrightarrow (x - 1989)(x - 11) = 0$$

$$\Leftrightarrow x = 1989 \text{ or } x = 11.$$

Hence, the numbers are 1989 and 11.

51. Let the numbers be x and $4x$.

$$\text{Then, } x \times 4x = 84 \times 21 \Leftrightarrow x^2 = \left(\frac{84 \times 21}{4}\right) \Leftrightarrow x = 21.$$

Hence, larger number = $4x = 84$.

52. Let the numbers be x and $(100 - x)$.

$$\text{Then, } x(100 - x) = 5 \times 495$$

$$\Leftrightarrow x^2 - 100x + 2475 = 0$$

$$\Leftrightarrow (x - 55)(x - 45) = 0$$

$$\Leftrightarrow x = 55 \text{ or } x = 45.$$

$\therefore$ The numbers are 45 and 55.

Required difference = $(55 - 45) = 10$.

53. Let the numbers be x and $(x + 2)$.

$$\text{Then, } x(x + 2) = 24$$

$$\Leftrightarrow x^2 + 2x - 24 = 0$$

$$\Leftrightarrow (x - 4)(x + 6) = 0 \Leftrightarrow x = 4.$$

So, the numbers are 4 and 6.

54. Let the numbers be a and b .

$$\text{Then, } a + b = 36 \text{ and } ab = 3 \times 105 = 315.$$

$$\therefore \text{ Required sum} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{36}{315} = \frac{4}{35}.$$

55. Let H.C.F. be h and L.C.M. be l .

$$\text{Then, } l = 12h \text{ and } l + h = 403.$$

$$\therefore 12h + h = 403 \text{ or } h = 31.$$

$$\text{So } l = (403 - 31) = 372.$$

$$\text{Hence, other number} = \left(\frac{31 \times 372}{93}\right) = 124.$$

56. First number = $(50 \times 2) = 100$.

$$\text{Second number} = \left(\frac{50 \times 250}{100} \right) = 125.$$

57. L.C.M. = $\frac{\text{Product of numbers}}{\text{H.C.F.}} = \frac{1320}{6} = 220$.

58. H.C.F. of co-prime numbers is 1. So, L.C.M. = $\frac{117}{1} = 117$.

59. Since H.C.F. is always a factor of L.C.M., we cannot have three numbers with H.C.F. 35 and L.C.M. 120.

60. H.C.F. of two numbers divides their L.C.M. exactly. Clearly, 8 is not a factor of 60.

61. The factors of 9570 are 2, 3, 5, 11 and 29. Clearly, H.C.F. can be any of these factors occurring not more than once and no number other than 2, 3, 5, 11 or 29 or having any factor other than these, can be the H.C.F. So, the only possibility is 11.

62. Clearly, the numbers are (23×13) and (23×14) .

$$\therefore \text{Larger number} = (23 \times 14) = 322.$$

63. Since 16 is not a factor of 136, it follows that there does not exist any pair of numbers with H.C.F. 16 and L.C.M. 136.

64. Product of numbers = $21 \times 4641 = 97461$.

Let the numbers be $21a$ and $21b$.

$$\text{Then, } 21a \times 21b = 97461 \Rightarrow ab = 221.$$

Now, co-primes with product 221 are (1, 221) and (13, 17).

So, the numbers are $(21 \times 1, 21 \times 221)$ and $(21 \times 13, 21 \times 17)$.

Since one number lies between 200 and 300, the suitable pair is (273, 357).

65. Product of numbers = 29×4147 .

Let the numbers be $29a$ and $29b$.

$$\text{Then, } 29a \times 29b = (29 \times 4147) \Rightarrow ab = 143.$$

Now, co-primes with product 143 are (1, 143) and (11, 13).

So, the numbers are $(29 \times 1, 29 \times 143)$ and $(29 \times 11, 29 \times 13)$.

Since both numbers are greater than 29, the suitable pair is $(29 \times 11, 29 \times 13)$, i.e., (319, 377).

$$\therefore \text{Required sum} = (319 + 377) = 696.$$

66. H.C.F. of two prime numbers is 1.

$$\text{Product of numbers} = (1 \times 161) = 161.$$

Let the numbers be a and b . Then, $ab = 161$.

Now, co-primes with product 161 are (1, 161) and (7, 23).

Since x and y are prime numbers and $x > y$, we have $x = 23$ and $y = 7$.

$$\therefore 3y - x = (3 \times 7) - 23 = -2.$$

67. H.C.F. of 2436 and 1001 is 7. Also, H.C.F. of 105 and 7 is 7.

$$\therefore \text{H.C.F. of } 105, 1001 \text{ and } 2436 \text{ is } 7.$$

68. For the minimum number of rows, the number of trees in each row must be the maximum.

$$\therefore \text{Number of trees in each row} = \text{H.C.F. of } 21, 42, 56 = 7.$$

Hence, number of rows

$$= \left(\frac{21 + 42 + 56}{7} \right) = \frac{119}{7} = 17.$$

69. Required length = H.C.F. of 700 cm, 385 cm and 1295 cm = 35 cm.

70. Required capacity

$$= \text{H.C.F. of } 120 \text{ litres and } 56 \text{ litres}$$

$$= 8 \text{ litres} = 8000 \text{ cc.} \quad [\because 1 \text{ litre} = 1000 \text{ cc}]$$

71. Maximum possible daily wage = H.C.F. of ₹ 5750 and ₹ 5000 = ₹ 250.

72. For the least number of bottles, the capacity of each bottle must be maximum.

$$\therefore \text{Capacity of each bottle} = \text{H.C.F. of } 403 \text{ litres, } 465 \text{ litres and } 496 \text{ litres} = 31 \text{ litres.}$$

Hence, required number of bottles

$$= \left(\frac{403 + 465 + 496}{31} \right) = \frac{1364}{31} = 44.$$

73. Required number of students = H.C.F. of 1001 and 910 = 91.

74. Largest size of the tile = H.C.F. of 378 cm and 525 cm = 21 cm.

75. For the least number of tiles, the size of the tile must be the maximum.

$$\text{Maximum size of the tile} = \text{H.C.F. of } 1517 \text{ cm and } 902 \text{ cm} = 41 \text{ cm.}$$

Hence, required number of tiles

$$= \frac{\text{Area of ceiling}}{\text{Area of each tile}} = \left(\frac{1517 \times 902}{41 \times 41} \right) = 814.$$

76. Number of books in each stack = H.C.F. of 336, 240 and 96 = 48.

$$\text{Hence, total number of stacks} = \left(\frac{336 + 240 + 96}{48} \right) = \frac{672}{48} = 14.$$

77. Maximum length of each part = H.C.F. of 78 cm, 104 cm, 117 cm, 169 cm = 13 cm.

$$\therefore \text{Number of pieces} =$$

$$\left(\frac{78 + 104 + 117 + 169}{13} \right) = \frac{468}{13} = 36.$$

78. Required number

$$= \text{H.C.F. of } (91 - 43), (183 - 91) \text{ and } (183 - 43)$$

$$= \text{H.C.F. of } 48, 92 \text{ and } 140 = 4.$$

79. $d = \text{H.C.F. of } (8506 - 7654),$

$$(9997 - 8506) \text{ and } (9997 - 7654)$$

$$= \text{H.C.F. of } 852, 1491 \text{ and } 2343 = 213.$$

Clearly, $r = 199$.

$$\therefore d - r = 213 - 199 = 14.$$

80. $N = \text{H.C.F. of } (4665 - 1305), (6905 - 4665) \text{ and } (6905 - 1305)$

$$= \text{H.C.F. of } 3360, 2240 \text{ and } 5600 = 1120.$$

$$\text{Sum of digits in } N = (1 + 1 + 2 + 0) = 4.$$

81. Required volume

$$= [\text{H.C.F. of } (129 - 57), (177 - 129)$$

$$\text{and } (177 - 57)] \text{ litres}$$

$$= (\text{H.C.F. of } 72, 48 \text{ and } 120) \text{ litres} = 24 \text{ litres.}$$

2	9570
3	4785
5	1595
11	319
	29

213	7654	35
	639	
	1264	
	1065	
	199	

82. Required number

= H.C.F. of $(1356 - 12)$, $(1868 - 12)$ and $(2764 - 12)$
 = H.C.F. of 1344, 1856 and 2752 = 64.

83. Required number

= H.C.F. of $(3026 - 11)$ and $(5053 - 13)$
 = H.C.F. of 3015 and 5040 = 45.

84. Required number

= H.C.F. of $(964 - 41)$, $(1238 - 31)$ and $(1400 - 51)$
 = H.C.F. of 923, 1207 and 1349 = 71.

85. L.C.M. of 8, 16, 40 and 80 = 80.

$$\frac{7}{8} = \frac{70}{80}; \frac{13}{16} = \frac{65}{80}; \frac{31}{40} = \frac{62}{80}.$$

Since, $\frac{70}{80} > \frac{65}{80} > \frac{62}{80}$, so $\frac{7}{8} > \frac{13}{16} > \frac{31}{40}$. So, $\frac{7}{8}$ is the largest.

86. Required number

= L.C.M. of 1, 2, 3, 4, 5, 6, 7, 8, 9
 = $2 \times 2 \times 3 \times 5 \times 7 \times 2 \times 3 = 2520$.

$$\begin{array}{r} 2 \overline{) 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9} \\ 2 \overline{) 1 - 1 - 3 - 2 - 5 - 3 - 7 - 4 - 9} \\ 3 \overline{) 1 - 1 - 3 - 1 - 5 - 3 - 7 - 2 - 9} \\ \hline 1 - 1 - 1 - 1 - 5 - 1 - 7 - 2 - 3 \end{array}$$

87. L.C.M. of 12, 18, 21, 30

= $2 \times 3 \times 2 \times 3 \times 7 \times 5$
 = 1260.

$\therefore$ Required number = $(1260 \div 2) = 630$.

88. Let the two numbers be a and b .

Then, $a + b = 45$... (i)

And, $a - b = 5$... (ii)

Adding (i) and (ii), we get: $2a = 50$ or $a = 25$.

Putting $a = 25$ in (i), we get: $b = 20$.

$\therefore$ L.C.M. = $5 \times 4 \times 5 = 100$.

89. Required fraction

= L.C.M. of $\frac{6}{7}, \frac{5}{14}, \frac{10}{21} = \frac{\text{L.C.M. of } 6, 5, 10}{\text{H.C.F. of } 7, 14, 21} = \frac{30}{7}$.

90. Least number of 5 digits is 10000. L.C.M. of 12, 15 and 18 is 180.

On dividing 10000 by 180, the remainder is 100.

$\therefore$ Required number = $10000 + (180 - 100) = 10080$.

91. Greatest number of 4 digits is 9999.

$$\begin{array}{r} 5 \overline{) 15 - 25 - 40 - 75} \\ 5 \overline{) 3 - 5 - 8 - 15} \\ 3 \overline{) 3 - 1 - 8 - 3} \\ \hline 1 - 1 - 8 - 1 \end{array}$$

L.C.M. of 15, 25, 40 and 75 = $5 \times 5 \times 3 \times 8 = 600$.

On dividing 9999 by 600, the remainder is 399.

$\therefore$ Required number = $(9999 - 399) = 9600$.

92. L.C.M. of 12, 18, 21 and 32 = $2 \times 2 \times 3 \times 3 \times 7 \times 8 = 2016$.

$$\begin{array}{r} 2 \overline{) 12 - 18 - 21 - 32} \\ 2 \overline{) 6 - 9 - 21 - 16} \\ 3 \overline{) 3 - 9 - 21 - 8} \\ \hline 1 - 3 - 7 - 8 \end{array}$$

So, the required number is a multiple of 2016 and lies between 4000 and 5000.

Hence, required number = 4032.

93. L.C.M. of 25, 50 and 75 = $5 \times 5 \times 2 \times 3 = 150$.

$$\begin{array}{r} 5 \overline{) 25 - 50 - 75} \\ 5 \overline{) 5 - 10 - 15} \\ \hline 1 - 2 - 3 \end{array}$$

On dividing 43582 by 150, the remainder is 82 and quotient is 290.

So, required number = $150 \times 291 = 43650$.

94. L.C.M. of 5, 6, 4 and 3 = 60. On dividing 2497 by 60, the remainder is 37.

$\therefore$ Number to be added = $(60 - 37) = 23$.

95. L.C.M. of 20, 28, 32, 35 = $2 \times 2 \times 5 \times 7 \times 8 = 1120$.

Required number = $(5834 - 1120) = 4714$.

$$\begin{array}{r} 2 \overline{) 20 - 28 - 32 - 35} \\ 2 \overline{) 10 - 14 - 16 - 35} \\ 5 \overline{) 5 - 7 - 8 - 35} \\ 7 \overline{) 1 - 7 - 8 - 7} \\ \hline 1 - 1 - 8 - 1 \end{array}$$

96. The least number divisible by 16, 20, 24

= L.C.M. of 16, 20, 24 = $240 = 2 \times 2 \times 2 \times 2 \times 3 \times 5$.

To make it a perfect square, it must be multiplied by 3×5 .

$\therefore$ Required number = $240 \times 3 \times 5 = 3600$.

97. Required number = (L.C.M. of 12, 16, 18, 21, 28) + 7

= $1008 + 7 = 1015$.

98. Required number = (L.C.M. of 24, 32, 36, 54) - 5

= $864 - 5 = 859$.

99. Required number = (L.C.M. of 12, 15, 20, 54) + 8

= $540 + 8 = 548$.

100. L.C.M. of 4, 5, 6, 7 = 420.

$\therefore$ Required number = $420 + 1 = 421$.

101. Greatest number of 3 digits is 999. L.C.M. of 6, 9 and 12 is 36.

On dividing 999 by 36, the remainder obtained is 27.

So, required number = $(999 - 27) + 3 = 975$.

102. Greatest number of 4 digits is 9999. L.C.M. of 4, 7 and 13 = 364.

On dividing 9999 by 364, the remainder obtained is 171.

$\therefore$ Greatest number of 4 digits divisible by 4, 7 and 13 = $(9999 - 171) = 9828$.

Hence, required number = $(9828 + 3) = 9831$.

103. Least number of 6 digits is 100000. L.C.M. of 4, 6, 10 and 15 = 60.

On dividing 100000 by 60, the remainder obtained is 40.

$\therefore$ Least number of 6 digits divisible by 4, 6, 10 and 15

$$= 100000 + (60 - 40) = 100020.$$

$$\therefore N = (100020 + 2) = 100022.$$

$$\text{Sum of digits in } N = (1 + 2 + 2) = 5.$$

104. L.C.M. of 4, 5, 6, 7 and 8 is 840.

Let the required number be $840k + 2$, which is a multiple of 13.

Least value of k for which $(840k + 2)$ is divisible by 13 is $k = 3$.

$$\therefore \text{Required number} = 840 \times 3 + 2 = 2522.$$

105. Here $(12 - 7) = 5$, $(15 - 10) = 5$ and $(16 - 11) = 5$.

$$\therefore \text{Required number} = (\text{L.C.M. of } 12, 15, 16) - 5 \\ = 240 - 5 = 235.$$

106. Here $(48 - 38) = 10$, $(60 - 50) = 10$, $(72 - 62) = 10$
 $(108 - 98) = 10$ & $(140 - 130) = 10$.

$$\therefore \text{Required number} = (\text{L.C.M. of } 48, 60, 72, 108, 140) \\ - 10 = 15120 - 10 = 15110.$$

107. Here $(18 - 7) = 11$, $(21 - 10) = 11$ and $(24 - 13) = 11$.
 L.C.M. of 18, 21 and 24 is 504.

Let the required number be $504k - 11$.

Least value of k for which $(504k - 11)$ is divisible by 23 is $k = 6$.

$$\therefore \text{Required number} = 504 \times 6 - 11 = 3024 - 11 = 3013.$$

108. Clearly, $(2 - 1) = 1$, $(3 - 2) = 1$ and $(4 - 3) = 1$.

L.C.M. of 2, 3, 4 = 12.

So, the sequence shall have numbers of the form $12k - 1$, where $k = 1, 2, 3, \dots$

$$\therefore \text{Third term of the sequence} = 12 \times 3 - 1 \\ = 36 - 1 = 35.$$

109. Clearly, $(4 - 1) = 3$, $(5 - 2) = 3$, $(6 - 3) = 3$, $(7 - 4) = 3$
 and $(8 - 5) = 3$.

L.C.M. of 4, 5, 6, 7, 8 = 840.

Greatest number of 4 digits = 9999.

On dividing 9999 by 840, the remainder is 759.

So, required number = $(9999 - 759) - 3 = 9237$.

110. L.C.M. of 5, 6, 7, 8 = 840.

$\therefore$ Required number is of the form $840k + 3$.

Least value of k for which $(840k + 3)$ is divisible by 9 is $k = 2$.

$$\therefore \text{Required number} = (840 \times 2 + 3) = 1683.$$

111. L.C.M. of 16, 18, 20, 25 = 3600.

Required number is of the form $3600k + 4$.

Least value of k for which $(3600k + 4)$ is divisible by 7 is $k = 5$.

$$\therefore \text{Required number} = (3600 \times 5 + 4) = 18004.$$

112. L.C.M. of 6, 8, 10, 12 = 120.

$\therefore$ Required number is of the form $120k + 5$.

Least value of k for which $(120k + 5)$ is divisible by 13 is $k = 7$.

$$\therefore \text{Required number} = (120 \times 7 + 5) = 845.$$

113. Required number of beads = L.C.M. of 16, 20, 36 = 720.

114. Interval of change = (L.C.M. of 60 and 62) sec = 1860 sec = 31 min.

So, the devices would beep together 31 min after 10 a.m., i.e., at 10.31 a.m.

115. L.C.M. of 2, 4, 6, 8, 10, 12 is 120.

So, the bells will toll together after every 120 seconds, i.e., 2 minutes.

$$\text{In 30 minutes, they will toll together } \left[\left(\frac{30}{2} \right) + 1 \right] = 16 \text{ times.}$$

116. Interval after which the bells will toll together

$$= (\text{L.C.M. of } 6, 7, 8, 9) \text{ sec} = 504 \text{ sec.}$$

$$\text{In 1.54 hours, they will toll together } \left[\left(\frac{1.54 \times 60 \times 60}{504} \right) + 1 \right]$$

times = 12 times.

117. Interval after which the devices will beep together

$$= (\text{L.C.M. of } 30, 60, 90, 105) \text{ min.}$$

$$= 1260 \text{ min} = 21 \text{ hrs.}$$

So, the devices will again beep together 21 hrs. after 12 noon i.e., at 9 a.m.

118. L.C.M. of 24, 36, 48 = 144.

So, the three girls will meet at one point in 144 seconds i.e., 2 min 24 sec.

119. L.C.M. of $\frac{91}{6}$, $\frac{65}{4}$ and $\frac{56}{3} = \frac{\text{L.C.M. of } 91, 65, 56}{\text{H.C.F. of } 6, 4, 3} = 3640$.

So, the three persons will be together at the starting point in 3640 sec

i.e., 1 hour 40 seconds.

120. L.C.M. of 252, 308 and 198 = 2772.

So, A, B and C will again meet at the starting point in 2772 sec. i.e., 46 min. 12 sec.

121. For one complete revolution, the first, second and third wheels take $\frac{60}{40}$, $\frac{60}{24}$, $\frac{60}{16}$ seconds i.e., $\frac{3}{2}$, $\frac{5}{2}$, $\frac{15}{4}$ seconds respectively.

$\therefore$ Time taken for all red spots to touch the ground again simultaneously.

$$= \left(\text{L.C.M. of } \frac{3}{2}, \frac{5}{2}, \frac{15}{4} \right) \text{ sec} = \left(\frac{\text{L.C.M. of } 3, 5, 15}{\text{H.C.F. of } 2, 2, 4} \right) \text{ sec} \\ = \frac{15}{2} \text{ sec} = 7\frac{1}{2} \text{ sec.}$$

122. First pendulum strikes once in $\frac{3}{5}$ seconds. Second pendulum strikes once in $\frac{4}{7}$ seconds.

$$\text{L.C.M. of } \frac{3}{5} \text{ and } \frac{4}{7} = \frac{\text{L.C.M. of } 3 \text{ and } 4}{\text{H.C.F. of } 5 \text{ and } 7} = 12.$$

So, they strike together after every 12 seconds.

$$\text{Thus, they strike together } \left(\frac{60}{12} + 1 \right) = 6 \text{ times in 1 minute.}$$

$\therefore$ Total number of clear strikes heard

$$= \left[\frac{60}{\left(\frac{3}{5} \right)} + \frac{60}{\left(\frac{4}{7} \right)} \right] - 6 = \left(60 \times \frac{5}{3} + 60 \times \frac{7}{4} \right) - 6 \\ = (100 + 105) - 6 = 199.$$

- 123.** H.C.F. of 132, 204 and 228.

$$132 = 2 \times 2 \times 3 \times 11;$$

$$204 = 2 \times 2 \times 3 \times 17;$$

$$228 = 2 \times 2 \times 3 \times 19$$

$\therefore$ HCF of 132, 204 and 228 is $2 \times 2 \times 3$ i.e. 12

$\therefore$ Required HCF = 12

Alternative method:

$$\begin{array}{r} 1 \\ 132 \overline{)204} \\ \underline{132} \\ 72 \\ 72 \overline{)132} \\ \underline{72} \\ 60 \\ 60 \overline{)72} \\ \underline{60} \\ 12 \\ 12 \overline{)60} \\ \underline{60} \\ \times \end{array}$$

Again HCF of 12 and 228

$$\begin{array}{r} 12 \overline{)228} \\ \underline{12} \\ 108 \\ 108 \\ \underline{108} \\ \times \end{array}$$

$\therefore$ Required HCF = 12

- 124.** The given three numbers are $2a$, $5a$ and $7a$. LCM of $2a$, $5a$ and $7a = 2 \times 5 \times 7 \times a = 70a$

- 125.** Product of two numbers = 1500, HCF = 10

$$\begin{aligned} \text{LCM} &= \frac{\text{Product of two numbers}}{\text{Their HCF}} \\ &= \frac{1500}{10} = 150 \end{aligned}$$

Required LCM is 150.

- 126.** LCM of 8, 12 and 16 = 48

$$\begin{array}{r} 2 \overline{)8-12-16} \\ 2 \overline{)4-6-8} \\ 2 \overline{)2-3-4} \\ 1-3-2 \end{array}$$

$$2 \times 2 \times 2 \times 2 \times 3 = 48$$

$\therefore$ Required number = $48a + 3$

Which is divisible by 7.

$$\therefore x = 48a + 3 = 7 \times 6a + 6a + 3$$

$$= (7 \times 6a) + (6a + 3) \text{ which is divisible by 7.}$$

i.e. $6a + 3$ is divisible by 7.

When $a = 3$, $6a + 3 = 18 + 3 = 21$ which is divisible by 7.

$$\therefore x = 48 \times 3 + 3 = 144 + 3 = 147$$

- 127.** Number of pair of positive integers whose sum is 99 and HCF is 9 is (9, 90); (18, 81); (27, 72); (36, 63) ; (45, 54).

- 128.** Let the numbers are $3x$ and $4x$

So, HCF = x

$\therefore$ HCF $\times$ LCM = Product of numbers

$$\Rightarrow x \times 120 = 3x \times 4x$$

$$\Rightarrow x \times 120 = 12x^2$$

$$\Rightarrow 120 = 12x$$

$$\Rightarrow x = 10$$

$\therefore$ Numbers are 30 and 40.

$\therefore$ Sum of two numbers = $30 + 40 = 70$

- 129.** The greatest 4 digit number is 9999

The LCM of 12, 18, 21, 28 is 252

On dividing 9999 by 252 the remainder comes out to be 171

$$\therefore \text{Required number} = 9999 - 171 = 9828$$

$$\begin{array}{r} 5 \overline{)45-75-90} \\ 3 \overline{)9-15-18} \\ 3 \overline{)3-5-6} \\ 1-5-2 \end{array}$$

$$\Rightarrow 5 \times 3 \times 3 \times 5 \times 2 = 450$$

LCM of 45, 75, 90 is 450

$\therefore$ Traffic lights will change simultaneously after 7 minutes 30 seconds i.e. at 7 : 27 : 45 hours.

3

Decimal Fractions

IMPORTANT FACTS AND FORMULAE

I. Decimal Fractions: Fractions in which denominators are powers of 10 are known as *decimal fractions*.

Thus, $\frac{1}{10} = 1 \text{ tenth} = .1$; $\frac{1}{100} = 1 \text{ hundredth} = .01$;

$\frac{99}{100} = 99 \text{ hundredths} = .99$; $\frac{7}{1000} = 7 \text{ thousandths} = .007$, etc.

II. Conversion of a Decimal into Vulgar Fraction: Put 1 in the denominator under the decimal point and annex with it as many zeros as is the number of digits after the decimal point. Now, remove the decimal point and reduce the fraction to its lowest terms.

Thus, $0.25 = \frac{25}{100} = \frac{1}{4}$; $2.008 = \frac{2008}{1000} = \frac{251}{125}$.

III. 1. Annexing zeros to the extreme right of a decimal fraction does not change its value.
Thus, $0.8 = 0.80 = 0.800$, etc.
2. If numerator and denominator of a fraction contain the same number of decimal places, then we remove the decimal sign.

Thus, $\frac{1.84}{2.99} = \frac{184}{299} = \frac{8}{13}$; $\frac{.365}{.584} = \frac{365}{584} = \frac{5}{8}$.

IV. Operations on Decimal Fractions:

1. **Addition and Subtraction of Decimal Fractions:** The given numbers are so placed under each other that the decimal points lie in one column. The numbers so arranged can now be added or subtracted in the usual way.

2. **Multiplication of a Decimal Fraction By a Power of 10:** Shift the decimal point to the right by as many places as is the power of 10.

Thus, $5.9632 \times 100 = 596.32$; $0.073 \times 10000 = 0.0730 \times 10000 = 730$.

3. **Multiplication of Decimal Fractions:** Multiply the given numbers considering them without the decimal point. Now, in the product, the decimal point is marked off to obtain as many places of decimal as is the sum of the number of decimal places in the given numbers.

Suppose we have to find the product $(.2 \times .02 \times .002)$.

Now, $2 \times 2 \times 2 = 8$. Sum of decimal places = $(1 + 2 + 3) = 6$.

$\therefore .2 \times .02 \times .002 = .000008$.

4. **Dividing a Decimal Fraction By a Counting Number:** Divide the given number without considering the decimal point, by the given counting number. Now, in the quotient, put the decimal point to give as many places of decimal as there are in the dividend.

Suppose we have to find the quotient $(0.0204 \div 17)$. Now, $204 \div 17 = 12$.

Dividend contains 4 places of decimal. So, $0.0204 \div 17 = 0.0012$.

5. **Dividing a Decimal Fraction By a Decimal Fraction:** Multiply both the dividend and the divisor by a suitable power of 10 to make divisor a whole number. Now, proceed as above.

Thus, $\frac{0.00066}{0.11} = \frac{0.00066 \times 100}{0.11 \times 100} = \frac{0.066}{11} = .006$.

V. Comparison of Fractions: Suppose some fractions are to be arranged in ascending or descending order of magnitude. Then, convert each one of the given fractions in the decimal form, and arrange them accordingly.

Suppose, we have to arrange the fractions $\frac{3}{5}$, $\frac{6}{7}$ and $\frac{7}{9}$ in descending order.

Now, $\frac{3}{5} = 0.6$, $\frac{6}{7} = 0.857$, $\frac{7}{9} = 0.777....$

Since $0.857 > 0.777 > 0.6$, so $\frac{6}{7} > \frac{7}{9} > \frac{3}{5}$.

VI. Recurring Decimal: If in a decimal fraction, a figure or a set of figures is repeated continuously, then such a number is called a *recurring decimal*.

In a recurring decimal, if a single figure is repeated, then it is expressed by putting a dot on it. If a set of figures is repeated, it is expressed by putting a bar on the set.

Thus, $\frac{1}{3} = 0.333..... = 0.\dot{3}$; $\frac{22}{7} = 3.142857142857..... = 3.\overline{142857}$.

Pure Recurring Decimal: A decimal fraction in which all the figures after the decimal point are repeated, is called a pure recurring decimal.

Converting a Pure Recurring Decimal into Vulgar Fraction: Write the repeated figures only once in the numerator and take as many nines in the denominator as is the number of repeating figures.

Thus, $0.\dot{5} = \frac{5}{9}$; $0.\overline{53} = \frac{53}{99}$; $0.\overline{067} = \frac{67}{999}$; etc.

Mixed Recurring Decimal: A decimal fraction in which some figures do not repeat and some of them are repeated, is called a mixed recurring decimal.

e.g., $0.173333 = 0.17\overline{3}$.

Converting a Mixed Recurring Decimal into Vulgar Fraction: In the numerator, take the difference between the number formed by all the digits after decimal point (taking repeated digits only once) and that formed by the digits which are not repeated. In the denominator, take the number formed by as many nines as there are repeating digits followed by as many zeros as is the number of non-repeating digits.

Thus, $0.1\dot{6} = \frac{16-1}{90} = \frac{15}{90} = \frac{1}{6}$; $0.22\overline{73} = \frac{2273-22}{9900} = \frac{2251}{9900}$.

VII. Some Basic Formulae:

- $(a + b)(a - b) = (a^2 - b^2)$
- $(a + b)^2 = (a^2 + b^2 + 2ab)$
- $(a - b)^2 = (a^2 + b^2 - 2ab)$
- $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
- $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
- $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
- $(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
- When $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.

SOLVED EXAMPLES

Ex. 1. Convert the following into vulgar fractions:

- (i) 0.75 (ii) 3.004 (iii) .0056

Sol. (i) $0.75 = \frac{75}{100} = \frac{3}{4}$. (ii) $3.004 = \frac{3004}{1000} = \frac{751}{250}$. (iii) $.0056 = \frac{56}{10000} = \frac{7}{1250}$.

Ex. 2. Arrange the fractions $\frac{5}{8}$, $\frac{7}{12}$, $\frac{13}{16}$, $\frac{16}{29}$ and $\frac{3}{4}$ in ascending order of magnitude.

Sol. Converting each of the given fractions into decimal form, we get:

$\frac{5}{8} = 0.625$, $\frac{7}{12} = 0.5833$, $\frac{13}{16} = 0.8125$, $\frac{16}{29} = 0.5517$ and $\frac{3}{4} = 0.75$.

Now, $0.5517 < 0.5833 < 0.625 < 0.75 < 0.8125$.

$\therefore \frac{16}{29} < \frac{7}{12} < \frac{5}{8} < \frac{3}{4} < \frac{13}{16}$.

Ex. 3. Arrange the fractions $\frac{3}{5}$, $\frac{4}{7}$, $\frac{8}{9}$ and $\frac{9}{11}$ in their descending order.

Sol. Clearly, $\frac{3}{5} = 0.6$, $\frac{4}{7} = 0.571$, $\frac{8}{9} = 0.88$, $\frac{9}{11} = 0.818$.

Now, $0.88 > 0.818 > 0.6 > 0.571$.

$$\therefore \frac{8}{9} > \frac{9}{11} > \frac{3}{5} > \frac{4}{7}.$$

Ex. 4. Evaluate: (i) $11.11 + 111.1 + 1111.11$

(Bank Recruitment, 2009)

(ii) $6202.5 + 620.25 + 62.025 + 6.2025 + 0.62025$

(iii) $5.064 + 3.98 + .7036 + 7.6 + .3 + 2$

Sol.

$$\begin{array}{r} \text{(i)} \quad 11.11 \\ 111.10 \\ + 1111.11 \\ \hline 1233.32 \end{array}$$

$$\begin{array}{r} \text{(ii)} \quad 6202.5 \\ 620.25 \\ 62.025 \\ 6.2025 \\ + 0.62025 \\ \hline 6891.59775 \end{array}$$

$$\begin{array}{r} \text{(iii)} \quad 5.064 \\ 3.98 \\ 0.7036 \\ 7.6 \\ 0.3 \\ + 2.0 \\ \hline 19.6476 \end{array}$$

Ex. 5. Evaluate: (i) $31.004 - 17.2386$

(ii) $13 - 5.1967$

Sol.

$$\begin{array}{r} \text{(i)} \quad 31.0040 \\ - 17.2386 \\ \hline 13.7654 \end{array}$$

$$\begin{array}{r} \text{(ii)} \quad 13.0000 \\ - 5.1967 \\ \hline 7.8033 \end{array}$$

Ex. 6. Evaluate: (i) $515.15 - 15.51 - 1.51 - 5.11 - 1.11$

(Bank P.O., 2009)

(ii) $43.231 - 12.779 - 6.542 - 0.669$

(Bank P.O., 2008)

Sol. (i) Given expression = $515.15 - (15.51 + 1.51 + 5.11 + 1.11) = 515.15 - 23.24 = 491.91$.

$$\begin{array}{r} 515.15 \\ 15.51 \\ 1.51 \\ 5.11 \\ + 1.11 \\ \hline 23.24 \end{array}$$

$$\begin{array}{r} 515.15 \\ - 23.24 \\ \hline 491.91 \end{array}$$

(ii) Given expression = $43.231 - (12.779 + 6.542 + 0.669) = 43.231 - 19.990 = 23.241$.

$$\begin{array}{r} 43.231 \\ 12.779 \\ 6.542 \\ + 0.669 \\ \hline 19.990 \end{array}$$

$$\begin{array}{r} 43.231 \\ - 19.990 \\ \hline 23.241 \end{array}$$

Ex. 7. What value will replace the question mark in the following equations?

(i) $5172.49 + 378.352 + ? = 9318.678$

(ii) $? - 7328.96 = 5169.38$

Sol.

(i) Let $5172.49 + 378.352 + x = 9318.678$.

Then, $x = 9318.678 - (5172.49 + 378.352) = 9318.678 - 5550.842 = 3767.836$.

(ii) Let $x - 7328.96 = 5169.38$. Then, $x = 5169.38 + 7328.96 = 12498.34$.

Ex. 8. Find the products: (i) 6.3204×100 (ii) $.069 \times 10000$

Sol. (i) $6.3204 \times 100 = 632.04$. (ii) $.069 \times 10000 = .0690 \times 10000 = 690$.

Ex. 9. Find the products:

(Bank P.O., 2008)

(i) 2.1693×1.4

(ii) $.4 \times .04 \times .004 \times 40$

(iii) $6.66 \times 66.6 \times 66$

Sol.

(i) $21693 \times 14 = 303702$. Sum of decimal places = $(4 + 1) = 5$.

$\therefore 2.1693 \times 1.4 = 3.03702$.

(ii) $4 \times 4 \times 4 \times 40 = 2560$. Sum of decimal places = $(1 + 2 + 3) = 6$.

$\therefore .4 \times .04 \times .004 \times 40 = .002560$.

(iii) $666 \times 666 \times 66 = 29274696$. Sum of decimal places = $(2 + 1) = 3$.

$\therefore 6.66 \times 66.6 \times 66 = 29274.696$.

Ex. 10. Given that $268 \times 74 = 19832$, find the value of $2.68 \times .74$.

Sol. Sum of decimal places = $(2 + 2) = 4$.
 $\therefore 2.68 \times .74 = 1.9832$.

Ex. 11. Find the quotient:

(i) $0.63 \div 9$ (ii) $0.0204 \div 17$ (iii) $3.1603 \div 13$

Sol. (i) $63 \div 9 = 7$. Dividend contains 2 places of decimal.
 $\therefore 0.63 \div 9 = .07$.
 (ii) $204 \div 17 = 12$. Dividend contains 4 places of decimal.
 $\therefore 0.0204 \div 17 = .0012$.
 (iii) $31603 \div 13 = 2431$. Dividend contains 4 places of decimal.
 $\therefore 3.1603 \div 13 = .2431$.

Ex. 12. Evaluate:

(i) $35 \div .07$ (ii) $2.5 \div 0.0005$ (iii) $136.09 \div 43.9$

Sol. (i) $\frac{35}{.07} = \frac{35 \times 100}{.07 \times 100} = \frac{3500}{7} = 500$.
 (ii) $\frac{2.5}{0.0005} = \frac{2.5 \times 10000}{0.0005 \times 10000} = \frac{25000}{5} = 5000$.
 (iii) $\frac{136.09}{43.9} = \frac{136.09 \times 10}{43.9 \times 10} = \frac{1360.9}{439} = 3.1$.

Ex. 13. What value will come in place of question mark in the following equations?

(i) $0.006 \div ? = 0.6$ (ii) $? \div .025 = 80$

Sol. (i) Let $\frac{0.006}{x} = 0.6$. Then, $x = \frac{0.006}{0.6} = \frac{0.006 \times 10}{0.6 \times 10} = \frac{0.06}{6} = 0.01$.
 (ii) Let $\frac{x}{.025} = 80$. Then, $x = 80 \times .025 = 2$.

Ex. 14. If $\frac{1}{3.718} = .2689$, then find the value of $\frac{1}{.0003718}$.

Sol. $\frac{1}{.0003718} = \frac{10000}{3.718} = \left(10000 \times \frac{1}{3.718} \right) = 10000 \times .2689 = 2689$.

Ex. 15. Evaluate:

(i) $0.5 \times 5.6 \div 0.5 \times 12$

(Bank Recruitment, 2010)

(ii) $25 \times 3.25 + 50.4 \div 24$

(Bank P.O., 2010)

(iii) $0.01 \times 0.1 - 0.001 \div 10 + 0.01$

(Bank P.O., 2009)

(iv) $12.28 \times 1.5 - 36 \div 2.4$

(Bank P.O., 2009)

Sol. (i) Given expression = $0.5 \times \frac{5.6}{0.5} \times 12 = \frac{5 \times 56 \times 12}{50} = \frac{3360}{50} = 67.2$.

(ii) Given expression = $25 \times 3.25 + \frac{50.4}{24} = 81.25 + 2.1 = 83.35$.

(iii) Given expression = $0.001 - \frac{0.001}{10} + 0.01 = 0.001 - 0.0001 + 0.01 = 0.011 - 0.0001 = 0.0109$.

(iv) Given expression = $12.28 \times 1.5 - \frac{36}{2.4} = 18.42 - 15 = 3.42$.

Ex. 16. The product of two numbers is 0.008 and one is $\frac{1}{5}$ of the other. Find the numbers.

(R.R.B., 2009)

Sol. Let the numbers be x and $\frac{x}{5}$. Then,

$$x \times \frac{x}{5} = 0.008 \Rightarrow \frac{x^2}{5} = 0.008 \Rightarrow x^2 = 0.008 \times 5 = 0.04 \Rightarrow x = \sqrt{0.04} = 0.2.$$

$$\therefore \text{Greater number} = 0.2. \text{ Smaller number} = \frac{0.2}{5} = 0.04.$$

Hence, the required numbers are 0.2 and 0.04.

Ex. 17. Express as vulgar fractions:

(S.S.C., 2010)

$$(i) \overline{0.37} \quad (ii) \overline{0.053} \quad (iii) \overline{0.001} \quad (iv) \overline{3.142857}$$

$$\text{Sol. } (i) \overline{0.37} = \frac{37}{99}. \quad (ii) \overline{0.053} = \frac{53}{999}. \quad (iii) \overline{0.001} = \frac{1}{999}. \quad (iv) \overline{3.142857} = 3 + \overline{0.142857} = 3 + \frac{142857}{999999} = 3\frac{142857}{999999}.$$

Ex. 18. Express as vulgar fractions:

$$(i) \overline{0.17} \quad (ii) \overline{0.1254} \quad (iii) \overline{2.536}$$

$$\text{Sol. } (i) \overline{0.17} = \frac{17-1}{90} = \frac{16}{90} = \frac{8}{45}. \quad (ii) \overline{0.1254} = \frac{1254-12}{9900} = \frac{1242}{9900} = \frac{69}{550}.$$

$$(iii) \overline{2.536} = 2 + \overline{0.536} = 2 + \frac{536-53}{900} = 2 + \frac{483}{900} = 2 + \frac{161}{300} = 2\frac{161}{300}.$$

Ex. 19. If $\frac{b}{a} = 0.25$, then what is the value of $\frac{2a-b}{2a+b} + \frac{2}{9}$?

(R.R.B., 2005)

$$\begin{aligned} \text{Sol. } \frac{2a-b}{2a+b} + \frac{2}{9} &= \frac{2 - \frac{b}{a}}{2 + \frac{b}{a}} + \frac{2}{9} && [\text{Dividing numerator and denominator by } a] \\ &= \frac{2 - 0.25}{2 + 0.25} + \frac{2}{9} = \frac{1.75}{2.25} + \frac{2}{9} = \frac{7}{9} + \frac{2}{9} = \frac{9}{9} = 1. \end{aligned}$$

Ex. 20. Find the value of $(1.99)^2$.

(Campus Recruitment, 2009)

$$\text{Sol. } (1.99)^2 = (2 - 0.01)^2 = 2^2 + (0.01)^2 - 2 \times 2 \times 0.01 = 4 + 0.0001 - 0.04 = 3.9601.$$

Ex. 21. Simplify: $\sqrt{\frac{0.009 \times 0.036 \times 0.016 \times 0.08}{0.002 \times 0.0008 \times 0.0002}}$

(S.S.C., 2010)

Sol. Since the sum of the decimal places in the numerator and denominator is the same, we have:

$$\text{Given expression} = \sqrt{\frac{9 \times 36 \times 16 \times 8}{2 \times 8 \times 2}} = \sqrt{1296} = 36.$$

Ex. 22. Simplify: $\frac{(0.35)^2 - (0.03)^2}{0.19}$.

(R.R.B., 2006)

$$\text{Sol. } \text{Given expression} = \frac{(0.35+0.03)(0.35-0.03)}{0.19} = \frac{0.38 \times 0.32}{0.19} = 2 \times 0.32 = 0.64.$$

Ex. 23. Find the value of $\frac{0.21 \times 0.21 \times 0.21 + 0.021 \times 0.021 \times 0.021}{0.63 \times 0.63 \times 0.63 + 0.063 \times 0.063 \times 0.063}$.

(S.S.C., 2006)

Sol. Let $a = 0.21$ and $b = 0.021$. Then,

$$\text{Given expression} = \frac{a^3 + b^3}{(3a)^3 + (3b)^3} = \frac{a^3 + b^3}{27a^3 + 27b^3} = \frac{a^3 + b^3}{27(a^3 + b^3)} = \frac{1}{27}.$$

Ex. 24. Simplify:

$$(i) \left(\frac{2.75 \times 2.75 \times 2.75 - 2.25 \times 2.25 \times 2.25}{2.75 \times 2.75 + 2.75 \times 2.25 + 2.25 \times 2.25} \right)$$

(C.P.O., 2007)

$$(ii) \left(\frac{0.0347 \times 0.0347 \times 0.0347 + 0.9653 \times 0.9653 \times 0.9653}{0.0347 \times 0.0347 - 0.0347 \times 0.9653 + 0.9653 \times 0.9653} \right)$$

(S.S.C., 2007)

Sol. (i) Given expression = $\left(\frac{a^3 - b^3}{a^2 + ab + b^2} \right)$, where $a = 2.75$, $b = 2.25$
 $= (a - b) = (2.75 - 2.25) = 0.5$.

(ii) Given expression = $\left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right)$, where $a = 0.0347$, $b = 0.9653$
 $= (a + b) = (0.0347 + 0.9653) = 1$.

Ex. 25. Find the value of: $\frac{(0.01)^2 + (0.22)^2 + (0.333)^2}{(0.001)^2 + (0.022)^2 + (0.0333)^2}$.

Sol. Given expression = $\frac{a^2 + b^2 + c^2}{\left(\frac{a}{10}\right)^2 + \left(\frac{b}{10}\right)^2 + \left(\frac{c}{10}\right)^2}$, where $a = 0.01$, $b = 0.022$, $c = 0.0333 = \frac{100(a^2 + b^2 + c^2)}{(a^2 + b^2 + c^2)} = 100$.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. The value of $\frac{42}{10000}$ in decimal fraction is

(L.I.C.A.D.O., 2008)

- (a) .0042 (b) .00042
(c) .0420 (d) 42000

2. The fraction $101\frac{27}{100000}$ in decimal form is

- (a) .01027 (b) .10127
(c) 101.00027 (d) 101.000027

3. When .36 is written in simplest fractional form, the sum of the numerator and the denominator is

- (a) 15 (b) 34
(c) 114 (d) 135

4. The place value of 9 in 0.06945 is (L.I.C.A.D.O., 2008)

- (a) 9 (b) $\frac{9}{10}$
(c) $\frac{9}{100}$ (d) $\frac{9}{1000}$

5. What decimal of an hour is a second?

- (a) .0025 (b) .0256
(c) .00027 (d) .000126

6. If $47.2506 = 4A + \frac{7}{B} + 2C + \frac{5}{D} + 6E$, then the value of $5A + 3B + 6C + D + 3E$ is

- (a) 53.6003 (b) 53.6031
(c) 153.6003 (d) 213.0003

7. Express $\frac{1999}{2111}$ in decimal.

(R.R.B., 2006)

- (a) 0.893 (b) 0.904
(c) 0.946 (d) 0.986

8. How many times is 0.1 greater than 0.01?

- (a) 1 time (b) 10 times
(c) 100 times (d) 1000 times

9. Out of the fractions $\frac{3}{7}, \frac{4}{11}, \frac{5}{9}, \frac{6}{13}, \frac{7}{12}$, which is the second largest? (Bank Recruitment, 2010)

- (a) $\frac{3}{7}$ (b) $\frac{4}{11}$
(c) $\frac{5}{9}$ (d) $\frac{6}{13}$
(e) $\frac{7}{12}$

10. Which of the following has fractions in ascending order?

- (a) $\frac{2}{3}, \frac{3}{5}, \frac{7}{9}, \frac{9}{11}, \frac{8}{9}$ (b) $\frac{3}{5}, \frac{2}{3}, \frac{9}{11}, \frac{7}{9}, \frac{8}{9}$
(c) $\frac{3}{5}, \frac{2}{3}, \frac{7}{9}, \frac{9}{11}, \frac{8}{9}$ (d) $\frac{8}{9}, \frac{9}{11}, \frac{7}{9}, \frac{2}{3}, \frac{3}{5}$
(e) $\frac{8}{9}, \frac{9}{11}, \frac{7}{9}, \frac{3}{5}, \frac{2}{3}$

11. Which of the following are in descending order of their values?

- (a) $\frac{5}{9}, \frac{7}{11}, \frac{8}{15}, \frac{11}{17}$ (b) $\frac{5}{9}, \frac{8}{15}, \frac{11}{17}, \frac{7}{11}$
(c) $\frac{11}{17}, \frac{7}{11}, \frac{8}{15}, \frac{5}{9}$ (d) $\frac{11}{17}, \frac{7}{11}, \frac{5}{9}, \frac{8}{15}$

12. What is the difference between the biggest and the smallest fraction among $\frac{2}{3}, \frac{3}{4}, \frac{4}{5}$ and $\frac{5}{6}$? (S.S.C., 2006)

- (a) $\frac{1}{6}$ (b) $\frac{1}{12}$
(c) $\frac{1}{20}$ (d) $\frac{1}{30}$

13. The least among the following fractions is (S.S.C., 2010)

(a) $\frac{15}{16}$ (b) $\frac{19}{20}$
(c) $\frac{24}{25}$ (d) $\frac{34}{35}$

14. Out of the fractions $\frac{9}{31}, \frac{3}{17}, \frac{6}{23}, \frac{4}{11}$ and $\frac{7}{25}$, which is the largest? (Bank Recruitment, 2010)

(a) $\frac{9}{31}$ (b) $\frac{3}{17}$
(c) $\frac{6}{23}$ (d) $\frac{4}{11}$
(e) None of these

15. If the fractions $\frac{2}{5}, \frac{3}{8}, \frac{4}{9}, \frac{5}{13}$ and $\frac{6}{11}$ are arranged in ascending order of their values, which one will be the fourth? (S.B.I.P.O., 2008)

(a) $\frac{2}{5}$ (b) $\frac{3}{8}$
(c) $\frac{4}{9}$ (d) $\frac{5}{13}$
(e) None of these

16. Which part contains the fractions in ascending order?

(a) $\frac{11}{14}, \frac{16}{19}, \frac{19}{21}$ (b) $\frac{16}{19}, \frac{11}{14}, \frac{19}{21}$
(c) $\frac{16}{19}, \frac{19}{21}, \frac{11}{14}$ (d) $\frac{19}{21}, \frac{11}{14}, \frac{16}{19}$

17. Which of the following fractions is the smallest?

(a) $\frac{13}{16}$ (b) $\frac{15}{19}$
(c) $\frac{17}{21}$ (d) $\frac{7}{8}$

18. Which of the following fractions is greater than $\frac{3}{4}$ and less than $\frac{5}{6}$?

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) $\frac{4}{5}$ (d) $\frac{9}{10}$

19. Which of the following fractions is less than $\frac{7}{8}$ and greater than $\frac{1}{3}$?

(a) $\frac{1}{4}$ (b) $\frac{23}{24}$
(c) $\frac{11}{12}$ (d) $\frac{17}{24}$

20. Which of the following numbers does not lie between $\frac{4}{5}$ and $\frac{7}{13}$?

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) $\frac{3}{4}$ (d) $\frac{5}{7}$

21. The arrangement of rational numbers $\frac{-7}{10}, \frac{5}{-8}, \frac{2}{-3}$ in ascending order is

(a) $\frac{2}{-3}, \frac{5}{-8}, \frac{-7}{10}$ (b) $\frac{5}{-8}, \frac{-7}{10}, \frac{2}{-3}$
(c) $\frac{-7}{10}, \frac{5}{-8}, \frac{2}{-3}$ (d) $\frac{-7}{10}, \frac{2}{-3}, \frac{5}{-8}$

22. Which of the following fractions lies between $\frac{2}{3}$ and $\frac{3}{5}$? (C.P.O., 2006)

(a) $\frac{2}{5}$ (b) $\frac{1}{3}$
(c) $\frac{1}{15}$ (d) $\frac{31}{50}$

23. Which of the following fractions is less than $\frac{1}{5}$?

(a) $\frac{8}{35}$ (b) $\frac{8}{37}$
(c) $\frac{8}{39}$ (d) $\frac{2}{11}$

24. Which of the following fractions is nearest to $\frac{2}{5}$?

(a) $\frac{4}{5}$ (b) $\frac{21}{50}$
(c) $\frac{20001}{50000}$ (d) $\frac{200001}{500000}$

25. The rational numbers lying between $\frac{1}{3}$ and $\frac{3}{4}$ are

(a) $\frac{117}{300}, \frac{287}{400}$ (b) $\frac{95}{300}, \frac{301}{400}$
(c) $\frac{99}{300}, \frac{301}{400}$ (d) $\frac{97}{300}, \frac{299}{500}$

26. Three rational numbers between $\frac{1}{5}$ and $\frac{1}{6}$ are

(a) $\frac{4}{25}, \frac{9}{50}, \frac{17}{100}$ (b) $\frac{4}{25}, \frac{17}{100}, \frac{167}{1000}$
(c) $\frac{4}{25}, \frac{9}{50}, \frac{167}{1000}$ (d) $\frac{9}{50}, \frac{17}{100}, \frac{167}{1000}$

27. $0.3 + 3 + 3.33 + 3.3 + 3.03 + 333 = ?$ (NABARD, 2009)
 (a) 345.99 (b) 355.96
 (c) 375.66 (d) 375.93
 (e) None of these
28. $636.66 + 366.36 + 363.33 = ?$ (Bank Recruitment, 2009)
 (a) 1336.35 (b) 1363.25
 (c) 1366.25 (d) 1636.25
 (e) None of these
29. $24.424 + 5.656 + 1.131 + 0.089 = ?$ (Bank P.O., 2009)
 (a) 31.003 (b) 31.3
 (c) 31.0003 (d) 31.03
 (e) None of these
30. $555.05 + 55.5 + 5.55 + 5 + 0.55 = ?$ (S.B.I.P.O., 2008)
 (a) 621.65 (b) 634.85
 (c) 647.35 (d) 655.75
 (e) None of these
31. $4 + 4.44 + 0.4 + 44.04 + 444 = ?$ (Bank P.O., 2008)
 (a) 469.88 (b) 487.66
 (c) 496.88 (d) 497.24
 (e) None of these
32. $999.99 + 99.99 + 9.99 = ?$ (Bank Recruitment, 2008)
 (a) 1019.89 (b) 1099.88
 (c) 1108.99 (d) 1109.99
 (e) None of these
33. $10.0001 + 9.9999 - 8.9995 = ?$ (E.P.F.O.S.S.A., 2004)
 (a) 9.0005 (b) 10.9995
 (c) 11.0001 (d) 11.0005
 (e) None of these
34. $48.95 - 32.006 = ?$
 (a) 16.089 (b) 16.35
 (c) 16.89 (d) 16.944
35. $58.621 - 13.829 - 7.302 - 1.214 = ?$ (Bank P.O., 2009)
 (a) 31.254 (b) 35.272
 (c) 36.276 (d) 37.281
 (e) None of these
36. $341.42 - 53.74 = ? - 62.86$ (Bank Recruitment, 2010)
 (a) 260.44 (b) 340.44
 (c) 350.54 (d) 450.54
 (e) None of these
37. $534.596 + 61.472 - 496.708 = ? + 27.271$ (Bank P.O., 2010)
 (a) 62.069 (b) 72.089
 (c) 126.631 (d) 132.788
 (e) None of these
38. $11.71 - 0.86 + 1.78 - 9.20 = ?$ (Bank Recruitment, 2010)
 (a) 2.43 (b) 3.13
 (c) 3.43 (d) 4.13
 (e) None of these
39. $726.34 + 888.12 - ? = 1001.88$ (Bank P.O., 2009)
 (a) 602.64 (b) 618.78
 (c) 621.58 (d) 654.54
 (e) None of these
40. $832.58 - 242.31 = 779.84 - ?$
 (a) 179.57 (b) 199.57
 (c) 295.05 (d) None of these
41. Which one of the following is wrong?
 (a) $\frac{9}{4} + 1.75 = 4$ (b) $\frac{9}{5} + 2.2 = 4$
 (c) $\frac{6}{5} + 2.8 = 4$ (d) $\frac{3}{2} + 1.5 = 4$
42. If $x = 10$ and $y = 0.1$, which of the following is the greatest?
 (a) $x^2 + y^2$ (b) $x^2 - y^2$
 (c) $x^2 y^2$ (d) $\frac{x^2}{y^2}$
43. Which of the following is closest to zero?
 (a) $(0.09)^2$ (b) 0.09
 (c) $(1 - 0.9)^2$ (d) $1 - (0.9)^2$
44. One hundredth of centimetre when written in fractions of kilometres, is equal to (M.C.A., 2005)
 (a) 0.0000001 (b) 0.000001
 (c) 0.0001 (d) 0.001
45. What will come in place of question mark in the following equation?
 $54.(?)3 + 543 + 5.43 = 603.26$
 (a) 5 (b) 6
 (c) 8 (d) None of these
46. Which of the following is equal to 3.14×10^6 ? (Hotel Management, 2003)
 (a) 314 (b) 3140
 (c) 3140000 (d) None of these
47. The number 518,000,000 when expressed in scientific notation, equals
 (a) 51.8×10^6 (b) 51.8×10^7
 (c) 5.18×10^8 (d) 5.18×10^9
48. 0.000006723 when expressed in scientific notation, is
 (a) 6723×10^{-5} (b) 67.23×10^{-7}
 (c) 6.723×10^{-6} (d) None of these
49. If $1.125 \times 10^k = 0.001125$, then the value of k is:
 (a) -4 (b) -3
 (c) -2 (d) -1
50. $(0.1 \times 0.01 \times 0.001 \times 10^7)$ is equal to (S.S.C., 2010)
 (a) $\frac{1}{10}$ (b) $\frac{1}{100}$
 (c) 10 (d) 100

- 51.** $383 \times 38 \times 3.8 = ?$ (Bank P.O., 2008)
 (a) 55305.2 (b) 56305.4
 (c) 57305.6 (d) 58305.8
 (e) None of these
- 52.** The product of 0.09 and 0.007 is (L.I.C.A.D.O., 2008)
 (a) 0.6300 (b) 0.00063
 (c) 0.00630 (d) 0.000063
- 53.** $0.1 \times 0.01 \times 0.001 = ?$ (M.C.A., 2005)
 (a) 0.0001 (b) 0.00001
 (c) 0.000001 (d) 0.0000001
- 54.** $3 \times 0.3 \times 0.03 \times 0.003 \times 30 = ?$
 (a) 0.0000243 (b) 0.000243
 (c) 0.00243 (d) 0.0243
- 55.** $14.4 \times 16.5 \times 8 = ?$ (Bank P.O., 2009)
 (a) 1908.0 (b) 1900.8
 (c) 1912.4 (d) 1924.8
- 56.** $47.7 \times 12.4 \times 8.6 = ?$ (Bank Recruitment, 2009)
 (a) 5086.728 (b) 5218.668
 (c) 5708.428 (d) 6180.656
 (e) None of these
- 57.** $32.4 \times 11.5 \times 8.5 = ?$ (Bank Recruitment, 2009)
 (a) 3149.5 (b) 3129.1
 (c) 3167.1 (d) 3162.5
 (e) None of these
- 58.** 0.04×0.0162 is equal to
 (a) 6.48×10^{-3} (b) 6.48×10^{-4}
 (c) 6.48×10^{-5} (d) 6.48×10^{-6}
- 59.** How many digits will be there to the right of the decimal point in the product of 95.75 and .02554?
 (a) 5 (b) 6
 (c) 7 (d) None of these
- 60.** $\left(.00625 \text{ of } \frac{23}{5} \right)$ when expressed as a vulgar fraction, equals
 (a) $\frac{23}{80}$ (b) $\frac{23}{800}$
 (c) $\frac{23}{8000}$ (d) $\frac{125}{23}$
- 61.** Which is the closest approximation to the following product?
 $0.3333 \times 0.25 \times 0.499 \times 0.125 \times 24$
 (a) $\frac{1}{8}$ (b) $\frac{3}{4}$
 (c) $\frac{3}{8}$ (d) $\frac{2}{5}$
- 62.** The number 0.0561 correct to two places of decimals is
 (a) 0.1 (b) 0.06
 (c) 0.05 (d) 0.056
- 63.** $(0.05 \times 5 - 0.005 \times 5)$ equals (S.S.C., 2005)
 (a) 0.0225 (b) 0.225
 (c) 0.250 (d) 0.275
- 64.** $19.99 \times 9.9 + 99.9 = ?$ (NABARD, 2008)
 (a) 129.79 (b) 297.801
 (c) 1009 (d) 296.910
 (e) None of these
- 65.** $345 + 25 \times 0.80 - 111 = ?$ (Bank P.O., 2008)
 (a) 254 (b) 324
 (c) 666 (d) 600
 (e) None of these
- 66.** $29.92 \times 2.4 + 21.28 \times 4.5 = ?$ (Bank P.O., 2009)
 (a) 167.568 (b) 167.658
 (c) 176.568 (d) 176.658
 (e) None of these
- 67.** $4.4 \times 5.8 \times 11.5 - 141.27 = ?$ (Bank Recruitment, 2009)
 (a) 121.17 (b) 147.51
 (c) 152.21 (d) 187.95
 (e) None of these
- 68.** Consider the following quotients:
 1. 368.39 divided by 17 2. 170.50 divided by 62
 3. 875.65 divided by 83
 Their correct sequence in decreasing order is:
 (a) 1, 3, 2 (b) 2, 1, 3
 (c) 2, 3, 1 (d) 3, 1, 2
- 69.** $0.213 \div 0.00213 = ?$
 (a) 1 (b) 10
 (c) 100 (d) None of these
- 70.** $\frac{25.025}{0.025}$ is equal to (R.R.B., 2004)
 (a) 1.01 (b) 10.1
 (c) 101 (d) 1001
- 71.** 4.036 divided by 0.04 gives (Hotel Management, 2003)
 (a) 1.009 (b) 10.09
 (c) 100.9 (d) None of these
- 72.** $\frac{1}{0.04}$ is equal to
 (a) $\frac{1}{40}$ (b) $\frac{2}{5}$
 (c) 2.5 (d) 25
- 73.** $(833.25 - 384.45) \div 24 = ?$ (Bank P.O., 2010)
 (a) 1.87 (b) 2.01
 (c) 18.7 (d) 20.1
 (e) None of these
- 74.** $6425 \div 125 \times 8 = ?$ (Bank Recruitment, 2009)
 (a) 41.12 (b) 64.25
 (c) 411.2 (d) 421.25
 (e) None of these

75. $7777 \div 77 \div 5 = ?$ (Bank Recruitment, 2009)
 (a) 15.2 (b) 18.5
 (c) 22.4 (d) 50.5
 (e) None of these
76. $(0.75 \times 4.4 \times 2.4) \div 0.6 = ?$ (Bank Recruitment, 2009)
 (a) 4.752 (b) 12
 (c) 13.2 (d) 15.84
 (e) None of these
77. $\left(\frac{0.05}{0.25} + \frac{0.25}{0.05}\right)^3 = ?$
 (a) 139.4 (b) 140
 (c) 140.6 (d) 143.9
78. The value of $0.0396 \div 2.51$ correct to 2 significant figures is
 (a) 0.015 (b) 0.0157
 (c) 0.016 (d) 0.017
79. $3927 + 5526 \div 12.5 = ?$ (L.I.C.A.D.O., 2007)
 (a) 750.24 (b) 756.24
 (c) 4369.08 (d) 4369.24
 (e) None of these
80. $(11.6 \div 0.8) (13.5 \div 2) = ?$ (R.R.B., 2008)
 (a) 98 (b) 99
 (c) 100 (d) None of these
81. $(0.05 \times 6.25) \div 2.5 = ?$ (Bank P.O., 2008)
 (a) 0.95 (b) 0.105
 (c) 0.115 (d) 1.25
 (e) None of these
82. $0.5 \times 0.5 + 0.5 \div 5$ is equal to (L.I.C.A.D.O., 2008)
 (a) 0.15 (b) 0.25
 (c) 0.35 (d) 0.45
83. $400 \div 0.1 \times 0.2 = ?$
 (a) 0.8 (b) 8
 (c) 80 (d) 8000
 (e) None of these
84. $5.5 - [6.5 - \{3.5 \div (6.5 - 5.5 - 2.5)\}]$ is equal to
 (a) -1 (b) 0
 (c) 0.1 (d) 1
85. $.04 \times ? = .000016$
 (a) 0.0004 (b) 0.04
 (c) 4 (d) None of these
86. $\frac{.009}{?} = .01$
 (a) .0009 (b) .09
 (c) .9 (d) 9
87. If $\frac{144}{0.144} = \frac{14.4}{x}$, then the value of x is
 (a) 0.0144 (b) 1.44
 (c) 14.4 (d) 144
88. $40.04 \div 0.4 = ? \times 0.05$
 (a) 20.02 (b) 20.2
 (c) 200.2 (d) 2002
89. $48 \div 75 \times 84.5 \div 20 = ?$ (Bank P.O., 2006)
 (a) 1.527 (b) 1.834
 (c) 2.704 (d) 2.914
 (e) None of these
90. $(5420 + 3312 + ?) \div 600 = 25.93$ (Bank Recruitment, 2009)
 (a) 6286 (b) 6584
 (c) 6826 (d) 6830
 (e) None of these
91. $[(?)^2 + (18)^2] \div 125 = 3.56$ (Bank Recruitment, 2009)
 (a) 11 (b) 12
 (c) 14 (d) 15
 (e) None of these
92. $786 \div 24 \times ? = 6.55$
 (a) 0.2 (b) 0.4
 (c) 4 (d) 5
 (e) None of these
93. A maximum of how many pieces of 12.6 cm can be cut from a 857 cm long rod?
 (a) 58 (b) 62
 (c) 64 (d) 68
94. Vishal donates blood thrice in 2 years – each time 350 ml. How many litres of blood will he donate in 6 years? (Bank Recruitment, 2010)
 (a) 1.2 (b) 3.15
 (c) 4.5 (d) 6.3
 (e) None of these
95. Terry consumes 1600 ml of milk everyday. How many litres of milk will she consume in 4 weeks? (Bank Recruitment, 2010)
 (a) 41.6 (b) 42.6
 (c) 43.4 (d) 44.8
 (e) None of these
96. A tailor has 37.5 metres of cloth and he has to make 8 pieces out of a metre of cloth. How many pieces can be make out of this cloth?
 (a) 320 (b) 360
 (c) 400 (d) None of these
97. The price of commodity X increases by 40 paise every year, while the price of commodity Y increases by 15 paise every year. If in 2004, the price of commodity X was ₹ 4.20 and that of Y was ₹ 6.30, in which year commodity X will cost 40 paise more than the commodity Y?
 (a) 2013 (b) 2014
 (c) 2015 (d) 2016

- 98.** A sum of ₹ 6.25 is made up of 80 coins which are either 10P or 5P. How many are there of each kind?
(A.T.M.A., 2006)
- (a) 45, 35 (b) 40, 40
(c) 35, 45 (d) 25, 55
- 99.** The number 0.121212 in the form $\frac{p}{q}$ is equal to
(S.S.C., 2010)
- (a) $\frac{2}{11}$ (b) $\frac{4}{11}$
(c) $\frac{2}{33}$ (d) $\frac{4}{33}$
- 100.** The rational number for the recurring decimal 0.125125 is
- (a) $\frac{63}{487}$ (b) $\frac{119}{993}$
(c) $\frac{125}{999}$ (d) None of these
- 101.** When $0.\overline{47}$ is converted into a fraction, the result is
(Section Officers', 2003)
- (a) $\frac{46}{90}$ (b) $\frac{46}{99}$
(c) $\frac{47}{90}$ (d) $\frac{47}{99}$
- 102.** $0.\overline{36}$ expressed in the form $\frac{p}{q}$ equals
- (a) $\frac{4}{11}$ (b) $\frac{4}{13}$
(c) $\frac{35}{90}$ (d) $\frac{35}{99}$
- 103.** The least among the following is
- (a) 0.2 (b) $1 \div 0.2$
(c) $0.\overline{2}$ (d) $(0.2)^2$
- 104.** $1.\overline{27}$ in the form $\frac{p}{q}$ is equal to
(S.S.C., 2010)
- (a) $\frac{127}{100}$ (b) $\frac{14}{11}$
(c) $\frac{73}{100}$ (d) $\frac{11}{14}$
- 105.** $0.\overline{423}$ is equivalent to the fraction
(C.P.O., 2005)
- (a) $\frac{94}{99}$ (b) $\frac{49}{99}$
(c) $\frac{491}{990}$ (d) $\frac{419}{990}$
- 106.** Express $0.29\overline{56}$ in the form $\frac{p}{q}$ (vulgar fraction)
(R.R.B., 2005)
- (a) $\frac{2956}{1000}$ (b) $\frac{2956}{10000}$
(c) $\frac{2927}{9900}$ (d) None of these
- 107.** Let $F = 0.841\overline{81}$. When F is written as a fraction in lowest terms, the denominator exceeds the numerator by
- (a) 13 (b) 14
(c) 29 (d) 87
- 108.** The value of $4.1\overline{2}$ is
- (a) $4\frac{11}{90}$ (b) $4\frac{11}{99}$
(c) $\frac{371}{900}$ (d) None of these
- 109.** $2.8\overline{768}$ is equal to
(C.P.O., 2006)
- (a) $2\frac{878}{999}$ (b) $2\frac{9}{10}$
(c) $2\frac{292}{333}$ (d) $2\frac{4394}{4995}$
- 110.** The value of $(0.\overline{2} + 0.\overline{3} + 0.\overline{32})$ is
(S.S.C., 2005)
- (a) $0.\overline{77}$ (b) $0.8\overline{2}$
(c) $0.8\overline{6}$ (d) $0.8\overline{7}$
- 111.** $0.1428\overline{57} \div 0.2857\overline{14}$ is equal to
(S.S.C., 2007)
- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) 2 (d) 10
- 112.** $3.8\overline{7} - 2.5\overline{9} = ?$
- (a) 1.20 (b) $1.\overline{2}$
(c) $1.\overline{27}$ (d) $1.\overline{28}$
- 113.** The simplification of $3.\overline{36} - 2.\overline{05} + 1.\overline{33}$ equals
- (a) 2.60 (b) 2.64
(c) $2.\overline{61}$ (d) $2.\overline{64}$
- 114.** $(0.\overline{09} \times 7.\overline{3})$ is equal to:
(S.S.C., 2003)
- (a) $\overline{.6}$ (b) $\overline{.657}$
(c) $\overline{.67}$ (d) $\overline{.657}$
- 115.** $(0.34\overline{67} + 0.13\overline{33})$ is equal to (Hotel Management, 2002)
- (a) $0.4\overline{8}$ (b) 0.48
(c) $0.48\overline{01}$ (d) 0.48
- 116.** $(8.3\overline{1} + 0.\overline{6} + 0.00\overline{2})$ is equal to
(S.S.C., 2005)
- (a) $8.9\overline{12}$ (b) $8.9\overline{12}$
(c) $8.9\overline{79}$ (d) $8.9\overline{79}$
- 117.** The sum of $\overline{2.75}$ and $\overline{3.78}$ is
- (a) $\overline{1.03}$ (b) $\overline{1.53}$
(c) $\overline{4.53}$ (d) $\overline{5.53}$

- 118.** If $\frac{547.527}{0.0082} = x$, then the value of $\frac{547527}{82}$ is
 (a) $\frac{x}{10}$ (b) $10x$
 (c) $100x$ (d) None of these
- 119.** If $2994 \div 14.5 = 172$, then $29.94 \div 1.45 = ?$
 (a) 0.172 (b) 1.72
 (c) 17.2 (d) 172
- 120.** If $213 \times 16 = 3408$, then 1.6×21.3 is equal to
 (a) 0.3408 (b) 3.408
 (c) 34.08 (d) 340.8
- 121.** If $\frac{1}{6.198} = 0.16134$, then the value of $\frac{1}{0.0006198}$ is
 (a) 0.016134 (b) 0.16134
 (c) 1613.4 (d) 16134
- 122.** When 52416 is divided by 312, the quotient is 168. What will be the quotient when 52.416 is divided by 0.0168?
 (a) 3.12 (b) 312
 (c) 3120 (d) None of these
- 123.** Given $168 \times 32 = 5376$, then $5.376 \div 16.8$ is equal to
 (a) 0.032 (b) 0.32
 (c) 3.2 (d) 32
- 124.** $54.327 \times 357.2 \times 0.0057$ is the same as
 (a) $5.4327 \times 3.572 \times 5.7$
 (b) $5.4327 \times 3.572 \times 0.57$
 (c) $54327 \times 3572 \times 0.0000057$
 (d) None of these
- 125.** $\frac{5.3472 \times 324.23}{3.489 \times 5.42}$ is the same as
 (a) $\frac{53472 \times 3.2423}{3.489 \times 54.2}$ (b) $\frac{53472 \times 32423}{3489 \times 542}$
 (c) $\frac{534.72 \times 324.23}{34.89 \times 5.42}$ (d) $\frac{53472 \times 3242.3}{3489 \times 542}$
- 126.** $\frac{96.54 - 89.63}{96.54 + 89.63} \div \frac{965.4 - 896.3}{9.654 + 8.963} = ?$
 (a) 10^{-2} (b) 10^{-1}
 (c) 10 (d) None of these
- 127.** If $1^3 + 2^3 + \dots + 9^3 = 2025$, then the value of $(0.11)^3 + (0.22)^3 + \dots + (0.99)^3$ is close to:
 (a) 0.2695 (b) 0.3695
 (c) 2.695 (d) 3.695
- 128.** The value of $\frac{1}{4} + \frac{1}{4 \times 5} + \frac{1}{4 \times 5 \times 6}$ correct to 4 decimal places is
 (a) 0.3075 (b) 0.3082
 (c) 0.3083 (d) 0.3085
- 129.** Find the value of the following expression upto four places of decimals.

$$\left[1 + \frac{1}{1 \times 2} + \frac{1}{1 \times 2 \times 4} + \frac{1}{1 \times 2 \times 4 \times 8} + \frac{1}{1 \times 2 \times 4 \times 8 \times 16} \right]$$

 (a) 1.6414 (b) 1.6415
 (c) 1.6416 (d) 1.6428
- 130.** The sum of the first 20 terms of the series $\frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \frac{1}{7 \times 8} + \dots$ is
 (a) 0.16 (b) 1.6
 (c) 16 (d) None of these
- 131.** The last digit in the decimal representation of $\left(\frac{1}{5}\right)^{2000}$ is (Hotel Management, 2010)
 (a) 2 (b) 4
 (c) 5 (d) 6
- 132.** Which of the following is equal to 1?
 (a) $\frac{(0.11)^2}{(1.1)^2 \times 0.1}$ (b) $\frac{(1.1)^2}{11^2 \times (0.01)^2}$
 (c) $\frac{(0.011)^2}{(1.1)^2 \times (0.01)^2}$ (d) $\frac{(0.11)^2}{11^2 \times 0.01}$
- 133.** The expression $\frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \dots + \frac{17}{5184} + \frac{19}{8100}$ is equal to (A.A.O. Exam, 2010)
 (a) 0.9 (b) 0.95
 (c) 0.99 (d) 1.91
- 134.** If $1.5x = 0.04y$, then the value of $\left(\frac{y-x}{y+x}\right)$ is (N.M.A.T. 2006; M.B.A. 2007; S.S.C. 2010)
 (a) $\frac{730}{77}$ (b) $\frac{73}{77}$
 (c) $\frac{7.3}{77}$ (d) None of these
- 135.** The value of $\left[35.7 - \left(3 + \frac{1}{3 + \frac{1}{3}} \right) - \left(2 + \frac{1}{2 + \frac{1}{2}} \right) \right]$ is
 (a) 30 (b) 34.8
 (c) 36.6 (d) 41.4
- 136.** $\frac{(0.1667)(0.8333)(0.3333)}{(0.2222)(0.6667)(0.1250)}$ is approximately equal to
 (a) 2 (b) 2.40
 (c) 2.43 (d) 2.50

137. $\frac{3.5 \times 1.4}{0.7} = ?$ (L.I.C.A.D.O., 2007)
 (a) 0.7 (b) 2.4
 (c) 3.5 (d) 7.1
 (e) None of these
138. $\frac{1.6 \times 3.2}{0.08} = ?$ (Bank Recruitment, 2007)
 (a) 0.8 (b) 6.4
 (c) 8 (d) 64
 (e) None of these
139. $\frac{4.41 \times 0.16}{2.1 \times 1.6 \times 0.21}$ is simplified to (S.S.C., 2010)
 (a) 1 (b) 0.1
 (c) 0.01 (d) 10
140. $\frac{.625 \times .0729 \times 28.9}{.0081 \times .025 \times 1.7} = ?$ (I.I.F.T. 2005)
 (a) 382.5 (b) 3625
 (c) 3725 (d) 3825
 (e) None of these
141. The value of $\frac{3.6 \times 0.48 \times 2.50}{0.12 \times 0.09 \times 0.5}$ is
 (a) 80 (b) 800
 (c) 8000 (d) 80000
142. $\frac{0.0203 \times 2.92}{0.0073 \times 14.5 \times 0.7} = ?$
 (a) 0.8 (b) 1.45
 (c) 2.40 (d) 3.25
143. The value of $\frac{3.157 \times 4126 \times 3.198}{63.972 \times 2835.121}$ is closest to
 (a) 0.002 (b) 0.02
 (c) 0.2 (d) 2
144. The value of $\frac{489.1375 \times 0.0483 \times 1.956}{0.0873 \times 92.581 \times 99.749}$ is closet to
 (a) 0.006 (b) 0.06
 (c) 0.6 (d) 6
145. The value of $\frac{241.6 \times 0.3814 \times 6.842}{0.4618 \times 38.25 \times 73.65}$ is close to
 (a) 0.2 (b) 0.4
 (c) 0.6 (d) 1
146. $(0.2 \times 0.2 + 0.01) (0.1 \times 0.1 + 0.02)^{-1}$ is equal to (Section Officers', 2003)
 (a) $\frac{5}{3}$ (b) $\frac{9}{5}$
 (c) $\frac{41}{4}$ (d) $\frac{41}{12}$
147. $\frac{5 \times 1.6 - 2 \times 1.4}{1.3} = ?$
 (a) 0.4 (b) 1.2
 (c) 1.4 (d) 4
148. The value of $(4.7 \times 13.26 + 4.7 \times 9.43 + 4.7 \times 77.31)$ is: (IGNOU, 2003)
 (a) 0.47 (b) 47
 (c) 470 (d) 4700
149. Simplify : $\frac{0.2 \times 0.2 + 0.2 \times 0.02}{0.044}$.
 (a) 0.004 (b) 0.4
 (c) 1 (d) 2
150. The value of $\left(\frac{8.6 \times 5.3 + 8.6 \times 4.7}{4.3 \times 9.7 - 4.3 \times 8.7} \right)$ is
 (a) 3.3 (b) 6.847
 (c) 13.9 (d) 20
151. The value of $\left(\frac{.896 \times .763 + .896 \times .237}{.7 \times .064 + .7 \times .936} \right)$ is
 (a) .976 (b) 9.76
 (c) 1.28 (d) 12.8
152. The value of $(1.25)^3 - 2.25 (1.25)^2 + 3.75 (0.75)^2 - (0.75)^3$ is
 (a) 1 (b) $\frac{1}{2}$
 (c) $\frac{1}{4}$ (d) $\frac{1}{8}$
153. $(78.95)^2 - (43.35)^2 = ?$ (S.B.I.P.O., 2008)
 (a) 4148 (b) 4235.78
 (c) 4305 (d) 4353.88
 (e) None of these
154. The value of $\frac{(75.8)^2 - (55.8)^2}{20}$ is
 (a) 20 (b) 40
 (c) 121.6 (d) 131.6
155. $\frac{(3.63)^2 - (2.37)^2}{3.63 + 2.37}$ is simplified to (C.P.O., 2006)
 (a) 1.26 (b) 1.36
 (c) 2.26 (d) 6
156. $\frac{(36.54)^2 - (3.46)^2}{?} = 40$
 (a) 3.308 (b) 4
 (c) 33.08 (d) 330.8
157. The value of $\frac{(67.542)^2 - (32.458)^2}{75.458 - 40.374}$ is
 (a) 1 (b) 10
 (c) 100 (d) None of these

158. $\left(\frac{1.49 \times 14.9 - 0.51 \times 5.1}{14.9 - 5.1}\right)$ is equal to (S.S.C., 2004)
 (a) 0.20 (b) 2.00
 (c) 20 (d) 22
159. $\frac{4.2 \times 4.2 - 1.9 \times 1.9}{2.3 \times 6.1} = ?$
 (a) 0.5 (b) 1
 (c) 1.9 (d) 4.2
160. Simplify: $\frac{5.32 \times 56 + 5.32 \times 44}{(7.66)^2 - (2.34)^2}$.
 (a) 7.2 (b) 8.5
 (c) 10 (d) 12
161. $\frac{(0.6)^4 - (0.5)^4}{(0.6)^2 + (0.5)^2}$ is equal to
 (a) 0.1 (b) 0.11
 (c) 1.1 (d) 11
162. $(7.5 \times 7.5 + 37.5 + 2.5 \times 2.5)$ is equal to
 (a) 30 (b) 60
 (c) 80 (d) 100
163. The simplification of $\frac{0.2 \times 0.2 + 0.02 \times 0.02 - 0.4 \times 0.02}{0.36}$ gives
 (a) 0.009 (b) 0.09
 (c) 0.9 (d) 9
164. $(99.75)^2 - 2250.0625 = ?$ (Bank P.O., 2008)
 (a) 6545.625 (b) 7700
 (c) 8875 (d) 9900.625
 (e) None of these
165. $(55.25)^2 - 637.5625 = ?$ (Bank P.O., 2008)
 (a) 25.25 (b) 625
 (c) 1375 (d) 2415
 (e) None of these
166. $\frac{3.25 \times 3.20 - 3.20 \times 3.05}{0.064}$ is equal to (S.S.C., 2010)
 (a) 1 (b) $\frac{1}{2}$
 (c) $\frac{1}{10}$ (d) 10
167. The value of $(.98)^3 + (.02)^3 + 3 \times .98 \times .02 - 1$ is (S.S.C., 2005)
 (a) 0 (b) 1
 (c) 1.09 (d) 1.98
168. The expression $(11.98 \times 11.98 + 11.98 \times x + 0.02 \times 0.02)$ will be a perfect square for x equal to:
 (a) 0.02 (b) 0.2
 (c) 0.04 (d) 0.4
169. The value of $\frac{(2.697 - 0.498)^2 + (2.697 + 0.498)^2}{2.697 \times 2.697 + 0.498 \times 0.498}$ is
 (a) 0.5 (b) 2
 (c) 2.199 (d) 3.195
170. The value of $\frac{(0.137 + 0.098)^2 - (0.137 - 0.098)^2}{0.137 \times 0.098}$ is
 (a) 0.039 (b) 0.235
 (c) 0.25 (d) 4
171. The value of $\left(\frac{0.051 \times 0.051 \times 0.051 + 0.041 \times 0.041 \times 0.041}{0.051 \times 0.051 - 0.051 \times 0.041 + 0.041 \times 0.041}\right)$ is (S.S.C., 2003)
 (a) 0.00092 (b) 0.0092
 (c) 0.092 (d) 0.92
172. The value of $\frac{5.71 \times 5.71 \times 5.71 - 2.79 \times 2.79 \times 2.79}{5.71 \times 5.71 + 5.71 \times 2.79 + 2.79 \times 2.79}$ is (S.S.C., 2005)
 (a) 2.82 (b) 2.92
 (c) 8.5 (d) 8.6
173. The value of $\left(\frac{0.943 \times 0.943 - 0.943 \times 0.057 + 0.057 \times 0.057}{0.943 \times 0.943 \times 0.943 + 0.057 \times 0.057 \times 0.057}\right)$ is (M.B.A., 2005)
 (a) 0.32 (b) 0.886
 (c) 1.1286 (d) None of these
174. The value of $\left(\frac{0.125 + 0.027}{0.5 \times 0.5 + 0.09 - 0.15}\right)$ is (S.S.C., 2010)
 (a) 0.08 (b) 0.2
 (c) 0.8 (d) 1
175. $\left(\frac{10.3 \times 10.3 \times 10.3 + 1}{10.3 \times 10.3 - 10.3 + 1}\right)$ is equal to: (S.S.C., 2004)
 (a) 9.3 (b) 10.3
 (c) 11.3 (d) 12.3
176. $\left[\frac{8(3.75)^3 + 1}{(7.5)^2 - 6.5}\right]$ is equal to: (S.S.C., 2003)
 (a) $\frac{9}{5}$ (b) 2.75
 (c) 4.75 (d) 8.5
177. The value of $\left(\frac{0.1 \times 0.1 \times 0.1 + 0.02 \times 0.02 \times 0.02}{0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04}\right)$ is (Hotel Management 2003; S.S.C., 2005)
 (a) 0.0125 (b) 0.125
 (c) 0.25 (d) 0.5

178. $\frac{(0.013)^3 + 0.000000343}{(0.013)^2 - 0.000091 + 0.000049} = ?$ (I.I.F.T., 2005)

- (a) 0.002 (b) 0.020
(c) 0.021 (d) 0.023
(e) None of these

179. The value of $\frac{(2.3)^3 - .027}{(2.3)^2 + .69 + .09}$ is (R.R.B., 2005)

- (a) 0 (b) 1.6
(c) 2 (d) 3.4

180. The value of $\frac{(0.06)^2 + (0.47)^2 + (0.079)^2}{(0.006)^2 + (0.047)^2 + (0.0079)^2}$ is

- (a) 0.1 (b) 10
(c) 100 (d) 1000
(N.M.A.T. 2006; G.B.O. 2007)

181. $\frac{(4.53 - 3.07)^2}{(3.07 - 2.15)(2.15 - 4.53)} + \frac{(3.07 - 2.15)^2}{(2.15 - 4.53)(4.53 - 3.07)} + \frac{(2.15 - 4.53)^2}{(4.53 - 3.07)(3.07 - 2.15)}$

is simplified to (C.P.O., 2004)

- (a) 0 (b) 1
(c) 2 (d) 3

182. The fraction equivalent to $\frac{2}{5}\%$ is

[Indian Railways Gr. 'D' Exam, 2014]

- (a) $\frac{1}{40}$ (b) $\frac{1}{125}$
(c) $\frac{1}{250}$ (d) $\frac{1}{500}$

183. The vulgar fraction of $0.\overline{3939}$ is

[SSC—CHSL (10+2) Exam, 2015]

- (a) $\frac{15}{33}$ (b) $\frac{11}{39}$
(c) $\frac{17}{39}$ (d) $\frac{13}{33}$

184. Solve $(41.99^2 - 18.04^2) \div ? = 13.11^2 - 138.99$

[IBPS—Bank Spl. Officers (IT) Exam, 2015]

- (a) 48 (b) 12
(c) 72 (d) 84
(e) 128

185. Solve $\frac{3}{5}$ of $\frac{4}{7}$ of $\frac{5}{12}$ of 1015 = ?

[United India Insurance Co. Ltd. (UIICL)

Assistant (Online) Exam, 2015]

- (a) 220 (b) 340
(c) 240 (d) 145
(e) 190

186. Solve $1\frac{1}{8} + 1\frac{6}{7} + 3\frac{3}{5} = ?$

[United India Insurance Co. Ltd. (UIICL)
Assistant (Online) Exam, 2015]

- (a) $8\frac{121}{140}$ (b) $6\frac{163}{280}$
(c) $9\frac{197}{280}$ (d) $7\frac{117}{140}$
(e) None of these

187. $2\frac{1.5}{5} + 2\frac{1}{6} - 1\frac{3.5}{15} = \left(\frac{(?)^{1/3}}{4}\right) + 1\frac{7}{30}$

[IDBI Bank Executive Officers Exam, 2015]

- (a) 2 (b) 8
(c) 512 (d) 324
(e) None of these

188. Solve $323.46 + 23.04 - 43.17 - ? = 303$

[NICL—AAO Exam, 2015]

- (a) 0.33 (b) 1.33
(c) 1.23 (d) 0.21
(e) 0.51

189. Solve $1\frac{1}{2} + 2\frac{2}{7} = 3\frac{1}{2} + ?$

[NICL—AAO Exam, 2015]

- (a) $\frac{2}{3}$ (b) $\frac{2}{7}$
(c) $\frac{5}{8}$ (d) $\frac{7}{2}$
(e) $\frac{1}{7}$

190. Solve $48.2 \times 2.5 \times 2.2 + ? = 270$

[NICL—AAO Exam, 2015]

- (a) 6.5 (b) 2.8
(c) 4.9 (d) 3.4
(e) 1.2

191. Solve $4\frac{2}{3} + 3\frac{1}{2} - 1\frac{2}{3} = ?$

[NICL—AAO Exam, 2015]

- (a) $2\frac{1}{5}$ (b) $2\frac{5}{3}$
(c) $1\frac{3}{4}$ (d) $1\frac{1}{2}$
(e) $6\frac{1}{2}$

192. Solve: $17292/33 \div 8 = ?$

[NICL—AAO Exam, 2015]

- (a) 23.5 (b) 53.5
(c) 65.5 (d) 33.5
(e) 45.5

193. Solve $1599 \div 39.99 + \frac{4}{5} \times 2449 - 120.05 = ?$

[IBPS—BANK PO/MT (Pre.) Exam, 2015]

- (a) 1680 (b) 1940
(c) 1640 (d) 1880
(e) 1780

194. Solve $1576 \div 45.02 + 23.99 \times \sqrt{255} = ?$

[IBPS—BANK PO/MT (Pre.) Exam, 2015]

- (a) 340 (b) 420
(c) 380 (d) 460
(e) 360

195. Solve $3899 \div 11.99 - 2379 \div 13.97 = ?$

[IBPS—BANK PO/MT (Pre.) Exam, 2015]

- (a) 125 (b) 250
(c) 155 (d) 135
(e) 225

196. Solve $2\frac{2}{9} + 4\frac{1}{18} - 1\frac{1}{2} = ?$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) $4\frac{5}{9}$ (b) $4\frac{7}{9}$
(c) $5\frac{8}{9}$ (d) $6\frac{1}{2}$
(e) $5\frac{2}{3}$

197. Solve $\frac{294 \div 14 \times 5 + 11}{?} = 8^2 \div 5 + 1.7$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) 8 (b) 6
(c) 12 (d) 5
(e) 10

198. Solve $2.5 \times 4.8 + 7.2 \times 1.5 - 1.2 \times 14 = ?$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) 1.2 (b) 6.5
(c) 4 (d) 4.8
(e) 6

199. Solve $\sqrt{197} \times 6.99 + 626.96 = ?$

[IBPS—Bank PO/PT Exam, 2015]

- (a) 885 (b) 725
(c) 825 (d) 650
(e) 675

200. Which of the following fractions is the largest?

$\frac{3}{2}, \frac{7}{3}, \frac{5}{4}, \frac{7}{2}$

[ESIC—UDC Exam, 2016]

- (a) $\frac{7}{3}$ (b) $\frac{5}{4}$
(c) $\frac{7}{2}$ (d) $\frac{3}{2}$

201. Solve $\frac{?}{529} = \frac{329}{?}$

[SBI Jr. Associates (Pre.) Exam, 2016]

- (a) 404 (b) 408
(c) 410 (d) 414
(e) 416

202. The numerator of a fraction is decreased by 25% and the denominator is increased by 250%. If the resultant fraction is $\frac{6}{5}$, what is the original fraction?

[SBI Jr. Associates (Pre.) Exam, 2016]

- (a) $\frac{22}{5}$ (b) $\frac{24}{5}$
(c) $\frac{27}{6}$ (d) $\frac{28}{5}$
(e) $\frac{30}{11}$

203. Solve $\frac{21.5}{5} + \frac{21}{6} - \frac{13.5}{15} = \left(\frac{(?)^{\frac{1}{3}}}{4} \right) + \frac{17}{30}$

[IBPS—Bank Sp. Officer (Marketing) Exam, 2016]

- (a) 2 (b) 8
(c) 512 (d) 324
(e) None of these

204. Solve $\left(\frac{18}{4} \right)^2 \times \left(\frac{455}{19} \right) \div \left(\frac{61}{799} \right) = ?$

[IBPS—Bank Sp. Officer (Marketing) Exam, 2016]

- (a) 6320 (b) 6400
(c) 6350 (d) 6430
(e) 6490

205. $\frac{5}{9}$ of a number is equal to twenty five percent of second number. Second number is equal to $\frac{1}{4}$ of third number. The value of third number is 2960. What is 30% of first number?

[DMRC—Customer Relationship Assistant (CRA) Exam, 2016]

- (a) 99.9 (b) 88.8
(c) 77.7 (d) None of these

206. Solve $7\frac{1}{2} - \left[2\frac{1}{4} + \left\{ 1\frac{1}{4} - \frac{1}{2} \left(1\frac{1}{2} - \frac{1}{3} - \frac{1}{6} \right) \right\} \right] = ?$

[UPSSSC—Lower Subordinate (Pre.) Exam, 2016]

- (a) $\frac{2}{9}$ (b) $4\frac{1}{2}$
(c) $9\frac{1}{2}$ (d) $1\frac{77}{228}$

ANSWERS

1. (a)	2. (c)	3. (b)	4. (d)	5. (c)	6. (c)	7. (c)	8. (b)	9. (c)	10. (c)
11. (d)	12. (a)	13. (a)	14. (d)	15. (c)	16. (a)	17. (b)	18. (c)	19. (d)	20. (a)
21. (d)	22. (d)	23. (d)	24. (d)	25. (a)	26. (d)	27. (e)	28. (e)	29. (b)	30. (a)
31. (c)	32. (e)	33. (d)	34. (d)	35. (c)	36. (c)	37. (b)	38. (c)	39. (e)	40. (d)
41. (d)	42. (d)	43. (a)	44. (a)	45. (c)	46. (c)	47. (c)	48. (c)	49. (b)	50. (c)
51. (a)	52. (b)	53. (c)	54. (c)	55. (b)	56. (a)	57. (c)	58. (b)	59. (b)	60. (b)
61. (a)	62. (b)	63. (b)	64. (b)	65. (a)	66. (a)	67. (c)	68. (a)	69. (c)	70. (d)
71. (c)	72. (d)	73. (c)	74. (c)	75. (e)	76. (c)	77. (c)	78. (c)	79. (c)	80. (d)
81. (e)	82. (c)	83. (e)	84. (b)	85. (a)	86. (c)	87. (a)	88. (d)	89. (c)	90. (c)
91. (a)	92. (a)	93. (d)	94. (b)	95. (d)	96. (d)	97. (b)	98. (a)	99. (d)	100. (c)
101. (d)	102. (a)	103. (d)	104. (b)	105. (d)	106. (c)	107. (d)	108. (a)	109. (c)	110. (d)
111. (a)	112. (d)	113. (d)	114. (a)	115. (c)	116. (d)	117. (c)	118. (a)	119. (c)	120. (c)
121. (c)	122. (c)	123. (b)	124. (a)	125. (d)	126. (a)	127. (c)	128. (c)	129. (c)	130. (a)
131. (d)	132. (c)	133. (c)	134. (b)	135. (a)	136. (d)	137. (e)	138. (d)	139. (a)	140. (d)
141. (b)	142. (a)	143. (c)	144. (b)	145. (b)	146. (a)	147. (d)	148. (c)	149. (c)	150. (d)
151. (c)	152. (d)	153. (d)	154. (d)	155. (a)	156. (c)	157. (c)	158. (b)	159. (b)	160. (c)
161. (b)	162. (d)	163. (b)	164. (b)	165. (d)	166. (d)	167. (a)	168. (c)	169. (b)	170. (d)
171. (c)	172. (b)	173. (d)	174. (c)	175. (c)	176. (d)	177. (b)	178. (b)	179. (c)	180. (c)
181. (d)	182. (c)	183. (d)	184. (a)	185. (d)	186. (b)	187. (c)	188. (a)	189. (b)	190. (c)
191. (e)	192. (c)	193. (d)	194. (b)	195. (c)	196. (b)	197. (a)	198. (e)	199. (b)	200. (c)
201. (e)	202. (d)	203. (e)	204. (c)	205. (a)	206. (b)				

SOLUTIONS

1. $\frac{42}{10000} = .0042$.

2. $101\frac{27}{100000} = 101 + \frac{27}{100000} = 101 + .00027 = 101.00027$.

3. $0.36 = \frac{36}{100} = \frac{9}{25}$.

Sum of numerator and denominator = $9 + 25 = 34$.

4. 9 is at thousandths place in 0.06945.

So, place value of 9 = $\frac{9}{1000}$.

5. Required decimal = $\frac{1}{60 \times 60} = \frac{1}{3600} = .00027$.

6. $4A + \frac{7}{B} + 2C + \frac{5}{D} + 6E = 47.2506$

$\Rightarrow 4A + \frac{7}{B} + 2C + \frac{5}{D} + 6E = 40 + 7 + 0.2 + 0.05 + 0.0006$

Comparing the terms on both sides, we get:

$4A = 40, \frac{7}{B} = 7, 2C = 0.2, \frac{5}{D} = 0.05, 6E = 0.0006$

or $A = 10, B = 1, C = 0.1, D = 100, E = 0.0001$.

$\therefore 5A + 3B + 6C + D + 3E$
 $= (5 \times 10) + (3 \times 1) + (6 \times 0.1) + 100 + (3 \times 0.0001)$
 $= 50 + 3 + 0.6 + 100 + 0.0003 = 153.6003$.

7. $2111 \overline{)19990} \quad 0.946 \quad \therefore \frac{1999}{2111} = 0.946$

$$\begin{array}{r} 18999 \\ \hline 9910 \\ \hline 8444 \\ \hline 14660 \\ \hline 12666 \\ \hline 1994 \end{array}$$

8. $0.1 = 10 \times 0.01$

9. Converting each of the given fractions into decimal form, we get:

$$\frac{3}{7} = 0.428, \frac{4}{11} = 0.363, \frac{5}{9} = 0.555, \frac{6}{13} = 0.461, \frac{7}{12} = 0.583$$

Clearly, $0.583 > 0.555 > 0.461 > 0.428 > 0.363$.

$$\text{So, } \frac{7}{12} > \frac{5}{9} > \frac{6}{13} > \frac{3}{7} > \frac{4}{11}.$$

10. Converting each of the given fractions into decimal form, we get:

$$\frac{2}{3} = 0.66, \frac{3}{5} = 0.6, \frac{7}{9} = 0.77, \frac{9}{11} = 0.81, \frac{8}{9} = 0.88.$$

Clearly, $0.6 < 0.66 < 0.77 < 0.81 < 0.88$.

$$\text{So, } \frac{3}{5} < \frac{2}{3} < \frac{7}{9} < \frac{9}{11} < \frac{8}{9}.$$

11. Converting each of the given fractions into decimal form, we get:

$$\frac{5}{9} = 0.55, \frac{7}{11} = 0.63, \frac{8}{15} = 0.533, \frac{11}{17} = 0.647.$$

Clearly, $0.647 > 0.63 > 0.55 > 0.533$.

$$\text{So, } \frac{11}{17} > \frac{7}{11} > \frac{5}{9} > \frac{8}{15}.$$

12. Converting each of the given fractions into decimal form, we get:

$$\frac{2}{3} = 0.66, \frac{3}{4} = 0.75, \frac{4}{5} = 0.8, \frac{5}{6} = 0.833.$$

Since $0.833 > 0.8 > 0.75 > 0.66$, so $\frac{5}{6} > \frac{4}{5} > \frac{3}{4} > \frac{2}{3}$.

$$\therefore \text{ Required difference} = \left(\frac{5}{6} - \frac{2}{3} \right) = \frac{1}{6}.$$

13. Converting each of the given fractions into decimal form, we get:

$$\frac{15}{16} = 0.9375, \frac{19}{20} = 0.95, \frac{24}{25} = 0.96, \frac{34}{35} = 0.971$$

Clearly, $0.9375 < 0.95 < 0.96 < 0.971$. So, $\frac{15}{16} < \frac{19}{20} < \frac{24}{25} < \frac{34}{35}$.

14. Converting each of the given fractions into decimal form, we get:

$$\frac{9}{31} = 0.29, \frac{3}{17} = 0.176, \frac{6}{23} = 0.26, \frac{4}{11} = 0.363, \frac{7}{25} = 0.28.$$

Clearly, $0.363 > 0.29 > 0.28 > 0.26 > 0.176$.

$$\text{So, } \frac{4}{11} > \frac{9}{31} > \frac{7}{25} > \frac{6}{23} > \frac{3}{17}.$$

15. Converting each of the given fractions into decimal form, we get:

$$\frac{2}{5} = 0.4, \frac{3}{8} = 0.375, \frac{4}{9} = 0.444, \frac{5}{13} = 0.384, \frac{6}{11} = 0.545$$

Clearly, $0.375 < 0.384 < 0.4 < 0.444 < 0.545$.

$$\text{So, } \frac{3}{8} < \frac{5}{13} < \frac{2}{5} < \frac{4}{9} < \frac{6}{11}.$$

16. Clearly, $\frac{11}{14} = 0.785, \frac{16}{19} = 0.842, \frac{19}{21} = 0.904$.

Now, $0.785 < 0.842 < 0.904$. So, $\frac{11}{14} < \frac{16}{19} < \frac{19}{21}$.

17. We have:

$$\frac{13}{16} = 0.8125, \frac{15}{19} = 0.7894, \frac{17}{21} = 0.8095 \text{ and } \frac{7}{8} = 0.875.$$

Since 0.7894 is the smallest, so $\frac{15}{19}$ is the smallest.

18. $\frac{3}{4} = 0.75, \frac{5}{6} = 0.833, \frac{1}{2} = 0.5, \frac{2}{3} = 0.66, \frac{4}{5} = 0.8, \frac{9}{10} = 0.9$.

Clearly, 0.8 lies between 0.75 and 0.833.

$$\therefore \frac{4}{5} \text{ lies between } \frac{3}{4} \text{ and } \frac{5}{6}.$$

19. $\frac{7}{8} = 0.875, \frac{1}{3} = 0.333, \frac{1}{4} = 0.25, \frac{23}{24} = 0.958,$

$$\frac{11}{12} = 0.916, \frac{17}{24} = 0.708.$$

Clearly, 0.708 lies between 0.333 and 0.875.

$$\therefore \frac{17}{24} \text{ lies between } \frac{1}{3} \text{ and } \frac{7}{8}.$$

20. $\frac{4}{5} = 0.8, \frac{7}{13} = 0.53, \frac{1}{2} = 0.5, \frac{2}{3} = 0.66, \frac{3}{4} = 0.75, \frac{5}{7} = 0.714$.

Clearly, 0.5 does not lie between 0.53 and 0.8.

$$\therefore \frac{1}{2} \text{ does not lie between } \frac{4}{5} \text{ and } \frac{7}{13}.$$

21. $\frac{-7}{10} = -0.7, \frac{5}{-8} = -\frac{5}{8} = -0.625, \frac{2}{-3} = -\frac{2}{3} = -0.66$.

Since $-0.7 < -0.66 < -0.625$, so $\frac{-7}{10} < \frac{2}{-3} < \frac{5}{-8}$.

22. $\frac{2}{3} = 0.666, \frac{3}{5} = 0.6, \frac{2}{5} = 0.4, \frac{1}{3} = 0.333, \frac{1}{15} = 0.066, \frac{31}{50} = 0.62$.

Clearly, 0.62 lies between 0.6 and 0.666. So, $\frac{31}{50}$ lies

between $\frac{2}{3}$ and $\frac{3}{5}$.

23. $\frac{1}{5} = 0.2, \frac{8}{35} = 0.228, \frac{8}{37} = 0.216, \frac{8}{39} = 0.205, \frac{2}{11} = 0.181$.

Clearly, $0.181 < 0.2$. So, $\frac{2}{11} < \frac{1}{5}$.

24. $\frac{2}{5} = 0.4, \frac{4}{5} = 0.8, \frac{21}{50} = 0.42, \frac{20001}{50000} = 0.40002,$
 $\frac{200001}{500000} = 0.400002$.

Clearly, 0.400002 is nearest to 0.4.

$$\text{So, } \frac{200001}{500000} \text{ is nearest to } \frac{2}{5}.$$

25. $\frac{1}{3} = 0.333, \frac{3}{4} = 0.75, \frac{117}{300} = 0.39, \frac{287}{400} = 0.7175,$
 $\frac{95}{300} = 0.316, \frac{301}{400} = 0.7525, \frac{99}{300} = 0.33,$

$$\frac{97}{300} = 0.323, \frac{299}{500} = 0.598.$$

Clearly, each one of 0.39 and 0.7175 lies between 0.333 and 0.75.

So, $\frac{117}{300}$ and $\frac{287}{400}$ lie between $\frac{1}{3}$ and $\frac{3}{4}$.

26. $\frac{1}{5} = 0.2$ and $\frac{1}{6} = 0.166$.

(a) $\frac{4}{25} = 0.16, \frac{9}{50} = 0.18, \frac{17}{100} = 0.17$

(b) $\frac{4}{25} = 0.16, \frac{17}{100} = 0.17, \frac{167}{1000} = 0.167$

(c) $\frac{4}{25} = 0.16, \frac{9}{50} = 0.18, \frac{167}{1000} = 0.167$

(d) $\frac{9}{50} = 0.18, \frac{17}{100} = 0.17, \frac{167}{1000} = 0.167$

27.	0.3	28.	636.66	29.	24.424
	3.0		366.36		5.656
	3.33		+ 363.33		1.131
	3.03		<u>1366.35</u>		+ 0.089
	+ 333.0				<u>31.300</u>
	<u>345.96</u>				

30.	555.05	31.	4.0	32.	9 9 9 9
	55.5		4.44		9 9 9 9
	5.55		0.4		+ 9 9 9
	5.0		44.04		+ 9 9 9
	+ 0.55		+ 444.0		<u>1 1 0 9 9 7</u>
	<u>621.65</u>		<u>496.88</u>		

33.	10.0 0 0 1		20.0000	34.	48.95
	+ 9.9 9 9 9		- 8.9995		- 32.0 0 6
	<u>20.0 0 0 0</u>		<u>11.0005</u>		<u>16.9 4 4</u>

35. Given expression = $58.621 - (13.829 + 7.302 + 1.214)$
 $= 58.621 - 22.345 = 36.276$.

36. Let $341.42 - 53.74 = x - 62.86$.

Then, $x = (341.42 + 62.86) - 53.74$
 $= 404.28 - 53.74 = 350.54$.

37. Let $534.596 + 61.472 - 496.708 = x + 27.271$

Then, $x = (534.596 + 61.472) - (496.708 + 27.271)$
 $= 596.068 - 523.979 = 72.089$.

38. Given expression = $(11.71 + 1.78) - (0.86 + 9.20)$
 $= 13.49 - 10.06 = 3.43$.

39. Let $726.34 + 888.12 - x = 1001.88$.

Then, $x = (726.34 + 888.12) - 1001.88$
 $= 1614.46 - 1001.88 = 612.58$.

40. Let $832.58 - 242.31 = 779.84 - x$.

Then, $x = (779.84 + 242.31) - 832.58$
 $= 1022.15 - 832.58 = 189.57$.

41. (a) $\frac{9}{4} + 1.75 = 2.25 + 1.75 = 4$.

(b) $\frac{9}{5} + 2.2 = 1.8 + 2.2 = 4$.

(c) $\frac{6}{5} + 2.8 = 1.2 + 2.8 = 4$.

(d) $\frac{3}{2} + 1.5 = 1.5 + 1.5 = 3$.

42. (a) $x^2 + y^2 = (10)^2 + (0.1)^2 = 100 + 0.01 = 100.01$.

(b) $x^2 - y^2 = (10)^2 - (0.1)^2 = 100 - 0.01 = 99.99$.

(c) $x^2 y^2 = (10)^2 \times (0.1)^2 = 100 \times 0.01 = 1$.

(d) $\frac{x^2}{y^2} = \frac{(10)^2}{(0.1)^2} = \frac{100}{0.01} = 10000$.

Clearly, $\frac{x^2}{y^2}$ is the greatest.

43. (a) $(0.09)^2 = 0.0081$

(b) 0.09

(c) $(1 - 0.9)^2 = (0.1)^2 = 0.01$

(d) $1 - (0.9)^2 = 1 - 0.81 = 0.19$

Clearly, $0.0081 < 0.01 < 0.09 < 0.19$.

So, 0.0081 is closest to zero.

44. Required fraction = $\frac{\frac{1}{100} \text{ cm}}{1 \text{ km}} = \frac{\left(\frac{1}{100}\right) \text{ cm}}{(1000 \times 100) \text{ cm}}$
 $= \frac{1}{100 \times 1000 \times 100}$
 $= \frac{1}{1000000} = 0.0000001$.

45. Let $x + 543 + 5.43 = 603.26$.

Then, $x = 603.26 - (543 + 5.43)$
 $= 603.26 - 548.43 = 54.83$.

$\therefore$ Missing digit = 8.

46. $3.14 \times 10^6 = 3.140000 \times 1000000 = 3140000$.

47. $518,000,000 = 5.18 \times 100000000 = 5.18 \times 10^8$.

48. $0.000006723 = \frac{0.000006723 \times 10^6}{10^6}$
 $= \frac{6.723}{10^6} = 6.723 \times 10^{-6}$.

49. $10^k = \frac{0.001125}{1.125} = \frac{1.125}{1125} = \frac{1.125 \times 10^3}{1125 \times 10^3} = \frac{1}{10^3} = 10^{-3}$.

$\therefore k = -3$.

50. Given expression = $\left(\frac{1}{10} \times \frac{1}{100} \times \frac{1}{1000} \times 10^7\right)$
 $= \left(\frac{10^7}{10^6}\right) = 10^{(7-6)} = 10$.

51. $383 \times 38 \times 38 = 553052$.

Number of decimal places = 1.

$\therefore 383 \times 38 \times 3.8 = 55305.2$.

52. $9 \times 7 = 63$.

Sum of decimal places = 5. $\therefore 0.09 \times 0.007 = 0.00063$.

53. Sum of decimal places = 6. $\therefore 0.1 \times 0.01 \times 0.001 = 0.000001$.

54. $3 \times 3 \times 3 \times 3 \times 30 = 2430$.

Sum of decimal places = 6.

$$\therefore 3 \times 0.3 \times 0.03 \times 0.003 \times 30 = 0.00243.$$

55. $144 \times 165 \times 8 = 190080.$

Sum of decimal places = 2.

$$\therefore 14.4 \times 16.5 \times 8 = 1900.80.$$

56. $477 \times 124 \times 86 = 5086728.$

Sum of decimal places = 3.

$$\therefore 47.7 \times 12.4 \times 8.6 = 5086.728.$$

57. $324 \times 115 \times 85 = 3167100.$

Sum of decimal places = 3.

$$\therefore 32.4 \times 11.5 \times 8.5 = 3167.1.$$

58. $4 \times 162 = 648.$

Sum of decimal places = 6.

$$\therefore 0.04 \times 0.0162 = 0.000648 = 6.48 \times 10^{-4}.$$

59. Sum of decimal places = 7.

Since the last digit to the extreme right will be zero

($\because 5 \times 4 = 20$), so there will be 6 significant digits to the right of the decimal point.

60. $\left(.00625 \text{ of } \frac{23}{5} \right) = \left(\frac{625}{100000} \times \frac{23}{5} \right) = \frac{23}{800}.$

61. Given product = $0.3 \times 0.25 \times 0.5 \times 0.125 \times 24$

$$= \left(\frac{3}{10} \times \frac{25}{100} \times \frac{5}{10} \times \frac{125}{1000} \times 24 \right)$$

$$= \frac{9}{80} = \frac{1}{8} \text{ (app.)}$$

62. The digit at the third decimal place is 6, which is greater than 5.

So, 0.0561, rounded to two decimal places, is 0.06.

63. Given expression = $0.25 - 0.025 = 0.225.$

64. Given expression = $197.901 + 99.9 = 297.801.$

65. Given expression = $345 + 20 - 111 = 365 - 111 = 254.$

66. Given expression = $71.808 + 95.76 = 167.568.$

67. Given expression = $293.48 - 141.27 = 152.21.$

68. 1. $36839 \div 17 = 2167.$

Dividend contains 2 places of decimal.

$$\therefore 368.39 \div 17 = 21.67.$$

2. $17050 \div 62 = 275.$ Dividend contains 2 places of decimal.

$$\therefore 170.50 \div 62 = 2.75.$$

3. $87565 \div 83 = 1055.$

Dividend contains 2 places of decimal.

$$\therefore 875.65 \div 83 = 10.55.$$

Since $21.67 > 10.55 > 2.75$, the desired order is 1, 3, 2.

69. $\frac{0.213}{0.00213} = \frac{0.213 \times 100000}{0.00213 \times 100000} = \frac{213 \times 100}{213} = 100.$

70. $\frac{25.025}{0.025} = \frac{25025}{25} = 1001.$

71. $\frac{4.036}{0.04} = \frac{403.6}{4} = 100.9.$

72. $\frac{1}{0.04} = \frac{100}{4} = 25.$

73. Given expression = $448.8 \div 24 = 18.7.$

74. Given expression = $51.4 \times 8 = 411.2.$

75. Given expression = $101 \div 5 = 20.2.$

76. Given expression = $7.92 \div 0.6 = 79.2 \div 6 = 13.2.$

77. $\left(\frac{0.05}{0.25} + \frac{0.25}{0.05} \right)^3 = \left(\frac{5}{25} + \frac{25}{5} \right)^3$

$$= \left(\frac{1}{5} + 5 \right)^3 = \left(\frac{26}{5} \right)^3 = (5.2)^3 = 140.608.$$

78. $\frac{0.0396}{2.51} = \frac{3.96}{251} = \left(\frac{396}{251 \times 100} \right) = \frac{1.577}{100} = 0.01577 \approx 0.016.$

79. Given expression = $3927 + \frac{5526}{12.5} = 3927 + \frac{55260}{125}$

$$= 3927 + 442.08 = 4369.08.$$

80. Given expression = $\frac{116}{8} \times \frac{13.5}{2} = 14.5 \times 6.75 = 97.875.$

81. Given expression = $\frac{0.05 \times 6.25}{2.5} = \frac{0.3125}{2.5} = \frac{3.125}{25} = 0.125.$

82. Given expression = $0.5 \times 0.5 + \frac{0.5}{5} = 0.25 + 0.1 = 0.35.$

83. Given expression = $\frac{400}{0.1} \times 0.2 = 4000 \times 0.2 = 800.$

84. Given expression = $5.5 - [6.5 - \{3.5 \div (6.5 - 3)\}]$

$$= 5.5 - [6.5 - \{3.5 \div 3.5\}]$$

$$= 5.5 - [6.5 - 1] = 5.5 - 5.5 = 0.$$

85. Let $.04 \times x = .000016.$

Then, $x = \frac{.000016}{.04} = \frac{.0016}{4} = .0004.$

86. Let $\frac{.009}{x} = .01.$ Then, $x = \frac{.009}{.01} = \frac{9}{1} = 9.$

87. $\frac{144}{0.144} = \frac{14.4}{x} \Leftrightarrow \frac{144 \times 1000}{144} = \frac{14.4}{x} \Leftrightarrow x = \frac{14.4}{1000} = 0.0144.$

88. Let $40.04 \div 0.4 = x \times 0.05.$

Then, $\frac{40.04}{0.4} = 0.05x \Rightarrow \frac{400.4}{4} = 0.05x$

$$\Rightarrow 0.05x = 100.1 \Rightarrow x = \frac{100.1}{0.05} = \frac{10010}{5} = 2002.$$

89. Given expression = $\frac{48}{75} \times \frac{84.5}{20} = 0.64 \times 4.225 = 2.704.$

90. Let $(5420 + 3312 + x) \div 600 = 25.93.$

Then, $\frac{8732 + x}{600} = 25.93$

$$\Rightarrow 8732 + x = 25.93 \times 600$$

$$\Rightarrow 8732 + x = 15558 \Rightarrow x = 6826.$$

91. Let $\frac{x^2 + (18)^2}{125} = 3.56.$

Then, $x^2 + 324 = 125 \times 3.56 = 445$

$$\Rightarrow x^2 = 121 \Rightarrow x = 11.$$

92. Let $786 \div 24 \times x = 6.55$.

Then, $\frac{786}{24} \times x = 6.55 \Rightarrow 32.75 x = 6.55$

$\Rightarrow x = \frac{6.55}{32.75} = \frac{655}{3275} = \frac{1}{5} = 0.2$.

93. Number of pieces = $\frac{857}{12.6} = \frac{8570}{126} = 68.01 \approx 68$.

94. Quantity of blood donated in 2 years = (350×3) ml
= 1050 ml = 1.05 litres.

$\therefore$ Quantity of blood donated in 6 years

= $\left(\frac{1.05}{2} \times 6\right)$ litres = 3.15 litres.

95. Daily consumption = 1600 ml = 1.6 litres.

Consumption in 4 weeks = $(1.6 \times 4 \times 7)$ litres = 44.8 litres.

96. Length of each piece = $\left(\frac{1}{8}\right)$ m = 0.125 m.

$\therefore$ Required number of pieces

= $\left(\frac{37.5}{0.125}\right) = \left(\frac{375 \times 100}{125}\right) = 300$.

97. Suppose commodity X will cost 40 paise more than Y after z years. Then,

$(4.20 + 0.40z) - (6.30 + 0.15z) = 0.40$

$\Leftrightarrow 0.25z = 0.40 + 2.10 \Leftrightarrow z = \frac{2.50}{0.25} = \frac{250}{25} = 10$.

$\therefore$ X will cost 40 paise more than Y 10 years after 2004 i.e., in 2014.

98. Let the number of 10P coins be x .

Then, number of 5P coins = $(80 - x)$.

$\therefore 0.1x + 0.05(80 - x) = 6.25$

$\Rightarrow 0.1x + 4 - 0.05x = 6.25 \Rightarrow 0.05x = 2.25$

$\Rightarrow x = \frac{2.25}{0.05} = \frac{225}{5} = 45$.

Hence, number of 10P coins = 45. And, number of 5P coins = $(80 - 45) = 35$.

99. $0.121212\dots = 0.\overline{12} = \frac{12}{99} = \frac{4}{33}$.

100. $0.125125\dots = 0.\overline{125} = \frac{125}{999}$.

101. $0.\overline{47} = \frac{47}{99}$.

102. $0.\overline{36} = \frac{36}{99} = \frac{4}{11}$.

103. $1 \div 0.2 = \frac{1}{0.2} = \frac{10}{2} = 5$; $0.\overline{2} = 0.222\dots$; $(0.2)^2 = 0.04$.

$0.04 < 0.2 < 0.22\dots < 5$.

Since 0.04 is the least, so $(0.2)^2$ is the least.

104. $1.\overline{27} = 1 + 0.\overline{27} = 1 + \frac{27}{99} = 1 + \frac{3}{11} = \frac{11+3}{11} = \frac{14}{11}$.

105. $0.\overline{423} = \frac{423-4}{990} = \frac{419}{990}$.

106. $0.29\overline{56} = \frac{2956-29}{9900} = \frac{2927}{9900}$.

107. $0.84\overline{181} = \frac{84181-841}{99000} = \frac{83340}{99000} = \frac{463}{550}$.

$\therefore$ Required difference = $(550 - 463) = 87$.

108. $4.\overline{12} = 4 + 0.\overline{12} = 4 + \frac{12-1}{90} = 4 + \frac{11}{90}$.

109. $2.\overline{8768} = 2 + 0.\overline{8768} = 2 + \frac{8768-8}{9990} = 2 + \frac{8760}{9990} = 2 + \frac{292}{333}$.

110. $0.\overline{2} + 0.\overline{3} + 0.\overline{32} = \left(\frac{2}{9} + \frac{3}{9} + \frac{32}{99}\right) = \left(\frac{22+33+32}{99}\right) = \frac{87}{99} = 0.\overline{87}$.

111. $0.\overline{142857} + 0.\overline{285714} = \frac{142857}{999999} + \frac{285714}{999999}$
= $\left(\frac{142857}{999999} \times \frac{999999}{285714}\right) = \frac{1}{2}$.

112. $3.\overline{87} - 2.\overline{59} = (3 + 0.\overline{87}) - (2 + 0.\overline{59}) = \left(3 + \frac{87}{99}\right) - \left(2 + \frac{59}{99}\right)$
= $1 + \left(\frac{87}{99} - \frac{59}{99}\right) = 1 + \frac{28}{99} = 1.\overline{28}$.

113. $3.\overline{36} - 2.\overline{05} + 1.\overline{33} = [(3 + 0.\overline{36}) + (1 + 0.\overline{33})] - (2 + 0.\overline{05})$
= $\left[4 + \left(\frac{36}{99} + \frac{33}{99}\right)\right] - \left[2 + \frac{5}{99}\right]$
= $2 + \left(\frac{36}{99} + \frac{33}{99} - \frac{5}{99}\right) = 2 + \frac{64}{99} = 2.\overline{64}$.

114. $0.\overline{09} \times 7.\overline{3} = \frac{9}{99} \times 7\frac{3}{9} = \frac{1}{11} \times \frac{66}{9} = \frac{2}{3} = 0.\overline{6}$.

115. $0.34\overline{67} + 0.13\overline{33} = \frac{3467-34}{9900} + \frac{1333-13}{9900} = \frac{3433+1320}{9900}$
= $\frac{4753}{9900} = \frac{4801-48}{9900} = 0.48\overline{01}$.

116. $(8.\overline{31} + 0.\overline{6} + 0.00\overline{2}) = 8 + \frac{31-3}{90} + \frac{6}{9} + \frac{2}{900}$
= $\frac{7200+280+600+2}{900}$
= $\frac{8082}{900} = 8\frac{882}{900} = 8 + \frac{979-97}{900} = 8.97\overline{9}$.

117. $\overline{2.75} + \overline{3.78} = (-2 + 0.75) + (-3 + 0.78)$
= $-5 + (0.75 + 0.78) = -5 + 1.53$
= $-5 + 1 + 0.53 = -4 + 0.53 = \overline{4.53}$.

118. $\frac{547527}{82} = \frac{54.7527}{0.0082} = \left(\frac{547.527}{0.0082} \times \frac{1}{10}\right) = \frac{x}{10}$.

119. $\frac{29.94}{1.45} = \frac{299.4}{14.5} = \left(\frac{2994}{14.5} \times \frac{1}{10}\right) = \frac{172}{10} = 17.2$.

120. $1.6 \times 21.3 = \left(\frac{16}{10} \times \frac{213}{10}\right) = \left(\frac{16 \times 213}{100}\right) = \frac{3408}{100} = 34.08$.

121. $\frac{1}{0.0006198} = \frac{10000}{6.198} = \left(10000 \times \frac{1}{6.198}\right)$
= $(10000 \times 0.16134) = 1613.4$.

122. Given, $\frac{52416}{312} = 168 \Leftrightarrow \frac{52416}{168} = 312$.

Now, $\frac{52.416}{0.0168} = \frac{524160}{168} = \left(\frac{52416}{168} \times 10\right) = (312 \times 10) = 3120$.

123. Given, $168 \times 32 = 5376$ or $5376 \div 168 = 32$.

Now, $\frac{5.376}{16.8} = \frac{53.76}{168} = \left(\frac{5376}{168} \times \frac{1}{100}\right) = \frac{32}{100} = 0.32$.

124. Number of decimal places in the given expression = 8.

Number of decimal places in (a) = 8.

Number of decimal places in (b) = 9.

Number of decimal places in (c) = 7.

Clearly, the expression in (a) is the same as the given expression.

125. For the expressions to be equivalent, the difference between the sum of the decimal places in the numerator and that in the denominator must be equal.

This difference is 1 in the given expression and 1 in (d).

So, (d) is the answer.

126. Given expression = $\frac{(96.54 - 89.63)}{(96.54 + 89.63)} \times \frac{(9.654 + 8.963)}{(9.654 - 8.963)}$

$$= \frac{(96.54 - 89.63)}{(96.54 + 89.63)} \times \frac{(9.654 + 8.963)}{(9.654 - 8.963)}$$

$$= \frac{(96.54 - 89.63)}{10(96.54 - 89.63)} \times \frac{(9.654 + 8.963)}{10(9.654 + 8.963)}$$

$$= \frac{1}{10} \times \frac{1}{10} = \frac{1}{100} = \frac{1}{10^2} = 10^{-2}$$

127. $(0.11)^3 + (0.22)^3 + \dots + (0.99)^3 = (0.11)^3 (1^3 + 2^3 + \dots + 9^3)$

$$= 0.001331 \times 2025 = 2.695275 \approx 2.695$$

128. $\frac{1}{4} + \frac{1}{4 \times 5} + \frac{1}{4 \times 5 \times 6} = \frac{1}{4} \left(1 + \frac{1}{5} + \frac{1}{30}\right)$

$$= \frac{1}{4} \left(\frac{30 + 6 + 1}{30}\right) = \frac{1}{4} \times \frac{37}{30}$$

$$= \frac{37}{120} = 0.3083$$

129. Given expression = $\frac{2 \times 4 \times 8 \times 16 + 4 \times 8 \times 16 + 8 \times 16 + 16 + 1}{2 \times 4 \times 8 \times 16}$

$$= \frac{1024 + 512 + 128 + 16 + 1}{1024}$$

$$= \frac{1681}{1024} = 1.6416$$

130. Given expression = $\frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \frac{1}{7 \times 8} + \dots + \frac{1}{24 \times 25}$

$$= \left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{8}\right) + \dots + \left(\frac{1}{24} - \frac{1}{25}\right)$$

$$= \left(\frac{1}{5} - \frac{1}{25}\right) = \frac{4}{25} = 0.16$$

131. $\frac{1}{5} = 0.2$; $\left(\frac{1}{5}\right)^2 = (0.2)^2 = 0.04$; $\left(\frac{1}{5}\right)^3 = (0.2)^3 = 0.008$;

$$\left(\frac{1}{5}\right)^4 = (0.2)^4 = 0.0016$$

Clearly, for every power which is a multiple of 4, the expression would have 6 as the last digit.

So, $\left(\frac{1}{5}\right)^{2000}$ would have 6 as the last digit.

132. To get 1 as the answer, the expression must have the same number of decimal places in the numerator and denominator.

The number of decimal places in the given expressions are as under:

(a) Numerator $\rightarrow$ 4; Denominator $\rightarrow$ 3

(b) Numerator $\rightarrow$ 2; Denominator $\rightarrow$ 4

(c) Numerator $\rightarrow$ 6; Denominator $\rightarrow$ 6

(d) Numerator $\rightarrow$ 4; Denominator $\rightarrow$ 2

$\therefore$ The correct answer is (c).

133. Given expression

$$= \left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{16}\right) + \dots + \left(\frac{1}{81} - \frac{1}{100}\right)$$

$$= 1 - \frac{1}{100} = \frac{99}{100} = 0.99$$

134. $\frac{x}{y} = \frac{0.04}{1.5} = \frac{4}{150} = \frac{2}{75} \Rightarrow \frac{y-x}{y+x} = \frac{1-\frac{x}{y}}{1+\frac{x}{y}} = \frac{1-\frac{2}{75}}{1+\frac{2}{75}} = \frac{73}{77}$

135. Given expression

$$= 35.7 - \left(3 + \frac{1}{10}\right) - \left(2 + \frac{1}{5}\right) = 35.7 - \left(3 + \frac{3}{10}\right) - \left(2 + \frac{2}{5}\right)$$

$$= 35.7 - \frac{33}{10} - \frac{12}{5} = 35.7 - \left(\frac{33}{10} + \frac{12}{5}\right)$$

$$= 35.7 - \frac{57}{10} = 35.7 - 5.7 = 30$$

136. Given expression = $\frac{(0.3333)}{(0.2222)} \times \frac{(0.1667)(0.8333)}{(0.6667)(0.1250)}$

$$= \frac{3333}{2222} \times \frac{\frac{1}{6} \times \frac{5}{6}}{\frac{2}{3} \times \frac{125}{1000}}$$

$$= \left(\frac{3}{2} \times \frac{1}{6} \times \frac{5}{6} \times \frac{3}{2} \times 8\right) = \frac{5}{2} = 2.50$$

137. Given expression = $\frac{35 \times 1.4}{7} = 5 \times 1.4 = 7$.

138. Given expression = $\frac{16 \times 32}{8} = 16 \times 4 = 64$.

139. Given expression = $\frac{441 \times 16}{21 \times 16 \times 21} = 1$.

$$140. \text{ Given expression} = \frac{625 \times 729 \times 289}{81 \times 25 \times 17} = 3825.$$

$$141. \frac{3.6 \times 0.48 \times 2.50}{0.12 \times 0.09 \times 0.5} = \frac{36 \times 48 \times 250}{12 \times 9 \times 5} = 800.$$

$$142. \frac{0.0203 \times 2.92}{0.0073 \times 14.5 \times 0.7} = \frac{203 \times 292}{73 \times 145 \times 7} = \frac{4}{5} = 0.8.$$

$$143. \frac{3.157 \times 4126 \times 3.198}{63.972 \times 2835.121} \approx \frac{3.2 \times 4126 \times 3.2}{64 \times 2835}$$

$$= \frac{32 \times 4126 \times 32}{64 \times 2835} \times \frac{1}{100}$$

$$= \frac{66016}{2835} \times \frac{1}{100} = \frac{23.28}{100} = 0.23 \approx 0.2.$$

$$144. \frac{489.1375 \times 0.0483 \times 1.956}{0.0873 \times 92.581 \times 99.749} \approx \frac{489 \times 0.05 \times 2}{0.09 \times 93 \times 100} = \frac{489}{9 \times 93 \times 10}$$

$$= \frac{163}{279} \times \frac{1}{10} = \frac{0.58}{10} = 0.058 \approx 0.06.$$

$$145. \frac{241.6 \times 0.3814 \times 6.842}{0.4618 \times 38.25 \times 73.65} \approx \frac{240 \times 0.38 \times 6.9}{0.46 \times 38 \times 75}$$

$$= \frac{240 \times 38 \times 69}{46 \times 38 \times 75} \times \frac{1}{10} = \left(\frac{24}{5} \times \frac{1}{10} \right) = \frac{4.8}{10} = 0.48.$$

So, the value is close to 0.4.

$$146. \text{ Given expression} = \frac{(0.2 \times 0.2 + 0.01)}{(0.1 \times 0.1 + 0.02)} = \frac{0.04 + 0.01}{0.01 + 0.02} = \frac{0.05}{0.03} = \frac{5}{3}.$$

$$147. \text{ Given expression} = \frac{8 - 2.8}{1.3} = \frac{5.2}{1.3} = \frac{52}{13} = 4.$$

$$148. \text{ Given expression} = 4.7 \times (13.26 + 9.43 + 77.31)$$

$$= 4.7 \times 100 = 470.$$

$$149. \text{ Given expression} = \frac{0.2(0.2 + 0.02)}{0.044} = \frac{0.2 \times 0.22}{0.044} = \frac{0.044}{0.044} = 1.$$

$$150. \text{ Given expression} = \frac{8.6 \times (5.3 + 4.7)}{4.3 \times (9.7 - 8.7)} = \frac{8.6 \times 10}{4.3 \times 1} = 20.$$

$$151. \text{ Given expression} = \frac{.896 \times (.763 + .237)}{.7 \times (.064 + .936)}$$

$$= \frac{.896 \times 1}{.7 \times 1} = \frac{8.96}{7} = 1.28.$$

$$152. \text{ Given expression} = (1.25)^3 - (0.75)^3 - 3 \times (1.25)^2 \times 0.75$$

$$+ 3 \times 1.25 \times (0.75)^2$$

$$= (1.25 - 0.75)^3 = (0.5)^3 = \left(\frac{1}{2} \right)^3 = \frac{1}{8}.$$

$$[\because (a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2]$$

$$153. (78.95)^2 - (43.35)^2 = (78.95 + 43.35)(78.95 - 43.35)$$

$$= 122.3 \times 35.6 = 4353.88.$$

$$[\because a^2 - b^2 = (a + b)(a - b)]$$

$$154. \text{ Given expression} = \frac{(a^2 - b^2)}{(a - b)}, \text{ where } a = 75.8, b = 55.8$$

$$= \frac{(a - b)(a + b)}{(a - b)} = a + b$$

$$= 75.8 + 55.8 = 131.6.$$

$$155. \text{ Given expression} = \frac{(a^2 - b^2)}{(a + b)}, \text{ where } a = 3.63, b = 2.37$$

$$= \frac{(a - b)(a + b)}{(a + b)} = (a - b)$$

$$= 3.63 - 2.37 = 1.26.$$

$$156. \text{ Let } \frac{(36.54)^2 - (3.46)^2}{x} = 40. \text{ Then,}$$

$$x = \frac{(36.54)^2 - (3.46)^2}{40} = \frac{(36.54)^2 - (3.46)^2}{36.54 + 3.46}$$

$$= \frac{a^2 - b^2}{a + b} = (a - b) = (36.54 - 3.46) = 33.08.$$

$$157. \text{ Given expression} = \frac{(67.542)^2 - (32.458)^2}{(67.542 + 7.196) - (32.458 + 7.916)}$$

$$= \frac{(67.542)^2 - (32.458)^2}{67.542 - 32.458}$$

$$= (67.542 + 32.458) = 100.$$

$$158. \text{ Given expression} = \left(\frac{1.49 \times 1.49 \times 10 - 0.51 \times 0.51 \times 10}{1.49 \times 10 - 0.51 \times 10} \right)$$

$$= \frac{10[(1.49)^2 - (0.51)^2]}{10(1.49 - 0.51)}$$

$$= (1.49 + 0.51) = 2.$$

$$159. \text{ Given expression} = \frac{(a^2 - b^2)}{(a + b)(a - b)} = \frac{(a^2 - b^2)}{(a^2 - b^2)} = 1.$$

$$160. \text{ Given expression} = \frac{5.32 \times (56 + 44)}{(7.66 + 2.34)(7.66 - 2.34)}$$

$$= \frac{5.32 \times 100}{10 \times 5.32} = 10.$$

$$161. \text{ Given expression} = \frac{[(0.6)^2]^2 - [(0.5)^2]^2}{(0.6)^2 + (0.5)^2}$$

$$= \frac{[(0.6)^2 + (0.5)^2][(0.6)^2 - (0.5)^2]}{(0.6)^2 + (0.5)^2}$$

$$= (0.6)^2 - (0.5)^2 = (0.6 + 0.5)(0.6 - 0.5)$$

$$= (1.1 \times 0.1) = 0.11.$$

$$162. \text{ Given expression} = (7.5 \times 7.5 + 2 \times 7.5 \times 2.5 + 2.5 \times 2.5)$$

$$= (a^2 + 2ab + b^2) = (a + b)^2$$

$$= (7.5 + 2.5)^2 = 10^2 = 100.$$

$$163. 0.2 \times 0.2 + 0.02 \times 0.02 - 0.4 \times 0.02$$

$$= 0.2 \times 0.2 + 0.02 \times 0.02 - 2 \times 0.2 \times 0.02$$

$$= (a^2 + b^2 - 2ab) = (a - b)^2 = (0.2 - 0.02)^2 = (0.18)^2.$$

$$\therefore \text{ Given expression} = \frac{(0.18 \times 0.18)}{0.36} = 0.09.$$

$$164. \text{ Given expression} = (100 - 0.25)^2 - 2250.0625$$

$$= (100)^2 + (0.25)^2 - 2 \times 100 \times 0.25 - 2250.0625$$

$$= 10000.0625 - 50 - 2250.0625 = 10000.0625 - 2300.0625$$

$$= 7700.$$

$$\begin{aligned}
 165. \text{ Given expression} &= (55.25)^2 - (25.25)^2 \\
 &= (55.25 + 25.25)(55.25 - 25.25) \\
 &= 80.5 \times 30 = 2415.
 \end{aligned}$$

$$\begin{aligned}
 166. \text{ Given expression} &= \frac{3.20(3.25 - 3.05)}{0.064} = \frac{3.20 \times 0.2}{0.064} \\
 &= \frac{0.64}{0.064} = \frac{64}{6.4} = 10.
 \end{aligned}$$

$$\begin{aligned}
 167. \text{ Given expression} &= [(.98)^3 + (.02)^3 + 3 \times .98 \times .02 \\
 & \quad (.98 + .02)] - 1 = (.98 + .02)^3 - 1 = (1)^3 - 1 = 0.
 \end{aligned}$$

$$\begin{aligned}
 168. \text{ Given expression} &= (11.98)^2 + (0.02)^2 + 11.98 \times x. \\
 \text{For the given expression to be a perfect square, we must have:} \\
 11.98 \times x &= 2 \times 11.98 \times 0.02 \text{ or } x = 0.04.
 \end{aligned}$$

$$169. \text{ Given expression} = \frac{(a-b)^2 + (a+b)^2}{a^2 + b^2} = \frac{2(a^2 + b^2)}{(a^2 + b^2)} = 2.$$

$$170. \text{ Given expression} = \frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} = 4.$$

$$\begin{aligned}
 171. \text{ Given expression} &= \frac{(0.051)^3 + (0.041)^3}{(0.051)^2 - (0.051 \times 0.041) + (0.041)^2} \\
 &= \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) \\
 &= (a + b) = (0.051 + 0.041) = 0.092.
 \end{aligned}$$

$$\begin{aligned}
 172. \text{ Given expression} &= \frac{(5.71)^3 - (2.79)^3}{(5.71)^2 + 5.71 \times 2.79 + (2.79)^2} \\
 &= \left(\frac{a^3 - b^3}{a^2 + ab + b^2} \right) \\
 &= (a - b) = (5.71 - 2.79) = 2.92.
 \end{aligned}$$

$$\begin{aligned}
 173. \text{ Given expression} &= \frac{(0.943)^2 - (0.943 \times 0.057) + (0.057)^2}{(0.943)^3 + (0.057)^3} \\
 &= \frac{a^2 - ab + b^2}{a^3 + b^3} = \frac{1}{a + b} \\
 &= \frac{1}{0.943 + 0.057} = 1.
 \end{aligned}$$

$$\begin{aligned}
 174. \text{ Given expression} &= \frac{(0.5)^3 + (0.3)^3}{(0.5)^2 + (0.3)^2 - (0.5 \times 0.3)} \\
 &= \left(\frac{a^3 + b^3}{a^2 + b^2 - ab} \right) \\
 &= (a + b) = (0.5 + 0.3) = 0.8.
 \end{aligned}$$

$$\begin{aligned}
 175. \text{ Given expression} &= \frac{(10.3)^3 + (1)^3}{(10.3)^2 - (10.3 \times 1) + (1)^2} = \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) \\
 &= (a + b) = (10.3 + 1) = 11.3.
 \end{aligned}$$

$$176. \text{ Given expression} = \frac{(2 \times 3.75)^3 + (1)^3}{(7.5)^2 - (7.5 \times 1) + (1)^2}$$

$$\begin{aligned}
 &= \frac{(7.5)^3 + (1)^3}{(7.5)^2 - (7.5 \times 1) + (1)^2} \\
 &= \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) = (a + b) = (7.5 + 1) = 8.5.
 \end{aligned}$$

$$177. \text{ Given expression} = \frac{(0.1)^3 + (0.02)^3}{2^3 [(0.1)^3 + (0.02)^3]} = \frac{1}{8} = 0.125.$$

$$\begin{aligned}
 178. \text{ Given expression} &= \frac{(0.013)^3 + (0.007)^3}{(0.013)^2 - (0.013 \times 0.007) + (0.007)^2} \\
 &= \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) \\
 &= a + b = 0.013 + 0.007 = 0.020.
 \end{aligned}$$

$$\begin{aligned}
 179. \text{ Given expression} &= \frac{(2.3)^3 - (0.3)^3}{(2.3)^2 + (2.3 \times 0.3) + (0.3)^2} \\
 &= \left(\frac{a^3 - b^3}{a^2 + ab + b^2} \right) \\
 &= (a - b) = (2.3 - 0.3) = 2.
 \end{aligned}$$

$$\begin{aligned}
 180. \text{ Given expression} &= \frac{a^2 + b^2 + c^2}{\left(\frac{a}{10}\right)^2 + \left(\frac{b}{10}\right)^2 + \left(\frac{c}{10}\right)^2}, \\
 \text{where } a &= 0.06, b = 0.47 \text{ and } c = 0.079. \\
 &= \frac{100(a^2 + b^2 + c^2)}{(a^2 + b^2 + c^2)} = 100.
 \end{aligned}$$

$$\begin{aligned}
 181. \text{ Given expression} &= \frac{(4.53 - 3.07)^3 + (3.07 - 2.15)^3 + (2.15 - 4.53)^3}{(4.53 - 3.07)(3.07 - 2.15)(2.15 - 4.53)} \\
 &= \frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3. \\
 &[\because \text{if } a + b + c = 0, a^3 + b^3 + c^3 = 3abc]
 \end{aligned}$$

$$182. \frac{2}{5}\% = \frac{2}{5} \times \frac{1}{100} = \frac{1}{250}$$

$$\begin{aligned}
 183. \text{ The given expression can be written in this form also} \\
 N &= 0.3939 \quad \dots(i) \\
 \text{Multiply equation (i) with 100 on both sides.} \\
 100N &= 39.39 \quad \dots(ii)
 \end{aligned}$$

$$\begin{aligned}
 \text{Subtracting equation (i) from (ii) we get} \\
 \Rightarrow 100N - N &= 39.39 - 0.39 \\
 99N &= 39 \\
 \Rightarrow N &= \frac{39}{99} = \frac{13}{33}
 \end{aligned}$$

$$\begin{aligned}
 184. \left\{ (41.99)^2 - (18.04)^2 \right\} \div ? &= (13.11)^2 - 138.99 \\
 \Rightarrow \left\{ (42)^2 - (18)^2 \right\} \div ? &= (13)^2 - 139 \quad \left\{ \because a^2 - b^2 = (a+b)(a-b) \right\} \\
 \Rightarrow \left\{ (42+18)(42-18) \right\} \div ? &= 169 - 139 \\
 \Rightarrow 60 \times 24 \div ? &= 30 \\
 \Rightarrow ? &= \frac{60 \times 24}{30} = 48
 \end{aligned}$$

185. Given $\frac{3}{5}$ of $\frac{4}{7}$ of $\frac{5}{12}$ of 1015

$$\Rightarrow x = \frac{3}{5} \times \frac{4}{7} \times \frac{5}{12} \times 1015 = 145$$

186. Given $1\frac{1}{8} + 1\frac{6}{7} + 3\frac{3}{5}$

$$\frac{9}{8} + \frac{13}{7} + \frac{18}{5} = x$$

By taking the LCM of 8, 7 and 5 is 280.

$$\frac{315 + 520 + 1008}{280} = \frac{1843}{280} = 6\frac{163}{280}$$

187. $2\frac{1.5}{5} + 2\frac{1}{6} - 1\frac{3.5}{15} = \frac{x^{1/3}}{4} + 1\frac{7}{30}$

$$\Rightarrow \frac{11.5}{5} + \frac{13}{6} - \frac{18.5}{15} = \frac{x^{1/3}}{4} + \frac{37}{30}$$

LCM of 5, 6 and 15 is 30.

$$\Rightarrow \frac{69 + 65 - 37}{30} = \frac{x^{1/3}}{4} + \frac{37}{30}$$

$$\Rightarrow \frac{97}{30} = \frac{x^{1/3}}{4} + \frac{37}{30}$$

$$\Rightarrow \frac{x^{1/3}}{4} = \frac{97}{30} - \frac{37}{30}$$

$$\Rightarrow x^{1/3} = \frac{60}{30} \times 4$$

$$\Rightarrow x^{1/3} = 8 \Rightarrow x = (8)^3$$

$$\Rightarrow x = 512$$

Hence, the number is 512.

188. Let the missing number be x .

$$\text{Given } 323.46 + 23.04 - 43.17 - x = 303$$

$$x = 323.46 + 23.04 - 43.17 - 303$$

$$= 0.33$$

189. Let the missing number be x .

$$\text{Given } 1\frac{1}{2} + 2\frac{2}{7} = 3\frac{1}{2} + x$$

$$x = 1\frac{1}{2} + 2\frac{2}{7} - 3\frac{1}{2}$$

$$x = \frac{3}{2} + \frac{16}{7} - \frac{7}{2}$$

$$x = \frac{21 + 32 - 49}{14}$$

$$x = \frac{4}{14} = \frac{2}{7}$$

Hence, the number is $\frac{2}{7}$.

190. Let the missing number be x .

$$\text{Given } 48.2 \times 2.5 \times 2.2 + x = 270$$

$$\Rightarrow x = 270 - 48.2 \times 2.5 \times 2.2$$

$$x = 270 - 265.1 = 4.9$$

Hence, the number is 4.9.

191. Given $4\frac{2}{3} + 3\frac{1}{2} + 1\frac{2}{3}$

$$= \frac{14}{3} + \frac{7}{2} + \frac{5}{3}$$

By taking LCM of 3, 2 and 3 is 6.

$$= \frac{28 + 21 + 10}{6}$$

$$= \frac{39}{6} = \frac{13}{2} = 6\frac{1}{2}$$

192. $\frac{17292}{33} \times \frac{1}{8}$

$$= \frac{17292}{33 \times 8} = 65.5$$

193. Given $= 1599 \div 39.99 + \frac{4}{5} \times 2449 - 120.05$

$$= 1600 \div 40 + \frac{4}{5} \times 2450 - 120$$

$$= 1600 \div 40 + 1960 - 120$$

$$= 40 + 1960 - 120 = 1880$$

194. Given expression

$$= 1576 \div 45.02 + 23.99 \times \sqrt{255}$$

$$= 1575 \div 45 + 24 \times \sqrt{256}$$

$$= 35 + 24 \times 16 = 35 + 384 = 419$$

195. $= 3899 \div 11.99 - 2379 \div 13.97$

$$= 3900 \div 12 - 2380 \div 14$$

$$\approx 325 - 170 \approx 155$$

196. $2\frac{2}{9} + 4\frac{1}{18} - 1\frac{1}{2} = ?$

Let the missing number be x

$$x = \frac{20}{9} + \frac{73}{18} - \frac{3}{2}$$

LCM of 9, 18 and 2 is 18.

$$= \frac{40 + 73 - 27}{18} = \frac{86}{18}$$

$$= \frac{43}{9} = 4\frac{7}{9}$$

197. Let the missing number be x .

$$\frac{294 \div 14 \times 5 + 11}{x} = 8^2 \div 5 + 1.7$$

$$\Rightarrow \frac{\frac{294}{14} \times 5 + 11}{x} = \frac{64}{5} + 1.7$$

$$\Rightarrow \frac{21 \times 5 + 11}{x} = 12.8 + 1.7$$

$$\Rightarrow \frac{105 + 11}{x} = 12.8 + 1.7$$

$$\Rightarrow \frac{116}{x} = 14.5$$

$$\Rightarrow x = \frac{116}{14.5} = \frac{116 \times 10}{145} = 8$$

Hence, the number is 8.

198. Let the missing number be x

$$x = 2.5 \times 4.8 + 7.2 \times 1.5 - 1.2 \times 14$$

$$= (12 + 10.8 - 16.8) = 6$$

199. Given $\sqrt{197} \times 6.99 + 626.96$

$$= \sqrt{196} \times 7 + 627$$

$$= 14 \times 7 + 627 = 98 + 627 = 725$$

200. Each fraction is equivalent to decimal

$$\frac{3}{2} = 1.5; \frac{7}{3} = 2.3$$

$$\frac{5}{4} = 1.25; \frac{7}{2} = 3.5$$

Hence, $\frac{7}{2}$ is largest fraction.

- 201.** Let the missing number be x ; $\frac{x}{529} = \frac{329}{x}$

$$x \times x = 529 \times 329$$

$$x = \sqrt{174041} = 417.18 \approx 416$$

Hence, the number is 416.

- 202.** Let original fraction be $\frac{a}{b}$.

Now, according to the question

$$\frac{a - a \times \frac{25}{100}}{b + b \times \frac{250}{100}} = \frac{6}{5}$$

$$\frac{0.75a}{3.50b} = \frac{6}{5}$$

$$\Rightarrow \frac{a}{b} = \frac{6}{5} \times \frac{3.50}{0.75} = \frac{6 \times 350 \times 100}{5 \times 75 \times 100} = \frac{28}{5}$$

- 203.** Let the missing number be x .

$$\frac{21.5}{5} + \frac{21}{6} - \frac{13.5}{15} = \frac{(x)^{\frac{1}{3}}}{4} + \frac{17}{30}$$

$$\frac{21.5}{5} + \frac{21}{6} - \frac{13.5}{15} - \frac{17}{30} = \frac{(x)^{\frac{1}{3}}}{4}$$

LCM of 5, 6, 15 and 30 is 30.

$$\frac{129 + 105 - 27 - 17}{30} = \frac{\sqrt[3]{x}}{4}$$

$$\sqrt[3]{x} = \frac{190 \times 4}{30} = 25.33 \approx 25$$

$$x = 25^3$$

$$x = 15625$$

Hence, the numbers 15625.

$$\begin{aligned} \text{204. } & \left(\frac{18}{4}\right)^2 \times \left(\frac{455}{19}\right) \div \left(\frac{61}{799}\right) \\ &= \frac{324}{16} \times \frac{455}{19} \times \frac{799}{61} = 6350 \end{aligned}$$

- 205.** Let the third number be 2960

$$\therefore \text{Second number} = \frac{1}{4} \text{ of third number} = \frac{1}{4} \times 2960 = 740$$

$$\frac{5}{9} \text{ of first number} = 25\% \text{ of second number}$$

$$\frac{5}{9} \text{ first number} = \frac{25 \times 740}{100} = 185$$

$$\Rightarrow \text{First number} = \frac{185 \times 9}{5} = 333$$

$$\therefore 30\% \text{ of } 333 = \frac{30}{100} \times 333 = 999$$

$$\begin{aligned} \text{206. } & 7\frac{1}{2} - \left[2\frac{1}{4} \div \left\{ 1\frac{1}{4} - \frac{1}{2} \left(1\frac{1}{2} - \frac{1}{3} - \frac{1}{6} \right) \right\} \right] \\ &= \frac{15}{2} - \left[\frac{9}{4} \div \left\{ \frac{5}{4} - \frac{1}{2} \left(\frac{3}{2} - \frac{1}{3} - \frac{1}{6} \right) \right\} \right] \\ &= \frac{15}{2} \left[\frac{9}{4} \div \left\{ \frac{5}{4} - \frac{1}{2} \left(\frac{9-2-1}{6} \right) \right\} \right] \\ &= \frac{15}{2} \left[\frac{9}{4} \div \left(\frac{5}{4} - \frac{1}{2} \right) \right] = \frac{15}{2} \left[\frac{9}{4} \div \left(\frac{5-2}{4} \right) \right] \\ &= \frac{15}{2} - \left[\frac{9}{4} \div \frac{3}{4} \right] = \frac{15}{2} - \left[\frac{9}{4} \times \frac{4}{3} \right] \\ &= \frac{15}{2} - 3 = \frac{15-6}{2} = \frac{9}{2} = 4\frac{1}{2} \end{aligned}$$

4

Simplification

IMPORTANT FACTS AND FORMULAE

I. 'BODMAS' Rule: This rule depicts the correct sequence in which the operations are to be executed, so as to find out the value of a given expression.

Here, 'B' stands for 'Bracket', 'O' for 'of', 'D' for 'Division', 'M' for 'Multiplication', 'A' for 'Addition' and 'S' for 'Subtraction'.

Thus, in simplifying an expression, first of all the brackets must be removed, strictly in the order (), { } and [].

After removing the brackets, we must use the following operations strictly in the order:

(i) of (ii) Division (iii) Multiplication (iv) Addition (v) Subtraction

II. Modulus of a Real Number: Modulus of a real number a is defined as

$$|a| = \begin{cases} a, & \text{if } a > 0 \\ -a, & \text{if } a < 0 \end{cases}$$

Thus, $|5| = 5$ and $|-5| = -(-5) = 5$.

III. Virnaculum (or Bar): When an expression contains Virnaculum, before applying the 'BODMAS' rule, we simplify the expression under the Virnaculum.

IV. Some Important Formulae:

$$(i) (a + b)^2 = (a^2 + b^2 + 2ab)$$

$$(ii) (a - b)^2 = (a^2 + b^2 - 2ab)$$

$$(iii) (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$$

$$(iv) (a + b)^2 - (a - b)^2 = 4ab$$

$$(v) (a^2 - b^2) = (a + b)(a - b)$$

$$(vi) (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$(vii) (a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$(viii) (a^3 + b^3) = (a + b)(a^2 - ab + b^2)$$

$$(ix) (a^3 - b^3) = (a - b)(a^2 + ab + b^2)$$

$$(x) (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(xi) a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

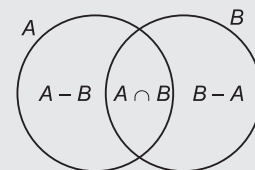
V. For any two sets A and B, we have:

$$(i) n(A - B) + n(A \cap B) = n(A)$$

$$(ii) n(B - A) + n(A \cap B) = n(B)$$

$$(iii) n(A \cup B) = n(A - B) + n(A \cap B) + n(B - A)$$

$$(iv) n(A \cup B) = n(A) + n(B) - n(A \cap B).$$



SOLVED EXAMPLES

Ex. 1. $4368 + 2158 - 596 - ? = 3421 + 1262$

(Bank P.O., 2009)

Sol. Let $4368 + 2158 - 596 - x = 3421 + 1262$

$$\Rightarrow x + 596 = (4368 + 2158) - (3421 + 1262)$$

$$\Rightarrow x + 596 = 6526 - 4683 = 1843$$

$$\Rightarrow x = 1843 - 596 = 1247.$$

Hence, required number = 1247.

Ex. 2. $3456 \div 12 \div 8 = ?$

(Bank Recruitment, 2008)

Sol. Given expression = $\frac{3456}{12} \div 8 = 288 \div 8 = 36$.

Ex. 3. $13 \times 252 \div 42 + 170 = ? + 47$

(Bank P.O., 2010)

Sol. Let $13 \times 252 \div 42 + 170 = x + 47$. Then,

$$13 \times \frac{252}{42} + 170 = x + 47 \Rightarrow 13 \times 6 + 170 = x + 47 \Rightarrow x + 47 = 78 + 170 = 248$$

$$\Rightarrow x = 248 - 47 = 201.$$

Hence, required number = 201.

Ex. 4. Simplify: (a) $460 \times 15 - 5 \times 20$

(Bank P.O., 2010)

(b) $1 \div [1 + 1 \div \{1 + 1 \div 1(1 + 1 \div 2)\}] + 1$

(A.A.O., 2009)

Sol. (a) Given expression = $6900 - 100 = 6800$.

$$\begin{aligned} \text{(b) Given expression} &= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \div 1 \left(1 + \frac{1}{2} \right) \right\} \right] + 1 \\ &= 1 \div \left[1 + 1 \div \left\{ 1 + 1 \div \frac{3}{2} \right\} \right] + 1 = 1 \div \left[1 + 1 \div \left\{ 1 + 1 \times \frac{2}{3} \right\} \right] + 1 \\ &= 1 \div \left[1 + 1 \div \left\{ 1 + \frac{2}{3} \right\} \right] + 1 = 1 \div \left[1 + 1 \div \frac{5}{3} \right] + 1 \\ &= 1 \div \left[1 + 1 \times \frac{3}{5} \right] + 1 = 1 \div \left[1 + \frac{3}{5} \right] + 1 = 1 \div \frac{8}{5} + 1 = 1 \times \frac{5}{8} + 1 = \frac{5}{8} + 1 = \frac{13}{8}. \end{aligned}$$

Ex. 5. Find the missing numeral:

(a) $(? - 2763) \div 86 \times 13 = 208$

(L.I.C.A.D.O., 2007)

(b) $3565 \div 23 + 4675 \div ? = 430$

(Bank Recruitment, 2008)

Sol. (a) Let $(x - 2763) \div 86 \times 13 = 208$.

$$\text{Then, } \frac{(x - 2763)}{86} \times 13 = 208 \Rightarrow \frac{(x - 2763)}{86} = \frac{208}{13} = 16$$

$$\Rightarrow (x - 2763) = 16 \times 86 = 1376$$

$$\Rightarrow x = 1376 + 2763 = 4139.$$

(b) Let $3565 \div 23 + 4675 \div x = 430$.

$$\text{Then, } \frac{3565}{23} + \frac{4675}{x} = 430 \Rightarrow 155 + \frac{4675}{x} = 430$$

$$\Rightarrow \frac{4675}{x} = 430 - 155 = 275 \Rightarrow x = \frac{4675}{275} = 17.$$

Ex. 6. Simplify: (a) $\frac{(6+6+6+6) \div 6}{4+4+4+4+4}$ (b) $\frac{(2+3) \times 5 + 3 \div \frac{1}{2}}{6+5 \times 4 \div \frac{4}{5}}$

(R.R.B., 2006)

Sol. (a) Given expression = $\frac{24 \div 6}{4+4+4+1} = \frac{4}{13}$.

$$\text{(b) Given expression} = \frac{5 \times 5 + 3 \times 2}{6 + 5 \times 4 \times \frac{5}{4}} = \frac{25 + 6}{6 + 25} = \frac{31}{31} = 1.$$

Ex. 7. What should come in place of both the question marks in the following equation?

$$\frac{128 + 16 \times ? - 7 \times 2}{7^2 - 8 \times 6 + ?^2} = 1$$

Sol. Let $\frac{128 + 16 \times x - 7 \times 2}{7^2 - 8 \times 6 + x^2} = 1$.

$$\text{Then, } 8x - 7 \times 2 = 49 - 48 + x^2 \Leftrightarrow 8x - 14 = 1 + x^2 \Leftrightarrow x^2 - 8x + 15 = 0$$

$$\Leftrightarrow x^2 - 3x - 5x + 15 = 0 \Leftrightarrow x(x - 3) - 5(x - 3) = 0$$

$$\Leftrightarrow (x - 3)(x - 5) = 0 \Leftrightarrow x = 3 \text{ or } x = 5.$$

Hence, missing number is 3 or 5.

Ex. 8. Simplify: (a) $108 \div 36$ of $\frac{1}{4} + \frac{2}{5} \times 3 \frac{1}{4}$

(b) $\frac{2}{3} \times \frac{5}{6} + \frac{4}{9} - \frac{3}{4} + \frac{2}{9} \times \frac{5}{9} \div \frac{2}{9}$

(P.C.S., 2009)

Sol. (a) Given expression = $108 \div 9 + \frac{2}{5} \times \frac{13}{4} = \frac{108}{9} + \frac{13}{10} = \left(12 + \frac{13}{10}\right) = \frac{133}{10} = 13 \frac{3}{10}$.

(b) Given expression = $\frac{2}{3} \times \frac{5}{6} + \frac{4}{9} - \frac{3}{4} + \frac{2}{9} \times \frac{5}{9} \times \frac{9}{2}$
 $= \frac{5}{9} + \frac{4}{9} - \frac{3}{4} + \frac{5}{9} = \frac{14}{9} - \frac{3}{4} = \frac{56 - 27}{36} = \frac{29}{36}$.

Ex. 9. What value will replace the question mark in the following equation?

$$4\frac{1}{2} + 3\frac{1}{6} + ? + 2\frac{1}{3} = 13\frac{2}{5}.$$

Sol. Let $\frac{9}{2} + \frac{19}{6} + x + \frac{7}{3} = \frac{67}{5}$.

Then, $x = \frac{67}{5} - \left(\frac{9}{2} + \frac{19}{6} + \frac{7}{3}\right) \Leftrightarrow x = \frac{67}{5} - \left(\frac{27 + 19 + 14}{6}\right) = \left(\frac{67}{5} - \frac{60}{6}\right)$
 $\Leftrightarrow x = \left(\frac{67}{5} - 10\right) = \frac{17}{5} = 3\frac{2}{5}$.

Hence, missing fraction = $3\frac{2}{5}$.

Ex. 10. Simplify: $\left[3\frac{1}{4} \div \left\{1\frac{1}{4} - \frac{1}{2}\left(2\frac{1}{2} - \frac{1}{4} - \frac{1}{6}\right)\right\}\right]$.

Sol. Given exp. = $\left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{1}{2}\left(\frac{5}{2} - \frac{3}{4} - \frac{1}{6}\right)\right\}\right] = \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{1}{2}\left(\frac{5}{2} - \frac{1}{12}\right)\right\}\right]$
 $= \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{1}{2}\left(\frac{30 - 1}{12}\right)\right\}\right] = \left[\frac{13}{4} \div \left\{\frac{5}{4} - \frac{29}{24}\right\}\right]$
 $= \left[\frac{13}{4} \div \left\{\frac{30 - 29}{24}\right\}\right] = \left[\frac{13}{4} \div \frac{1}{24}\right] = \left[\frac{13}{4} \times 24\right] = 78$.

Ex. 11. Simplify: $\frac{\frac{7}{2} + \frac{5}{2} \times \frac{3}{2}}{\frac{7}{2} + \frac{5}{2} \text{ of } \frac{3}{2}} + 5.25$

Sol. Given exp. = $\frac{\frac{7}{2} \times \frac{2}{5} \times \frac{3}{2}}{\frac{7}{2} \div \frac{4}{15}} + 5.25 = \frac{\frac{21}{10}}{\frac{7}{2} \times \frac{15}{4}} \div \frac{525}{100} = \frac{21}{10} \times \frac{15}{14} \times \frac{100}{525} = \frac{6}{14} = \frac{3}{7}$.

Ex. 12. Simplify: $b - [b - (a + b) - \{b - (b - a - b)\} + 2a]$.

Sol. Given exp. = $b - [b - (a + b) - \{b - (b - a + b)\} + 2a]$
 $= b - [b - a - b - \{b - (2b - a)\} + 2a]$
 $= b - [-a - \{b - 2b + a\} + 2a]$
 $= b - [-a - \{-b + a\} + 2a]$
 $= b - [-a + b - a + 2a] = b - b = 0$.

Ex. 13. If $x + y = 23$ and $xy = 126$, what is the value of $x^2 + y^2$?

(Bank Recruitment, 2010)

Sol. $(x + y) = 23 \Rightarrow (x + y)^2 = (23)^2 = 529 \Rightarrow x^2 + y^2 + 2xy = 529$
 $\Rightarrow x^2 + y^2 + 2 \times 126 = 529 \Rightarrow x^2 + y^2 = 529 - 252 = 277.$

Ex. 14. If $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$, find the value of $\frac{c^2 - a^2}{c^2 + a^2}$.

(M.B.A., 2007)

Sol. $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16} \Rightarrow \frac{a}{b} \times \frac{b}{c} = \frac{4}{5} \times \frac{15}{16} = \frac{3}{4} \Rightarrow \frac{a}{c} = \frac{3}{4} \Rightarrow \frac{c}{a} = \frac{4}{3}.$

$$\therefore \frac{c^2 - a^2}{c^2 + a^2} = \frac{\frac{c^2}{a^2} - 1}{\frac{c^2}{a^2} + 1} = \frac{\left(\frac{c}{a}\right)^2 - 1}{\left(\frac{c}{a}\right)^2 + 1} = \frac{\left(\frac{4}{3}\right)^2 - 1}{\left(\frac{4}{3}\right)^2 + 1} = \frac{\left(\frac{16}{9} - 1\right)}{\left(\frac{16}{9} + 1\right)} = \frac{7}{9} \times \frac{9}{25} = \frac{7}{25}.$$

Ex. 15. If $a = \frac{4xy}{x+y}$, find the value of $\frac{a+2x}{a-2x} + \frac{a+2y}{a-2y}$.

(M.C.A., 2007)

Sol. $\frac{a+2x}{a-2x} + \frac{a+2y}{a-2y} = \frac{\frac{4xy}{x+y} + 2x}{\frac{4xy}{x+y} - 2x} + \frac{\frac{4xy}{x+y} + 2y}{\frac{4xy}{x+y} - 2y} = \frac{4xy + 2x(x+y)}{4xy - 2x(x+y)} + \frac{4xy + 2y(x+y)}{4xy - 2y(x+y)}$
 $= \frac{2x(2y + x + y)}{2x(2y - x - y)} + \frac{2y(2x + x + y)}{2y(2x - x - y)} = \frac{(3y + x)}{(y - x)} + \frac{(3x + y)}{(x - y)}$
 $= \frac{(3x + y)}{(x - y)} - \frac{(3y + x)}{(x - y)} = \frac{(3x + y) - (3y + x)}{(x - y)} = \frac{2x - 2y}{x - y} = 2.$

Ex. 16. Find the value of $4 - \frac{5}{1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}}$.

Sol. Given exp. $= 4 - \frac{5}{1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}} = 4 - \frac{5}{1 + \frac{1}{3 + \frac{4}{9}}} = 4 - \frac{5}{1 + \frac{1}{(31/9)}}$
 $= 4 - \frac{5}{1 + \frac{9}{31}} = 4 - \frac{5}{(40/31)} = 4 - \frac{5 \times 31}{40} = 4 - \frac{31}{8} = \frac{1}{8}.$

Ex. 17. If $\frac{2x}{1 + \frac{1}{1 + \frac{x}{1-x}}} = 1$, then find the value of x .

Sol. We have: $\frac{2x}{1 + \frac{1}{(1-x) + x}} = 1 \Leftrightarrow \frac{2x}{1 + \frac{1}{[1/(1-x)]}} = 1 \Leftrightarrow \frac{2x}{1 + (1-x)} = 1$
 $\Leftrightarrow 2x = 2 - x \Leftrightarrow 3x = 2 \Leftrightarrow x = \frac{2}{3}.$

Ex. 18. (i) If $4x + 5y = 83$ and $\frac{3x}{2y} = \frac{21}{22}$, then what is the value of $y - x$?

(Bank P.O., 2008)

(ii) If $\frac{x}{4} - \frac{x-3}{6} = 1$, then find the value of x .

Sol. (i) $\frac{3x}{2y} = \frac{21}{22} \Rightarrow \frac{x}{y} = \frac{21}{22} \times \frac{2}{3} = \frac{7}{11} \Rightarrow x = \frac{7}{11}y.$

$$4x + 5y = 83 \Rightarrow 4 \times \frac{7}{11}y + 5y = 83 \Rightarrow \frac{28}{11}y + 5y = 83$$

$$\Rightarrow \frac{83}{11}y = 83 \Rightarrow y = 83 \times \frac{11}{83} = 11.$$

$$\therefore x = \left(\frac{7}{11} \times 11 \right) = 7. \text{ So, } y - x = 11 - 7 = 4.$$

(ii) $\frac{x}{4} - \frac{x-3}{6} = 1 \Leftrightarrow \frac{3x - 2(x-3)}{12} = 1 \Leftrightarrow 3x - 2x + 6 = 12 \Leftrightarrow x = 6.$

Ex. 19. If $2x + 3y = 34$ and $\frac{x+y}{y} = \frac{13}{8}$, then find the value of $5y + 7x$.

Sol. The given equations are:

$$2x + 3y = 34 \quad \dots(i) \text{ and, } \frac{x+y}{y} = \frac{13}{8} \Rightarrow 8x + 8y = 13y \Rightarrow 8x - 5y = 0 \quad \dots(ii)$$

Multiplying (i) by 5, (ii) by 3 and adding, we get: $34x = 170$ or $x = 5$.

Putting $x = 5$ in (i), we get: $y = 8$.

$$\therefore 5y + 7x = (5 \times 8 + 7 \times 5) = 40 + 35 = 75.$$

Ex. 20. The cost of 4 bags and 3 boxes is ₹ 555 and the cost of 3 bags and 4 boxes is ₹ 460. What is the cost of one bag? (Bank P.O., 2009)

Sol. Let the cost of 1 bag be ₹ x and that of 1 box be ₹ y .

$$\text{Then, } 4x + 3y = 555 \quad \dots(i) \text{ and } 3x + 4y = 460 \quad \dots(ii)$$

$$\text{Adding (i) and (ii), we get: } 7x + 7y = 1015 \text{ or } x + y = 145 \quad \dots(iii)$$

$$\text{Subtracting (ii) from (i), we get: } x - y = 95 \quad \dots(iv)$$

$$\text{Adding (iii) and (iv), we get: } 2x = 240 \text{ or } x = 120.$$

Hence, cost of 1 bag = ₹ 120.

Ex. 21. If $2x + 3y + z = 55$, $x + z - y = 4$ and $y - x + z = 12$, then what are the values of x , y and z ?

Sol. The given equations are:

$$2x + 3y + z = 55 \quad \dots(i); \quad x + z - y = 4 \quad \dots(ii); \quad y - x + z = 12 \quad \dots(iii)$$

$$\text{Subtracting (ii) from (i), we get: } x + 4y = 51 \quad \dots(iv)$$

$$\text{Subtracting (iii) from (i), we get: } 3x + 2y = 43 \quad \dots(v)$$

$$\text{Multiplying (v) by 2 and subtracting (iv) from it, we get: } 5x = 35 \text{ or } x = 7.$$

$$\text{Putting } x = 7 \text{ in (iv), we get: } 4y = 44 \text{ or } y = 11.$$

$$\text{Putting } x = 7, y = 11 \text{ in (i), we get: } z = 8.$$

Ex. 22. If $x^2 - 7x = -12$, what is the value of x ? (M.B.A., 2006)

Sol. $x^2 - 7x = -12 \Leftrightarrow x^2 - 7x + 12 = 0 \Leftrightarrow x^2 - 3x - 4x + 12 = 0$

$$\Leftrightarrow x(x-3) - 4(x-3) = 0 \Leftrightarrow (x-3)(x-4) = 0 \Leftrightarrow x = 3 \text{ or } x = 4.$$

Ex. 23. Find the value of $\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right) \dots \left(1 - \frac{1}{100}\right)$.

Sol. Given expression $= \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \dots \times \frac{99}{100} = \frac{2}{100} = \frac{1}{50}.$

Ex. 24. Find the value of $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \dots + \frac{1}{9 \times 10}.$

Sol. Given expression $= \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \dots + \left(\frac{1}{9} - \frac{1}{10}\right)$
 $= \left(\frac{1}{2} - \frac{1}{10}\right) = \frac{4}{10} = \frac{2}{5}.$

Ex. 25. Simplify : $99\frac{48}{49} \times 245$.

Sol. Given expression = $\left(100 - \frac{1}{49}\right) \times 245 = \frac{4899}{49} \times 245 = 4899 \times 5 = 24495$.

Ex. 26. Find the value of: $\frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110}$.

(A.A.O. Exam, 2009)

Sol. Given expression = $\left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{8}\right) + \left(\frac{1}{8} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{10}\right) + \left(\frac{1}{10} - \frac{1}{11}\right)$
 $= \left(\frac{1}{5} - \frac{1}{11}\right) = \frac{6}{55}$.

Ex. 27. A board 7 ft. 9 inches long is divided into 3 equal parts. What is the length of each part?

Sol. Length of board = 7 ft. 9 inches = $(7 \times 12 + 9)$ inches = 93 inches.

$\therefore$ Length of each part = $\left(\frac{93}{3}\right)$ inches = 31 inches = 2 ft. 7 inches.

Ex. 28. Ram went to a shop to buy 50 kg of rice. He buys two varieties of rice which cost him ₹ 4.50 per kg and ₹ 5 per kg. He spends a total of ₹ 240. What was the quantity of rice bought which cost him ₹ 4.50 per kg?

(M.A.T., 2007)

Sol. Let the quantity of rice bought at ₹ 4.50 per kg be x kg.

Then, quantity of rice bought at ₹ 5 per kg = $(50 - x)$ kg.

$\therefore 4.50x + 5(50 - x) = 240 \Rightarrow 250 - 0.5x = 240 \Rightarrow 0.5x = 10 \Rightarrow x = 20$.

Hence, quantity of rice bought at ₹ 4.50 per kg = 20 kg.

Ex. 29. A boy was asked to multiply a certain number by 53. He multiplied it by 35 and got his answer less than the correct one by 1206. Find the number to be multiplied.

(SNAP, 2005)

Sol. Let the required number be x .

Then, $53x - 35x = 1206 \Leftrightarrow 18x = 1206 \Leftrightarrow x = \frac{1206}{18} = 67$.

Hence, number to be multiplied = 67.

Ex. 30. $\frac{4}{15}$ of $\frac{5}{7}$ of a number is greater than $\frac{4}{9}$ of $\frac{2}{5}$ of the same number by 8. What is half of that number?

Sol. Let the number be x .

Then, $\frac{4}{15}$ of $\frac{5}{7}$ of $x - \frac{4}{9}$ of $\frac{2}{5}$ of $x = 8 \Leftrightarrow \frac{4}{21}x - \frac{8}{45}x = 8$

$$\Leftrightarrow \left(\frac{4}{21} - \frac{8}{45}\right)x = 8 \Leftrightarrow \left(\frac{60 - 56}{315}\right)x = 8 \Leftrightarrow \frac{4}{315}x = 8$$

$$\Leftrightarrow x = \left(\frac{8 \times 315}{4}\right) = 630 \Leftrightarrow \frac{1}{2}x = 315.$$

Hence, required number = 315.

Ex. 31. A man owns $\frac{2}{3}$ of the market research bureau business and sells $\frac{3}{4}$ of his shares for ₹ 75000. What is the value of business?

(Campus Recruitment, 2010)

Sol. Let the total value be ₹ x .

Then, $\frac{3}{4}$ of $\frac{2}{3}$ of $x = 75000 \Leftrightarrow \frac{x}{2} = 75000 \Leftrightarrow x = 150000$.

$\therefore$ Value of business = ₹ 150000.

Ex. 32. A man spends $\frac{2}{5}$ of his salary on house rent, $\frac{3}{10}$ of his salary on food and $\frac{1}{8}$ of his salary on conveyance.

If he has ₹ 1400 left with him, find his expenditure on food and conveyance.

Sol. Part of the salary left = $1 - \left(\frac{2}{5} + \frac{3}{10} + \frac{1}{8}\right) = 1 - \frac{33}{40} = \frac{7}{40}$.

Let the monthly salary be ₹ x .

$$\text{Then, } \frac{7}{40} \text{ of } x = 1400 \Leftrightarrow x = \left(\frac{1400 \times 40}{7}\right) = 8000.$$

$$\therefore \text{Expenditure on food} = ₹ \left(\frac{3}{10} \times 8000\right) = ₹ 2400.$$

$$\text{Expenditure on conveyance} = ₹ \left(\frac{1}{8} \times 8000\right) = ₹ 1000.$$

Ex. 33. A third of Arun's marks in Mathematics exceeds a half of his marks in English by 30. If he got 240 marks in the two subjects together, how many marks did he get in English?

Sol. Let Arun's marks in Mathematics and English be x and y respectively.

$$\text{Then, } \frac{1}{3}x - \frac{1}{2}y = 30 \Leftrightarrow 2x - 3y = 180 \quad \dots(i) \text{ and } x + y = 240 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 180$ and $y = 60$.

$\therefore$ Arun's marks in English = 60.

Ex. 34. A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out and four bottles of oil were poured into it,

it was $\frac{3}{4}$ full. How many bottles of oil can the tin contain?

Sol. Suppose x bottles can fill the tin completely.

$$\text{Then, } \frac{4}{5}x - \frac{3}{4}x = (6 - 4) \Leftrightarrow \frac{x}{20} = 2 \Leftrightarrow x = 40.$$

$\therefore$ Required number of bottles = 40.

Ex. 35. If $\frac{1}{8}$ of a pencil is black, $\frac{1}{2}$ of the remaining is white and the remaining $3\frac{1}{2}$ cm is blue, find the total length of the pencil.

Sol. Let the total length of the pencil be x cm. Then,

$$\text{Black part} = \left(\frac{x}{8}\right) \text{ cm. Remaining part} = \left(x - \frac{x}{8}\right) \text{ cm} = \left(\frac{7x}{8}\right) \text{ cm.}$$

$$\text{White part} = \left(\frac{1}{2} \times \frac{7x}{8}\right) \text{ cm} = \left(\frac{7x}{16}\right) \text{ cm. Remaining part} = \left(\frac{7x}{8} - \frac{7x}{16}\right) \text{ cm} = \frac{7x}{16} \text{ cm.}$$

$$\therefore \frac{7x}{16} = \frac{7}{2} \text{ or } x = \frac{16}{2} = 8 \text{ cm.}$$

Hence, total length of the pencil = 8 cm.

Ex. 36. At a college football game, $\frac{4}{5}$ of the seats in the lower deck of the stadium were sold. If $\frac{1}{4}$ of all the seating in the stadium is located in the lower deck, and if $\frac{2}{3}$ of all the seats in the stadium were sold, what fraction of the unsold seats in the stadium was in the lower deck?

Sol. Let the total number of seats in the stadium be x .

$$\text{Then, number of seats in the lower deck} = \frac{x}{4}.$$

$$\text{Number of seats sold} = \frac{2x}{3}. \text{ Number of seats unsold} = \left(x - \frac{2x}{3}\right) = \frac{x}{3}.$$

$$\text{Number of sold seats in lower deck} = \frac{4}{5} \text{ of } \frac{x}{3} = \frac{x}{5}.$$

$$\text{Number of unsold seats in lower deck} = \left(\frac{x}{3} - \frac{x}{5}\right) = \frac{x}{15}.$$

$$\therefore \text{ Required fraction} = \frac{x/15}{x/3} = \frac{1}{5}.$$

Ex. 37. In a certain office, $\frac{1}{3}$ of the workers are women, $\frac{1}{2}$ of the women are married and $\frac{1}{3}$ of the married women have children. If $\frac{3}{4}$ of the men are married and $\frac{2}{3}$ of the married men have children, what part of workers are without children?

Sol. Let the total number of workers be x . Then,

$$\text{Number of women} = \frac{x}{3} \text{ and number of men} = \left(x - \frac{x}{3}\right) = \frac{2x}{3}.$$

$$\text{Number of women having children} = \frac{1}{3} \text{ of } \frac{1}{2} \text{ of } \frac{x}{3} = \frac{x}{18}.$$

$$\text{Number of men having children} = \frac{2}{3} \text{ of } \frac{3}{4} \text{ of } \frac{2x}{3} = \frac{x}{3}.$$

$$\text{Number of workers having children} = \left(\frac{x}{18} + \frac{x}{3}\right) = \frac{7x}{18}.$$

$$\therefore \text{ Workers having no children} = \left(x - \frac{7x}{18}\right) = \frac{11x}{18} = \frac{11}{18} \text{ of all workers.}$$

Ex. 38. A crate of mangoes contains one bruised mango for every 30 mangoes in the crate. If 3 out of every 4 bruised mangoes are considered unsaleable, and there are 12 unsaleable mangoes in the crate, then how many mangoes are there in the crate?

Sol. Let the total number of mangoes in the crate be x . Then,

$$\text{Number of bruised mangoes} = \frac{1}{30}x.$$

$$\text{Number of unsaleable mangoes} = \left(\frac{3}{4} \times \frac{1}{30}x\right) = \frac{1}{40}x.$$

$$\therefore \frac{1}{40}x = 12 \text{ or } x = (12 \times 40) = 480.$$

Hence, total number of mangoes in the crate = 480.

Ex. 39. The cost of 4 rings and 2 bangles is ₹ 57200. What is the cost of 6 rings and 3 bangles? (Bank Recruitment, 2009)

$$\text{Sol. We have: } 4R + 2B = 57200 \Rightarrow \frac{3}{2}(4R + 2B) = \frac{3}{2} \times 57200 \Rightarrow 6R + 3B = 85800.$$

Hence, cost of 6 rings and 3 bangles = ₹ 85800.

Ex. 40. The cost of 15 kg of sugar is ₹ 255, the cost of 17 kg of tea is ₹ 1615 and the cost of 22 kg of rice is ₹ 572. What is the total cost of 18 kg of sugar, 21 kg of tea and 27 kg of rice? (Bank P.O., 2008)

$$\text{Sol. Cost of 1 kg sugar} = ₹ \left(\frac{255}{15}\right) = ₹ 17; \text{ Cost of 1 kg tea} = ₹ \left(\frac{1615}{17}\right) = ₹ 95;$$

$$\text{Cost of 1 kg rice} = ₹ \left(\frac{572}{22}\right) = ₹ 26.$$

$$\therefore \text{ Required cost} = ₹ (17 \times 18 + 95 \times 21 + 26 \times 27) = ₹ (306 + 1995 + 702) = ₹ 3003.$$

Ex. 41. If $*$ is an operation such that $x * y = 3x + 2y$, find the value of $2 * 3 + 3 * 4$.

Sol. $2 * 3 + 3 * 4 = (3 \times 2 + 2 \times 3) + (3 \times 3 + 2 \times 4) = (6 + 6) + (9 + 8)$
 $= 12 + 17 = 29$.

Ex. 42. If $a^2 + b^2 = 117$ and $ab = 54$, then find the value of $\frac{a+b}{a-b}$.

Sol. $(a+b)^2 = a^2 + b^2 + 2ab = 117 + 2 \times 54 = 225 \Rightarrow a+b = 15$.
 $(a-b)^2 = a^2 + b^2 - 2ab = 117 - 2 \times 54 = 9 \Rightarrow a-b = 3$.
 $\therefore \frac{a+b}{a-b} = \frac{15}{3} = 5$.

Ex. 43. Find the value of $\left(\frac{75983 \times 75983 - 45983 \times 45983}{30000} \right)$.

Sol. Given expression $= \frac{(75983)^2 - (45983)^2}{(75983 - 45983)} = \frac{(a^2 - b^2)}{(a-b)}$, where $a = 75983, b = 45983$
 $= \frac{(a+b)(a-b)}{(a-b)} = (a+b) = (75983 + 45983) = 121966$.

Ex. 44. Find the value of $\left(\frac{343 \times 343 \times 343 - 113 \times 113 \times 113}{343 \times 343 + 343 \times 113 + 113 \times 113} \right)$.

(B.Ed. Entrance, 2010)

Sol. Given expression $= \frac{(a^3 - b^3)}{(a^2 + ab + b^2)}$, where $a = 343, b = 113$
 $= (a-b) = (343 - 113) = 230$.

Ex. 45. If $(x-4)^3 + (x-9)^3 + (x-8)^3 = 3(x-4)(x-9)(x-8)$, find the value of x .

Sol. If $a^3 + b^3 + c^3 = 3abc$, then $a + b + c = 0$.

So, we have: $(x-4) + (x-9) + (x-8) = 0 \Rightarrow 3x - 21 = 0 \Rightarrow 3x = 21 \Rightarrow x = 7$.

Ex. 46. There are certain number of benches in a classroom. If four students sit on each bench then three benches remain unoccupied. If, however, three students sit on each bench then three students remain standing in the class. Find the number of students in the class.

(P.C.S., 2009)

Sol. Let the total number of benches in the class be x .

Case I. When 4 students sit on each bench

In this case, total number of students $= 4(x-3)$.

Case II. When 3 students sit on each bench

In this case, total number of students $= 3x + 3$.

$\therefore 4(x-3) = 3x + 3 \Leftrightarrow 4x - 12 = 3x + 3 \Leftrightarrow x = 15$.

Hence, number of students in the class $= 4(x-3) = 4 \times 12 = 48$.

Ex. 47. A man divides ₹ 8600 among 5 sons, 4 daughters and 2 nephews. If each daughter receives four times as much as each nephew, and each son receives five times as much as each nephew, how much does each daughter receive?

Sol. Let the share of each nephew be ₹ x .

Then, share of each daughter = ₹ $(4x)$; share of each son = ₹ $(5x)$.

So, $5 \times 5x + 4 \times 4x + 2 \times x = 8600 \Leftrightarrow 25x + 16x + 2x = 8600$

$\Leftrightarrow 43x = 8600 \Leftrightarrow x = 200$.

$\therefore$ Share of each daughter = ₹ $(4 \times 200) = ₹ 800$.

Ex. 48. ₹ 6500 were divided equally among a certain number of persons. Had there been 15 more persons, each would have got ₹ 30 less. Find the original number of persons.

(M.A.T., 2005)

Sol. Let the original number of persons be x .

Then, $\frac{6500}{x} - \frac{6500}{(x+15)} = 30 \Leftrightarrow 6500 \left[\frac{x+15-x}{x(x+15)} \right] = 30 \Leftrightarrow 30x(x+15) = 6500 \times 15$

$$\Leftrightarrow x^2 + 15x = 3250 \Leftrightarrow x^2 + 15x - 3250 = 0$$

$$\Leftrightarrow x^2 + 65x - 50x - 3250 = 0$$

$$\Leftrightarrow x(x + 65) - 50(x + 65) = 0$$

$$\Leftrightarrow (x + 65)(x - 50) = 0 \Leftrightarrow x = 50.$$

Hence, original number of persons = 50.

Ex. 49. Village X has a population of 68000, which is decreasing at the rate of 1200 per year. Village Y has a population of 42000, which is increasing at the rate of 800 per year. In how many years will the population of the two villages be equal?

Sol. Let the population of villages X and Y be equal after p years.

$$\text{Then, } 68000 - 1200p = 42000 + 800p \Rightarrow 2000p = 26000 \Rightarrow p = 13.$$

So, their population will be equal after 13 years.

Ex. 50. From a group of boys and girls, 15 girls leave. There are then left 2 boys for each girl. After this, 45 boys leave. There are then 5 girls for each boy. Find the number of girls in the beginning.

Sol. Let at present there be x boys. Then, number of girls at present = $5x$.

Before the boys had left: Number of boys = $x + 45$ and number of girls = $5x$.

$$\therefore x + 45 = 2 \times 5x \Leftrightarrow 9x = 45 \Leftrightarrow x = 5.$$

Hence, number of girls in the beginning = $5x + 15 = 25 + 15 = 40$.

Ex. 51. If x is an integer such that $x + \frac{1}{x} = \frac{17}{4}$, find the value of $x - \frac{1}{x}$.

(M.B.A., 2008)

$$\text{Sol. } x + \frac{1}{x} = \frac{17}{4} \Leftrightarrow \left(x + \frac{1}{x}\right)^2 = \left(\frac{17}{4}\right)^2 \Leftrightarrow x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = \frac{289}{16}$$

$$\Leftrightarrow x^2 + \frac{1}{x^2} = \frac{289}{16} - 2 = \frac{257}{16} \Leftrightarrow x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x} = \frac{257}{16} - 2$$

$$\Leftrightarrow \left(x - \frac{1}{x}\right)^2 = \frac{225}{16} = \left(\frac{15}{4}\right)^2 \Leftrightarrow x - \frac{1}{x} = \frac{15}{4}.$$

Ex. 52. A zoo keeper counted the heads of the animals in a zoo and found it to be 80. When he counted the legs of the animals he found it to be 260. If the zoo had either pigeons or horses, how many horses were there in the zoo?

(R.R.B., 2009)

Sol. Let the number of pigeons be x and the number of horses be y .

Then, total number of heads = $x + y$.

total number of legs = $2x + 4y$.

$$\therefore x + y = 80 \quad \dots(i)$$

$$\text{And, } 2x + 4y = 260 \text{ or } x + 2y = 130 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $y = 50$.

Putting $y = 50$ in (i), we get: $x = 30$.

Hence, number of horses in the zoo = 50.

Ex. 53. In a caravan, in addition to 50 hens there are 45 goats and 8 camels with some keepers. If the total number of feet be 224 more than the number of heads, find the number of keepers.

Sol. Let the number of keepers be x . Then,

Total number of heads = $(50 + 45 + 8 + x) = (103 + x)$.

Total number of feet = $(45 + 8) \times 4 + (50 + x) \times 2 = (312 + 2x)$.

$$\therefore (312 + 2x) - (103 + x) = 224 \Leftrightarrow x = 15.$$

Hence, number of keepers = 15.

Ex. 54. When an amount was distributed among 14 boys, each of them got ₹ 80 more than the amount received by each boy when the same amount is distributed equally among 18 boys. What was the amount?

Sol. Let the total amount be ₹ x . Then,

$$\frac{x}{14} - \frac{x}{18} = 80 \Leftrightarrow \frac{2x}{126} = 80 \Leftrightarrow \frac{x}{63} = 80 \Leftrightarrow x = 63 \times 80 = 5040.$$

Hence, total amount = ₹ 5040.

Ex. 55. Mr. Bhaskar is on tour and he has ₹ 360 for his expenses. If he exceeds his tour by 4 days, he must cut down his daily expenses by ₹ 3. For how many days is Mr. Bhaskar on tour?

Sol. Suppose Mr. Bhaskar is on tour for x days. Then,

$$\frac{360}{x} - \frac{360}{x+4} = 3 \Leftrightarrow \frac{1}{x} - \frac{1}{x+4} = \frac{1}{120} \Leftrightarrow x(x+4) = 4 \times 120 = 480$$

$$\Leftrightarrow x^2 + 4x - 480 = 0 \Leftrightarrow (x+24)(x-20) = 0 \Leftrightarrow x = 20.$$

Hence, Mr. Bhaskar is on tour for 20 days.

Ex. 56. A railway half-ticket costs half the full fare. But the reservation charge on half-ticket is the same as that on full ticket. One reserved first-class ticket for a journey between two stations is ₹ 525 and the cost of one full and one half reserved first-class tickets is ₹ 850. What is the reservation charge? (Bank P.O., 2008)

Sol. Let the full fare be ₹ x and the reservation charge per ticket be ₹ y .

$$\text{Then, } x + y = 525 \quad \dots(i)$$

$$\text{And, } \frac{3x}{2} + 2y = 850 \text{ or } 3x + 4y = 1700 \quad \dots(ii)$$

Multiplying (i) by 3 and subtracting from (ii), we get: $y = 125$.

Hence, reservation charge = ₹ 125.

Ex. 57. Tom reads at an average rate of 30 pages per hour, while Jane reads at an average rate of 40 pages per hour. If Tom starts reading a novel at 4: 30 and Jane begins reading an identical copy of the same book at 5: 20, then at what time will they be reading the same page? (M.A.T., 2007)

Sol. Suppose they are reading the same page x hours after 5: 20.

$$\text{Time from 4: 30 to 5: 20} = 50 \text{ min} = \frac{5}{6} \text{ hr.}$$

$$\text{Then, } 30\left(x + \frac{5}{6}\right) = 40x \Leftrightarrow 30x + 25 = 40x \Leftrightarrow 10x = 25 \Leftrightarrow x = 2.5.$$

Hence, Tom and Jane will be reading the same page $2\frac{1}{2}$ hrs i.e., 2 hr 30 min after 5: 20 i.e., at 7: 50.

Ex. 58. In an objective examination of 90 questions, 5 marks are allotted for every correct answer and 2 marks are deducted for every wrong answer. After attempting all the 90 questions a student got a total of 387 marks. Find the number of questions that he attempted wrong. (M.A.T. 2007)

Sol. Let the number of questions attempted correctly be x .

$$\text{Then, number of questions attempted wrong} = (90 - x)$$

$$\therefore 5x - 2(90 - x) = 387 \Leftrightarrow 7x = 387 + 180 = 567 \Leftrightarrow x = 81.$$

$$\text{Hence, number of questions attempted wrong} = (90 - 81) = 9.$$

Ex. 59. Kiran had 85 currency notes in all, some of which were of ₹ 100 denomination and the remaining of ₹ 50 denomination. The total amount of all these currency notes was ₹ 5000. How much amount did she have in the denomination of ₹ 50?

Sol. Let the number of 50-rupee notes be x .

$$\text{Then, the number of 100-rupee notes} = (85 - x).$$

$$\therefore 50x + 100(85 - x) = 5000 \Leftrightarrow x + 2(85 - x) = 100 \Leftrightarrow x = 70.$$

$$\text{So, required amount} = ₹ (50 \times 70) = ₹ 3500.$$

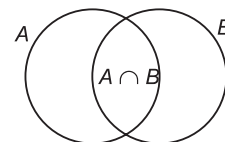
Ex. 60. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. Find the number of people who like both coffee and tea. (P.C.S., 2007)

Sol. Let A and B represent the set of people who like coffee and tea respectively.

$$\text{Then, } n(A) = 37, n(B) = 52, n(A \cup B) = 70.$$

$$\begin{aligned} \therefore n(A \cap B) &= n(A) + n(B) - n(A \cup B) \\ &= 37 + 52 - 70 = 89 - 70 = 19. \end{aligned}$$

Hence, number of people who like both coffee and tea = 19.



Ex. 61. A train starts full of passengers. At the first station, it drops one-third of the passengers and takes 280 more. At the second station, it drops one-half of the new total and takes 12 more. On arriving at the third station, it is found to have 248 passengers. Find the number of passengers in the beginning.

Sol. Let the number of passengers in the beginning be x .

$$\text{After 1st station, number of passengers} = \left(x - \frac{x}{3}\right) + 280 = \left(\frac{2x}{3} + 280\right).$$

$$\text{After 2nd station, number of passengers} = \frac{1}{2} \left(\frac{2x}{3} + 280\right) + 12.$$

$$\therefore \frac{1}{2} \left(\frac{2x}{3} + 280\right) + 12 = 248 \Leftrightarrow \frac{2x}{3} + 280 = 2 \times 236 \Leftrightarrow \frac{2x}{3} = 192$$

$$\Leftrightarrow x = \left(192 \times \frac{3}{2}\right) = 288.$$

$\therefore$ Required number of passengers = 288.

Ex. 62. Arun and Sajal are friends. Each has some money. If Arun gives ₹ 30 to Sajal, then Sajal will have twice the money left with Arun. But, if Sajal gives ₹ 10 to Arun, then Arun will have thrice as much as is left with Sajal. How much money does each have?

Sol. Suppose Arun has ₹ x and Sajal has ₹ y . Then,

$$2(x - 30) = y + 30 \Rightarrow 2x - y = 90 \quad \dots(i)$$

$$\text{and } x + 10 = 3(y - 10) \Rightarrow x - 3y = -40 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 62$ and $y = 34$.

$\therefore$ Arun has ₹ 62 and Sajal has ₹ 34.

Ex. 63. An executive goes on a business trip. His daily schedule has a definite pattern. If he is busy with a meeting in the morning, he is free in the afternoon. When he returns, he realises that he attended 15 meetings altogether. There were 12 free mornings and 13 free afternoons. What was the duration of his trip?

Sol. Let M and A represent the number of busy mornings and busy afternoons respectively.

Then, total number of mornings in the trip = $12 + M$.

And, total number of afternoons in the trip = $13 + A$.

$$\text{Clearly, } 12 + M = 13 + A \text{ or } M - A = 1 \quad \dots(i)$$

Total number of meetings = $M + A$

$$\text{So, } M + A = 15 \quad \dots(ii)$$

Adding (i) and (ii), we get: $2M = 16$ or $M = 8$.

Hence, duration of the trip = $(12 + 8) = 20$ days.

Ex. 64. A man received a cheque. The rupees had been transposed for paise and vice-versa. After spending 5 rupees 42 paise, he discovered that he now had exactly six times the value of the correct cheque amount. What amount should he have received? (M.A.T., 2005)

Sol. Let the amount received be x rupees and y paise.

Then, amount written on cheque = $(100x + y)$ paise.

Actual amount to be received = $(100y + x)$ paise.

$$\therefore (100x + y) - 542 = 6(100y + x)$$

$$\Leftrightarrow 100x + y - 542 = 600y + 6x \Leftrightarrow 94x - 599y = 542$$

$$\Leftrightarrow 94x = 542 + 599y$$

$$\Leftrightarrow x = \frac{542 + 599y}{94} = \frac{470 + 470y + 72 + 129y}{94} = 5(1 + y) + \frac{72 + 129y}{94}.$$

For an integral value of x , $(72 + 129y)$ must be divisible by 94. This happens when $y = 6$.

$$\therefore x = 5(1 + 6) + \frac{72 + 129 \times 6}{94} = 35 + 9 = 44.$$

Hence, the man should have received ₹ 6.44.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. What is 304 times 141? (Bank Recruitment, 2009)
 - (a) 39640
 - (b) 38760
 - (c) 42864
 - (d) 45942
 - (e) None of these
2. If $A + B = 96$ and A is half of B , then the value of A will be
 - (a) 22
 - (b) 32
 - (c) 48
 - (d) 64
 - (e) None of these
3. $1888 \div 32 \div 8 = ?$ (Bank P.O., 2008)
 - (a) 7.375
 - (b) 9.485
 - (c) 29.5
 - (d) 472
 - (e) None of these
4. $-76 \times 33 + 221 = ?$ (NABARD, 2009)
 - (a) -2287
 - (b) 2287
 - (c) -19304
 - (d) 19304
 - (e) None of these
5. $4848 \div 24 \times 11 - 222 = ?$ (Bank P.O., 2010)
 - (a) 200
 - (b) 2444
 - (c) 2000
 - (d) $115\frac{3}{8}$
 - (e) None of these
6. $(425 \times 4000) \div 16000 \times 12 = ?$ (Bank Recruitment, 2009)
 - (a) 8.85
 - (b) 925
 - (c) 1275
 - (d) 1700
 - (e) None of these
7. $[(84)^2 \div 28 \times 12] \div 24 = 7 \times ?$ (Bank P.O., 2009)
 - (a) 15
 - (b) 17
 - (c) 19
 - (d) 21
 - (e) None of these
8. $(98764 + 89881 + 99763 + 66342) \div (1186 + ? + 1040 + 1870) = 55$ (Bank Recruitment, 2008)
 - (a) 2254
 - (b) 2354
 - (c) 2368
 - (d) 2404
 - (e) None of these
9. $1148 \div 28 \times 1408 \div 32 = ?$ (S.B.I.P.O., 2008)
 - (a) 1800
 - (b) 1804
 - (c) 1814
 - (d) 1822
 - (e) None of these
10. $-224 + (-314) \times (-9) = ?$
 - (a) -547
 - (b) -2602
 - (c) 547
 - (d) 2602
 - (e) None of these
11. $853 + ? \div 17 = 1000$
 - (a) 2482
 - (b) 2499
 - (c) 2516
 - (d) 16147
 - (e) None of these
12. $(? - 968) \div 79 \times 4 = 512$ (Bank Recruitment, 2007)
 - (a) 10185
 - (b) 10190
 - (c) 11075
 - (d) 11080
 - (e) None of these
13. $[(125)^2 \div 50 \times 20] \div 25 = ?$
 - (a) 11
 - (b) 100
 - (c) 150
 - (d) 250
 - (e) None of these
14. $999 \times 99 \times 9 \div 99 \div 9 \div 3 = ?$ (Bank P.O., 2009)
 - (a) 99
 - (b) 111
 - (c) 333
 - (d) 999
 - (e) None of these
15. If $a, b, c, \dots, x, y, z$ are 26 natural numbers, then the value of $(x - a)(x - b)(x - c) \dots (x - y)(x - z)$ is
 - (a) 0
 - (b) 1
 - (c) 13
 - (d) 26
16. Simplify: $1 - [1 - \{1 - (1 - \overline{1 - 1})\}]$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
17. What mathematical operation should come at the place of '?' in the equation:
 $2? 6 - 12 \div 4 + 2 = 11.$
 - (a) +
 - (b) -
 - (c) $\times$
 - (d) $\div$
18. If $45 - [28 - \{37 - (15 - *)\}] = 58$, then $*$ is equal to:
 - (a) -29
 - (b) -19
 - (c) 19
 - (d) 29
19. $\frac{343 \times 49}{216 \times 16 \times 81} = ?$ (Bank P.O., 2010)
 - (a) $\frac{7^5}{6^7}$
 - (b) $\frac{7^5}{6^8}$
 - (c) $\frac{7^4}{6^8}$
 - (d) $\frac{7^6}{6^7}$
 - (e) None of these
20. The value of x in the equation $\frac{113 \times 4 - x \times 2}{13 \times 9 - 5 \times 7} = 5$ is (P.C.S., 2008)
 - (a) 21
 - (b) 27
 - (c) 35
 - (d) 42
21. $\frac{18 \times 14 - 6 \times 8}{488 \div 4 - 20} = ?$ (L.I.C.A.D.O., 2007)
 - (a) $\frac{1}{2}$
 - (b) $\frac{3}{4}$
 - (c) 2
 - (d) 4
 - (e) None of these

22. $\frac{5+5+5}{7+7+7} - \frac{5+5}{7+7} + \frac{5}{7} = ?$

(a) $\frac{1}{7}$

(b) $\frac{3}{7}$

(c) $\frac{5}{7}$

(d) $\frac{13}{14}$

23. $\frac{3+4 \div 2 \times 3}{4+3 \times 2 \div 3} = ?$

(a) $\frac{7}{36}$

(b) $\frac{11}{18}$

(c) $\frac{3}{2}$

(d) $\frac{9}{4}$

24. $\frac{4+4 \times 18-6-8}{123 \times 6-146 \times 5} = ?$

(a) 1

(b) 2

(c) 6.65

(d) 7.75

25. $\frac{180 \times 15 - 12 \times 20}{140 \times 8 + 2 \times 55} = ?$

(a) $\frac{1}{7}$

(b) $\frac{4}{5}$

(c) 2

(d) 4

(e) None of these

26. Evaluate: $\frac{8 - [5 - (-3 + 2)] \div 2}{|5 - 3| - |5 - 8| \div 3}$

(a) 2

(b) 3

(c) 4

(d) 5

27. Given that $(1^2 + 2^2 + 3^2 + \dots + 10^2) = 385$, the value of $(2^2 + 4^2 + 6^2 + \dots + 20^2)$ is equal to (M.B.A., 2006)

(a) 770

(b) 1155

(c) 1540

(d) $(385)^2$

28. $2\frac{1}{4} + 1\frac{1}{3} - 4\frac{1}{2} = ?$

(Bank Recruitment, 2010)

(a) $-1\frac{1}{12}$

(b) $\frac{11}{12}$

(c) $-\frac{11}{12}$

(d) $1\frac{1}{12}$

(e) None of these

29. $4\frac{3}{7} - 1\frac{3}{14} = ? + 2\frac{3}{28}$

(Bank P.O., 2010)

(a) $1\frac{3}{28}$

(b) $3\frac{1}{14}$

(c) $3\frac{3}{14}$

(d) $3\frac{1}{28}$

(e) None of these

30. $1\frac{3}{4} - 1\frac{1}{5} + 1\frac{5}{8} = ?$

(Bank Recruitment, 2010)

(a) $2\frac{7}{40}$

(b) $3\frac{7}{40}$

(c) $9\frac{5}{8}$

(d) $10\frac{7}{8}$

(e) None of these

31. $12\frac{1}{3} + 10\frac{5}{6} - 7\frac{2}{3} - 1\frac{4}{7} = ?$

(Bank P.O., 2008)

(a) $11\frac{13}{14}$

(b) $13\frac{11}{14}$

(c) $13\frac{13}{14}$

(d) $14\frac{11}{13}$

(e) None of these

32. $6\frac{9}{13} - 4\frac{4}{11} + 2\frac{2}{5} = ?$

(Bank P.O., 2006)

(a) $4\frac{596}{715}$

(b) $4\frac{521}{715}$

(c) $9\frac{324}{715}$

(d) $9\frac{386}{715}$

(e) None of these

33. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{7} + \frac{1}{14} + \frac{1}{28}$ is equal to

(a) 2

(b) 2.5

(c) 3

(d) 3.5

34. $\frac{1}{\left(2\frac{1}{3}\right)} + \frac{1}{\left(1\frac{3}{4}\right)}$ is equal to

(a) $\frac{7}{14}$

(b) $\frac{12}{49}$

(c) $4\frac{1}{12}$

(d) None of these

35. $5\frac{5}{6} - 3\frac{8}{9} - ? = 1$

(a) $\frac{2}{3}$

(b) $\frac{3}{2}$

(c) $\frac{17}{18}$

(d) 3

36. If $\frac{1}{3} + \frac{1}{2} + \frac{1}{x} = 4$, then $x = ?$

(a) $\frac{5}{18}$

(b) $\frac{6}{19}$

(c) $\frac{18}{5}$

(d) $\frac{24}{11}$

37. $\frac{532}{648} \times \frac{432}{588} = ?$

(L.I.C.A.D.O., 2007)

(a) $\frac{2}{21}$

(b) $\frac{38}{63}$

(c) $\frac{21}{64}$

(d) $\frac{19}{21}$

(e) None of these

38. $1\frac{10}{11} \times 17\frac{2}{7} \times 36\frac{1}{6} = ?$

(a) $397\frac{5}{6}$

(b) $397\frac{1}{2}$

(c) $1193\frac{5}{6}$

(d) $1193\frac{1}{2}$

(e) None of these

39. $\frac{3}{7}$ of 455 + $\frac{5}{8}$ of 456 = ?

(Bank Recruitment, 2009)

(a) 448

(b) 464

(c) 476

(d) 480

(e) None of these

40. $\frac{3}{5}$ of $\frac{4}{7}$ of $\frac{5}{9}$ of $\frac{21}{24}$ of 504 = ?

(a) 63

(b) 69

(c) 96

(d) 109

(e) None of these

41. $6\frac{5}{6} \times 5\frac{1}{3} + 17\frac{2}{3} \times 4\frac{1}{2} = ?$

(a) $112\frac{1}{3}$

(b) $116\frac{2}{3}$

(c) 240

(d) 663

(e) None of these

42. $4\frac{4}{5} \div 6\frac{2}{5} = ?$

(Bank Recruitment, 2009)

(a) $\frac{3}{4}$

(b) $\frac{5}{7}$

(c) $\frac{5}{8}$

(d) $\frac{7}{11}$

(e) None of these

43. $1\frac{1}{4} + 1\frac{5}{9} \times 1\frac{5}{8} \div 6\frac{1}{2} = ?$

(Bank P.O., 2009)

(a) 17

(b) 27

(c) 18

(d) 42

(e) None of these

44. $\frac{225}{836} \times \frac{152}{245} \div 1\frac{43}{77} = ?$

(Bank Recruitment, 2009)

(a) $\frac{6}{49}$

(b) $\frac{6}{11}$

(c) $\frac{3}{28}$

(d) $\frac{1}{7}$

(e) None of these

45. $\left(16\frac{2}{5} - 12\frac{1}{15}\right) \div 3\frac{4}{81} = ?$

(a) $1\frac{9}{19}$

(b) $1\frac{9}{13}$

(c) $2\frac{8}{13}$

(d) $3\frac{2}{19}$

(e) None of these

46. $18\frac{3}{4} \times ? \div \frac{6}{37} = 1480$

(a) $11\frac{7}{12}$

(b) $11\frac{4}{5}$

(c) $12\frac{5}{12}$

(d) $12\frac{4}{5}$

(e) None of these

47. $\frac{3}{2} \times \frac{11}{5} \div \left(\frac{25}{44} \times \frac{11}{5}\right) \div \frac{33}{15} = ?$

(a) $\frac{1}{2}$

(b) $\frac{2}{3}$

(c) $\frac{126}{125}$

(d) $5\frac{101}{125}$

(e) None of these

48. If $3\frac{x}{7} \times 2\frac{y}{5} = 10$, then the values of x and y respectively, would be

(a) 3, 4

(b) 5, 7

(c) 4, 3

(d) 4, 4

49. $1 + 2 \div \left\{1 + 2 \div \left(1 + \frac{1}{3}\right)\right\}$ is equal to

(a) $1\frac{4}{5}$

(b) $2\frac{1}{4}$

(c) $4\frac{1}{5}$

(d) $5\frac{1}{4}$

50. Simplify: $10\frac{1}{8}$ of $\frac{12}{15} \div \frac{35}{36}$ of $\frac{20}{49}$.

(a) $17\frac{5}{12}$

(b) $17\frac{8}{17}$

(c) $20\frac{3}{25}$

(d) $20\frac{103}{250}$

51. $-\frac{1}{2} - \frac{2}{3} + \frac{4}{5} - \frac{1}{3} + \frac{1}{5} + \frac{3}{4}$ is simplified to

(a) $-\frac{3}{10}$

(b) $-\frac{10}{3}$

(c) -2

(d) 1

52. When $\left(\frac{1}{2} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6}\right)$ is divided by

$\left(\frac{2}{5} - \frac{5}{9} + \frac{3}{5} - \frac{7}{18}\right)$, the result is

- (a) $2\frac{1}{18}$ (b) $3\frac{1}{6}$
(c) $3\frac{3}{10}$ (d) $5\frac{1}{10}$

53. Which of the following can be used to compute $\left(34 \times 4\frac{1}{2}\right)$?

- (a) $(30 \times 4) + \left(4 \times 4\frac{1}{2}\right)$
(b) $(34 \times 40) + \left(34 \times \frac{1}{2}\right)$
(c) $\left(30 \times 4\frac{1}{2}\right) + (4 \times 4)$
(d) $\left(34 \times \frac{1}{2}\right) + (30 \times 4) + (4 \times 4)$

54. $\frac{3}{8}$ of $168 \times 15 \div 5 + ? = 549 \div 9 + 235$

- (a) 107 (b) 174
(c) 189 (d) 296
(e) None of these

55. Find the value of * in the following

$$1\frac{2}{3} \div \frac{2}{7} \times \frac{*}{7} = 1\frac{1}{4} \times \frac{2}{3} \div \frac{1}{6}$$

(S.S.C., 2002)

- (a) 0.006 (b) $\frac{1}{6}$
(c) 0.6 (d) 6

56. $5\frac{2}{3} \div ? \frac{5}{6} = 2$

- (a) 2 (b) 3
(c) 4 (d) None of these

57. Supply the two missing figures in order indicated by x and y in the given equation, the fractions being in their lowest terms.

(IGNOU, 2003)

$$5\frac{1}{x} \times y\frac{3}{4} = 20$$

- (a) 3, 1 (b) 3, 3
(c) 4, 1 (d) 5, 3

58. The difference of $1\frac{3}{16}$ and its reciprocal is equal to

(M.A.T., 2002)

- (a) $1\frac{1}{8}$ (b) $\frac{4}{3}$
(c) $\frac{15}{16}$ (d) None of these

59. How many $\frac{1}{8}s$ are there in $37\frac{1}{2}$?

- (a) 300 (b) 400
(c) 500 (d) Cannot be determined

60. Let $a = (4 \div 3) \div 3 \div 4$, $b = 4 \div (3 \div 3) \div 4$, $c = 4 \div 3 \div (3 \div 4)$. The maximum value among the above three is

(I.A.M., 2007)

- (a) a (b) b
(c) c (d) All equal

61. If $I = \frac{3}{4} \div \frac{5}{6}$, $II = 3 \div [(4 \div 5) \div 6]$, $III = [3 \div (4 \div 5)]$

$\div 6$, $IV = 3 \div 4 \div (5 \div 6)$, then

(A.A.O., 2009)

- (a) I and II are equal (b) I and III are equal
(c) I and IV are equal (d) All are equal

62. The value of $\left(\frac{5}{7} \text{ of } 1\frac{6}{13}\right) \div \left(2\frac{5}{7} \div 3\frac{1}{4}\right)$ is

- (a) $\frac{20}{169}$ (b) 1
(c) $\frac{5}{4}$ (d) $1\frac{119}{180}$

63. $2\frac{3}{4} \div 2\frac{2}{3} \div 1\frac{1}{12} = ?$

- (a) $\frac{39}{48}$ (b) $1\frac{1}{4}$
(c) $\frac{169}{144}$ (d) None of these

64. The value of $\frac{2}{3} \times \frac{3}{\frac{5}{6} \div \frac{2}{3} \text{ of } 1\frac{1}{4}}$ is

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) 1 (d) 2

65. $\frac{4335}{4(?)24} \div 1\frac{7}{8} = \frac{289}{528}$

- (a) 1 (b) 2
(c) 8 (d) None of these

66. $5\frac{1}{3} - 3\frac{2}{3} \div 1\frac{1}{3} \div ? + 3\frac{1}{5} \div 1\frac{1}{5} = 7$

- (a) $1\frac{1}{2}$ (b) $2\frac{1}{3}$
(c) $3\frac{1}{4}$ (d) None of these

67. $9 - 1\frac{2}{9} \text{ of } 3\frac{3}{11} \div 5\frac{1}{7} \text{ of } \frac{7}{9} = ?$

(S.S.C., 2002)

- (a) $\frac{5}{4}$ (b) 8
(c) $8\frac{32}{81}$ (d) 9

68. $\frac{5}{6} \div \frac{6}{7} \times ? - \frac{8}{9} \div 1\frac{3}{5} + \frac{3}{4} \times 3\frac{1}{3} = 2\frac{7}{9}$

- (a) $\frac{7}{6}$ (b) $\frac{6}{7}$
(c) 1 (d) None of these

69. $\frac{3}{4} \div 2\frac{1}{4}$ of $\frac{2}{3} - \frac{\frac{1}{2} - \frac{1}{3}}{\frac{1}{2} + \frac{1}{3}} \times 3\frac{1}{3} + \frac{5}{6} = ?$

- (a) $\frac{7}{18}$ (b) $\frac{49}{54}$
(c) $\frac{2}{3}$ (d) $\frac{1}{6}$

70. A student was asked to solve the fraction $\frac{\frac{7}{3} + 1\frac{1}{2} \text{ of } \frac{5}{3}}{2 + 1\frac{2}{3}}$ and his answer was $\frac{1}{4}$. By how much was his answer wrong?

- (a) 1 (b) $\frac{1}{55}$
(c) $\frac{1}{220}$ (d) None of these

71. Simplify: $\frac{\frac{1}{3} + \frac{3}{4} \left(\frac{2}{5} - \frac{1}{3} \right)}{1\frac{2}{3} \text{ of } \frac{3}{4} - \frac{1}{4} \text{ of } \frac{4}{5}}$

- (a) $\frac{1}{63}$ (b) $\frac{23}{40}$
(c) $\frac{23}{55}$ (d) $\frac{23}{63}$

72. Find the value of $1 - \left(\frac{1\frac{2}{3}}{3\frac{1}{2}} + \frac{1\frac{1}{6}}{3\frac{1}{2}} \right)$

- (a) $\frac{3}{20}$ (b) $\frac{4}{21}$
(c) $\frac{8}{21}$ (d) $\frac{13}{21}$

73. If the expression $2\frac{1}{2}$ of $\frac{3}{4} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \frac{3}{2} \left[\frac{2}{3} - \frac{1}{2} \text{ of } \frac{2}{3} \right]$ is simplified, we get (R.R.B., 2006)

- (a) $\frac{1}{2}$ (b) $\frac{7}{8}$
(c) $1\frac{5}{8}$ (d) $2\frac{3}{5}$

74. The value of $\frac{1\frac{1}{7} - \frac{2}{3} + \frac{\frac{2}{5}}{1 - \frac{1}{25}}}{1 - \frac{1}{7} \left[\frac{1}{3} + \frac{\frac{2}{5}}{1 - \frac{2}{5}} \right]}$ is (I.I.F.T., 2005)

- (a) $\frac{3}{4}$ (b) $\frac{24}{25}$
(c) 1 (d) $1\frac{1}{24}$

75. $\frac{6}{5 - \frac{5}{3}} \div \frac{4 - \frac{2}{1}}{5 - \frac{3}{2}} - \frac{2}{5} \text{ of } \left\{ \frac{6}{9} + \frac{2}{3} \text{ of } \frac{1}{2} \right\} = ?$ (M.C.A., 2005)

- (a) $1\frac{1}{3}$ (b) $2\frac{13}{49}$
(c) $1\frac{7}{16}$ (d) $2\frac{3}{5}$

76. The expression $\frac{5\frac{5}{8}}{6\frac{3}{7}}$ of $\frac{6\frac{7}{11}}{9\frac{1}{8}} \div \frac{8}{9} \left(2\frac{3}{11} + \frac{13}{22} \right)$ of $\frac{3}{5}$ equals

- (a) 1 (b) $\frac{1}{2}$
(c) $\frac{7}{9}$ (d) $\frac{5}{12}$

77. The simplified value of $\frac{\frac{1}{3} \div \frac{1}{3} \times \frac{1}{3}}{\frac{1}{3} \div \frac{1}{3} \text{ of } \frac{1}{3}} - \frac{1}{9}$ is

- (a) 0 (b) $\frac{1}{9}$
(c) $\frac{1}{3}$ (d) 1

78. The value of $\frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \text{ of } \frac{1}{2}}$ is

- (a) 1 (b) $1\frac{1}{3}$
(c) $2\frac{2}{3}$ (d) 3

79. $\frac{3\frac{1}{4} - \frac{4}{5} \text{ of } \frac{5}{6}}{4\frac{1}{3} \div \frac{1}{5} - \left(\frac{3}{10} + 21\frac{1}{5} \right)}$ is equal to

- (a) $\frac{1}{6}$ (b) $2\frac{7}{12}$
 (c) $15\frac{1}{2}$ (d) $21\frac{1}{2}$
80. $\frac{7\frac{1}{2} - 5\frac{3}{4}}{3\frac{1}{2} + ?} \div \frac{\frac{1}{2} + 1\frac{1}{4}}{1\frac{1}{5} + 3\frac{1}{2}} = 0.6$
 (a) $4\frac{1}{3}$ (b) $4\frac{1}{2}$
 (c) $4\frac{2}{3}$ (d) None of these
81. Which of the following pairs of fractions adds up to a number greater than 5?
 (a) $\frac{5}{3}, \frac{3}{4}$ (b) $\frac{7}{3}, \frac{11}{5}$
 (c) $\frac{11}{4}, \frac{8}{3}$ (d) $\frac{13}{5}, \frac{11}{6}$
82. $5 - \left[\frac{3}{4} + \left\{ 2\frac{1}{2} - \left(0.5 + \frac{1}{6} - \frac{1}{7} \right) \right\} \right]$ is equal to
 (a) $1\frac{19}{84}$ (b) $2\frac{61}{84}$
 (c) $2\frac{23}{84}$ (d) $2\frac{47}{84}$
83. If $[p]$ means the greatest integer less than or equal to p , then $\left[-\frac{1}{4} \right] + \left[4\frac{1}{4} \right] + [3]$ is equal to:
 (a) 4 (b) 5
 (c) 6 (d) 7
84. If $x = -3$, then $x^3 - x^2 - x$ will be equal to (U.P.G.I.C., 2009)
 (a) -33 (b) -27
 (c) 15 (d) 54
85. Let $R = gS - 4$, when $S = 8$ and $R = 16$. If $S = 10$, then R is equal to
 (a) 11 (b) 14
 (c) 20 (d) 21
86. If $x = 5$, $y = 3$ and $z = 2$, then $\frac{x(y-z)}{y(x+y+z)} = ?$
 (a) $\frac{1}{6}$ (b) $\frac{1}{30}$
 (c) 1 (d) 5
87. If $a = 7$, $b = 5$, then the value of $a^3 - b^3 + 3a^2b$ is
 (a) 218 (b) 307
 (c) 735 (d) 953
88. $-7m - [3n - \{8m - (4n - 10m)\}]$ simplifies to (R.R.B., 2006)
 (a) $11m - 5n$ (b) $11m - 7n$
 (c) $11n - 7m$ (d) $13n - 11m$

89. If $a + 2b = 6$ and $ab = 4$, then what is $\frac{2}{a} + \frac{1}{b}$? (M.B.A., 2006)
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{3}{2}$ (d) 2
 (e) $\frac{5}{2}$
90. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{7}$, the value of $(a + b + c) \div c$ is
 (a) $\frac{1}{7}$ (b) $\frac{1}{2}$
 (c) 2 (d) 7
91. The value of $2a^3 - [3a^3 + 4b^3 - \{2a^3 + (-7a^3)\} + 5a^3 - 7b^3]$ is (Teacher's Exam, 2007)
 (a) $-11a^3 + 3b^3$ (b) $7b^3 + 3a^3$
 (c) $11a^3 - 3b^3$ (d) $-(11a^3 + 3b^3)$
92. If $\frac{x}{y+z} = a$; $\frac{y}{z+x} = b$ and $\frac{z}{x+y} = c$, then $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$ is equal to
 (a) 2 (b) 3
 (c) 6 (d) $x + y + z$
93. If $y = (x + 3)^2$, then $(-2x - 6)^2$ is equal to
 (a) $-4y^2$ (b) $-2y^2$
 (c) $-4y$ (d) $2y$
 (e) $4y$
94. Which among the following is the correct value of $(x + a)(x + b)$? (R.R.B., 2007)
 (a) $x^2 + abx + ab$ (b) $x^2 - (a + b)x + ab$
 (c) $x^2 + (a + b)x + ab$ (d) $x^2 + (a - b)x + ab$
95. The value of $\left[1 + \frac{1}{x+1} \right] \left[1 + \frac{1}{x+2} \right] \left[1 + \frac{1}{x+3} \right] \left[1 + \frac{1}{x+4} \right]$ is
 (a) $\frac{x+5}{x+1}$ (b) $\frac{x+1}{x+5}$
 (c) $1 + \frac{1}{x+5}$ (d) $\frac{1}{x+5}$
96. If $(a - b)$ is 6 more than $(c + d)$ and $(a + b)$ is 3 less than $(c - d)$, then $(a - c)$ is
 (a) 0.5 (b) 1
 (c) 1.5 (d) None of these
97. If $\frac{a}{b} = \frac{4}{3}$, then $\frac{3a+2b}{3a-2b} = ?$ (M.B.A., 2007)
 (a) -1 (b) 3
 (c) 5 (d) 6

98. If $\frac{x}{y} = \frac{6}{5}$, then the value of $\left(\frac{6}{7} - \frac{5x-y}{5x+y}\right)$ is equal to
- (a) $\frac{1}{7}$ (b) $\frac{2}{7}$
 (c) $\frac{3}{7}$ (d) $\frac{4}{7}$
99. If $\frac{x}{y} = \frac{1}{3}$, then $\frac{x^2 + y^2}{x^2 - y^2} = ?$ (R.R.B., 2006)
- (a) $-\frac{10}{9}$ (b) $\frac{5}{4}$
 (c) $-\frac{5}{4}$ (d) $-\frac{5}{3}$
100. If $\frac{x}{2y} = \frac{6}{7}$, the value of $\frac{x-y}{x+y} + \frac{14}{19}$ is
- (a) $\frac{13}{19}$ (b) $\frac{15}{19}$
 (c) 1 (d) $1\frac{1}{19}$
101. If $\frac{a}{b} = \frac{4}{5}$ and $\frac{b}{c} = \frac{15}{16}$, then $\frac{c^2 - a^2}{c^2 + a^2}$ is (M.B.A., 2007)
- (a) $\frac{1}{7}$ (b) $\frac{7}{25}$
 (c) $\frac{3}{4}$ (d) None of these
102. If $x = 1 - q$ and $y = 2q + 1$, then for what value of q , x is equal to y ?
- (a) -1 (b) 0
 (c) $\frac{1}{2}$ (d) 2
103. Find x if $\frac{x}{5} - \frac{x}{6} = 4$.
- (a) -120 (b) -100
 (c) 100 (d) 120
104. If $4x + 5y = 83$ and $\frac{3x}{2y} = \frac{21}{22}$, then $y - x = ?$
- (a) 3 (b) 4
 (c) 7 (d) 11
105. If $a = \frac{x}{x+y}$ and $b = \frac{y}{x-y}$, then $\frac{ab}{a+b}$ is equal to
- (a) $\frac{xy}{x^2 + y^2}$ (b) $\frac{x^2 + y^2}{xy}$
 (c) $\frac{x}{x+y}$ (d) $\left(\frac{y}{x+y}\right)^2$
106. If $\frac{a}{b} = \frac{1}{3}$, $\frac{b}{c} = 2$, $\frac{c}{d} = \frac{1}{2}$, $\frac{d}{e} = 3$ and $\frac{e}{f} = \frac{1}{4}$, then what is the value of $\frac{abc}{def}$? (C.A.T., 2006)
- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$
 (c) $\frac{3}{8}$ (d) $\frac{27}{4}$
 (e) $\frac{27}{8}$
107. If $\frac{m}{n} = \frac{4}{3}$ and $\frac{r}{t} = \frac{9}{14}$, the value of $\frac{3mr - nt}{4nt - 7mr}$ is (M.B.A. 2011)
- (a) $-5\frac{1}{2}$ (b) $-\frac{11}{14}$
 (c) $-1\frac{1}{4}$ (d) $\frac{11}{14}$
108. If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$, then $c + \frac{1}{a}$ is equal to (S.S.C., 2007)
- (a) 0 (b) $\frac{1}{2}$
 (c) 1 (d) 2
109. If $y + \frac{1}{z} = 1$ and $x + \frac{1}{y} = 1$, then what is the value of xyz ?
- (a) -1 (b) 0
 (c) $\frac{1}{2}$ (d) 1
110. If $\frac{x}{(2x+y+z)} = \frac{y}{(x+2y+z)} = \frac{z}{(x+y+2z)} = a$, then find a if $x + y + z \neq 0$. (M.A.T., 2005)
- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$
 (c) $\frac{1}{2}$ (d) $\frac{1}{8}$
111. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = K$, then the value of K is
- (a) $\pm \frac{1}{2}$ (b) $\frac{1}{2}$ or -1
 (c) -1 (d) $\frac{1}{2}$
112. If $a - 8 = b$, then determine the value of $|a - b| - |b - a|$.
- (a) 0 (b) 2
 (c) 4 (d) 16

113. If $x = \frac{a}{a-1}$ and $y = \frac{1}{a-1}$, then

- (a) x is equal to y
- (b) x is equal to y only if $a < 1$
- (c) x is greater than y
- (d) x is greater than y only if $a < 1$
- (e) y is greater than x only if $a < 1$

114. If $0 < a < 1$, then the value of $a + \frac{1}{a}$ is

- (a) less than 2
- (b) greater than 2
- (c) less than 4
- (d) greater than 4

115. If $\frac{a}{x} + \frac{y}{b} = 1$ and $\frac{b}{y} + \frac{z}{c} = 1$, then $\frac{x}{a} + \frac{c}{z}$ will be equal to:

- (a) 0
- (b) $\frac{b}{y}$
- (c) 1
- (d) $\frac{y}{b}$

116. If a, b, c are integers; $a^2 + b^2 = 45$ and $b^2 + c^2 = 40$, then the values of a, b and c respectively are

- (a) 2, 6, 3
- (b) 3, 2, 6
- (c) 5, 4, 3
- (d) None of these

117. If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$? (L.I.C.A.D.O., 2007)

- (a) 110
- (b) 113
- (c) 121
- (d) Cannot be determined
- (e) None of these

118. If $(a - b) = 4$ and $ab = 2$, then $(a^2 + b^2) = ?$

- (a) 18
- (b) 20
- (c) 25
- (d) None of these

119. If $a = bc$ and $c = a - b$, then the value of a is

(R.R.B., 2008)

- (a) $b^2 - 1$
- (b) $\frac{b^2}{b-1}$
- (c) $\frac{b}{b-1}$
- (d) None of these

120. If $a + b + c = 0$, find the value of

$$\frac{a^2}{(a^2 - bc)} + \frac{b^2}{(b^2 - ca)} + \frac{c^2}{(c^2 - ab)}.$$

(M.A.T., 2005)

- (a) 0
- (b) 1
- (c) 2
- (d) 4

121. If $\frac{x}{y} = \frac{a+2}{a-2}$, then $\frac{x^2 - y^2}{x^2 + y^2}$ is equal to

- (a) $\frac{8a}{a^2 + 4}$
- (b) $\frac{4a}{a^2 - 4}$
- (c) $\frac{4}{a^2}$
- (d) $\frac{4a}{a^2 + 4}$

122. If $3x + 7 = x^2 + P = 7x + 5$, what is the value of P ?

- (a) $\frac{1}{2}$
- (b) $8\frac{1}{4}$
- (c) $8\frac{1}{2}$
- (d) Cannot be determined

123. If $\frac{2a+b}{a+4b} = 3$, then find the value of $\frac{a+b}{a+2b}$.

- (a) $\frac{2}{7}$
- (b) $\frac{5}{9}$
- (c) $\frac{10}{7}$
- (d) $\frac{10}{9}$

124. If $(2a + 3b)(2c - 3d) = (2a - 3b)(2c + 3d)$, then

- (a) $\frac{a}{b} = \frac{c}{d}$
- (b) $\frac{a}{d} = \frac{c}{b}$
- (c) $\frac{a}{b} = \frac{d}{c}$
- (d) $\frac{b}{a} = \frac{c}{d}$

125. If $(a + b + 2c + 3d)(a - b - 2c + 3d) = (a - b + 2c - 3d)(a + b - 2c - 3d)$, then $2bc$ is equal to

- (a) $\frac{3}{2}$
- (b) $\frac{3a}{2d}$
- (c) $3ad$
- (d) a^2d^2

126. The expression $\frac{x+y}{x-y} \div \frac{(x+y)^2}{(x^2 - y^2)}$ is equal to

- (a) -1
- (b) $x - y$
- (c) 1
- (d) $x + y$

127. $\frac{a^2 - b^2 - 2bc - c^2}{a^2 + b^2 + 2ab - c^2}$ is equivalent to

- (a) $\frac{a-b+c}{a+b+c}$
- (b) $\frac{a-b-c}{a-b+c}$
- (c) $\frac{a-b-c}{a+b-c}$
- (d) $\frac{a+b+c}{a-b+c}$

128. If $x + y + z = 0$, then $x^2 + xy + y^2$ equals

- (a) $y^2 + yz + z^2$
- (b) $y^2 - yz + z^2$
- (c) $z^2 - zx + x^2$
- (d) $z^2 + zx + x^2$

129. If $a + b + c = 2s$, then the value of $(s - a)^2 + (s - b)^2 + (s - c)^2 + s^2$ will be

- (a) $s^2 - a^2 - b^2 - c^2$
- (b) $s^2 + a^2 + b^2 + c^2$
- (c) $a^2 + b^2 + c^2$
- (d) $4s^2 - a^2 - b^2 - c^2$

130. If $(x + y)^2 - z^2 = 4$, $(y + z)^2 - x^2 = 9$, $(z + x)^2 - y^2 = 36$, what is/are the value(s) of $x + y + z$?

- (a) 0
- (b) ± 1
- (c) ± 3
- (d) ± 7

- 131.** If $a + b = 2c$, then the value of $\frac{a}{a-c} + \frac{b}{b-c}$ is
 (a) $\frac{1}{2}$ (b) 1
 (c) 2 (d) 3
- 132.** $\frac{x}{x-1} + \frac{1}{x+1} + \frac{2x}{1-x^2} = ?$
 (a) 1 (b) 2
 (c) 3 (d) 4
- 133.** The value of $\frac{(a+b)^2}{(a^2-b^2)}$ is (R.R.B., 2006)
 (a) $\frac{ab}{a+b}$ (b) $\frac{2ab}{a-b}$
 (c) $\frac{a+b}{a-b}$ (d) None of these
- 134.** If $ab + bc + ca = 0$, then what is the value of $\left(\frac{1}{a^2-bc} + \frac{1}{b^2-ca} + \frac{1}{c^2-ab}\right)$?
 (a) 0 (b) 1
 (c) 3 (d) $a + b + c$
- 135.** How many boxes are required for filling 15 kg of sweet if each box is filled with 250 grams of sweet? (Bank Recruitment, 2010)
 (a) 30 (b) 70
 (c) 80 (d) 120
 (e) None of these
- 136.** The bus fare for one person is ₹ 420 from Agra to Aligarh and the train fare between the same places for one person is equal to three-fourths the bus fare for two persons between the same places. What is the total fare paid by 3 persons travelling by bus and 4 persons travelling by train between the two places? (Bank P.O., 2010)
 (a) ₹ 3360 (b) ₹ 3406
 (c) ₹ 3440 (d) ₹ 3460
 (e) None of these
- 137.** The cost of 6 pens and 3 pencils is ₹ 84. One-third of the cost of one pen is equal to the cost of one pencil. What is the total cost of 4 pens and 5 pencils? (Bank Recruitment, 2010)
 (a) ₹ 66 (b) ₹ 68
 (c) ₹ 72 (d) ₹ 78
 (e) None of these
- 138.** If an amount of ₹ 4,36,563 is distributed equally amongst 69 persons, how much amount would each person get? (Bank Recruitment, 2009)
 (a) ₹ 5876 (b) ₹ 5943
 (c) ₹ 6148 (d) ₹ 6327
 (e) None of these
- 139.** 32 shirt pieces of 120 cm each can be cut from a reel of cloth. After cutting these pieces 80 cm of cloth remains. What is the length of reel of cloth in metres? (Bank Recruitment, 2009)
 (a) 38.70 metres (b) 39.20 metres
 (c) 3870 metres (d) 3920 metres
 (e) None of these
- 140.** A canteen requires 798 bananas for a week. Total how many bananas did it require for the months of January, February and March 2008? (Bank Recruitment, 2008)
 (a) 10277 (b) 10374 (c) 10480
 (d) 10586 (e) None of these
- 141.** Ram has ₹ 6 more than Mohan and ₹ 9 more than Sohan. All the three have ₹ 33 in all. Ram has a share of (R.R.B., 2006)
 (a) ₹ 7 (b) ₹ 10
 (c) ₹ 13 (d) ₹ 16
- 142.** What is the maximum number of half-pint bottles of cream that can be filled with a 4-gallon can of cream? (2pt. = 1qt. and 4qt. = 1gal.) (Campus Recruitment, 2010)
 (a) 16 (b) 24
 (c) 30 (d) 64
- 143.** The sum of the weights of A and B is 80 kg. Half of the weight of A is equal to $\frac{5}{6}$ times the weight of B. Find the weight of B.
 (a) 20 kg (b) 30 kg
 (c) 40 kg (d) 60 kg
- 144.** a is greater than b by 2 and b is greater than c by 10. If $a + b + c = 130$, then $(b + c) - a = ?$
 (a) 34 (b) 38
 (c) 42 (d) 44
 (e) None of these
- 145.** The price of item X rises by ₹ 40 per year and that of item Y by ₹ 15 per year. If the price of item X and Y in the year 2002 was ₹ 420 and ₹ 630 respectively, in which year the price of item X will be ₹ 40 more than the price of item Y?
 (a) 2010 (b) 2011
 (c) 2012 (d) 2013
- 146.** How many pieces of 85 cm length can be cut from a rod 42.5 metres long?
 (a) 30 (b) 40
 (c) 60 (d) None of these
- 147.** On Sports Day, if 30 children were made to stand in a column, then 16 columns could be formed. If 24 children were made to stand in a column, then how many columns could be formed?
 (a) 20 (b) 22
 (c) 29 (d) 45

- 148.** The number of students in each section of a school is 24. After admitting new students, three new sections were started. Now, the total number of sections is 16 and there are 21 students in each section. The number of new students admitted is
 (a) 14 (b) 24
 (c) 48 (d) 114
- 149.** A man earns ₹ 20 on the first day and spends ₹ 15 on the next day. He again earns ₹ 20 on the third day and spends ₹ 15 on the fourth day. If he continues to save like this, how soon will he have ₹ 60 in hand? (M.A.T., 2006)
 (a) On 17th day (b) On 27th day
 (c) On 30th day (d) On 40th day
- 150.** A car company sold 150 cars in a special 6-day sale. Each day, the company sold 6 more than the previous day. How many cars were sold on the 6th day?
 (a) 35 (b) 40
 (c) 50 (d) 60
 (e) 70
- 151.** A group of 1200 persons consisting of captains and soldiers is travelling in a train. If for every 15 soldiers there is one captain, then the number of captains in the group is
 (a) 70 (b) 75
 (c) 80 (d) 82
- 152.** It costs ₹ x each to make the first thousand copies of a compact disc and ₹ y to make each subsequent copy. If z is greater than 1000, how much will it cost to make z copies of the compact disc?
 (a) $zx - zy$ (b) $1000x + yz$
 (c) $1000(x - y) + yz$ (d) $1000(z - y) + xz$
- 153.** The total monthly salary of 4 men and 2 women is ₹ 46,000. If a woman earns ₹ 500 more than a man, what is the monthly salary of a woman?
 (a) ₹ 6500 (b) ₹ 7500
 (c) ₹ 8000 (d) ₹ 9000
- 154.** A pineapple costs ₹ 7 each. A watermelon costs ₹ 5 each. X spends ₹ 38 on these fruits. The number of pineapples purchased is
 (a) 2 (b) 3
 (c) 4 (d) Data inadequate
- 155.** Water boils at 212°F or 100°C and melts at 32°F or 0°C. If the temperature of a particular day is 35°C, it is equivalent to
 (a) 85°F (b) 90°F
 (c) 95°F (d) 99°F
- 156.** 74 is divided into two parts so that 5 times one part and 11 times the other part are together equal to 454. The parts are (R.R.B., 2006)
 (a) 14, 60 (b) 60, 14
 (c) 30, 44 (d) 44, 30
- 157.** A sink contains exactly 12 litres of water. If water is drained from the sink until it holds exactly 6 litres of water less than the quantity drained away, then how many litres of water were drained away? (M.A.T., 2006)
 (a) 2 (b) 3
 (c) 6 (d) 9
- 158.** A family has several children. Each boy in the family has as many sisters as brothers and each girl has twice as many brothers as sisters.. How many brothers and sisters are there? (SNAP, 2005)
 (a) 4 brothers, 3 sisters (b) 4 brothers, 4 sisters
 (c) 3 brothers, 4 sisters (d) Cannot say
- 159.** If 1 Japanese Yen = 0.01 US Dollars, 100 US Dollars = 5000 Indian Rupees (INR), how many Japanese Yens are 100 INR? (JMET, 2008)
 (a) 20 (b) 200
 (c) 500 (d) 2000
- 160.** 12 buckets of water fill a tank when the capacity of each bucket is 13.5 litres. How many buckets will be needed to fill the same tank, if the capacity of each bucket is 9 litres?
 (a) 8 (b) 15
 (c) 16 (d) 18
- 161.** $\left[1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{3}}}\right] \div 1\frac{4}{7}$ is equal to (M.B.A., 2007)
 (a) 1 (b) $1\frac{1}{3}$
 (c) $1\frac{1}{4}$ (d) $1\frac{1}{7}$
- 162.** $\frac{13}{48}$ is equal to (C.P.O., 2006)
 (a) $\frac{1}{3 + \frac{1}{1 + \frac{1}{16}}}$ (b) $\frac{1}{2 + \frac{1}{1 + \frac{1}{8}}}$
 (c) $\frac{1}{3 + \frac{1}{1 + \frac{1}{1 + \frac{1}{8}}}}$ (d) $\frac{1}{3 + \frac{1}{1 + \frac{1}{2 + \frac{1}{4}}}}$
- 163.** Find the value of $\frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{1}{3 - \frac{1}{2}}}}} + \frac{3}{3 - \frac{4}{3 + \frac{1}{2 - \frac{1}{2}}}}$.
 (a) $\frac{13}{7}$ (b) $\frac{15}{7}$
 (c) $\frac{11}{21}$ (d) $\frac{17}{28}$

164. If $2 = x + \frac{1}{1 + \frac{1}{3 + \frac{1}{4}}}$, then the value of x is

- (a) $\frac{12}{17}$ (b) $\frac{13}{17}$
(c) $\frac{18}{17}$ (d) $\frac{21}{17}$

165. If $\frac{2 + \frac{1}{3 + \frac{4}{5}}}{2 + \frac{1}{3 + \frac{1}{1 + \frac{1}{4}}}} = x$, then the value of x is

- (a) $\frac{1}{7}$ (b) $\frac{3}{7}$
(c) 1 (d) $\frac{8}{7}$

166. $8 - 8 \times \frac{2\frac{1}{5} - 1\frac{2}{7}}{2 - \frac{1}{6 - \frac{1}{6}}}$ is equal to

- (a) 2 (b) 4
(c) 6 (d) 8

167. $\frac{2}{2 + \frac{2}{3 + \frac{2}{3 + \frac{2}{3}}}} \times 0.39$ is simplified to

- (a) $\frac{1}{3}$ (b) 2
(c) 6 (d) None of these

168. Simplify: $\frac{1}{1 + \frac{\frac{2}{3}}{1 + \frac{\frac{8}{9}}{1 + \frac{2}{3} + \frac{9}{1 - \frac{2}{3}}}}}$

- (a) $\frac{11}{13}$ (b) $\frac{13}{15}$
(c) $\frac{13}{11}$ (d) $\frac{15}{13}$

169. If $\frac{2}{2 + \frac{2}{2 + \frac{2}{x}}} = 3$, then x is equal to

- (a) $-\frac{4}{7}$ (b) $\frac{4}{7}$
(c) $-\frac{4}{9}$ (d) $\frac{4}{9}$

170. If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y, z are natural

numbers, then x, y, z are

- (a) 1, 2, 5 (b) 1, 5, 2
(c) 5, 2, 11 (d) 11, 2, 5

171. Let $x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots \infty}}}$. Which of the following

is correct?

- (a) $x^2 + x + 1 = 0$ (b) $x^2 - x + 1 = 0$
(c) $x^2 + x - 1 = 0$ (d) $x^2 - x - 1 = 0$

172. Let $P = \frac{1}{1 + \frac{1}{1 + \frac{1}{3 + \frac{1}{4 + \dots}}}}$ and $Q = \frac{1}{2 + \frac{1}{3 + \frac{1}{4 + \frac{1}{\dots}}}}$.

Then, $P + Q$ when calculated, gives

- (a) 1 (b) 2
(c) 3 (d) 4

173. $\frac{\left[\left(1 + \frac{1}{10 + \frac{1}{10}} \right) \left(1 + \frac{1}{10 + \frac{1}{10}} \right) - \left(1 - \frac{1}{10 + \frac{1}{10}} \right) \left(1 - \frac{1}{10 + \frac{1}{10}} \right) \right]}{\left[\left(1 + \frac{1}{10 + \frac{1}{10}} \right) + \left(1 - \frac{1}{10 + \frac{1}{10}} \right) \right]}$

simplifies to

- (a) $\frac{20}{101}$ (b) $\frac{90}{101}$
(c) $\frac{100}{101}$ (d) $\frac{101}{100}$

174. Which of the following values of x and y satisfy the following equations I and II?

- I. $3x + y = 19$ II. $x - y = 9$
(a) -7, -2 (b) -7, 2
(c) 7, -2 (d) 7, 2

175. If $4x = p(x + 3) + q(x - 1)$ is an identity, then the values of p and q are

(I.A.M., 2007)

- (a) 1, -3 (b) 1, 3
(c) 1, 1 (d) 3, 1

- 176.** If $x = y = 2z$ and $xyz = 256$, then $x = ?$
(Campus Recruitment, 2005)
(a) 2 (b) 4
(c) 8 (d) None of these
- 177.** If $3y + 9x = 54$ and $\frac{28x}{13y} = \frac{140}{39}$, then what is the value of $y - x$?
(Bank P.O., 2009)
(a) -2 (b) -1
(c) 1 (d) 2 (e) None of these
- 178.** If $3x + 7y = 75$ and $5x - 5y = 25$, then what is the value of $x + y$?
(Bank P.O., 2007)
(a) 14 (b) 15
(c) 16 (d) 17
(e) None of these
- 179.** If $a + b = 5$ and $3a + 2b = 20$, then $(3a + b)$ will be:
(a) 10 (b) 15
(c) 20 (d) 25
- 180.** If $2p + 3q = 18$ and $2p - q = 2$, then $2p + q = ?$
(a) 6 (b) 7
(c) 10 (d) 20
- 181.** If $2x + y = 5$ and $3x - 4y = 2$, then the value of $2xy$ is
(a) 4 (b) 6
(c) 8 (d) 10
- 182.** If $3x - 5y = 5$ and $\frac{x}{x+y} = \frac{5}{7}$, then what is the value of $2xy$ is
(a) 3 (b) 4
(c) 6 (d) 9
(e) None of these
- 183.** If $4x + 3y = 18xy$ and $2x - 5y + 4xy = 0$, then the values of x and y will be respectively
(a) $-\frac{1}{2}$ and $-\frac{1}{3}$ (b) -1 and -3
(c) $\frac{1}{2}$ and $\frac{1}{3}$ (d) $\frac{1}{4}$ and $\frac{1}{3}$
- 184.** If $2x + y = 17$; $y + 2z = 15$ and $x + y = 9$, then what is the value of $4x + 3y + z$?
(a) 41 (b) 43
(c) 45 (d) 55
(e) None of these
- 185.** If $3x - 4y + z = 7$; $2x - z + 3y = 19$; $x + 2y + 2z = 24$, then what is the value of z ?
(a) 4 (b) 5
(c) 6 (d) 8
- 186.** If $2x + y = 15$, $2y + z = 25$ and $2z + x = 26$, what is the value of z ?
(a) 4 (b) 7
(c) 9 (d) 11
- 187.** If $2x + 3y = 31$, $y - z = 4$ and $x + 2z = 11$, then what is the value of $x + y + z$?
(a) 12 (b) 13
(c) 15 (d) 16
- 188.** The price of 10 chairs is equal to that of 4 tables. The price of 15 chairs and 2 tables together is ₹ 4000. The total price of 12 chairs and 3 tables is
(a) ₹ 3500 (b) ₹ 3750
(c) ₹ 3840 (d) ₹ 3900
- 189.** If two jeans and three shirts cost ₹ 4000 and three jeans and two shirts cost ₹ 3500, how much does a jean cost?
(P.C.S., 2009)
(a) ₹ 500 (b) ₹ 1000
(c) ₹ 1500 (d) ₹ 2000
- 190.** Cost of 8 pens and 4 pencils is ₹ 176 and the cost of 2 pens and 2 pencils is ₹ 48. What is the cost of one pen?
(Bank P.O., 2009)
(a) ₹ 12 (b) ₹ 14
(c) ₹ 16 (d) ₹ 18
(e) None of these
- 191.** The cost of two dozen apples and three dozen bananas is ₹ 136. The cost of 5 dozen bananas and one dozen apples is ₹ 110. What is the price of one dozen bananas?
(Bank Recruitment, 2009)
(a) ₹ 16 (b) ₹ 18
(c) ₹ 20 (d) ₹ 24
(e) None of these
- 192.** The cost of one pencil, two pens and four erasers is ₹ 22 while the cost of five pencils, four pens and two erasers is ₹ 32. How much will three pencils, three pens and three erasers cost?
(Campus Recruitment, 2010)
(a) ₹ 21 (b) ₹ 24
(c) ₹ 27 (d) ₹ 30
- 193.** Ram Singh goes to Pushkar Mela with ₹ 10000 to buy exactly 100 animals. He finds that cows are sold at ₹ 1000, horses at ₹ 300 and chicken at ₹ 50. How many chicken should he buy to meet his target of 100 animals?
(SNAP, 2005)
(a) 88 (b) 90
(c) 92 (d) 94
- 194.** 3 men and 4 boys can earn ₹ 756 in 7 days. 11 men and 13 boys can earn ₹ 3008 in 8 days. In what time will 7 men with 9 boys earn ₹ 2480?
(P.C.S., 2006)
(a) 8 days (b) 9 days
(c) 10 days (d) 11 days
- 195.** At a certain fast food restaurant, Brian can buy 3 burgers, 7 shakes and one order of fries for ₹ 120 exactly. At the same place, it would cost ₹ 164.50 for 4 burgers, 10 shakes and one order of fries. How much would it cost for an ordinary meal of one burger, one shake and one fries?
(a) ₹ 21 (b) ₹ 31
(c) ₹ 41 (d) Cannot be determined

196. If $2x^2 + 12x + 18 = 0$, what is the value of x ?

(Bank P.O., 2006)

- (a) -3 (b) -2
(c) 2 (d) 3
(e) More than one answer

197. If $x^2 - 7x = -12$, what is the value of x ? (C.E.T., 2006)

- (a) -3 or -4 (b) 3 or 4
(c) 3 or -4 (d) Cannot be determined
(e) None of these

198. The roots of the equation $2x^2 - 11x + 15 = 0$ are

(Campus Recruitment, 2008)

- (a) $3, \frac{5}{2}$ (b) $5, \frac{3}{2}$
(c) $-3, -\frac{5}{2}$ (d) None of these

199. The value of $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{m}\right)$ is

(P.C.S., 2008)

- (a) 1 (b) $\frac{1}{m}$
(c) $\frac{1}{2m}$ (d) $\frac{1}{1 \cdot 2 \cdot 3 \cdots (m-1)m}$

200. The expression $\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right) \cdots \left(1 + \frac{1}{n}\right)$

simplifies to

(M.B.A. 2006, 2008)

- (a) $\frac{n+1}{3}$ (b) $\frac{n}{n+1}$
(c) $\frac{3}{n}$ (d) $1 + \frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{5} \cdots \frac{1}{n}$

201. When simplified, the product $\left(2 - \frac{1}{3}\right)\left(2 - \frac{3}{5}\right)$

$\left(2 - \frac{5}{7}\right) \cdots \left(2 - \frac{997}{999}\right)$ is equal to (R.R.B., 2007)

- (a) $\frac{5}{999}$ (b) $\frac{1001}{999}$
(c) $\frac{1}{1001}$ (d) $\frac{1001}{3}$

202. $\frac{3}{4}\left(1 + \frac{1}{3}\right)\left(1 + \frac{2}{3}\right)\left(1 - \frac{2}{5}\right)\left(1 + \frac{6}{7}\right)\left(1 - \frac{12}{13}\right) = ?$

- (a) $\frac{1}{5}$ (b) $\frac{1}{6}$
(c) $\frac{1}{7}$ (d) None of these

203. The value of $\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right)\left(1 - \frac{1}{5^2}\right) \cdots \left(1 - \frac{1}{11^2}\right)$

$\left(1 - \frac{1}{12^2}\right)$ is

(a) $\frac{11}{20}$ (b) $\frac{13}{15}$

(c) $\frac{13}{18}$ (d) $\frac{15}{16}$

(e) None of these

204. What is the value of the following expression?

$$\frac{1}{(2^2 - 1)} + \frac{1}{(4^2 - 1)} + \frac{1}{(6^2 - 1)} + \cdots + \frac{1}{(20^2 - 1)}$$

(a) $\frac{9}{19}$ (b) $\frac{10}{19}$

(c) $\frac{10}{21}$ (d) $\frac{11}{21}$

205. Find the sum : $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72}$

$+ \frac{1}{90} + \frac{1}{110} + \frac{1}{132}$.

(a) $\frac{7}{8}$ (b) $\frac{11}{12}$

(c) $\frac{15}{16}$ (d) $\frac{17}{18}$

206. The sum of the first 35 terms of the series

$$\frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \cdots$$

(a) $-\frac{1}{2}$ (b) $-\frac{1}{4}$

(c) $\frac{1}{4}$ (d) None of these

207. $\left(1\frac{1}{2} + 11\frac{1}{2} + 111\frac{1}{2} + 1111\frac{1}{2}\right)$ is equal to (S.S.C., 2010)

(a) 617 (b) 618

(c) $1234\frac{1}{2}$ (d) 1236

208. $\left(999\frac{999}{1000} \times 7\right)$ is equal to (C.P.O., 2007)

(a) $6633\frac{7}{1000}$ (b) $6993\frac{7}{1000}$

(c) $6999\frac{993}{1000}$ (d) $7000\frac{7}{1000}$

209. The value of $999\frac{995}{999} \times 999$ is

(a) 990809 (b) 998996

(c) 998999 (d) 999824

210. $\left(999\frac{1}{7} + 999\frac{2}{7} + 999\frac{3}{7} + 999\frac{4}{7} + 999\frac{5}{7} + 999\frac{6}{7}\right)$

is simplified to

(a) 2997 (b) 5979

(c) 5994 (d) 5997

211. The value of $998\frac{2}{17} + 998\frac{3}{17} + 998\frac{5}{17} + 998\frac{8}{17}$
 $+ 998\frac{16}{17}$ is (I.A.M., 2007)

- (a) 4990 (b) 4992
 (c) 9998 (d) 10000

212. The simplest value of $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots$
 $+ \frac{1}{9 \times 10}$ is (P.C.S., 2008)

- (a) $\frac{1}{10}$ (b) $\frac{9}{10}$
 (c) 1 (d) 10

213. $\left(\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \frac{1}{10.13} + \frac{1}{13.16}\right)$ is equal to (S.S.C., 2007)

- (a) $\frac{1}{3}$ (b) $\frac{3}{8}$
 (c) $\frac{5}{16}$ (d) $\frac{41}{7280}$

214. When simplified, the sum $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \dots + \frac{1}{n(n+1)}$ is equal to (S.S.C., 2006)

- (a) $\frac{1}{n}$ (b) $\frac{1}{n+1}$
 (c) $\frac{n}{n+1}$ (d) $\frac{2(n-1)}{n}$

215. The value of $1 + \frac{1}{4 \times 3} + \frac{1}{4 \times 3^2} + \frac{1}{4 \times 3^3}$ is

- (a) $\frac{121}{108}$ (b) $\frac{3}{2}$
 (c) $\frac{31}{2}$ (d) None of these

216. $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{4 \cdot 5 \cdot 6}$ is equal to

- (a) $\frac{7}{30}$ (b) $\frac{11}{30}$
 (c) $\frac{13}{30}$ (d) $\frac{17}{30}$

217. The value of $\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \frac{9}{4^2 \cdot 5^2} + \frac{11}{5^2 \cdot 6^2} + \frac{13}{6^2 \cdot 7^2} + \frac{15}{7^2 \cdot 8^2} + \frac{17}{8^2 \cdot 9^2} + \frac{19}{9^2 \cdot 10^2}$ is (S.S.C., 2004)

- (a) $\frac{1}{100}$ (b) $\frac{99}{100}$
 (c) 1 (d) $\frac{101}{100}$

218. The sum of the first 99 terms of the series $\frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \frac{9}{400} + \dots$

- (a) $\frac{99}{100}$ (b) $\frac{999}{1000}$
 (c) $\frac{9999}{10000}$ (d) 1

219. Mayank, Mirza, Little and Jaspal bought a motorbike for ₹ 60000. Mayank paid one-half of the sum of the amounts paid by the other boys, Mirza paid one-third of the sum of the amounts paid by the other boys and Little paid one-fourth of the sum of the amounts paid by the other boys. How much did Jaspal have to pay?

- (a) ₹ 13000 (b) ₹ 15000
 (c) 17000 (d) None of these

220. Income of a company doubles after every one year. If the initial income was ₹ 4 lakhs, what would be the income after 5 years? (Bank P.O., 2003)

- (a) ₹ 1.24 crores (b) ₹ 1.28 crores
 (c) ₹ 2.52 crores (d) ₹ 2.56 crores
 (e) None of these

221. Breeding of a certain species of insects is incredible everyday the total number of such insects in a closed glass jar is double the number on the previous day. There was just one insect in the jar on 1/2/2007, and the jar was full to the brim with these insects on 28/2/2007. On which date of February was the jar quarter-full? (P.C.S., 2008)

- (a) 7 (b) 14
 (c) 26 (d) None of these

222. The flowers in a basket double every minute and the basket gets full in one hour. In how much time, the basket was $\frac{1}{32}$ full?

- (a) 12 minutes (b) 32 minutes
 (c) 45 minutes (d) 55 minutes

223. A man's investment doubles in every 5 years. If he invested ₹ 5000 in each of the years 1990, 1995, 2000 and 2005, then what was the total amount received by him in 2010?

- (a) ₹ 30000 (b) ₹ 70000
 (c) ₹ 140000 (d) ₹ 150000

224. After measuring 120 metres of a rope, it was discovered that the measuring metre rod was 3 cm longer. The true length of the rope measured is (S.S.C., 2007)

- (a) 116 m 40 cm (b) 121 m 20 cm
 (c) 123 m (d) 123 m 60 cm

225. Smita was asked to multiply a certain number by 36. She multiplied it by 63 instead and got an answer 3834 more than the correct one. What was the number to be multiplied? (Bank P.O., 2008)

- (a) 126 (b) 142
(c) 148 (d) 152
(e) None of these
- 226.** A boy multiplied 423 by a number and obtained 65589 as his answer. If both the fives in the answer are wrong and all other figures are correct, the correct answer is:
(a) 60489 (b) 61189
(c) 62189 (d) 62389
- 227.** A candidate was asked to find $\frac{7}{8}$ of a positive number. He found $\frac{7}{18}$ of the same by mistake. If his answer was 770 less than the correct one, then the original given number was
(a) 1260 (b) 1548
(c) 6160 (d) None of these
(Campus Recruitment, 2010)
- 228.** A student was asked to divide a number by $\frac{17}{8}$. Instead, he actually multiplied it by $\frac{17}{8}$ and hence got 225 more than the expected answer. What was the expected answer?
(a) 64 (b) 126
(c) 136 (d) None of these
- 229.** One junior student is asked to divide half a number by 6 and the other half by 4 and then add the quantities. Instead of doing so, the student divides the given number by 5. If the answer is 4 short of the correct answer, then the number is
(a) 320 (b) 360
(c) 400 (d) 480 (M.B.A., 2008)
- 230.** If $3\frac{2}{3}$ is subtracted from $9\frac{1}{9}$ and the difference is multiplied by 450, what is the final answer?
(a) 2045 (b) 2250
(c) 2540 (d) Cannot be determined
(e) None of these
(Bank P.O., 2009)
- 231.** By how much is $\frac{3}{4}$ of 968 less than $\frac{7}{8}$ of 1008?
(a) 146 (b) 154
(c) 158 (d) 165
(e) None of these
(Bank P.O., 2010)
- 232.** If $x = y - \frac{y}{10}$, where y is a positive integer which increases in value, then x
(a) first increases in value then decreases
(b) decreases in value
(c) increases in value
(d) first decreases then increases in value
(P.C.S., 2009)
- 233.** The difference between $\frac{3}{4}$ th of $\frac{4}{5}$ th of a number and $\frac{1}{6}$ th of $\frac{2}{5}$ th of the same number is 648. What is the number?
(a) 1110 (b) 1215
(c) 1325 (d) 1440
(e) None of these
(Bank P.O., 2009)
- 234.** Which number gives the same result when added to $1\frac{1}{2}$ and when multiplied by $1\frac{1}{2}$?
(a) 1 (b) 3
(c) 5 (d) 7
- 235.** $\frac{5}{12}$ of which sum is equal to $3\frac{3}{4}$ of ₹ 100?
(a) ₹ 750 (b) ₹ 800
(c) ₹ 900 (d) ₹ 1000
- 236.** $\frac{3}{8}$ is what part of $\frac{1}{12}$?
(a) $\frac{3}{7}$ (b) $\frac{1}{12}$
(c) $\frac{4}{3}$ (d) None of these
- 237.** The smallest fraction which should be subtracted from the sum of $1\frac{3}{4}$, $2\frac{1}{2}$, $5\frac{7}{12}$, $3\frac{1}{3}$ and $2\frac{1}{4}$ to make the result a whole number is
(a) $\frac{5}{12}$ (b) $\frac{7}{12}$
(c) $\frac{1}{2}$ (d) 7
- 238.** What fraction should be added to the sum of $5\frac{3}{4}$, $4\frac{4}{5}$ and $7\frac{3}{8}$ to make the result a whole number?
(a) $\frac{1}{40}$ (b) $\frac{2}{40}$
(c) $\frac{3}{40}$ (d) $\frac{4}{40}$
(M.B.A., 2006)
- 239.** If x is a positive number, then which of the following fractions has the greatest value?
(a) $\frac{x}{x}$ (b) $\frac{x}{x+1}$
(c) $\frac{x+1}{x}$ (d) $\frac{x+2}{x+3}$

240. By how much does $\frac{6}{7/8}$ exceed $\frac{6/7}{8}$?

- (a) $6\frac{1}{8}$ (b) $6\frac{3}{4}$
(c) $7\frac{3}{4}$ (d) $7\frac{5}{6}$

241. If $\frac{4}{5}$ of an estate be worth ₹ 16,800, then the value of $\frac{3}{7}$ of the estate is

- (a) ₹ 9000 (b) ₹ 21,000
(c) ₹ 72,000 (d) ₹ 90,000

242. Two-fifth of one-fourth of three-seventh of a number is 15. What is half of that number?

- (a) 94 (b) 96
(c) 188 (d) 196
(e) None of these

243. What fraction of an hour is a second?

- (a) $\frac{1}{24}$ (b) $\frac{1}{60}$
(c) $\frac{1}{120}$ (d) $\frac{1}{3600}$

244. When a ball bounces, it rises to $\frac{3}{4}$ of the height from which it fell. If the ball is dropped from a height of 32 m, how high will it rise at the third bounce?

- (a) 13 m (b) $13\frac{1}{2}$ m
(c) $14\frac{1}{2}$ m (d) None of these

245. Sanket earns twice as much in the month of March as in each of the other months of the year. What part of his entire annual earnings was earned in March?

- (a) $\frac{1}{7}$ (b) $\frac{1}{6}$
(c) $\frac{2}{11}$ (d) $\frac{2}{13}$

246. If one-third of a tank holds 80 litres of water, then the quantity of water that half of the tank holds is

- (a) $\frac{80}{3}$ litres (b) 100 litres
(c) 120 litres (d) 240 litres

247. A person travels 3.5 km from place A to place B. Out of this distance, he travels $1\frac{2}{3}$ km on bicycle,

$1\frac{1}{6}$ km on scooter and the rest on foot. What portion of the whole distance does he cover on foot?

- (a) $\frac{3}{19}$ (b) $\frac{4}{11}$
(c) $\frac{4}{21}$ (d) $\frac{5}{6}$

248. What fraction of $\frac{4}{7}$ must be added to itself to make the sum $1\frac{1}{14}$?

- (a) $\frac{1}{2}$ (b) $\frac{4}{7}$
(c) $\frac{7}{8}$ (d) $\frac{15}{14}$

249. Express $\frac{2}{3}$ of $\frac{1}{4}$ of ₹ 25.20 as a fraction of $1\frac{1}{2}$ of ₹ 36.

- (a) $\frac{5}{8}$ (b) $\frac{5}{42}$
(c) $\frac{7}{90}$ (d) $\frac{11}{90}$

250. A 70 cm long wire is to be cut into two pieces such that one piece will be $\frac{2}{5}$ as long as the other. How many centimetres will the shorter piece be?

- (a) 10 (b) 14
(c) 20 (d) 28

251. A certain amount is distributed among A, B and C. A gets $\frac{3}{16}$ and B gets $\frac{1}{4}$ of the whole amount. If C gets ₹ 81, then B gets

- (a) ₹ 30 (b) ₹ 32
(c) ₹ 36 (d) ₹ 40

252. $\frac{1}{10}$ of a pole is coloured red, $\frac{1}{20}$ white, $\frac{1}{30}$ blue, $\frac{1}{40}$ black, $\frac{1}{50}$ violet, $\frac{1}{60}$ yellow and the rest is green. If the length of the green portion of the pole is 12.08 metres, then the length of the pole is

(S.S.C., 2004)

- (a) 16 m (b) 18 m
(c) 20 m (d) 30 m

253. If we multiply a fraction by itself and divide the product by its reciprocal, the fraction thus obtained is $18\frac{26}{27}$. The original fraction is

- (a) $\frac{8}{27}$ (b) $1\frac{1}{3}$
(c) $2\frac{2}{3}$ (d) None of these

- 254.** The marks scored in an examination are converted from 50 to 10 for the purpose of internal assessment. The highest marks were 47 and the lowest were 14. The difference between the maximum and the minimum internal assessment scores is
 (a) 3.3 (b) 4.8
 (c) 6.6 (d) 7.4
- 255.** One-third of Rahul's savings in National Savings Certificate is equal to one-half of his savings in Public Provident Fund. If he has ₹ 1,50,000 as total savings, how much has he saved in Public Provident Fund?
 (a) ₹ 30,000 (b) ₹ 50,000
 (c) ₹ 60,000 (d) ₹ 90,000
- 256.** In a family, the father took $\frac{1}{4}$ of the cake and he had 3 times as much as each of the other members had. The total number of family members is
 (a) 3 (b) 7
 (c) 10 (d) 12
- 257.** A waiter's salary consists of his salary and tips. During one week his tips were $\frac{5}{4}$ of his salary. What fraction of his income came from tips?
 (a) $\frac{4}{9}$ (b) $\frac{5}{4}$
 (c) $\frac{5}{8}$ (d) $\frac{5}{9}$
- 258.** A sum of ₹ 1360 has been divided among A, B and C such that A gets $\frac{2}{3}$ of what B gets and B gets $\frac{1}{4}$ of what C gets. B's share is (M.A.T., 2002)
 (a) ₹ 120 (b) ₹ 160
 (c) ₹ 240 (d) ₹ 300
- 259.** Three friends had dinner at a restaurant. When the bill was received, Amita paid $\frac{2}{3}$ as much as Veena paid and Veena paid $\frac{1}{2}$ as much as Tanya paid. What fraction of the bill did Veena pay? (SNAP, 2005)
 (a) $\frac{1}{3}$ (b) $\frac{3}{11}$
 (c) $\frac{12}{31}$ (d) $\frac{5}{8}$
- 260.** $\frac{1}{4}$ of a tank holds 135 litres of water. What part of the tank is full if it contains 180 litres of water?
 (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
 (c) $\frac{2}{3}$ (d) $\frac{2}{5}$
- 261.** A tank is $\frac{2}{5}$ full. If 16 litres of water is added to the tank, it becomes $\frac{6}{7}$ full. The capacity of the tank is
 (a) 28 litres (b) 32 litres
 (c) 35 litres (d) 42 litres
- 262.** A drum of kerosene is $\frac{3}{4}$ full. When 30 litres of kerosene is drawn from it, it remains $\frac{7}{12}$ full. The capacity of the drum is (S.S.C., 2010)
 (a) 120 litres (b) 135 litres
 (c) 150 litres (d) 180 litres
- 263.** A tin of oil was $\frac{5}{8}$ full. When 10 bottles of oil was taken out and 8 bottles of oil was poured into it, it was $\frac{3}{5}$ full. How many bottles of oil can the tin contain?
 (a) 20 (b) 30
 (c) 40 (d) 80
- 264.** 1 m of an object is to be represented by 10 cm on the drawing scale. The representative fraction (R.F.) of the scale is
 (a) $\frac{1}{10}$ (b) $\frac{1}{20}$
 (c) $\frac{1}{100}$ (d) None of these
- 265.** An area of 100 sq. cm on the map represents an area of 49 sq. km on the field. The R.F. of the scale to be used is
 (a) $\frac{1}{10000}$ (b) $\frac{1}{20000}$
 (c) $\frac{1}{49000}$ (d) $\frac{1}{70000}$
- 266.** The fluid contained in a bucket can fill four large bottles or seven small bottles. A full large bottle is used to fill an empty small bottle. What fraction of the fluid is left over in the large bottle when the small one is full?
 (a) $\frac{2}{7}$ (b) $\frac{3}{7}$
 (c) $\frac{4}{7}$ (d) $\frac{5}{7}$
- 267.** To fill a tank, 25 buckets of water is required. How many buckets of water will be required to fill the same tank if the capacity of the bucket is reduced to two-fifth of its present?
 (a) 10 (b) 35
 (c) $62\frac{1}{2}$ (d) Cannot be determined
 (e) None of these

- 268.** A stationary engine has enough fuel to run 12 hours when its tank is $\frac{4}{5}$ full. How long will it run when the tank is $\frac{1}{3}$ full? (Campus Recruitment, 2009)
- (a) Less than 2 hours (b) 2 hours
(c) 3 hours (d) 4 hours
(e) 5 hours
- 269.** A tree grows only $\frac{3}{5}$ as fast as the one beside it. In four years the combined growth of the trees is eight feet. How much does the shorter tree grow in 2 years? (Campus Recruitment, 2008)
- (a) Less than 2 feet (b) 2 feet
(c) $2\frac{1}{2}$ feet (d) 3 feet
(e) More than 3 feet
- 270.** A car is filled with four and a half gallons of fuel for a round trip. If the amount of fuel taken while going is $\frac{1}{4}$ more than the amount taken for coming, what is the amount of fuel consumed while coming back? (Campus Recruitment, 2009)
- (a) Less than 2 gallons (b) 2 gallons
(c) $2\frac{1}{2}$ gallons (d) 3 gallons
(e) More than 3 gallons
- 271.** The lowest temperature in the night in a city is one-third more than $\frac{1}{2}$ the highest during the day. Sum of the lowest temperature and the highest temperature is 100 degrees. Then what is the lowest temperature? (Campus Recruitment, 2010)
- (a) 30 degrees (b) 40 degrees
(c) 36 degrees (d) None of these
- 272.** In a class there are 18 boys who are over 160 cm tall. If these boys constitute three-fourths of the boys and the total number of boys is two-thirds of the number of students in the class, then what is the number of girls in the class?
- (a) 6 (b) 12
(c) 18 (d) 24
- 273.** Peter gave one-fourth of the amount he had to Michael. Michael in turn gave half of what he received from Peter to Sam. If the difference between the remaining amount with Peter and the amount received by Sam is ₹ 500, how much money did Michael receive from Peter?
- (a) ₹ 100 (b) ₹ 200
(c) ₹ 400 (d) Data inadequate
(e) None of these
- 274.** Four children A, B, C and D divide a bag of sweets. A takes $\frac{1}{3}$ of them, B $\frac{2}{5}$ th of the remainder and the rest is equally shared between C and D. What fraction of the sweets did C or D get?
- (a) $\frac{1}{4}$ (b) $\frac{1}{5}$
(c) $\frac{1}{6}$ (d) $\frac{1}{17}$
- 275.** Rita travelled 1200 km by air, which formed $\frac{2}{5}$ th of her journey. One-third of the whole journey she travelled by car and the remaining journey by train. Find the distance travelled by the train. (R.R.B., 2006)
- (a) 480 km (b) 800 km
(c) 1600 km (d) 1800 km
- 276.** A man spends one-quarter of his income on food, one-fifth of it on rent and remaining, which is ₹ 231, on other commodities. Find his total income.
- (a) ₹ 400 (b) ₹ 410
(c) ₹ 420 (d) ₹ 460
- 277.** A boy read $\frac{3}{8}$ th of a book on one day and $\frac{4}{5}$ th of the remainder on another day. If there were 30 pages unread, how many pages did the book contain? (I.M.T. 2002; XAT, 2006)
- (a) 240 (b) 300
(c) 600 (d) None of these
- 278.** An institute organised a fete and $\frac{1}{5}$ of the girls and $\frac{1}{8}$ of the boys participated in the same. What fraction of the total number of students took part in the fete?
- (a) $\frac{2}{13}$ (b) $\frac{13}{40}$
(c) Data inadequate (d) None of these
- 279.** At an International Dinner, $\frac{1}{5}$ of the people attending were French men. If the number of French women at the dinner was $\frac{2}{3}$ greater than the number of French men, and there were no other French people at the dinner, then what fraction of the people at the dinner were not French? (M.B.A., 2003)
- (a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{2}{3}$ (d) $\frac{7}{15}$

- 280.** In a class, $\frac{3}{5}$ of the students are girls and rest are boys. If $\frac{2}{9}$ of the girls and $\frac{1}{4}$ of the boys are absent, what part of the total number of students is present?
 (a) $\frac{17}{25}$ (b) $\frac{18}{49}$
 (c) $\frac{23}{30}$ (d) $\frac{23}{36}$
- 281.** One-third of the boys and one-half of the girls of a college participated in a social work project. If the number of participating students is 300 out of which 100 are boys, what is the total number of students in the college?
 (a) 500 (b) 600
 (c) 700 (d) 800
- 282.** To win an election, a candidate needs $\frac{3}{4}$ of the votes cast. If after $\frac{2}{3}$ of the votes have been counted, a candidate has $\frac{5}{6}$ of what he needs, then what part of the remaining votes does he still need?
 (M.A.T., 2008)
 (a) $\frac{1}{8}$ (b) $\frac{3}{8}$
 (c) $\frac{1}{10}$ (d) $\frac{1}{4}$
- 283.** A fires 5 shots to B's 3 but A kills only once in 3 shots while B kills once in 2 shots. When B has missed 27 times, A has killed
 (a) 30 birds (b) 60 birds
 (c) 72 birds (d) 90 birds
- 284.** If every 2 out of 3 readymade shirts need alterations in the collar, every 3 out of 4 need alterations in the sleeves, and every 4 out of 5 need it in the body, how many alterations will be required for 60 shirts?
 (a) 24 (b) 123
 (c) 133 (d) 143
- 285.** The sum of three fractions is $2\frac{11}{24}$. When the largest fraction is divided by the smallest, the fraction thus obtained is $\frac{7}{6}$ which is $\frac{1}{3}$ more than the middle one. The fractions are
 (a) $\frac{3}{5}, \frac{4}{7}, \frac{2}{3}$ (b) $\frac{7}{8}, \frac{5}{6}, \frac{3}{4}$
 (c) $\frac{7}{9}, \frac{2}{3}, \frac{3}{5}$ (d) None of these
- 286.** A millionaire bought a lot of hats $\frac{1}{4}$ of which were brown. The millionaire sold $\frac{2}{3}$ of the hats including $\frac{4}{5}$ of the brown hats. What fraction of the unsold hats were brown?
 (Campus Recruitment, 2010)
 (a) $\frac{1}{60}$ (b) $\frac{1}{15}$
 (c) $\frac{3}{20}$ (d) $\frac{3}{5}$
 (e) $\frac{3}{4}$
- 287.** Equal amounts of water were poured into two empty jars of different capacities, which made one jar $\frac{1}{4}$ full and the other jar $\frac{1}{3}$ full. If the water in the jar with lesser capacity is, then poured into the jar with the greater capacity, what fraction of the larger jar will be filled with water?
 (M.A.T., 2005)
 (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
 (c) $\frac{3}{4}$ (d) $\frac{2}{7}$
- 288.** Mr. X spends $\frac{1}{4}$ of his income on food and $\frac{1}{3}$ less than what he spent on food, on the education of his children. What fraction of his income did he spend on food and education?
 (a) $\frac{2}{7}$ (b) $\frac{1}{2}$
 (c) $\frac{5}{12}$ (d) $\frac{7}{12}$
- 289.** A body of 7300 troops is formed of 4 battalions so that $\frac{1}{2}$ of the first, $\frac{2}{3}$ of the second, $\frac{3}{4}$ of the third and $\frac{4}{5}$ of the fourth are all composed of the same number of men. How many men are there in the second battalion?
 (M.A.T., 2008)
 (a) 1500 (b) 1600
 (c) 1800 (d) 2400
- 290.** After reading $\frac{3}{5}$ of the biology homework on Monday night, Sanjay read $\frac{1}{3}$ of the remaining homework on Tuesday night. What fraction of the original homework would Sanjay have to read on Wednesday night to complete the assignment?

- (a) $\frac{1}{5}$ (b) $\frac{1}{15}$
 (c) $\frac{2}{15}$ (d) $\frac{4}{15}$

291. The highest score in an inning was $\frac{3}{11}$ of the total and the next highest was $\frac{3}{11}$ of the remainder. If

the scores differed by 9, the total score was

- (a) 110 (b) 121
 (c) 132 (d) 143

292. Mechanics are paid twice the hourly wages of sales people. Custodial workers are paid one-third the hourly wages of mechanics. What fraction of the hourly wages of custodial workers are sales people paid? (M.B.A., 2006)

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
 (c) $\frac{2}{3}$ (d) $\frac{3}{2}$
 (e) $\frac{4}{3}$

293. Ganeshi's monthly income is twice Jassi's monthly income. Two-third of Jassi's monthly income is equal to Sukhvinder's monthly income. If Sukhvinder's annual income is ₹ 2.34 lakhs, what is Ganeshi's monthly income? (Bank Recruitment, 2010)

- (a) ₹ 14625 (b) ₹ 29250
 (c) ₹ 28230 (d) ₹ 58500
 (e) None of these

294. The cost of 36 kg of rice last year was ₹ 1044 and the cost of 24 kg of rice this year is ₹ 768. What is the difference between the cost per kg of rice last year and the cost per kg of rice this year? (Bank Recruitment, 2008)

- (a) ₹ 3 (b) ₹ 4
 (c) ₹ 5 (d) ₹ 6
 (e) None of these

295. The cost of 12 belts and 30 wallets is ₹ 8940. What is the cost of 4 belts and 10 wallets? (Bank Recruitment, 2009)

- (a) ₹ 2780 (b) ₹ 2870
 (c) ₹ 2890 (d) ₹ 2980
 (e) None of these

296. The cost of 5 pendants and 8 chains is ₹ 1,45,785. What would be the cost of 15 pendants and 24 chains? (Bank P.O., 2009)

- (a) ₹ 3,25,285 (b) ₹ 4,39,355
 (c) ₹ 5,50,000 (d) Cannot be determined
 (e) None of these

297. The cost of 21 tables and 35 chairs is ₹ 41825. What is the cost of 9 tables and 15 chairs?

- (a) ₹ 17775 (b) ₹ 17925
 (c) ₹ 18075 (d) ₹ 18725
 (e) None of these

298. The cost of 13 kg of sugar is ₹ 195, the cost of 17 kg of rice is ₹ 544 and the cost of 21 kg of wheat is ₹ 336. What is the total cost of 21 kg of sugar, 26 kg of rice and 19 kg of wheat? (M.A.T., 2009)

- (a) ₹ 1306 (b) ₹ 1451
 (c) ₹ 1500 (d) ₹ 1636

299. The most economical price among the following is (Campus Recruitment, 2008)

- (a) 10 kilo for ₹ 160 (b) 2 kilo for ₹ 30
 (c) 4 kilo for ₹ 70 (d) 20 kilo for ₹ 340
 (e) 8 kilo for ₹ 130

300. There are 200 questions on a 3 hour examination. Among these questions are 50 Mathematics problems. It is suggested that twice as much time be spent on each Maths problem as for each other question. How many minutes should be spent on Mathematics problems? (Campus Recruitment, 2010)

- (a) 36 (b) 60
 (c) 72 (d) 100

301. What part of an hour elapses from 4.56 P.M. to 5.32 P.M.? (R.R.B., 2006)

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
 (c) $\frac{3}{5}$ (d) $\frac{3}{4}$

302. 72 hours 6 minutes $\div$ 14 =?

- (a) 59 minutes (b) 5 hours 9 minutes
 (c) 6 hours 9 minutes (d) 7 hours 8 minutes

303. A class starts at 10 a.m. and lasts till 1.27 p.m. Four periods are held during this interval. After every period, 5 minutes are given free to the students. The exact duration of each period is

- (a) 42 minutes (b) 48 minutes
 (c) 51 minutes (d) 53 minutes

304. A light was seen at intervals of 13 seconds. It was seen for the first time at 1 hr. 54 min. 50 secs. a.m. and the last time at 3 hrs. 17 min. 49 secs. a.m. How many times was the light seen? (A.A.O. Exam., 2003)

- (a) 360 (b) 375
 (c) 378 (d) 384

305. A woman works continuously for 7 days in the kitchen and refuses to work on the 8th day when her husband takes over. If she starts her work on a Sunday, then the 10th time she rests will be on which day of the week?

- (a) Sunday (b) Monday
 (c) Tuesday (d) Friday

306. If $x * y = x^2 + y^2 - xy$, then the value of $9 * 11$ is

- (a) 93 (b) 103
(c) 113 (d) 121

307. If $a * b = \frac{ab}{a+b}$, find the value of $3 * (3 * -1)$.

- (a) -3 (b) -1.5
(c) -1 (d) $\frac{2}{3}$

308. If $a * b = 2a - 3b + ab$, then $3 * 5 + 5 * 3$ is equal to

- (a) 22 (b) 24
(c) 26 (d) 28

309. If $x \oplus y = x^2 + 2y$, what is the value of p if $4 \oplus (3 \oplus p) = 50$?

- (a) 4 (b) 7
(c) 8 (d) 12.5

310. If $a * b * c$ means $\frac{a+b}{c}$ for all numbers except 0,

then $(a * b * c) * a * b$ is equal to

- (a) 0 (b) 1
(c) $\frac{a+b+c}{ab}$ (d) $\frac{a+b+ac}{bc}$
(e) $\frac{ab+bc+ca}{a+b+c}$

Directions (Questions 311 to 313): Each question below consists of a number. Your task is to follow the following steps in order to find the result from the given alternatives. (M.C.A., 2005)

Step I: Multiply the number by 3 and add the square of the number.

Step II: Divide the result obtained after Step I by 5 and add 6.

Step III: Divide the result obtained after Step II by 3 and then square the resulting number.

311. 6 (a) 13 (b) 30

(c) 31.03 (d) 31.36

312. 2.5 (a) 8.10 (b) 8.51 (approx.)
(c) 18.05 (d) 80.50 (approx.)

313. 5 (a) 2.80 (b) 12.80
(c) 20.08 (approx.) (d) 21.80 (approx.)

314. If the operation $\wedge$ is defined by the equation $x \wedge y = 2x + y$, what is the value of a in $2 \wedge a = a \wedge 3$?

(Campus Recruitment, 2009)

- (a) 0 (b) -1
(c) 1 (d) 4

315. If $\oplus$ is an operation such that $a + b = \begin{cases} 2a, & \text{when } a > b \\ a + b, & \text{when } a < b \\ a^2, & \text{when } a = b \end{cases}$

then $\left[\frac{(5 \oplus 7) + (4 \oplus 4)}{3(5 \oplus 5) - (15 \oplus 11) - 3} \right]$ is equal to (C.P.O., 2007)

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$

(c) $\frac{14}{13}$ (d) $\frac{14}{23}$

316. At Srinagar, starting at 9 a.m. on a certain day, snow began to fall at a rate of $1\frac{1}{4}$ inches every two hours until 3 p.m. If there was already $2\frac{1}{4}$ inches of snow on the ground at 9 a.m., how many inches of snow was on the ground at 3 p.m. that day?

(a) $3\frac{3}{4}$ (b) 6

(c) 7 (d) $7\frac{1}{2}$

317. A squirrel starts climbing up a tree at the speed of 6 metres a minute but after every 6 metres it slips down 4 metres in the next minute. It will be able to reach the top 120 metres high in (R.R.B., 2006)

(a) 20 minutes (b) 60 minutes

(c) 115 minutes (d) $1\frac{1}{2}$ hours

318. Ram had ₹ 1000 in his Savings Bank Account. Every month in the first week he needs money, so he withdraws ₹ 500, but by the end of the month, he deposits ₹ 750. After how many months the original amount will grow three times? (R.R.B., 2005)

(a) 6 months (b) 7 months

(c) 8 months (d) 9 months

319. The growth of a medicinal plant of height 1 metre, is 1 cm/day and it is growing straight vertically upward. A man cuts 6 cm from the top, at a regular interval of every 11 days. Its height will be 2 metres after (I.A.M., 2007)

(a) 220 days (b) 250 days

(c) 275 days (d) None of these

320. A stairway 10 ft. high is such that each step accounts for half a foot upward and one foot forward. What distance will an ant travel if it starts from ground level to reach the top of the stairway? (R.R.B., 2009)

(a) 30 ft. (b) 33 ft.

(c) 10 ft. (d) 29 ft.

321. The sum of all proper fractions whose denominators are less than or equal to 100, is

(a) $22\frac{1}{4}$ (b) 1925

(c) 2475 (d) None of these

322. The value of $\frac{1-x^4}{1+x} \div \frac{1+x^2}{x} \times \frac{1}{x(1-x)}$ is

(a) 1 (b) $1 - x^2$

(c) $\frac{1}{x}$ (d) $1 + x$

(e) None of these

323. $\frac{17}{15} \times \frac{17}{15} + \frac{2}{15} \times \frac{2}{15} - \frac{17}{15} \times \frac{4}{15}$ is equal to

- (a) 0 (b) 1
(c) 10 (d) 11

324. $2 \times 13\frac{3}{4} \times 5\frac{1}{4} + 5\frac{1}{4} \times 5\frac{1}{4} + 13\frac{3}{4} \times 13\frac{3}{4} = ?$

- (a) 311 (b) 316
(c) 361 (d) 380

325. The simplification of $\left(\frac{75983 \times 75983 - 45983 \times 45983}{30000} \right)$

yields the result

- (a) 121796 (b) 121866
(c) 121956 (d) 121966

326. Simplify: $\frac{13\frac{4}{7} \times 13\frac{4}{7} + 8\frac{3}{7} \times 8\frac{3}{7}}{13\frac{4}{7} + 8\frac{3}{7}}$ (R.R.B., 2006)

- (a) 18 (b) 20
(c) 22 (d) 24

327. $\frac{(a-b)^2 - (a+b)^2}{-4a} = \frac{x}{y}$. On simplifying the given

equation, which of the following equations will be obtained? (Bank P.O., 2009)

- (a) $xy = b$ (b) $bx = y$
(c) $ab = x$ (d) $yb = x$
(e) $ay = x$

328. What is $\frac{\frac{7}{8} \times \frac{7}{8} + \frac{5}{6} \times \frac{5}{6} + \frac{7}{8} \times \frac{5}{6}}{\frac{7}{8} \times \frac{7}{8} - \frac{5}{6} \times \frac{5}{6}}$ equal to? (S.S.C., 2007)

- (a) $\frac{41}{24}$ (b) $\frac{1}{24}$
(c) 41 (d) None of these

329. The simplified value of $1998^2 - 1997^2 + 1996^2 - 1995^2 + 1994^2 - 1993^2$ is

- (a) 11953 (b) 11958
(c) 11963 (d) 11973

330. $\frac{(856 + 167)^2 + (856 - 167)^2}{856 \times 856 + 167 \times 167} = ?$

- (a) 1 (b) 2
(c) 689 (d) 1023

331. $\frac{(469 + 174)^2 - (469 - 174)^2}{469 \times 174} = ?$ (M.B.A., 2002)

- (a) 2 (b) 4
(c) 295 (d) 643

332. If $a - b = 3$ and $a^2 + b^2 = 29$, find the value of ab . (R.R.B., 2003)

- (a) 10 (b) 12
(c) 15 (d) 18

333. If $\frac{x^2 - 1}{x + 1} = 4$, $x = ?$

- (a) 0 (b) 1
(c) 5 (d) Cannot be determined
(e) None of these

334. $\frac{\left(3\frac{2}{3}\right)^2 - \left(2\frac{1}{2}\right)^2}{\left(4\frac{3}{4}\right)^2 - \left(3\frac{1}{3}\right)^2} \div \frac{3\frac{2}{3} - 2\frac{1}{2}}{4\frac{3}{4} - 3\frac{1}{3}} = ?$

- (a) $\frac{37}{97}$ (b) $\frac{74}{97}$
(c) $1\frac{23}{74}$ (d) None of these

335. The simplified value of

$$\frac{\left(1 + \frac{1}{1 + \frac{1}{100}}\right) \left(1 + \frac{1}{1 + \frac{1}{100}}\right) - \left(1 - \frac{1}{1 + \frac{1}{100}}\right) \left(1 - \frac{1}{1 + \frac{1}{100}}\right)}{\left(1 + \frac{1}{1 + \frac{1}{100}}\right) + \left(1 - \frac{1}{1 + \frac{1}{100}}\right)}$$
 is

- (a) 100 (b) $\frac{200}{101}$
(c) 200 (d) $\frac{202}{100}$

336. The value of $\frac{769435 \times 770001 - 566}{769435 + 770001 \times 769434}$ is

- (a) -1 (b) -2
(c) 1 (d) 2

337. If $a - b = 1$, then the value of $a^3 - b^3 - 3ab$ will be

- (a) -3 (b) -1
(c) 1 (d) 3

338. If $x - y = 1$ and $x^2 + y^2 = 41$, then the value of $x + y$ will be

- (a) 5 or 4 (b) -5 or -4
(c) ± 9 (d) ± 1

339. If $x^4 + y^4 = 17$ and $x + y = 1$, then what is the value of $x^2y^2 - 2xy$?

- (a) 8 (b) 10
(c) 12 (d) 16

340. If $x = a + m$, $y = b + m$, $z = c + m$, the value of $\frac{x^2 + y^2 + z^2 - yz - zx - xy}{a^2 + b^2 + c^2 - ab - bc - ca}$ is

- (a) 1 (b) $\frac{x + y + z}{a + b + c}$
(c) $\frac{a + b + c}{x + y + z}$ (d) not possible to find

341. If $a + b + c = 13$, $a^2 + b^2 + c^2 = 69$, then find $ab + bc + ca$.

- (a) -50 (b) 50
(c) 69 (d) 75

342. If $\frac{x^2 + y^2 + z^2 - 64}{xy - yz - zx} = -2$ and $x + y = 3z$, then the value of z is

- (a) 2 (b) 3
(c) 4 (d) None of these

343. $\left(\frac{785 \times 785 \times 785 + 435 \times 435 \times 435}{785 \times 785 + 435 \times 435 - 785 \times 435} \right)$ simplifies to

- (a) 350 (b) 785
(c) 1220 (d) 1320

344. $\left(\frac{147 \times 147 + 147 \times 143 + 143 \times 143}{147 \times 147 \times 147 - 143 \times 143 \times 143} \right) = ?$

- (a) $\frac{1}{4}$ (b) 290
(c) $\frac{1}{290}$ (d) 4

345. $\frac{(13)^3 + 7^3}{(13)^2 + 7^2 - ?} = 20$

- (a) 6 (b) 20
(c) 91 (d) None of these

346. The value of $\left(\frac{\left(\frac{3}{5} \right)^3 - \left(\frac{2}{5} \right)^3}{\left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2} \right)$ is (S.S.C., 2003)

- (a) $\frac{1}{5}$ (b) $\frac{19}{25}$
(c) $\frac{21}{25}$ (d) 1

347. $\frac{38 \times 38 \times 38 + 34 \times 34 \times 34 + 28 \times 28 \times 28 - 38 \times 34 \times 84}{38 \times 38 + 34 \times 34 + 28 \times 28 - 38 \times 34 - 34 \times 28 - 38 \times 28}$

is equal to

- (a) 24 (b) 32
(c) 44 (d) 100

348. The value of $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{9(x-y)(y-z)(z-x)}$ is equal to

- (a) 0 (b) $\frac{1}{9}$
(c) $\frac{1}{3}$ (d) 1

349. If $x = 5$, $y = 3$, the value of $\frac{x^3 - y^3}{x^2 - y^2} - \frac{3xy}{x + y}$ will be

(P.C.S., 2009)

- (a) 1 (b) $\frac{1}{2}$
(c) $\frac{3}{2}$ (d) $\frac{5}{2}$

350. If $a + b + c = 11$ and $ab + bc + ca = 20$, then the value of the expression $a^3 + b^3 + c^3 - 3abc$ will be

- (a) 121 (b) 341
(c) 671 (d) 781

351. The value of $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{(x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3}$ is

- (a) 0
(b) 1
(c) $[2(x+y+z)]^{-1}$
(d) $[(x+y)(y+z)(z+x)]^{-1}$

352. If $a + b + c = 0$, the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$ is

- (a) $3abc$ (b) $\frac{1}{3}$
(c) 1 (d) 3

353. If $a = 29$, $b = 24$, $c = 27$, the value of $a^3 + b^3 + c^3 - 3abc$ is

- (a) 1420 (b) 1520
(c) 1620 (d) 1920

354. If $\frac{x^2 - 1}{x} = 4$, then $\frac{x^6 - 1}{x^3}$ is

- (a) 63 (b) 66
(c) 75 (d) 76

355. $(32)^3 + (79)^3 - (111)^3 + 3 \times 32 \times 79 \times 111$ is equal to

- (a) 0 (b) 1
(c) 10000 (d) 30007

356. If $(x + y) = 3$, $xy = 2$, then what is the value of $x^3 + y^3$?

- (a) 6 (b) 7
(c) 8 (d) 9

(e) None of these

357. The value of $\frac{x^2 - (y-z)^2}{(x+z)^2 - y^2} + \frac{y^2 - (x-z)^2}{(x+y)^2 - z^2} + \frac{z^2 - (x-y)^2}{(y+z)^2 - x^2}$ is

(M.B.A., 2006)

- (a) -1 (b) 0
(c) 1 (d) None of these

358. If $\frac{p}{a} + \frac{q}{b} + \frac{r}{c} = 1$ and $\frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 0$ where $a, b, c, p,$

q, r are non-zero real numbers, then $\frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2}$ is

- equal to
(a) 0 (b) 1
(c) 3 (d) 9

- 359.** If $x^4 - y^4 = 15$, then find the value of $x^4 + y^4$, where x and y are natural numbers.
 (a) 17 (b) 31
 (c) 71 (d) 113
- 360.** Along a yard 225 metres long, 26 trees are planted at equal distances, one tree being at each end of the yard. What is the distance between two consecutive trees?
 (a) 8 metres (b) 9 metres
 (c) 10 metres (d) 15 metres
- 361.** In a garden, there are 10 rows and 12 columns of mango trees. The distance between the two trees is 2 metres and a distance of one metre is left from all sides of the boundary of the garden. The length of the garden is
 (a) 20 m (b) 22 m
 (c) 24 m (d) 26 m
- 362.** A farmer has decided to build a wire fence along one straight side of his property. For this, he planned to place several fence-posts at 6 m intervals, with posts fixed at both ends of the side. After he bought the posts and wire, he found that the number of posts he had bought was 5 less than required. However, he discovered that the number of posts he had bought would be just sufficient if he spaced them 8 m apart. What is the length of the side of his property and how many posts did he buy? (M.A.T., 2007)
 (a) 100 m, 15 (b) 100 m, 16
 (c) 120 m, 15 (d) 120 m, 16
- 363.** On a 2 km road, a total number of 201 trees are planted on each side of the road at equal distances. How many such trees in all will be planted on both sides of a 50 km road such that the distance between two consecutive trees is the same as that of the consecutive trees on the 2 km road?
 (a) 501 (b) 5000
 (c) 5001 (d) 5025
 (e) 5050
- 364.** When all the students in a school are made to stand in rows of 54, 30 such rows are formed. If the students are made to stand in rows of 45, how many such rows will be formed? (Bank P.O., 2010)
 (a) 25 (b) 32
 (c) 36 (d) 42
 (e) None of these
- 365.** There are x trees and y parrots. If in each tree only one parrot is sitting, then one parrot remains without a tree. If in each tree, two parrots are sitting, then one tree remains without a parrot. Therefore, x and y are respectively
 (a) 3 and 4 (b) 4 and 5
 (c) 5 and 6 (d) 6 and 7
- 366.** In a classroom, there are certain number of benches. If 6 students are made to sit on a bench, then to accommodate all of them, one more bench is needed. However, if 7 students are made to sit on a bench, then after accommodating all of them, space for 5 students is left. What is the total number of students in the class?
 (a) 30 (b) 42
 (c) 72 (d) None of these
- 367.** There are benches in a classroom. If 4 students sit on each bench, then three benches are left vacant and if 3 students sit on each bench, then 3 students are left standing. The total number of students in the class is
 (a) 36 (b) 42
 (c) 48 (d) 54
- 368.** Students of a class are preparing for a drill and are made to stand in rows. If 4 students are extra in a row, then there would be 2 rows less. But there would be 4 more rows if 4 students are less in a row. The number of students in the class is (M.A.T., 2006)
 (a) 56 (b) 65
 (c) 69 (d) 96
- 369.** If a sum of ₹ 275 is to be divided between Ram and Shyam so that Ram gets $\frac{3}{4}$ th more of what Shyam gets, then the share of Ram will be
 (a) ₹ 100 (b) ₹ 160
 (c) ₹ 175 (d) ₹ 200
- 370.** Sweets were distributed equally among 64 children. After giving 7 sweets to each child 15 sweets were left out. Total how many sweets were there? (Bank P.O., 2009)
 (a) 436 (b) 446
 (c) 448 (d) 463
 (e) None of these
- 371.** X gives $\frac{1}{2}$ of his property to his wife and $\frac{1}{2}$ of the rest to his son. The remainder is divided equally between his two daughters. The share of each daughter is
 (a) $\frac{1}{4}$ (b) $\frac{1}{6}$
 (c) $\frac{1}{8}$ (d) $\frac{2}{3}$
- 372.** If each child is given 10 sweets, there are 3 sweets left over. But if each is given 11, then the number of sweets is 4 less. Find the number of sweets. (R.R.B., 2009)
 (a) 37 (b) 57
 (c) 73 (d) 75

- 373.** 180 oranges are distributed among 70 boys and girls such that each boy gets 2 and each girl gets 3 oranges. The number of boys are
 (a) 25 (b) 30
 (c) 40 (d) 70
- 374.** A farmer divides his herd of cows among his four sons so that first son gets one-half of the herd, the second son gets one-fourth, the third son one-fifth and the fourth son 7 cows. The total number of cows in the herd is (R.R.B., 2006)
 (a) 100 (b) 140
 (c) 180 (d) 240
- 375.** A man had two sons. To the elder, he left $\frac{5}{11}$ of his property, to the younger $\frac{5}{11}$ of the remainder, the rest to the widow. Find the share of the sons if the widow gets ₹ 3600. (M.B.A. 2007)
 (a) ₹ 1200, ₹ 1000 (b) ₹ 6600, ₹ 2000
 (c) ₹ 7500, ₹ 1000 (d) None of these
- 376.** If ₹ 370 are divided among 10 men, 12 women and 20 boys such that each man gets an amount equal to that received by one woman and one boy together and each woman gets twice the amount received by a boy, then the amount received by 10 men would be
 (a) ₹ 100 (b) ₹ 120
 (c) ₹ 130 (d) ₹ 150
- 377.** A sum of ₹ 750 is distributed among A, B, C and D in such a manner that A gets as much as B and C together, B gets ₹ 125 more than C and D gets as much as C. What is A's share?
 (a) ₹ 100 (b) ₹ 225
 (c) ₹ 275 (d) ₹ 325
- 378.** A bonus of ₹ 1000 is to be divided among three people so that Rohit receives twice as much as Sachin, who receives one-fifth as much as Gagan. How much money should Gagan receive?
 (a) ₹ 100 (b) ₹ 250
 (c) ₹ 375 (d) ₹ 625
- 379.** Three boys agree to divide a bag of marbles in the following manner. The first boy takes one more than half the marbles. The second takes a third of the number remaining. The third boy finds that he is left with twice as many marbles as the second boy. The original number of marbles (M.B.A., 2011)
 (a) is 8 or 38
 (b) is 14 or 32
 (c) is 20 or 26
 (d) Cannot be determined from the given data
- 380.** A man has divided his total money in his will in such a way that half of it goes to his wife, $\frac{2}{3}$ rd of the remaining among his three sons equally and the rest among his four daughters equally. If each daughter gets ₹ 20,000, how much money will each son get?
 (a) ₹ 48,233.33 (b) ₹ 50,333.33
 (c) ₹ 53,333.33 (d) Data inadequate
 (e) None of these
- 381.** Anand, Bijoy, Chetan and Dharma together have ₹ 47 with them. Anand and Bijoy together have ₹ 27; Chetan and Anand have ₹ 25 and Dharma and Anand have ₹ 23. How much money does Bijoy have?
 (a) ₹ 9 (b) ₹ 11
 (c) ₹ 13 (d) ₹ 28
- 382.** When a sum of money was distributed to 12 persons instead of 16 persons, each person got ₹ 400 more. What was the sum?
 (a) ₹ 14400 (b) ₹ 16000
 (c) ₹ 19200 (d) Data inadequate
 (e) None of these
- 383.** After distributing the sweets equally among 25 children, 8 sweets remain. Had the number of children been 28, 22 sweets would have been left after equally distributing. What was the total number of sweets?
 (a) 328 (b) 348
 (c) 358 (d) Data inadequate
- 384.** On a school's Annual Day sweets were to be equally distributed amongst 112 children. But on that particular day, 32 children were absent. Thus the remaining children got 6 extra sweets. How many sweets was each child originally supposed to get? (Bank P.O., 2009)
 (a) 15 (b) 18
 (c) 24 (d) Cannot be determined
 (e) None of these
- 385.** A sum of money is equally divided among a number of children. Had there been 16 children more, each would have received ₹ 2 less and had there been 16 fewer, each would have received ₹ 3 more. Find the sum of the money distributed.
 (a) ₹ 880 (b) ₹ 896
 (c) ₹ 928 (d) ₹ 960
- 386.** Vidushi and Sanya distributed ₹ 100 each in charity. Vidushi distributes money to 5 more people than Sanya and Sanya gives each ₹ 1 more than Vidushi. How many people are recipients of the charity?
 (a) 45 (b) 60
 (c) 90 (d) None of these
- 387.** In a classroom the number of boys is three times the number of girls. Which of the following numbers does not represent the total number of students in the classroom?
 (a) 40 (b) 42
 (c) 44 (d) 48

- 388.** There are some boys and girls in a room. The square of the number of girls is less than the square of the number of boys by 28. If there were two more girls, the number of boys would have been the same as that of the girls. The total number of boys and girls in the room is (C.P.O., 2007)
- (a) 7 (b) 10
(c) 14 (d) 56
- 389.** A classroom has equal number of boys and girls. Eight girls left to play kho-kho, leaving twice as many boys as girls in the classroom. What was the total number of girls and boys present initially?
- (a) 16 (b) 24
(c) 32 (d) Cannot be determined
(e) None of these
- 390.** What is the value of the expression $(1+x)(1+x^2)(1+x^4)(1+x^8)(1-x)$?
- (a) $x^8 + 1$ (b) $x^{16} - 1$
(c) $1 + x^{16}$ (d) $1 - x^{16}$
- 391.** The expression $\frac{1}{x-1} - \frac{1}{x+1} - \frac{2}{x^2+1} - \frac{4}{x^4+1}$ is equal to
- (a) $\frac{8}{x^8+1}$ (b) $\frac{8}{x^8-1}$
(c) $\frac{8}{x^7-1}$ (d) $\frac{8}{x^7+1}$
- 392.** $\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - 1\right)\left(x^2 + \frac{1}{x^2} + 1\right)$ is equal to (S.S.C., 2006)
- (a) $x^6 - \frac{1}{x^6}$ (b) $x^8 - \frac{1}{x^8}$
(c) $x^6 + \frac{1}{x^6}$ (d) $x^8 + \frac{1}{x^8}$
- 393.** If $\left(x + \frac{1}{x}\right) = 3$, then $\left(x^2 + \frac{1}{x^2}\right) = ?$
- (a) 7 (b) 9
(c) 10 (d) 27
- 394.** If $\left(a - \frac{1}{a}\right) = p$, then $\left(a^2 + \frac{1}{a^2}\right)$ is equal to
- (a) p (b) $p - 2$
(c) $p + 2$ (d) None of these
- 395.** If $\frac{x-1}{x} = 3$, then the value of $1 + \frac{1}{x^2}$ is
- (a) 9 (b) 10
(c) 11 (d) None of these
- 396.** If $\left(x^2 + \frac{1}{x^2}\right) = 34$, then $\left(x + \frac{1}{x}\right)$ is equal to
- (a) 3 (b) 4
(c) 5 (d) None of these
- 397.** If $\left(x + \frac{1}{x}\right) = 2$, then $\left(x - \frac{1}{x}\right)$ is equal to (R.R.B., 2006)
- (a) 0 (b) 1
(c) 2 (d) 5
- 398.** If $\left(a + \frac{1}{a}\right) = 6$, then $\left(a^4 + \frac{1}{a^4}\right) = ?$ (R.R.B., 2008)
- (a) 1154 (b) 1158
(c) 1160 (d) 1164
- 399.** If $\left(x - \frac{1}{x}\right) = 2$, then the value of $\left(x^4 + \frac{1}{x^4}\right)$ is
- (a) 4 (b) 8
(c) 12 (d) 34
- 400.** If $\left(x - \frac{1}{x}\right) = \sqrt{21}$, the value of $\left(x^2 + \frac{1}{x^2}\right)\left(x + \frac{1}{x}\right)$ is
- (a) 42 (b) 63
(c) 115 (d) 120
(e) 125
- 401.** If $\left(x^4 + \frac{1}{x^4}\right) = 322$, the value of $\left(x - \frac{1}{x}\right)$ is
- (a) 4 (b) $3\sqrt{2}$
(c) 6 (d) 8
- 402.** If $x + \frac{1}{x} = p$, the value of $x^3 + \frac{1}{x^3}$ is
- (a) $p^3 + \frac{1}{p^3}$ (b) $p^3 - 3p$
(c) p^3 (d) $p^3 + 3p$
- 403.** If $x - \frac{1}{x} = 1$, then the value of $x^3 - \frac{1}{x^3}$ is equal to
- (a) 2 (b) 3
(c) 4 (d) 8
- 404.** If $\left(a^4 + \frac{1}{a^4}\right) = 1154$, then the value of $\left(a^3 + \frac{1}{a^3}\right) = ?$
- (a) 198 (b) 200
(c) 216 (d) None of these
- 405.** If $\left(x + \frac{1}{x}\right) = 3$, then the value of $\left(x^6 + \frac{1}{x^6}\right)$ is
- (a) 322 (b) 364
(c) 414 (d) 927
- 406.** If $\left(x + \frac{1}{x}\right) = \sqrt{13}$, then the value of $\left(x^3 - \frac{1}{x^3}\right)$ is (Delhi Police, 2010)
- (a) 26 (b) 27
(c) 30 (d) 36
- 407.** If $\left(4b^2 + \frac{1}{b^2}\right) = 2$, then $\left(8b^3 + \frac{1}{b^3}\right) = ?$ (S.S.C., 2008)
- (a) 0 (b) 1
(c) 2 (d) 5

- 408.** If $\left(2p + \frac{1}{p}\right) = 4$, the value of $\left(p^3 + \frac{1}{8p^3}\right)$ is (S.S.C., 2010)
- (a) 4 (b) 5
(c) 8 (d) 15
- 409.** In a herd of cows, the number of legs are 14 more than twice the number of heads. The number of cows in the herd is (P.C.S., 2009)
- (a) 5 (b) 7
(c) 10 (d) 12
- 410.** In a group of buffaloes and ducks, the number of legs is 24 more than twice the number of heads. What is the number of buffaloes in the group?
- (a) 6 (b) 8
(c) 10 (d) 12
- 411.** Krishan has some hens and some cows. If the total number of animal heads is 59 and the total number of feet is 190, how many cows does Krishan have? (M.A.T., 2009)
- (a) 23 (b) 32
(c) 36 (d) Cannot be determined
- 412.** There are some parrots and some tigers in a forest. If the total number of animal heads in the forest is 858 and the total number of animal legs is 1846, what is the number of parrots in the forest? (Bank P.O., 2010)
- (a) 800 (b) 833
(c) 845 (d) Cannot be determined
(e) None of these
- 413.** A certain number of horses and an equal number of men are going somewhere. Half of the owners are on their horses' back while the remaining ones are walking along leading their horses. If the number of legs walking on the ground is 70, then how many horses are there?
- (a) 10 (b) 12
(c) 14 (d) 16
- 414.** A railway half-ticket costs half the full fare and the reservation charge is the same on half ticket as on full ticket. One reserved first class ticket from Chennai to Trivandrum costs ₹ 216 and one full reserved and one half reserved first class tickets cost ₹ 327. What is the basic first class full fare and what is the reservation charge? (R.R.B., 2009)
- (a) ₹ 105, ₹ 6 (b) ₹ 216, ₹ 12
(c) ₹ 210, ₹ 6 (d) ₹ 210, ₹ 12
- 415.** An employee may claim ₹ 7 for each km when he travels by taxi and ₹ 6 for each km if he drives his own car. If in one week he claimed ₹ 675 for travelling 90 km, how many kms did he travel by taxi? (L.I.C.A.A.O., 2007)
- (a) 135 (b) 155
(c) 162 (d) 170
- 416.** In a party 15 people shake their hands with each other. How many times did the hand-shakes take place? (R.R.B., 2009)
- (a) 105 (b) 120
(c) 135 (d) 165
- 417.** In a group of children, each child exchanges a gift with every other child. If the number of gifts is 132, then the number of children in the group is
- (a) 10 (b) 11
(c) 12 (d) 13
- 418.** A man has 1044 candles. After burning, he can make a new candle from 9 stubs left behind. Find the maximum number of candles that can be made.
- (a) 116 (b) 120
(c) 130 (d) 140
- 419.** A computer printed 176400 lines in a given day. If the printer was in operation for 7 hours during the day, how many lines did it print per minute? (Campus Recruitment, 2008)
- (a) 280 (b) 360
(c) 420 (d) 440
- 420.** A man takes 8 minutes to type a page. If 1710 pages are to be typed in the afternoon between 1 o' clock to 2 o' clock, how many men are required? (R.R.B., 2009)
- (a) 221 (b) 249
(c) 256 (d) None of these
- 421.** Ashu reads at an average rate of 30 pages per hour while Neeru reads at an average rate of 40 pages per hour. If Ashu starts reading a novel at 2: 30 and Neeru begins reading an identical copy of the same book at 3: 20, at what time will they be reading the same page?
- (a) 5: 00 (b) 5: 50
(c) 6: 40 (d) 7: 00
(e) 7: 30
- 422.** In a test, a candidate secured 336 marks out of maximum marks x . If the maximum marks x had been converted into 400 marks, he would have secured 192 marks. What was the maximum marks of the test? (Bank P.O., 2009)
- (a) 500 (b) 650
(c) 700 (d) 750
(e) 800
- 423.** The marks scored in an examination are converted from 50 to 10 for the purpose of internal assessment. The highest marks were 47 and the lowest were 14. The difference between the maximum and the minimum internal assessment scores is
- (a) 3.3 (b) 4.8
(c) 6.6 (d) 7.4

- 424.** In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. A student attempted all the 200 questions and scored in all 200 marks. The number of questions he answered correctly was (S.S.C., 2010)
- (a) 60 (b) 68
(c) 80 (d) 82
- 425.** On a 26 question test, five points were deducted for each wrong answer and eight points were credited for each correct answer. If all the questions were answered, how many were correct if the score was zero?
- (a) 6 (b) 9
(c) 10 (d) 13
- 426.** In an examination, a student attempted 15 questions correctly and secured 40 marks. If there were two types of questions (2 marks and 4 marks questions), how many questions of 2 marks did he attempt correctly?
- (a) 5 (b) 10
(c) 20 (d) 40
- 427.** In an examination there are 30 questions. 1 mark is given for each correct answer and 0.25 is deducted for every incorrect answer. Ankur attempted all the questions and scored 13.75. How many incorrect answers did he have? (XAT, 2009)
- (a) 10 (b) 11
(c) 12 (d) 15
(e) None of these
- 428.** For aiming at a target a person gets one rupee each time when he hits it and loses one rupee when he misses it. If he gets ₹ 30 after aiming at the target one hundred times, then how many times did he miss the target? (P.C.S., 2009)
- (a) 25 (b) 35
(c) 40 (d) 45
- 429.** There are some cows, bulls and 45 hens in a group. One caretaker looks after 15 animals. The number of bulls is twice the number of cows. If the number of heads is less than the total number of feet by 186 (including the caretakers), how many caretakers are there?
- (a) 5 (b) 6
(c) 8 (d) 9
(e) None of these
- 430.** A man's basic pay for a 40 hours' week is ₹ 200. Overtime is paid at 25% above the basic rate. In a certain week, he worked overtime and his total was ₹ 300. He, therefore, worked for a total of (in hours) (L.I.C.A.A.O., 2007)
- (a) 52 (b) 56
(c) 58 (d) 62
- 431.** In a regular week, there are 5 working days and for each day, the working hours are 8. A man gets ₹ 24 per hour for regular work and ₹ 32 per hour for overtime. If he earns ₹ 4320 in 4 weeks, then how many hours does he work for?
- (a) 160 (b) 175
(c) 180 (d) 195
- 432.** David gets on the elevator at the 11th floor of a building and rides up at the rate of 57 floors per minute. At the same time, Albert gets on an elevator at the 51st floor of the same building and rides down at the rate of 63 floors per minute. If they continue travelling at these rates, then at which floor will their paths cross? (M.B.A., 2008)
- (a) 19th (b) 28th
(c) 30th (d) 37th
- 433.** Mr. Shah decided to walk down the escalator of a tube station. He found that if he walks down 26 steps, he requires 30 seconds to reach the bottom. However, if he steps down 34 stairs he would only require 18 seconds to get to the bottom. If the time is measured from the moment the top step begins to descend to the time he steps off the last step at the bottom, find out the height of the stairway in steps. (Campus Recruitment, 2009)
- (a) 42 (b) 46
(c) 52 (d) 54
- 434.** Mohan engaged a servant on the condition that he would pay him ₹ 200 and a uniform after 10 days. The servant served only for 5 days and got ₹ 20 and a uniform. Find the price of the uniform. (R.R.B. 2006)
- (a) ₹ 80 (b) ₹ 120
(c) ₹ 140 (d) ₹ 160
- 435.** A labourer was engaged for 20 days on the condition that he will receive ₹ 60 for each day he works and he will be fined ₹ 5 for each day he is absent. If he received ₹ 745 in all, then the number of days on which he remained absent is
- (a) 3 (b) 5
(c) 7 (d) 9
- 436.** The taxi charges in a city comprise of a fixed charge, together with the charge of the distance covered. For a journey of 16 km, the charges paid are ₹ 156 and for a journey of 24 km, the charges paid are ₹ 204. What will a person have to pay for travelling a distance of 30 km?
- (a) ₹ 236 (b) ₹ 240
(c) ₹ 248 (d) ₹ 252
- 437.** M men agree to purchase a gift for ₹ D. If three men drop out how much more will each have to contribute towards the purchase of the gift? (Campus Recruitment, 2008)
- (a) $\frac{D}{M-3}$ (b) $\frac{MD}{3}$
(c) $\frac{M}{D-3}$ (d) $\frac{3D}{M^2-3M}$

- 438.** N number of persons decided to raise ₹ 3 lakhs by equal contributions from each. Had they contributed ₹ 50 each extra, the contribution would have been ₹ 3.25 lakhs. How many persons are there?
 (a) 400 (b) 450
 (c) 600 (d) Cannot be determined
 (e) None of these
- 439.** Each boy contributed rupees equal to the number of girls and each girl contributed rupees equal to the number of boys in a class of 60 students. If the total contribution thus collected is ₹ 1600, how many boys are there in the class? (M.B.A., 2006)
 (a) 25 (b) 30
 (c) 50 (d) Data inadequate
- 440.** Eight people are planning to share equally the cost of a rental car. If one person withdraws from the arrangement and the others share equally the entire cost of the car, then the share of each of the remaining persons increased by
 (a) $\frac{1}{7}$ (b) $\frac{1}{8}$
 (c) $\frac{1}{9}$ (d) $\frac{7}{8}$
- 441.** A person on tour has ₹ 360 for his daily expenses. If he extends his tour for 4 days, he has to cut his daily expenses by ₹ 3. Find the original duration of the tour.
 (a) 15 days (b) 20 days
 (c) 24 days (d) 30 days
- 442.** Some students planned a picnic. The budget for food was ₹ 500. But, 5 of them failed to go and thus the cost of food for each member increased by ₹ 5. How many students attended the picnic? (M.A.T., 2006)
 (a) 15 (b) 20
 (c) 25 (d) 30
- 443.** A class decided to have a party at a total cost of ₹ 720. Four students decided to stay out of the party. To meet the expenses the remaining students had to increase their share by ₹ 9. What is the original cost per student? (M.A.T., 2010)
 (a) ₹ 18 (b) ₹ 20
 (c) ₹ 24 (d) ₹ 36
- 444.** A group of boys decided to buy a few CDs whose total price was between ₹ 200 and ₹ 250. But at the time of purchase, two of the boys declined to contribute as a result of which the remaining boys had to pay Re 1 more than they had originally planned. What was the price of the CDs if the boys contributed equally and in whole number of rupees?
 (a) ₹ 210 (b) ₹ 220
 (c) ₹ 230 (d) ₹ 240
- 445.** Rahul owes ₹ X and gives a ₹ 50 note in payment. He receives the following change: $3X$ fifty-paise coins, 14 ten-paise coins and $4X$ five-paise coins. X is equal to:
 (a) 12 (b) 16
 (c) 18 (d) 22
- 446.** A total of 324 coins of 20 paise and 25 paise make a sum of ₹ 71. The number of 25-paise coins is
 (a) 120 (b) 124
 (c) 144 (d) 200
- 447.** A man has one-rupee, 5-rupee and 10-rupee currency notes of worth ₹ 480. If the number of notes of each denomination be same, then the total number of notes will be (R.R.B., 2007)
 (a) 45 (b) 60
 (c) 75 (d) 90
- 448.** The total value of a collection of coins of denominations Re 1, 50-paise, 25-paise, 10-paise and 5-paise is ₹ 380. If the number of coins of each denomination is the same, then the number of one-rupee coins is
 (a) 160 (b) 180
 (c) 200 (d) 220
- 449.** A bag contains three types of coins — 1-rupee coins, 50 p-coins and 25 p-coins totalling 175 coins. If the total value of the coins of each kind be the same, the total amount in the bag is
 (a) ₹ 75 (b) ₹ 126
 (c) ₹ 175 (d) ₹ 300
- 450.** In a box, there are a certain number of coins which amount to ₹ 25. There are 2-rupee coins, 50-paise coins and 25-paise coins. If there are at least one 2-rupee coin, two 50-paise coins and two 25-paise coins, then what could be the minimum number of coins in the box?
 (a) 12 (b) 14
 (c) 16 (d) 18
- Directions (Questions 451 to 452):** These questions are based on the following information:
 (Campus Recruitment, 2008)
- 85 children went to an amusement park where they could ride on merry-go-round, roller coaster and Ferris wheel. It was known that 20 of them took all the three rides and 55 of them have taken at least two of the three rides. Each ride costs Re 1, and the total receipt of the amusement park was ₹ 145.
- 451.** How many children took exactly one ride?
 (a) 5 (b) 10
 (c) 15 (d) 20
- 452.** How many children did not try any of the rides?
 (a) 5 (b) 10
 (c) 15 (d) 20
- 453.** In a class of 50 students, 25 take Bengali, 16 take Hindi, 12 students take no language. How many take both Bengali and Hindi? (P.C.S., 2008)
 (a) 3 (b) 4
 (c) 9 (d) 13

- 454.** Out of 100 students in a class, 60 take tea, 40 take coffee and 25 take both. The number of students not taking either tea or coffee is (P.C.S., 2009)
 (a) 25 (b) 28
 (c) 30 (d) 32
- 455.** In a class of 25 students, 12 have taken Mathematics, 8 have taken Mathematics but not Biology. The number of students who have taken both Mathematics and Biology is (R.R.B., 2006)
 (a) 4 (b) 8
 (c) 24 (d) 36
- 456.** In a group of 52 persons, 16 drink tea but not coffee and 33 drink tea. How many drink coffee but not tea? (M.B.A., 2008)
 (a) 3 (b) 7
 (c) 17 (d) 19
- 457.** In a town with population of 4000, 3000 people are egg eaters, 2000 meat eaters and 1500 eat both eggs and meat. How many are pure vegetarians?
 (a) 400 (b) 500
 (c) 1000 (d) 1500
- 458.** There are 50 students admitted to a nursery class. Some students can speak only English and some can speak only Hindi. Ten students can speak both English and Hindi. If the number of students who can speak English is 21, then how many students can speak Hindi, how many can speak only Hindi and how many can speak only English?
 (a) 21, 11 and 29 respectively
 (b) 28, 18 and 22 respectively
 (c) 37, 27 and 13 respectively
 (d) 39, 29 and 11 respectively
- 459.** In an office, $\frac{3}{4}$ of the staff can neither type nor take shorthand. However, $\frac{1}{5}$ th can type and $\frac{1}{3}$ rd can take shorthand. What part of the whole staff can do both?
 (a) $\frac{1}{5}$ (b) $\frac{3}{40}$
 (c) $\frac{13}{40}$ (d) $\frac{17}{60}$
- 460.** In a group of players in a college, 20 are in the basketball team, 24 in the hockey team and 27 in the cricket team. If 12 play hockey and basketball, 10 play cricket and basketball, 14 play hockey and cricket and 7 play all the three games, then the total number of players in the group is
 (a) 42 (b) 43
 (c) 45 (d) 49
- 461.** In the cinema set of a movie, 125 mechanical aliens were created. Some of these aliens had peculiar features. 40 had two noses, 30 had three legs, 20

had four ears, 10 had two noses and three legs, 12 had three legs and four ears, 5 had two noses and four ears and 3 had all the three unusual features. How many were there without any of these unusual features?

- (a) 5 (b) 35
 (c) 80 (d) None of these

Directions (Questions 462 to 466): These questions are based on the following information: (S.B.I.P.O., 2007)

Children in a class play only one or two or all of the three games – badminton, football and cricket. 5 children play only cricket, 8 children play only football and 7 children play only badminton. 3 children play only two games – badminton and football, 4 children play only two games – cricket and football, and another 4 children play only two games – badminton and cricket. 2 children play all the three games.

- 462.** How many children play football as well as cricket?
 (a) 4 (b) 6
 (c) 7 (d) 15
 (e) None of these
- 463.** In all, how many children play badminton?
 (a) 12 (b) 13
 (c) 14 (d) 17
 (e) None of these
- 464.** In all, how many children play football?
 (a) 8 (b) 14
 (c) 15 (d) 17
 (e) None of these
- 465.** How many children play badminton as well as cricket?
 (a) 4 (b) 6
 (c) 9 (d) 10
 (e) None of these
- 466.** Total how many children are there in the class?
 (a) 31 (b) 33
 (c) 35 (d) 36
 (e) None of these
- 467.** A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of the three popular options – air-conditioning, radio and power windows – were already installed.

(M.A.T., 2006)

The survey found:

- 15 had air-conditioning
- 2 had air-conditioning and power windows but not radios
- 12 had radio
- 6 had air-conditioning and radio but no power-windows
- 11 had power-windows
- 4 had radio and power windows
- 3 had all three options

What is the number of cars that had none of the options?

- (a) 1 (b) 2
(c) 3 (d) 4

468. In a class of 50 students, 23 speak English, 15 speak Hindi and 18 speak Punjabi. 3 speak only English and Hindi, 6 speak only Hindi and Punjabi and 6 speak only English and Punjabi. If 9 speak only English, then how many speak all the three?

- (a) 1 (b) 2
(c) 3 (d) 5

469. In a hospital, there were 200 Diabetes, 150 Hyperglycaemia and 150 Gastro-enteritis patients. Of these, 80 patients were treated for both Diabetes and Hyperglycaemia. Sixty patients were treated for Gastro-enteritis and Hyperglycaemia, while 70 were treated for Diabetes and Gastro-enteritis. Some of these patients have all the three diseases. Doctor Dennis treats patients with only Diabetes. Doctor Hormis treats patients with only Hyperglycaemia and Doctor Gerard treats patients with only Gastro-enteritis. Doctor Paul is a Generalist. Therefore, he can treat patients with multiple diseases. Patients always prefer a specialist for their disease. If Dr. Dennis had 80 patients, then the other three doctors can be arranged in terms of the number of patients treated as

- (a) Paul > Hormis > Gerard
(b) Gerard > Paul > Hormis
(c) Paul > Gerard > Hormis
(d) None of these

470. In a joint family of 18 members, 10 take milk, 8 take tea and 6 take coffee; 5 take milk as well as tea but none takes milk and coffee. The number of members who take tea as well as coffee is

- (a) None (b) 1
(c) 2 (d) 3

Directions (Questions 471 to 475): Study the following information carefully to answer the given questions:

(Bank P.O., 2006)

The teachers' colony has 2800 members, out of which 650 members read only English newspaper. 550 members read only Hindi newspaper and 450 members read only Marathi newspaper. The number of members reading all the 3 newspapers is 100. Members reading Hindi as well as English newspaper are 200. 400 members read Hindi as well as Marathi newspaper and 300 members read English as well as Marathi newspaper.

471. Find the number of members reading Hindi newspaper.

- (a) 750 (b) 980 (c) 1000
(d) 1020 (e) None of these

472. How many members read only one newspaper?

- (a) 1540 (b) 1560
(c) 1640 (d) 1650
(e) None of these

473. Find the number of members reading no newspaper.

- (a) 150 (b) 460
(c) 550 (d) 750 (e) None of these

474. How many members read at least two newspapers?

- (a) 500 (b) 600
(c) 800 (d) 1000

- (e) None of these

475. Find the difference between the number of members reading English as well as Marathi newspaper and the number of members reading English as well as Hindi newspaper.

- (a) 50 (b) 100
(c) 200 (d) 300

- (e) None of these

476. In a garrison, there was some pilferage of ration everyday. If the pilferage was 12 kg a day, the ration would have lasted for 75 days. If the pilferage was restricted to 8 kg per day, the ration would have lasted for 90 days. For how many days would the ration have lasted, if there was no pilferage?

- (a) 105 (b) 120
(c) 150 (d) 180

477. Two candles having the same lengths are such that one burns out completely in 3 hours at a uniform rate and the other in 4 hours. At what time should both the candles be lighted together so that at 4 P.M. the length of one is twice the length of the other?

(A.A.O. Exam, 2010)

- (a) 1: 24 P.M. (b) 1: 30 P.M.
(c) 1: 36 P.M. (d) 1: 42 P.M.

478. A man purchased 40 fruits – apples and oranges – for ₹ 17. Had he purchased as many oranges as apples and as many apples as oranges, he would have paid ₹ 15. Find the cost of one pair of an apple and an orange.

(SNAP, 2007)

- (a) 60 paise (b) 70 paise
(c) 80 paise (d) 1 rupee

479. A buyer purchased few books of equal value for a total cost of ₹ 720. If the value of each were ₹ 2 less than the price at which the buyer originally bought, she could have purchased 4 more books than what she had bought. How many books did she originally purchase?

(M.A.T., 2006)

- (a) 18 (b) 20
(c) 34 (d) 36

480. A small confectioner bought a certain number of pastries flavoured pineapple, mango and black-forest from the bakery, giving for each pastry as many rupees as there were pastry of that kind; altogether he bought 23 pastries and spent ₹ 211. Find the number of each kind of pastry that he bought, if mango pastry is cheaper than pineapple pastry and dearer than black-forest pastry.

(I.I.F.T., 2010)

- (a) (10, 9, 4) (b) (11, 9, 3)
(c) (10, 8, 5) (d) (11, 8, 4)

- 481.** One test tube contains some acid and another test tube contains an equal quantity of water. To prepare a solution, 20 grams of the acid is poured into the second test tube. Then, two-thirds of the so-formed solution is poured from the second tube into the first. If the fluid in first test tube is four times that in the second, what quantity of water was taken initially? (M.A.T. 2009; N.M.A.T., 2005)
- (a) 40 grams (b) 60 grams
(c) 80 grams (d) 100 grams
- 482.** One bottle is half-full of milk and another bottle with twice the capacity is one-quarter full of milk. If water is added so that both the bottles are full and the contents of both are then poured into a third bottle that is empty and large enough to hold the contents of both, what fraction of the contents in the third bottle is milk? (Campus Recruitment, 2008)
- (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{8}$
- 483.** A train started with 540 passengers. At the first stop $\frac{1}{9}$ of them got down and 24 got up. On its second stop $\frac{1}{8}$ of the passengers then existing got down and 9 got up. With how many passengers did it reach the third stop? (R.R.B., 2008)
- (a) 450 (b) 500
(c) 540 (d) 550
- 484.** A train started from a station with a certain number of passengers. At the first halt, half of the passengers got down and 125 passengers got in. At the second halt, half of the passengers left and 100 entered. Then the train left for its destination with 250 passengers. The number of passengers in the train at the start was
- (a) 250 (b) 350
(c) 450 (d) 550
- 485.** The total number of digits used in numbering the pages of a book having 366 pages, is (R.R.B., 2008)
- (a) 732 (b) 990
(c) 1098 (d) 1305
- 486.** A printer numbers the pages of a book starting with 1 and uses 3189 digits in all. How many pages does the book have?
- (a) 1000 (b) 1074
(c) 1075 (d) 1080
- 487.** In a class, there are two sections A and B. If 10 students of section B shift over to section A, the strength of A becomes three times the strength of B. But if 10 students shift over from A to B, both A and B become equal in strength. How many students are there in sections A and B? (R.R.B., 2006)
- (a) 80, 40 (b) 90, 40
(c) 45, 15 (d) 50, 30
- 488.** Recently my brother and I played chess for chocolates. Whoever lost the game gave the other a chocolate. After the last game we counted the chocolates. I had 20 more chocolates than I started with, although he won 7 games. There were no draws. How many games did we play?
- (a) 27 (b) 34
(c) 37 (d) 54
- 489.** Robin says, "If Jai gives me ₹ 40, he will have half as much as Atul, but if Atul gives me ₹ 40, then the three of us will have the same amount." What is the total amount of money that Robin, Jai and Atul have between them?
- (a) ₹ 240 (b) ₹ 320
(c) ₹ 360 (d) ₹ 420
- 490.** A, B, C and D play a game of cards. A says to B, "If I give you 8 cards, you will have as many as C has and I shall have 3 less than what C has. Also, if I take 6 cards from C, I shall have twice as many as D has." If B and D together have 50 cards, how many cards has A got?
- (a) 23 (b) 27
(c) 37 (d) 40
- 491.** A, B, C, D and E play a game of cards. A says to B, "If you give me 3 cards, you will have as many as E has and if I give you 3 cards, you will have as many as D has." A and B together have 10 cards more than what D and E together have. If B has 2 cards more than what C has and the total number of cards be 133, how many cards does B have?
- (a) 22 (b) 23
(c) 25 (d) 35
- 492.** A gives B as many rupees as B has and C as many rupees as C has. Similarly, B then gives A and C as many rupees as each then has. C, similarly, then gives A and B as many rupees as each then has. If each finally has ₹ 16, with how many rupees does A start? (M.B.A., 2010)
- (a) 26 (b) 28
(c) 30 (d) 32
- 493.** Iqbal dealt some cards to Mushtaq and himself from a pack of playing cards and laid the rest aside. Iqbal then said to Mushtaq, "If you give me a certain number of your cards, I will have four times as many cards as you have. If I give you the same number of cards, I will have thrice as many cards as you." How many cards did Iqbal have?

- (a) 9 (b) 12
(c) 31 (d) 35
(e) None of these
- 494.** While out on picnic, a group of boys came upon an apple tree. One of the boys climbed the tree and picked enough apples for each boy to have three, with none left over. Then along came three boys, making it impossible to divide the picked apples evenly. However, after picking one more apple and adding it to the total, every boy had two apples, with none left over. How many apples were finally divided?
- (a) 10 (b) 12
(c) 16 (d) 18
- 495.** A teacher bought a certain number of friendship bands for the students in the classroom for Friendship Day celebrations. Each student gives a friendship band to every other student. However, on the Friendship Day, 2 students were absent as a result of which 122 friendship bands were not utilized. But one out of the two absent students came late. Find the number of friendship bands which finally remained unutilized.
- (a) 57 (b) 59
(c) 62 (d) 64
- 496.** Reena bought chocolates to distribute among her friends on her 18th birthday. If she gives 3 chocolates to each friend, one friend will get only 2 chocolates. Also, if she gives 2 chocolates to each friend, she will be left with 15 chocolates. How many chocolates did she buy?
- (a) 44 (b) 47
(c) 50 (d) None of these
- 497.** A certain organisation has three committees. Only two persons are members of all committees, but every pair of committees have three members in common. What is the least possible number of members on any one committee? (Campus Recruitment, 2008)
- (a) 4 (b) 5
(c) 6 (d) None of these
- 498.** In a certain game, each player scores either 2 points or 5 points. If n players score 2 points and m players score 5 points, and the total number of points scored is 50, what is the least possible difference between m and n ?
- (a) 1 (b) 3
(c) 5 (d) 7
- 499.** A school has to buy at least 15 chairs within a budgetary ceiling of ₹ 2000. A chair with arms costs ₹ 160 and one without arms costs ₹ 100. What is the maximum number of chairs with arms that the school can buy?
- (a) 7 (b) 8
(c) 9 (d) 12
- 500.** On Monday, a certain animal shelter housed 55 cats and dogs. By Friday, exactly $\frac{1}{5}$ of the cats and $\frac{1}{4}$ of the dogs had been adopted; no new cats or dogs were brought to the shelter during this period. What is the greatest possible number of pets that could have been adopted from the animal shelter between Monday and Friday?
- (a) 11 (b) 12
(c) 13 (d) 14
- 501.** Out of two-thirds of the total number of basketball matches, a team has won 17 matches and lost 3 of them. What is the maximum number of matches that the team can lose and still win more than three-fourth of the total number of matches if it is true that no match can end in a tie?
- (a) 3 (b) 4
(c) 5 (d) 6
- 502.** Manick visited his cousin Aniket during the summer vacation. In the mornings, they both would go for swimming. In the evenings, they would play tennis. They would engage in at most one activity per day i.e., either they went swimming or played tennis each day. There were days when they took rest and stayed home all day long. There were 32 mornings when they did nothing, 18 evenings when they stayed at home, and a total of 28 days when they swam or played tennis. What duration of the summer vacation did Manick stay with Aniket? (M.B.A. 2011)
- (a) 36 days (b) 39 days
(c) 46 days (d) 58 days
- 503.** 10 cows can graze in a field for 15 days and 20 cows can graze in the same field for 10 days. For how many days can 30 cows graze in the field?
- (a) 5 days (b) $7\frac{1}{2}$ days
(c) $7\frac{2}{3}$ days (d) $8\frac{1}{3}$ days
(e) Cannot be determined
- 504.** Grass in a lawn grows equally thick and at a uniform rate. It takes 24 days for 70 cows and 60 days for 30 cows to eat the whole of the grass. How many cows are needed to eat the grass in 96 days?
- (a) 18 (b) 20
(c) 24 (d) 36
(e) None of these

Directions (Questions 505-506): These questions are based on the following information:

In a holy city, there are ten shrines and a certain number of holy lakes. A group of pilgrims stayed in the city for a few days and visited the shrines and lakes during their stay. At the end of the stay it turned out that in all each shrine was visited exactly by 4 pilgrims and each lake was visited exactly by 6 pilgrims. Each pilgrim visited exactly 5 shrines and 3 lakes.

505. The number of lakes in the holy city is

- (a) 4 (b) 6
(c) 8 (d) 10

506. The number of pilgrims in the group is

- (a) 4 (b) 6
(c) 8 (d) 10

507. From a number of apples, a man sells half the number of existing apples plus 1 to the first customer, sells $\frac{1}{3}$ rd of the remaining apples plus 1 to the second customer and $\frac{1}{5}$ th of the remaining apples plus 1

to the third customer. He then finds that he has 3 apples left. How many apples did he have originally?

- (a) 15 (b) 18
(c) 20 (d) 25

508. Ravi has two examinations on Wednesday – Engineering Mathematics in the morning and Engineering Drawing in the afternoon. He has a fixed amount of time to read the textbooks of both these subjects on Tuesday. During this time he can read 80 pages of Engineering Mathematics and 100 pages of Engineering Drawing. Alternatively, he can also read 50 pages of Engineering Mathematics and 250 pages of Engineering Drawing. Assume that the amount of time it takes to read one page of the textbook of either subject is constant. Ravi is confident about Engineering Drawing and wants to devote full time to reading Engineering Mathematics. The number of Engineering Mathematics text book pages he can read on Tuesday is (M.B.A., 2007)

- (a) 60 (b) 100
(c) 300 (d) 500

509. Rohan went to the post-office to buy five-rupee, two-rupee and one-rupee stamps. He paid the clerk ₹ 20, and since he had no change, he gave Rohan three more one-rupee stamps. If the number of stamps of each type that he had ordered initially was more than one, what was the total number of stamps that he bought?

- (a) 8 (b) 9
(c) 10 (d) 12

510. Aditya went to a stationery shop to buy some Parker pens. Gel pens cost ₹ 300 each while fountain pens cost ₹ 400 each. Aditya spent a total of ₹ 3600 on pens. If he had bought as many fountain pens as the number of gel pens he actually bought and vice versa, he would have saved an amount equal to half the cost of one pen of one of the two types. Find the total number of pens he bought.

- (a) 8 (b) 9
(c) 10 (d) 12

Directions (Questions 511-513): Study the following information carefully to answer these questions:

A young girl Roopa leaves home with x flowers and goes to the bank of a nearby river. On the bank of the river, there are four places of worship, standing in a row. She dips all the x flowers into the river, the number of flowers doubles. Then, she enters the first place of worship and offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the second place of worship and offers y flowers to the deity. She dips the remaining flowers into the river and again the number of flowers doubles. She goes to the third place of worship and offers y flowers to the deity. She dips the remaining flowers into the river and again the number of flowers doubles. She goes to the fourth place of worship and offers y flowers to the deity. Now she is left with no flowers in hand.

511. If Roopa leaves home with 30 flowers, the number of flowers she offers to each deity is

- (a) 30 (b) 31
(c) 32 (d) 33

512. The minimum number of flowers that could be offered to each deity is

- (a) 0 (b) 15
(c) 16 (d) Cannot be determined

513. The minimum number of flowers with which Roopa leaves home is

- (a) 0 (b) 15
(c) 16 (d) Cannot be determined

514. The owner of a local jewellery store hired 3 watchmen to guard his diamonds, but a thief still got in and stole some diamonds. On the way out the thief met each watchman, one at a time. To each he gave $\frac{1}{2}$ of the diamonds he had then, and 2 more besides.

He escaped with one diamond. How many did he steal originally?

- (a) 25 (b) 36
(c) 40 (d) None of these

515. Three friends returning from a movie, stopped to eat at a restaurant. After dinner, they paid their bill

and noticed a bowl of mints, at the front counter. Divya took $\frac{1}{3}$ of the mints, but returned four because

she had a momentary pang of guilt. Reema then took one-fourth of what was left but returned three for similar reasons. Shweta then took half of the remainder but threw two back into the bowl. The bowl had only 17 mints left when the raid was over. How many mints were originally in the bowl?

- (a) 31 (b) 38
(c) 41 (d) None of these

516. At a stationery shop it costs ₹ 185 for 4 gel-pens, 8 ball-point pens and one marker pen and ₹ 315 for 7 gel-pens, 15 ball-point pens and one marker pen. Then what would be the cost of one gel-pen, one ball-point pen and one marker pen?

- (a) ₹ 45 (b) ₹ 55
(c) ₹ 60 (d) ₹ 70

517. In a cricket match, Team A scored 232 runs without losing a wicket. The score consisted of byes, wides and runs scored by two opening batsmen, Ram and Shyam. The runs scored by the two batsmen are 26 times wides. There are 8 more byes than wides. If the ratio of runs scored by Ram and Shyam is 6: 7, then the runs scored by Ram is (M.B.A., 2008)

- (a) 88 (b) 96
(c) 102 (d) 112
(e) None of these

518. Balls are arranged in rows to form an equilateral triangle. The first row consists of one ball, the second row consists of two balls and so on. If 669 more balls are added, then all the balls can be arranged in the shape of a square and each of the sides then contains 8 balls less than each side of the triangle had. The initial number of balls is

- (a) 1500 (b) 1540
(c) 1600 (d) 1690

Directions (Questions 519 and 520): Refer to the data below and answer these questions:

It took a group of men 5 days to shift 545 crates. Every day after the first, 6 more men than the previous day were put on the job. Also, every day after the first, each man, by arrangement, shifted 5 fewer crates than the earlier day. The result was that during the latter part of the period, the number of crates shifted per day began to go down.

519. What was the number of crates shifted on the third day?

- (a) 26 (b) 137
(c) 152 (d) 169
(e) None of these

520. What was the number of men on the fifth day?

- (a) 1 (b) 7
(c) 13 (d) 19
(e) None of these

521. $\frac{20+8 \times 0.5}{20-?} = 12$. Find the value in place of (?)

[Indian Railway Group 'D' Exam, 2014]

- (a) 2 (b) 8
(c) 18 (d) None of these

522. If $\frac{a}{b} + \frac{b}{a} = 2$, then the value of $(a - b)$ is

[SSC—CHSL (10 + 2) Exam, 2015]

- (a) 1 (b) 2
(c) -1 (d) 0

523. $24.96^2 \div (34.11 \div 20.05) + 67.96 - 89.11 = ?$

[IBPS—Bank Spl. Officers (IT) Exam, 2015]

- (a) 884 (b) 346
(c) 252 (d) 424
(e) 366

524. If $\frac{x+1}{x-1} = \frac{a}{b}$ and $\frac{1-y}{1+y} = \frac{b}{a}$, then the value of $\frac{x-y}{1+xy}$ is [SSC—CHSL (10+2) Exam, 2015]

- (a) $\frac{2ab}{a^2 - b^2}$ (b) $\frac{a^2 - b^2}{2ab}$
(c) $\frac{a^2 + b^2}{2ab}$ (d) $\frac{a^2 - b^2}{ab}$

525. $200 \div 25 \times 4 + 12 - 3 = ?$

[United India Insurance Co. Ltd. (UIICL)

Assistant (Online) Exam, 2015]

- (a) 35 (b) 40
(c) 30 (d) 41
(e) 50

526. $14 \times 627 \div \sqrt{1089} = (?)^3 + 141$

[IDBI Bank Executive Officers Exam, 2015]

- (a) $5\sqrt{5}$ (b) $(125)^3$
(c) 25 (d) 5
(e) None of these

527. Evaluate: $(923 - 347) / ? = 32$ [NICL—AAO Exam, 2015]

- (a) 35 (b) 20
(c) 18 (d) 15
(e) 40

Direction: What approximate value will come in place of the question mark (?) in the following question? (You are not expected to calculate the exact value).

528. $1559.95 - 7.99 \times 24.96 - ?^2 = 1154$

[IBPS Bank PO/MT (Pre.) Exam, 2015]

- (a) 14 (b) 24
(c) 32 (d) 18
(e) 8

529. Solve $1\frac{4}{5} + 20 - 280 \div 25 = ?$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) $8\frac{1}{5}$ (b) $9\frac{1}{2}$
(c) $11\frac{1}{2}$ (d) $10\frac{3}{5}$
(e) $12\frac{1}{5}$

530. $((64 - 38) \times 4) \div 13 = ?$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) 4 (b) 1
(c) 8 (d) 2
(e) 5

531. If $x + y = 2a$ then the value of $\frac{a}{x-a} + \frac{a}{y-a}$ is

[SSC—CHSL (10+2) Exam, 2015]

- (a) 2 (b) 0
(c) -1 (d) 1

Directions (Questions 532 to 533): What approximate value should come in place of question mark(?) in the following questions? (NOTE: You are not expected to calculate the exact value).

532. $421 \div 35 \times 299.97 \div 25.05 = ?^2$

[IBPS—Bank PO/MT Exam, 2015]

- (a) 22 (b) 24
(c) 28 (d) 12
(e) 18

533. $19.99 \times 15.98 + 224.98 + 125.02 = ?$

[IBPS—Bank PO/MT Exam, 2015]

- (a) 620 (b) 580
(c) 670 (d) 560
(e) 520

534. $3625 \times ? = 1450$

[United India Insurance (UIICL) Assistant (Online) Exam, 2015]

- (a) $\frac{1}{3}$ (b) $\frac{2}{5}$
(c) $\frac{1}{5}$ (d) $\frac{4}{5}$
(e) $\frac{3}{5}$

535. Solve: $128.43 + 30.21 + ? = 173$

[NICL—AAO Exam, 2015]

- (a) 35.66 (b) 29.66
(c) 43.66 (d) 24.66
(e) 14.36

536. Evaluate: $123 \times 999 + 123$

- (a) 246999 (b) 123000
(c) 246000 (d) 123999

[ESIC—UDC Exam, 2016]

537. Simplify $\frac{(359+256)^2 + (359-256)^2}{359 \times 359 + 256 \times 256}$

[ESIC—UDC Exam, 2016]

- (a) 1089 (b) 615
(c) 516 (d) 2

538. $84368 + 65466 - 72009 - 13964 = ?$

[SBI—Jr. Associates (Pre.) Exam, 2016]

- (a) 61481 (b) 62921
(c) 63861 (d) 64241
(e) None of these

539. Solve $4376 + 3209 - 1784 + 97 = 3125 + ?$

[SBI—Jr. Associates (Pre.) Exam, 2016]

- (a) 2713 (b) 2743
(c) 2773 (d) 2793
(e) 2737

540. Solve $14 \times 627 \div \sqrt{(1089)} = (?)^3 + 141$

[IBPS—Bank Spl. Officer (Marketing) Exam, 2016]

- (a) $5\sqrt{5}$ (b) $(125)^3$
(c) 25 (d) 5
(e) None of these

541. If $\left(x + \frac{1}{x}\right) = 3$, then $\left(x^2 + \frac{1}{x^2}\right)$ is

[UPSSSC—Lower Subordinate (Pre.) Exam, 2016]

- (a) $\frac{10}{3}$ (b) $\frac{82}{9}$
(c) 7 (d) 11

542. If $x + y + z = 0$, then, $x^3 + y^3 + z^3 + 3xyz$ is equal to

[CDS 2016]

- (a) 0 (b) $6xyz$
(c) $12xyz$ (d) xyz

543. What is the remainder when 4^{96} is divided by 6?

[CDS 2016]

- (a) 4 (b) 3
(c) 2 (d) 1

Direction: In the question given below, if the given mathematical symbols are changed from '+' to '÷', '-' to '×', '÷', to '-' and from '×' to '+', then choose your answers from the following options.

544. Solve $67 \times 119 + 17 - 27 \div 259 = ?$

[DMRC—Customer Relationship Assistant (CRA) Exam, 2016]

- (a) -13 (b) -3
(c) 4 (d) 7

545. Solve $\frac{(0.73)^3 + (0.27)^3}{(0.73)^2 + (0.27)^2 - 0.73 \times 0.27} = ?$

[UPSSSC—Lower Subordinate (Pre.) Exam, 2016]

- (a) 0.27 (b) 0.4087
(c) 0.73 (d) 1

ANSWERS

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (c) | 2. (b) | 3. (a) | 4. (a) | 5. (c) | 6. (c) | 7. (e) | 8. (b) | 9. (b) | 10. (d) |
| 11. (b) | 12. (d) | 13. (d) | 14. (c) | 15. (a) | 16. (a) | 17. (c) | 18. (c) | 19. (a) | 20. (a) |
| 21. (c) | 22. (c) | 23. (c) | 24. (d) | 25. (c) | 26. (d) | 27. (c) | 28. (c) | 29. (a) | 30. (a) |
| 31. (c) | 32. (b) | 33. (a) | 34. (d) | 35. (c) | 36. (b) | 37. (b) | 38. (d) | 39. (d) | 40. (e) |
| 41. (e) | 42. (a) | 43. (e) | 44. (c) | 45. (e) | 46. (d) | 47. (e) | 48. (d) | 49. (a) | 50. (d) |
| 51. (a) | 52. (d) | 53. (d) | 54. (a) | 55. (d) | 56. (a) | 57. (b) | 58. (d) | 59. (a) | 60. (c) |
| 61. (c) | 62. (c) | 63. (d) | 64. (d) | 65. (b) | 66. (d) | 67. (b) | 68. (b) | 69. (c) | 70. (d) |
| 71. (d) | 72. (b) | 73. (c) | 74. (d) | 75. (c) | 76. (d) | 77. (a) | 78. (c) | 79. (c) | 80. (a) |
| 81. (c) | 82. (c) | 83. (c) | 84. (a) | 85. (d) | 86. (a) | 87. (d) | 88. (b) | 89. (c) | 90. (c) |
| 91. (a) | 92. (a) | 93. (e) | 94. (c) | 95. (a) | 96. (c) | 97. (b) | 98. (a) | 99. (c) | 100. (c) |
| 101. (b) | 102. (b) | 103. (d) | 104. (b) | 105. (a) | 106. (c) | 107. (b) | 108. (c) | 109. (a) | 110. (b) |
| 111. (d) | 112. (a) | 113. (c) | 114. (b) | 115. (c) | 116. (d) | 117. (b) | 118. (b) | 119. (b) | 120. (c) |
| 121. (d) | 122. (b) | 123. (d) | 124. (a) | 125. (c) | 126. (c) | 127. (c) | 128. (d) | 129. (c) | 130. (d) |
| 131. (c) | 132. (a) | 133. (c) | 134. (a) | 135. (e) | 136. (e) | 137. (b) | 138. (d) | 139. (b) | 140. (b) |
| 141. (d) | 142. (d) | 143. (b) | 144. (a) | 145. (c) | 146. (d) | 147. (a) | 148. (b) | 149. (a) | 150. (b) |
| 151. (b) | 152. (c) | 153. (c) | 154. (c) | 155. (c) | 156. (b) | 157. (d) | 158. (a) | 159. (b) | 160. (d) |
| 161. (a) | 162. (d) | 163. (b) | 164. (d) | 165. (c) | 166. (b) | 167. (d) | 168. (b) | 169. (a) | 170. (b) |
| 171. (d) | 172. (a) | 173. (a) | 174. (c) | 175. (b) | 176. (c) | 177. (a) | 178. (d) | 179. (d) | 180. (c) |
| 181. (a) | 182. (a) | 183. (c) | 184. (e) | 185. (b) | 186. (d) | 187. (c) | 188. (d) | 189. (a) | 190. (e) |
| 191. (e) | 192. (c) | 193. (d) | 194. (c) | 195. (b) | 196. (a) | 197. (b) | 198. (a) | 199. (b) | 200. (a) |
| 201. (d) | 202. (c) | 203. (c) | 204. (c) | 205. (b) | 206. (b) | 207. (d) | 208. (c) | 209. (b) | 210. (d) |
| 211. (b) | 212. (b) | 213. (c) | 214. (c) | 215. (a) | 216. (a) | 217. (b) | 218. (c) | 219. (a) | 220. (b) |
| 221. (c) | 222. (d) | 223. (d) | 224. (d) | 225. (b) | 226. (a) | 227. (d) | 228. (a) | 229. (d) | 230. (e) |
| 231. (e) | 232. (c) | 233. (b) | 234. (b) | 235. (c) | 236. (d) | 237. (a) | 238. (c) | 239. (c) | 240. (b) |
| 241. (a) | 242. (e) | 243. (d) | 244. (b) | 245. (d) | 246. (c) | 247. (c) | 248. (c) | 249. (c) | 250. (c) |
| 251. (c) | 252. (a) | 253. (c) | 254. (c) | 255. (c) | 256. (c) | 257. (d) | 258. (c) | 259. (b) | 260. (b) |
| 261. (c) | 262. (d) | 263. (d) | 264. (a) | 265. (d) | 266. (b) | 267. (c) | 268. (e) | 269. (a) | 270. (b) |
| 271. (b) | 272. (b) | 273. (b) | 274. (b) | 275. (b) | 276. (c) | 277. (a) | 278. (a) | 279. (d) | 280. (c) |
| 281. (c) | 282. (b) | 283. (a) | 284. (c) | 285. (b) | 286. (c) | 287. (a) | 288. (c) | 289. (c) | 290. (d) |

291. (b)	292. (d)	293. (d)	294. (a)	295. (d)	296. (e)	297. (b)	298. (b)	299. (b)	300. (c)
301. (c)	302. (b)	303. (b)	304. (d)	305. (c)	306. (b)	307. (a)	308. (a)	309. (a)	310. (d)
311. (d)	312. (b)	313. (d)	314. (c)	315. (b)	316. (b)	317. (c)	318. (c)	319. (d)	320. (d)
321. (c)	322. (a)	323. (b)	324. (c)	325. (d)	326. (c)	327. (d)	328. (c)	329. (d)	330. (b)
331. (b)	332. (a)	333. (c)	334. (b)	335. (b)	336. (c)	337. (c)	338. (c)	339. (a)	340. (a)
341. (b)	342. (c)	343. (c)	344. (a)	345. (c)	346. (b)	347. (d)	348. (c)	349. (b)	350. (c)
351. (d)	352. (d)	353. (b)	354. (d)	355. (a)	356. (d)	357. (c)	358. (b)	359. (a)	360. (b)
361. (c)	362. (d)	363. (c)	364. (c)	365. (a)	366. (c)	367. (c)	368. (d)	369. (c)	370. (d)
371. (c)	372. (c)	373. (b)	374. (b)	375. (d)	376. (d)	377. (d)	378. (d)	379. (d)	380. (c)
381. (c)	382. (c)	383. (c)	384. (a)	385. (d)	386. (a)	387. (b)	388. (c)	389. (c)	390. (d)
391. (b)	392. (a)	393. (a)	394. (d)	395. (d)	396. (d)	397. (a)	398. (a)	399. (d)	400. (c)
401. (a)	402. (b)	403. (c)	404. (a)	405. (a)	406. (d)	407. (a)	408. (b)	409. (b)	410. (d)
411. (c)	412. (e)	413. (c)	414. (c)	415. (a)	416. (a)	417. (c)	418. (c)	419. (c)	420. (d)
421. (b)	422. (c)	423. (c)	424. (c)	425. (c)	426. (b)	427. (e)	428. (b)	429. (b)	430. (b)
431. (b)	432. (c)	433. (b)	434. (d)	435. (c)	436. (b)	437. (d)	438. (e)	439. (d)	440. (a)
441. (b)	442. (b)	443. (d)	444. (b)	445. (c)	446. (b)	447. (d)	448. (c)	449. (a)	450. (d)
451. (c)	452. (c)	453. (a)	454. (a)	455. (a)	456. (d)	457. (b)	458. (d)	459. (d)	460. (a)
461. (d)	462. (b)	463. (e)	464. (d)	465. (b)	466. (b)	467. (b)	468. (d)	469. (c)	470. (b)
471. (e)	472. (d)	473. (e)	474. (e)	475. (b)	476. (c)	477. (c)	478. (c)	479. (d)	480. (b)
481. (d)	482. (b)	483. (a)	484. (b)	485. (b)	486. (b)	487. (d)	488. (b)	489. (c)	490. (d)
491. (c)	492. (a)	493. (c)	494. (c)	495. (c)	496. (b)	497. (a)	498. (b)	499. (b)	500. (c)
501. (b)	502. (b)	503. (b)	504. (b)	505. (a)	506. (c)	507. (c)	508. (b)	509. (c)	510. (c)
511. (c)	512. (c)	513. (b)	514. (b)	515. (d)	516. (b)	517. (b)	518. (b)	519. (d)	520. (e)
521. (c)	522. (d)	523. (b)	524. (a)	525. (d)	526. (d)	527. (c)	528. (a)	529. (d)	530. (c)
531. (b)	532. (d)	533. (c)	534. (b)	535. (e)	536. (b)	537. (d)	538. (c)	539. (c)	540. (d)
541. (c)	542. (b)	543. (a)	544. (b)	545. (a)					

SOLUTIONS

1. $304 \times 141 = 42864$.

2. $A + B = 96 \Rightarrow \frac{1}{2}B + B = 96 \Rightarrow \frac{3}{2}B = 96 \Rightarrow B = \left(96 \times \frac{2}{3}\right) = 64$.

$\therefore A = \left(\frac{1}{2} \times 64\right) = 32$.

3. $1888 \div 32 \div 8 = \frac{1888}{32} \div 8 = 59 \div 8 = \frac{59}{8} = 7.375$.

4. Given exp. $= -2508 + 221 = -2287$.

5. Given exp. $= \frac{4848}{24} \times 11 - 222 = 202 \times 11 - 222$
 $= 2222 - 222 = 2000$.

6. Given exp. $= \frac{425 \times 4000}{16000} \times 12 = \frac{425}{4} \times 12 = 425 \times 3 = 1275$.

7. Let $[(84)^2 \div 28 \times 12] \div 24 = 7 \times x$.

Then, $\left[\frac{84 \times 84}{28} \times 12\right] \div 24 = 7x \Leftrightarrow [84 \times 3 \times 12] \div 24 = 7x$

$\Leftrightarrow 7x = \frac{3024}{24} = 126 \Leftrightarrow x = 18$.

8. Let the missing number be x .

Then, $\frac{354750}{4096 + x} = 55 \Leftrightarrow 55(4096 + x) = 354750$

$\Leftrightarrow 55x = 354750 - 225280$

$\Leftrightarrow 55x = 129470 \Leftrightarrow x = 2354$.

9. Given expression = $41 \times 44 = 1804$.
10. Given expression = $(-224) + 2826 = 2602$.
11. Let $853 + x \div 17 = 1000$.
Then, $853 + \frac{x}{17} = 1000 \Leftrightarrow \frac{x}{17} = 1000 - 853 = 147$
 $\Leftrightarrow x = 147 \times 17 = 2499$.
12. Let $(x - 968) \div 79 \times 4 = 512$.
Then, $\frac{x - 968}{79} \times 4 = 512 \Leftrightarrow x - 968 = \frac{512 \times 79}{4} = 10112$
 $\Leftrightarrow x = 10112 + 968 = 11080$.
13. Given exp. = $\left[\frac{125 \times 125}{50} \times 20 \right] \div 25 = 6250 \div 25 = 250$.
14. Given exp. = $999 \times 99 \times \frac{9}{99} \div 9 \div 3 = 999 \times 99 \times \frac{9}{99} \times \frac{1}{9} \div 3$
 $= 999 \times 99 \times \frac{9}{99} \times \frac{1}{9} \times \frac{1}{3} = 333$.
15. $(x - a)(x - b)(x - c) \dots (x - y)(x - z)$
 $= (x - a)(x - b)(x - c) \dots (x - x)(x - y)(x - z)$
 $= \{(x - a)(x - b)(x - c) \dots (x - w)\} \times 0 \times [(x - y)(x - z)]$
 $= 0$.
16. Given exp. = $1 - [1 - \{1 - (1 - 0)\}] = 1 - [1 - \{1 - 1\}]$
 $= 1 - [1 - 0] = 1 - 1 = 0$.
17. Let $2x6 - 12 \div 4 + 2 = 11$.
Then, $2x6 - 3 + 2 = 11 \Leftrightarrow 2x6 = 11 + 3 - 2 = 12$.
So 'x' must be replaced by 'x'.
18. Let $45 - [28 - \{37 - (15 - x)\}] = 58$.
Then, $45 - [28 - \{37 - 15 + x\}] = 58$
 $\Leftrightarrow 45 - [28 - \{22 + x\}] = 58$
 $\Leftrightarrow 45 - [28 - 22 - x] = 58 \Leftrightarrow 45 - [6 - x] = 58$
 $\Leftrightarrow 45 - 6 + x = 58$
 $\Leftrightarrow 39 + x = 58 \Leftrightarrow x = 58 - 39 = 19$.
19. Given exp. = $\frac{7^3 \times 7^2}{6^3 \times 2^4 \times 3^4} = \frac{7^{(3+2)}}{6^3 \times (2 \times 3)^4} = \frac{7^5}{6^3 \times 6^4}$
 $= \frac{7^5}{6^{(3+4)}} = \frac{7^5}{6^7}$.
20. $\frac{113 \times 4 - x \times 2}{13 \times 9 - 5 \times 7} = 5 \Leftrightarrow \frac{452 - 2x}{117 - 35} = 5 \Leftrightarrow \frac{452 - 2x}{82} = 5$
 $\Leftrightarrow 452 - 2x = 410$
 $\Leftrightarrow 2x = 452 - 410 = 42 \Leftrightarrow x = 21$.
21. Given expression = $\frac{252 - 48}{122 - 20} = \frac{204}{102} = 2$.
22. Given expression = $\frac{15}{21} - \frac{10}{14} + \frac{5}{7} = \frac{5}{7} - \frac{5}{7} + \frac{5}{7} = \frac{5}{7}$.
23. Given expression = $\frac{3 + 2 \times 3}{4 + 3 \times \frac{2}{3}} = \frac{3 + 6}{4 + 2} = \frac{9}{6} = \frac{3}{2}$.
24. Given expression = $\frac{4 + 72 - 14}{738 - 730} = \frac{76 - 14}{8} = \frac{62}{8} = 7.75$.
25. Given expression = $\frac{2700 - 240}{1120 + 110} = \frac{2460}{1230} = 2$.
26. Given exp. = $\frac{8 - [5 - (-1)] \div 2}{|2| - |-3| \div 3} = \frac{8 - [5 + 1] \div 2}{2 - 3 \div 3} = \frac{8 - 6 \div 2}{2 - 1}$
 $= \frac{8 - 3}{1} = 5$.
27. $(2^2 + 4^2 + 6^2 + \dots + 20^2) = 2^2(1^2 + 2^2 + 3^2 + \dots + 10^2)$
 $= 4 \times 385 = 1540$.
28. Given exp. = $\frac{9}{4} + \frac{4}{3} - \frac{9}{2} = \frac{27 + 16 - 54}{12} = -\frac{11}{12}$.
29. Let $4\frac{3}{7} - 1\frac{3}{14} = x + 2\frac{3}{28}$.
Then, $\frac{31}{7} - \frac{17}{14} = x + \frac{59}{28} \Leftrightarrow x = \frac{31}{7} - \frac{17}{14} - \frac{59}{28}$
 $\Leftrightarrow x = \frac{124 - 34 - 59}{28} = \frac{31}{28} = 1\frac{3}{28}$.
30. Given exp. = $\frac{7}{4} - \frac{6}{5} + \frac{13}{8} = \frac{70 - 48 + 65}{40} = \frac{87}{40} = 2\frac{7}{40}$.
31. Given exp. = $\frac{37}{3} + \frac{65}{6} - \frac{23}{3} - \frac{11}{7} = \frac{518 + 455 - 322 - 66}{42}$
 $= \frac{585}{42} = \frac{195}{14} = 13\frac{13}{14}$.
32. Given exp. = $\frac{87}{13} - \frac{48}{11} + \frac{12}{5} = \frac{4785 - 3120 + 1716}{715}$
 $= \frac{3381}{715} = 4\frac{521}{715}$.
33. Given exp. = $\frac{28 + 14 + 7 + 4 + 2 + 1}{28} = \frac{56}{28} = 2$.
34. Given exp. = $\frac{1}{(7/3)} + \frac{1}{(7/4)} = \frac{3}{7} + \frac{4}{7} = \frac{7}{7} = 1$.
35. Let $\frac{35}{6} - \frac{35}{9} - x = 1$.
Then, $x = \frac{35}{6} - \frac{35}{9} - 1 = \frac{35}{6} - \left(\frac{35}{9} + 1\right)$
 $= \frac{35}{6} - \frac{44}{9} = \frac{105 - 88}{18} = \frac{17}{18}$.
36. $\frac{1}{x} = 4 - \left(\frac{1}{3} + \frac{1}{2}\right) = 4 - \left(\frac{2+3}{6}\right) = 4 - \frac{5}{6}$
 $= \frac{24 - 5}{6} = \frac{19}{6} \Rightarrow x = \frac{6}{19}$.
37. $\frac{\overset{19}{\cancel{133}}\overset{54}{\cancel{2}}}{\underset{81}{\cancel{532}}\underset{447}{\cancel{21}}} \times \frac{\overset{54}{\cancel{432}}}{\underset{447}{\cancel{588}}} = \frac{38}{63}$.
38. Given expression = $\frac{\overset{3}{\cancel{21}}}{\underset{11}{\cancel{41}}} \times \frac{\overset{11}{\cancel{121}}}{\underset{7}{\cancel{7}}} \times \frac{217}{\underset{2}{\cancel{6}}} = \frac{2387}{2} = 1193\frac{1}{2}$.
39. Given expression = $\left(\frac{3}{7} \times 455 + \frac{5}{8} \times 456\right) = (3 \times 65 + 5 \times 57)$
 $= 195 + 285 = 480$.
40. Given expression = $\left(\frac{3}{5} \times \frac{4}{7} \times \frac{5}{9} \times \frac{21}{24} \times 504\right) = 84$.

$$41. \text{ Given exp. } = \left(\frac{41}{6} \times \frac{16}{3} + \frac{53}{3} \times \frac{9}{2} \right) = \left(\frac{328}{9} + \frac{159}{2} \right) \\ = \frac{656 + 1431}{18} = \frac{2087}{18} = 115 \frac{17}{18}.$$

$$42. \text{ Given exp. } = \frac{24}{5} \div \frac{32}{5} = \frac{24}{5} \times \frac{5}{32} = \frac{3}{4}.$$

$$43. \text{ Given exp. } = \frac{5}{4} + \frac{14}{9} \times \frac{13}{8} \times \frac{2}{13} = \frac{5}{4} + \frac{7}{18} = \frac{45 + 14}{36} = \frac{59}{36} = 1 \frac{23}{36}.$$

$$44. \text{ Given exp. } = \frac{225}{836} \times \frac{152}{245} \div \frac{120}{77} = \frac{225}{836} \times \frac{152}{245} \times \frac{77}{120} = \frac{3}{28}.$$

$$45. \text{ Given exp. } = \left(\frac{82}{5} - \frac{181}{15} \right) \div \frac{247}{81} = \left(\frac{246 - 181}{15} \right) \times \frac{81}{247} \\ = \frac{65}{15} \times \frac{81}{247} = \frac{27}{19} = 1 \frac{8}{19}.$$

$$46. \text{ Let } 18 \frac{3}{4} \times x \div \frac{6}{37} = 1480.$$

$$\text{Then, } \frac{75}{4} \times x \times \frac{37}{6} = 1480 \Leftrightarrow x = \frac{1480 \times 4 \times 6}{75 \times 37} = \frac{64}{5} = 12 \frac{4}{5}.$$

$$47. \text{ Given exp. } = \frac{3}{2} \times \frac{11}{5} \div \frac{5}{4} \div \frac{33}{15} = \frac{3}{2} \times \frac{11}{5} \times \frac{4}{5} \div \frac{33}{15} \\ = \frac{3}{2} \times \frac{11}{5} \times \frac{4}{5} \times \frac{15}{33} = \frac{6}{5} = 1 \frac{1}{5}.$$

$$48. \quad 3 \frac{x}{7} \times 2 \frac{y}{5} = 10 \Leftrightarrow \frac{21+x}{7} \times \frac{10+y}{5} = 10 \quad \dots(i) \\ \Leftrightarrow (21+x)(10+y) = 350 \\ \text{Clearly, } x = 4, y = 4 \text{ satisfy the equation (i).}$$

$$49. \text{ Given exp. } = 1 + 2 \div \left\{ 1 + 2 \div \frac{4}{3} \right\} = 1 + 2 \div \left\{ 1 + 2 \times \frac{3}{4} \right\} \\ = 1 + 2 \div \left\{ 1 + \frac{3}{2} \right\} = 1 + 2 \div \frac{5}{2} = 1 + 2 \times \frac{2}{5} \\ = 1 + \frac{4}{5} = \frac{9}{5} = 1 \frac{4}{5}.$$

$$50. \text{ Given exp. } = \frac{27}{8} \text{ of } \frac{3}{15} \div \frac{5}{36} \text{ of } \frac{5}{49} = \frac{81}{10} \div \frac{25}{63} \\ = \frac{81}{10} \times \frac{63}{25} = \frac{5103}{250} = 20 \frac{103}{250}.$$

$$51. \text{ Given exp. } = \left(-\frac{2}{3} - \frac{1}{3} \right) + \left(\frac{4}{5} + \frac{1}{5} \right) + \left(\frac{3}{4} - \frac{1}{2} \right) \\ = \left(\frac{2}{3} - \frac{4}{3} + \frac{1}{3} \right) - \left(\frac{1}{5} + \frac{4}{5} \right) + \frac{1}{2} \\ = \frac{-1+1+\frac{1}{4}}{-\frac{1}{3}-1+\frac{1}{2}} = \frac{\frac{1}{4}}{\frac{-2-6+3}{6}} = \frac{\frac{1}{4}}{-\frac{5}{6}} \\ = \frac{1}{4} \times \left(-\frac{6}{5} \right) = -\frac{3}{10}.$$

$$52. \quad \left(\frac{\frac{1}{2} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6}}{\frac{2}{5} - \frac{5}{9} + \frac{3}{5} - \frac{7}{18}} \right) = \left(\frac{\frac{30-15+12-10}{60}}{\frac{2}{5} + \frac{3}{5} - \left(\frac{5}{9} + \frac{7}{18} \right)} \right)$$

$$= \frac{\left(\frac{17}{60} \right)}{1 - \frac{17}{18}} = \left(\frac{17}{60} \times 18 \right) = \frac{51}{10} = 5 \frac{1}{10}.$$

$$53. \quad \left(34 \times 4 \frac{1}{2} \right) = 34 \times \left(4 + \frac{1}{2} \right) = (34 \times 4) + \left(34 \times \frac{1}{2} \right) \\ = (30+4) \times 4 + \left(34 \times \frac{1}{2} \right) \\ = (30 \times 4) + (4 \times 4) + \left(34 \times \frac{1}{2} \right).$$

$$54. \text{ Let } \frac{3}{8} \text{ of } 168 \times 15 \div 5 + x = 549 \div 9 + 235. \\ \text{Then, } 63 \times 15 \div 5 + x = 61 + 235 \Leftrightarrow 63 \times 3 + x = 296 \\ \Leftrightarrow 189 + x = 296 \\ \Leftrightarrow x = 107.$$

$$55. \text{ Let } \frac{5}{3} \div \frac{2}{7} \times \frac{x}{7} = \frac{5}{4} \times \frac{2}{3} \div \frac{1}{6}.$$

$$\text{Then, } \frac{5}{3} \times \frac{7}{2} \times \frac{x}{7} = \frac{5}{4} \times \frac{2}{3} \times 6$$

$$\Leftrightarrow \frac{5}{6} x = 5 \Leftrightarrow x = \left(\frac{5 \times 6}{5} \right) = 6.$$

$$56. \text{ Let } 5 \frac{2}{3} \div x \frac{5}{6} = 2.$$

$$\text{Then, } \frac{17}{3} \div x \frac{5}{6} = 2 \Leftrightarrow x \frac{5}{6} = \frac{17}{3} \times \frac{1}{2} = \frac{17}{6} \Leftrightarrow x \frac{5}{6} = 2 \frac{5}{6}.$$

$$\therefore x = 2.$$

$$57. \text{ Given equation is: } \frac{(5x+1)}{x} \times \frac{(4y+3)}{4} = 20 \quad \dots(i) \\ \Leftrightarrow (5x+1)(4y+3) = 80x \\ \text{Clearly, } x = 3 \text{ and } y = 3 \text{ satisfy (i).}$$

$$58. \text{ Required difference } = \frac{19}{16} - \frac{16}{19} = \frac{19^2 - 16^2}{304} \\ = \frac{(19+16)(19-16)}{304} = \frac{35 \times 3}{304} = \frac{105}{304}.$$

$$59. \text{ Required number } = \frac{37 \frac{1}{2}}{1/8} = \frac{75/2}{1/8} = \frac{75}{2} \times 8 = 300.$$

$$60. \quad a = (4 \div 3) \div 3 \div 4 = \frac{4}{3} \times \frac{1}{3} \times \frac{1}{4} = \frac{1}{9}.$$

$$b = 4 \div (3 \div 3) \div 4 = 4 \div 1 \div 4 = 1.$$

$$c = 4 \div 3 \div (3 \div 4) = 4 \div 3 \div \frac{3}{4} = \frac{4}{3} \times \frac{4}{3} = \frac{16}{9}.$$

Clearly, c is the greatest.

$$61. \quad \text{I. } \frac{3}{4} \div \frac{5}{6} = \frac{3}{4} \times \frac{6}{5} = \frac{9}{10}.$$

$$\text{II. } 3 \div [(4 \div 5) \div 6] = 3 \div \left(\frac{4}{5} \times \frac{1}{6} \right) = 3 \div \frac{4}{30} = 3 \times \frac{30}{4} = \frac{45}{2}.$$

$$\text{III. } [3 \div (4 \div 5)] \div 6 = \left[3 \div \frac{4}{5} \right] \div 6 = \left[3 \times \frac{5}{4} \right] \div 6 = \frac{15}{4} \times \frac{1}{6} = \frac{5}{8}.$$

$$\text{IV. } 3 \div 4 \div (5 \div 6) = 3 \div 4 \div \frac{5}{6} = \frac{3}{4} \times \frac{6}{5} = \frac{9}{10}.$$

So, I and IV are equal.

$$62. \text{ Given exp. } = \left(\frac{5}{7} \times \frac{19}{13}\right) \div \left(\frac{19}{7} \times \frac{4}{13}\right) = \frac{5 \times 19}{7 \times 13} \times \frac{7 \times 13}{19 \times 4} = \frac{5}{4}.$$

$$63. \text{ Given exp. } = \frac{11}{4} \div \frac{8}{3} \div \frac{13}{12} = \frac{11}{4} \times \frac{3}{8} \times \frac{12}{13} = \frac{99}{104}.$$

$$64. \text{ Given exp. } = \frac{2}{3} \times \frac{3}{\frac{5}{6} \div \frac{2}{3} \text{ of } \frac{5}{4}} = \frac{2}{3} \times \frac{3}{\frac{5}{6} \div \frac{5}{6}} = \frac{2}{3} \times 3 = 2.$$

$$65. \text{ Let } \frac{4335}{x} \div \frac{15}{8} = \frac{289}{528}.$$

$$\begin{aligned} \text{Then, } \frac{4335}{x} &= \frac{289}{528} \times \frac{15}{8} \Leftrightarrow \frac{4335}{x} = \frac{289 \times 5}{176 \times 8} \\ \Leftrightarrow x &= \left(\frac{4335 \times 176 \times 8}{289 \times 5}\right) = 4224. \end{aligned}$$

$\therefore$ Missing digit = 2.

$$66. \text{ Let } \frac{16}{3} - \frac{11}{3} \div \frac{4}{3} \div x + \frac{16}{5} \div \frac{6}{5} = 7. \text{ Then,}$$

$$\frac{16}{3} - \frac{11}{3} \times \frac{3}{4} \times \frac{1}{x} + \frac{16}{5} \times \frac{5}{6} = 7$$

$$\Leftrightarrow \frac{16}{3} - \frac{11}{4x} + \frac{8}{3} = 7$$

$$\Leftrightarrow \frac{24}{3} - \frac{11}{4x} = 7$$

$$\Leftrightarrow \frac{11}{4x} = 8 - 7 = 1$$

$$\Leftrightarrow 4x = 11 \Leftrightarrow x = \frac{11}{4} = 2\frac{3}{4}.$$

$$67. \text{ Given exp. } = 9 - \frac{11}{9} \text{ of } \frac{36}{11} \div \frac{36}{7} \text{ of } \frac{7}{9} = 9 - 4 \div 4 = 9 - 1 = 8.$$

$$68. \text{ Let } \frac{5}{6} \div \frac{6}{7} \times x - \frac{8}{9} \div \frac{8}{5} + \frac{3}{4} \times \frac{10}{3} = \frac{25}{9}. \text{ Then,}$$

$$\frac{5}{6} \times \frac{7}{6} \times x - \frac{8}{9} \times \frac{5}{8} + \frac{3}{4} \times \frac{10}{3} = \frac{25}{9} \Leftrightarrow \frac{35}{36}x - \frac{5}{9} + \frac{5}{2} = \frac{25}{9}$$

$$\Leftrightarrow \frac{35}{36}x = \frac{25}{9} + \frac{5}{9} - \frac{5}{2} = \frac{10}{3} - \frac{5}{2}$$

$$\Leftrightarrow \frac{35}{36}x = \frac{5}{6} \Leftrightarrow x = \left(\frac{5}{6} \times \frac{36}{35}\right) = \frac{6}{7}.$$

$$69. \text{ Given exp. } = \frac{3}{4} \div \frac{9}{4} \text{ of } \frac{2}{3} - \left(\frac{3-2}{6}\right) \times \frac{10}{3} + \frac{5}{6}$$

$$= \frac{3}{4} \div \frac{9}{4} - \frac{1}{6} \times \frac{6}{5} \times \frac{10}{3} + \frac{5}{6}$$

$$= \frac{3}{4} \times \frac{2}{3} - \frac{2}{3} + \frac{5}{6} = \left(\frac{1}{2} - \frac{2}{3} + \frac{5}{6}\right)$$

$$= \left(\frac{3-4+5}{6}\right) = \frac{4}{6} = \frac{2}{3}.$$

$$70. \frac{\frac{7}{3} + 1\frac{1}{2} \text{ of } \frac{5}{3}}{2 + 1\frac{2}{3}} = \frac{\frac{7}{3} + \frac{3}{2} \text{ of } \frac{5}{3}}{2 + \frac{5}{3}} = \frac{\frac{7}{3} + \frac{5}{2}}{\frac{11}{3}} = \frac{29}{6} \times \frac{3}{11} = \frac{29}{22}.$$

$$\therefore \text{ Required answer } = \frac{29}{22} - \frac{1}{4} = \frac{58-11}{44} = \frac{47}{44} = 1\frac{3}{44}.$$

$$\begin{aligned} 71. \text{ Given exp. } &= \frac{\frac{1}{3} + \frac{3}{4} \left(\frac{6-5}{15}\right)}{\frac{5}{3} \text{ of } \frac{3}{4} - \frac{1}{5}} = \frac{\frac{1}{3} + \frac{3}{4} \times \frac{1}{15}}{\frac{5}{4} - \frac{1}{5}} \\ &= \frac{\frac{1}{3} + \frac{1}{20}}{\frac{25-4}{20}} = \frac{23}{60} \times \frac{20}{21} = \frac{23}{63}. \end{aligned}$$

$$\begin{aligned} 72. \text{ Given exp. } &= 1 - \left(\frac{\frac{5}{3} + \frac{7}{6}}{\frac{7}{2} - \frac{7}{2}}\right) = 1 - \left(\frac{5}{3} \times \frac{2}{7} + \frac{7}{6} \times \frac{2}{7}\right) \\ &= 1 - \left(\frac{10}{21} + \frac{1}{3}\right) = 1 - \frac{17}{21} = \frac{4}{21}. \end{aligned}$$

$$\begin{aligned} 73. \text{ Given exp. } &= \frac{5}{2} \text{ of } \frac{3}{4} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \frac{3}{2} \left[\frac{2}{3} - \frac{1}{3}\right] \\ &= \frac{5}{2} \text{ of } \frac{3}{4} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \left(\frac{3}{2} \times \frac{1}{3}\right) \\ &= \frac{15}{8} \times \frac{1}{2} \div \frac{3}{2} + \frac{1}{2} \div \frac{1}{2} \\ &= \frac{15}{8} \times \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times 2 = \frac{5}{8} + 1 = 1\frac{5}{8}. \end{aligned}$$

$$\begin{aligned} 74. \text{ Given exp. } &= \frac{\frac{8}{7} - \frac{2}{3} + \frac{2}{5} \times \frac{25}{24}}{1 - \frac{1}{7} \left[\frac{1}{3} + \frac{2}{5} \times \frac{5}{3}\right]} = \frac{\frac{8}{7} - \frac{2}{3} + \frac{5}{12}}{1 - \frac{1}{7} \left[\frac{1}{3} + \frac{2}{3}\right]} \\ &= \frac{96-56+35}{84} = \frac{75}{84} \times \frac{7}{6} = \frac{25}{24} = 1\frac{1}{24}. \end{aligned}$$

$$\begin{aligned} 75. \text{ Given exp. } &= \frac{6}{\left(\frac{10}{3}\right)} \div \frac{4 - \frac{2}{\left(\frac{7}{2}\right)}}{\left(\frac{7}{2}\right)} - \frac{2}{5} \text{ of } \left\{\frac{6}{9} + \frac{1}{3}\right\} \\ &= \frac{6 \times 3}{10} \div \frac{4 - \frac{2 \times 2}{7}}{\left(\frac{7}{2}\right)} - \frac{2}{5} \text{ of } 1 \\ &= \frac{9}{5} \div \frac{\left(4 - \frac{4}{7}\right)}{\left(\frac{7}{2}\right)} - \frac{2}{5} = \frac{9}{5} \div \left(\frac{24}{7} \times \frac{2}{7}\right) - \frac{2}{5} \\ &= \frac{9}{5} \times \frac{49}{24} - \frac{2}{5} = \frac{147}{80} - \frac{2}{5} = \frac{147-32}{80} = \frac{115}{80} = \frac{23}{16} = 1\frac{7}{16}. \end{aligned}$$

$$\begin{aligned} 76. \text{ Given exp. } &= \left(\frac{45}{8} \times \frac{7}{45}\right) \text{ of } \left(\frac{73}{11} \times \frac{8}{73}\right) + \frac{8}{9} \left(\frac{25}{11} + \frac{13}{22}\right) \text{ of } \frac{3}{5} \\ &= \frac{7}{8} \text{ of } \frac{8}{11} \div \left(\frac{8}{9} \times \frac{63}{22}\right) \text{ of } \frac{3}{5} \\ &= \frac{7}{11} \div \frac{28}{11} \text{ of } \frac{3}{5} = \frac{7}{11} \div \frac{84}{55} = \frac{7}{11} \times \frac{55}{84} = \frac{5}{12}. \end{aligned}$$

$$77. \text{ Given exp.} = \frac{\frac{1}{3} \times 3 \times \frac{1}{3}}{\frac{1}{3} \div \frac{1}{9}} - \frac{1}{9} = \frac{\frac{1}{3}}{\frac{1}{3} \times 9} - \frac{1}{9} = \frac{1}{3} \times \frac{1}{3} - \frac{1}{9} = \frac{1}{9} - \frac{1}{9} = 0.$$

$$78. \text{ Given exp.} = \frac{\frac{1}{2} \div \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = \frac{\frac{1}{2} \times 4}{\frac{2+1}{4}} = 2 \times \frac{4}{3} = \frac{8}{3} = 2\frac{2}{3}.$$

$$79. \text{ Given exp.} = \frac{\frac{13}{4} - \frac{4}{5} \text{ of } \frac{5}{6}}{\frac{13}{3} \div \frac{1}{5} - \left(\frac{3}{10} + \frac{106}{5}\right)} = \frac{\frac{13}{4} - \frac{2}{3}}{\frac{13}{3} \times 5 - \frac{215}{10}}$$

$$= \frac{\frac{31}{12} - \frac{43}{30}}{\frac{65}{3} - \frac{43}{2}} = \left(\frac{31}{12} \times 6\right) = \frac{31}{2} = 15\frac{1}{2}.$$

$$80. \text{ Let } \frac{\frac{15}{7} - \frac{23}{4}}{\frac{7}{2} + x} \div \frac{\frac{1}{6} + \frac{5}{7}}{\frac{5}{5} + \frac{2}{2}} = \frac{6}{10}.$$

$$\text{Then, } \left[\frac{7}{4} \times \frac{2}{(7+2x)}\right] \div \left[\frac{7}{4} \times \frac{10}{47}\right] = \frac{3}{5}$$

$$\Leftrightarrow \frac{7}{2(7+2x)} = \frac{3}{5} \times \frac{7}{4} \times \frac{10}{47} = \frac{21}{94}$$

$$\Leftrightarrow 7+2x = \left(\frac{7}{2} \times \frac{94}{21}\right) = \frac{47}{3}$$

$$\Leftrightarrow 2x = \frac{47}{3} - 7 = \frac{26}{3} \Leftrightarrow x = \left(\frac{26}{3} \times \frac{1}{2}\right) = \frac{13}{3} = 4\frac{1}{3}.$$

$$81. \frac{5}{3} + \frac{3}{4} = \frac{20+9}{12} = \frac{29}{12} = 2\frac{5}{12} < 5; \frac{7}{3} + \frac{11}{5}$$

$$= \frac{35+33}{15} = \frac{68}{15} = 4\frac{8}{15} < 5;$$

$$\frac{11}{4} + \frac{8}{3} = \frac{33+32}{12} = \frac{65}{12} = 5\frac{5}{12} > 5; \frac{13}{5} + \frac{11}{6}$$

$$= \frac{78+55}{30} = \frac{133}{30} = 4\frac{13}{30} < 5.$$

$$82. \text{ Given exp.} = 5 - \left[\frac{3}{4} + \left\{\frac{5}{2} - \left(\frac{1}{2} + \frac{7-6}{42}\right)\right\}\right]$$

$$= 5 - \left[\frac{3}{4} + \left\{\frac{5}{2} - \left(\frac{1}{2} + \frac{1}{42}\right)\right\}\right]$$

$$= 5 - \left[\frac{3}{4} + \left\{\frac{5}{2} - \frac{22}{42}\right\}\right] = 5 - \left[\frac{3}{4} + \frac{83}{42}\right] = 5 - \frac{229}{84}$$

$$= \left(\frac{420-229}{84}\right) = \frac{191}{84} = 2\frac{23}{84}.$$

$$83. \text{ Given exp.} = -1 + 4 + 3 = 6.$$

$$84. x^3 - x^2 - x = (-3)^3 - (-3)^2 - (-3) = -27 - 9 + 3 = -33.$$

$$85. \text{ Putting } S = 8 \text{ and } R = 16, \text{ we have:}$$

$$R = gS - 4 \Leftrightarrow 16 = 8g - 4$$

$$\Leftrightarrow 8g = 20 \Leftrightarrow g = \frac{20}{8} = \frac{5}{2}.$$

$$\therefore \text{ When } S = 10, R = gS - 4$$

$$\Leftrightarrow R = \frac{5}{2} \times 10 - 4 = 25 - 4 = 21.$$

$$86. \frac{x(y-z)}{y(x+y+z)} = \frac{5(3-2)}{3(5+3+2)} = \frac{5}{3 \times 10} = \frac{1}{6}.$$

$$87. a^3 - b^3 + 3a^2b = 7^3 - 5^3 + 3 \times 7^2 \times 5$$

$$= 343 - 125 + 735 = 953.$$

$$88. \text{ Given exp.} = -7m - \{3n - \{8m - 4n + 10m\}\}$$

$$= -7m - \{3n - \{18m - 4n\}\}$$

$$= -7m - \{3n - 18m + 4n\} = -7m - \{7n - 18m\}$$

$$= -7m - 7n + 18m = 11m - 7n.$$

$$89. \frac{2}{a} + \frac{1}{b} = \frac{2b+a}{ab} = \frac{6}{4} = \frac{3}{2}.$$

$$90. \text{ Let } \frac{a}{3} = \frac{b}{4} = \frac{c}{7} = k. \text{ Then, } a = 3k, b = 4k, c = 7k.$$

$$\therefore \frac{(a+b+c)}{c} = \frac{3k+4k+7k}{7k} = \frac{14k}{7k} = 2.$$

$$91. \text{ Given exp.} = 2a^3 - \{3a^3 + 4b^3 - \{-5a^3\} + 5a^3 - 7b^3\}$$

$$= 2a^3 - \{3a^3 + 4b^3 + 5a^3 + 5a^3 - 7b^3\}$$

$$= 2a^3 - \{13a^3 - 3b^3\} = 2a^3 - 13a^3 + 3b^3$$

$$= -11a^3 + 3b^3.$$

$$92. \frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} = \frac{1}{1+\frac{x}{y+z}} + \frac{1}{1+\frac{y}{z+x}} + \frac{1}{1+\frac{z}{x+y}}$$

$$= \frac{y+z}{x+y+z} + \frac{z+x}{x+y+z} + \frac{x+y}{x+y+z}$$

$$= \frac{2(x+y+z)}{(x+y+z)} = 2.$$

$$93. (-2x-6)^2 = [-2(x+3)]^2 = 4(x+3)^2 = 4y.$$

$$94. (x+a)(x+b) = x^2 + ax + bx + ab = x^2 + (a+b)x + ab.$$

$$95. \left[1 + \frac{1}{x+1}\right] \left[1 + \frac{1}{x+2}\right] \left[1 + \frac{1}{x+3}\right] \left[1 + \frac{1}{x+4}\right]$$

$$= \left[\frac{(x+1)+1}{x+1}\right] \left[\frac{(x+2)+1}{x+2}\right] \left[\frac{(x+3)+1}{x+3}\right] \left[\frac{(x+4)+1}{x+4}\right]$$

$$= \left(\frac{x+2}{x+1}\right) \left(\frac{x+3}{x+2}\right) \left(\frac{x+4}{x+3}\right) \left(\frac{x+5}{x+4}\right) = \frac{x+5}{x+1}.$$

$$96. (a-b) - (c+d) = 6 \text{ and } (c-d) - (a+b) = 3$$

$$\Rightarrow (a-c) - (b+d) = 6 \text{ and } (c-a) - (b+d) = 3$$

$$\Rightarrow (b+d) = (a-c) - 6 \text{ and } (b+d) = (c-a) - 3$$

$$\Rightarrow (a-c) - 6 = (c-a) - 3 \Rightarrow 2(a-c) = 3$$

$$\Rightarrow (a-c) = \frac{3}{2} = 1.5.$$

$$97. \frac{3a+2b}{3a-2b} = \frac{3\frac{a}{b}+2}{3\frac{a}{b}-2} = \frac{3 \times \frac{4}{3} + 2}{3 \times \frac{4}{3} - 2} = \frac{6}{2} = 3.$$

$$98. \left(\frac{6}{7} - \frac{5x-y}{5x+y}\right) = \frac{6}{7} - \left(\frac{\frac{5x}{y} - 1}{\frac{5x}{y} + 1}\right) = \frac{6}{7} - \left(\frac{5 \times \frac{6}{5} - 1}{5 \times \frac{6}{5} + 1}\right)$$

$$= \frac{6}{7} - \left(\frac{6-1}{6+1}\right) = \frac{6}{7} - \frac{5}{7} = \frac{1}{7}.$$

$$99. \frac{x^2 + y^2}{x^2 - y^2} = \frac{\frac{x^2}{y^2} + 1}{\frac{x^2}{y^2} - 1} = \frac{\left(\frac{x}{y}\right)^2 + 1}{\left(\frac{x}{y}\right)^2 - 1} = \frac{\left(\frac{1}{3}\right)^2 + 1}{\left(\frac{1}{3}\right)^2 - 1}$$

$$= \frac{\left(\frac{1}{9} + 1\right)}{\left(\frac{1}{9} - 1\right)} = \frac{10}{9} \times \left(-\frac{9}{8}\right) = -\frac{5}{4}.$$

$$100. \frac{x}{2y} = \frac{6}{7} \Rightarrow \frac{x}{y} = \left(2 \times \frac{6}{7}\right) = \frac{12}{7}.$$

$$\therefore \frac{x-y}{x+y} + \frac{14}{19} = \frac{\frac{x}{y} - 1}{\frac{x}{y} + 1} + \frac{14}{19} = \frac{\frac{12}{7} - 1}{\frac{12}{7} + 1} + \frac{14}{19} = \frac{(5/7)}{(19/7)} + \frac{14}{19}$$

$$= \left(\frac{5}{7} \times \frac{7}{19}\right) + \frac{14}{19} = \frac{5}{19} + \frac{14}{19} = \frac{19}{19} = 1.$$

$$101. \frac{a}{b} = \frac{4}{5} \text{ and } \frac{b}{c} = \frac{15}{16} \Rightarrow \left(\frac{a}{b} \times \frac{b}{c}\right) = \left(\frac{4}{5} \times \frac{15}{16}\right) \Rightarrow \frac{a}{c} = \frac{3}{4}.$$

$$\therefore \frac{c^2 - a^2}{c^2 + a^2} = \frac{1 - \left(\frac{a^2}{c^2}\right)}{1 + \left(\frac{a^2}{c^2}\right)} = \frac{1 - \left(\frac{a}{c}\right)^2}{1 + \left(\frac{a}{c}\right)^2} = \frac{1 - \frac{9}{16}}{1 + \frac{9}{16}} = \frac{(7/16)}{(25/16)} = \frac{7}{25}.$$

$$102. x = y \Leftrightarrow 1 - q = 2q + 1 \Leftrightarrow 3q = 0 \Leftrightarrow q = 0.$$

$$103. \frac{x}{5} - \frac{x}{6} = 4 \Leftrightarrow \frac{6x - 5x}{30} = 4 \Leftrightarrow x = 120.$$

$$104. \frac{3x}{2y} = \frac{21}{22} \Rightarrow \frac{x}{y} = \left(\frac{21}{22} \times \frac{2}{3}\right) = \frac{7}{11} \Rightarrow x = \frac{7}{11}y.$$

$$4x + 5y = 83 \Rightarrow 4 \times \frac{7}{11}y + 5y = 83 \Rightarrow \frac{28}{11}y + 5y = 83$$

$$\Rightarrow 83y = 83 \times 11 \Rightarrow y = 11.$$

$$\therefore x = \frac{7}{11}y = \left(\frac{7}{11} \times 11\right) = 7.$$

$$\text{So, } y - x = 11 - 7 = 4.$$

$$105. \frac{ab}{a+b} = \frac{\frac{x}{x+y} \cdot \frac{y}{x-y}}{\frac{x}{x+y} + \frac{y}{x-y}} = \frac{\frac{xy}{x^2 - y^2}}{\frac{x(x-y) + y(x+y)}{x^2 - y^2}} = \frac{xy}{x^2 + y^2}.$$

$$106. \frac{a}{b} = \frac{1}{3} \Rightarrow b = 3a; \frac{b}{c} = 2 \Rightarrow c = \frac{b}{2} = \frac{3a}{2};$$

$$\frac{c}{d} = \frac{1}{2} \Rightarrow d = 2c = 2\left(\frac{3a}{2}\right) = 3a;$$

$$\frac{d}{e} = 3 \Rightarrow e = \frac{d}{3} = \left(\frac{3a}{3}\right) = a; \frac{e}{f} = \frac{1}{4} \Rightarrow f = 4e = 4a.$$

$$\therefore \frac{abc}{def} = \frac{(a)(3a)\left(\frac{3a}{2}\right)}{(3a)(a)(4a)} = \frac{9}{2}a^3 \times \frac{1}{12a^3} = \frac{3}{8}.$$

$$107. \frac{m}{n} = \frac{4}{3} \text{ and } \frac{r}{t} = \frac{9}{14} \Rightarrow \frac{mr}{nt} = \frac{4}{3} \times \frac{9}{14} = \frac{6}{7}.$$

$$\therefore \frac{3mr - nt}{4nt - 7mr} = \frac{3\frac{mr}{nt} - 1}{4 - 7\frac{mr}{nt}} = \frac{3 \times \frac{6}{7} - 1}{4 - 7 \times \frac{6}{7}}$$

$$= \frac{\frac{18}{7} - 1}{4 - 6} = \frac{11}{7} \times \left(-\frac{1}{2}\right) = -\frac{11}{14}.$$

$$108. a + \frac{1}{b} = 1 \Rightarrow ab + 1 = b \Rightarrow ab - b = -1$$

$$\Rightarrow b(a - 1) = -1 \Rightarrow b = \frac{1}{(1-a)}.$$

$$b + \frac{1}{c} = 1 \Rightarrow bc + 1 = c \Rightarrow bc - c = -1$$

$$\Rightarrow c(b - 1) = -1 \Rightarrow c = \frac{1}{(1-b)}.$$

$$\therefore c + \frac{1}{a} = \frac{1}{a} + \frac{1}{a} = \frac{1}{1 - \left(\frac{1}{1-a}\right)} + \frac{1}{a} = \frac{1}{\frac{(1-a)-1}{(1-a)}} + \frac{1}{a}$$

$$= \frac{(1-a)}{-a} + \frac{1}{a} = \frac{(a-1)}{a} + \frac{1}{a} = \frac{a-1+1}{a} = \frac{a}{a} = 1.$$

$$109. y + \frac{1}{z} = 1 \Rightarrow yz + 1 = z \Rightarrow yz - z = -1$$

$$\Rightarrow z(y - 1) = -1 \Rightarrow z = \frac{1}{(1-y)}.$$

$$x + \frac{1}{y} = 1 \Rightarrow xy + 1 = y \Rightarrow xy - y = -1$$

$$\Rightarrow y(x - 1) = -1 \Rightarrow y = \frac{1}{(1-x)}.$$

$$\therefore xyz = x \left(\frac{1}{1-x}\right) \left(\frac{1}{1-y}\right) = x \left(\frac{1}{1-x}\right) \left(\frac{1}{1 - \frac{1}{1-x}}\right)$$

$$= x \left(\frac{1}{1-x}\right) \left(\frac{1-x}{-x}\right) = -1.$$

$$110. \frac{x}{(2x+y+z)} = a \Rightarrow x = a(2x+y+z) \quad \dots(i)$$

$$\frac{y}{(x+2y+z)} = a \Rightarrow y = a(x+2y+z) \quad \dots(ii)$$

$$\frac{z}{(x+y+2z)} = a \Rightarrow z = a(x+y+2z) \quad \dots(iii)$$

Adding (i), (ii) and (iii), we get: $x + y + z$

$$= a(4x + 4y + 4z) \Rightarrow a = \frac{x+y+z}{4(x+y+z)} = \frac{1}{4}.$$

$$111. \frac{a}{b+c} = K \Rightarrow a = K(b+c) \quad \dots(i)$$

$$\frac{b}{c+a} = K \Rightarrow b = K(c+a) \quad \dots(ii)$$

$$\frac{c}{a+b} = K \Rightarrow c = K(a+b) \quad \dots(iii)$$

Adding (i), (ii), and (iii), we get: $a + b + c$

$$= K(2a + 2b + 2c) \Rightarrow K = \frac{a + b + c}{2(a + b + c)} = \frac{1}{2}.$$

112. $a - 8 = b \Rightarrow a - b = 8$ and $b - a = -8$.

$$\therefore |a - b| - |b - a| = |8| - |-8| = 8 - 8 = 0.$$

113. $x = \frac{a}{a-1} = 1 + \frac{1}{a-1} = 1 + y. \quad \therefore x > y.$

114. a is positive and $a < 1 \Rightarrow \frac{1}{a} > 1. \quad \therefore \left(a + \frac{1}{a}\right) > 2.$

115. $\frac{a}{x} + \frac{y}{b} = 1 \Rightarrow \frac{a}{x} = 1 - \frac{y}{b} = \frac{b-y}{b} \Rightarrow \frac{x}{a} = \frac{b}{b-y}.$

$$\frac{b}{y} + \frac{z}{c} = 1 \Rightarrow \frac{z}{c} = 1 - \frac{b}{y} = \frac{y-b}{y} \Rightarrow \frac{c}{z} = \frac{y}{y-b} = \frac{-y}{(b-y)}.$$

$$\therefore \frac{x}{a} + \frac{c}{z} = \frac{b}{(b-y)} - \frac{y}{(b-y)} = \frac{(b-y)}{(b-y)} = 1.$$

116. $a^2 + b^2 = 45 \quad \dots(i)$
and $b^2 + c^2 = 40 \quad \dots(ii)$

Subtracting, we get: $a^2 - c^2 = 5 \Rightarrow (a + c)(a - c) = 5.$

$$\therefore (a + c) = 5 \text{ and } (a - c) = 1.$$

Solving, we get: $a = 3, c = 2$. Putting $c = 2$ in (ii), we get: $b = 6$.

117. $x + y = 15 \Rightarrow (x + y)^2 = (15)^2 = 225$
 $\Rightarrow x^2 + y^2 + 2xy = 225$
 $\Rightarrow x^2 + y^2 = 225 - 2xy = 225 - 2 \times 56$
 $= 225 - 112 = 113.$

118. $(a - b) = 4 \Rightarrow (a - b)^2 = 4^2 = 16 \Rightarrow a^2 + b^2 - 2ab = 16$
 $\Rightarrow a^2 + b^2 = 16 + 2ab = 16 + 2 \times 2$
 $= 16 + 4 = 20.$

119. $a = bc$ and $c = a - b \Rightarrow a = b(a - b) \Rightarrow a = ab - b^2$
 $\Rightarrow b^2 = ab - a = a(b - 1) \Rightarrow a = \frac{b^2}{b-1}.$

120. $a + b + c = 0 \Rightarrow a = -(b + c) \Rightarrow a^2 = (b + c)^2.$

$$\therefore \frac{a^2}{(a^2 - bc)} + \frac{b^2}{(b^2 - ca)} + \frac{c^2}{(c^2 - ab)}$$

$$= \frac{(b+c)^2}{(b+c)^2 - bc} + \frac{b^2}{b^2 + c(b+c)} + \frac{c^2}{c^2 + b(b+c)}$$

$$= \frac{(b+c)^2}{b^2 + c^2 + bc} + \frac{b^2}{b^2 + c^2 + bc} + \frac{c^2}{b^2 + c^2 + bc}$$

$$= \frac{b^2 + c^2 + 2bc + b^2 + c^2}{b^2 + c^2 + bc} = \frac{2(b^2 + c^2 + bc)}{b^2 + c^2 + bc} = 2.$$

121. $\frac{x^2 - y^2}{x^2 + y^2} = \frac{\left(\frac{x^2}{y^2}\right) - 1}{\left(\frac{x^2}{y^2}\right) + 1} = \frac{\left(\frac{x}{y}\right)^2 - 1}{\left(\frac{x}{y}\right)^2 + 1} = \frac{\left(\frac{a+2}{a-2}\right)^2 - 1}{\left(\frac{a+2}{a-2}\right)^2 + 1}$
 $= \frac{(a+2)^2 - (a-2)^2}{(a+2)^2 + (a-2)^2} = \frac{4 \times a \times 2}{2(a^2 + 4)} = \frac{4a}{a^2 + 4}.$

122. $3x + 7 = 7x + 5 \Rightarrow 7x - 3x = 2 \Rightarrow 4x = 2 \Rightarrow x = \frac{1}{2}.$

Now, $3x + 7 = x^2 + P \Rightarrow \frac{3}{2} + 7 = \frac{1}{4} + P$
 $\Rightarrow P = \frac{17}{2} - \frac{1}{4} = \frac{33}{4} = 8\frac{1}{4}.$

123. $\frac{2a+b}{a+4b} = 3 \Rightarrow 2a+b = 3(a+4b) \Rightarrow a = -11b.$

$$\frac{a+b}{a+2b} = \frac{-11b+b}{-11b+2b} = \frac{-10b}{-9b} = \frac{10}{9}.$$

124. $(2a + 3b)(2c - 3d) = (2a - 3b)(2c + 3d)$
 $\Rightarrow \frac{(2a + 3b)}{(2a - 3b)} = \frac{(2c + 3d)}{(2c - 3d)}$

$$\Rightarrow \frac{2\left(\frac{a}{b}\right) + 3}{2\left(\frac{a}{b}\right) - 3} = \frac{2\left(\frac{c}{d}\right) + 3}{2\left(\frac{c}{d}\right) - 3} \Rightarrow \frac{a}{b} = \frac{c}{d}.$$

125. $(a + b + 2c + 3d)(a - b - 2c + 3d)$
 $= (a - b + 2c - 3d)(a + b - 2c - 3d)$
 $\Rightarrow [(a + b) + (2c + 3d)][(a - b) - (2c - 3d)]$
 $= [(a - b) + (2c - 3d)][(a + b) - (2c + 3d)]$
 $\Rightarrow (a + b)(a - b) - (a + b)(2c - 3d) + (a - b)(2c + 3d)$
 $- (2c + 3d)(2c - 3d)$
 $= (a - b)(a + b) - (a - b)(2c + 3d) + (a + b)(2c - 3d)$
 $- (2c + 3d)(2c - 3d)$
 $\Rightarrow (a + b)(2c - 3d) = (a - b)(2c + 3d)$
 $\Rightarrow 2ac - 3ad + 2bc - 3bd = 2ac + 3ad - 2bc - 3bd$
 $\Rightarrow 4bc = 6ad \Rightarrow 2bc = 3ad.$

126. $\frac{x+y}{x-y} + \frac{(x+y)^2}{x^2 - y^2} = \frac{x+y}{x-y} \times \frac{(x^2 - y^2)}{(x+y)^2}$
 $= \frac{(x^2 - y^2)}{(x-y)(x+y)} = \frac{(x^2 - y^2)}{(x^2 - y^2)} = 1.$

127. $\frac{a^2 - b^2 - 2bc - c^2}{a^2 + b^2 + 2ab - c^2} = \frac{a^2 - (b^2 + 2bc + c^2)}{(a^2 + b^2 + 2ab) - c^2} = \frac{a^2 - (b+c)^2}{(a+b)^2 - c^2}$
 $= \frac{(a+b+c)(a-b-c)}{(a+b+c)(a+b-c)} = \frac{a-b-c}{a+b-c}.$

128. $x + y + z = 0 \Rightarrow y = -(x + z).$
 $\therefore x^2 + xy + y^2 = x^2 + x(-x - z) + (x + z)^2$
 $= x^2 - x^2 - xz + x^2 + z^2 + 2xz$
 $= x^2 + z^2 + zx.$

129. $(s - a)^2 + (s - b)^2 + (s - c)^2 + s^2$
 $= (s^2 + a^2 - 2sa) + (s^2 + b^2 - 2sb) + (s^2 + c^2 - 2sc) + s^2$
 $= 4s^2 + (a^2 + b^2 + c^2) - 2s(a + b + c)$
 $= (2s)^2 + (a^2 + b^2 + c^2) - (a + b + c)(a + b + c)$
 $= (a + b + c)^2 + (a^2 + b^2 + c^2) - (a + b + c)^2$
 $= a^2 + b^2 + c^2.$

130. $[(x + y)^2 - z^2] + [(y + z)^2 - x^2] + [(z + x)^2 - y^2]$
 $= 4 + 9 + 36$
 $\Rightarrow (x + y + z)(x + y - z) + (x + y + z)$
 $(y + z - x) + (x + y + z)(z + x - y) = 49$
 $\Rightarrow (x + y + z)[(x + y - z) + (y + z - x) + (z + x - y)] = 49$
 $\Rightarrow (x + y + z)(x + y + z) = 49 \Rightarrow (x + y + z)^2 = 49$
 $\Rightarrow (x + y + z) = \pm 7.$

131. $a + b = 2c \Rightarrow a = 2c - b$.

$$\begin{aligned}\therefore \frac{a}{a-c} + \frac{b}{b-c} &= \frac{2c-b}{(2c-b)-c} + \frac{b}{(b-c)} = \frac{2c-b}{c-b} + \frac{b}{b-c} \\ &= \frac{b-2c}{b-c} + \frac{b}{b-c} = \frac{b-2c+b}{b-c} = \frac{2(b-c)}{b-c} = 2.\end{aligned}$$

132. $\frac{x}{x-1} + \frac{1}{x+1} + \frac{2x}{1-x^2} = \frac{x}{x-1} + \frac{1}{x+1} - \frac{2x}{x^2-1}$

$$\begin{aligned}&= \frac{x(x+1) + (x-1) - 2x}{(x^2-1)} \\ &= \frac{x^2 + x + x - 1 - 2x}{x^2-1} = \frac{x^2-1}{x^2-1} = 1.\end{aligned}$$

133. $\frac{(a+b)^2}{(a^2-b^2)} = \frac{(a+b)(a+b)}{(a+b)(a-b)} = \frac{a+b}{a-b}$.

134. $ab + bc + ca = 0 \Rightarrow ab = -bc - ca, bc = -ab - ca, ca = -ab - bc$.

$$\begin{aligned}\therefore \frac{1}{a^2-bc} + \frac{1}{b^2-ca} + \frac{1}{c^2-ab} \\ &= \frac{1}{a^2+ab+ac} + \frac{1}{b^2+ab+bc} + \frac{1}{c^2+bc+ca} \\ &= \frac{1}{a(a+b+c)} + \frac{1}{b(a+b+c)} + \frac{1}{c(a+b+c)} \\ &= \frac{bc+ca+ab}{abc(a+b+c)} = 0. [\because ab+bc+ca=0]\end{aligned}$$

135. Number of boxes required = $\frac{15 \times 1000}{250} = 60$.

136. Bus fare for 1 person = ₹ 420.
Train fare for 1 person = ₹ $\left(\frac{3}{4} \times 840\right)$ = ₹ 630.

$$\therefore \text{Total fare} = ₹ (3 \times 420 + 4 \times 630) = ₹ (1260 + 2520) = ₹ 3780.$$

137. Let the cost of one pen be ₹ x . Then, cost of 1 pencil = ₹ $\left(\frac{x}{3}\right)$.

$$\therefore 6x + 3 \times \frac{x}{3} = 84 \Rightarrow 7x = 84 \Rightarrow x = 12.$$

So, total cost of 4 pens and 5 pencils

$$= ₹ \left(4x + \frac{5x}{3}\right) = ₹ \left(\frac{17x}{3}\right) = ₹ \left(\frac{17 \times 12}{3}\right) = ₹ 68.$$

138. Amount received by each person = ₹ $\left(\frac{436563}{69}\right)$ = ₹ 6327.

$$\begin{array}{r} 69 \overline{)436563} \quad 6327 \\ \underline{414} \\ 225 \\ \underline{207} \\ 186 \\ \underline{138} \\ 483 \\ \underline{483} \\ \times \end{array}$$

139. Length of reel = $(32 \times 120 + 80)$ cm = 3920 cm = 39.20 m.

140. Number of bananas required daily = $\frac{798}{7} = 114$.

$$\begin{aligned}\therefore \text{Required number} &= [114 \times (\overset{\text{Jan}}{31} + \overset{\text{Feb}}{29} + \overset{\text{Mar}}{31})] \\ &= (114 \times 91) = 10374.\end{aligned}$$

141. Suppose Mohan has ₹ x . Then, Ram has ₹ $(x + 6)$ and Sohan has ₹ $(x - 3)$.

$$\begin{aligned}\therefore x + (x + 6) + (x - 3) &= 33 \Leftrightarrow 3x = 30 \Leftrightarrow x = 10. \\ \text{So, Ram's share} &= ₹ (10 + 6) = ₹ 16.\end{aligned}$$

142. 4 gal = (4×4) qt = 16 qt = (16×2) pt = 32 pt.

$$\therefore \text{Number of bottles} = \frac{32}{\left(\frac{1}{2}\right)} = 64.$$

143. $\frac{1}{2}A = \frac{5}{6}B \Rightarrow A = \left(2 \times \frac{5}{6}\right)B = \frac{5}{3}B$.

$$\begin{aligned}\therefore A + B = 80 &\Rightarrow \frac{5}{3}B + B = 80 \Rightarrow \frac{8}{3}B = 80 \\ &\Rightarrow B = \left(\frac{80 \times 3}{8}\right) = 30 \text{ kg}.\end{aligned}$$

144. $a = b + 2 \Rightarrow b = a - 2$

$$\begin{aligned}\text{And, } b = c + 10 &\Rightarrow c = b - 10 = (a - 2) - 10 = a - 12. \\ a + b + c = 130 &\Rightarrow a + (a - 2) + (a - 12) = 130 \\ &\Rightarrow 3a = 144 \Rightarrow a = 48. \\ \therefore (b + c) - a &= [(a - 2) + (a - 12)] - a \\ &= (2a - 14) - a = (a - 14) = 48 - 14 = 34.\end{aligned}$$

145. Let the given condition be fulfilled n years after 2002.

$$\begin{aligned}\text{Then, price of item X after } n \text{ years} &= ₹ (420 + 40n). \\ \text{price of item Y after } n \text{ years} &= ₹ (630 + 15n). \\ \therefore 420 + 40n &= (630 + 15n) + 40 \\ \Rightarrow 420 + 40n &= 670 + 15n \Rightarrow 25n = 250 \Rightarrow n = 10. \\ \text{So, the required year is } &10 \text{ years after 2002 i.e., 2012.}\end{aligned}$$

146. Number of pieces = $\left(\frac{42.5 \times 100}{85}\right) = \frac{4250}{85} = 50$.

147. Total number of children = $(30 \times 16) = 480$.

$$\therefore \text{Number of columns of 24 children each} = \left(\frac{480}{24}\right) = 20.$$

148. Original number of sections = $(16 - 3) = 13$.

Original number of students = $(24 \times 13) = 312$.

Present number of students = $(21 \times 16) = 336$.

Number of new students admitted = $(336 - 312) = 24$.

149. Money earned in 2 days = ₹ $(20 - 15) = ₹ 5$.

$$\text{Money earned in 16 days} = ₹ \left(\frac{5}{2} \times 16\right) = ₹ 40.$$

On 17th day, money in hand = ₹ $(40 + 20) = ₹ 60$.

150. Let the number of cars sold on the first day be x .

$$\begin{aligned}\text{Then, } x + (x + 6) + (x + 12) + (x + 18) + (x + 24) \\ + (x + 30) &= 150 \\ \Leftrightarrow 6x + 90 &= 150 \Leftrightarrow 6x = 60 \Leftrightarrow x = 10.\end{aligned}$$

$$\therefore \text{Number of cars sold on 6th day} = (10 + 30) = 40.$$

151. Let the number of captains in the group be x .

Then, number of soldiers = $15x$.

$$\therefore x + 15x = 1200 \Leftrightarrow 16x = 1200 \Leftrightarrow x = 75.$$

152. Required cost = ₹ $[1000 \times x + (z - 1000) \times y]$

$$\begin{aligned}&= ₹ (1000x + zy - 1000y) \\ &= ₹ [1000(x - y) + yz].\end{aligned}$$

153. Let the monthly salary of a man be ₹ x .
Then, monthly salary of a woman = ₹ $(x + 500)$.
 $\therefore 4x + 2(x + 500) = 46000 \Leftrightarrow 6x = 45000 \Leftrightarrow x = 7500$.
Monthly salary of a woman = $x + 500 = ₹ 8000$.

154. Let the number of pineapples and watermelons be x and y respectively.

$$\text{Then, } 7x + 5y = 38 \text{ or } 5y = (38 - 7x) \text{ or } y = \frac{38 - 7x}{5}.$$

Clearly, y is a whole number, only when $(38 - 7x)$ is divisible by 5. This happens when $x = 4$.

155. Let F and C denote the temperatures in Fahrenheit and Celsius respectively.

$$\text{Then, } \frac{F - 32}{212 - 32} = \frac{C - 0}{100 - 0} \Leftrightarrow \frac{F - 32}{180} = \frac{C}{100}.$$

$$\text{If } C = 35, \text{ then } F = \left(\frac{35}{100} \times 180 \right) + 32 = 63 + 32 = 95.$$

156. Let the two parts be x and $(74 - x)$.
Then, $5x + 11(74 - x) = 454 \Leftrightarrow 6x = 360 \Leftrightarrow x = 60$.
So, the two parts are 60 and 14.

157. Let the quantity of water drained away be x litres.
Then, $12 - x = x - 6 \Rightarrow 2x = 18 \Rightarrow x = 9$.

158. Let the number of boys in the family be x .
Then each boy has $(x - 1)$ brothers and $(x - 1)$ sisters.
So, number of girls = $(x - 1)$.
Thus each girl has x brothers and $(x - 2)$ sisters.
 $\therefore x = 2(x - 2) \Leftrightarrow x = 2x - 4 \Leftrightarrow x = 4$.
So, there are 4 boys and 3 girls i.e., 4 brothers and 3 sisters in the family.

159. $5000 \text{ INR} = 100 \text{ USD} = \left(100 \times \frac{1}{0.01} \right) \text{ Japanese Yen}$
 $= 10000 \text{ Japanese Yen}$
 $\Rightarrow 1 \text{ INR} = 2 \text{ Japanese Yen}$
 $\Rightarrow 100 \text{ INR} = 200 \text{ Japanese Yen}.$

160. Capacity of the tank = $(13.5 \times 12) \text{ litres} = 162 \text{ litres}$.
 $\therefore$ Required number of buckets = $\left(\frac{162}{9} \right) = 18$.

161. Given exp. = $\left[1 + \frac{1}{1 + \frac{1}{(4/3)}} \right] \div \frac{11}{7} = \left[1 + \frac{1}{1 + \frac{3}{4}} \right] \div \frac{11}{7}$
 $= \left[1 + \frac{1}{(7/4)} \right] \div \frac{11}{7} = \left(1 + \frac{4}{7} \right) \div \frac{11}{7} = \frac{11}{7} \div \frac{11}{7} = 1.$

162. (a) $\frac{1}{3 + \frac{1}{1 + \frac{1}{16}}} = \frac{1}{3 + \frac{1}{(17/16)}} = \frac{1}{3 + \frac{16}{17}} = \frac{1}{(67/17)} = \frac{17}{67}.$

(b) $\frac{1}{2 + \frac{1}{1 + \frac{1}{8}}} = \frac{1}{2 + \frac{1}{(9/8)}} = \frac{1}{2 + \frac{8}{9}} = \frac{1}{(26/9)} = \frac{9}{26}.$

(c) $\frac{1}{3 + \frac{1}{1 + \frac{1}{1 + \frac{1}{8}}}} = \frac{1}{3 + \frac{1}{1 + \frac{1}{(9/8)}}} = \frac{1}{3 + \frac{1}{1 + \frac{8}{9}}}$

$$= \frac{1}{3 + \frac{1}{(17/9)}} = \frac{1}{3 + \frac{9}{17}} = \frac{1}{\left(\frac{60}{17} \right)} = \frac{17}{60}.$$

(d) $\frac{1}{3 + \frac{1}{1 + \frac{1}{2 + \frac{1}{4}}}} = \frac{1}{3 + \frac{1}{1 + \frac{1}{(9/4)}}} = \frac{1}{3 + \frac{1}{1 + \frac{4}{9}}}$
 $= \frac{1}{3 + \frac{1}{(13/9)}} = \frac{1}{3 + \frac{9}{13}} = \frac{1}{(48/13)} = \frac{13}{48}.$

163. Given exp. = $\frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{1}{(5/2)}}}} + \frac{3}{3 - \frac{4}{3 + \frac{1}{(3/2)}}}$
 $= \frac{1}{1 + \frac{1}{3 - \frac{4}{2 + \frac{2}{5}}}} + \frac{3}{3 - \frac{4}{3 + \frac{2}{3}}}$
 $= \frac{1}{1 + \frac{1}{3 - \frac{4}{(12/5)}}} + \frac{3}{3 - \frac{4}{(11/3)}}$
 $= \frac{1}{1 + \frac{1}{3 - \frac{20}{12}}} + \frac{3}{3 - \frac{12}{11}} = \frac{1}{1 + \frac{1}{3 - \frac{5}{3}}} + \frac{3}{(21/11)}$
 $= \frac{1}{1 + \frac{1}{(4/3)}} + \frac{33}{21}$
 $= \frac{1}{1 + \frac{3}{4}} + \frac{11}{7} = \frac{1}{(7/4)} + \frac{11}{7} = \frac{4}{7} + \frac{11}{7} = \frac{15}{7}.$

164. $x = 2 - \frac{1}{1 + \frac{1}{(13/4)}} = 2 - \frac{1}{1 + \frac{4}{13}} = 2 - \frac{1}{(17/13)} = 2 - \frac{13}{17} = \frac{21}{17}.$

165. $x = \frac{2 + \frac{1}{(19/5)}}{2 + \frac{1}{3 + \frac{1}{(5/4)}}} = \frac{2 + \frac{5}{19}}{2 + \frac{1}{3 + \frac{4}{5}}} = \frac{2 + \frac{5}{19}}{2 + \frac{1}{(19/5)}} = \frac{2 + \frac{5}{19}}{2 + \frac{5}{19}} = 1.$

166. Given exp. = $8 - 8 \times \frac{\frac{11}{5} - \frac{9}{7}}{2 - \frac{1}{(35/6)}}$
 $= 8 - 8 \times \frac{\frac{32}{35}}{2 - \frac{6}{35}} = 8 - 8 \times \frac{32}{35} \times \frac{35}{64} = 8 - 4 = 4.$

167. Given exp. = $\frac{2}{2 + \frac{2}{3 + \frac{2}{(11/3)}} \times 0.39} = \frac{2}{2 + \frac{2}{3 + \frac{6}{11}} \times 0.39}$

$$\begin{aligned}
 &= \frac{2}{2 + \frac{2}{(39/11)} \times 0.39} \\
 &= \frac{2}{2 + \frac{22}{39} \times \frac{39}{100}} = \frac{2}{2 + \frac{22}{100}} \\
 &= \frac{2}{2 + \frac{11}{50}} = \frac{2}{(111/50)} = \frac{100}{111}.
 \end{aligned}$$

168. Given exp. = $\frac{1}{1 + \frac{\frac{2}{3}}{\frac{5}{3} + \frac{8}{9} \times 3}} = \frac{1}{1 + \frac{2/3}{5/3 + 8/3}} = \frac{1}{1 + \frac{2/3}{13/3}} = \frac{13}{15}.$

169. $\frac{2}{2 + \frac{2}{2 + \frac{2}{x}}} = 3 \Leftrightarrow \frac{2}{2 + \frac{2}{\left(\frac{2x+2}{x}\right)}} = 3$

$$\begin{aligned}
 &\Leftrightarrow \frac{2}{2 + \frac{2x}{2x+2}} = 3 \Leftrightarrow \frac{2}{\frac{2(2x+2)+2x}{2x+2}} = 3 \\
 &\Leftrightarrow \frac{2(2x+2)}{2(3x+2)} = 3 \\
 &\Leftrightarrow \frac{(2x+2)}{(3x+2)} = 3 \Leftrightarrow 2x+2 = 9x+6 \\
 &\Leftrightarrow 7x = -4 \Leftrightarrow x = -\frac{4}{7}.
 \end{aligned}$$

170. $2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}} = \frac{37}{13} = 2\frac{11}{13} = 2 + \frac{11}{13} \Rightarrow \frac{1}{x + \frac{1}{y + \frac{1}{z}}} = \frac{11}{13}$

$$\Rightarrow x + \frac{1}{y + \frac{1}{z}} = 1 + \frac{2}{11}$$

$$\Rightarrow x + \frac{1}{y + \frac{1}{z}} = 1 + \frac{2}{11}$$

$$\Rightarrow x = 1, y + \frac{1}{z} = \frac{11}{2} = 5\frac{1}{2} = 5 + \frac{1}{2}$$

$$\Rightarrow x = 1, y = 5, z = 2.$$

171. Clearly, we have: $x = 1 + \frac{1}{x} = \frac{x+1}{x}.$

$$\Rightarrow x^2 = x + 1 \Rightarrow x^2 - x - 1 = 0.$$

172. Let $x = \frac{1}{3 + \frac{1}{4 + \dots}}$

$$\begin{aligned}
 &\dots \\
 &+ \frac{1}{1008}
 \end{aligned}$$

Then, $P = \frac{1}{1 + \frac{1}{1+x}} = \frac{1+x}{2+x}, Q = \frac{1}{2+x}.$

$$\therefore P + Q = \frac{1+x}{2+x} + \frac{1}{2+x} = \frac{2+x}{2+x} = 1.$$

173. Given exp. = $\frac{a^2 - b^2}{a+b}$, where $a = \left(1 + \frac{1}{10 + \frac{1}{10}}\right)$ and

$$\begin{aligned}
 b &= \left(1 - \frac{1}{10 + \frac{1}{10}}\right) \\
 &= (a-b) = \left(1 + \frac{1}{10 + \frac{1}{10}}\right) - \left(1 - \frac{1}{10 + \frac{1}{10}}\right) \\
 &= 2 \times \frac{1}{(101/10)} = \frac{20}{101}.
 \end{aligned}$$

174. $3x + y = 19 \dots(i)$ and $x - y = 9 \dots(ii)$

Adding (i) and (ii), we get: $4x = 28$ or $x = 7.$

Putting $x = 7$ in (i), we get: $y = -2.$

175. $4x = p(x+3) + q(x-1) \Rightarrow 4x = px + 3p + qx - q$

$$\begin{aligned}
 &\Rightarrow 4x = (p+q)x + (3p-q) \\
 &\Rightarrow p+q = 4 \text{ and } 3p-q = 0 \\
 &\Rightarrow p+q = 4 \text{ and } p = \frac{q}{3} \\
 &\Rightarrow \frac{q}{3} + q = 4 \Rightarrow \frac{4q}{3} = 4 \Rightarrow q = 3.
 \end{aligned}$$

$$p + q = 4 \text{ and } q = 3 \Rightarrow p = 1.$$

176. $xyz = 256 \Rightarrow (2z)(2z)z = 256 \Rightarrow 4z^3$

$$= 256 \Rightarrow z^3 = 64 \Rightarrow z = 4.$$

$$\therefore x = 2z = (2 \times 4) = 8.$$

177. $\frac{28x}{13y} = \frac{140}{39} \Rightarrow \frac{x}{y} = \left(\frac{140}{39} \times \frac{13}{28}\right) = \frac{5}{3} \Rightarrow x = \frac{5}{3}y.$

$$3y + 9x = 54 \Rightarrow y + 3x = 18$$

$$\Rightarrow y + 3 \times \frac{5}{3}y = 18 \Rightarrow 6y = 18 \Rightarrow y = 3.$$

$$\therefore x = \left(\frac{5}{3} \times 3\right) = 5.$$

$$\text{So, } y - x = (3 - 5) = -2.$$

178. $3x + 7y = 75 \dots(i)$

$$5x - 5y = 25 \Rightarrow x - y = 5 \Rightarrow 7x - 7y = 35 \dots(ii)$$

Adding (i) and (ii), we get: $10x = 110$ or $x = 11.$

Putting $x = 11$ in (i), we get: $7y = 42$ or $y = 6.$

$$\therefore x + y = (11 + 6) = 17.$$

179. $a + b = 5 \dots(i)$ and $3a + 2b = 20 \dots(ii)$

Multiplying (i) by 2 and subtracting from (ii), we get: $a = 10.$

Putting $a = 10$ in (i), we get: $b = -5.$

$$\therefore (3a + b) = 3 \times 10 + (-5) = 30 - 5 = 25.$$

180. $(2p + 3q) + (2p - q) = 18 + 2 \Rightarrow 4p + 2q = 20$

$$\Rightarrow 2(2p + q) = 20 \Rightarrow 2p + q = 10.$$

181. $2x + y = 5$...(i) and $3x - 4y = 2$...(ii)
 Multiplying (i) by 4 and adding (ii) to it, we get:
 $11x = 22$ or $x = 2$.
 Putting $x = 2$ in (i), we get: $y = 1$.
 So, $2xy = 2 \times 2 \times 1 = 4$.

182. $3x - 5y = 5$...(i) and $\frac{x}{x+y} = \frac{5}{7} \Rightarrow 7x = 5x + 5y$
 $\Rightarrow 2x - 5y = 0$...(ii)
 Subtracting (ii) from (i), we get: $x = 5$.
 Putting $x = 5$ in (i), we get: $y = 2$. So, $x - y = 5 - 2 = 3$.

183. $4x + 3y = 18xy$...(i) and $2x - 5y = -4xy$...(ii)
 Dividing (i) and (ii) by xy , we get:
 $\frac{3}{x} + \frac{4}{y} = 18$...(iii) and $\frac{5}{x} - \frac{2}{y} = 4$...(iv)

Multiplying (iv) by 2 and adding (iii) to it, we get:
 $\frac{13}{x} = 26$ or $x = \frac{1}{2}$.

Putting $x = \frac{1}{2}$ in (iii), we get: $y = \frac{1}{3}$.

184. $2x + y = 17$...(i)
 $y + 2z = 15$...(ii)
 and $x + y = 9$...(iii)
 Subtracting (iii) from (i), we get: $x = 8$.
 Putting $x = 8$ in (i), we get: $y = 1$. Putting $y = 1$ in (ii),
 we get: $2z = 14$ or $z = 7$.
 $\therefore 4x + 3y + z = 4 \times 8 + 3 \times 1 + 7 = 42$.

185. $3x - 4y + z = 7$...(i)
 $2x + 3y - z = 19$...(ii)
 and $x + 2y + 2z = 24$...(iii)
 Adding (i) and (ii), we get: $5x - y = 26$...(iv)
 Subtracting (i) from (ii) and adding to (iii), we get:
 $9y = 36$ or $y = 4$.
 Putting $y = 4$ in (iv), we get: $5x = 30$ or $x = 6$.
 Putting $x = 6, y = 4$ in (iii), we get: $2z = 10$ or $z = 5$.

186. $2x + y = 15$...(i)
 $2y + z = 25$...(ii)
 and $2z + x = 26$...(iii)
 Adding (i), (ii) and (iii), we get: $3(x + y + z) = 66$ or $x + y + z = 22$...(iv)
 From (ii), we have: $y = \frac{25 - z}{2}$. From (iii), we have:
 $x = 26 - 2z$.

$$\therefore (26 - 2z) + \left(\frac{25 - z}{2}\right) + z = 22 \Leftrightarrow 77 - 3z = 44$$

$$\Leftrightarrow 3z = 33 \Leftrightarrow z = 11.$$

187. $2x + 3y = 31$...(i)
 $y - z = 4$...(ii)
 and $x + 2z = 11$...(iii)
 Multiplying (iii) by 2 and subtracting from (i), we get:
 $3y - 4z = 9$...(iv)
 Solving (ii) and (iv), we get: $y = 7, z = 3$.
 Putting $y = 7$ in (i), we get: $x = 5$.
 $\therefore x + y + z = (5 + 7 + 3) = 15$.

188. Let the cost of a chair and that of a table be ₹ x and ₹ y respectively.

$$\text{Then, } 10x = 4y \text{ or } y = \frac{5}{2}x.$$

$$\therefore 15x + 2y = 4000 \Leftrightarrow 15x + 2 \times \frac{5}{2}x = 4000 \Leftrightarrow 20x = 4000 \Leftrightarrow x = 200.$$

$$\text{So, } y = \left(\frac{5}{2} \times 200\right) = 500.$$

$$\therefore \text{Cost of 12 chairs and 3 tables} = 12x + 3y \\ = ₹ (12 \times 200 + 3 \times 500) = ₹ (2400 + 1500) \\ = ₹ 3900.$$

189. Let the price of a jean be ₹ x and that of a shirt be ₹ y .
 Then, $2x + 3y = 4000$...(i) and $3x + 2y = 3500$...(ii)

Adding (i) and (ii),

$$\text{we get: } 5x + 5y = 7500 \text{ or } x + y = 1500 \text{ ...(iii)}$$

$$\text{Subtracting (ii) from (i), we get: } -x + y = 500 \text{ ...(iv)}$$

$$\text{Adding (iii) and (iv), we get: } 2y = 2000 \text{ or } y = 1000.$$

$$\text{Putting } y = 1000 \text{ in (iii), we get: } x = 500.$$

Hence, cost of a jean = ₹ 500.

190. Let the cost of one pen be ₹ x and that of one pencil be ₹ y .

$$\text{Then, } 8x + 4y = 176 \text{ or } 2x + y = 44 \text{ ...(i)}$$

$$\text{And, } 2x + 2y = 48 \text{ or } x + y = 24 \text{ ...(ii)}$$

$$\text{Subtracting (ii) from (i), we get: } x = 20.$$

Hence, cost of 1 pen = ₹ 20.

191. Let the cost of 1 dozen apples be ₹ x and that of 1 dozen bananas be ₹ y . Then,

$$2x + 3y = 136 \text{ ...(i) and, } x + 5y = 110 \text{ or } 2x + 10y = 220 \text{ ...(ii)}$$

$$\text{Subtracting (i) from (ii), we get: } 7y = 84 \text{ or } y = 12.$$

192. Let the cost of 1 pencil, 1 pen and 1 eraser be ₹ x , ₹ y and ₹ z respectively.

$$\text{Then, } x + 2y + 4z = 22 \text{ ...(i)}$$

$$\text{and } 5x + 4y + 2z = 32 \text{ ...(ii)}$$

Adding (i) and (ii), we get:

$$6(x + y + z) = 54 \text{ or } x + y + z = 9.$$

$$\therefore \text{Cost of 3 pencils, 3 pens and 3 erasers} \\ = 3(x + y + z) = ₹ (3 \times 9) = ₹ 27.$$

193. Let the number of cows and horses bought be x and y respectively.

$$\text{Then, number of chicken bought} = 100 - (x + y).$$

$$1000x + 300y + 50 [100 - (x + y)] = 10000$$

$$\Rightarrow 1000x + 300y + 5000 - 50(x + y) = 10000$$

$$\Rightarrow 950x + 250y = 5000 \Rightarrow 50(19x + 5y) = 5000$$

$$\Rightarrow 19x + 5y = 100 \Rightarrow 5y = 100 - 19x \Rightarrow y = \frac{100 - 19x}{5}.$$

Clearly, for y to be a whole number, $(100 - 19x)$ must be divisible by 5 and this is possible only when $x = 5$.

$$\text{When } x = 5, y = \frac{100 - 19 \times 5}{5} = \frac{5}{5} = 1.$$

$$\therefore \text{Number of chicken} = [100 - (5 + 1)] = 94.$$

- 194.** Let 1 man's daily earning be ₹ x and 1 boy's daily earning be ₹ y .

$$\text{Then, } 3x + 4y = \frac{756}{7} = 108 \quad \dots(i)$$

$$\text{And, } 11x + 13y = \frac{3008}{8} = 376 \quad \dots(ii)$$

Multiplying (i) by 11 and (ii) by 3, we get:

$$33x + 44y = 1188 \quad \dots(iii)$$

$$33x + 39y = 1128 \quad \dots(iv)$$

Subtracting (iv) from (iii), we get: $5y = 60$ or $y = 12$.

Putting $y = 12$ in (i), we get: $3x = 60$ or $x = 20$.

$\therefore$ Required time

$$= \frac{2480}{7x + 9y} = \frac{2480}{7 \times 20 + 9 \times 12} = \frac{2480}{248} = 10 \text{ days.}$$

- 195.** Let the price of 1 burger, 1 shake and 1 fries be ₹ x , ₹ y and ₹ z respectively. Then,

$$3x + 7y + z = 120 \quad \dots(i)$$

$$4x + 10y + z = 164.50 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $x + 3y = 44.50$

$$\text{or } 2x + 6y = 89 \quad \dots(iii)$$

Subtracting (iii) from (i), we get: $x + y + z = (120 - 89) = 31$.

$\therefore$ Cost of 1 burger, 1 shake and 1 fries = ₹ 31.

- 196.** $2x^2 + 12x + 18 = 0 \Leftrightarrow x^2 + 6x + 9 = 0$

$$\Leftrightarrow x^2 + 3x + 3x + 9 = 0$$

$$\Leftrightarrow x(x + 3) + 3(x + 3) = 0$$

$$\Leftrightarrow (x + 3)^2 = 0 \Leftrightarrow x = -3.$$

- 197.** $x^2 - 7x = -12 \Leftrightarrow x^2 - 7x + 12 = 0$

$$\Leftrightarrow x^2 - 3x - 4x + 12 = 0$$

$$\Leftrightarrow x(x - 3) - 4(x - 3) = 0$$

$$\Leftrightarrow (x - 3)(x - 4) = 0$$

$$\Leftrightarrow x = 3 \text{ or } x = 4.$$

- 198.** $2x^2 - 11x + 15 = 0 \Leftrightarrow 2x^2 - 6x - 5x + 15 = 0$

$$\Leftrightarrow 2x(x - 3) - 5(x - 3) = 0$$

$$\Leftrightarrow (x - 3)(2x - 5) = 0$$

$$\Leftrightarrow x = 3 \text{ or } x = \frac{5}{2}.$$

- 199.** Given exp. = $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{m-1}{m} = \frac{1}{m}$.

- 200.** Given exp. = $\frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \dots \times \frac{n+1}{n} = \frac{n+1}{3}$.

- 201.** Given exp. = $\frac{5}{3} \times \frac{7}{5} \times \frac{9}{7} \times \dots \times \frac{1001}{999} = \frac{1001}{3}$.

- 202.** Given exp. = $\frac{3}{4} \times \frac{4}{3} \times \frac{5}{3} \times \frac{3}{5} \times \frac{13}{7} \times \frac{1}{13} = \frac{1}{7}$.

- 203.** Given exp.

$$= \left(\frac{3^2 - 1}{3^2} \right) \left(\frac{4^2 - 1}{4^2} \right) \left(\frac{5^2 - 1}{5^2} \right) \dots \left(\frac{11^2 - 1}{11^2} \right) \left(\frac{12^2 - 1}{12^2} \right)$$

$$= \left[\frac{(3+1)(3-1)}{(2+1)(4-1)} \right] \left[\frac{(4+1)(4-1)}{(3+1)(5-1)} \right] \left[\frac{(5+1)(5-1)}{(4+1)(6-1)} \right]$$

$$\dots \left[\frac{(11+1)(11-1)}{(10+1)(12-1)} \right] \left[\frac{(12+1)(12-1)}{(11+1)(13-1)} \right]$$

$$= \frac{(3-1)}{(2+1)} \times \frac{(12+1)}{(13-1)} = \frac{2}{3} \times \frac{13}{12} = \frac{13}{18}.$$

- 204.** Given exp. = $\frac{1}{(2-1)(2+1)} + \frac{1}{(4-1)(4+1)} + \frac{1}{(6-1)(6+1)} + \dots + \frac{1}{(20-1)(20+1)}$

$$= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{19 \times 21}$$

$$= \frac{1}{2} \left(1 - \frac{1}{3} \right) + \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right) + \frac{1}{2} \left(\frac{1}{5} - \frac{1}{7} \right) + \dots + \frac{1}{2} \left(\frac{1}{19} - \frac{1}{21} \right)$$

$$= \frac{1}{2} \left(1 - \frac{1}{21} \right) = \frac{1}{2} \times \frac{20}{21} = \frac{10}{21}.$$

- 205.** Given exp. = $\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right)$

$$+ \left(\frac{1}{4} - \frac{1}{5} \right) + \dots + \left(\frac{1}{11} - \frac{1}{12} \right) = \left(1 - \frac{1}{12} \right) = \frac{11}{12}.$$

- 206.** Clearly, sum of first 6 terms is zero. So, sum of first 30 terms = 0.

$$\therefore \text{ Required sum} = \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \frac{1}{2} - \frac{1}{3} \right) = -\frac{1}{4}.$$

- 207.** Given exp. = $(1 + 11 + 111 + 1111) + \left(\frac{1}{2} \times 4 \right)$

$$= 1234 + 2 = 1236.$$

- 208.** Given exp. = $\left(1000 - \frac{1}{1000} \right) \times 7 = \left(7000 - \frac{7}{1000} \right)$

$$= 6999 \frac{993}{1000}.$$

- 209.** Given exp. = $\left(1000 - \frac{4}{999} \right) \times 999 = 999000 - 4 = 998996.$

- 210.** Given exp. = $\left(1000 - \frac{6}{7} \right) + \left(1000 - \frac{5}{7} \right) + \left(1000 - \frac{4}{7} \right)$

$$+ \left(1000 - \frac{3}{7} \right) + \left(1000 - \frac{2}{7} \right) + \left(1000 - \frac{1}{7} \right)$$

$$= 6000 - \left(\frac{6}{7} + \frac{5}{7} + \frac{4}{7} + \frac{3}{7} + \frac{2}{7} + \frac{1}{7} \right)$$

$$= 6000 - \frac{21}{7} = 6000 - 3 = 5997.$$

Another Method:

$$\text{Given exp.} = (999 \times 6) + \left(\frac{1}{7} + \frac{2}{7} + \frac{3}{7} + \frac{4}{7} + \frac{5}{7} + \frac{6}{7} \right)$$

$$= 5994 + \frac{21}{7} = 5994 + 3 = 5997.$$

- 211.** Given exp. = $(998 \times 5) + \left(\frac{2}{17} + \frac{3}{17} + \frac{5}{17} + \frac{8}{17} + \frac{16}{17} \right)$

$$= 4990 + \frac{34}{17} = 4990 + 2 = 4992.$$

- 212.** Given exp. = $\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{9} - \frac{1}{10} \right)$

$$= \left(1 - \frac{1}{10} \right) = \frac{9}{10}.$$

$$\begin{aligned}
 213. \text{ Given exp. } &= \frac{1}{3} \left(1 - \frac{1}{4} \right) + \frac{1}{3} \left(\frac{1}{4} - \frac{1}{7} \right) + \frac{1}{3} \left(\frac{1}{7} - \frac{1}{10} \right) \\
 &+ \frac{1}{3} \left(\frac{1}{10} - \frac{1}{13} \right) + \frac{1}{3} \left(\frac{1}{13} - \frac{1}{16} \right) \\
 &= \frac{1}{3} \left[\left(1 - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{10} \right) \right. \\
 &\quad \left. + \left(\frac{1}{10} - \frac{1}{13} \right) + \left(\frac{1}{13} - \frac{1}{16} \right) \right] \\
 &= \frac{1}{3} \left(1 - \frac{1}{16} \right) = \frac{1}{3} \times \frac{15}{16} = \frac{5}{16}.
 \end{aligned}$$

$$\begin{aligned}
 214. \text{ Given exp. } &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) \\
 &+ \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\
 &= \left(1 - \frac{1}{n+1} \right) = \frac{n}{n+1}.
 \end{aligned}$$

$$215. \text{ Given exp. } = \frac{4 \times 3^3 + 3^2 + 3 + 1}{4 \times 3^3} = \frac{108 + 9 + 3 + 1}{108} = \frac{121}{108}.$$

$$\begin{aligned}
 216. \text{ Given exp. } &= \frac{4 \cdot 5 \cdot 6 + 5 \cdot 6 + 2 \cdot 6 + 2 \cdot 3}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \frac{120 + 30 + 12 + 6}{720} \\
 &= \frac{168}{720} = \frac{7}{30}.
 \end{aligned}$$

$$\begin{aligned}
 217. \text{ Given exp. } &= \left(\frac{1}{1^2} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{3^2} \right) + \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \\
 &+ \left(\frac{1}{4^2} - \frac{1}{5^2} \right) + \dots + \left(\frac{1}{9^2} - \frac{1}{10^2} \right) \\
 &= \left(\frac{1}{1^2} - \frac{1}{10^2} \right) = \left(1 - \frac{1}{100} \right) = \frac{99}{100}.
 \end{aligned}$$

$$\begin{aligned}
 218. \text{ Given exp. } &= \frac{4-1}{4 \times 1} + \frac{9-4}{9 \times 4} + \frac{16-9}{16 \times 9} + \dots \\
 &= \left(1 - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{9} \right) + \left(\frac{1}{9} - \frac{1}{16} \right) + \dots \\
 &= \left(\frac{1}{1^2} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{3^2} \right) + \left(\frac{1}{3^2} - \frac{1}{4^2} \right) + \dots
 \end{aligned}$$

$$\therefore 99\text{th term of the series} = \left(\frac{1}{99^2} - \frac{1}{100^2} \right).$$

$$\begin{aligned}
 \therefore \text{ Given exp. } &= \left(\frac{1}{1^2} - \frac{1}{2^2} \right) + \left(\frac{1}{2^2} - \frac{1}{3^2} \right) + \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \\
 &+ \dots + \left(\frac{1}{98^2} - \frac{1}{99^2} \right) + \left(\frac{1}{99^2} - \frac{1}{100^2} \right) \\
 &= \left(1 - \frac{1}{100^2} \right) = \left(1 - \frac{1}{10000} \right) = \frac{9999}{10000}.
 \end{aligned}$$

219. Suppose Mayank, Mirza and Little paid ₹ x , ₹ y and ₹ z respectively.

$$\text{Then, } x = \frac{1}{2} (60000 - x), y = \frac{1}{3} (60000 - y), z = \frac{1}{4} (60000 - z).$$

$$\Rightarrow 2x = 60000 - x, 3y = 60000 - y, 4z = 60000 - z$$

$$\Rightarrow 3x = 60000, 4y = 60000, 5z = 60000$$

$$\Rightarrow x = 20000, y = 15000, z = 12000.$$

Amount paid by Jaspal

$$= ₹ [60000 - (20000 + 15000 + 12000)] = ₹ 13000.$$

220. Income after 1 year = ₹ (4×2^1) lakhs.

Income after 2 years = ₹ $(4 \times 2 \times 2)$ lakhs

$$= ₹ (4 \times 2^2) \text{ lakhs}$$

$\therefore$ Income after 5 years = ₹ (4×2^5) lakhs = ₹ 128 lakhs

$$= ₹ 1.28 \text{ crores.}$$

221. Since the population doubles every day, so it was half-full on 27/2/2007 and quarter-full on 26/2/2007.

222. Since the number doubles every minute, so the basket was $\frac{1}{2^n}$ full in $(60 - n)$ minutes.

Thus, it was $\frac{1}{32} \left(= \frac{1}{2^5} \right)$ full in $(60 - 5)$ i.e., 55 minutes.

223. Total amount received by the man in 2010

$$= ₹ [(5000 \times 2^4) + (5000 \times 2^3) + (5000 \times 2^2) + (5000 \times 2)]$$

$$= ₹ (80000 + 40000 + 20000 + 10000) = ₹ 150000.$$

224. True length of the rope = 120 m + (3×120) cm

$$= 120 \text{ m} + 360 \text{ cm}$$

$$= 120 \text{ m} + 3 \text{ m } 60 \text{ cm} = 123 \text{ m } 60 \text{ cm.}$$

225. Let the number be x . Then, $63x - 36x = 3834$

$$\Leftrightarrow 27x = 3834$$

$$\Leftrightarrow x = 142.$$

226. Among the given numbers, only 60489 is a multiple of 423.

227. Let the given number be x .

$$\text{Then, } \frac{7}{8}x - \frac{7}{18}x = 770 \Leftrightarrow \frac{63x - 28x}{72} = 770 \Leftrightarrow \frac{35x}{72} = 770$$

$$\Leftrightarrow x = \left(\frac{770 \times 72}{35} \right) = 1584.$$

228. Let the number be x .

$$\text{Then, } \frac{17}{8}x - \frac{8}{17}x = 225 \Leftrightarrow \frac{289x - 64x}{136} = 225$$

$$\Leftrightarrow \frac{225}{136}x = 225 \Leftrightarrow x = 136.$$

$$\therefore \text{ Expected answer} = 136 + \frac{17}{8} = \left(136 \times \frac{8}{17} \right) = 64.$$

229. Let the number be $2x$.

$$\text{Then, } \left(\frac{x}{6} + \frac{x}{4} \right) - \frac{2x}{5} = 4 \Leftrightarrow \frac{10x + 15x - 24x}{60} = 4 \Leftrightarrow x = 240.$$

Hence, required number = $2x = 480$.

230. Final answer = $\left(9\frac{1}{9} - 3\frac{2}{3} \right) \times 450 = \left(\frac{82}{9} - \frac{11}{3} \right) \times 450$

$$= \left(\frac{49}{9} \times 450 \right) = 2450.$$

231. $\frac{7}{8}$ of 1008 - $\frac{3}{4}$ of 968 = $\left(\frac{7}{8} \times 1008 \right) - \left(\frac{3}{4} \times 968 \right)$

$$= 882 - 726 = 156.$$

232. $x = y - \frac{y}{10} \Rightarrow x = \frac{9y}{10} \Rightarrow x = \frac{9}{10} \times y \Rightarrow x \propto y$.

Thus, as y increases, x also increases.

233. Let the number be x . Then,

$$\frac{3}{4} \text{ of } \frac{4}{5} \text{ of } x - \frac{1}{6} \text{ of } \frac{2}{5} \text{ of } x = 648$$

$$\Rightarrow \frac{3x}{5} - \frac{x}{15} = 648 \Rightarrow \frac{8x}{15} = 648 \Rightarrow x = \frac{648 \times 15}{8} = 1215.$$

234. Let the required number be x .

$$\text{Then, } x + 1\frac{1}{2} = x \times 1\frac{1}{2} \Rightarrow x + \frac{3}{2} = \frac{3}{2}x \Rightarrow \frac{x}{2} = \frac{3}{2} \Rightarrow x = 3.$$

235. Let the required sum be ₹ x .

$$\text{Then, } \frac{5}{12} \text{ of } x = 3\frac{3}{4} \text{ of } 100 \Leftrightarrow \frac{5x}{12} = \left(\frac{15}{4} \times 100\right) = 375$$

$$\Leftrightarrow x = \left(\frac{375 \times 12}{5}\right) = 900.$$

236. Let x of $\frac{1}{12} = \frac{3}{8}$. Then, $\frac{x}{12} = \frac{3}{8} \Leftrightarrow x = \left(\frac{3}{8} \times 12\right) = \frac{9}{2}$.

$$\text{237. Sum of given fractions} = \frac{7}{4} + \frac{5}{2} + \frac{67}{12} + \frac{10}{3} + \frac{9}{4}$$

$$= \left(\frac{21 + 30 + 67 + 40 + 27}{12}\right) = \frac{185}{12}.$$

The whole number just less than $\frac{185}{12}$ is 15.

$$\text{Let } \frac{185}{12} - x = 15. \text{ Then, } x = \left(\frac{185}{12} - 15\right) = \frac{5}{12}.$$

$$\text{238. Sum of given fractions} = 5\frac{3}{4} + 4\frac{4}{5} + 7\frac{3}{8} = \frac{23}{4} + \frac{24}{5} + \frac{59}{8}$$

$$= \frac{230 + 192 + 295}{40} = \frac{717}{40} = 17\frac{37}{40}.$$

$$\therefore \text{ Required fraction} = \left(18 - 17\frac{37}{40}\right) = \frac{3}{40}.$$

239. Clearly, $\frac{x+1}{x}$ is the only fraction in which the numerator is greater than the denominator. So, it is the greatest fraction.

$$\text{240. } \frac{6}{7/8} - \frac{6/7}{8} = 6 \times \frac{8}{7} - \frac{6}{7} \times \frac{1}{8} = \frac{48}{7} - \frac{6}{56}$$

$$= \frac{384 - 6}{56} = \frac{378}{56} = \frac{27}{4} = 6\frac{3}{4}.$$

241. Let the value of the estate be ₹ x .

$$\text{Then, } \frac{4}{5} \text{ of } x = 16800 \Leftrightarrow x = \left(\frac{16800 \times 5}{4}\right) = 21000$$

$$\Leftrightarrow \frac{3}{7}x = \left(\frac{3}{7} \times 21000\right) = 9000.$$

242. Let the number be x . Then,

$$\frac{2}{5} \text{ of } \frac{1}{4} \text{ of } \frac{3}{7} \text{ of } x = 15 \Leftrightarrow x = \left(15 \times \frac{7}{3} \times 4 \times \frac{5}{2}\right) = 350$$

$$\Leftrightarrow \frac{1}{2}x = 175.$$

$$\text{243. Required fraction} = \frac{1 \text{ sec.}}{1 \text{ hr.}} = \frac{1 \text{ sec.}}{(1 \times 60 \times 60) \text{ sec.}} = \frac{1}{3600}.$$

$$\text{244. Height at the third bounce} = \left[32 \times \left(\frac{3}{4}\right)^3\right] \text{ m} = \left(32 \times \frac{27}{64}\right) \text{ m}$$

$$= \frac{27}{2} \text{ m} = 13\frac{1}{2} \text{ m}.$$

245. Suppose Sanket earns ₹ x in each of the other eleven months.

Then, Sanket's earning in March = ₹ $(2x)$.

Sanket's annual earning = ₹ $(11x + 2x) = ₹ (13x)$.

$$\therefore \text{ Required fraction} = \frac{2x}{13x} = \frac{2}{13}.$$

246. Let the capacity of the tank be x litres.

$$\text{Then, } \frac{1}{3}x = 80 \Leftrightarrow x = 240 \Leftrightarrow \frac{1}{2}x = 120.$$

$$\text{247. Distance travelled on foot} = \left[\frac{7}{2} - \left(\frac{5}{3} + \frac{7}{6}\right)\right] \text{ km}$$

$$= \left(\frac{7}{2} - \frac{17}{6}\right) \text{ km} = \frac{2}{3} \text{ km}.$$

$$\therefore \text{ Required fraction} = \frac{(2/3)}{(7/2)} = \left(\frac{2}{3} \times \frac{2}{7}\right) = \frac{4}{21}.$$

248. Let the required fraction be x . Then,

$$\frac{4}{7}x + \frac{4}{7} = \frac{15}{14} \Leftrightarrow \frac{4}{7}x = \left(\frac{15}{14} - \frac{4}{7}\right) = \frac{7}{14} = \frac{1}{2}$$

$$\Leftrightarrow x = \left(\frac{1}{2} \times \frac{7}{4}\right) = \frac{7}{8}.$$

$$\text{249. Required fraction} = \frac{\frac{2}{3} \text{ of } \frac{1}{4} \text{ of ₹ } 25.20}{\frac{3}{2} \text{ of ₹ } 36} = \frac{\text{₹ } 4.20}{\text{₹ } 54} = \frac{42}{540} = \frac{7}{90}.$$

250. Let the length of longer piece be x cm.

$$\text{Then, length of shorter piece} = \left(\frac{2}{5}x\right) \text{ cm}.$$

$$\therefore x + \frac{2}{5}x = 70 \Leftrightarrow \frac{7x}{5} = 70 \Leftrightarrow x = \left(\frac{70 \times 5}{7}\right) = 50.$$

$$\text{Hence, length of shorter piece} = \frac{2}{5}x = \left(\frac{2}{5} \times 50\right) \text{ cm} = 20 \text{ cm}.$$

251. Let the whole amount be ₹ x . Then,

$$A's \text{ share} = ₹ \left(\frac{3}{16}x\right); B's \text{ share} = ₹ \left(\frac{x}{4}\right);$$

$$\text{and } C's \text{ share} = ₹ \left[x - \left(\frac{3x}{16} + \frac{x}{4}\right)\right] = ₹ \left(\frac{9x}{16}\right).$$

$$\therefore \frac{9x}{16} = 81 \Leftrightarrow x = \left(\frac{81 \times 16}{9}\right) = 144.$$

$$\text{Hence, } B's \text{ share} = ₹ \left(\frac{144}{4}\right) = ₹ 36.$$

$$\text{252. Green portion} = \left[1 - \left(\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{40} + \frac{1}{50} + \frac{1}{60}\right)\right]$$

$$= \left[1 - \frac{1}{10} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)\right]$$

$$= 1 - \frac{1}{10} \times \frac{147}{60} = 1 - \frac{147}{600} = \frac{453}{600}.$$

Let the length of the pole be x metres.

$$\text{Then, } \frac{453}{600}x = 12.08 \Leftrightarrow x = \left(\frac{12.08 \times 600}{453} \right) = 16.$$

- 253.** Let the fraction be $\frac{a}{b}$. Then,

$$\begin{aligned} \left(\frac{a}{b} \times \frac{a}{b} \right) \div \frac{b}{a} &= \frac{512}{27} \Leftrightarrow \frac{a}{b} \times \frac{a}{b} \times \frac{a}{b} = \frac{512}{27} \\ \Leftrightarrow \left(\frac{a}{b} \right)^3 &= \left(\frac{8}{3} \right)^3 \Leftrightarrow \frac{a}{b} = \frac{8}{3} = 2\frac{2}{3}. \end{aligned}$$

- 254.** Maximum internal assessment score = $\left(\frac{47}{50} \times 10 \right) = 9.4$.

$$\text{Minimum internal assessment score} = \left(\frac{14}{50} \times 10 \right) = 2.8.$$

$$\therefore \text{Required difference} = (9.4 - 2.8) = 6.6.$$

- 255.** Let savings in N.S.C. and P.P.F. be ₹ x and ₹ $(150000 - x)$ respectively. Then,

$$\frac{1}{3}x = \frac{1}{2}(150000 - x) \Leftrightarrow \frac{x}{3} + \frac{x}{2} = 75000$$

$$\Leftrightarrow \frac{5x}{6} = 75000$$

$$\Leftrightarrow x = \left(\frac{75000 \times 6}{5} \right) = 90000.$$

$$\therefore \text{Savings in Public Provident Fund} = ₹ (150000 - 90000) = ₹ 60000.$$

- 256.** Let there be $(x + 1)$ members. Then,

$$\text{Father's share} = \frac{1}{4}, \text{ share of each other member} = \frac{3}{4x}.$$

$$\therefore 3\left(\frac{3}{4x}\right) = \frac{1}{4} \Leftrightarrow 4x = 36 \Leftrightarrow x = 9.$$

Hence, total number of family members = 10.

- 257.** Let salary = ₹ x . Then, tips = ₹ $\left(\frac{5}{4}x\right)$.

$$\text{Total income} = ₹ \left(x + \frac{5}{4}x \right) = ₹ \left(\frac{9x}{4} \right).$$

$$\therefore \text{Required fraction} = \left(\frac{5x}{4} \times \frac{4}{9x} \right) = \frac{5}{9}.$$

- 258.** Let C's share = ₹ x . Then, B's share = ₹ $\left(\frac{x}{4}\right)$, A's share

$$= ₹ \left(\frac{2}{3} \times \frac{x}{4} \right) = ₹ \frac{x}{6}.$$

$$\therefore \frac{x}{6} + \frac{x}{4} + x = 1360 \Leftrightarrow \frac{17x}{12} = 1360$$

$$\Leftrightarrow x = \left(\frac{1360 \times 12}{17} \right) = ₹ 960.$$

$$\text{Hence, B's share} = ₹ \left(\frac{960}{4} \right) = ₹ 240.$$

- 259.** Let Tanya's share = ₹ x . Then, Veena's share

$$= ₹ \left(\frac{x}{2} \right), \text{ Amita's share} = ₹ \left(\frac{2}{3} \times \frac{x}{2} \right) = ₹ \left(\frac{x}{3} \right).$$

$$\text{Total bill} = ₹ \left(x + \frac{x}{2} + \frac{x}{3} \right) = ₹ \left(\frac{11x}{6} \right).$$

$$\therefore \text{Required fraction} = \left(\frac{x}{2} \times \frac{6}{11x} \right) = \frac{3}{11}.$$

- 260.** Let the capacity of the tank be x litres.

$$\text{Then, } \frac{1}{4}x = 135 \Leftrightarrow x = 135 \times 4 = 540.$$

$$\therefore \text{Required fraction} = \left(\frac{180}{540} \right) = \frac{1}{3}.$$

- 261.** Let the capacity of the tank be x litres.

$$\text{Then, } \frac{6}{7}x - \frac{2}{5}x = 16$$

$$\Leftrightarrow 30x - 14x = 16 \times 35$$

$$\Leftrightarrow 16x = 560$$

$$\Leftrightarrow x = 35.$$

- 262.** Let the capacity of the drum be x litres.

$$\text{Then, } \frac{3}{4}x - \frac{7}{12}x = 30$$

$$\Leftrightarrow 9x - 7x = 12 \times 30$$

$$\Leftrightarrow 2x = 360$$

$$\Leftrightarrow x = 180.$$

- 263.** Let x be the number of bottles of oil that the tin can contain.

$$\text{Then, } \frac{5}{8}x - \frac{3}{5}x = 10 - 8 \Leftrightarrow 25x - 24x = 2 \times 40 \Leftrightarrow x = 80.$$

- 264.** R.F. = $\frac{10 \text{ cm}}{1 \text{ m}} = \frac{10 \text{ cm}}{100 \text{ cm}} = \frac{1}{10}.$

- 265.** An area of 49 sq. km is represented by an area of 100 sq. cm on the map.

So, a distance of 7 km is represented by a length of 10 cm on the map.

$$\therefore \text{R.F.} = \frac{10 \text{ cm}}{7 \text{ km}} = \frac{10 \text{ cm}}{7 \times 10^5 \text{ cm}} = \frac{1}{70000}.$$

- 266.** Let the capacity of the bucket be x litres. Then,

$$\text{Capacity of 1 large bottle} = \frac{x}{4}; \text{ Capacity of 1 small bottle} = \frac{x}{7}.$$

$$\text{Fluid left in large bottle} = \left(\frac{x}{4} - \frac{x}{7} \right) = \frac{3x}{28}.$$

$$\therefore \text{Required fraction} = \left(\frac{3x/28}{x/4} \right) = \left(\frac{3x}{28} \times \frac{4}{x} \right) = \frac{3}{7}.$$

- 267.** Let the capacity of 1 bucket = x .

$$\text{Then, capacity of tank} = 25x.$$

$$\text{New capacity of bucket} = \frac{2}{5}x.$$

$$\begin{aligned} \therefore \text{Required number of buckets} &= \frac{25x}{(2x/5)} \\ &= \left(25x \times \frac{5}{2x} \right) = \frac{125}{2} = 62\frac{1}{2}. \end{aligned}$$

268. Clearly, more fuel, the longer will the engine run.

$$\therefore \frac{4}{5} : \frac{1}{3} :: 12 : x \text{ or } \frac{4}{5}x = \frac{1}{3} \times 12 = 4 \text{ or } x = 4 \times \frac{5}{4} = 5.$$

269. Suppose the taller tree grows x feet in 1 year.

Then, the shorter tree grows $\frac{3x}{5}$ ft in 1 year.

$$\begin{aligned} \therefore 4x + 4 \times \frac{3}{5}x &= 8 \Leftrightarrow 4x + \frac{12}{5}x = 8 \Leftrightarrow \frac{32}{5}x = 8 \\ \Leftrightarrow x &= \frac{8 \times 5}{32} = \frac{5}{4}. \end{aligned}$$

$$\begin{aligned} \text{Growth of shorter tree in 2 years} &= \left(2 \times \frac{3x}{5}\right) \text{ ft} \\ &= \left(2 \times \frac{3}{5} \times \frac{5}{4}\right) \text{ ft} = \frac{3}{2} \text{ ft} = 1\frac{1}{2} \text{ ft}. \end{aligned}$$

270. Let the amount of fuel consumed while coming back be x gallons.

Then, amount of fuel consumed while going

$$= \left(x + \frac{x}{4}\right) \text{ gallons} = \frac{5x}{4} \text{ gallons}.$$

$$\therefore x + \frac{5x}{4} = 4\frac{1}{2} \Leftrightarrow \frac{9x}{4} = \frac{9}{2} \Leftrightarrow x = \frac{9}{2} \times \frac{4}{9} = 2.$$

271. Let the highest temperature be x degrees.

$$\begin{aligned} \text{Then, lowest temperature} &= \left[\left(1 + \frac{1}{3}\right)\frac{x}{2}\right] \text{ degrees} \\ &= \left(\frac{4}{3} \times \frac{x}{2}\right) \text{ degrees} = \frac{2x}{3} \text{ degrees}. \end{aligned}$$

$$\therefore x + \frac{2x}{3} = 100 \Leftrightarrow \frac{5x}{3} = 100 \Leftrightarrow x = \frac{100 \times 3}{5} = 60.$$

$$\text{So, lowest temperature} = \left(\frac{2}{3} \times 60\right) \text{ degrees} = 40 \text{ degrees}.$$

272. Let the total number of students in the class be x .

$$\text{Then, number of boys} = \frac{2x}{3}; \text{ number of girls} = \frac{x}{3}.$$

$$\text{Number of boys who are over 160 cm tall} = \left(\frac{3}{4} \times \frac{2x}{3}\right) = \frac{x}{2}.$$

$$\text{So, } \frac{x}{2} = 18 \text{ or } x = 36.$$

$$\therefore \text{Number of girls} = \frac{36}{3} = 12.$$

273. Suppose initially Peter had ₹ x . Then,

$$\text{Amount received by Michael} = ₹ \left(\frac{x}{4}\right). \text{ Amount remain-}$$

$$\text{ing with Peter} = ₹ \left(x - \frac{x}{4}\right) = ₹ \left(\frac{3x}{4}\right).$$

$$\text{Amount received by Sam} = ₹ \left(\frac{1}{2} \times \frac{x}{4}\right) = ₹ \left(\frac{x}{8}\right).$$

$$\therefore \frac{3x}{4} - \frac{x}{8} = 500 \Leftrightarrow 5x = 4000 \Leftrightarrow x = 800.$$

$$\text{Hence, amount received by Michael} = (x/4) = ₹ 200.$$

$$\mathbf{274.} \text{ A's share} = \frac{1}{3}. \text{ Remainder} = \left(1 - \frac{1}{3}\right) = \frac{2}{3}.$$

$$\text{B's share} = \frac{2}{5} \text{ of } \frac{2}{3} = \frac{4}{15}. \text{ Rest} = \left(\frac{2}{3} - \frac{4}{15}\right) = \frac{6}{15} = \frac{2}{5}.$$

$$\text{C's share} = \text{D's share} = \frac{1}{2} \text{ of } \frac{2}{5} = \frac{1}{5}.$$

275. Let the total length of the journey be x km.

$$\text{Then, } \frac{2}{5}x = 1200 \Leftrightarrow x = \frac{1200 \times 5}{2} = 3000.$$

$$\text{Distance travelled by car} = \left(\frac{1}{3} \times 3000\right) \text{ km} = 1000 \text{ km}.$$

$$\therefore \text{Distance travelled by train} = [3000 - (1200 + 1000)] \text{ km} = 800 \text{ km}.$$

276. Let the total income be ₹ x .

Then, remaining money

$$= ₹ \left[x - \left(\frac{x}{4} + \frac{x}{5}\right)\right] = ₹ \left(x - \frac{9x}{20}\right) = ₹ \frac{11x}{20}.$$

$$\text{So, } \frac{11x}{20} = 231 \Leftrightarrow x = \frac{231 \times 20}{11} = 420.$$

277. Part read on first day = $\frac{3}{8}$. Remaining part = $\left(1 - \frac{3}{8}\right) = \frac{5}{8}$.

$$\text{Part read on second day} = \frac{4}{5} \text{ of } \frac{5}{8} = \frac{1}{2}.$$

$$\text{Unread part} = \left[1 - \left(\frac{3}{8} + \frac{1}{2}\right)\right] = \frac{1}{8}.$$

Let the number of pages be x .

$$\text{Then, } \frac{1}{8}x = 30 \text{ or } x = 30 \times 8 = 240.$$

278. Out of 5 girls, 1 took part in fete. Out of 8 boys, 1 took part in fete.

$\therefore$ Out of 13 students, 2 took part in fete.

$$\text{Hence, } \frac{2}{13} \text{ of the total number took part in fete}.$$

279. French men = $\frac{1}{5}$; French women = $\left(\frac{1}{5} + \frac{2}{3} \times \frac{1}{5}\right) = \frac{5}{15} = \frac{1}{3}$.

$$\text{French people} = \left(\frac{1}{5} + \frac{1}{3}\right) = \frac{8}{15}.$$

$$\therefore \text{Not French} = \left(1 - \frac{8}{15}\right) = \frac{7}{15}.$$

280. Girls = $\frac{3}{5}$; Boys = $\left(1 - \frac{3}{5}\right) = \frac{2}{5}$.

$$\begin{aligned} \text{Fraction of students absent} &= \frac{2}{9} \text{ of } \frac{3}{5} + \frac{1}{4} \text{ of } \frac{2}{5} \\ &= \frac{6}{45} + \frac{1}{10} = \frac{21}{90} = \frac{7}{30}. \end{aligned}$$

$$\therefore \text{Fraction of students present} = \left(1 - \frac{7}{30}\right) = \frac{23}{30}.$$

281. Number of boys who participate = 100.

$$\therefore \frac{1}{3} \text{ of boys} = 100 \text{ or total number of boys} = 300.$$

Number of girls who participate = 200.

$\therefore \frac{1}{2}$ of girls = 200 or total number of girls = 400.

Hence, total number of students = (300 + 400) = 700.

- 282.** Let the number of votes cast be x . Then, number of votes required = $\frac{3x}{4}$.

Counted votes = $\frac{2x}{3}$. Uncounted votes = $\left(x - \frac{2x}{3}\right) = \frac{x}{3}$.

Votes won by the candidate = $\frac{5}{6}$ of $\frac{3x}{4} = \frac{5x}{8}$.

Remaining votes required = $\left(\frac{3x}{4} - \frac{5x}{8}\right) = \frac{x}{8}$.

$\therefore$ Required fraction = $\frac{(x/8)}{(x/3)} = \left(\frac{x}{8} \times \frac{3}{x}\right) = \frac{3}{8}$.

- 283.** Let the total number of shots be x . Then,

Shots fired by A = $\frac{5}{8}x$; Shots fired by B = $\frac{3}{8}x$.

Killing shots by A = $\frac{1}{3}$ of $\frac{5}{8}x = \frac{5x}{24}$;

Shots missed by B = $\frac{1}{2}$ of $\frac{3}{8}x = \frac{3}{16}x$.

$\therefore \frac{3x}{16} = 27$ or $x = \left(\frac{27 \times 16}{3}\right) = 144$.

Birds killed by A = $\frac{5x}{24} = \left(\frac{5}{24} \times 144\right) = 30$.

- 284.** Number of alterations required in 1 shirt = $\left(\frac{2}{3} + \frac{3}{4} + \frac{4}{5}\right)$
 $= \frac{133}{60}$.

$\therefore$ Number of alterations required in 60 shirts = $\left(\frac{133}{60} \times 60\right)$
 $= 133$.

- 285.** Let the largest fraction be x and the smallest be y .

Then, $\frac{x}{y} = \frac{7}{6}$ or $y = \frac{6}{7}x$.

Let the middle one be z .

Then, $x + \frac{6}{7}x + z = \frac{59}{24}$ or $z = \left(\frac{59}{24} - \frac{13x}{7}\right)$.

$\therefore \frac{59}{24} - \frac{13x}{7} + \frac{1}{3} = \frac{7}{6} \Leftrightarrow \frac{13x}{7} = \frac{59}{24} + \frac{1}{3} - \frac{7}{6} = \frac{39}{24}$

$\Leftrightarrow x = \left(\frac{39}{24} \times \frac{7}{13}\right) = \frac{7}{8}$.

So, $x = \frac{7}{8}$, $y = \frac{6}{7} \times \frac{7}{8} = \frac{3}{4}$ and $z = \frac{59}{24} - \frac{13}{7} \times \frac{7}{8} = \frac{20}{24} = \frac{5}{6}$.

Hence, the fractions are $\frac{7}{8}$, $\frac{5}{6}$ and $\frac{3}{4}$.

- 286.** Let the number of hats purchased be x . Then, number of brown hats = $\frac{x}{4}$.

Number of hats sold = $\frac{2x}{3}$. Number of hats left unsold

$= \left(x - \frac{2x}{3}\right) = \frac{x}{3}$.

Number of brown hats sold = $\frac{4}{5}$ of $\frac{x}{4} = \frac{x}{5}$. Number of

brown hats left unsold = $\left(\frac{x}{4} - \frac{x}{5}\right) = \frac{x}{20}$.

$\therefore$ Required fraction = $\frac{\left(\frac{x}{20}\right)}{\left(\frac{x}{3}\right)} = \frac{x}{20} \times \frac{3}{x} = \frac{3}{20}$.

- 287.** Let the capacities of the bigger and smaller jars be x litres and y litres respectively.

Then, $\frac{x}{4} = \frac{y}{3} \Rightarrow y = \frac{3}{4}x$.

Quantity of water in bigger jar = $\left(\frac{x}{4} + \frac{y}{3}\right)$ litres

$= \left(\frac{x}{4} + \frac{1}{3} \times \frac{3}{4}x\right)$ litres = $\frac{x}{2}$ litres.

Hence, $\frac{1}{2}$ of the larger jar is filled with water.

- 288.** Let Mr. X's total income be ₹ x . Then, expenditure on food = ₹ $\frac{x}{4}$.

Expenditure on education = ₹ $\left[\left(1 - \frac{1}{3}\right)\frac{x}{4}\right]$
 $= ₹ \left(\frac{2}{3} \times \frac{x}{4}\right) = ₹ \frac{x}{6}$.

$\therefore$ Required fraction = $\frac{\left(\frac{x}{4} + \frac{x}{6}\right)}{x} = \frac{5x}{12} \times \frac{1}{x} = \frac{5}{12}$.

- 289.** Let the number of men in the 1st, 2nd, 3rd and 4th battalions be x , y , z and t respectively.

Then, $\frac{1}{2}x = \frac{2}{3}y = \frac{3}{4}z = \frac{4}{5}t \Rightarrow x = \frac{4}{3}y, z = \frac{8}{9}y, t = \frac{5}{6}y$.

Now, $x + y + z + t = 7300$

$\Rightarrow \frac{4}{3}y + y + \frac{8}{9}y + \frac{5}{6}y = 7300$

$\Rightarrow \frac{24y + 18y + 16y + 15y}{18} = 7300$

$\Rightarrow 73y = 7300 \times 18 \Rightarrow y = 1800$.

- 290.** Required fraction = Fraction of work left after Tuesday

$= 1 - \left(\frac{3}{5} + \frac{1}{3} \text{ of } \frac{2}{5}\right) = 1 - \left(\frac{3}{5} + \frac{2}{15}\right)$

$= 1 - \frac{11}{15} = \frac{4}{15}$.

- 291.** Let the total score be x . Then, highest score = $\frac{3x}{11}$.

Remainder = $\left(x - \frac{3x}{11}\right) = \frac{8x}{11}$. Next highest score

$$= \frac{3}{11} \text{ of } \frac{8x}{11} = \frac{24x}{121}.$$

$$\therefore \frac{3x}{11} - \frac{24x}{121} = 9 \Leftrightarrow 33x - 24x = 9 \times 121$$

$$\Leftrightarrow 9x = 9 \times 121 \Leftrightarrow x = 121.$$

292. Let the hourly wage of each sales person be ₹ x .

Then, hourly wage of each mechanic = ₹ $(2x)$.

Hourly wage of each custodial worker = ₹ $\left(\frac{2x}{3}\right)$.

$$\therefore \text{ Required fraction} = \frac{x}{\left(\frac{2x}{3}\right)} = \frac{3}{2}.$$

293. Sukhvinder's monthly income = ₹ $\left(\frac{234000}{12}\right)$ = ₹ 19500.

Jassi's monthly income = ₹ $\left(\frac{3}{2} \times 19500\right)$ = ₹ 29250.

$\therefore$ Ganeshi's monthly income = ₹ (2×29250) = ₹ 58500.

294. Cost per kg of rice last year = ₹ $\left(\frac{1044}{36}\right)$ = ₹ 29.

Cost per kg of rice this year = ₹ $\left(\frac{768}{24}\right)$ = ₹ 32.

$\therefore$ Required difference = ₹ $(32 - 29)$ = ₹ 3.

295. $12B + 30W = 8940 \Rightarrow \frac{1}{3}(12B + 30W) = \frac{1}{3} \times 8940$

$$\Rightarrow 4B + 10W = 2980.$$

296. $5P + 8C = 145785 \Rightarrow 3(5P + 8C) = 3 \times 145785$

$$\Rightarrow 15P + 24C = 437355.$$

297. $21T + 35C = 41825 \Rightarrow \frac{3}{7}(21T + 35C) = \frac{3}{7} \times 41825$

$$\Rightarrow 9T + 15C = 17925.$$

298. Cost of 1 kg of sugar = ₹ $\left(\frac{195}{13}\right)$ = ₹ 15.

Cost of 1 kg of rice = ₹ $\left(\frac{544}{17}\right)$ = ₹ 32.

Cost of 1 kg of wheat = ₹ $\left(\frac{336}{21}\right)$ = ₹ 16.

$\therefore$ Required cost = ₹ $(15 \times 21 + 32 \times 26 + 16 \times 19)$

$$= ₹ (315 + 832 + 304) = ₹ 1451.$$

299. (a) Price per kg = ₹ $\left(\frac{160}{10}\right)$ = ₹ 16;

(b) Price per kg = ₹ $\left(\frac{30}{2}\right)$ = ₹ 15;

(c) Price per kg = ₹ $\left(\frac{70}{4}\right)$ = ₹ 17.50;

(d) Price per kg = ₹ $\left(\frac{340}{20}\right)$ = ₹ 17;

(e) Price per kg = ₹ $\left(\frac{130}{8}\right)$ = ₹ 16.25.

Clearly, the most economical price is 2 kilo for ₹ 30.

300. Let the time spent on each Maths problem be $(2x)$ minutes and that spent on each other question be x minutes.

Then, $50 \times 2x + 150x = 3 \times 60 \Leftrightarrow 250x = 180$

$$\Leftrightarrow x = \frac{180}{250} = \frac{18}{25}.$$

$\therefore$ Total time spent on Mathematics problems

$$= \left(50 \times 2 \times \frac{18}{25}\right) \text{ min} = 72 \text{ min}.$$

301. Duration from 4.56 P.M. to 5.32 P.M. = 36 min.

$$= \left(\frac{36}{60}\right) \text{ hour} = \frac{3}{5} \text{ hour.}$$

$$14 \overline{) 4326}$$

302. 72 hr 6 min = $(72 \times 60 + 6)$ min = 4326 min.

$$\therefore 72 \text{ hr 6 min} \div 14 = 4326 \text{ min} \div 14$$

$$= 309 \text{ min}$$

$$= 5 \text{ hr 9 min.}$$

303. Time between 10 a.m. and 13.27 hours = 3 hrs. 27 min. = 207 min.

Total duration of free time = (5×3) min = 15 min.

Remaining time = $(207 - 15)$ min. = 192 min.

$\therefore$ Duration of each of the 4 periods

$$= \left(\frac{192}{4}\right) \text{ min.} = 48 \text{ min.}$$

	Hrs.	Min.	Sec.
	3	17	49
(-)	1	54	50
	1	22	59

Total time = $(1 \times 60 + 22)$ min. + 59 sec.

$$= (82 \times 60 + 59) \text{ sec.}$$

$$= 4979 \text{ sec.}$$

$\therefore$ Number of times the light is seen = $\left(\frac{4979}{13} + 1\right) = 384.$

305. The woman works from Sunday to Saturday and rests for the first time on Sunday. She then works from Monday to Sunday and rests for the second time on Monday. Likewise, she rests for the third time on Tuesday. Thus, starting from Sunday, for the 10th time she would rest on Tuesday. (as the cycle repeats after every 7 days)

306. $9 * 11 = 9^2 + (11)^2 - 9 \times 11 = 81 + 121 - 99 = 103.$

307. $(3 * -1) = \frac{3 \times (-1)}{3 + (-1)} = \frac{-3}{2}$. So, $3 * (3 * -1)$

$$= 3 * \left(\frac{-3}{2}\right) = \frac{3 \times \left(\frac{-3}{2}\right)}{3 + \left(\frac{-3}{2}\right)} = \frac{-9}{\frac{3}{2}} \times \frac{2}{3} = -3.$$

308. $3 * 5 + 5 * 3 = (2 \times 3 - 3 \times 5 + 3 \times 5) + (2 \times 5 - 3 \times 3 + 5 \times 3)$

$$= (6 + 10 - 9 + 15) = 22.$$

309. $4 \oplus (3 \oplus p) = 4 \oplus (3^2 + 2p) = 4 \oplus (9 + 2p) = 4^2 + 2(9 + 2p) = 34 + 4p.$

$$\therefore 34 + 4p = 50 \Rightarrow 4p = 50 - 34 = 16 \Rightarrow p = 4.$$

310. $(a * b * c) * a * b = \left(\frac{a+b}{c}\right) * a * b = \frac{\left(\frac{a+b}{c}\right) + a}{b} = \frac{a+b+ac}{bc}.$

311. Step I: $6 \times 3 + 6^2 = 18 + 36 = 54.$

Step II: $(54 \div 5) + 6 = 10.8 + 6 = 16.8.$

Step III: $(16.8 \div 3)^2 = (5.6)^2 = 31.36.$

312. Step I: $2.5 \times 3 + (2.5)^2 = 7.5 + 6.25 = 13.75.$

Step II: $(13.75 \div 5) + 6 = 2.75 + 6 = 8.75.$

Step III: $(8.75 \div 3)^2 = (2.917)^2 = 8.508 \approx 8.51.$

313. Step I: $5 \times 3 + 5^2 = 15 + 25 = 40.$

Step II: $(40 \div 5) + 6 = 8 + 6 = 14.$

Step III: $(14 \div 3)^2 = (4.67)^2 \approx 21.80.$

314. $2 \wedge a = a \wedge 3 \Rightarrow 2 \times 2 + a = 2 \times a + 3$
 $\Rightarrow 4 + a = 2a + 3 \Rightarrow a = 1.$

315.
$$\left[\frac{(5 \oplus 7) + (4 \oplus 4)}{3(5 \oplus 5) - (15 \oplus 11) - 3} \right] = \left[\frac{(5+7) + 4^2}{3 \times 5^2 - 2 \times 15 - 3} \right]$$

$$= \left[\frac{12+16}{75-30-3} \right] = \frac{28}{42} = \frac{2}{3}.$$

316. Thickness of snow at 3 p.m. = $\left[2\frac{1}{4} + \left(3 \times 1\frac{1}{4} \right) \right]$ inches
 $= \left(\frac{9}{4} + 3 \times \frac{5}{4} \right)$ inches
 $= \left(\frac{9}{4} + \frac{15}{4} \right)$ inches
 $= \left(\frac{24}{4} \right)$ inches = 6 inches.

317. In 2 min, the squirrel climbs $(6 - 4)$ m = 2 m.

In (2×57) , i.e., 114 min, the squirrel climbs (2×57) m = 114 m.

In 115 min, the squirrel climbs $(114 + 6)$ m = 120 m to reach the top.

318. For the money to grow 3 times, Ram needs to deposit ₹ $(3000 - 1000) = ₹ 2000.$

A net amount of ₹ $(750 - 500) = ₹ 250$ is deposited in 1 month.

So, a net amount of ₹ 2000 will be deposited in $\left(\frac{2000}{250} \right) = 8$ months.

319. In 11 days, the height of the plant increases by $(11 - 6)$ cm i.e., 5 cm.

In (11×18) i.e., 198 days, the height of the plant increases by (5×18) i.e., 90 cm.

In the next 10 days, the height of the plant increases by another 10 cm.

So, in 208 days, the height of the plant will increase by $(90 + 10) = 100$ cm or 1 m and become 2 m.

320. Number of steps = $\left(10 \div \frac{1}{2} \right) = 20.$

To reach the top, the ant covers the height of 20 steps and the width of 19 steps because the moment it ascends the height of 20th step, it reaches the top.

$\therefore$ Distance travelled by the ant

$$= \left[\left(\frac{1}{2} \times 20 \right) + (1 \times 19) \right] \text{ ft} = (10 + 19) \text{ ft} = 29 \text{ ft}.$$

321. Required sum = $\frac{1}{2} + \left(\frac{1}{3} + \frac{2}{3} \right) + \left(\frac{1}{4} + \frac{2}{4} + \frac{3}{4} \right) + \left(\frac{1}{5} + \frac{2}{5} + \frac{3}{5} + \frac{4}{5} \right) + \dots$
 $+ \left(\frac{1}{100} + \frac{2}{100} + \frac{3}{100} + \dots + \frac{99}{100} \right)$

$$= \frac{1}{2} + \frac{2}{2} + \frac{3}{2} + \frac{4}{2} + \dots + \frac{99}{2}$$

$$= \frac{1}{2} \times (1 + 2 + 3 + \dots + 99)$$

$$= \frac{1}{2} \times \frac{99 \times 100}{2} = 2475.$$

322. Given exp. = $\frac{(1-x^2)(1+x^2)}{(1+x)} \times \frac{x}{1+x^2} \times \frac{1}{x(1-x)}$
 $= \frac{(1-x)(1+x)(1+x^2)x}{(1+x)(1+x^2)x(1-x)} = 1.$

323. Given exp. = $\left(\frac{17}{15} \right)^2 + \left(\frac{2}{15} \right)^2 - 2 \times \frac{17}{15} \times \frac{2}{15} = \left(\frac{17}{15} - \frac{2}{15} \right)^2 = 1.$

324. Given exp. = $\left(13\frac{3}{4} \right)^2 + \left(5\frac{1}{4} \right)^2 + 2 \times 13\frac{3}{4} \times 5\frac{1}{4}$
 $= \left(13\frac{3}{4} + 5\frac{1}{4} \right)^2 = \left(\frac{55}{4} + \frac{21}{4} \right)^2 = \left(\frac{76}{4} \right)^2 = (19)^2 = 361.$

325. Given exp. = $\frac{(75983)^2 - (45983)^2}{75983 - 45983}$
 $= \frac{(75983 - 45983)(75983 + 45983)}{(75983 - 45983)}$
 $= 75983 + 45983 = 121966.$

326. Given exp. = $\frac{a^2 - b^2}{a - b}$, (where $a = 13\frac{4}{7}$, $b = 8\frac{3}{7}$)
 $= a + b = 13\frac{4}{7} + 8\frac{3}{7} = \frac{95}{7} + \frac{59}{7} = \frac{154}{7} = 22.$

327. $\frac{(a-b)^2 - (a+b)^2}{-4a} = \frac{x}{y} \Leftrightarrow \frac{(a^2 + b^2 - 2ab) - (a^2 + b^2 + 2ab)}{-4a}$
 $= \frac{x}{y}$
 $\Leftrightarrow \frac{-4ab}{-4a} = \frac{x}{y} \Leftrightarrow b = \frac{x}{y} \Leftrightarrow x = yb.$

328. Given exp. = $\frac{\left(\frac{7}{8} + \frac{5}{6} \right)^2}{\left(\frac{7}{8} \right)^2 - \left(\frac{5}{6} \right)^2} = \frac{\left(\frac{7}{8} + \frac{5}{6} \right)^2}{\left(\frac{7}{8} + \frac{5}{6} \right) \left(\frac{7}{8} - \frac{5}{6} \right)}$
 $= \frac{\left(\frac{7}{8} + \frac{5}{6} \right)}{\left(\frac{7}{8} - \frac{5}{6} \right)} = \left(\frac{41}{24} \times 24 \right) = 41.$

329. Given exp. = $(1998 + 1997)(1998 - 1997) + (1996 + 1995)(1996 - 1995) + (1994 + 1993)(1994 - 1993)$
 $= 1998 + 1997 + 1996 + 1995 + 1994 + 1993 = 11773.$

$$330. \text{ Given exp.} = \frac{(a+b)^2 + (a-b)^2}{a^2 + b^2} \text{ (where } a = 856, b = 167\text{)}$$

$$= \frac{2(a^2 + b^2)}{(a^2 + b^2)} = 2.$$

$$331. \text{ Given exp.} = \frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} \\ = 4 \text{ (where } a = 469, b = 174\text{)}.$$

$$332. 2ab = (a^2 + b^2) - (a - b)^2 = 29 - 9 = 20 \Rightarrow ab = 10.$$

$$333. \frac{x^2 - 1}{x + 1} = 4 \Leftrightarrow \frac{(x+1)(x-1)}{x+1} = 4 \Leftrightarrow x-1 = 4 \Leftrightarrow x = 5.$$

$$334. \text{ If } a = 3\frac{2}{3}, b = 2\frac{1}{2}, c = 4\frac{3}{4}, d = 3\frac{1}{3}, \text{ then}$$

$$\text{Given exp.} = \frac{(a^2 - b^2)}{(c^2 - d^2)} \div \frac{(a-b)}{(c-d)} = \frac{(a^2 - b^2)}{(c^2 - d^2)} \times \frac{(c-d)}{(a-b)} \\ = \frac{(a+b)}{(c+d)} = \frac{3\frac{2}{3} + 2\frac{1}{2}}{4\frac{3}{4} + 3\frac{1}{3}} = \frac{\frac{11}{3} + \frac{5}{2}}{\frac{19}{4} + \frac{10}{3}} \\ = \frac{\frac{37}{6} \times \frac{12}{97}}{\frac{74}{97}}.$$

$$335. \text{ Given exp.} = \frac{a^2 - b^2}{a + b} = a - b$$

$$= \left(1 + \frac{1}{1 + \frac{1}{100}}\right) - \left(1 - \frac{1}{1 + \frac{1}{100}}\right) \\ = 2 \times \frac{1}{(101/100)} = 2 \times \frac{100}{101} = \frac{200}{101}.$$

$$336. \text{ Given exp.} = \frac{769435 \times 770001 - (770001 - 769435)}{769435 + 770001 \times (769435 - 1)} \\ = \frac{769435 \times 770001 - 770001 + 769435}{769435 + 770001 \times 769435 - 770001} = 1.$$

$$337. a - b = 1 \Rightarrow (a - b)^3 = 1^3 = 1 \Rightarrow a^3 - b^3 - 3ab(a - b) = 1 \\ \Rightarrow a^3 - b^3 - 3ab = 1 [\because a - b = 1]$$

$$338. x - y = 1 \Rightarrow (x - y)^2 = 1 \Rightarrow x^2 + y^2 - 2xy = 1 \\ \Rightarrow 41 - 2xy = 1 \Rightarrow 2xy = 40. \\ \therefore (x + y)^2 = x^2 + y^2 + 2xy = 41 + 40 = 81 \\ \Rightarrow (x + y) = \pm 9.$$

$$339. x + y = 1 \Rightarrow (x + y)^2 = 1^2 = 1 \Rightarrow x^2 + y^2 + 2xy = 1 \\ \Rightarrow x^2 + y^2 = 1 - 2xy \\ \Rightarrow (x^2 + y^2)^2 = (1 - 2xy)^2 \\ \Rightarrow x^4 + y^4 + 2x^2y^2 = 1 + 4x^2y^2 - 4xy \\ \Rightarrow 2x^2y^2 - 4xy = 17 - 1 = 16 [\because x^4 + y^4 = 17] \\ \Rightarrow x^2y^2 - 2xy = 8.$$

$$340. \frac{x^2 + y^2 + z^2 - yz - zx - xy}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = \frac{(a+m)^2 + (b+m)^2 + (c+m)^2 - (b+m)(c+m) \\ - (c+m)(a+m) - (a+m)(b+m)}{a^2 + b^2 + c^2 - ab - bc - ca}$$

$$\frac{a^2 + m^2 + 2am + b^2 + m^2 + 2bm + c^2 + m^2 + 2cm \\ - bc - bm - cm - m^2 - ca - cm - am - m^2 \\ - ab - am - bm - m^2}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = \frac{a^2 + b^2 + c^2 - ab - bc - ca}{a^2 + b^2 + c^2 - ab - bc - ca} = 1.$$

$$341. (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \\ \Rightarrow 2(ab + bc + ca) = (a + b + c)^2 - (a^2 + b^2 + c^2) \\ = 169 - 69 = 100$$

$$\Rightarrow ab + bc + ca = 50.$$

$$342. \text{ Given: } x^2 + y^2 + z^2 - 64 = -2(xy - yz - zx) \quad \dots(i)$$

$$\text{Now, } [x + y + (-z)]^2 = x^2 + y^2 + z^2 + 2(xy - yz - zx)$$

$$\Rightarrow (3z - z)^2 = x^2 + y^2 + z^2 + 2(xy - yz - zx)$$

$$\Rightarrow -2(xy - yz - zx) = (x^2 + y^2 + z^2) - (2z)^2 \quad \dots(ii)$$

$$\text{From (i) and (ii), we get: } (2z)^2 = 64 \Leftrightarrow 4z^2 = 64$$

$$\Leftrightarrow z^2 = 16$$

$$\Leftrightarrow z = 4.$$

$$343. \text{ Given exp.} = \left(\frac{a^3 + b^3}{a^2 + b^2 - ab}\right) = (a + b) \text{ (where } a = 785, b = 435\text{)} \\ = (785 + 435) = 1220.$$

$$344. \text{ Given exp.} = \left(\frac{a^2 + ab + b^2}{a^3 - b^3}\right) = \left(\frac{1}{a - b}\right) \\ \text{(where } a = 147, b = 143\text{)} = \left(\frac{1}{147 - 143}\right) = \frac{1}{4}.$$

$$345. \text{ Let } \frac{13^3 + 7^3}{13^2 + 7^2 - x} = 20.$$

$$\text{Then, } \frac{13^3 + 7^3}{13 + 7} = 13^2 + 7^2 - x$$

$$\Leftrightarrow 13^2 + 7^2 - 13 \times 7 = 13^2 + 7^2 - x$$

$$\Leftrightarrow x = 13 \times 7 = 91.$$

$$346. \text{ Given exp.} = \frac{a^3 - b^3}{a^2 - b^2} = \frac{(a-b)(a^2 + ab + b^2)}{(a-b)(a+b)} = \frac{(a^2 + ab + b^2)}{(a+b)}$$

$$= \frac{\left(\frac{3}{5}\right)^2 + \left(\frac{3}{5} \times \frac{2}{5}\right) + \left(\frac{2}{5}\right)^2}{\left(\frac{3}{5} + \frac{2}{5}\right)}$$

$$= \frac{\frac{9}{25} + \frac{6}{25} + \frac{4}{25}}{\frac{5}{5}} = \frac{19}{25}.$$

$$347. \text{ Given exp.} = \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = a + b + c = (38 + 34 + 28) = 100.$$

$$348. \text{ Since } (x - y) + (y - z) + (z - x) = 0, \\ \text{so } (x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x).$$

$$\therefore \text{ Given exp.} = \frac{3(x - y)(y - z)(z - x)}{9(x - y)(y - z)(z - x)} = \frac{1}{3}.$$

$$\begin{aligned}
 349. \quad \frac{x^3 - y^3}{x^2 - y^2} - \frac{3xy}{x + y} &= \frac{(x - y)(x^2 + y^2 + xy)}{(x - y)(x + y)} - \frac{3xy}{(x + y)} \\
 &= \frac{(x^2 + y^2 + xy) - 3xy}{(x + y)} = \frac{(x^2 + y^2 - 2xy)}{(x + y)} \\
 &= \frac{(x - y)^2}{(x + y)} = \frac{(5 - 3)^2}{(5 + 3)} = \frac{4}{8} = \frac{1}{2}.
 \end{aligned}$$

$$\begin{aligned}
 350. \quad (a + b + c) &= 11 \Rightarrow (a + b + c)^2 = (11)^2 = 121 \\
 &\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 121 \\
 &\Rightarrow a^2 + b^2 + c^2 + 2 \times 20 = 121 \\
 &\Rightarrow a^2 + b^2 + c^2 = 121 - 40 = 81. \\
 \therefore a^3 + b^3 + c^3 - 3abc &= (a + b + c) \\
 &\quad (a^2 + b^2 + c^2 - ab - bc - ca) \\
 &= 11(81 - 20) = 11 \times 61 = 671.
 \end{aligned}$$

$$\begin{aligned}
 351. \quad \text{Since } (x - y) + (y - z) + (z - x) &= 0, \\
 \text{so } (x - y)^3 + (y - z)^3 + (z - x)^3 &= 3(x - y)(y - z)(z - x). \\
 \text{Since } (x^2 - y^2) + (y^2 - z^2) + (z^2 - x^2) &= 0, \\
 \text{so } (x^2 - y^2)^3 + (y^2 - z^2)^3 + (z^2 - x^2)^3 &= 3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2). \\
 \therefore \text{Given exp.} &= \frac{3(x - y)(y - z)(z - x)}{3(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)} \\
 &= \frac{1}{(x + y)(y + z)(z + x)} \\
 &= [(x + y)(y + z)(z + x)]^{-1}.
 \end{aligned}$$

$$\begin{aligned}
 352. \quad \frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} &= \frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3. \\
 [\because a + b + c = 0 \Rightarrow a^3 + b^3 + c^3 &= 3abc]
 \end{aligned}$$

$$\begin{aligned}
 353. \quad a^3 + b^3 + c^3 - 3abc &= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \\
 &= (29 + 24 + 27)[(29)^2 + (24)^2 + (27)^2 \\
 &\quad - 29 \times 24 - 24 \times 27 - 27 \times 29] \\
 &= 80(841 + 576 + 729 - 696 - 648 - 783) = 80 \times 19 = 1520.
 \end{aligned}$$

$$\begin{aligned}
 354. \quad \frac{x^2 - 1}{x} = 4 &\Rightarrow \left(\frac{x^2 - 1}{x}\right)^3 = 4^3 = 64 \\
 &\Rightarrow \frac{x^6 - 1 - 3x^2(x^2 - 1)}{x^3} = 64 \\
 &\Rightarrow \frac{x^6 - 1 - 3x^2 \cdot 4x}{x^3} = 64 \\
 &\Rightarrow \frac{x^6 - 1 - 12x^3}{x^3} = 64x^3 \\
 &\Rightarrow x^6 - 1 = 76x^3 \Rightarrow \frac{x^6 - 1}{x^3} = 76.
 \end{aligned}$$

$$\begin{aligned}
 355. \quad \text{Let } a = 32, b = 79, c = -111. \text{ Then, } a + b + c &= 0. \\
 \text{So, } a^3 + b^3 + c^3 = 3abc &\Rightarrow (32)^3 + (79)^3 - (111)^3 \\
 &= 3 \times 32 \times 79 \times (-111) \\
 \therefore \text{Given exp.} &= -(3 \times 32 \times 79 \times 111) + 3 \times 32 \times 79 \\
 &\quad \times 111 = 0.
 \end{aligned}$$

$$\begin{aligned}
 356. \quad (x + y) = 3 &\Rightarrow (x + y)^2 = 9 \\
 &\Rightarrow x^2 + y^2 + 2xy = 9 \\
 &\Rightarrow x^2 + y^2 = 9 - 2xy = 9 - 4 = 5. \\
 \therefore x^3 + y^3 &= (x + y)(x^2 + y^2 - xy) = 3(5 - 2) \\
 &= 3 \times 3 = 9. [\because x + y = 3, x^2 + y^2 = 5, xy = 2]
 \end{aligned}$$

$$\begin{aligned}
 357. \quad \text{Given expression} &= \frac{(x + y - z)(x - y + z)}{(x + z + y)(x + z - y)} \\
 &\quad + \frac{(y + x - z)(y - x + z)}{(x + y + z)(x + y - z)} + \frac{(z + x - y)(z - x + y)}{(y + z + x)(y + z - x)} \\
 &= \frac{(x + y - z)}{(x + y + z)} + \frac{(y - x + z)}{(x + y + z)} + \frac{(x - y + z)}{(x + y + z)} \\
 &= \frac{(x + y - z) + (y - x + z) + (x - y + z)}{(x + y + z)} \\
 &= \frac{x + y + z}{x + y + z} = 1.
 \end{aligned}$$

$$358. \quad \frac{a}{p} + \frac{b}{q} + \frac{c}{r} = 0 \Rightarrow aqr + bpr + cpq = 0 \quad \dots(i)$$

$$\begin{aligned}
 \frac{p}{a} + \frac{q}{b} + \frac{r}{c} &= 1 \Rightarrow \left(\frac{p}{a} + \frac{q}{b} + \frac{r}{c}\right)^2 = 1 \\
 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2} + 2\left(\frac{pq}{ab} + \frac{pr}{ac} + \frac{qr}{bc}\right) &= 1 \\
 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2} + \frac{2(pqc + prb + qra)}{abc} &= 1 \\
 \Rightarrow \frac{p^2}{a^2} + \frac{q^2}{b^2} + \frac{r^2}{c^2} &= 1 \quad [\text{Using (i)}]
 \end{aligned}$$

$$\begin{aligned}
 359. \quad x^4 - y^4 = 15 &\Rightarrow (x^2 - y^2)(x^2 + y^2) = 15 = 1 \times 15 \text{ or } 3 \times 5 \\
 \Rightarrow (x^2 - y^2) = 3 \text{ and } (x^2 + y^2) &= 5 \\
 [\because x \text{ and } y \text{ are natural numbers}] \\
 \Rightarrow 2x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x &= 2.
 \end{aligned}$$

$$\begin{aligned}
 [\text{Adding above two equations}] \\
 \text{Putting } x = 2 \text{ in } x^2 + y^2 = 5, \text{ we get: } y &= 1. \\
 \therefore (x^4 + y^4) &= (2^4 + 1^4) = (16 + 1) = 17.
 \end{aligned}$$

$$\begin{aligned}
 360. \quad 26 \text{ trees have 25 gaps between them. Hence, required} \\
 \text{distance} &= \left(\frac{225}{25}\right) \text{ m} = 9 \text{ m}.
 \end{aligned}$$

$$\begin{aligned}
 361. \quad \text{Each row contains 12 plants.} \\
 12 \text{ plants have 11 gaps between them. So, total length of} \\
 \text{gaps} &= (11 \times 2) \text{ metres and 1 metre is left on each side.} \\
 \therefore \text{Length} &= (22 + 2) \text{ m} = 24 \text{ m}.
 \end{aligned}$$

$$\begin{aligned}
 362. \quad \text{Let the length of the side be } x \text{ metres.} \\
 \text{Then, number of posts required when placed 6 m apart} \\
 &= \left(\frac{x}{6} + 1\right).
 \end{aligned}$$

$$\begin{aligned}
 \text{And, number of posts required when placed 8 m apart} \\
 &= \left(\frac{x}{8} + 1\right).
 \end{aligned}$$

$$\therefore \left(\frac{x}{6} + 1\right) - 5 = \left(\frac{x}{8} + 1\right) \Leftrightarrow \frac{x}{6} - \frac{x}{8} = 5 \Leftrightarrow \frac{x}{24} = 5 \Leftrightarrow x = 120 \text{ m}.$$

$$\text{So, number of posts bought} = \left(\frac{120}{8} + 1\right) = 16.$$

$$363. \quad \text{Gap between 2 consecutive trees} = \left(\frac{2000}{200}\right) \text{ m} = 10 \text{ m}.$$

$$[\because 201 \text{ trees have 200 gaps between them}]$$

$$\begin{aligned}
 \therefore \text{Required number of trees} &= \left(\frac{50 \times 1000}{10} + 1\right) \\
 &= (5000 + 1) = 5001.
 \end{aligned}$$

364. Total number of students = 54×30 .

$$\therefore \text{Required number of rows} = \left(\frac{54 \times 30}{45} \right) = 36.$$

365. Clearly, we have:

$$x + 1 = y \Leftrightarrow x - y = -1 \quad \dots(i)$$

$$\text{And, } 2(x - 1) = y \Leftrightarrow 2x - y = 2 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $x = 3$. Putting $x = 3$ in (i), we get: $y = 4$.

366. Let the number of benches in the class be x .

$$\text{Then, } 6(x + 1) = 7x - 5 \Leftrightarrow x = 11.$$

Hence, number of students in the class

$$= 6(x + 1) = 6 \times 12 = 72.$$

367. Let the number of benches in the class be x .

$$\text{Then, } 4(x - 3) = 3x + 3 \Leftrightarrow x = 15.$$

$$\therefore \text{Total number of students} = 3 \times 15 + 3 = 48.$$

368. Let the number of students in a row be x and the total number of rows be y . Then,

$$\text{total number of students in the class} = xy.$$

$$\therefore (x + 4)(y - 2) = xy \Rightarrow xy - 2x + 4y - 8 = xy$$

$$\Rightarrow 2x - 4y = -8 \Rightarrow x - 2y = -4 \quad \dots(i)$$

$$\text{And, } (x - 4)(y + 4) = xy \Rightarrow xy + 4x - 4y - 16 = xy$$

$$\Rightarrow 4x - 4y = 16 \Rightarrow x - y = 4 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $y = 8$.

Putting $y = 8$ in (ii), we get: $x = 12$.

$$\text{Hence, total number of students in the class} = xy = (12 \times 8) = 96.$$

369. Let Shyam's share be ₹ x . Then, Ram's share

$$= ₹ \left(x + \frac{3}{4}x \right) = ₹ \frac{7x}{4}.$$

$$\therefore x + \frac{7x}{4} = 275 \Leftrightarrow \frac{11x}{4} = 275 \Leftrightarrow x = \frac{275 \times 4}{11} = 100.$$

$$\text{So, Ram's share} = ₹ \left(\frac{7 \times 100}{4} \right) = ₹ 175.$$

370. Total number of sweets = $(64 \times 7 + 15) = 448 + 15 = 463$.

371. Let the whole property be worth ₹ x . Then,

$$\text{wife's share} = ₹ \frac{x}{2}; \text{ Son's share} = ₹ \left(\frac{1}{2} \text{ of } \frac{x}{2} \right) = ₹ \frac{x}{4}.$$

$$\text{Remaining share} = ₹ \left[x - \left(\frac{x}{2} + \frac{x}{4} \right) \right] = ₹ \frac{x}{4}.$$

$$\therefore \text{Share of each daughter} = ₹ \left(\frac{1}{2} \text{ of } \frac{x}{4} \right) = ₹ \frac{x}{8}.$$

372. Let the number of children be x . Then, $10x + 3 = 11x - 4 \Rightarrow x = 7$.

$$\therefore \text{Number of sweets} = (10 \times 7 + 3) = 73.$$

373. Let the number of boys be x . Then, number of girls = $(70 - x)$.

$$\therefore 2x + 3(70 - x) = 180 \Leftrightarrow x = 30.$$

374. Let the total number of cows in the herd be x .

$$\text{Then, share of fourth son} = x - \left(\frac{x}{2} + \frac{x}{4} + \frac{x}{5} \right) = x - \frac{19x}{20} = \frac{x}{20}.$$

$$\therefore \frac{x}{20} = 7 \text{ or } x = 140.$$

375. Let the total property be worth ₹ x .

$$\text{Then, elder son's share} = ₹ \left(\frac{5x}{11} \right). \text{ Balance} = ₹ \left(x - \frac{5x}{11} \right)$$

$$= ₹ \frac{6x}{11}.$$

$$\text{Younger son's share} = ₹ \left(\frac{5}{11} \text{ of } \frac{6x}{11} \right) = ₹ \left(\frac{30x}{121} \right).$$

$$\therefore \text{Widow's share} = ₹ \left[x - \left(\frac{5x}{11} + \frac{30x}{121} \right) \right]$$

$$= ₹ \left(x - \frac{85x}{121} \right) = ₹ \left(\frac{36x}{121} \right).$$

$$\frac{36x}{121} = 3600 \Leftrightarrow x = \frac{3600 \times 121}{36} = 12100.$$

$$\text{So, elder son's share} = ₹ \left(\frac{5}{11} \times 12100 \right) = ₹ 5500;$$

$$\text{Younger son's share} = ₹ \left(\frac{30}{121} \times 12100 \right) = ₹ 3000.$$

376. Suppose each boy gets ₹ x . Then, each woman gets ₹ $2x$ and each man gets ₹ $(x + 2x)$ i.e., ₹ $3x$.

$$\text{So, } 10 \times 3x + 12 \times 2x + 20x = 370 \Leftrightarrow 30x + 24x + 20x = 370 \Leftrightarrow 74x$$

$$= 370 \Leftrightarrow x = 5.$$

$$\therefore \text{Amount received by 10 men} = 30x = ₹ (30 \times 5) = ₹ 150.$$

377. Let D's share = ₹ x . Then, C's share = ₹ x .

$$\text{B's share} = ₹ (x + 125). \text{ A's share} = ₹ (x + x + 125) = ₹ (2x + 125).$$

$$\therefore (2x + 125) + (x + 125) + x + x = 750$$

$$\Leftrightarrow 5x = 500$$

$$\Leftrightarrow x = 100.$$

$$\text{Hence, A's share} = 2x + 125 = ₹ (2 \times 100 + 125) = ₹ 325.$$

378. Let Gagan's share = ₹ x .

$$\text{Then, Sachin's share} = ₹ \left(\frac{x}{5} \right) \text{ and Rohit's share} = ₹ \left(\frac{2x}{5} \right).$$

$$\therefore \frac{2x}{5} + \frac{x}{5} + x = 1000 \Leftrightarrow 8x = 5000 \Leftrightarrow x = 625.$$

379. If the second boy takes one-third of a certain number of marbles, the third boy is left with two-thirds of that number, which is twice the number with the second boy, regardless of the total number of marbles. Hence, the total number of marbles cannot be determined.

380. Wife's share = $\frac{1}{2}$. Remaining part = $\left(1 - \frac{1}{2} \right) = \frac{1}{2}$.

$$\text{Share of 3 sons} = \left(\frac{2}{3} \text{ of } \frac{1}{2} \right) = \frac{1}{3}. \text{ Remaining part}$$

$$= \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{6}.$$

$$\text{Each daughter's share} = \frac{1}{4} \times \frac{1}{6} = \frac{1}{24}.$$

$$\text{Let the total money be ₹ } x. \text{ Then, } \frac{1}{24} x = 20000$$

$$\Leftrightarrow x = 20000 \times 24 = 480000.$$

$$\therefore \text{Each son's share} = ₹ \left[\frac{1}{3} \times \left(\frac{1}{3} \times 480000 \right) \right] = ₹ 53,333.33.$$

- 381.** $A + B + C + D = 47$... (i)
 $A + B = 27$... (ii)
 $A + C = 25$... (iii)
 $A + D = 23$... (iv)
 Adding (ii), (iii) and (iv), we get:
 $2A + (A + B + C + D) = 75 \Leftrightarrow 2A + 47 = 75 \Leftrightarrow 2A = 28 \Leftrightarrow A = 14$.
 Putting $A = 14$ in (ii), we get: $B = 13$.
- 382.** Let the sum be ₹ x .
 Then, $\frac{x}{12} - \frac{x}{16} = 400 \Leftrightarrow \frac{x}{48} = 400 \Leftrightarrow x = 400 \times 48 = 19200$.
- 383.** Let the total number of sweets be $(25x + 8)$.
 Then, $(25x + 8) - 22$ is divisible by 28
 $\Leftrightarrow (25x - 14)$ is divisible by 28
 $\Leftrightarrow 28x - (3x + 14)$ is divisible by 28
 $\Leftrightarrow (3x + 14)$ is divisible by 28 $\Leftrightarrow x = 14$.
 $\therefore$ Total number of sweets = $(25 \times 14 + 8) = 358$.
- 384.** Let the total number of sweets be x . Then,
 $\frac{x}{80} - \frac{x}{112} = 6 \Leftrightarrow \frac{7x - 5x}{560} = 6 \Leftrightarrow \frac{x}{280} = 6 \Leftrightarrow x = 1680$.
 $\therefore$ Number of sweets each child was originally supposed to get = $\left(\frac{1680}{112}\right) = 15$.
- 385.** Let the total number of children be n and the amount received by each child be ₹ x .
 Then, total sum distributed = ₹ (nx) .
 So, $(n + 16)(x - 2) = nx$
 $\Leftrightarrow nx - 2n + 16x - 32 = nx$
 $\Leftrightarrow 16x - 2n = 32$... (i)
 And, $(n - 16)(x + 3) = nx$
 $\Leftrightarrow nx + 3n - 16x - 48 = nx$
 $\Leftrightarrow 16x - 3n = -48$... (ii)
 Subtracting (ii) from (i), we get: $n = 80$.
 Putting $n = 80$ in (i), we get: $16x = 192$ or $x = 12$.
 Hence, total sum distributed = $nx = ₹ (80 \times 12) = ₹ 960$.
- 386.** Suppose, Sanya and Vidushi donate money to x and $(x + 5)$ people respectively.
 Then, $\frac{100}{x} - \frac{100}{x+5} = 1 \Leftrightarrow 100(x + 5) - 100x = x(x + 5)$
 $\Leftrightarrow x^2 + 5x - 500 = 0$
 $\Leftrightarrow (x - 20)(x + 25) = 0 \Leftrightarrow x = 20$.
 $\therefore$ Total number of recipients of charity = $x + (x + 5) = (2x + 5) = 45$.
- 387.** Let the number of girls be x . Then, number of boys = $3x$.
 Total number of students = $x + 3x = 4x$.
 Clearly, the total number of students must be a multiple of 4.
- 388.** Let the number of boys be x . Then, number of girls = $(x - 2)$.
 $\therefore x^2 - (x - 2)^2 = 28$
 $\Leftrightarrow x^2 - (x^2 + 4 - 4x) = 28$
 $\Leftrightarrow 4x - 4 = 28$
 $\Leftrightarrow 4x = 32$
 $\Leftrightarrow x = 8$.
 So, number of boys = 8; number of girls = 6.
 $\therefore$ Total number of boys and girls = $8 + 6 = 14$.

- 389.** Let number of boys = x . Then, number of girls = x .
 Now, $2(x - 8) = x$ or $x = 16$.
 $\therefore$ Total number of students = $2x = (2 \times 16) = 32$.
- 390.** Given exp. = $(1 + x)(1 - x)(1 + x^2)(1 + x^4)(1 + x^8)$
 $= (1 - x^2)(1 + x^2)(1 + x^4)(1 + x^8) = (1 - x^4)(1 + x^4)(1 + x^8)$
 $= (1 - x^8)(1 + x^8) = (1 - x^{16})$.
- 391.** Given exp. = $\left(\frac{1}{x-1} - \frac{1}{x+1}\right) - \frac{2}{x^2+1} - \frac{4}{x^4+1}$
 $= \left[\frac{(x+1) - (x-1)}{(x-1)(x+1)}\right] - \frac{2}{x^2+1} - \frac{4}{x^4+1}$
 $= \left(\frac{2}{x^2-1} - \frac{2}{x^2+1}\right) - \frac{4}{x^4+1}$
 $= \left[\frac{2(x^2+1) - 2(x^2-1)}{(x^2-1)(x^2+1)}\right] - \frac{4}{x^4+1}$
 $= \frac{4}{x^4-1} - \frac{4}{x^4+1} = \frac{4(x^4+1) - 4(x^4-1)}{(x^4-1)(x^4+1)}$
 $= \frac{8}{(x^8-1)}$.
- 392.** Given exp. = $\left[\left(x + \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - x \cdot \frac{1}{x}\right)\right] \left[\left(x - \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} + x \cdot \frac{1}{x}\right)\right]$
 $= \left(x^3 + \frac{1}{x^3}\right)\left(x^3 - \frac{1}{x^3}\right) = \left(x^6 - \frac{1}{x^6}\right)$.
- 393.** $x + \frac{1}{x} = 3 \Rightarrow \left(x + \frac{1}{x}\right)^2 = 3^2 \Rightarrow x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = 9$
 $\Rightarrow x^2 + \frac{1}{x^2} = 9 - 2 = 7$.
- 394.** $a - \frac{1}{a} = p \Rightarrow \left(a - \frac{1}{a}\right)^2 = p^2 \Rightarrow a^2 + \frac{1}{a^2} - 2 \cdot a \cdot \frac{1}{a} = p^2$
 $\Rightarrow a^2 + \frac{1}{a^2} = p^2 + 2$.
- 395.** $\frac{x-1}{x} = 3 \Rightarrow 1 - \frac{1}{x} = 3 \Rightarrow \left(1 - \frac{1}{x}\right)^2 = 9$
 $\Rightarrow 1 + \frac{1}{x^2} - \frac{2}{x} = 9$
 $\Rightarrow 1 + \frac{1}{x^2} = 9 + \frac{2}{x} = 9 + \frac{2}{\left(-\frac{1}{2}\right)} = 9 - 4 = 5$.
 $\left[\because \frac{x-1}{x} = 3 \Rightarrow 3x = x-1 \Rightarrow 2x = -1 \Rightarrow x = -\frac{1}{2}\right]$
- 396.** $x^2 + \frac{1}{x^2} = 34 \Rightarrow x^2 + \frac{1}{x^2} + 2 \cdot x^2 \cdot \frac{1}{x^2} = 34 + 2 = 36$
 $\Rightarrow \left(x + \frac{1}{x}\right)^2 = 36 \Rightarrow \left(x + \frac{1}{x}\right) = 6$.

$$397. \left(x + \frac{1}{x}\right) = 2 \Rightarrow \left(x + \frac{1}{x}\right)^2 = 2^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 4 \Rightarrow x^2 + \frac{1}{x^2} = 2$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x} = 2 - 2 = 0$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 0 \Rightarrow x - \frac{1}{x} = 0.$$

$$398. \left(a + \frac{1}{a}\right) = 6 \Rightarrow \left(a + \frac{1}{a}\right)^2 = 6^2 = 36$$

$$\Rightarrow a^2 + \frac{1}{a^2} + 2 = 36 \Rightarrow \left(a^2 + \frac{1}{a^2}\right) = 34$$

$$\Rightarrow \left(a^2 + \frac{1}{a^2}\right)^2 = (34)^2$$

$$\Rightarrow a^4 + \frac{1}{a^4} + 2 = 1156 \Rightarrow \left(a^4 + \frac{1}{a^4}\right) = 1154.$$

$$399. \left(x - \frac{1}{x}\right) = 2 \Rightarrow \left(x - \frac{1}{x}\right)^2 = 2^2 = 4$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 4 \Rightarrow x^2 + \frac{1}{x^2} = 6$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = 6^2 = 36$$

$$\Rightarrow x^4 + \frac{1}{x^4} + 2 = 36 \Rightarrow x^4 + \frac{1}{x^4} = 34.$$

$$400. \left(x - \frac{1}{x}\right) = \sqrt{21} \Rightarrow \left(x - \frac{1}{x}\right)^2 = (\sqrt{21})^2 = 21$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 21 \Rightarrow x^2 + \frac{1}{x^2} = 23$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 25$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 5^2 \Rightarrow x + \frac{1}{x} = 5.$$

$$\therefore \left(x^2 + \frac{1}{x^2}\right) \left(x + \frac{1}{x}\right) = 23 \times 5 = 115.$$

$$401. x^4 + \frac{1}{x^4} = 322 \Rightarrow x^4 + \frac{1}{x^4} + 2 = 324$$

$$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 = (18)^2 \Rightarrow x^2 + \frac{1}{x^2} = 18$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 16$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 16 \Rightarrow \left(x - \frac{1}{x}\right) = 4.$$

$$402. \left(x + \frac{1}{x}\right) = p \Rightarrow \left(x + \frac{1}{x}\right)^3 = p^3$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) = p^3$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3p = p^3 \Rightarrow x^3 + \frac{1}{x^3} = p^3 - 3p.$$

$$403. \left(x - \frac{1}{x}\right) = 1 \Rightarrow \left(x - \frac{1}{x}\right)^3 = 1$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \cdot x \cdot \frac{1}{x} \left(x - \frac{1}{x}\right) = 1$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 = 1 \Rightarrow x^3 - \frac{1}{x^3} = 1 + 3 = 4.$$

$$404. \left(a^4 + \frac{1}{a^4}\right) = 1154 \Rightarrow a^4 + \frac{1}{a^4} + 2 = 1156$$

$$\Rightarrow \left(a^2 + \frac{1}{a^2}\right)^2 = 1156 \Rightarrow \left(a^2 + \frac{1}{a^2}\right) = 34$$

$$\Rightarrow a^2 + \frac{1}{a^2} + 2 = 36$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = 36 \Rightarrow a + \frac{1}{a} = 6$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^3 = 6^3 = 216$$

$$\Rightarrow a^3 + \frac{1}{a^3} + 3 \cdot a \cdot \frac{1}{a} \left(a + \frac{1}{a}\right) = 216$$

$$\Rightarrow a^3 + \frac{1}{a^3} + 3 \times 6 = 216$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 216 - 18 = 198.$$

$$405. x + \frac{1}{x} = 3 \Rightarrow \left(x + \frac{1}{x}\right)^3 = 3^3 = 27$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \cdot x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right) = 27$$

$$\Rightarrow x^3 + \frac{1}{x^3} + 3 \times 3 = 27 \Rightarrow x^3 + \frac{1}{x^3} = 18$$

$$\Rightarrow \left(x^3 + \frac{1}{x^3}\right)^2 = (18)^2 = 324$$

$$\Rightarrow x^6 + \frac{1}{x^6} + 2 = 324 \Rightarrow x^6 + \frac{1}{x^6} = 322.$$

$$406. x + \frac{1}{x} = \sqrt{13} \Rightarrow \left(x + \frac{1}{x}\right)^2 - 4 = (\sqrt{13})^2 - 4 = 13 - 4 = 9$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 = 9$$

$$\Rightarrow \left(x - \frac{1}{x}\right) = 3$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^3 = 3^3 = 27$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \cdot x \cdot \frac{1}{x} \left(x - \frac{1}{x}\right) = 27$$

$$\Rightarrow x^3 - \frac{1}{x^3} - 3 \times 3 = 27 \Rightarrow x^3 - \frac{1}{x^3} = 27 + 9 = 36.$$

$$\begin{aligned}
 407. \quad \left(4b^2 + \frac{1}{b^2}\right) &= 2 \Rightarrow \left(2b + \frac{1}{b}\right)^2 - 4 = 2 \\
 &\Rightarrow \left(2b + \frac{1}{b}\right)^2 = 6 \Rightarrow \left(2b + \frac{1}{b}\right) = \sqrt{6} \\
 &\Rightarrow \left(2b + \frac{1}{b}\right)^3 = (\sqrt{6})^3 = 6\sqrt{6} \\
 &\Rightarrow 8b^3 + \frac{1}{b^3} + 3 \times 2b \times \frac{1}{b} \cdot \left(2b + \frac{1}{b}\right) = 6\sqrt{6} \\
 &\Rightarrow \left(8b^3 + \frac{1}{b^3}\right) + 6\sqrt{6} = 6\sqrt{6} \\
 &\Rightarrow \left(8b^3 + \frac{1}{b^3}\right) = 0.
 \end{aligned}$$

$$\begin{aligned}
 408. \quad \left(2p + \frac{1}{p}\right) &= 4 \Rightarrow \frac{1}{2} \left(2p + \frac{1}{p}\right) = \frac{1}{2} \times 4 \Rightarrow \left(p + \frac{1}{2p}\right) = 2 \\
 &\Rightarrow \left(p + \frac{1}{2p}\right)^3 = 2^3 = 8 \\
 &\Rightarrow p^3 + \frac{1}{8p^3} + 3p \cdot \frac{1}{2p} \left(p + \frac{1}{2p}\right) = 8 \\
 &\Rightarrow p^3 + \frac{1}{8p^3} + 3 \times \frac{1}{2} \times 2 = 8 \\
 &\Rightarrow p^3 + \frac{1}{8p^3} = 8 - 3 = 5.
 \end{aligned}$$

409. Let the number of cows in the herd be x .
Then, number of legs = $4x$; number of heads = x .
 $\therefore 4x - 2x = 14$ or $2x = 14$ or $x = 7$.

410. Let the number of buffaloes be x and the number of ducks be y .
Then, $4x + 2y = 2(x + y) + 24 \Leftrightarrow 2x = 24 \Leftrightarrow x = 12$.

411. Let the number of hens be x and the number of cows be y .
Then, $x + y = 59$... (i)
and $2x + 4y = 190$ or $x + 2y = 95$... (ii)
Subtracting (i) from (ii), we get: $y = 36$.

412. Let the number of parrots be x and the number of tigers be y .
Then, $x + y = 858$... (i)
and $2x + 4y = 1846$ or $x + 2y = 923$... (ii)
Subtracting (i) from (ii), we get: $y = 65$.
Putting $y = 65$ in (i), we get: $x = 858 - 65 = 793$.

413. Let the number of horses be x . Then, number of men = x .
 $\therefore 4x + \frac{x}{2} \times 2 = 70 \Leftrightarrow 5x = 70 \Leftrightarrow x = 14$.

414. Let the full fare be ₹ x and the reservation charge be ₹ y per ticket.
Then, $x + y = 216$... (i)
and $\frac{3}{2}x + 2y = 327$ or $3x + 4y = 654$... (ii)
Multiplying (i) by 3 and subtracting from (ii), we get: $y = 6$.
Putting $y = 6$ in (i), we get: $x = 210$.

415. Let the distance travelled by taxi be x km.
Then, distance travelled by own car = $(90 - x)$ km.
 $\therefore 7x + 6(90 - x) = 675 \Leftrightarrow x = 675 - 540 = 135$ km.

416. Clearly, first man shakes hand with all other 14 men; second man shakes hand with 13 men (other than first man); third shakes hand with 12 men and so on.
 $\therefore$ Total number of hand-shakes = $(14 + 13 + 12 + \dots + 1)$
 $= \frac{14 \times 15}{2} = 105$.

417. Let there be n children in the group.
Then, first child exchanges gifts with other $(n - 1)$ children; second child exchanges gifts with other $(n - 2)$ children; and so on.
So, number of gifts = $2 [1 + 2 + \dots + (n - 1)]$
 $= 2 \times \frac{n(n - 1)}{2} = n(n - 1)$.
 $\therefore n(n - 1) = 132 \Leftrightarrow n^2 - n - 132 = 0$
 $\Leftrightarrow n^2 - 12n + 11n - 132 = 0$
 $\Leftrightarrow (n - 12)(n + 11) = 0 \Leftrightarrow n = 12$.

418. 1044 candles produce 1044 butts, from which $\frac{1044}{9} = 116$ candles can be formed.

The resultant 116 butts give $\frac{116}{9}$ i.e., 12 candles and 8 butts.
12 resultant butts plus 8 remaining butts = 20 butts from which $\frac{20}{9}$ i.e., 2 candles and 2 butts will be obtained.

In the end, 4 butts will be left behind.
 $\therefore$ Total number of candles formed

$$= (116 + 12 + 2) = 130.$$

419. Number of lines printed in 1 hour = $\frac{176400}{7} = 25200$.
Number of lines printed per minute = $\frac{25200}{60} = 420$.

420. Number of pages typed by 1 man in 1 hour = $\frac{60}{8} = 7.5$.
 $\therefore$ Number of men required = $\frac{1710}{7.5} = 228$.

421. Number of pages read by Ashu from 2: 30 to 3: 20 i.e., in 50 min or $\frac{5}{6}$ hr = $\left(\frac{5}{6} \times 30\right) = 25$.

Let the two read the same page x hours after 3: 20.

Then, pages read by Ashu = $(25 + 30x)$. And, pages read by Neeru = $40x$.

$$\therefore 25 + 30x = 40x \Rightarrow x = \frac{5}{2} = 2\frac{1}{2}.$$

Hence, the two read the same page $2\frac{1}{2}$ hours after 3: 20 i.e., at 5: 50.

$$422. \quad \frac{336}{x} = \frac{192}{400} \Rightarrow x = \frac{336 \times 400}{192} = 700.$$

423. Maximum internal assessment score = $\left(\frac{47}{50} \times 10\right) = 9.4$.

Minimum internal assessment score = $\left(\frac{14}{50} \times 10\right) = 2.8$.

$\therefore$ Required difference = $(9.4 - 2.8) = 6.6$.

424. Let the number of questions answered correctly be x .

Then, number of wrong answers = $(200 - x)$.

$\therefore 4x - (200 - x) = 200 \Rightarrow 5x = 400 \Rightarrow x = 80$.

425. Let the number of correct answers be x . Then, number of wrong answers = $(26 - x)$.

$\therefore 8x - 5(26 - x) = 0 \Rightarrow 13x = 130 \Rightarrow x = 10$.

426. Suppose the student attempted x questions of 2 marks correctly.

Then, number of 4 marks questions attempted correctly = $(15 - x)$.

$\therefore 2x + 4(15 - x) = 40 \Leftrightarrow 2x + 60 - 4x = 40$

$\Leftrightarrow 2x = 20 \Leftrightarrow x = 10$.

427. Let the number of correct answers be x . Then, number of incorrect answers = $(30 - x)$.

$\therefore x - 0.25(30 - x) = 13.75$

$\Leftrightarrow x - 7.5 + 0.25x = 13.75$

$\Leftrightarrow 1.25x = 21.25 \Leftrightarrow x = \frac{2125}{125} = 17$.

So, number of incorrect answers = $30 - 17 = 13$.

428. Suppose he hits the target x times.

Then, he misses it $(100 - x)$ times.

$\therefore x - (100 - x) = 30 \Rightarrow 2x = 130 \Rightarrow x = 65$.

So, number of times he misses the target = $100 - 65 = 35$.

429. Let the number of cows be x . Then, number of bulls = $2x$.

Number of caretakers = $\frac{x + 2x + 45}{15} = \frac{3x + 45}{15} = \frac{x + 15}{5}$.

Number of heads = $x + 2x + 45 + \frac{x + 15}{5} = \frac{16x + 240}{5}$.

Number of feet = $4x + 4 \times 2x + 2 \times 45 + 2\left(\frac{x + 15}{5}\right)$
 $= 4x + 8x + 90 + \left(\frac{2x + 30}{5}\right) = \frac{62x + 480}{5}$.

$\therefore \frac{62x + 480}{5} - \frac{16x + 240}{5} = 186 \Rightarrow 46x + 240 = 930$
 $\Rightarrow 46x = 690 \Rightarrow x = 15$.

Hence, number of caretakers = $\frac{x + 15}{5} = \frac{30}{5} = 6$.

430. Man's basic pay per hour = $\text{₹} \left(\frac{200}{40}\right) = \text{₹} 5$.

Overtime charges per hour = 125% of $\text{₹} 5 = \text{₹} 6.25$.

Suppose the man worked overtime for x hours.

Then, $200 + 6.25x = 300 \Rightarrow 6.25x = 100 \Rightarrow x = 16$.

$\therefore$ Total number of working hours = $(40 + 16) = 56$.

431. Suppose the man works overtime for x hours.

Now, working hours in 4 weeks = $(5 \times 8 \times 4) = 160$.

$\therefore 160 \times 24 + x \times 32 = 4320 \Rightarrow 32x = 480 \Rightarrow x = 15$.

Hence, total hours of work = $(160 + 15) = 175$.

432. Suppose their paths cross after x minutes.

Then, $11 + 57x = 51 - 63x \Leftrightarrow 120x = 40 \Leftrightarrow x = \frac{1}{3}$.

Number of floors covered by David in

$\frac{1}{3} \text{ min} = \left(\frac{1}{3} \times 57\right) = 19$.

So, their paths cross at $(11 + 19)$ i.e., 30th floor.

433. Let the total number of steps be n .

Then, $\frac{30}{n - 26} = \frac{18}{n - 34} \Leftrightarrow 30(n - 34) = 18(n - 26)$

$\Leftrightarrow 30n - 18n = 552 \Leftrightarrow 12n = 552 \Leftrightarrow n = 46$.

434. Let the price of the uniform be $\text{₹} x$.

Then, servant's pay per day = $\text{₹} \left(\frac{200 + x}{10}\right)$.

$\therefore 5\left(\frac{200 + x}{10}\right) = 20 + x \Leftrightarrow 200 + x = 40 + 2x$

$\Leftrightarrow x = 160$.

435. Suppose he remained absent for x days.

Then, number of days on which he was present = $(20 - x)$.

$\therefore 60(20 - x) - 5x = 745 \Leftrightarrow 65x = 455 \Rightarrow x = 7$.

436. Let the fixed charge be $\text{₹} x$ and variable charge be $\text{₹} y$ per km.

Then, $x + 16y = 156$... (i) and $x + 24y = 204$... (ii)

Subtracting (i) from (ii), we get: $8y = 48$ or $y = 6$.

Putting $y = 6$ in (i), we get: $x = 60$.

$\therefore$ Cost of travelling 30 km = $\text{₹} (60 + 30 \times 6) = \text{₹} 240$.

437. Original contribution = $\text{₹} \left(\frac{D}{M}\right)$.

New contribution = $\text{₹} \left(\frac{D}{M - 3}\right)$.

$\therefore$ Required contribution = $\text{₹} \left[\frac{D}{(M - 3)} - \frac{D}{M} \right]$
 $= \text{₹} \frac{DM - D(M - 3)}{M(M - 3)}$
 $= \text{₹} \left(\frac{3D}{M^2 - 3M} \right)$.

438. $N \times 50 = (325000 - 300000) = 25000 \Rightarrow N = 500$.

439. Let number of boys = x . Then, number of girls = $(60 - x)$.

$\therefore x(60 - x) + (60 - x)x = 1600$

$\Leftrightarrow 2x^2 - 120x + 1600 = 0$

$\Leftrightarrow x^2 - 60x + 800 = 0$

$\Leftrightarrow (x - 40)(x - 20) = 0$

$\Leftrightarrow x = 40$ or $x = 20$.

So, we are not definite, Hence, data is inadequate.

440. Original share of 1 person = $\frac{1}{8}$. New share = $\frac{1}{7}$.

Increase = $\left(\frac{1}{7} - \frac{1}{8}\right) = \frac{1}{56}$.

$\therefore$ Required fraction = $\frac{(1/56)}{(1/8)} = \left(\frac{1}{56} \times 8\right) = \frac{1}{7}$.

441. Let the original duration of the tour be n days.

$$\begin{aligned}\text{Then, } \frac{360}{n} - \frac{360}{n+4} &= 3 \Leftrightarrow 360 \left[\frac{n+4-n}{n(n+4)} \right] = 3 \\ &\Leftrightarrow n(n+4) = 120 \times 4 = 480 \\ &\Leftrightarrow n^2 + 4n - 480 = 0 \Leftrightarrow (n+24)(n-20) = 0 \Leftrightarrow n = 20.\end{aligned}$$

442. Let the original number of students be x .

$$\begin{aligned}\text{Then, } \frac{500}{x-5} - \frac{500}{x} &= 5 \Leftrightarrow \frac{1}{x-5} - \frac{1}{x} = \frac{1}{100} \Leftrightarrow x(x-5) = 500 \\ &\Leftrightarrow x^2 - 5x - 500 = 0 \\ &\Leftrightarrow (x-25)(x+20) = 0 \Leftrightarrow x = 25.\end{aligned}$$

Hence, number of students who attended the picnic = $(25 - 5) = 20$.

443. Let the original cost per student be ₹ x .

$$\begin{aligned}\text{Then, } \frac{720}{x} - \frac{720}{x+9} &= 4 \Leftrightarrow 720 \left[\frac{x+9-x}{x(x+9)} \right] = 4 \\ &\Leftrightarrow x(x+9) = 1620 \\ &\Leftrightarrow x^2 + 9x - 1620 = 0 \\ &\Leftrightarrow (x+45)(x-36) = 0 \Leftrightarrow x = 36.\end{aligned}$$

444. Let the total price of the CDs be ₹ x and the number of boys be n .

$$\begin{aligned}\text{Then, } \frac{x}{n-2} - \frac{x}{n} &= 1 \Leftrightarrow \frac{nx - x(n-2)}{n(n-2)} = 1 \\ &\Leftrightarrow 2x = n^2 - 2n \Leftrightarrow x = \frac{n^2 - 2n}{2}.\end{aligned}$$

Since x is a whole number, so n must be even.

For $n = 20$, we have: $x = 180 < 200$.

For $n = 22$, we have: $x = 220$.

For $n = 24$, we have: $x = 264 > 250$.

So, $n = 22$, $x = ₹ 220$.

445. $X + 3X \times 0.50 + 14 \times 0.10 + 4X \times 0.05 = 50$
 $\therefore X + 1.5X + 1.40 + 0.2X = 50 \Leftrightarrow 2.7X = 48.60$
 $\Leftrightarrow X = 18$.

446. Let the number of 20-paise coins be x . Then, number of 25-paise coins = $(324 - x)$.

$$\begin{aligned}\therefore 0.20 \times x + 0.25(324 - x) &= 71 \\ \Leftrightarrow 20x + 25(324 - x) &= 7100 \\ \Leftrightarrow 5x = 1000 \Leftrightarrow x &= 200.\end{aligned}$$

Hence, number of 25-paise coins = $(324 - x) = 124$.

447. Let number of notes of each denomination be x .
 Then, $x + 5x + 10x = 480 \Leftrightarrow 16x = 480 \Leftrightarrow x = 30$.
 Hence, total number of notes = $3x = 90$.

448. Let the number of coins of each denomination be x .

$$\begin{aligned}\text{Then, } x + 0.5x + 0.25x + 0.10x + 0.05x &= 380 \\ \Rightarrow 1.9x = 380 \Rightarrow x &= \frac{380}{1.9} = 200.\end{aligned}$$

Hence, number of one-rupee coins = 200.

449. Let the number of 50 p -coins and 25 p -coins be x and y respectively.

Then, number of 1 rupee coins = $175 - (x + y)$.

$$\therefore 0.5x = 0.25y \text{ or } 2x = y.$$

So, number of 50 p -coins = x , number of 25 p -coins = $2x$, number of 1 rupee coins = $(175 - 3x)$.

$$\therefore 175 - 3x = 0.5x \Leftrightarrow 3.5x = 175 \Leftrightarrow x = \frac{175}{3.5} = 50.$$

$$\begin{aligned}\text{Hence, total amount} &= (175 - 3x) + 0.5x + 0.25 \times 2x \\ &= 175 - 3x + 0.5x + 0.5x = 175 - 3x + x \\ &= 175 - 2x = ₹ (175 - 2 \times 50) = ₹ 75.\end{aligned}$$

450. Total value of one 2-rupee coin, two 50- p coins and two 25- p coins

$$= ₹ (2 + 2 \times 0.50 + 2 \times 0.25) = ₹ 3.50.$$

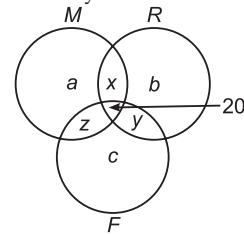
Remaining amount = ₹ $(25 - 3.50) = ₹ 21.50$.

For the minimum number of coins, we must have maximum amount in denominations of 2-rupee and 50- p which shall be ten 2-rupee and three 50- p coins i.e., 13 more coins.

$$\therefore \text{Required number of coins} = (1 + 2 + 2) + 13 = 18.$$

Questions 451 and 452

Let the circles M , R and F represent the number of children who took ride on merry-go-round, roller coaster and Ferris wheel only.



Let a = number of children who took ride on merry-go-round only;

b = number of children who took ride on roller coaster only;

c = number of children who took ride on Ferris wheel only;

x = number of children who took ride on both merry-go-round and roller coaster;

y = number of children who took ride on both roller coaster and Ferris wheel;

z = number of children who took ride on both merry-go-round and Ferris wheel.

Then, number of children who took exactly one ride = $a + b + c$;

number of children who took exactly two rides

$$= x + y + z;$$

number of children who took all three rides = 20;

number of children who took at least 2 rides

$$= x + y + z + 20.$$

$$\therefore \text{We have: } x + y + z + 20 = 55 \Leftrightarrow x + y + z = 35$$

$$\text{Total receipt} = ₹ [a + b + c + 2(x + y + z) + 3 \times 20]$$

$$= ₹ (a + b + c + 2 \times 35 + 60)$$

$$= ₹ (a + b + c + 130).$$

$$\therefore a + b + c + 130 = 145 \text{ or } a + b + c = 15.$$

451. Number of children who took exactly one ride

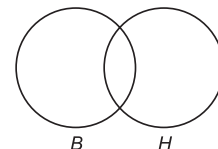
$$= a + b + c = 15.$$

452. Number of children who did not try any ride

$$= 85 - (a + b + c + x + y + z + 20)$$

$$= 85 - (15 + 35 + 20) = 15.$$

453. Number of students who took any one or both the languages

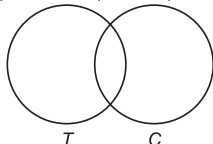


$$= 50 - 12 = 38.$$

$$n(B) = 25, n(H) = 16, n(B \cup H) = 38.$$

$$\begin{aligned} \therefore \text{Number of students who take both Bengali and Hindi} \\ &= n(B \cap H) = n(B) + n(H) - n(B \cup H) \\ &= (25 + 16) - 38 = (41 - 38) = 3. \end{aligned}$$

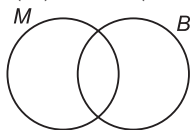
$$454. n(T) = 60, n(C) = 40, n(T \cap C) = 25.$$



$$\begin{aligned} \therefore \text{Number of students who take either tea or coffee or both} \\ &= n(T \cup C) = n(T) + n(C) - n(T \cap C) \\ &= 60 + 40 - 25 = 100 - 25 = 75. \end{aligned}$$

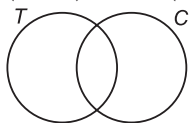
$$\therefore \text{Required number of students} = (100 - 75) = 25.$$

$$455. n(M \cup B) = 25, n(M) = 12, n(M - B) = 8.$$



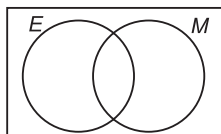
$$\begin{aligned} \text{Number students who have taken both Mathematics and} \\ \text{Biology} &= n(M \cap B) = n(M) - n(M - B) = 12 - 8 = 4. \end{aligned}$$

$$456. n(T \cup C) = 52, n(T - C) = 16, n(T) = 33.$$



$$\begin{aligned} \text{Number of persons who drink coffee but not tea} \\ &= n(C - T) = n(T \cup C) - n(T) = 52 - 33 = 19. \end{aligned}$$

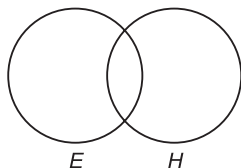
$$457. n(E) = 3000, n(M) = 2000, n(E \cap M) = 1500.$$



$$\begin{aligned} n(E \cup M) &= n(E) + n(M) - n(E \cap M) = 3000 + 2000 - 1500 \\ &= 3500. \end{aligned}$$

$$\begin{aligned} \text{Number of pure vegetarians} &= 4000 - n(E \cup M) \\ &= 4000 - 3500 = 500. \end{aligned}$$

$$458. n(E \cup H) = 50; n(E \cap H) = 10; n(E) = 21.$$



$$\begin{aligned} \text{Number of students who can speak Hindi} &= n(H) \\ &= n(E \cap H) + n(H - E) = 10 + 29 = 39. \end{aligned}$$

$$\begin{aligned} \text{Number of students who can speak only Hindi} \\ &= n(H - E) = n(H) - n(E \cap H) = 39 - 10 = 29. \end{aligned}$$

$$\begin{aligned} \text{Number of students who can speak only English} \\ &= n(E - H) = n(E) - n(E \cap H) = 21 - 10 = 11. \end{aligned}$$

$$459. \text{Let the total number of staff members be } x.$$

Then, the number who can type or take shorthand

$$= \left(x - \frac{3x}{4} \right) = \frac{x}{4}.$$

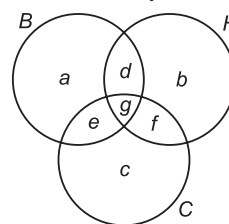
Let A and B represent the sets of persons who can type and take shorthand respectively.

$$\text{Then, } n(A \cup B) = \frac{x}{4}, n(A) = \frac{x}{5} \text{ and } n(B) = \frac{x}{3}.$$

$$n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

$$= \left(\frac{x}{5} + \frac{x}{3} - \frac{x}{4} \right) = \left(\frac{12x + 20x - 15x}{60} \right) = \frac{17x}{60}.$$

460. Let circles B , H and C represent the number of players who play basketball, hockey and cricket respectively.



$$\text{Then, } a + d + e + g = 20 \quad \dots(i)$$

$$b + d + f + g = 24 \quad \dots(ii)$$

$$c + e + f + g = 27 \quad \dots(iii)$$

$$d + g = 12 \quad \dots(iv)$$

$$e + g = 10 \quad \dots(v)$$

$$f + g = 14 \quad \dots(vi)$$

$$g = 7 \quad \dots(vii)$$

From (iv), (v), (vi) and (vii), we have:

$$d = 12 - g = 12 - 7 = 5;$$

$$e = 10 - g = 10 - 7 = 3;$$

$$f = 14 - g = 14 - 7 = 7.$$

From (iii), we have: $c = 27 - (e + f + g)$

$$= 27 - (3 + 7 + 7) = 10.$$

From (ii), we have: $b = 24 - (d + f + g)$

$$= 24 - (5 + 7 + 7) = 5.$$

From (i), we have: $a = 20 - (d + e + g)$

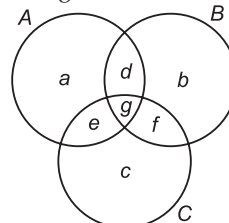
$$= 20 - (5 + 3 + 7) = 5.$$

Hence, total number of players = $a + b + c + d + e + f + g$

$$= (5 + 5 + 10 + 5 + 3 + 7 + 7)$$

$$= 42.$$

461. Let the circles A , B and C represent the aliens having two noses, three legs and four ears respectively.



$$\text{Then, } a + d + e + g = 40 \quad \dots(i)$$

$$b + d + f + g = 30 \quad \dots(ii)$$

$$c + e + f + g = 20 \quad \dots(iii)$$

$$d + g = 10 \quad \dots(iv)$$

$$f + g = 12 \quad \dots(v)$$

$$e + g = 5 \quad \dots(vi)$$

$$g = 3 \quad \dots(vii)$$

From (iv), (v), (vi) and (vii), we have:

$$d = 10 - g = 10 - 3 = 7;$$

$$e = 5 - g = 5 - 3 = 2;$$

$$f = 12 - g = 12 - 3 = 9.$$

From (iii), we have: $c = 20 - (e + f + g)$
 $= 20 - (2 + 9 + 3) = 20 - 14 = 6.$

From (ii), we have: $b = 30 - (d + f + g)$
 $= 30 - (7 + 9 + 3) = 30 - 19 = 11.$

From (i), we have: $a = 40 - (d + e + g)$
 $= 40 - (7 + 2 + 3) = 40 - 12 = 28.$

$\therefore$ Number of aliens who had one or more of the three unusual features

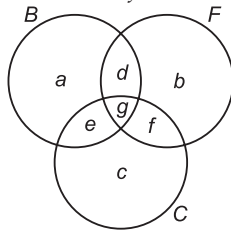
$$= a + b + c + d + e + f + g$$

$$= (28 + 11 + 6 + 7 + 2 + 9 + 3) = 66.$$

Hence, number of aliens without any of the unusual features $= (100 - 66) = 34.$

Questions 462 to 466

Let the circles B, F and C represent the children who play badminton, football and cricket respectively. Then, $c = 5, b = 8, a = 7, d = 3, f = 4, e = 4, g = 2.$



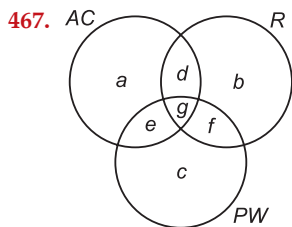
462. Required number $= f + g = 4 + 2 = 6.$

463. Required number $= a + d + e + g = 7 + 3 + 4 + 2 = 16.$

464. Required number $= b + d + f + g = 8 + 3 + 4 + 2 = 17.$

465. Required number $= e + g = 4 + 2 = 6.$

466. Required number $= a + b + c + d + e + f + g$
 $= 7 + 8 + 5 + 3 + 4 + 4 + 2 = 33.$



$$a + d + e + g = 15 \quad \dots(i)$$

$$e = 2 \quad \dots(ii)$$

$$b + d + f + g = 12 \quad \dots(iii)$$

$$d = 6 \quad \dots(iv)$$

$$c + e + f + g = 11 \quad \dots(v)$$

$$f + g = 4 \quad \dots(vi)$$

$$g = 3 \quad \dots(vii)$$

From (i), (ii), (iv) and (vii), we have: $a = 15 - (d + e + g)$
 $= 15 - (6 + 2 + 3)$
 $= 15 - 11 = 4.$

From (vi), we have: $f = 4 - g = 4 - 3 = 1.$

From (iii), we have: $b = 12 - (d + f + g)$
 $= 12 - (6 + 1 + 3) = 2.$

From (v), we have: $c = 11 - (e + f + g)$
 $= 11 - (2 + 1 + 3) = 5.$

$\therefore$ Number of cars which had at least one option
 $= a + b + c + d + e + f + g$
 $= 4 + 2 + 5 + 6 + 2 + 1 + 3 = 23.$

So, number of cars that had none of the options
 $= 25 - 23 = 2.$

468. $a + d + e + g = 23 \quad \dots(i)$

$$b + d + f + g = 15 \quad \dots(ii)$$

$$c + e + f + g = 18 \quad \dots(iii)$$

$$d = 3 \quad \dots(iv)$$

$$f = 6 \quad \dots(v)$$

$$e = 6 \quad \dots(vi)$$

$$a = 9 \quad \dots(vii)$$

From (i), we have: $g = 23 - (a + d + e)$
 $= 23 - (9 + 3 + 6) = 5.$

$\therefore$ Number of students who speak all the three languages $= g = 5.$

469. $a + d + e + g = 200 \quad \dots(i)$

$$b + d + f + g = 150 \quad \dots(ii)$$

$$c + e + f + g = 150 \quad \dots(iii)$$

$$d + g = 80 \quad \dots(iv)$$

$$f + g = 60 \quad \dots(v)$$

$$e + g = 70 \quad \dots(vi)$$

$$a = 80 \quad \dots(vii)$$

From (i), (iv) and (vii), we get:

$$a + e = 200 - (d + g) = 200 - 80 = 120.$$

$$\Rightarrow e = 120 - a = 120 - 80 = 40.$$

From (vi), we have: $g = 70 - e = 70 - 40 = 30.$

From (v), we have: $f = 60 - g = 60 - 30 = 30.$

From (iv), we have: $d = 80 - g = 80 - 30 = 50.$

From (ii), we have: $b = 150 - (d + f + g) = 150 - (50 + 30 + 30) = 40.$

From (iii), we have: $c = 150 - (e + f + g)$
 $= 150 - (40 + 30 + 30) = 50.$

Number of Dr Hormis' patients $= b = 40.$

Number of Dr Gerard's patients $= c = 50.$

Number of Dr Paul's patients $= d + e + f + g$
 $= 50 + 40 + 30 + 30 = 150.$

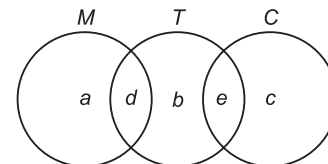
470. $a + b + c + d + e = 18 \quad \dots(i)$

$$a + d = 10 \quad \dots(ii)$$

$$b + d + e = 8 \quad \dots(iii)$$

$$c + e = 6 \quad \dots(iv)$$

$$d = 5 \quad \dots(v)$$



Adding (ii) and (iv) and subtracting from (i), we get:

$$b = 18 - (10 + 6) = 18 - 16 = 2.$$

From (iii), we have:

$$e = 8 - (b + d) = 8 - (2 + 5) = 8 - 7 = 1.$$

$\therefore$ Number of members who take tea as well as coffee
 $= e = 1.$

Questions 471 to 475

We have: $a = 650, b = 550,$

$c = 450, g = 100, d + g = 200,$

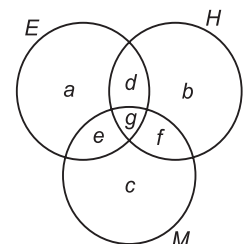
$f + g = 400, e + g = 300.$

So,

$e = 300 - g = 300 - 100 = 200; f$

$= 400 - g = 400 - 100 = 300;$

$d = 200 - g = 200 - 100 = 100.$



471. Number of members reading Hindi newspaper
 $= b + d + f + g = 550 + 100 + 300 + 100 = 1050$.
472. Number of members who read only one newspaper
 $= a + b + c = 650 + 550 + 450 = 1650$.
473. Number of members reading no newspaper
 $= 2800 - (a + b + c + d + e + f + g)$
 $= 2800 - (650 + 550 + 450 + 100 + 200 + 300 + 100)$
 $= 2800 - 2350 = 450$.
474. Number of members who read at least 2 newspapers
 $= d + e + f + g = 100 + 200 + 300 + 100 = 700$.
475. Required difference $= (e + g) - (d + g) = e - d$
 $= 200 - 100 = 100$.
476. Let the consumption of ration be x kg per day. Then,
 $75(x + 12) = 90(x + 8)$
 $\Leftrightarrow 75x + 900 = 90x + 720$
 $\Leftrightarrow 15x = 180 \Leftrightarrow x = 12$.
 $\therefore$ Total quantity of ration $= 75(x + 12)$
 $= (75 \times 24) \text{ kg} = 1800 \text{ kg}$.
Hence, it would last for $\left(\frac{1800}{12}\right)$ days i.e., 150 days.
477. Let l be the length of each candle and x be the number of hours before 4 P.M. when the candles should be lighted.
Length of first candle burnt $= \frac{l}{3} \times x$. Length of second candle burnt $= \frac{l}{4} \times x$.
Remaining length of first candle $= l - \frac{xl}{3}$.
Remaining length of second candle $= l - \frac{xl}{4}$.
 $\therefore l - \frac{xl}{4} = 2 \left(l - \frac{xl}{3} \right) \Leftrightarrow l \left(1 - \frac{x}{4} \right) = 2l \left(1 - \frac{x}{3} \right)$
 $\Leftrightarrow 1 - \frac{x}{4} = 2 \left(1 - \frac{x}{3} \right) \Leftrightarrow \frac{2x}{3} - \frac{x}{4} = 1$
 $\Leftrightarrow \frac{5x}{12} = 1 \Leftrightarrow x = \frac{12}{5} = 2\frac{2}{5}$.
So, the candles should be lighted $2\frac{2}{5}$ hrs i.e., 2 hr 24 min before 4 P.M. i.e., at 1: 36 P.M.
478. Suppose the man bought x apples. Then, number of oranges bought $= (40 - x)$.
Let the cost of each apple be ₹ a and that of each orange be ₹ b .
Then, $ax + b(40 - x) = 17 \Leftrightarrow ax - bx = 17 - 40b \dots(i)$
And, $a(40 - x) + bx = 15 \Leftrightarrow ax - bx = 40a - 15 \dots(ii)$
From (i) and (ii), we get:
 $40a - 15 = 17 - 40b \Leftrightarrow 40a + 40b$
 $= 32 \Leftrightarrow 40(a + b) = 32 \Leftrightarrow a + b = \frac{4}{5}$.
Hence, cost of 1 apple and 1 orange
 $= ₹ \frac{4}{5} = \left(\frac{4}{5} \times 100 \right) p = 80 p$.
479. Suppose she purchased n books for ₹ 720. Then, cost of each book $= ₹ \left(\frac{720}{n} \right)$.

$$\therefore \frac{720}{n} - \frac{720}{n+4} = 2 \Rightarrow \frac{(n+4) - n}{n(n+4)} = \frac{1}{360} \Rightarrow n^2 + 4n - 1440 = 0$$

$$\Rightarrow (n+40)(n-36) = 0 \Rightarrow n = 36.$$

480. Let the number of pineapple, mango and black-forest pastries be x , y and z respectively.
Then, cost of each pineapple pastry = ₹ x ; cost of each mango pastry = ₹ y ;
cost of each black-forest pastry = ₹ z .
Thus, $x + y + z = 23$. And, $x^2 + y^2 + z^2 = 211$.
So, we have to find 3 numbers whose sum is 23 and the sum of whose squares is 211. Such a triplet is (11, 9, 3).
481. Suppose each tube contains x grams initially. Then,
 $4 \left[\frac{1}{3}(x+20) \right] = (x-20) + \frac{2}{3}(x+20)$
 $\Leftrightarrow \frac{2}{3}(x+20) = (x-20)$
 $\Leftrightarrow \frac{2}{3}x + \frac{40}{3} = x - 20$
 $\Leftrightarrow \frac{x}{3} = \frac{40}{3} + 20 = \frac{100}{3} \Leftrightarrow x = 100$.
482. Let the capacity of the first bottle be x litres.
Then, capacity of the second bottle $= 2x$ litres.
And, capacity of the third bottle $= (x + 2x)$ litres
 $= 3x$ litres.
Quantity of milk in the first bottle $= \frac{x}{2}$ litres.
Quantity of milk in the second bottle $= \left(\frac{2x}{4} \right)$ litres $= \frac{x}{2}$ litres.
Quantity of milk in the third bottle $= \left(\frac{x}{2} + \frac{x}{2} \right)$ litres $= x$ litres.
 $\therefore$ Required fraction $= \left(\frac{x}{3x} \right) = \frac{1}{3}$.
483. Number of passengers after the first stop
 $= \left[540 - \left(\frac{1}{9} \text{ of } 540 \right) + 24 \right] = 504$.
Number of passengers after the second stop
 $= \left[504 - \left(\frac{1}{8} \text{ of } 504 \right) + 9 \right] = 450$.
484. Let the number of passengers at the start be x .
Then, number of passengers after the first stop
 $= \left(\frac{x}{2} + 125 \right)$.
Number of passengers after the second stop
 $= \frac{1}{2} \left(\frac{x}{2} + 125 \right) + 100 = \left(\frac{x}{4} + \frac{325}{2} \right)$.
 $\therefore \frac{x}{4} + \frac{325}{2} = 250 \Rightarrow \frac{x}{4} = 250 - \frac{325}{2} = \frac{175}{2}$
 $\Rightarrow x = \left(\frac{175}{2} \times 4 \right) = 350$.

- 485.** Total number of digits =
(No. of digits in 1-digit page nos. + No. of digits in 2-digit page nos. + No. of digits in 3-digit page nos.)
 $= (1 \times 9 + 2 \times 90 + 3 \times 267)$
 $= (9 + 180 + 801) = 990.$
- 486.** No. of digits in 1-digit page nos. = $1 \times 9 = 9.$
 No. of digits in 2-digit page nos. = $2 \times 90 = 180.$
 No. of digits in 3-digit page nos. = $3 \times 900 = 2700.$
 No. of digits in 4-digit page nos. = $3189 - (9 + 180 + 2700) = 3189 - 2889 = 300.$
 $\therefore$ No. of pages with 4-digit page nos. = $\left(\frac{300}{4}\right) = 75.$
 Hence, total number of pages = $(999 + 75) = 1074.$
- 487.** Let the number of students in sections A and B be x and y respectively. Then,
 $x + 10 = 3(y - 10) \Rightarrow 3y - x = 40 \quad \dots(i)$
 And, $x - 10 = y + 10 \Rightarrow x - y = 20 \quad \dots(ii)$
 Adding (i) and (ii), we get: $2y = 60$ or $y = 30.$
 Putting $y = 30$ in (ii), we get: $x = 50.$
- 488.** Clearly, the boy lost 7 chocolates, regained the 7 lost chocolates and won 20 more.
 $\therefore$ Number of games played = $(7 + 7 + 20) = 34.$
- 489.** Clearly, we have:
 $J - 40 = \frac{1}{2}A \quad \dots(i)$
 $A - 40 = J \quad \dots(ii)$
 $A - 40 = R + 40 \quad \dots(iii)$
 Solving (i) and (ii), we get: $J = 120$ and $A = 160.$
 Putting $A = 160$ in (iii), we get: $R = 80.$
 $\therefore$ Total money = $R + J + A = ₹(80 + 120 + 160) = ₹360.$
- 490.** We have:
 $B + 8 = C \quad \dots(i)$
 $A - 8 = C - 3 \quad \dots(ii)$
 $A + 6 = 2D \quad \dots(iii)$
 $B + D = 50 \quad \dots(iv)$
 Putting $C = A - 5$ from (ii) into (i), we have:
 $B + 8 = A - 5$ or $A - B = 13 \quad \dots(v)$
 Putting $D = 50 - B$ from (iv) into (iii), we have:
 $A + 6 = 100 - 2B$ or $A + 2B = 94 \quad \dots(vi)$
 Solving (v) and (vi), we get: $B = 27, A = 40.$
- 491.** We have:
 $B - 3 = E \quad \dots(i)$
 $B + 3 = D \quad \dots(ii)$
 $A + B = D + E + 10 \quad \dots(iii)$
 $B = C + 2 \quad \dots(iv)$
 $A + B + C + D + E = 133 \quad \dots(v)$
 From (i) and (ii), we have: $2B = D + E \quad \dots(vi)$
 Using (iii), (iv) and (vi) in (v), we get:
 $(B + 10) + B + (B - 2) + 2B = 133$
 $\Leftrightarrow 5B = 125$
 $\Leftrightarrow B = 25.$
- 492.** Since A and B each has ₹ 16 in the end, so C gave each of them ₹ 8. So, before third transaction, C had ₹ $(16 + 8 \times 2) = ₹ 32$ while A and B each had ₹ 8. Thus, in second transaction, B gave ₹ 4 to A and ₹ 16 to C. So, before second transaction, A had ₹ 4, B had ₹ $(8 + 4 + 16) = ₹ 28$ and C had ₹ 16. Thus, in first transaction, A gave ₹ 14 to B and ₹ 8 to C. So, before first transaction, A had ₹ $(4 + 14 + 8) = ₹ 26$, B had ₹ 14 and C had ₹ 8.

- 493.** Suppose Iqbal has x cards and Mushtaq has y cards.
 Let the number of cards given be $z.$
 Then, $x + z = 4(y - z) \Rightarrow x - 4y = -5z \quad \dots(i)$
 And, $x - z = 3(y + z) \Rightarrow x - 3y = 4z \quad \dots(ii)$
 Subtracting (i) from (ii), we get: $y = 9z.$ Putting $y = 9z$ in (i), we get: $x = 31z.$
 Clearly, since a pack of playing cards has 52 cards, z cannot be greater than 1.
 So, $x = 31$ and $y = 9 \quad [\because z = 1]$
 Hence, Iqbal had 31 cards.
- 494.** Let the original number of boys be $x.$ Then, number of apples first picked = $3x.$
 Final number of boys = $x + 3.$ Final number of apples = $3x + 1.$
 $\therefore \frac{3x + 1}{x + 3} = 2 \Leftrightarrow 3x + 1 = 2x + 6 \Leftrightarrow x = 5.$
 Hence, required number of apples = $3 \times 5 + 1 = 16.$
- 495.** Let there be n students in the classroom.
 Total number of friendship bands bought = $n(n - 1).$
 If two students were absent, number of bands utilized = $(n - 2)(n - 3).$
 $\therefore n(n - 1) - (n - 2)(n - 3)$
 $= 122 \Leftrightarrow n^2 - n - n^2 + 5n - 6$
 $= 122 \Leftrightarrow 4n = 128 \Leftrightarrow n = 32.$
 So, number of bands finally unutilized = $32 \times 31 - 31 \times 30 = 992 - 930 = 62.$
- 496.** Let the total number of chocolates she bought be x and the total number of her friends be $y.$
 Then, $3(y - 1) + 2 = x \Leftrightarrow x - 3y = -1 \quad \dots(i)$
 And, $2y + 15 = x \Leftrightarrow x - 2y = 15 \quad \dots(ii)$
 Subtracting (i) from (ii), we get: $y = 16.$ Putting $y = 16$ in (ii), we get: $x = 47.$
- 497.** Clearly, each committee has two members which are common to all three committees. In addition, it has a third member common to the second committee and a fourth one common to the third committee.
- 498.** $2n + 5m = 50 \Leftrightarrow 2n = 50 - 5m \Leftrightarrow n = \frac{50 - 5m}{2}.$
 Since n is a natural number, so $(50 - 5m)$ must be even and so, m must be even
 i.e., $m = 2, 4, 6$ or 8
 $[\because m < 10, \text{ as for } m = 10, n = 0]$
 If $m = 2, n = 20;$
 If $m = 4, n = 15;$
 If $m = 6, n = 10;$
 If $m = 8, n = 5.$
 $\therefore$ Least possible difference = $8 - 5 = 3.$
- 499.** Let the number of chairs with arms bought be x and those without arms be $y.$
 Then, $x + y \geq 15$ or $y \geq 15 - x$
 And, $160x + 100y \leq 2000$
 $\Rightarrow 160x + 100(15 - x) \leq 2000$
 $[\because \text{For maximum value of } x, y \text{ has to be minimum}]$
 $\Rightarrow 160x + 1500 - 100x \leq 2000 \Rightarrow 60x \leq 500 \Rightarrow x \leq 8.$
 Hence, maximum value of x is 8.

500. Let the number of cats be x . Then, number of dogs = $55 - x$.

Clearly, x is a multiple of 5 i.e., $x = 5n$ where $n \leq 10$.

And, $(55 - x)$ is a multiple of 4 $\Rightarrow 55 - 5n = 4m$

$$\Rightarrow m = \frac{55 - 5n}{4}.$$

Since m is a natural number, $(55 - 5n)$ must be a multiple of 4 which happens for $n = 3$ or 7 .

Case I. If $n = 3$, $m = 10$.

Number of cats = 15, Number of dogs = 40.

Number of pets adopted = $\left(\frac{1}{5} \times 15 + \frac{1}{4} \times 40\right) = 13$.

Case II. If $n = 7$, $m = 5$.

Number of cats = 35, Number of dogs = 20.

Number of pets adopted = $\left(\frac{1}{5} \times 35 + \frac{1}{4} \times 20\right) = 12$.

$\therefore$ Required number = 13.

501. The team has played a total of $(17 + 3) = 20$ matches.

This constitutes $\frac{2}{3}$ of the matches. Hence, total number

of matches played = 30. To win $\frac{3}{4}$ of them, a team has

to win at least 23 of them. The team has thus to win a minimum of 6 matches out of remaining 10. So, it can lose a maximum of 4 of them.

502. Let M represent the number of mornings when they swam.

Then, total number of mornings in the vacation
= $32 + M$.

Let E represent the number of evenings when they played tennis.

Then, total number of evenings in the vacation = $18 + E$.

$\therefore 32 + M = 18 + E \Rightarrow E - M = 14$ (i)

Total number of days when they swam or played
= $E + M$.

$\therefore E + M = 28$... (ii)

Adding (i) and (ii), we get: $2E = 42$ or $E = 21$.

Hence, duration of vacations = $18 + E = 18 + 21 = 39$.

503. Note: $10 \times 15 \neq 20 \times 10$. So, this is not a question of Chain Rule and the original quantity of grass and its growth rate are to be considered.

Let g be the total units of grass in the field originally, r be the units of grass grown per day and c be the units of grass each cow eats per day.

Then, $g + 15r = (10 \times 15) c = 150c$... (i)

$g + 10r = (20 \times 10) c = 200c$... (ii)

Subtracting (ii) from (i), we get: $5r = -50c$ or $r = -10c$.

Putting $r = -10c$ in (i), we get: $g = 300c$.

Let the required number of days be x .

Then, $g + xr = 300c \Rightarrow 300c - 10xc = 30xc \Rightarrow 40xc = 300c$

$$\Rightarrow 40x = 300 \Rightarrow x = 7\frac{1}{2}.$$

504. Let g be the total units of grass in the field originally, r be the units of grass grown per day and c be the units of grass each cow eats per day.

Then, units of grass consumed by 70 cows in 24 days
= $(70 \times 24 \times c) = 1680c$.

Units of grass consumed by 30 cows in 60 days

$$= (30 \times 60 \times c) = 1800c.$$

We have: $g + 24r = 1680c$... (i)

$g + 60r = 1800c$... (ii)

Subtracting (i) from (ii), we get: $36r = 120c \Rightarrow r = \frac{10}{3}c$.

Putting $r = \frac{10}{3}c$ in (i), we get: $g = 1600c$.

Let the required number of cows be x .

Then, $g + 96r = x \times 96 \times c \Rightarrow 1600c + \frac{96 \times 10}{3}c = 96xc$

$$\Rightarrow 1920c = 96xc \Rightarrow 96x = 1920 \Rightarrow x = 20.$$

Questions 505 and 506

Let the total number of pilgrims in the group be x .

Then, total number of visits to shrines

= Number of shrines $\times$ Number of pilgrims visiting each shrine = $10 \times 4 = 40$.

So, $5x = 40$ or $x = 8$.

Let the number of lakes be n .

Then, total number of visits to lakes

= Number of lakes $\times$ Number of pilgrims visiting each lake = $6n$.

$$\therefore 6n = 8 \times 3 = 24 \text{ or } n = 4.$$

505. Number of lakes in the city = 4.

506. Number of pilgrims in the group = 8.

507. Let the total number of apples be x . Then,

Apples sold to 1st customer = $\left(\frac{x}{2} + 1\right)$. Remaining apples

$$= x - \left(\frac{x}{2} + 1\right) = \left(\frac{x}{2} - 1\right).$$

Apples sold to 2nd customer = $\frac{1}{3}\left(\frac{x}{2} - 1\right) + 1$

$$= \frac{x}{6} - \frac{1}{3} + 1 = \left(\frac{x}{6} + \frac{2}{3}\right).$$

Remaining apples = $\left(\frac{x}{2} - 1\right) - \left(\frac{x}{6} + \frac{2}{3}\right)$

$$= \left(\frac{x}{2} - \frac{x}{6}\right) - \left(1 + \frac{2}{3}\right) = \left(\frac{x}{3} - \frac{5}{3}\right).$$

Apples sold to 3rd customer = $\frac{1}{5}\left(\frac{x}{3} - \frac{5}{3}\right) + 1 = \left(\frac{x}{15} + \frac{2}{3}\right)$.

Remaining apples = $\left(\frac{x}{3} - \frac{5}{3}\right) - \left(\frac{x}{15} + \frac{2}{3}\right)$

$$= \left(\frac{x}{3} - \frac{x}{15}\right) - \left(\frac{5}{3} + \frac{2}{3}\right) = \left(\frac{4x}{15} - \frac{7}{3}\right).$$

$$\therefore \frac{4x}{15} - \frac{7}{3} = 3 \Leftrightarrow \frac{4x}{15} = \frac{16}{3} \Leftrightarrow x = \left(\frac{16}{3} \times \frac{15}{4}\right) = 20.$$

508. Let the time taken to read one page of Engineering Mathematics be M units and that taken to read one page of Engineering Drawing be D units.

Then, $80M + 100D = 50M + 250D \Rightarrow 30M = 150D$

$$D \Rightarrow D = \frac{M}{5}.$$

$$\begin{aligned}\therefore 80M + 100D &= 80M + 100 \times \frac{M}{5} \\ &= 80M + 20M = 100M.\end{aligned}$$

So, Ravi can read 100 pages of Engineering Mathematics.

- 509.** Since Rohan paid ₹ 20 and because of lack of change, the clerk gave him stamps worth ₹ 3, it can be concluded that the total value of the stamps that he wanted to buy is ₹ 17. Since he ordered initially a minimum of 2 stamps of each denomination, if he bought exactly 2 stamps each, his total value is ₹ $[2(5 + 2 + 1)] = ₹ 16$. To make it ₹ 17, he must have bought 1 one-rupee stamp more.

So, total number of stamps ordered = $(2 + 2 + 3) = 7$.

Since the clerk gave him 3 more, so total number of stamps bought = $(7 + 3) = 10$.

- 510.** Suppose Aditya bought x gel pens and y fountain pens. Then.

$$300x + 400y = 3600 \Rightarrow 3x + 4y = 36 \quad \dots(i)$$

Also, $(300x + 400y) - (400x + 300y) = 150$ or 200

$$\Rightarrow 100y - 100x = 150 \text{ or } 200$$

$$\Rightarrow y - x = 1.5 \text{ or } y - x = 2.$$

If $y - x = 1.5$, we don't get integral values for x and y .

$$\therefore y - x = 2 \quad \dots(ii)$$

Solving (i) and (ii), we get: $x = 4, y = 6$.

Hence, total number of pens bought = $4 + 6 = 10$.

Questions 511 to 513

Flowers left after first place of worship = $2x - y$.

Flowers left after second place of worship = $2(2x - y) - y$
 $= 4x - 3y$.

Flowers left after third place of worship = $2(4x - 3y) - y$
 $= 8x - 7y$.

Flowers left after fourth place of worship = $2(8x - 7y) - y$
 $= 16x - 15y$.

$$\therefore 16x - 15y = 0 \Rightarrow 16x = 15y.$$

- 511.** If $x = 30$, then $y = \frac{16 \times 30}{15} = 32$.

- 512.** For minimum y , we find the L.C.M. of 16 and 15 which is 240.

$$\therefore 16x = 15y = 240 \Rightarrow x = 15, y = 16.$$

- 513.** Clearly, required number = minimum value of $x = 15$.

- 514.** Suppose the thief stole x diamonds.

$$\text{Diamonds given to first watchman} = \frac{x}{2} + 2 = \frac{x+4}{2}.$$

$$\text{Remaining diamonds} = x - \left(\frac{x+4}{2}\right) = \frac{x-4}{2}.$$

$$\text{Diamonds given to second watchman} = \frac{x-4}{4} + 2 = \frac{x+4}{4}.$$

$$\text{Remaining diamonds} = \left(\frac{x-4}{2}\right) - \left(\frac{x+4}{4}\right) = \frac{x-12}{4}.$$

$$\text{Diamonds given to third watchman} = \frac{x-12}{8} + 2 = \frac{x+4}{8}.$$

$$\text{Remaining diamonds} = \left(\frac{x-12}{4}\right) - \left(\frac{x+4}{8}\right) = \frac{x-28}{8}.$$

$$\therefore \frac{x-28}{8} = 1 \text{ or } x-28=8 \text{ or } x=36.$$

Another Method: The thief escaped with 1 diamond.

Before third watchman, he had $(1 + 2) \times 2 = 6$ diamonds.

Before second watchman, he had $(6 + 2) \times 2 = 16$ diamonds.

Before first watchman, he had $(16 + 2) \times 2 = 36$ diamonds.

Thus, he stole 36 diamonds originally.

- 515.** Let there be x mints in the bowl originally.

Then, mints left after Divya had taken from the bowl

$$= \left(\frac{2}{3}x + 4\right).$$

Mints left after Reema had taken from the bowl

$$= \frac{3}{4} \left(\frac{2}{3}x + 4\right) + 3 = \left(\frac{x}{2} + 6\right).$$

Mints left after Shweta had taken from the bowl

$$= \frac{1}{2} \left(\frac{x}{2} + 6\right) + 2 = \left(\frac{x}{4} + 5\right).$$

$$\therefore \frac{x}{4} + 5 = 17 \Rightarrow \frac{x}{4} = 12 \Rightarrow x = 48.$$

- 516.** Let the cost of one gel-pen, one ball-point pen and one marker pen be ₹ x , ₹ y and ₹ z respectively.

$$\text{Then, } 4x + 8y + z = 185 \quad \dots(i)$$

$$7x + 15y + z = 315 \quad \dots(ii)$$

$$\text{Subtracting (i) from (ii), we get: } 3x + 7y = 130 \quad \dots(iii)$$

Multiplying (iii) by 2 and subtracting from (ii), we get:
 $x + y + z = 55$

Hence, required cost = ₹ 55.

- 517.** Let the number of byes, wides and runs be x, y and z respectively.

Then, $x + y + z = 232$. Also, $z = 26y$ and $x = y + 8$.
 $\dots(i)$

$$\therefore (y + 8) + y + 26y = 232 \Leftrightarrow 28y = 224$$

$$\Leftrightarrow y = 8. \text{ So, } z = 26y = 208.$$

Let the number of runs scored by Ram and Shyam be $6a$ and $7a$ respectively.

$$\text{Then, } 6a + 7a = 208 \Leftrightarrow 13a = 208 \Leftrightarrow a = 16.$$

Hence, runs scored by Ram = $6a = 6 \times 16 = 96$.

- 518.** Let there be n rows of balls in the equilateral triangle.

$$\text{So, total number of balls} = 1 + 2 + \dots + n = n = \frac{n(n+1)}{2}.$$

Since each row in the square has 8 balls less than that in the equilateral triangle, so the square has $(n - 8)$ rows with $(n - 8)$ balls in each row.

So, total number of balls = $(n - 8)^2$.

$$\therefore \frac{n(n+1)}{2} + 669 = (n-8)^2$$

$$\Rightarrow n^2 + n + 1338 = 2(n^2 + 64 - 16n)$$

$$\Rightarrow n^2 - 33n - 1210 = 0$$

$$\Rightarrow n^2 - 55n + 22n - 1210 = 0$$

$$\Rightarrow (n - 55)(n + 22) = 0 \Rightarrow n = 55 \quad [\because n \neq -22]$$

So, initial number of balls

$$= \frac{n(n+1)}{2} = \frac{55 \times 56}{2} = 55 \times 28 = 1540.$$

Questions 519 and 520

Let the number of men in the beginning be m .

Suppose each man shifts c crates on the first day. Then,

$$mc + (m + 6)(c - 5) + (m + 12)(c - 10) + (m + 18)(c - 15) + (m + 24)(c - 20) = 545$$

$$\begin{aligned} \Rightarrow mc + mc - 5m + 6c - 30 + mc - 10m + 12c - 120 + \\ mc - 15m + 18c - 270 + mc \\ - 20m + 24c - 480 = 545 \\ \Rightarrow 5mc - 50m + 60c = 1445 \\ \Rightarrow mc - 10m + 12c = 289 \\ \Rightarrow mc - 10m = 289 - 12c \\ \Rightarrow m = \frac{289 - 12c}{c - 10}. \end{aligned}$$

Clearly, $c > 20$

[$\because$ if $c < 20$, then number of crates shifted on 5th day i.e., $(c - 20) < 0$ which is not possible]

Thus, we have to find $c > 20$ such that m is an integer. This is not so for $c = 21$ and $c = 22$.

$$\text{For } c = 23, \text{ we have: } m = \frac{289 - 276}{13} = \frac{13}{13} = 1.$$

$$\begin{aligned} \text{519. Number of crates shifted on third day} &= (m + 12)(c - 10) \\ &= (1 + 12)(23 - 10) \\ &= 13 \times 13 = 169. \end{aligned}$$

$$\text{520. Number of men on 5th day} = m + 24 = 25.$$

521. Let the missing number be x .

$$\text{Given } \frac{20 + 8 \times 0.5}{20 - x} = 12$$

$$\Rightarrow \frac{20 + 4}{20 - x} = 12 \Rightarrow \frac{24}{20 - x} = 12$$

$$\Rightarrow 20 - x = \frac{24}{12} = 2$$

$$\Rightarrow x = 20 - 2 = 18$$

Hence, the number is 18.

$$\text{522. Given } \frac{a}{b} + \frac{b}{a} = 2$$

$$\Rightarrow \frac{a^2 + b^2}{ab} = 2$$

$$\Rightarrow a^2 + b^2 = 2ab$$

$$\Rightarrow a^2 + b^2 - 2ab = 0$$

$$\Rightarrow (a - b)^2 = 0$$

$$\Rightarrow a - b = 0$$

$$\begin{aligned} \text{523. Given } (24.96)^2 \div (34.11 \div 20.05) + 67.96 - 89.11 \\ (25)^2 \div (34 \div 20) + 68 - 89 \end{aligned}$$

$$\approx (25)^2 \div \left(\frac{34}{20}\right) + 68 - 89$$

$$\approx 625 \div 1.7 + 68 - 89$$

$$\approx 367.6 + 68 - 89$$

$$\approx 367 + 68 - 89 \approx 346$$

$$\text{524. } \frac{x+1}{x-1} = \frac{a}{b}$$

By componendo and dividendo

$$\frac{x+1+x-1}{x+1-x+1} = \frac{a+b}{a-b}$$

$$\Rightarrow \frac{2x}{2} = \frac{a+b}{a-b}$$

$$\Rightarrow x = \frac{a+b}{a-b} \quad \dots(i)$$

$$\text{Again, } \frac{1-y}{1+y} = \frac{b}{a}$$

$$\Rightarrow \frac{1+y}{1-y} = \frac{a}{b}$$

$$\Rightarrow \frac{1+y+1-y}{1+y-1+y} = \frac{a+b}{a-b}$$

$$\Rightarrow \frac{2}{2y} = \frac{a+b}{a-b}$$

$$\Rightarrow \frac{1}{y} = \frac{a+b}{a-b}$$

$$\Rightarrow y = \frac{a-b}{a+b} \quad \dots(ii)$$

Subtracting equation (ii) from (i) we get

$$\begin{aligned} \therefore x - y &= \frac{a+b}{a-b} - \frac{a-b}{a+b} \\ \left\{ \because (a+b)^2 - (a-b)^2 &= 4ab \text{ and } a^2 - b^2 = (a-b)(a+b) \right\} \\ &= \frac{(a+b)^2 - (a-b)^2}{(a+b)(a-b)} = \frac{4ab}{a^2 - b^2} \end{aligned}$$

Multiply equation (i) and (ii) we get

$$xy = \frac{a+b}{a-b} \times \frac{a-b}{a+b} = 1$$

$\therefore$ Expression

$$= \frac{x-y}{1+xy} = \frac{4ab}{a^2 - b^2} \cdot \frac{1}{1+1}$$

$$= \frac{4ab}{2(a^2 - b^2)} = \frac{2ab}{a^2 - b^2}$$

$$\text{525. Given } 200 \div 25 \times 4 + 12 - 3$$

$$= \frac{200}{25} \times 4 + 12 - 3$$

$$= 8 \times 4 + 12 - 3$$

$$= 32 + 12 - 3 = 44 - 3 = 41$$

$$\text{526. Let the number be } a.$$

$$\text{Given } 14 \times 627 \div \sqrt{1089} = (a)^3 + 141$$

$$14 \times 627 \div \sqrt{1089} = a^3 + 141$$

$$\Rightarrow 14 \times 627 \div 33 = a^3 + 141$$

$$\Rightarrow 14 \times 19 = a^3 + 141$$

$$\Rightarrow 266 = a^3 + 141$$

$$\Rightarrow a^3 = 125$$

$$\Rightarrow a = \sqrt[3]{125}$$

$$\Rightarrow a = 5$$

$$\text{527. Let the number be } x.$$

$$\text{Given } (923 - 347)/x = 32$$

$$\Rightarrow x = \frac{576}{32} = 18$$

$$\Rightarrow x = 18$$

$$\text{528. Let the number be } a.$$

$$\text{Given } 1559.95 - 7.99 \times 24.96 - a^2 = 1154$$

$$\Rightarrow 1560 - 8 \times 25 - a^2 = 1154$$

$$\Rightarrow 1360 - 200 - a^2 = 1154$$

$$\Rightarrow a^2 = 1360 - 1154 = 206$$

$$\Rightarrow a \approx \sqrt{206} \approx 14$$

529. Given $1\frac{4}{5} + 20 - 280 \div 25$

$$= 1\frac{4}{5} + 20 - \frac{280}{25}$$

$$= \frac{9}{5} + 20 - \frac{56}{5}$$

$$= \frac{9 + 100 - 56}{5}$$

$$= \frac{53}{5} = 10\frac{3}{5}$$

530. $= \{(64 - 38) \times 4\} \div 13$

$$= \frac{26 \times 4}{13} = 8$$

531. Given $x + y = 2a$

$$\therefore \frac{a}{x-a} + \frac{a}{y-a} = \frac{a(y-a) + a(x-a)}{(x-a)(y-a)}$$

$$= \frac{ay - a^2 + ax - a^2}{(x-a)(y-a)}$$

$$= \frac{a(x+y) - 2a^2}{(x-a)(y-a)}$$

$$= \frac{a \cdot 2a - 2a^2}{(x-a)(y-a)}$$

$$= \frac{2a^2 - 2a^2}{(x-a)(y-a)} = 0$$

532. Let the number be x .

$$\text{Given } 421 \div 35 \times 299.97 \div 25.05 = x^2$$

$$(x)^2 \approx 420 \div 35 \times 300 \div 25$$

$$= 12 \times 300 \div 25 = 12 \times 12$$

$$\Rightarrow x^2 = 12 \times 12$$

$$\therefore x = \sqrt{12 \times 12} = 12$$

Hence, the number is 12.

533. Given

$$= 19.99 \times 15.98 + 224.98 + 125.02$$

$$\approx 20 \times 16 + 225 + 125 = 320 + 225 + 125 = 670$$

534. Given $3625 \times ? = 1450$

$$? = \frac{1450}{3625} = \frac{725 \times 2}{725 \times 5} = \frac{2}{5}$$

535. Given: $128.43 + 30.21 + ? = 173$

$$? = 173 - 128.43 - 30.21$$

$$= 173 - 158.64 = 14.36$$

536. Given expression

$$= 123 \times 999 + 123$$

$$= 123 \times (999 + 1)$$

$$= 123 \times 1000$$

$$= 123000$$

537. Given expression $\frac{(359+256)^2 + (359-256)^2}{359 \times 359 + 256 \times 256}$

$$= \frac{(359+256)^2 + (359-256)^2}{(359)^2 + (256)^2}$$

Let, $a = 359$ and $b = 256$

$$\therefore \text{expression} = \frac{(a+b)^2 + (a-b)^2}{a^2 + b^2}$$

$$= \frac{2(a^2 + b^2)}{a^2 + b^2} = 2$$

538. $84368 + 65466 - 72009 - 13964$

$$= 149834 - 85973 = 63861$$

539. Let the number be x .

$$4376 + 3209 - 1784 + 97 = 3125 + x$$

$$\Rightarrow 7682 - 1784 = 3125 + x$$

$$\text{or, } x = 7682 - 1784 - 3125$$

$$\Rightarrow x = 2773$$

Hence, the number is 2773.

540. Let the missing number is a .

$$\text{Given } 14 \times 627 \div \sqrt{(1089)} = (a)^3 + 141$$

$$\frac{14 \times 627}{\sqrt{1089}} = (a)^3 + 141$$

$$\frac{14 \times 627}{33} = a^3 + 141$$

$$266 = a^3 + 141$$

$$a^3 = 266 - 141$$

$$a^3 = 125$$

$$a = \sqrt[3]{125} = 5$$

541. Given $\left(x + \frac{1}{x}\right) = 3$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = 3^2$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 9$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 7$$

542. As we know

$$a^3 + b^3 + c^3 - 3abc = (a^2 + b^2 + c^2 - ab - bc - ca)(a + b + c)$$

When $a + b + c = 0$

Then $a^3 + b^3 + c^3 - 3abc = 0$

When $x + y + z = 0$

$$\Rightarrow x^3 + y^3 + z^3 = 3xyz$$

$$\Rightarrow x^3 + y^3 + z^3 + 3xyz = 3xyz + 3xyz = 6xyz$$

543. Unit digit of $4^1 = 4$

Unit digit of $4^2 = 16$

Unit digit of $4^3 = 64$

Unit digit $4^{96} = 4^{2 \times 48} = 4^2 = 16$

$\therefore$ each unit digit of 4 repeats itself after two places

$\therefore 16 \div 6$

$\therefore$ Remainder = 4

544. $67 \times 119 + 17 - 27 \div 259$

$$= 67 + 119 \div 17 \times 27 - 259$$

$$= 67 + 7 \times 27 - 259$$

$$= 67 + 189 - 259$$

$$= 256 - 259 = -3$$

545. Given = $\frac{(0.73)^3 + (0.27)^3}{(0.73)^2 + (0.27)^2 - 0.73 \times 0.27}$

Let $a = 0.73$ and $b = 0.27$

$$= \frac{(0.73 + 0.27) \left[(0.73)^2 + (0.27)^2 - 0.73 \times 0.27 \right]}{(0.73)^2 + (0.27)^2 - 0.73 \times 0.27}$$

$$\left[\because a^3 + b^3 = (a + b)(a^2 + b^2 - ab) \right]$$

$$= 0.73 + 0.27 = 1$$

5

Square Roots and Cube Roots

IMPORTANT FACTS AND FORMULAE

I. Square Root: If $x^2 = y$, we say that the square root of y is x and we write, $\sqrt{y} = x$.

Thus, $\sqrt{4} = 2$, $\sqrt{9} = 3$, $\sqrt{196} = 14$.

II. Cube Root: The cube root of a given number x is the number whose cube is x . We denote the cube root of x by $\sqrt[3]{x}$.

Thus, $\sqrt[3]{8} = \sqrt[3]{2 \times 2 \times 2} = 2$, $\sqrt[3]{343} = \sqrt[3]{7 \times 7 \times 7} = 7$ etc.

Note:

$$1. \sqrt{xy} = \sqrt{x} \times \sqrt{y} \quad 2. \sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}} = \frac{\sqrt{x}}{\sqrt{y}} \times \frac{\sqrt{y}}{\sqrt{y}} = \frac{\sqrt{xy}}{y}$$

SOLVED EXAMPLES

Ex. 1. Evaluate $\sqrt{6084}$ by factorization method.

Sol. Method: Express the given number as the product of prime factors. Now, take the product of these prime factors choosing one out of every pair of the same primes. This product gives the square root of the given number.

Thus, resolving 6084 into prime factors, we get:

$$6084 = 2^2 \times 3^2 \times 13^2$$

$$\therefore \sqrt{6084} = (2 \times 3 \times 13) = 78.$$

Ex. 2. Find the square root of 1471369.

Sol. Method: In the given number, mark off the digits in pairs starting from the unit's digit. Each pair and the remaining one digit is called a period.

Now, $1^2 = 1$. On subtracting, we get 0 as remainder.

Now, bring down the next period i.e., 47.

Now, trial divisor is $1 \times 2 = 2$ and trial dividend is 47.

So, we take 22 as divisor and put 2 as quotient.

The remainder is 3.

Next, we bring down the next period which is 13.

Now, trial divisor is $12 \times 2 = 24$ and trial dividend is 313. So, we take 241 as dividend and 1 as quotient.

The remainder is 72.

Bring down the next period i.e., 69.

Now, the trial divisor is $121 \times 2 = 242$ and the trial dividend is 7269. So, we take 3 as quotient and 2423 as divisor. The remainder is then zero.

$$\text{Hence, } \sqrt{1471369} = 1213.$$

2	6084
2	3042
3	1521
3	507
13	169
	13

1	1471369 (1213
	1
22	47
	44
241	313
	241
2423	7269
	7269
	x

Ex. 3. Evaluate: $\sqrt{248 + \sqrt{52 + \sqrt{144}}}$.

(P.C.S., 2009)

Sol. Given expression = $\sqrt{248 + \sqrt{52 + 12}} = \sqrt{248 + \sqrt{64}} = \sqrt{248 + 8} = \sqrt{256} = 16$.

Ex. 4. Simplify : $\frac{112}{\sqrt{196}} \times \frac{\sqrt{576}}{12} \times \frac{\sqrt{256}}{8}$.

Sol. Given expression = $\frac{112}{14} \times \frac{24}{12} \times \frac{16}{8} = 8 \times 2 \times 2 = 32$.

Ex. 5. If $a * b * c = \frac{\sqrt{(a+2)(b+3)}}{c+1}$, then find the value of $6 * 15 * 3$.

Sol. $6 * 15 * 3 = \frac{\sqrt{(6+2)(15+3)}}{3+1} = \frac{\sqrt{8 \times 18}}{4} = \frac{\sqrt{144}}{4} = \frac{12}{4} = 3$.

Ex. 6. Find the value of $\sqrt{1\frac{9}{16}}$.

Sol. $\sqrt{1\frac{9}{16}} = \sqrt{\frac{25}{16}} = \frac{\sqrt{25}}{\sqrt{16}} = \frac{5}{4} = 1\frac{1}{4}$.

Ex. 7. What is the square root of 0.0009?

Sol. $\sqrt{0.0009} = \sqrt{\frac{9}{10000}} = \frac{\sqrt{9}}{\sqrt{10000}} = \frac{3}{100} = 0.03$.

Ex. 8. Evaluate $\sqrt{175.2976}$.

Sol. Method: We make even number of decimal places by affixing a zero, if necessary. Now, we mark off periods and extract the square root as shown.

$\therefore \sqrt{175.2976} = 13.24$.

Ex. 9. What will come in place of question mark in each of the following questions?

$$(i) \sqrt{\frac{32.4}{?}} = 2 \quad (ii) \sqrt{86.49} + \sqrt{5 + (?)^2} = 12.3$$

Sol. (i) Let $\sqrt{\frac{32.4}{x}} = 2$. Then, $\frac{32.4}{x} = 4 \Leftrightarrow 4x = 32.4 \Leftrightarrow x = 8.1$.

(ii) Let $\sqrt{86.49} + \sqrt{5 + x^2} = 12.3$.

$$\text{Then, } 9.3 + \sqrt{5 + x^2} = 12.3 \Leftrightarrow \sqrt{5 + x^2} = 12.3 - 9.3 = 3$$

$$\Leftrightarrow 5 + x^2 = 9 \Leftrightarrow x^2 = 9 - 5 = 4 \Leftrightarrow x = \sqrt{4} = 2.$$

Ex. 10. Find the value of $\sqrt{\frac{0.289}{0.00121}}$.

(IGNOU, 2003)

Sol. $\sqrt{\frac{0.289}{0.00121}} = \sqrt{\frac{0.28900}{0.00121}} = \sqrt{\frac{28900}{121}} = \frac{170}{11}$.

Ex. 11. If $\sqrt{841} = 29$, then find the value of $\sqrt{841} + \sqrt{8.41} + \sqrt{0.0841} + \sqrt{0.000841}$.

(M.B.A., 2006)

Sol. Given expression = $\sqrt{841} + \sqrt{\frac{841}{10^2}} + \sqrt{\frac{841}{10^4}} + \sqrt{\frac{841}{10^6}} = \sqrt{841} + \frac{\sqrt{841}}{10} + \frac{\sqrt{841}}{10^2} + \frac{\sqrt{841}}{10^3}$
 $= 29 + \frac{29}{10} + \frac{29}{100} + \frac{29}{1000} = 29 + 2.9 + 0.29 + 0.029 = 32.219$.

Ex. 12. If $\sqrt{1 + \frac{x}{144}} = \frac{13}{12}$, then find the value of x .

1	175.2976 (13.24
	1
23	75
	69
262	629
	524
2644	10576
	10576
	×

Sol. $\sqrt{1 + \frac{x}{144}} = \frac{13}{12} \Rightarrow \left(1 + \frac{x}{144}\right) = \left(\frac{13}{12}\right)^2 = \frac{169}{144} \Rightarrow \frac{x}{144} = \frac{169}{144} - 1$
 $\Rightarrow \frac{x}{144} = \frac{25}{144} \Rightarrow x = 25.$

Ex. 13. Simplify : $\frac{1}{\sqrt{100} - \sqrt{99}} - \frac{1}{\sqrt{99} - \sqrt{98}} + \frac{1}{\sqrt{98} - \sqrt{97}} - \frac{1}{\sqrt{97} - \sqrt{96}} + \dots + \frac{1}{\sqrt{2} - \sqrt{1}}.$ (Section Officers', 2005)

Sol. Given expression

$$\begin{aligned} &= \frac{1}{\sqrt{100} - \sqrt{99}} \times \frac{\sqrt{100} + \sqrt{99}}{\sqrt{100} + \sqrt{99}} - \frac{1}{\sqrt{99} - \sqrt{98}} \times \frac{\sqrt{99} + \sqrt{98}}{\sqrt{99} + \sqrt{98}} + \frac{1}{\sqrt{98} - \sqrt{97}} \times \frac{\sqrt{98} + \sqrt{97}}{\sqrt{98} + \sqrt{97}} \\ &\quad - \frac{1}{\sqrt{97} - \sqrt{96}} \times \frac{\sqrt{97} + \sqrt{96}}{\sqrt{97} + \sqrt{96}} + \dots + \frac{1}{\sqrt{2} - \sqrt{1}} \times \frac{\sqrt{2} + \sqrt{1}}{\sqrt{2} + \sqrt{1}} \\ &= \frac{\sqrt{100} + \sqrt{99}}{(100 - 99)} - \frac{\sqrt{99} + \sqrt{98}}{(99 - 98)} + \frac{\sqrt{98} + \sqrt{97}}{(98 - 97)} - \frac{\sqrt{97} + \sqrt{96}}{(97 - 96)} + \dots + \frac{\sqrt{2} + \sqrt{1}}{(2 - 1)} \\ &= (\sqrt{100} + \sqrt{99}) - (\sqrt{99} + \sqrt{98}) + (\sqrt{98} + \sqrt{97}) - (\sqrt{97} + \sqrt{96}) + \dots + (\sqrt{2} + \sqrt{1}) \\ &= \sqrt{100} + \sqrt{1} = 10 + 1 = 11. \end{aligned}$$

Ex. 14. Find the sum : $3 + \frac{1}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} - \frac{1}{3 - \sqrt{3}}.$

(S.S.C., 2007)

Sol. $3 + \frac{1}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} - \frac{1}{3 - \sqrt{3}} = 3 + \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} \times \frac{3 - \sqrt{3}}{3 - \sqrt{3}} - \frac{1}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}}$
 $= 3 + \frac{\sqrt{3}}{3} + \frac{3 - \sqrt{3}}{(9 - 3)} - \frac{3 + \sqrt{3}}{(9 - 3)} = 3 + \frac{\sqrt{3}}{3} + \frac{3 - \sqrt{3}}{6} - \frac{3 + \sqrt{3}}{6}$
 $= \frac{18 + 2\sqrt{3} + 3 - \sqrt{3} - 3 - \sqrt{3}}{6} = \frac{18}{6} = 3.$

Ex. 15. Find the value of $\sqrt{3}$ upto three places of decimal.

Sol.

1	3.000000 (1.732
	1
27	200
	189
343	1100
	1029
3462	7100
	6924

$\therefore \sqrt{3} = 1.732.$

Ex. 16. If $\sqrt{3} = 1.732$, find the value of $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75}$ correct to 3 places of decimal.

(S.S.C., 2004)

Sol. $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75} = \sqrt{64 \times 3} - \frac{1}{2}\sqrt{16 \times 3} - \sqrt{25 \times 3} = 8\sqrt{3} - \frac{1}{2} \times 4\sqrt{3} - 5\sqrt{3}$
 $= 3\sqrt{3} - 2\sqrt{3} = \sqrt{3} = 1.732.$

Ex. 17. If $\sqrt{0.05 \times 0.5 \times a} = 0.5 \times 0.05 \times \sqrt{b}$, then find the value of $\frac{a}{b}$.

(M.B.A., 2006)

Sol. Clearly, we have: $\frac{\sqrt{a}}{\sqrt{b}} = \frac{0.5 \times 0.05}{\sqrt{0.05 \times 0.5}} \Rightarrow \frac{a}{b} = \frac{0.5 \times 0.05 \times 0.5 \times 0.05}{0.05 \times 0.5} = 0.5 \times 0.05 = 0.025.$

Ex. 18. Evaluate : $\sqrt{\frac{9.5 \times .0085 \times 18.9}{.0017 \times 1.9 \times 0.021}}$.

Sol. Given exp. = $\sqrt{\frac{9.5 \times .0085 \times 18.900}{.0017 \times 1.9 \times 0.021}}$.

Now, since the sum of decimal places in the numerator and denominator under the radical sign is the same, we remove the decimal.

$$\therefore \text{Given exp.} = \sqrt{\frac{95 \times 85 \times 18900}{17 \times 19 \times 21}} = \sqrt{5 \times 5 \times 900} = 5 \times 30 = 150.$$

Ex. 19. Simplify : $\sqrt{[(12.1)^2 - (8.1)^2] \div [(0.25)^2 + (0.25)(19.95)]}$.

(C.B.I., 2003)

Sol. Given exp. = $\sqrt{\frac{(12.1+8.1)(12.1-8.1)}{(0.25)(0.25+19.95)}} = \sqrt{\frac{20.2 \times 4}{0.25 \times 20.2}} = \sqrt{\frac{4}{0.25}} = \sqrt{\frac{400}{25}} = \sqrt{16} = 4.$

Ex. 20. If $x = 1 + \sqrt{2}$ and $y = 1 - \sqrt{2}$, find the value of $(x^2 + y^2)$.

Sol. $x^2 + y^2 = (1 + \sqrt{2})^2 + (1 - \sqrt{2})^2 = 2[(1)^2 + (\sqrt{2})^2] = 2 \times 3 = 6.$

Ex. 21. Evaluate $\sqrt{0.9}$ upto 3 places of decimal.

Sol.

$$\begin{array}{r|l} 9 & 0.900000 \text{ (0.948)} \\ & 81 \\ \hline 184 & 900 \\ & 736 \\ \hline 1888 & 16400 \\ & 15104 \\ \hline \end{array} \quad \therefore \sqrt{0.9} = 0.948.$$

Ex. 22. Find the square root of $0.\dot{1}$.

Sol. $\sqrt{0.\dot{1}} = \sqrt{\frac{1}{9}} = \frac{1}{3} = 0.333 \dots = 0.\dot{3}.$

Ex. 23. If $\sqrt{15} = 3.88$, find the value of $\sqrt{\frac{5}{3}}$.

Sol. $\sqrt{\frac{5}{3}} = \sqrt{\frac{5 \times 3}{3 \times 3}} = \frac{\sqrt{15}}{3} = \frac{3.88}{3} = 1.2933 \dots = 1.29\bar{3}.$

Ex. 24. Find the least square number which is exactly divisible by 10, 12, 15 and 18.

Sol. L.C.M. of 10, 12, 15, 18 = 180. Now, $180 = 2 \times 2 \times 3 \times 3 \times 5 = 2^2 \times 3^2 \times 5.$

To make it a perfect square, it must be multiplied by 5.

$$\therefore \text{Required number} = (2^2 \times 3^2 \times 5^2) = 900.$$

Ex. 25. Find the greatest number of five digits which is a perfect square.

Sol. Greatest number of 5 digits is 99999.

$$\begin{array}{r|l} 3 & 99999 \text{ (316)} \\ & 9 \\ \hline 61 & 99 \\ & 61 \\ \hline 626 & 3899 \\ & 3756 \\ \hline & 143 \end{array}$$

$$\therefore \text{Required number} = (99999 - 143) = 99856.$$

Ex. 26. Find the smallest number that must be added to 1780 to make it a perfect square.

Sol.

$$\begin{array}{r} 4 \overline{) 1780} \quad (42 \\ \underline{16} \\ 82 \\ \underline{164} \\ 16 \end{array}$$

$$\therefore \text{Number to be added} = (43)^2 - 1780 = 1849 - 1780 = 69.$$

Ex. 27. If $\sqrt{2} = 1.4142$, find the value of $\frac{\sqrt{2}}{(2+\sqrt{2})}$.

Sol.
$$\frac{\sqrt{2}}{(2+\sqrt{2})} = \frac{\sqrt{2}}{(2+\sqrt{2})} \times \frac{(2-\sqrt{2})}{(2-\sqrt{2})} = \frac{2\sqrt{2}-2}{(4-2)} = \frac{2(\sqrt{2}-1)}{2} = (\sqrt{2}-1) = (1.4142-1) = 0.4142.$$

Ex. 28. If $x = \left(\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} \right)$ and $y = \left(\frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} \right)$, find the value of $(x^2 + y^2)$.

Sol.
$$x = \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} = \frac{(\sqrt{5}+\sqrt{3})^2}{(5-3)} = \frac{5+3+2\sqrt{15}}{2} = 4+\sqrt{15}.$$

$$y = \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} = \frac{(\sqrt{5}-\sqrt{3})^2}{(5-3)} = \frac{5+3-2\sqrt{15}}{2} = 4-\sqrt{15}.$$

$$\therefore x^2 + y^2 = (4+\sqrt{15})^2 + (4-\sqrt{15})^2 = 2[(4)^2 + (\sqrt{15})^2] = 2 \times 31 = 62.$$

Ex. 29. Find the value of: $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}}$

(S.S.C., 2006)

Sol. Let $\sqrt{6+\sqrt{6+\sqrt{6+\dots}}} = x.$

$$\text{Then, } \sqrt{6+x} = x \Leftrightarrow 6+x = x^2 \Rightarrow x^2 - x - 6 = 0 \Leftrightarrow x^2 - 3x + 2x - 6 = 0$$

$$\Leftrightarrow x(x-3) + 2(x-3) = 0 \Leftrightarrow (x-3)(x+2) = 0 \Leftrightarrow x = 3.$$

$[\because x \neq -2]$

$$\text{Hence, } \sqrt{6+\sqrt{6+\sqrt{6+\dots}}} = 3.$$

Ex. 30. Find the cube root of 2744.

Sol. Method : Resolve the given number as the product of prime factors and take the product of prime factors, choosing one out of three of the same prime factors.

Resolving 2744 as the product of prime factors, we get :

$$2744 = 2^3 \times 7^3.$$

$$\therefore \sqrt[3]{2744} = 2 \times 7 = 14.$$

2	2744
2	1372
2	686
7	343
7	49
	7

Ex. 31. By what least number 4320 be multiplied to obtain a number which is a perfect cube?

Sol. Clearly, $4320 = 2^3 \times 3^3 \times 2^2 \times 5.$

To make it a perfect cube, it must be multiplied by 2×5^2 i.e., 50.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer:

1. $\sqrt{53824} = ?$
 (a) 202 (b) 232
 (c) 242 (d) 332
2. The square root of 41209 is equal to (L.I.C.A.D.O., 2008)
 (a) 103 (b) 203
 (c) 303 (d) 403
3. The square root of 123454321 is (P.C.S., 2008)
 (a) 111111 (b) 12341
 (c) 11111 (d) 11211
4. The number of digits in the square root of 625685746009 is (S.S.C., 2007)
 (a) 4 (b) 5
 (c) 6 (d) 7
5. $\sqrt{\sqrt{17956} + \sqrt{24025}} = ?$ (Bank P.O., 2008)
 (a) 19 (b) 155
 (c) 256 (d) 289
 (e) None of these
6. $\sqrt{\sqrt{44944} + \sqrt{52441}} = ?$ (Bank P.O., 2008)
 (a) 17 (b) 312
 (c) 441 (d) 485
 (e) None of these
7. One-fourth of the sum of prime numbers, greater than 4 but less than 16, is the square of (P.C.S., 2009)
 (a) 3 (b) 4
 (c) 5 (d) 7
8. The value of $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$ is
 (a) 4 (b) 6
 (c) 8 (d) 10
9. Evaluate $\sqrt{41 - \sqrt{21 + \sqrt{19 - \sqrt{9}}}}$.
 (a) 3 (b) 5
 (c) 6 (d) 6.4
10. $\sqrt{176 + \sqrt{2401}}$ is equal to
 (a) 14 (b) 15
 (c) 18 (d) 24
11. $\frac{\sqrt{196}}{7} \times \frac{\sqrt{441}}{7} \times \frac{120}{\sqrt{225}} = ?$ (R.R.B., 2008)
 (a) 48 (b) 54
 (c) 58 (d) 84
12. $\left(\sqrt{\frac{225}{729}} - \sqrt{\frac{25}{144}} \right) \div \sqrt{\frac{16}{81}} = ?$
 (a) $\frac{1}{48}$ (b) $\frac{5}{48}$
 (c) $\frac{5}{16}$ (d) None of these
13. $(15)^2 + (18)^2 - 20 = \sqrt{?}$ (Bank P.O., 2006)
 (a) 22 (b) 23
 (c) 529 (d) 279841
 (e) None of these
14. $\sqrt{?} \times \sqrt{484} = 1034$ (L.I.C.A.D.O., 2007)
 (a) 2025 (b) 2209
 (c) 2304 (d) 2401
 (e) None of these
15. $\sqrt{11881} \times \sqrt{?} = 10137$ (Bank Recruitment, 2008)
 (a) 8281 (b) 8649
 (c) 9216 (d) 9409
 (e) None of these
16. In the equation $\frac{4050}{\sqrt{x}} = 450$, the value of x is (L.I.C.A.D.O., 2008)
 (a) 9 (b) 49
 (c) 81 (d) 100
17. $\sqrt{\frac{16}{25}} \times \sqrt{\frac{?}{25}} \times \frac{16}{25} = \frac{256}{625}$ (R.R.B. 2008)
 (a) 5 (b) 8
 (c) 16 (d) None of these
18. The square root of $(272^2 - 128^2)$ is
 (a) 144 (b) 200
 (c) 240 (d) 256
19. If $x * y = x + y + \sqrt{xy}$, the value of $6 * 24$ is
 (a) 41 (b) 42
 (c) 43 (d) 44
20. If $y = 5$, then what is the value of $10y\sqrt{y^3 - y^2}$?
 (a) $50\sqrt{2}$ (b) 100
 (c) $200\sqrt{5}$ (d) 500
21. $\sqrt{110\frac{1}{4}} = ?$
 (a) 10.25 (b) 10.5
 (c) 11.5 (d) 19.5

22. $\sqrt{\frac{25}{81} - \frac{1}{9}} = ?$ (Hotel Management, 2002)
- (a) $\frac{2}{3}$ (b) $\frac{4}{9}$
- (c) $\frac{16}{81}$ (d) $\frac{25}{81}$
23. $[(\sqrt{81})^2]^2 = (?)^2$ (Specialist Officers, 2006)
- (a) 8 (b) 9
- (c) 4096 (d) 6561
- (e) None of these
24. The digit in the unit's place in the square root of 15876 is
- (a) 2 (b) 4
- (c) 6 (d) 8
25. Which of the following is closest to $\sqrt{3}$? (S.S.C., 2005)
- (a) 1.69 (b) $\frac{173}{100}$
- (c) 1.75 (d) $\frac{9}{5}$
26. How many two-digit numbers satisfy this property: The last digit (unit's digit) of the square of the two-digit number is 8?
- (a) 1 (b) 2
- (c) 3 (d) None of these
27. What percentage of the numbers from 1 to 50 have squares that end in the digit 1? (M.B.A., 2006)
- (a) 1 (b) 5
- (c) 10 (d) 11
- (e) 20
28. While solving a mathematical problem, Samidha squared a number and then subtracted 25 from it rather than the required i.e., first subtracting 25 from the number and then squaring it. But she got the right answer. What was the given number? (Bank P.O., 2006)
- (a) 13 (b) 38
- (c) 48
- (d) Cannot be determined (e) None of these
29. How many perfect squares lie between 120 and 300? (S.S.C., 2010)
- (a) 5 (b) 6
- (c) 7 (d) 8
30. The number of perfect square numbers between 50 and 1000 is
- (a) 21 (b) 22
- (c) 23 (d) 24
- (Section Officers', 2003)
31. A man born in the first half of the nineteenth century was x years old in the year x^2 . He was born in (M.B.A., 2011)
- (a) 1806 (b) 1812
- (c) 1825 (d) 1836
32. R is a positive number. It is multiplied by 8 and then squared. The square is now divided by 4 and the square root is taken. The result of the square root is Q . What is the value of Q ? (SNAP, 2010)
- (a) $3R$ (b) $4R$
- (c) $7R$ (d) $9R$
33. The smallest natural number which is a perfect square and which ends in 3 identical digits lies between
- (a) 1000 and 2000 (b) 2000 and 3000
- (c) 3000 and 4000 (d) 4000 and 5000
34. $\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right)^2$ is equal to (S.S.C., 2005)
- (a) $2\frac{1}{2}$ (b) $3\frac{1}{2}$
- (c) $4\frac{1}{2}$ (d) $5\frac{1}{2}$
35. If the product of four consecutive natural numbers increased by a natural number p , is a perfect square, then the value of p is (C.P.O., 2006)
- (a) 1 (b) 2
- (c) 4 (d) 8
36. What is the square root of 0.16?
- (a) 0.004 (b) 0.04
- (c) 0.4 (d) 4
37. The value of $\sqrt{0.000441}$ is (S.S.C., 2002)
- (a) 0.00021 (b) 0.0021
- (c) 0.021 (d) 0.21
38. $\sqrt{0.00004761}$ equals (C.B.I., 2003)
- (a) 0.00069 (b) 0.0069
- (c) 0.0609 (d) 0.069
39. $1.5^2 \times \sqrt{0.0225} = ?$ (Bank P.O., 2002)
- (a) 0.0375 (b) 0.3375
- (c) 3.275 (d) 32.75
40. $\sqrt{0.01 + \sqrt{0.0064}} = ?$
- (a) 0.03 (b) 0.3
- (c) 0.42 (d) None of these
41. The value of $\sqrt{0.01} + \sqrt{0.81} + \sqrt{1.21} + \sqrt{0.0009}$ is (S.S.C., 2002)
- (a) 2.03 (b) 2.1
- (c) 2.11 (d) 2.13

42. $\sqrt{.0025} \times \sqrt{2.25} \times \sqrt{.0001} = ?$
 (a) .000075 (b) .0075
 (c) .075 (d) None of these
43. $\sqrt{1.5625} = ?$ (S.B.I.P.O., 2003)
 (a) 1.05 (b) 1.25
 (c) 1.45 (d) 1.55
44. If $\sqrt{.00000676} = .0026$, the square root of 67,60,000 is:
 (a) $\frac{1}{26}$ (b) 26
 (c) 260 (d) 2600
45. If $\sqrt{18225} = 135$, then the value of $(\sqrt{182.25} + \sqrt{1.8225} + \sqrt{0.018225} + \sqrt{0.00018225})$ is
 (a) 1.49985 (b) 14.9985
 (c) 149.985 (d) 1499.85
46. If $\sqrt{4096} = 64$, then the value of $\sqrt{40.96} + \sqrt{0.4096} + \sqrt{0.004096} + \sqrt{0.00004096}$ up to two places of decimals is (S.S.C., 2005)
 (a) 7.09 (b) 7.10
 (c) 7.11 (d) 7.12
47. Given that $\sqrt{13} = 3.605$ and $\sqrt{130} = 11.40$, find the value of $\sqrt{1.3} + \sqrt{1300} + \sqrt{0.013}$.
 (a) 36.164 (b) 36.304
 (c) 37.164 (d) 37.304
48. If $\frac{52}{x} = \sqrt{\frac{169}{289}}$, the value of x is
 (a) 52 (b) 58
 (c) 62 (d) 68
49. For what value of * the statement $\left(\frac{*}{15}\right)\left(\frac{*}{135}\right) = 1$ is true? (S.S.C., 2002)
 (a) 15 (b) 25
 (c) 35 (d) 45
50. Which number should replace both the question marks in the following equation? (Bank P.O., 2008)
 $\frac{?}{1776} = \frac{111}{?}$
 (a) 343 (b) 414
 (c) 644 (d) 543
 (e) None of these
51. Which number can replace both the question marks in the equation $\frac{4\frac{1}{2}}{?} = \frac{?}{32}$.
 (a) 1 (b) 7
 (c) $7\frac{1}{2}$ (d) None of these
52. What should come in place of both the question marks in the equation $\frac{?}{\sqrt{128}} = \frac{\sqrt{162}}{?}$.
 (a) 12 (b) 14
 (c) 144 (d) 196
53. If $\sqrt{x + \frac{x}{y}} = x\sqrt{\frac{x}{y}}$, where x and y are positive real numbers, then y is equal to (Hotel Mgmt, 2010)
 (a) $x + 1$ (b) $x - 1$
 (c) $x^2 + 1$ (d) $x^2 - 1$
54. The number $25^{64} \times 64^{25}$ is the square of a natural number n . The sum of the digits of n is (A.A.O. Exam., 2010)
 (a) 7 (b) 14
 (c) 21 (d) 28
55. If $0.13 \div p^2 = 13$, then p equals
 (a) 0.01 (b) 0.1
 (c) 10 (d) 100
56. What number should be divided by $\sqrt{0.25}$ to give the result as 25? (C.B.I., 2003)
 (a) 12.5 (b) 25
 (c) 50 (d) 125
57. If $\sqrt{3^n} = 729$, then the value of n is (Section Officers, 2003)
 (a) 6 (b) 8
 (c) 10 (d) 12
58. If $\sqrt{18 \times 14 \times x} = 84$, then x equals
 (a) 22 (b) 24
 (c) 28 (d) 32
59. $28\sqrt{?} + 1426 = \frac{3}{4}$ of 2872
 (a) 576 (b) 676
 (c) 1296 (d) 1444
60. $\sqrt{\frac{?}{169}} = \frac{54}{39}$
 (a) 108 (b) 324
 (c) 2916 (d) 4800
61. If $\sqrt{x} \div \sqrt{441} = 0.02$, then the value of x is
 (a) 0.1764 (b) 1.764
 (c) 1.64 (d) 2.64
62. $\sqrt{\frac{.0196}{?}} = 0.2$
 (a) 0.49 (b) 0.7
 (c) 4.9 (d) None of these
63. $\sqrt{0.0169 \times ?} = 1.3$ (Hotel Management, 2001)
 (a) 10 (b) 100
 (c) 1000 (d) None of these

64. If $\sqrt{1369} + \sqrt{.0615 + x} = 37.25$, then x is equal to

- (a) 10^{-1} (b) 10^{-2}
(c) 10^{-3} (d) None of these

65. If $\sqrt{(x-1)(y+2)} = 7$, x and y being positive whole numbers, then the values of x and y respectively are

- (a) 8, 5 (b) 15, 12
(c) 22, 19 (d) None of these

66. If $\sqrt{.04 \times .4 \times a} = .004 \times .4 \times \sqrt{b}$, then $\frac{a}{b}$ is

- (a) 16×10^{-3} (b) 16×10^{-4}
(c) 16×10^{-5} (d) None of these

67. Three-fifth of the square of a certain number is 126.15. What is the number?

- (a) 14.5 (b) 75.69
(c) 145 (d) 210.25

68. $\sqrt{\frac{0.361}{0.00169}} = ?$

- (a) $\frac{1.9}{13}$ (b) $\frac{19}{13}$
(c) $\frac{1.9}{130}$ (d) $\frac{190}{13}$

69. $\sqrt{\frac{48.4}{0.289}}$ is equal to (S.S.C., 2004)

- (a) $1\frac{5}{17}$ (b) $12\frac{1}{17}$
(c) $12\frac{16}{17}$ (d) $129\frac{7}{17}$

70. If $\sqrt{1 + \frac{x}{169}} = \frac{14}{13}$, then x is equal to

- (a) 1 (b) 13
(c) 27 (d) None of these

71. $\sqrt{1 + \frac{55}{729}} = 1 + \frac{x}{27}$, then the value of x is

- (a) 1 (b) 3
(c) 5 (d) 7

72. $\sqrt{\frac{4}{3}} - \sqrt{\frac{3}{4}} = ?$

- (a) $\frac{4\sqrt{3}}{6}$ (b) $\frac{1}{2\sqrt{3}}$
(c) 1 (d) $-\frac{1}{2\sqrt{3}}$

73. The value of $\sqrt{2}$ upto three places of decimal is

- (a) 1.410 (b) 1.412
(c) 1.413 (d) 1.414

74. $(2\sqrt{27} - \sqrt{75} + \sqrt{12})$ is equal to

- (a) $\sqrt{3}$ (b) $2\sqrt{3}$
(c) $3\sqrt{3}$ (d) $4\sqrt{3}$

75. By how much does $\sqrt{12} + \sqrt{18}$ exceed $\sqrt{3} + \sqrt{2}$?

- (a) $\sqrt{2} - 4\sqrt{3}$ (b) $\sqrt{3} + 2\sqrt{2}$
(c) $2(\sqrt{3} - \sqrt{2})$ (d) $3(\sqrt{3} - \sqrt{2})$

76. $\frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} = ?$

- (a) $2\sqrt{6}$ (b) 2
(c) $6\sqrt{2}$ (d) $\frac{2}{\sqrt{6}}$

77. The value of $\frac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}}$ is

- (a) $\frac{3}{4}$ (b) $1\frac{1}{3}$
(c) $1\frac{7}{9}$ (d) $1\frac{3}{4}$

78. If $3\sqrt{5} + \sqrt{125} = 17.88$, then what will be the value of $\sqrt{80} + 6\sqrt{5}$?

- (a) 13.41 (b) 20.46
(c) 21.66 (d) 22.35

79. $\sqrt{50} \times \sqrt{98}$ is equal to

- (a) 63.75 (b) 65.95
(c) 70 (d) 70.25

80. Given $\sqrt{2} = 1.414$. The value of

$\sqrt{8} + 2\sqrt{32} - 3\sqrt{128} + 4\sqrt{50}$ is: (S.S.C., 2003)

- (a) 8.426 (b) 8.484
(c) 8.526 (d) 8.876

81. The approximate value of $\frac{3\sqrt{12}}{2\sqrt{28}} \div \frac{2\sqrt{21}}{\sqrt{98}}$ is

(Section Officers, 2003)

- (a) 1.0605 (b) 1.0727
(c) 1.6007 (d) 1.6026

82. $\sqrt{110.25} \times \sqrt{0.01} \div \sqrt{0.0025} - \sqrt{420.25}$ equals

(SNAP, 2010)

- (a) 0.50 (b) 0.64
(c) 0.73 (d) 0.75

83. $\sqrt{\frac{.081 \times .484}{.0064 \times 6.25}}$ is equal to

- (a) 0.9 (b) 0.99
(c) 9 (d) 99

84. $\sqrt{\frac{0.204 \times 42}{0.07 \times 3.4}}$ is equal to
 (a) $\frac{1}{6}$ (b) 0.06
 (c) 0.6 (d) 6
85. $\sqrt{\frac{0.081 \times 0.324 \times 4.624}{1.5625 \times 0.0289 \times 72.9 \times 64}}$ is equal to
 (a) 0.024 (b) 0.24
 (c) 2.4 (d) 24
86. $\sqrt{\frac{9.5 \times .085}{.0017 \times .19}}$ equals
 (a) .05 (b) 5
 (c) 50 (d) 500
87. The value of $\sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{(0.003)^2 + (0.021)^2 + (0.0065)^2}}$ is
 (a) 0.1 (b) 10
 (c) 10^2 (d) 10^3 (S.S.C., 2002)
88. The square root of $(7 + 3\sqrt{5})(7 - 3\sqrt{5})$ is (S.S.C., 2004)
 (a) $\sqrt{5}$ (b) 2
 (c) 4 (d) $3\sqrt{5}$
89. $\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)^2$ simplifies to
 (a) $\frac{3}{4}$ (b) $\frac{4}{\sqrt{3}}$
 (c) $\frac{4}{3}$ (d) None of these
90. If $a = 0.1039$, then the value of $\sqrt{4a^2 - 4a + 1} + 3a$ is
 (a) 0.1039 (b) 0.2078
 (c) 1.1039 (d) 2.1039 (C.B.I., 2003)
91. The square root of $\frac{(0.75)^3}{1 - 0.75} + [0.75 + (0.75)^2 + 1]$ is
 (a) 1 (b) 2
 (c) 3 (d) 4
92. If $3a = 4b = 6c$ and $a + b + c = 27\sqrt{29}$, then $\sqrt{a^2 + b^2 + c^2}$ is
 (a) $3\sqrt{29}$ (b) 81
 (c) 87 (d) None of these
93. The square root of $0.\bar{4}$ is (S.S.C., 2004)
 (a) $0.\bar{6}$ (b) $0.\bar{7}$
 (c) $0.\bar{8}$ (d) $0.\bar{9}$
94. Which one of the following numbers has rational square root?
 (a) 0.4 (b) 0.09
 (c) 0.9 (d) 0.025
95. The value of $\sqrt{0.4}$ is
 (a) 0.02 (b) 0.2
 (c) 0.51 (d) 0.63
96. $\sqrt{0.2} = ?$ (R.R.B., 2007)
 (a) 0.02 (b) 0.2
 (c) 0.447 (d) 0.632
97. The value of $\sqrt{0.121}$ is
 (a) 0.011 (b) 0.11
 (c) 0.347 (d) 1.1
98. The value of $\sqrt{0.064}$ is
 (a) 0.008 (b) 0.08
 (c) 0.252 (d) 0.8
99. The value of $\sqrt{\frac{0.16}{0.4}}$ is (IGNOU, 2003)
 (a) 0.02 (b) 0.2
 (c) 0.63 (d) None of these
100. The value of $\frac{1 + \sqrt{0.01}}{1 - \sqrt{0.1}}$ is close to
 (a) 0.6 (b) 1.1
 (c) 1.6 (d) 1.7
101. The square root of 535.9225 is (R.R.B., 2006)
 (a) 23.15 (b) 23.45
 (c) 24.15 (d) 28.25
102. If $\sqrt{5} = 2.236$, then the value of $\frac{1}{\sqrt{5}}$ is
 (a) .367 (b) .447
 (c) .745 (d) None of these
103. If $\sqrt{24} = 4.899$, the value of $\sqrt{\frac{8}{3}}$ is
 (a) 0.544 (b) 1.333
 (c) 1.633 (d) 2.666
104. If $\sqrt{6} = 2.449$, then the value of $\frac{3\sqrt{2}}{2\sqrt{3}}$ is
 (a) 0.6122 (b) 0.8163
 (c) 1.223 (d) 1.2245
105. If $\sqrt{5} = 2.236$, then the value of $\frac{\sqrt{5}}{2} - \frac{10}{\sqrt{5}} + \sqrt{125}$ is equal to
 (a) 5.59 (b) 7.826
 (c) 8.944 (d) 10.062

- 106.** If $2 * 3 = \sqrt{13}$ and $3 * 4 = 5$, then the value of $5 * 12$ is
 (a) $\sqrt{17}$ (b) $\sqrt{29}$
 (c) 12 (d) 13
- 107.** If 1537* is a perfect square, then the digit which replaces * is (Hotel Management, 2007)
 (a) 2 (b) 4
 (c) 5 (d) 6
- 108.** The smallest perfect square that is divisible by 7! is (I.I.F.T., 2010)
 (a) 19600 (b) 44100
 (c) 176400 (d) 705600
- 109.** The least perfect square number divisible by 3, 4, 5, 6 and 8 is
 (a) 900 (b) 1200
 (c) 2500 (d) 3600
- 110.** The least perfect square, which is divisible by each of 21, 36 and 66, is
 (a) 213444 (b) 214344
 (c) 214434 (d) 231444
- 111.** The least number by which 294 must be multiplied to make it a perfect square, is
 (a) 2 (b) 3
 (c) 6 (d) 24
- 112.** Find the smallest number by which 5808 should be multiplied so that the product becomes a perfect square.
 (a) 2 (b) 3
 (c) 7 (d) 11
- 113.** The least number by which 1470 must be divided to get a number which is a perfect square, is
 (a) 5 (b) 6
 (c) 15 (d) 30
- 114.** What is the smallest number to be subtracted from 549162 in order to make it a perfect square?
 (a) 28 (b) 36
 (c) 62 (d) 81
- 115.** What is the least number which should be subtracted from 0.000326 to make it a perfect square?
 (a) 0.000002 (b) 0.000004
 (c) 0.02 (d) 0.04
- 116.** What is the least number to be added to 7700 to make it a perfect square?
 (a) 77 (b) 98
 (c) 131 (d) 221
 (e) None of these (Bank Recruitment, 2008)
- 117.** The smallest number to be added to 680621 to make the sum a perfect square is (S.S.C., 2005)
 (a) 4 (b) 5
 (c) 6 (d) 8
- 118.** The greatest four-digit perfect square number is
 (a) 9000 (b) 9801
 (c) 9900 (d) 9981
- 119.** The least number of 4 digits which is a perfect square is
 (a) 1000 (b) 1016
 (c) 1024 (d) 1036
- 120.** The sum of 18 consecutive natural numbers is a perfect square. What is the smallest possible value of this sum? (A.A.O. Exam, 2009)
 (a) 169 (b) 225
 (c) 289 (d) 441
- 121.** $\sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}}$
 $\cdot \sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}$ is equal to
 (a) 1 (b) 2
 (c) 4 (d) $\sqrt{6}$
- 122.** Given $\sqrt{5} = 2.2361$, $\sqrt{3} = 1.7321$, then $\frac{1}{\sqrt{5}-\sqrt{3}}$ is equal to
 (a) 1.98 (b) 1.984
 (c) 1.9841 (d) 2
- 123.** $\frac{1}{(\sqrt{9}-\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} - \frac{1}{(\sqrt{6}-\sqrt{5})}$
 $+ \frac{1}{(\sqrt{5}-\sqrt{4})}$ is equal to (M.B.A., 2007)
 (a) 0 (b) $\frac{1}{3}$
 (c) 1 (d) 5
- 124.** Determine the value of
 $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots + \frac{1}{\sqrt{120}+\sqrt{121}}$.
 (a) 8 (b) 10
 (c) $\sqrt{120}$ (d) $12\sqrt{2}$ (M.A.T., 2005)
- 125.** The expression $1 - \frac{1}{1+\sqrt{3}} + \frac{1}{1-\sqrt{3}}$ equals
 (a) $1-\sqrt{3}$ (b) 1
 (c) $-\sqrt{3}$ (d) $\sqrt{3}$ (M.B.A., 2011)
- 126.** $\left(2 + \sqrt{2} + \frac{1}{2+\sqrt{2}} + \frac{1}{\sqrt{2}-2}\right)$ simplifies to
 (a) $2-\sqrt{2}$ (b) 2
 (c) $2+\sqrt{2}$ (d) $2\sqrt{2}$ (M.B.A., 2007)

127. What is the value of $\frac{1}{\sqrt{5} + \sqrt{3}} + \frac{2}{3 + \sqrt{5}} - \frac{3}{3 + \sqrt{3}}$?

(I.I.F.T., 2005)

- (a) $-\frac{1}{2}$ (b) 0
(c) $\frac{1}{2}$ (d) 1

128. If $\sqrt{2} = 1.4142$, the value of $\frac{7}{(3 + \sqrt{2})}$ is (R.R.B., 2005)

- (a) 1.5858 (b) 3.4852
(c) 3.5858 (d) 4.4142

129. $\left[\frac{3\sqrt{2}}{\sqrt{6} - \sqrt{3}} - \frac{4\sqrt{3}}{\sqrt{6} - \sqrt{2}} - \frac{6}{\sqrt{8} - \sqrt{12}} \right] = ?$

(Teachers' Exam, 2010)

- (a) $\sqrt{3} - \sqrt{2}$ (b) $\sqrt{3} + \sqrt{2}$
(c) $5\sqrt{3}$ (d) 1

130. $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$ is equal to

- (a) 1 (b) 2
(c) $6 - \sqrt{35}$ (d) $6 + \sqrt{35}$

131. If $\frac{5 + 2\sqrt{3}}{7 + 4\sqrt{3}} = a + b\sqrt{3}$, then (R.R.B., 2010)

- (a) $a = -11, b = -6$ (b) $a = -11, b = 6$
(c) $a = 11, b = -6$ (d) $a = 6, b = 11$

132. If $\sqrt{2} = 1.414$, the square root of $\frac{\sqrt{2} - 1}{\sqrt{2} + 1}$ is nearest to

- (a) 0.172 (b) 0.414
(c) 0.586 (d) 1.414

133. Given that $\sqrt{3} = 1.732$, the value of

$\frac{3 + \sqrt{6}}{5\sqrt{3} - 2\sqrt{12} - \sqrt{32} + \sqrt{50}}$ is (S.S.C., 2007)

- (a) 1.414 (b) 1.732
(c) 2.551 (d) 4.899

134. $\left(\frac{2 + \sqrt{3}}{2 - \sqrt{3}} + \frac{2 - \sqrt{3}}{2 + \sqrt{3}} + \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \right)$ simplifies to (S.S.C., 2008)

- (a) $16 - \sqrt{3}$ (b) $4 - \sqrt{3}$
(c) $2 - \sqrt{3}$ (d) $2 + \sqrt{3}$

135. If $x = \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}}$ and $y = \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$, then $(x + y)$ equals (S.S.C., 2005)

- (a) $2(\sqrt{5} + \sqrt{3})$ (b) $2\sqrt{15}$
(c) 8 (d) 16

136. $\frac{\sqrt{2}(2 + \sqrt{3})}{\sqrt{3}(\sqrt{3} + 1)} \times \frac{\sqrt{2}(2 - \sqrt{3})}{\sqrt{3}(\sqrt{3} - 1)}$ is equal to (C.P.O., 2006)

- (a) $3\sqrt{2}$ (b) $\frac{\sqrt{2}}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{3}$

137. $\frac{12}{3 + \sqrt{5} + 2\sqrt{2}}$ is equal to (S.S.C., 2007)

- (a) $1 - \sqrt{5} + \sqrt{2} + \sqrt{10}$
(b) $1 + \sqrt{5} + \sqrt{2} - \sqrt{10}$
(c) $1 + \sqrt{5} - \sqrt{2} + \sqrt{10}$
(d) $1 - \sqrt{5} - \sqrt{2} + \sqrt{10}$

138. $\left[\frac{1}{\sqrt{2} + \sqrt{3} - \sqrt{5}} + \frac{1}{\sqrt{2} - \sqrt{3} - \sqrt{5}} \right]$ in simplified form equals (S.S.C., 2005)

- (a) 0 (b) $\frac{1}{\sqrt{2}}$
(c) 1 (d) $\sqrt{2}$

139. If $x = (7 - 4\sqrt{3})$, then the value of $\left(x + \frac{1}{x} \right)$ is

- (a) $3\sqrt{3}$ (b) $8\sqrt{3}$
(c) 14 (d) $14 + 8\sqrt{3}$

140. If $x = 3 + \sqrt{8}$, then $x^2 + \frac{1}{x^2}$ is equal to (S.S.C., 2007)

- (a) 30 (b) 34
(c) 36 (d) 38

141. If $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$, $b = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$, then the value of $a^2 + b^2$ would be (M.B.A., 2008)

- (a) 10 (b) 98
(c) 99 (d) 100

142. If $a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ and $b = \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$, the value

of $\left(\frac{a^2 + ab + b^2}{a^2 - ab + b^2} \right)$ is

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$

- 143.** If $x = \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots \infty}}}$ then the positive value of x is (M.B.A., 2006)
- (a) $\frac{\sqrt{7} + 1}{2}$ (b) $\frac{\sqrt{6} + 1}{2}$
 (c) $\frac{\sqrt{5} + 1}{2}$ (d) $\frac{\sqrt{3} + 1}{2}$
- 144.** $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$ is equal to (S.S.C., 2005)
- (a) 1 (b) 1.5
 (c) 2 (d) 2.5
- 145.** If $a = \sqrt{3 + \sqrt{3 + \sqrt{3 + \dots}}}$, then which of the following is true? (M.B.A., 2007)
- (a) $2 < a < 3$ (b) $a > 3$
 (c) $3 < a < 4$ (d) $a = 3$
- 146.** $\frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} - \sqrt{3-x}} = 2$. Then x is equal to (S.S.C., 2010)
- (a) $\frac{5}{7}$ (b) $\frac{7}{5}$
 (c) $\frac{5}{12}$ (d) $\frac{12}{5}$
- 147.** One-fourth of a herd of camels was seen in the forest. Twice the square root of the herd had gone to mountains and the remaining 15 camels were seen on the bank of a river. Find the total number of camels. (M.A.T., 2005)
- (a) 32 (b) 34
 (c) 35 (d) 36
- 148.** A gardener plants 17956 trees in such a way that there are as many rows as there are trees in a row. The number of trees in a row are (M.B.A., 2006)
- (a) 134 (b) 136
 (c) 144 (d) 154
- 149.** The number of trees in each row of a garden is equal to the total number of rows in the garden. After 111 trees have been uprooted in a storm, there remain 10914 trees in the garden. The number of rows of trees in the garden is (C.P.O., 2007)
- (a) 100 (b) 105
 (c) 115 (d) 125
- 150.** 1250 oranges were distributed among a group of girls of a class. Each girl got twice as many oranges as the number of girls in that group. The number of girls in the group was (P.C.S., 2006)
- (a) 25 (b) 45
 (c) 50 (d) 100
- 151.** A General wishes to draw up his 36581 soldiers in the form of a solid square. After arranging them, he found that some of them are left over. How many are left?
- (a) 65 (b) 81
 (c) 100 (d) None of these
- 152.** A group of students decided to collect as many paise from each member of the group as is the number of members. If the total collection amounts to ₹ 59.29, the number of members in the group is
- (a) 57 (b) 67
 (c) 77 (d) 87
- 153.** A mobile company offered to pay the Indian Cricket Team as much money per run scored by the side as the total number it gets in a one-dayer against Australia. Which one of the following cannot be the total amount to be spent by the company in this deal? (P.C.S., 2008)
- (a) 21,904 (b) 56,169
 (c) 1,01,761 (d) 1,21,108
- 154.** $\sqrt[3]{148877} = ?$ (L.I.C.A.D.O., 2007)
- (a) 43 (b) 49
 (c) 53 (d) 59
- 155.** $\sqrt[3]{681472} = ?$ (Bank P.O., 2009)
- (a) 76 (b) 88
 (c) 96 (d) 98
- 156.** $1728 \div \sqrt[3]{262144} \times ? - 288 = 4491$ (Bank P.O., 2008)
- (a) 148 (b) 156
 (c) 173 (d) 177
- 157.** $99 \times 21 - \sqrt[3]{?} = 1968$ (NABARD, 2008)
- (a) 1367631 (b) 111
 (c) 1366731 (d) 1367
- 158.** The cube root of .000216 is
- (a) .6 (b) .06
 (c) .006 (d) None of these
- 159.** $\sqrt[3]{4\frac{12}{125}} = ?$
- (a) $1\frac{2}{5}$ (b) $1\frac{3}{5}$
 (c) $1\frac{4}{5}$ (d) $2\frac{2}{5}$
- 160.** $\sqrt[3]{\sqrt{.000064}} = ?$
- (a) .02 (b) .2
 (c) 2 (d) None of these
- 161.** The smallest positive integer n , for which $864n$ is a perfect cube, is (C.P.O., 2007)
- (a) 1 (b) 2
 (c) 3 (d) 4
- 162.** Value of $\sqrt{.01} \times \sqrt[3]{.008} - .02$ is (P.C.S., 2006)
- (a) 0 (b) 1
 (c) 2 (d) 3

163. The value of $\sqrt[3]{\frac{0.2 \times 0.2 \times 0.2 + 0.04 \times 0.04 \times 0.04}{0.4 \times 0.4 \times 0.4 + 0.08 \times 0.08 \times 0.08}}$ is (S.S.C., 2005)

- (a) 0.125 (b) 0.25
(c) 0.5 (d) 0.75

164. A rationalising factor of $(\sqrt[3]{9} - \sqrt[3]{3} + 1)$ is (S.S.C., 2007)

- (a) $\sqrt[3]{3} - 1$ (b) $\sqrt[3]{3} + 1$
(c) $\sqrt[3]{9} - 1$ (d) $\sqrt[3]{9} + 1$

165. The largest four-digit number which is a perfect cube, is

- (a) 8000 (b) 9261
(c) 9999 (d) None of these

166. By what least number must 21600 be multiplied so as to make it a perfect cube? (M.A.T., 2002)

- (a) 6 (b) 10
(c) 20 (d) 30

167. What is the smallest number by which 3600 be divided to make it a perfect cube?

- (a) 9 (b) 50
(c) 300 (d) 450

168. Which smallest number must be added to 710 so that the sum is a perfect cube? (S.S.C., 2005)

- (a) 11 (b) 19
(c) 21 (d) 29

169. Solve $\sqrt{7921} = ?$ [Indian Railway Gr. 'D' Exam, 2014]

- (a) 89 (b) 87
(c) 37 (d) 47

170. Solve $\sqrt[4]{(625)^3} = ?$ [Indian Railway Gr. 'D' Exam, 2014]

- (a) $\sqrt[3]{1875}$ (b) 25
(c) 125 (d) None of these

171. If $\sqrt{y} = 4x$, then $\frac{x^2}{y}$ is [SSC—CHSL (10+2) Exam, 2015]

- (a) 2 (b) $\frac{1}{16}$
(c) $\frac{1}{4}$ (d) 4

Direction (Q. No. 172): What approximate value will come in place of question mark(?) in the given question? (You are not expected to calculate the exact value)

172. $\sqrt{2025.11} \times \sqrt{256.04} + \sqrt{399.95} \times \sqrt{?} = 33.98 \times 40.11$

[IBPS—Bank Spl. Officer (IT) Exam, 2015]

- (a) 1682 (b) 1024
(c) 1582 (d) 678

173. If $\sqrt{7} = 2.645$, then the value of $\frac{1}{\sqrt{28}}$ up to three places of decimal is [SSC—CHSL (10+2) Exam, 2015]

- (a) 0.183 (b) 0.185
(c) 0.187 (d) 0.189

174. Solve: $\left(\sqrt{\frac{25}{9}} - \sqrt{\frac{64}{81}}\right) \div \sqrt{\frac{16}{324}} = ?$

- (a) 4.5 (b) 2.5
(c) 1.5 (d) 3.5

[United India Insurance Co. Ltd. (UIICL) Assistant (Online) Exam, 2015]

175. Solve $1728 \div \sqrt[3]{262144} \times ? - 288 = 4491$

- (a) 148 (b) 156
(c) 173 (d) 177
(e) 185

[United India Insurance Co. Ltd. (UIICL) Assistant (Online) Exam, 2015]

176. Solve: $(\sqrt{7} + 11)^2 = (?)^{\frac{1}{3}} + 2\sqrt{847} + 122$

- (a) $36 + 44\sqrt{7}$ (b) 6
(c) 216 (d) 36

[IDBI Bank Executive Officers Exam, 2015]

Directions (Q. No. 177 & 178): What will come in place of question mark in these questions?

177. $? - \sqrt{(784)} = 6 \times \sqrt{(324)}$

(NICT—AAO Exam, 2015)

- (a) 128 (b) 160
(c) 236 (d) 136

178. $\sqrt{(2116)} - \sqrt{1600} = \sqrt{(?)}$

[NICT—AAO Exam, 2015]

- (a) 20 (b) 64
(c) 81 (d) 36

179. Solve $\sqrt{(27 \div 5 \times ?)} + 15 = 5.4 \div 6 + 0.3$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) 2 (b) 6
(c) 10 (d) 4

180. $\sqrt{24 \div 0.5 + 1} + \sqrt{18 \div 0.6 + 6} = ?$

- (a) 19 (b) 13
(c) 12 (d) 15

[IBPS—RRB Office Assistant (Online) Exam, 2015]

181. $(\sqrt{63} + \sqrt{252}) \times (\sqrt{175} + \sqrt{28}) = ?$

[IBPS—RRB Office Assistant (Online) Exam, 2015]

- (a) $16\sqrt{7}$ (b) 441
(c) 16 (d) $7\sqrt{7}$

182. $9x^2 + 25 - 30x$ can be expressed as the square of

[SSC—CHSL (10+2) Exam, 2015]

- (a) $3x^2 - 25$ (b) $3x - 5$
(c) $-3x - 5$ (d) $3x - 5$

183. If $\sqrt{33} = 5.745$, then the value of the following is

approximately $\sqrt{\frac{3}{11}}$

[SSC—CHSL (10+2) Exam, 2015]

- (a) 1 (b) 6.32
(c) 0.5223 (d) 2.035

184. $\sqrt{?} + 14 = \sqrt{2601}$ [SBI Jr. Associates (Pre.) Exam, 2016]

- (a) 1521 (b) 1369
(c) 1225 (d) 961

185. If $a = \frac{\sqrt{3}}{2}$, then $\sqrt{1+a} + \sqrt{1-a} = ?$

[DMRC—Train Operator (Station Controller) Exam, 2016]

- (a) $(2 - \sqrt{3})$ (b) $(2 + \sqrt{3})$
(c) $\left(\frac{\sqrt{3}}{2}\right)$ (d) $\sqrt{3}$

186. What is $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$ equal to? [CDS 2016]

- (a) 5 (b) $5\sqrt{2}$
(c) $5\sqrt{5}$ (d) $\sqrt{5}$

187. The square root of $\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$ is

- (a) 1 (b) 2 [CDS 2016]
(c) 3 (d) 4

188. $\sqrt{10 + 2\sqrt{6} + 2\sqrt{10} + 2\sqrt{15}}$ is equal to

- (a) $(\sqrt{2} + \sqrt{3} + \sqrt{5})$ (b) $(\sqrt{2} + \sqrt{3} - \sqrt{5})$
(c) $(\sqrt{2} + \sqrt{5} - \sqrt{3})$ (d) None of these

[DMRC—Customer Relationship Assistant (CRA) Exam, 2016]

ANSWERS

1. (b)	2. (b)	3. (c)	4. (c)	5. (e)	6. (e)	7. (a)	8. (a)	9. (c)	10. (b)
11. (a)	12. (c)	13. (d)	14. (b)	15. (b)	16. (c)	17. (c)	18. (c)	19. (b)	20. (d)
21. (b)	22. (b)	23. (e)	24. (c)	25. (b)	26. (d)	27. (e)	28. (a)	29. (c)	30. (d)
31. (a)	32. (b)	33. (a)	34. (c)	35. (a)	36. (c)	37. (c)	38. (b)	39. (b)	40. (b)
41. (d)	42. (d)	43. (b)	44. (d)	45. (b)	46. (c)	47. (d)	48. (d)	49. (d)	50. (e)
51. (d)	52. (a)	53. (d)	54. (b)	55. (b)	56. (a)	57. (d)	58. (c)	59. (b)	60. (b)
61. (a)	62. (a)	63. (b)	64. (c)	65. (a)	66. (c)	67. (a)	68. (d)	69. (c)	70. (c)
71. (a)	72. (b)	73. (d)	74. (c)	75. (b)	76. (b)	77. (b)	78. (d)	79. (c)	80. (b)
81. (a)	82. (a)	83. (b)	84. (d)	85. (a)	86. (c)	87. (b)	88. (b)	89. (c)	90. (c)
91. (b)	92. (c)	93. (a)	94. (b)	95. (d)	96. (c)	97. (c)	98. (c)	99. (c)	100. (c)
101. (a)	102. (b)	103. (c)	104. (d)	105. (b)	106. (d)	107. (d)	108. (c)	109. (d)	110. (a)
111. (c)	112. (b)	113. (d)	114. (d)	115. (a)	116. (e)	117. (a)	118. (b)	119. (c)	120. (b)
121. (a)	122. (c)	123. (d)	124. (b)	125. (a)	126. (b)	127. (b)	128. (a)	129. (c)	130. (d)
131. (c)	132. (b)	133. (b)	134. (a)	135. (c)	136. (d)	137. (b)	138. (b)	139. (c)	140. (b)
141. (b)	142. (b)	143. (c)	144. (c)	145. (a)	146. (d)	147. (d)	148. (a)	149. (b)	150. (a)
151. (c)	152. (c)	153. (d)	154. (c)	155. (b)	156. (d)	157. (a)	158. (b)	159. (b)	160. (b)
161. (b)	162. (a)	163. (c)	164. (b)	165. (b)	166. (b)	167. (d)	168. (b)	169. (a)	170. (c)
171. (b)	172. (b)	173. (d)	174. (d)	175. (d)	176. (c)	177. (d)	178. (d)	179. (d)	180. (b)
181. (b)	182. (d)	183. (c)	184. (b)	185. (d)	186. (d)	187. (b)	188. (a)		

SOLUTIONS

1.

$$\begin{array}{r} 2 \overline{) 5 \ 38 \ 24} \ (232) \\ \underline{4} \\ 43 \\ \underline{1 \ 38} \\ 1 \ 29 \\ \underline{9 \ 24} \\ 9 \ 24 \\ \underline{ \times} \end{array}$$

$$\therefore \sqrt{53824} = 232.$$

2.

$$\begin{array}{r} 2 \overline{) 4 \ 12 \ 09} \ (203) \\ \underline{4} \\ 40 \\ \underline{12} \\ 0 \\ 403 \\ \underline{12 \ 09} \\ 12 \ 09 \\ \underline{ \times} \end{array}$$

$$\therefore \sqrt{41209} = 203.$$

3.

$$\begin{array}{r} 1 \overline{) 1 \ 23 \ 45 \ 43 \ 21} \ (11111) \\ \underline{1} \\ 21 \\ \underline{23} \\ 21 \\ \underline{21} \\ 221 \\ \underline{2 \ 45} \\ 2 \ 21 \\ \underline{22 \ 43} \\ 22 \ 21 \\ \underline{222 \ 21} \\ 222 \ 21 \\ \underline{ \times} \end{array}$$

$$\therefore \sqrt{123454321} = 11111.$$

4. The number of digits in the square root of a perfect square number of n digits is

(i) $\frac{n}{2}$, if n is even

(ii) $\frac{n+1}{2}$, if n is odd

Here, $n = 12$. So, required number of digits = $\frac{n}{2} = \frac{12}{2} = 6$.

5. $\sqrt{\sqrt{17956} + \sqrt{24025}} = \sqrt{134 + 155} = \sqrt{289} = 17$.

6. $\sqrt{\sqrt{44944} + \sqrt{52441}} = \sqrt{212 + 229} = \sqrt{441} = 21$.

7. Sum of prime numbers greater than 4 but less than 16 = $(5 + 7 + 11 + 13) = 36$.

$\therefore \frac{1}{4} \times 36 = 9 = 3^2$.

8. Given exp. $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{169}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{121}}}$
 $= \sqrt{10 + \sqrt{25 + 11}} = \sqrt{10 + \sqrt{36}}$
 $= \sqrt{10 + 6} = \sqrt{16} = 4$.

9. Given exp. $= \sqrt{41 - \sqrt{21 + \sqrt{19 - 3}}}$
 $= \sqrt{41 - \sqrt{21 + \sqrt{16}}} = \sqrt{41 - \sqrt{21 + 4}}$
 $= \sqrt{41 - \sqrt{25}} = \sqrt{41 - 5} = \sqrt{36} = 6$.

10. Given exp. $= \sqrt{176 + 49} = \sqrt{225} = 15$.

11. Given exp. $= \frac{14}{7} \times \frac{21}{7} \times \frac{120}{15} = 2 \times 3 \times 8 = 48$.

12. Given exp. $= \left(\frac{\sqrt{225}}{\sqrt{729}} - \frac{\sqrt{25}}{\sqrt{144}} \right) \div \frac{\sqrt{16}}{\sqrt{81}}$
 $= \left(\frac{15}{27} - \frac{5}{12} \right) \div \frac{4}{9} = \left(\frac{15}{108} \times \frac{9}{4} \right) = \frac{5}{16}$.

13. Let $(15)^2 + (18)^2 - 20 = \sqrt{x}$.

Then, $\sqrt{x} = 225 + 324 - 20 = 529$

$\Leftrightarrow x = (529)^2 = 279841$.

14. Let $\sqrt{x} \times \sqrt{484} = 1034$. Then, $\sqrt{x} \times 22 = 1034$

$\Leftrightarrow \sqrt{x} = \frac{1034}{22} = 47$

$\Leftrightarrow x = (47)^2 = 2209$.

15. Let $\sqrt{11881} \times \sqrt{x} = 10137$. Then, $109 \times \sqrt{x} = 10137$

$\Leftrightarrow \sqrt{x} = \frac{10137}{109} = 93$

$\Leftrightarrow x = (93)^2 = 8649$.

16. $\frac{4050}{\sqrt{x}} = 450 \Leftrightarrow \sqrt{x} = \frac{4050}{450} = 9$

$\Leftrightarrow x = 9^2 = 81$.

17. Let $\sqrt{\frac{16}{25}} \times \sqrt{\frac{x}{25}} \times \frac{16}{25} = \frac{256}{625}$.

Then, $\frac{4}{5} \times \frac{\sqrt{x}}{2} \times \frac{16}{25} = \frac{256}{625}$

$\Leftrightarrow \frac{64\sqrt{x}}{625} = \frac{256}{625}$

$\Leftrightarrow \sqrt{x} = \frac{256}{625} \times \frac{625}{64}$

$\Leftrightarrow x = 4^2 = 16$.

18. $\sqrt{(272)^2 - (128)^2} = \sqrt{(272 + 128)(272 - 128)}$
 $= \sqrt{400 \times 144} = \sqrt{57600} = 240$.

19. $6 * 24 = 6 + 24 + \sqrt{6 \times 24} = 30 + \sqrt{144}$
 $= 30 + 12 = 42$.

20. $10y\sqrt{y^3 - y^2} = 10 \times 5 \sqrt{5^3 - 5^2}$
 $= 50 \times \sqrt{125 - 25} = 50 \times \sqrt{100}$
 $= 50 \times 10 = 500$.

21. $\sqrt{110\frac{1}{4}} = \sqrt{\frac{441}{4}} = \frac{\sqrt{441}}{\sqrt{4}} = \frac{21}{2} = 10.5$.

22. $\sqrt{\frac{25}{81} - \frac{1}{9}} = \sqrt{\frac{25-9}{81}} = \sqrt{\frac{16}{81}} = \frac{\sqrt{16}}{\sqrt{81}} = \frac{4}{9}$.

23. Let $[(\sqrt{81})^2]^2 = x^2$. Then, $x^2 = (81)^2$ or $x = 81$.

24.

1	1	58	76	(126
22	58			
246	14	76		
	14	76		
	×			

 $\therefore \sqrt{15876} = 126$.

25.

1	3	00	00	00	(1.732
27	2	00			
343	11	00			
3492	71	00			
	69	84			
	1	16			

 $\therefore \sqrt{3} = 1.73 = \frac{173}{100}$.

26. A number ending in 8 can never be a perfect square.

27. The squares of numbers having 1 and 9 as the unit's digit end in the digit 1.

Such numbers are: 1, 9, 11, 19, 21, 29, 31, 39, 41, 49 i.e., there are 10 such numbers.

$$\therefore \text{Required percentage} = \left(\frac{10}{50} \times 100 \right) \% = 20\%.$$

28. Let the given number be x .

$$\text{Then, } x^2 - 25 = (x - 25)^2 \Leftrightarrow x^2 - 25 = x^2 + 625 - 50x \Leftrightarrow 50x = 650 \Leftrightarrow x = 13.$$

29. $(11)^2 = 121$ and $(17)^2 = 289$.

So, the perfect squares between 120 and 300 are the squares of numbers from 11 to 17. Clearly, these are 7 in number.

30. The first perfect square number after 50 is 64 ($= 8^2$) and the last perfect square number before 1000 is 961 [$= (31)^2$].

So, the perfect squares between 50 and 1000 are the squares of numbers from 8 to 31. Clearly, these are 24 in number.

31. Clearly, the man was born between 1800 and 1850.

The only perfect square number between 1800 and 1850 is 1849. And, $1849 = (43)^2$.

So, the man was 43 years old in 1849. Thus, he was born in $(1849 - 43) = 1806$.

$$32. Q = \sqrt{\frac{(8R)^2}{4}} = \frac{\sqrt{(8R)^2}}{\sqrt{4}} = \frac{8R}{2} = 4R.$$

33. The smallest such number is 1444 [$= (38)^2$]. It lies between 1000 and 2000.

$$34. \left(\sqrt{2} + \frac{1}{\sqrt{2}} \right)^2 = (\sqrt{2})^2 + \left(\frac{1}{\sqrt{2}} \right)^2 + 2 \times \sqrt{2} \times \frac{1}{\sqrt{2}} \\ = 2 + \frac{1}{2} + 2 = 4\frac{1}{2}.$$

35. We have:

$$1 \times 2 \times 3 \times 4 = 24 \text{ and } 24 + 1 = 25 [= 5^2].$$

$$2 \times 3 \times 4 \times 5 = 120 \text{ and } 120 + 1 = 121 [= 11^2]$$

$$3 \times 4 \times 5 \times 6 = 360 \text{ and } 360 + 1 = 361 [= 19^2]$$

$$4 \times 5 \times 6 \times 7 = 840 \text{ and } 840 + 1 = 841 [= 29^2].$$

$$\therefore p = 1.$$

$$36. \sqrt{0.16} = \sqrt{\frac{16}{100}} = \frac{\sqrt{16}}{\sqrt{100}} = \frac{4}{10} = 0.4.$$

$$37. \sqrt{0.000441} = \sqrt{\frac{441}{10^6}} = \frac{\sqrt{441}}{\sqrt{10^6}} = \frac{21}{10^3} = \frac{21}{1000} = 0.021.$$

$$38. \sqrt{0.00004761} = \sqrt{\frac{4761}{10^8}} = \frac{\sqrt{4761}}{\sqrt{10^8}} = \frac{69}{10^4} = \frac{69}{10000} = 0.0069.$$

$$39. 1.5^2 \times \sqrt{0.0225} = 1.5^2 \times \sqrt{\frac{225}{10000}} = 2.25 \times \frac{15}{100} \\ = 2.25 \times 0.15 = 0.3375.$$

$$40. \sqrt{0.01} + \sqrt{0.0064} = \sqrt{0.01} + \sqrt{\frac{64}{10000}} \\ = \sqrt{0.01} + \frac{8}{100} = \sqrt{0.01 + 0.08} = \sqrt{0.09} = 0.3.$$

$$41. \text{Given exp.} = \sqrt{\frac{1}{100}} + \sqrt{\frac{81}{100}} + \sqrt{\frac{121}{100}} + \sqrt{\frac{9}{10000}} \\ = \frac{1}{10} + \frac{9}{10} + \frac{11}{10} + \frac{3}{100} \\ = 0.1 + 0.9 + 1.1 + 0.03 = 2.13.$$

$$42. \text{Given exp.} = \sqrt{\frac{25}{10000}} \times \sqrt{\frac{225}{100}} \times \sqrt{\frac{1}{10000}} \\ = \frac{5}{100} \times \frac{15}{10} \times \frac{1}{100} = \frac{75}{10000} = 0.00075.$$

$$43. \begin{array}{r|l} 1 & 1.5\overline{6} \ 25 \ (1.25) \\ & 1 \\ \hline 22 & 56 \\ & 44 \\ \hline 245 & 12 \ 25 \\ & 12 \ 25 \\ \hline & \times \end{array} \quad \therefore \sqrt{1.5625} = 1.25.$$

$$44. \sqrt{6760000} = \sqrt{0.00000676 \times 10^{12}} = \sqrt{0.00000676} \times \sqrt{10^{12}} \\ = .0026 \times 10^6 = 2600.$$

$$45. \text{Given exp.} = \sqrt{\frac{18225}{10^2}} + \sqrt{\frac{18225}{10^4}} + \sqrt{\frac{18225}{10^6}} + \sqrt{\frac{18225}{10^8}} \\ = \frac{\sqrt{18225}}{10} + \frac{\sqrt{18225}}{10^2} + \frac{\sqrt{18225}}{10^3} + \frac{\sqrt{18225}}{10^4} \\ = \frac{135}{10} + \frac{135}{100} + \frac{135}{1000} + \frac{135}{10000} \\ = 13.5 + 1.35 + 0.135 + 0.0135 = 14.9985.$$

$$46. \text{Given exp.} = \sqrt{\frac{4096}{10^2}} + \sqrt{\frac{4096}{10^4}} + \sqrt{\frac{4096}{10^6}} + \sqrt{\frac{4096}{10^8}} \\ = \frac{\sqrt{4096}}{10} + \frac{\sqrt{4096}}{10^2} + \frac{\sqrt{4096}}{10^3} + \frac{\sqrt{4096}}{10^4} \\ = \frac{64}{10} + \frac{64}{100} + \frac{64}{1000} + \frac{64}{10000} \\ = 6.4 + 0.64 + 0.064 + 0.0064 \\ = 7.1104 \approx 7.11.$$

$$47. \text{Given exp.} = \sqrt{1.30} + \sqrt{1300} + \sqrt{0.0130} \\ = \sqrt{\frac{130}{100}} + \sqrt{13 \times 100} + \sqrt{\frac{130}{10000}} \\ = \frac{\sqrt{130}}{10} + \sqrt{13} \times 10 + \frac{\sqrt{130}}{100} \\ = \frac{11.40}{10} + 3.605 \times 10 + \frac{11.40}{100} \\ = 1.14 + 36.05 + 0.114 = 37.304.$$

$$48. \frac{52}{x} = \sqrt{\frac{169}{289}} \\ \Leftrightarrow \frac{52}{x} = \frac{13}{17} \\ \Leftrightarrow x = \left(\frac{52 \times 17}{13} \right) = 68.$$

49. Let the missing number be x .

$$\text{Then, } x^2 = 15 \times 135 \Leftrightarrow x = \sqrt{15 \times 135} \\ = \sqrt{15^2 \times 3^2} = 15 \times 3 = 45.$$

50. Let $\frac{x}{1776} = \frac{111}{x}$. Then, x^2
 $= 111 \times 1776 = 111 \times 111 \times 16$
 $\Rightarrow x = \sqrt{(111)^2 \times (4)^2} = 111 \times 4 = 444.$
51. Let $\frac{4\frac{1}{2}}{x} = \frac{x}{32}$. Then, $x^2 = 32 \times \frac{9}{2} = 144 \Leftrightarrow x = \sqrt{144} = 12.$
52. Let $\frac{x}{\sqrt{128}} = \frac{\sqrt{162}}{x}$. Then, $x^2 = \sqrt{128 \times 162} = \sqrt{64 \times 2 \times 18 \times 9}$
 $= \sqrt{8^2 \times 6^2 \times 3^2} = 8 \times 6 \times 3 = 144.$
 $\therefore x = \sqrt{144} = 12.$
53. $\sqrt{x + \frac{x}{y}} = x \sqrt{\frac{x}{y}} \Rightarrow x + \frac{x}{y} = x^2 \cdot \frac{x}{y} \Rightarrow \frac{xy + x}{y} = \frac{x^3}{y}$
 $\Rightarrow xy + x = x^3$
 $\Rightarrow y + 1 = x^2$
 $\Rightarrow y = x^2 - 1.$
54. $n^2 = (25)^{64} \times (64)^{25} = (5^2)^{64} \times (2^6)^{25}$
 $= 5^{128} \times 2^{150} = 5^{128} \times 2^{128} \times 2^{22}$
 $\Rightarrow n = 5^{64} \times 2^{64} \times 2^{11} = (5 \times 2)^{64} \times 2^{11} = 10^{64} \times 2048.$
 $\therefore$ Sum of digits of $n = 2 + 0 + 4 + 8 = 14.$
55. $\frac{0.13}{p^2} = 13$
 $\Leftrightarrow p^2 = \frac{0.13}{13} = \frac{1}{100}$
 $\Leftrightarrow p = \sqrt{\frac{1}{100}} = \frac{1}{10} = 0.1.$
56. Let the required number be x . Then, $\frac{x}{\sqrt{0.25}} = 25$
 $\Leftrightarrow \frac{x}{0.5} = 25$
 $\Leftrightarrow x = 25 \times 0.5 = 12.5.$
57. $\sqrt{3^n} = 729 = 3^6 \Leftrightarrow (\sqrt{3^n})^2 = (3^6)^2 \Leftrightarrow 3^n = 3^{12} \Leftrightarrow n = 12.$
58. $\sqrt{18 \times 14 \times x} = 84 \Leftrightarrow 18 \times 14 \times x = 84 \times 84$
 $\Leftrightarrow x = \frac{84 \times 84}{18 \times 14} = 28.$
59. Let $28\sqrt{x} + 1426 = 3 \times 718.$
 Then, $28\sqrt{x} = 2154 - 1426 \Leftrightarrow 28\sqrt{x} = 728 \Leftrightarrow \sqrt{x} = 26$
 $\Leftrightarrow x = (26)^2 = 676.$
60. Let $\sqrt{\frac{x}{169}} = \frac{54}{39}$. Then, $\frac{\sqrt{x}}{13} = \frac{54}{39} \Leftrightarrow \sqrt{x} = \left(\frac{54}{39} \times 13\right) = 18$
 $\Leftrightarrow x = (18)^2 = 324.$
61. $\frac{\sqrt{x}}{\sqrt{441}} = 0.02 \Leftrightarrow \frac{\sqrt{x}}{21} = 0.02$
 $\Leftrightarrow \sqrt{x} = 0.02 \times 21 = 0.42 \Leftrightarrow x$
 $= (0.42)^2 = 0.1764.$

62. Let $\sqrt{\frac{.0196}{x}} = 0.2$. Then, $\frac{.0196}{x} = 0.04$
 $\Leftrightarrow x = \frac{.0196}{.04} = \frac{1.96}{4} = .49.$
63. Let $\sqrt{0.0169 \times x} = 1.3$. Then, $0.0169x = (1.3)^2 = 1.69$
 $\Rightarrow x = \frac{1.69}{0.0169} = 100.$
64. $37 + \sqrt{.0615 + x} = 37.25 \Leftrightarrow \sqrt{.0615 + x} = 0.25$
 $\Leftrightarrow .0615 + x = (0.25)^2 = 0.0625$
 $\Leftrightarrow x = .001 = \frac{1}{10^3} = 10^{-3}.$
65. $\sqrt{(x-1)(y+2)} = 7 \Rightarrow (x-1)(y+2) = (7)^2 \Rightarrow (x-1) = 7$
 and $(y+2) = 7 \Rightarrow x = 8$ and $y = 5.$
66. $\frac{\sqrt{a}}{\sqrt{b}} = \frac{.004 \times .4}{\sqrt{.04 \times .4}} \Rightarrow \frac{a}{b} = \frac{.004 \times .4 \times .004 \times .4}{.04 \times .4} = \frac{.0000064}{.04}$
 $\Rightarrow \frac{a}{b} = \frac{.00064}{4} = .00016 = \frac{16}{10^5} = 16 \times 10^{-5}.$
67. Let the number be x . Then,
 $\frac{3}{5}x^2 = 126.15 \Leftrightarrow x^2 = \left(126.15 \times \frac{5}{3}\right) = 210.25$
 $\Leftrightarrow x = \sqrt{210.25} = 14.5.$
68. $\sqrt{\frac{0.361}{0.00169}} = \sqrt{\frac{0.36100}{0.00169}} = \sqrt{\frac{36100}{169}} = \frac{190}{13}.$
69. $\sqrt{\frac{48.4}{0.289}} = \sqrt{\frac{48.400}{0.289}} = \sqrt{\frac{48400}{289}} = \frac{220}{17} = 12\frac{16}{17}.$
70. $\sqrt{1 + \frac{x}{169}} = \frac{14}{13} \Rightarrow 1 + \frac{x}{169} = \frac{196}{169}$
 $\Rightarrow \frac{x}{169} = \left(\frac{196}{169} - 1\right) = \frac{27}{169} \Rightarrow x = 27.$
71. $\sqrt{1 + \frac{55}{729}} = 1 + \frac{x}{27} \Rightarrow \sqrt{\frac{784}{729}} = \frac{27+x}{27}$
 $\Rightarrow \frac{28}{27} = \frac{27+x}{27} \Rightarrow 27+x = 28 \Rightarrow x = 1.$
72. $\sqrt{\frac{4}{3}} - \sqrt{\frac{3}{4}} = \frac{\sqrt{4}}{\sqrt{3}} - \frac{\sqrt{3}}{\sqrt{4}} = \frac{\sqrt{4} \times \sqrt{4} - \sqrt{3} \times \sqrt{3}}{\sqrt{12}} = \frac{4-3}{2\sqrt{3}} = \frac{1}{2\sqrt{3}}.$
- 73.
- | | |
|------|-------------------|
| 1 | 2.00 00 00 (1.414 |
| | 1 |
| 24 | 100 |
| | 96 |
| 281 | 400 |
| | 281 |
| 2824 | 11900 |
| | 11296 |
- $\therefore \sqrt{2} = 1.414$
74. $2\sqrt{27} - \sqrt{75} + \sqrt{12} = 2\sqrt{9 \times 3} - \sqrt{25 \times 3} + \sqrt{4 \times 3}$
 $= 6\sqrt{3} - 5\sqrt{3} + 2\sqrt{3} = 3\sqrt{3}.$

$$\begin{aligned}
 75. (\sqrt{12} + \sqrt{18}) - (\sqrt{3} + \sqrt{2}) &= (\sqrt{4 \times 3} + \sqrt{9 \times 2}) - (\sqrt{3} + \sqrt{2}) \\
 &= (2\sqrt{3} + 3\sqrt{2}) - (\sqrt{3} + \sqrt{2}) \\
 &= (2\sqrt{3} - \sqrt{3}) + (3\sqrt{2} - \sqrt{2}) = \sqrt{3} + 2\sqrt{2}.
 \end{aligned}$$

$$76. \frac{\sqrt{24} + \sqrt{216}}{\sqrt{96}} = \frac{\sqrt{4 \times 6} + \sqrt{36 \times 6}}{\sqrt{16 \times 6}} = \frac{2\sqrt{6} + 6\sqrt{6}}{4\sqrt{6}} = \frac{8\sqrt{6}}{4\sqrt{6}} = 2.$$

$$\begin{aligned}
 77. \frac{\sqrt{80} - \sqrt{112}}{\sqrt{45} - \sqrt{63}} &= \frac{\sqrt{16 \times 5} - \sqrt{16 \times 7}}{\sqrt{9 \times 5} - \sqrt{9 \times 7}} \\
 &= \frac{4\sqrt{5} - 4\sqrt{7}}{3\sqrt{5} - 3\sqrt{7}} = \frac{4(\sqrt{5} - \sqrt{7})}{3(\sqrt{5} - \sqrt{7})} = \frac{4}{3} = 1\frac{1}{3}.
 \end{aligned}$$

$$\begin{aligned}
 78. 3\sqrt{5} + \sqrt{125} &= 17.88 \Rightarrow 3\sqrt{5} + \sqrt{25 \times 5} = 17.88 \\
 \Rightarrow 3\sqrt{5} + 5\sqrt{5} &= 17.88 \Rightarrow 8\sqrt{5} = 17.88 \Rightarrow \sqrt{5} = 2.235. \\
 \therefore \sqrt{80} + 6\sqrt{5} &= \sqrt{16 \times 5} + 6\sqrt{5} = 4\sqrt{5} + 6\sqrt{5} \\
 &= 10\sqrt{5} = (10 \times 2.235) = 22.35.
 \end{aligned}$$

$$79. \sqrt{50} \times \sqrt{98} = \sqrt{50 \times 98} = \sqrt{4900} = 70.$$

$$\begin{aligned}
 80. \text{Given exp.} &= \sqrt{4 \times 2} + 2\sqrt{16 \times 2} - 3\sqrt{64 \times 2} + 4\sqrt{25 \times 2} \\
 &= 2\sqrt{2} + 8\sqrt{2} - 24\sqrt{2} + 20\sqrt{2} = 6\sqrt{2} = 6 \times 1.414 = 8.484.
 \end{aligned}$$

$$\begin{aligned}
 81. \text{Given exp.} &= \frac{3\sqrt{12}}{2\sqrt{28}} \times \frac{\sqrt{98}}{2\sqrt{21}} = \frac{3\sqrt{4 \times 3}}{2\sqrt{4 \times 7}} \times \frac{\sqrt{49 \times 2}}{2\sqrt{21}} \\
 &= \frac{6\sqrt{3}}{4\sqrt{7}} \times \frac{7\sqrt{2}}{2\sqrt{21}} = \frac{21\sqrt{6}}{4\sqrt{7 \times 21}} = \frac{21\sqrt{6}}{28\sqrt{3}} \\
 &= \frac{3}{4}\sqrt{2} = \frac{3}{4} \times 1.414 = 3 \times 0.3535 = 1.0605.
 \end{aligned}$$

$$\begin{aligned}
 82. \text{Given exp.} &= \sqrt{\frac{11025}{100}} \times \sqrt{\frac{1}{100}} \div \sqrt{\frac{25}{10000}} - \sqrt{\frac{42025}{100}} \\
 &= \frac{105}{10} \times \frac{1}{10} \div \frac{5}{100} - \frac{205}{10} \\
 &= \frac{105}{100} \times \frac{100}{5} - \frac{205}{10} \\
 &= 21 - \frac{205}{10} = \frac{5}{10} = \frac{1}{2} = 0.50.
 \end{aligned}$$

83. Sum of decimal places in the numerator and denominator under the radical sign being the same, we remove the decimal.

$$\therefore \text{Given exp.} = \sqrt{\frac{81 \times 484}{64 \times 625}} = \frac{9 \times 22}{8 \times 25} = 0.99.$$

$$84. \text{Given exp.} = \sqrt{\frac{204 \times 42}{7 \times 34}} = \sqrt{36} = 6.$$

$$\begin{aligned}
 85. \text{Given exp.} &= \sqrt{\frac{81 \times 324 \times 4624}{15625 \times 289 \times 729 \times 64}} \\
 &= \frac{9 \times 18 \times 68}{125 \times 17 \times 27 \times 8} = \frac{3}{125} = 0.024.
 \end{aligned}$$

$$\begin{aligned}
 86. \text{Given exp.} &= \sqrt{\frac{9.5 \times .08500}{.19 \times .0017}} = \sqrt{\frac{95 \times 8500}{19 \times 17}} \\
 &= \sqrt{5 \times 500} = \sqrt{2500} = 50.
 \end{aligned}$$

$$\begin{aligned}
 87. \text{Given exp.} &= \sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{\left(\frac{0.03}{10}\right)^2 + \left(\frac{0.21}{10}\right)^2 + \left(\frac{0.065}{10}\right)^2}} \\
 &= \sqrt{\frac{100[(0.03)^2 + (0.21)^2 + (0.065)^2]}{(0.03)^2 + (0.21)^2 + (0.065)^2}} = \sqrt{100} = 10.
 \end{aligned}$$

$$\begin{aligned}
 88. \sqrt{(7 + 3\sqrt{5})(7 - 3\sqrt{5})} &= \sqrt{(7)^2 - (3\sqrt{5})^2} \\
 &= \sqrt{49 - 45} = \sqrt{4} = 2.
 \end{aligned}$$

$$\begin{aligned}
 89. \left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)^2 &= (\sqrt{3})^2 + \left(\frac{1}{\sqrt{3}}\right)^2 - 2 \times \sqrt{3} \times \frac{1}{\sqrt{3}} \\
 &= 3 + \frac{1}{3} - 2 = 1 + \frac{1}{3} = \frac{4}{3}.
 \end{aligned}$$

$$\begin{aligned}
 90. \sqrt{4a^2 - 4a + 1 + 3a} \\
 &= \sqrt{(1)^2 + (2a)^2 - 2 \times 1 \times 2a + 3a} = \sqrt{(1 - 2a)^2 + 3a} \\
 &= (1 - 2a) + 3a = (1 + a) = (1 + 0.1039) = 1.1039.
 \end{aligned}$$

$$\begin{aligned}
 91. \sqrt{\frac{(0.75)^3}{(1 - 0.75)} + [0.75 + (0.75)^2 + 1]} \\
 &= \sqrt{\frac{(0.75)^3 + (1 - 0.75)[(1)^2 + (0.75)^2 + 1 \times 0.75]}{1 - 0.75}} \\
 &= \sqrt{\frac{(0.75)^3 + [(1)^3 - (0.75)^3]}{1 - 0.75}} \\
 &= \sqrt{\frac{1}{0.25}} = \sqrt{\frac{100}{25}} = \sqrt{4} = 2.
 \end{aligned}$$

$$\begin{aligned}
 92. a + b + c &= 27\sqrt{29} \Rightarrow 2c + \frac{3}{2}c + c = 27\sqrt{29} \\
 \Rightarrow \frac{9}{2}c &= 27\sqrt{29} \Rightarrow c = 6\sqrt{29}.
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{a^2 + b^2 + c^2} &= \sqrt{(a + b + c)^2 - 2(ab + bc + ca)} \\
 &= \sqrt{(27\sqrt{29})^2 - 2\left(2c \times \frac{3}{2}c + \frac{3}{2}c \times c + c \times 2c\right)} \\
 &= \sqrt{(729 \times 29) - 2\left(3c^2 + \frac{3}{2}c^2 + 2c^2\right)} \\
 &= \sqrt{(729 \times 29) - 2 \times \frac{13}{2}c^2} \\
 &= \sqrt{(729 \times 29) - 13 \times (6\sqrt{29})^2} = \sqrt{29(729 - 468)} \\
 &= \sqrt{29 \times 261} = \sqrt{29 \times 29 \times 9} = 29 \times 3 = 87.
 \end{aligned}$$

$$93. \sqrt{0.\bar{4}} = \sqrt{\frac{4}{9}} = \frac{2}{3} = 0.666..... = 0.\bar{6}.$$

$$94. \sqrt{0.09} = \sqrt{\frac{9}{100}} = \frac{3}{10} = 0.3, \text{ which is rational.}$$

95.

$$\begin{array}{r} 6 \overline{) 0.40 \ 00 \ 00} \ (.63 \\ \underline{36} \\ 123 \underline{4 \ 00} \\ 3 \ 69 \end{array}$$

$$\therefore \sqrt{0.4} = 0.63$$

96.

$$\begin{array}{r} 4 \overline{) 0.20 \ 00 \ 00} \ (.447 \\ \underline{16} \\ 84 \underline{4 \ 00} \\ 3 \ 36 \\ 887 \underline{64 \ 00} \\ 62 \ 09 \end{array}$$

$$\therefore \sqrt{0.2} = 0.447$$

97.

$$\begin{array}{r} 3 \overline{) 0.12 \ 10 \ 00} \ (.347 \\ \underline{9} \\ 64 \underline{3 \ 10} \\ 2 \ 56 \\ 687 \underline{54 \ 00} \\ 48 \ 00 \end{array}$$

$$\therefore \sqrt{0.121} = 0.347$$

98.

$$\begin{array}{r} 2 \overline{) 0.06 \ 40 \ 00} \ (.252 \\ \underline{4} \\ 45 \underline{2 \ 40} \\ 2 \ 25 \\ 502 \underline{1500} \\ 1004 \end{array}$$

$$\therefore \sqrt{0.064} = 0.252$$

99. $\sqrt{\frac{0.16}{0.4}} = \sqrt{\frac{0.16}{0.40}} = \sqrt{\frac{16}{40}} = \sqrt{\frac{4}{10}} = \sqrt{0.4} = 0.63.$

100. $\frac{1+\sqrt{0.01}}{1-\sqrt{0.1}} = \frac{1+0.1}{1-0.316} = \frac{1.1}{0.684} = \frac{1100}{684} = 1.6.$

$$\begin{array}{r} 3 \overline{) 0.10 \ 00 \ 00} \ (.0316 \\ \underline{9} \\ 61 \underline{1 \ 00} \\ 61 \\ 626 \underline{39 \ 00} \\ 37 \ 56 \end{array}$$

101. $\sqrt{535.9225} = \sqrt{\frac{5359225}{10000}} = \frac{2315}{100} = 23.15.$

$$\begin{array}{r} 2 \overline{) 5 \ 35 \ 92 \ 25} \ (2315 \\ \underline{4} \\ 43 \underline{1 \ 35} \\ 1 \ 29 \\ 461 \underline{6 \ 92} \\ 4 \ 61 \\ 4625 \underline{231 \ 25} \\ 231 \ 25 \\ \times \end{array}$$

102. $\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5} = \frac{2.236}{5} = 0.447.$

103. $\sqrt{\frac{8}{3}} = \sqrt{\frac{8 \times 3}{3 \times 3}} = \frac{\sqrt{24}}{3} = \frac{4.899}{3} = 1.633.$

104. $\frac{3\sqrt{2}}{2\sqrt{3}} = \frac{3\sqrt{2}}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{3\sqrt{6}}{2 \times 3} = \frac{\sqrt{6}}{2} = \frac{2.449}{2} = 1.2245.$

105. $\frac{\sqrt{5}}{2} - \frac{10}{\sqrt{5}} + \sqrt{125} = \frac{(\sqrt{5})^2 - 20 + 2\sqrt{5} \times 5\sqrt{5}}{2\sqrt{5}}$
 $= \frac{5 - 20 + 50}{2\sqrt{5}} = \frac{35}{2\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{35\sqrt{5}}{10} = \frac{7}{2} \times 2.236$
 $= 7 \times 1.118 = 7.826.$

106. Clearly, $a * b = \sqrt{a^2 + b^2}$. $\therefore 5 * 12$
 $= \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$

107. Let the missing digit be x .

$$\begin{array}{r} 1 \overline{) 1 \ 53 \ 7x} \ (124 \\ \underline{1} \\ 22 \underline{53} \\ 44 \\ 244 \underline{9 \ 7x} \\ 9 \ 76 \\ \times \end{array}$$

Then, $x = 6$.

108. $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$
 $= 7 \times 2 \times 3 \times 5 \times 2^2 \times 3 \times 2$
 $= 2^4 \times 3^2 \times 5 \times 7.$

Thus, the smallest perfect square number which is divisible by $7!$ is

$$(2^4 \times 3^2 \times 5 \times 7) \times (5 \times 7)$$

$$= 5040 \times 35 = 176400.$$

109. L.C.M. of 3, 4, 5, 6, 8 is 120. Now, 120
 $= 2 \times 2 \times 2 \times 3 \times 5.$

To make it a perfect square, it must be multiplied by $2 \times 3 \times 5$.

So, required number $= 2^2 \times 2^2 \times 3^2 \times 5^2 = 3600.$

110. L.C.M. of 21, 36, 66
 $= 2772$. Now, 2772
 $= 2 \times 2 \times 3 \times 3 \times 7 \times 11.$

To make it a perfect square, it must be multiplied by 7×11 .

So, required number $= 2^2 \times 3^2 \times 7^2 \times 11^2 = 213444.$

111. $294 = 7 \times 7 \times 2 \times 3.$

To make it a perfect square, it must be multiplied by 2×3 i.e., 6.

$\therefore$ Required number = 6.

112. $5808 = 2 \times 2 \times 2 \times 2 \times 3 \times 11 \times 11 = 2^2 \times 2^2 \times 3 \times 11^2.$

To make it a perfect square, it must be multiplied by 3.

113. $1470 = 7 \times 7 \times 5 \times 6$. To make it a perfect square, it must be divided by 5×6 , i.e., 30.

$$\begin{array}{r}
 7 \overline{54 \ 91 \ 62} \ (741) \\
 \underline{49} \\
 144 \overline{5 \ 91} \\
 \underline{5 \ 76} \\
 1481 \overline{15 \ 62} \\
 \underline{14 \ 81} \\
 81
 \end{array}$$

∴ Required number to be subtracted = 81.

$$115. 0.000326 = \frac{326}{10^6}$$

$$\begin{array}{r}
 1 \overline{3 \ 26} \ (18) \\
 \underline{1} \\
 28 \overline{2 \ 26} \\
 \underline{2 \ 24} \\
 2
 \end{array}$$

∴ Required number to be subtracted = $\frac{2}{10^6} = 0.000002$.

$$\begin{array}{r}
 8 \overline{77 \ 00} \ (87) \\
 \underline{64} \\
 167 \overline{13 \ 00} \\
 \underline{11 \ 69} \\
 1 \ 31
 \end{array}$$

∴ Number to be added
 $= (88)^2 - 7700 = 7744 - 7700 = 44$.

$$\begin{array}{r}
 8 \overline{68 \ 06 \ 21} \ (824) \\
 \underline{64} \\
 162 \overline{4 \ 06} \\
 \underline{3 \ 24} \\
 1644 \overline{82 \ 21} \\
 \underline{65 \ 76} \\
 16 \ 45
 \end{array}$$

∴ Number to be added = $(825)^2 - 680621$
 $= 680625 - 680621 = 4$.

118. Greatest number of four digits is 9999.

$$\begin{array}{r}
 9 \overline{99 \ 99} \ (99) \\
 \underline{81} \\
 189 \overline{18 \ 99} \\
 \underline{17 \ 01} \\
 1 \ 98
 \end{array}$$

∴ Reqd. number = $(9999 - 198) = 9801$.

119. Least number of 4 digits is 1000.

$$\begin{array}{r}
 3 \overline{10 \ 00} \ (31) \\
 \underline{9} \\
 61 \overline{1 \ 00} \\
 \underline{61} \\
 39
 \end{array}$$

∴ $(31)^2 < 1000 < (32)^2$.

Hence, required number = $(32)^2 = 1024$.

120. Let the 18 consecutive natural numbers be $x, (x+1), (x+2), (x+3), \dots, (x+17)$. Then, $x + (x+1) + (x+2) + \dots + (x+17) = 18x + (1+2+3+\dots+17) = 18x + 153$.

Putting $x = 1, 2, 3, 4, \dots$ we find that the smallest value of x for which $(18x + 153)$ becomes a perfect square is $x = 4$.

∴ Required value = $18 \times 4 + 153 = 72 + 153 = 225$.

121. Given expression

$$\begin{aligned}
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2^2 - \left(\sqrt{2+\sqrt{2+\sqrt{3}}}\right)^2} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{4 - (2+\sqrt{2+\sqrt{3}})} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2+\sqrt{2+\sqrt{3}}} \cdot \sqrt{2-\sqrt{2+\sqrt{3}}} \\
 &= \sqrt{2+\sqrt{3}} \cdot \sqrt{2^2 - \left(\sqrt{2+\sqrt{3}}\right)^2} = \sqrt{2+\sqrt{3}} \cdot \sqrt{2-\sqrt{3}} \\
 &= \sqrt{2^2 - (\sqrt{3})^2} = \sqrt{4-3} = \sqrt{1} = 1.
 \end{aligned}$$

$$\begin{aligned}
 122. \frac{1}{(\sqrt{5}-\sqrt{3})} &= \frac{1}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} = \frac{(\sqrt{5}+\sqrt{3})}{(5-3)} \\
 &= \frac{(2.2361+1.7321)}{2} = \frac{3.9682}{2} = 1.9841.
 \end{aligned}$$

$$\begin{aligned}
 123. \text{ Given expression } &= \frac{1}{(\sqrt{9}-\sqrt{8})} \times \frac{(\sqrt{9}+\sqrt{8})}{(\sqrt{9}+\sqrt{8})} - \frac{1}{(\sqrt{8}-\sqrt{7})} \\
 &\quad \times \frac{(\sqrt{8}+\sqrt{7})}{(\sqrt{8}+\sqrt{7})} + \frac{1}{(\sqrt{7}-\sqrt{6})} \times \frac{(\sqrt{7}+\sqrt{6})}{(\sqrt{7}+\sqrt{6})} \\
 &\quad - \frac{1}{(\sqrt{6}-\sqrt{5})} \times \frac{(\sqrt{6}+\sqrt{5})}{(\sqrt{6}+\sqrt{5})} + \frac{1}{(\sqrt{5}-\sqrt{4})} \times \frac{(\sqrt{5}+\sqrt{4})}{(\sqrt{5}+\sqrt{4})} \\
 &= \frac{(\sqrt{9}+\sqrt{8})}{(9-8)} - \frac{(\sqrt{8}+\sqrt{7})}{(8-7)} + \frac{(\sqrt{7}+\sqrt{6})}{(7-6)} \\
 &\quad - \frac{(\sqrt{6}+\sqrt{5})}{(6-5)} + \frac{(\sqrt{5}+\sqrt{4})}{(5-4)} \\
 &= (\sqrt{9}+\sqrt{8}) - (\sqrt{8}+\sqrt{7}) + (\sqrt{7}+\sqrt{6}) - (\sqrt{6}+\sqrt{5}) \\
 &\quad + (\sqrt{5}+\sqrt{4}) = (\sqrt{9}+\sqrt{4}) = 3+2=5.
 \end{aligned}$$

$$\begin{aligned}
 124. \text{ Given exp. } &= \frac{1}{\sqrt{2}+\sqrt{1}} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} \\
 &\quad + \dots + \frac{1}{\sqrt{121}+\sqrt{120}} \\
 &= \frac{1}{\sqrt{2}+\sqrt{1}} \times \frac{\sqrt{2}-\sqrt{1}}{\sqrt{2}-\sqrt{1}} + \frac{1}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} \\
 &\quad + \frac{1}{\sqrt{4}+\sqrt{3}} \times \frac{\sqrt{4}-\sqrt{3}}{\sqrt{4}-\sqrt{3}} + \dots + \frac{1}{\sqrt{121}+\sqrt{120}} \times \frac{\sqrt{121}-\sqrt{120}}{\sqrt{121}-\sqrt{120}} \\
 &= \frac{\sqrt{2}-\sqrt{1}}{2-1} + \frac{\sqrt{3}-\sqrt{2}}{3-2} + \frac{\sqrt{4}-\sqrt{3}}{4-3} + \dots + \frac{\sqrt{121}-\sqrt{120}}{121-120} \\
 &= \sqrt{2}-\sqrt{1} + \sqrt{3}-\sqrt{2} + \sqrt{4}-\sqrt{3} + \dots + \sqrt{121}-\sqrt{120} \\
 &= -1 + \sqrt{121} = -1 + 11 = 10.
 \end{aligned}$$

$$\begin{aligned}
 125. \text{ Given exp. } &= \frac{(1+\sqrt{3})(1-\sqrt{3})-(1-\sqrt{3})+(1+\sqrt{3})}{(1+\sqrt{3})(1-\sqrt{3})} \\
 &= \frac{1-(\sqrt{3})^2-1+\sqrt{3}+1+\sqrt{3}}{1^2-(\sqrt{3})^2} \\
 &= \frac{2\sqrt{3}-2}{1-3} = \frac{2(\sqrt{3}-1)}{(-2)} = 1-\sqrt{3}.
 \end{aligned}$$

$$\begin{aligned}
 126. \text{ Given exp. } &= (2+\sqrt{2}) + \frac{1}{(2+\sqrt{2})} \times \frac{(2-\sqrt{2})}{(2-\sqrt{2})} \\
 &\quad - \frac{1}{(2-\sqrt{2})} \times \frac{(2+\sqrt{2})}{(2+\sqrt{2})} \\
 &= (2+\sqrt{2}) + \frac{(2-\sqrt{2})}{(4-2)} - \frac{(2+\sqrt{2})}{(4-2)} \\
 &= (2+\sqrt{2}) + \frac{1}{2}(2-\sqrt{2}) - \frac{1}{2}(2+\sqrt{2}) = 2.
 \end{aligned}$$

$$\begin{aligned}
 127. \text{ Given exp. } &= \frac{1}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} + \frac{2}{(3+\sqrt{5})} \\
 &\quad \times \frac{(3-\sqrt{5})}{(3-\sqrt{5})} - \frac{3}{(3+\sqrt{3})} \times \frac{(3-\sqrt{3})}{(3-\sqrt{3})} \\
 &= \frac{(\sqrt{5}-\sqrt{3})}{(5-3)} + \frac{2(3-\sqrt{5})}{(9-5)} - \frac{3(3-\sqrt{3})}{(9-3)} \\
 &= \frac{\sqrt{5}-\sqrt{3}}{2} + \frac{2(3-\sqrt{5})}{4} - \frac{3(3-\sqrt{3})}{6} \\
 &= \frac{6(\sqrt{5}-\sqrt{3})+6(3-\sqrt{5})-6(3-\sqrt{3})}{12} = 0.
 \end{aligned}$$

$$\begin{aligned}
 128. \frac{7}{(3+\sqrt{2})} &= \frac{7}{(3+\sqrt{2})} \times \frac{(3-\sqrt{2})}{(3-\sqrt{2})} = \frac{7(3-\sqrt{2})}{(9-2)} \\
 &= (3-\sqrt{2}) = (3-1.4142) = 1.5858.
 \end{aligned}$$

$$\begin{aligned}
 129. \text{ Given exp. } &= \frac{3\sqrt{2}}{(\sqrt{6}-\sqrt{3})} \times \frac{(\sqrt{6}+\sqrt{3})}{(\sqrt{6}+\sqrt{3})} - \frac{4\sqrt{3}}{(\sqrt{6}-\sqrt{2})} \\
 &\quad \times \frac{(\sqrt{6}+\sqrt{2})}{(\sqrt{6}+\sqrt{2})} - \frac{6}{2(\sqrt{2}-\sqrt{3})} \\
 &= \frac{3\sqrt{2}(\sqrt{6}+\sqrt{3})}{(6-3)} - \frac{4\sqrt{3}(\sqrt{6}+\sqrt{2})}{(6-2)} \\
 &\quad + \frac{3}{(\sqrt{3}-\sqrt{2})} \times \frac{(\sqrt{3}+\sqrt{2})}{(\sqrt{3}+\sqrt{2})} \\
 &= \sqrt{2}(\sqrt{6}+\sqrt{3}) - \sqrt{3}(\sqrt{6}+\sqrt{2}) + 3(\sqrt{3}+\sqrt{2}) \\
 &= \sqrt{12} + \sqrt{6} - \sqrt{18} - \sqrt{6} + 3\sqrt{3} + 3\sqrt{2} \\
 &= 2\sqrt{3} - 3\sqrt{2} + 3\sqrt{3} + 3\sqrt{2} = 5\sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 130. \frac{\sqrt{7}+\sqrt{5}}{\sqrt{7}-\sqrt{5}} &= \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}-\sqrt{5})} \times \frac{(\sqrt{7}+\sqrt{5})}{(\sqrt{7}+\sqrt{5})} = \frac{(\sqrt{7}+\sqrt{5})^2}{(7-5)} \\
 &= \frac{7+5+2\sqrt{35}}{2} = \frac{12+2\sqrt{35}}{2} = 6+\sqrt{35}.
 \end{aligned}$$

$$131. a+b\sqrt{3} = \frac{(5+2\sqrt{3})}{(7+4\sqrt{3})} \times \frac{(7-4\sqrt{3})}{(7-4\sqrt{3})} = \frac{35-20\sqrt{3}+14\sqrt{3}-24}{(7)^2-(4\sqrt{3})^2}$$

$$= \frac{11-6\sqrt{3}}{49-48} = 11-6\sqrt{3}. \quad \therefore a = 11, b = -6.$$

$$\begin{aligned}
 132. \frac{\sqrt{2}-1}{\sqrt{2}+1} &= \frac{(\sqrt{2}-1)}{(\sqrt{2}+1)} \times \frac{(\sqrt{2}-1)}{(\sqrt{2}-1)} = (\sqrt{2}-1)^2. \\
 \therefore \sqrt{\frac{\sqrt{2}-1}{\sqrt{2}+1}} &= (\sqrt{2}-1) = (1.414-1) = 0.414.
 \end{aligned}$$

$$\begin{aligned}
 133. \text{ Given exp. } &= \frac{3+\sqrt{6}}{5\sqrt{3}-4\sqrt{3}-4\sqrt{2}+5\sqrt{2}} = \frac{(3+\sqrt{6})}{(\sqrt{3}+\sqrt{2})} \\
 &= \frac{(3+\sqrt{6})}{(\sqrt{3}+\sqrt{2})} \times \frac{(\sqrt{3}-\sqrt{2})}{(\sqrt{3}-\sqrt{2})} \\
 &= \frac{3\sqrt{3}-3\sqrt{2}+3\sqrt{2}-2\sqrt{3}}{(3-2)} = \sqrt{3} = 1.732.
 \end{aligned}$$

$$\begin{aligned}
 134. \text{ Given exp. } &= \frac{(2+\sqrt{3})}{(2-\sqrt{3})} \times \frac{(2+\sqrt{3})}{(2+\sqrt{3})} + \frac{(2-\sqrt{3})}{(2+\sqrt{3})} \times \frac{(2-\sqrt{3})}{(2-\sqrt{3})} \\
 &\quad + \frac{(\sqrt{3}-1)}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} \\
 &= \frac{(2+\sqrt{3})^2}{(4-3)} + \frac{(2-\sqrt{3})^2}{(4-3)} + \frac{(\sqrt{3}-1)^2}{(3-1)} \\
 &= [(2+\sqrt{3})^2 + (2-\sqrt{3})^2] + \frac{4-2\sqrt{3}}{2} \\
 &= 2(4+3) + 2 - \sqrt{3} = 16 - \sqrt{3}.
 \end{aligned}$$

$$\begin{aligned}
 135. x+y &= \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} = \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \times \frac{(\sqrt{5}+\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \\
 &\quad + \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})} \\
 &= \frac{(\sqrt{5}+\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2} + \frac{(\sqrt{5}-\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2} \\
 &= \frac{(\sqrt{5}+\sqrt{3})^2 + (\sqrt{5}-\sqrt{3})^2}{5-3} \\
 &= \frac{2[(\sqrt{5})^2 + (\sqrt{3})^2]}{2} = 5+3 = 8.
 \end{aligned}$$

$$\begin{aligned}
 136. \text{ Given exp. } &= \frac{(\sqrt{2})^2}{(\sqrt{3})^2} \cdot \frac{(2+\sqrt{3})(2-\sqrt{3})}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{2}{3} \left[\frac{2^2-(\sqrt{3})^2}{(\sqrt{3})^2-1^2} \right] \\
 &= \frac{2}{3} \left(\frac{4-3}{3-1} \right) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}.
 \end{aligned}$$

$$\begin{aligned}
 137. \text{ Given exp. } &= \frac{12}{3+(\sqrt{5}+2\sqrt{2})} \times \frac{3-(\sqrt{5}+2\sqrt{2})}{3-(\sqrt{5}+2\sqrt{2})} \\
 &= \frac{12(3-\sqrt{5}-2\sqrt{2})}{3^2-(\sqrt{5}+2\sqrt{2})^2} = \frac{12(3-\sqrt{5}-2\sqrt{2})}{9-(5+8+4\sqrt{10})} \\
 &= \frac{12(3-\sqrt{5}-2\sqrt{2})}{(-4-4\sqrt{10})} = \frac{3(\sqrt{5}+2\sqrt{2}-3)}{\sqrt{10}+1} \\
 &= \frac{3(\sqrt{5}+2\sqrt{2}-3)}{\sqrt{10}+1} \times \frac{\sqrt{10}-1}{\sqrt{10}-1}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{3\sqrt{50} - 3\sqrt{5} + 6\sqrt{20} - 6\sqrt{2} - 9\sqrt{10} + 9}{10 - 1} \\
 &= \frac{15\sqrt{2} - 3\sqrt{5} + 12\sqrt{5} - 6\sqrt{2} - 9\sqrt{10} + 9}{9} \\
 &= \frac{9\sqrt{2} + 9\sqrt{5} - 9\sqrt{10} + 9}{9} = 1 + \sqrt{2} + \sqrt{5} - \sqrt{10}.
 \end{aligned}$$

138. Given exp. = $\frac{1}{(\sqrt{2}-\sqrt{5})+\sqrt{3}} + \frac{1}{(\sqrt{2}-\sqrt{5})-\sqrt{3}}$

$$\begin{aligned}
 &= \frac{[(\sqrt{2}-\sqrt{5})-\sqrt{3}] + [(\sqrt{2}-\sqrt{5})+\sqrt{3}]}{[(\sqrt{2}-\sqrt{5})+\sqrt{3}][(\sqrt{2}-\sqrt{5})-\sqrt{3}]} \\
 &= \frac{2(\sqrt{2}-\sqrt{5})}{(\sqrt{2}-\sqrt{5})^2 - (\sqrt{3})^2} = \frac{2(\sqrt{2}-\sqrt{5})}{(2+5-2\sqrt{10})-3} \\
 &= \frac{2(\sqrt{2}-\sqrt{5})}{4-2\sqrt{10}} = \frac{\sqrt{2}-\sqrt{5}}{2-\sqrt{10}} \\
 &= \frac{\sqrt{2}-\sqrt{5}}{\sqrt{2}(\sqrt{2}-\sqrt{5})} = \frac{1}{\sqrt{2}}.
 \end{aligned}$$

139. $x + \frac{1}{x} = (7-4\sqrt{3}) + \frac{1}{(7-4\sqrt{3})} \times \frac{(7+4\sqrt{3})}{(7+4\sqrt{3})}$

$$\begin{aligned}
 &= (7-4\sqrt{3}) + \frac{(7+4\sqrt{3})}{(49-48)} \\
 &= (7-4\sqrt{3}) + (7+4\sqrt{3}) = 14.
 \end{aligned}$$

140. $x = 3 + \sqrt{8} \Rightarrow x^2 = (3 + \sqrt{8})^2 = 3^2 + (\sqrt{8})^2 + 2 \times 3 \times \sqrt{8}$

$$\begin{aligned}
 &= 9 + 8 + 6\sqrt{8} = 17 + 12\sqrt{2}.
 \end{aligned}$$

$$\begin{aligned}
 x^2 + \frac{1}{x^2} &= (17 + 12\sqrt{2}) + \frac{1}{(17 + 12\sqrt{2})} \times \frac{(17 - 12\sqrt{2})}{(17 - 12\sqrt{2})} \\
 &= (17 + 12\sqrt{2}) + \frac{(17 - 12\sqrt{2})}{289 - 288} \\
 &= (17 + 12\sqrt{2}) + (17 - 12\sqrt{2}) = 34.
 \end{aligned}$$

141. $a = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} \times \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} + \sqrt{2}}$

$$\begin{aligned}
 &= \frac{(\sqrt{3} + \sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} = \frac{3 + 2 + 2\sqrt{6}}{3 - 2} = 5 + 2\sqrt{6}.
 \end{aligned}$$

$$\begin{aligned}
 b &= \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} = \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} \\
 &= \frac{(\sqrt{3} - \sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} = \frac{3 + 2 - 2\sqrt{6}}{3 - 2} = 5 - 2\sqrt{6}.
 \end{aligned}$$

$$\begin{aligned}
 a^2 + b^2 &= (5 + 2\sqrt{6})^2 + (5 - 2\sqrt{6})^2 = 2[(5)^2 + (2\sqrt{6})^2] \\
 &= 2(25 + 24) = 2 \times 49 = 98.
 \end{aligned}$$

142. $a = \frac{(\sqrt{5}+1)}{(\sqrt{5}-1)} \times \frac{(\sqrt{5}+1)}{(\sqrt{5}+1)}$

$$\begin{aligned}
 &= \frac{(\sqrt{5}+1)^2}{(5-1)} = \frac{5+1+2\sqrt{5}}{4} = \left(\frac{3+\sqrt{5}}{2}\right).
 \end{aligned}$$

$$\begin{aligned}
 b &= \frac{(\sqrt{5}-1)}{(\sqrt{5}+1)} \times \frac{(\sqrt{5}-1)}{(\sqrt{5}-1)} = \frac{(\sqrt{5}-1)^2}{(5-1)} \\
 &= \frac{5+1-2\sqrt{5}}{4} = \left(\frac{3-\sqrt{5}}{2}\right).
 \end{aligned}$$

$$\begin{aligned}
 \therefore a^2 + b^2 &= \frac{(3+\sqrt{5})^2}{4} + \frac{(3-\sqrt{5})^2}{4} \\
 &= \frac{(3+\sqrt{5})^2 + (3-\sqrt{5})^2}{4} = \frac{2(9+5)}{4} = 7.
 \end{aligned}$$

Also, $ab = \frac{(3+\sqrt{5})}{2} \cdot \frac{(3-\sqrt{5})}{2} = \frac{(9-5)}{4} = 1.$

$$\therefore \frac{a^2 + ab + b^2}{a^2 - ab + b^2} = \frac{(a^2 + b^2) + ab}{(a^2 + b^2) - ab} = \frac{7+1}{7-1} = \frac{8}{6} = \frac{4}{3}.$$

143. $x = \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots \infty}}} \Leftrightarrow x = \sqrt{1+x}$

$$\Leftrightarrow x^2 = 1 + x \Leftrightarrow x^2 - x - 1 = 0$$

$$\Leftrightarrow x = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times (-1)}}{2}$$

$$\Leftrightarrow x = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}.$$

Hence, positive value of x is $\frac{1+\sqrt{5}}{2}.$

144. Let $x = \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$

Then, $x = \sqrt{2+x} \Leftrightarrow x^2 = 2+x \Leftrightarrow x^2 - x - 2 = 0$

$$\Leftrightarrow x^2 - 2x + x - 2 = 0$$

$$\Leftrightarrow x(x-2) + (x-2) = 0 \Leftrightarrow (x-2)(x+1) = 0$$

$$\Leftrightarrow x = 2. [\because x \neq -1]$$

145. $a = \sqrt{3 + \sqrt{3 + \sqrt{3 + \dots}}} \Leftrightarrow a = \sqrt{3+a} \Leftrightarrow a^2 = 3+a$

$$\Leftrightarrow a^2 - a - 3 = 0$$

$$\Leftrightarrow a = \frac{1 \pm \sqrt{(-1)^2 - 4 \times 1 \times (-3)}}{2}$$

$$\Leftrightarrow a = \frac{1 \pm \sqrt{1+12}}{2} = \frac{1 \pm \sqrt{13}}{2} \Leftrightarrow a = \frac{1 + \sqrt{13}}{2} [\because a > 0]$$

$$\Leftrightarrow a = \frac{1+3.6}{2} = \frac{4.6}{2} = 2.3.$$

$$\therefore 2 < a < 3.$$

146. $\frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} - \sqrt{3-x}} = 2$

$$\Leftrightarrow \frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} - \sqrt{3-x}} \times \frac{\sqrt{3+x} + \sqrt{3-x}}{\sqrt{3+x} + \sqrt{3-x}} = 2$$

$$\Leftrightarrow \frac{(\sqrt{3+x} + \sqrt{3-x})^2}{(\sqrt{3+x})^2 - (\sqrt{3-x})^2} = 2$$

$$\Leftrightarrow \frac{(3+x) + (3-x) + 2\sqrt{(3+x)(3-x)}}{(3+x) - (3-x)} = 2$$

$$\Leftrightarrow 6 + 2\sqrt{9-x^2} = 4x \Leftrightarrow 2\sqrt{9-x^2} = 4x - 6$$

$$162. \sqrt{.01} \times \sqrt[3]{.008} - .02 = \sqrt{(.1)^2} \times \sqrt[3]{(.2)^3} - .02 \\ = .1 \times .2 - .02 = .02 - .02 = 0.$$

$$163. \text{ Given exp. } = \sqrt[3]{\frac{0.008 + 0.000064}{0.064 + 0.000512}} = \sqrt[3]{\frac{0.008064}{0.064512}} \\ = \sqrt[3]{\frac{8064}{64512}} = \sqrt[3]{\frac{1}{8}} = \frac{1}{2} = 0.5.$$

$$164. \text{ Let } \sqrt[3]{3} = x.$$

$$\text{Then, } (\sqrt[3]{9} - \sqrt[3]{3} + 1) = (x^2 - x + 1) = \frac{x^3 + 1}{x + 1} = \frac{(\sqrt[3]{3})^3 + 1}{(\sqrt[3]{3} + 1)}.$$

$$\Rightarrow (\sqrt[3]{9} - \sqrt[3]{3} + 1)(\sqrt[3]{3} + 1) = (\sqrt[3]{3})^3 + 1 \\ = 3 + 1 = 4, \text{ which is rational.}$$

$$165. \text{ Clearly, } (21)^3 = 9261 \text{ and } (22)^3 = 10648.$$

So, 9261 is the largest four-digit number which is a perfect cube.

$$166. 21600 = 2^5 \times 3^3 \times 5^2.$$

To make it a perfect cube, it must be multiplied by (2×5) , i.e., 10.

$$167. 3600 = 2^3 \times 5^2 \times 3^2 \times 2.$$

To make it a perfect cube, it must be divided by $5^2 \times 3^2 \times 2$, i.e. 450.

$$168. \text{ Required number to be added} \\ = 9^3 - 710 = 729 - 710 = 19.$$

$$169. \begin{array}{r|l} 8 & \overline{79 \ 21} \\ 8 & 64 \\ \hline 169 & 1521 \\ \hline 9 & 1521 \\ & \times \end{array} \quad \begin{array}{l} 89 \\ \\ \\ \end{array} \\ \Rightarrow \sqrt{7921} = 89$$

$$170. \text{ Given } ? = \sqrt[4]{(625)^3} \\ = (625)^{\frac{3}{4}} = (5 \times 5 \times 5 \times 5)^{\frac{3}{4}} \\ = (5^4)^{\frac{3}{4}} = 5^3 = 125$$

$$171. \text{ Given } \sqrt{y} = 4x$$

$$\Rightarrow y = (4x)^2 = 16x^2$$

$$\Rightarrow y = 16x^2$$

$$\therefore \frac{x^2}{y} = \frac{1}{16}$$

$$172. \text{ Let the number be } x.$$

$$\text{Given: } \sqrt{2025.11} \times \sqrt{256.04} + \sqrt{399.95} \times \sqrt{7} = 33.98 \times 40$$

$$\sqrt{2025} \times \sqrt{256} + \sqrt{400} \times \sqrt{7} = 34 \times 40$$

$$45 \times 16 + 20 \times \sqrt{7} = 1360$$

$$20 \times \sqrt{7} = 1360 - 720 = 640$$

$$\sqrt{7} = \frac{640}{20} = 32$$

$$\Rightarrow ? = (32)^2$$

$$\therefore ? = 32 \times 32 = 1024$$

Hence, the number is 1024.

$$173. \frac{1}{\sqrt{28}} = \frac{1}{\sqrt{2 \times 2 \times 7}} = \frac{1}{2\sqrt{7}} \\ = \frac{\sqrt{7}}{2\sqrt{7} \times \sqrt{7}} = \frac{\sqrt{7}}{14} \\ = \frac{2.645}{14} = 0.189$$

$$174. = \left(\sqrt{\frac{25}{9}} - \sqrt{\frac{64}{81}} \right) \div \sqrt{\frac{16}{324}} \\ = \left(\frac{5}{3} - \frac{8}{9} \right) \div \frac{4}{18}$$

LCM of 3 and 9 is 9.

$$= \left(\frac{15-8}{9} \right) \div \frac{2}{9} = \frac{7}{9} \div \frac{2}{9} \\ = \frac{7}{9} \times \frac{9}{2} = \frac{7}{2} = 3.5$$

$$175. \text{ Given: } 1728 \div \sqrt[3]{262144} \times ? - 288 = 4491$$

$$\Rightarrow 1728 \div \sqrt[3]{64 \times 64 \times 64} \times ?$$

$$= 4491 + 288$$

$$\Rightarrow \frac{1728}{64} \times ? = 4779$$

$$\Rightarrow 27 \times ? = 4779$$

$$\Rightarrow ? = \frac{4779}{27} = 177$$

$$176. \text{ Let the number be } a.$$

$$(\sqrt{7} + 11)^2 \\ = a^{\frac{1}{3}} + 2\sqrt{847} + 122 \\ \Rightarrow 7 + 121 + 22\sqrt{7} \\ = a^{\frac{1}{3}} + 22\sqrt{7} + 122 \\ \Rightarrow 128 - 122 = a^{\frac{1}{3}}$$

$$\Rightarrow a^{\frac{1}{3}} = 6$$

$$\Rightarrow a = (6)^3 = 216$$

Hence, the number is 216.

$$177. \text{ Let the number be } x.$$

$$x - \sqrt{784} = 6 \times \sqrt{324}$$

$$x - 28 = 6 \times 18$$

$$x - 28 = 108$$

$$x = 108 + 28 = 136$$

$$178. \text{ Let the number be } x.$$

$$\text{Given } \sqrt{2116} - \sqrt{1600} = \sqrt{(x)}$$

$$46 - 40 = \sqrt{(x)}$$

$$\Rightarrow 6 = \sqrt{x}$$

$$\Rightarrow x = (6)^2 = 36$$

Hence, the number is 36.

179. Let the number be x .

$$\sqrt{(27 \div 5 \times x) + 15} = 5.4 \div 6 + 0.3$$

$$\Rightarrow \sqrt{\frac{(27 \div 5 \times x)}{15}} = \frac{5.4}{6} + 0.3$$

$$\Rightarrow \sqrt{\frac{27 \times x}{5 \times 15}} = \frac{5.4}{6} + 0.3$$

$$\Rightarrow \sqrt{\frac{9 \times x}{5 \times 5}} = 0.9 + 0.3$$

$$\Rightarrow \frac{3}{5} \sqrt{x} = 1.2$$

$$\Rightarrow \sqrt{x} = \frac{1.2 \times 5}{3} = 2 \Rightarrow x = (2)^2$$

$$\Rightarrow x = 2 \times 2 = 4^2$$

180. Given $\sqrt{24 \div 0.5 + 1} + \sqrt{18 \div 0.6 + 6}$

$$= \sqrt{24 \times \frac{1}{0.5} + 1} + \sqrt{18 \times \frac{1}{0.6} + 6}$$

$$= \sqrt{24 \times \frac{10}{5} + 1} + \sqrt{\frac{18 \times 10}{6} + 6}$$

$$= \sqrt{48 + 1} + \sqrt{30 + 6}$$

$$\sqrt{49} + \sqrt{36} = 7 + 6 = 13$$

181. Given $(\sqrt{63} + \sqrt{252}) \times (\sqrt{175} + \sqrt{28})$

$$= (\sqrt{63} + \sqrt{4 \times 63}) \times (\sqrt{25 \times 7} + \sqrt{4 \times 7})$$

$$= (\sqrt{7 \times 9} + \sqrt{4 \times 9 \times 7}) \times (\sqrt{5 \times 5 \times 7} + \sqrt{2 \times 2 \times 7})$$

$$= (3\sqrt{7} + 6\sqrt{7}) \times (5\sqrt{7} + 2\sqrt{7})$$

$$= 9\sqrt{7} \times 7\sqrt{7} = 441$$

182. Given $9x^2 + 25 - 30x$

$$\text{We have to find } \sqrt{9x^2 + 25 - 30x}$$

$$= \sqrt{(3x)^2 - 2 \cdot 3x \cdot 5 + (-5)^2} \quad \left\{ \because a^2 - 2ab + b^2 = (a - b)^2 \right\}$$

$$= \sqrt{(3x - 5)^2} = 3x - 5$$

183. $\sqrt{\frac{3}{11}} = \sqrt{\frac{3 \times 11}{11 \times 11}} = \frac{\sqrt{33}}{11}$

$$= \frac{5.745}{11} = 0.5223$$

184. Let the number be x .

$$\text{Given } \sqrt{x} + 14 = \sqrt{2601}$$

$$\text{or, } \sqrt{x} = 51 - 14 = 37$$

$$\text{or } x = 37^2 = 1369$$

185. $a = \frac{\sqrt{3}}{2}$ (given)

$$\therefore \sqrt{1+a} + \sqrt{1-a}$$

$$= \sqrt{1 + \frac{\sqrt{3}}{2}} + \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2+\sqrt{3}}{2}} + \sqrt{\frac{2-\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2(2+\sqrt{3})}{4}} + \sqrt{\frac{2(2-\sqrt{3})}{4}}$$

$$= \sqrt{\frac{4+2\sqrt{3}}{4}} + \sqrt{\frac{4-2\sqrt{3}}{4}}$$

$$= \sqrt{\frac{3+1+2 \times \sqrt{3} \times 1}{2}} + \sqrt{\frac{3+1-2 \times \sqrt{3} \times 1}{2}}$$

$$\left\{ \begin{array}{l} \sqrt{3}^2 + (1)^2 - 2 \times \sqrt{3} \times 1 = (\sqrt{3} - 1)^2 \\ \therefore (\sqrt{3})^2 + (1)^2 + 2 \times \sqrt{3} \times 1 = (\sqrt{3} + 1)^2 \\ a^2 + b^2 + 2ab = (a + b)^2 \\ a^2 + b^2 - 2ab = (a - b)^2 \end{array} \right\}$$

$$= \frac{\sqrt{(\sqrt{3}+1)^2}}{2} + \frac{\sqrt{(\sqrt{3}-1)^2}}{2}$$

$$= \frac{\sqrt{3}+1+\sqrt{3}-1}{2}$$

$$= \frac{2\sqrt{3}}{2} = \sqrt{3}$$

186. Given $\frac{5+\sqrt{10}}{5\sqrt{5}-2\sqrt{20}-\sqrt{32}+\sqrt{50}}$

$$= \frac{5+\sqrt{10}}{5\sqrt{5}-2 \times 2\sqrt{5}-2 \times 2\sqrt{2}+5\sqrt{2}}$$

$$= \frac{5+\sqrt{10}}{5\sqrt{5}-4\sqrt{5}-4\sqrt{2}+5\sqrt{2}}$$

$$= \frac{5+\sqrt{10}}{\sqrt{5}+\sqrt{2}} = \frac{\sqrt{5}(\sqrt{5}+\sqrt{2})}{\sqrt{5}+\sqrt{2}} = \sqrt{5}$$

187. Given: $\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$

$$= \frac{(0.75)^2 \times 0.75}{0.25} + [0.75 + 0.5625 + 1]$$

$$= 0.5625 \times 3 + [0.75 + 0.5625 + 1]$$

$$= 1.6875 + 2.3125 = 4$$

$$\text{Square root of } 4 = 2$$

188. Given $\sqrt{10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15}}$

$$= \sqrt{10+2 \times \sqrt{3} \times \sqrt{2}+2 \times \sqrt{2} \times \sqrt{5}+2 \times \sqrt{3} \times \sqrt{5}}$$

$$= \sqrt{2+3+5+2 \times \sqrt{2} \times \sqrt{3}+2 \times \sqrt{5} \times \sqrt{2}+2 \times \sqrt{5} \times \sqrt{3}}$$

$$= \sqrt{(\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{5})^2 + 2 \times \sqrt{2} \times \sqrt{3} + 2 \times \sqrt{5} \times \sqrt{2} + 2 \times \sqrt{5} \times \sqrt{3}}$$

$$\left\{ \because a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = (a + b + c)^2 \right\}$$

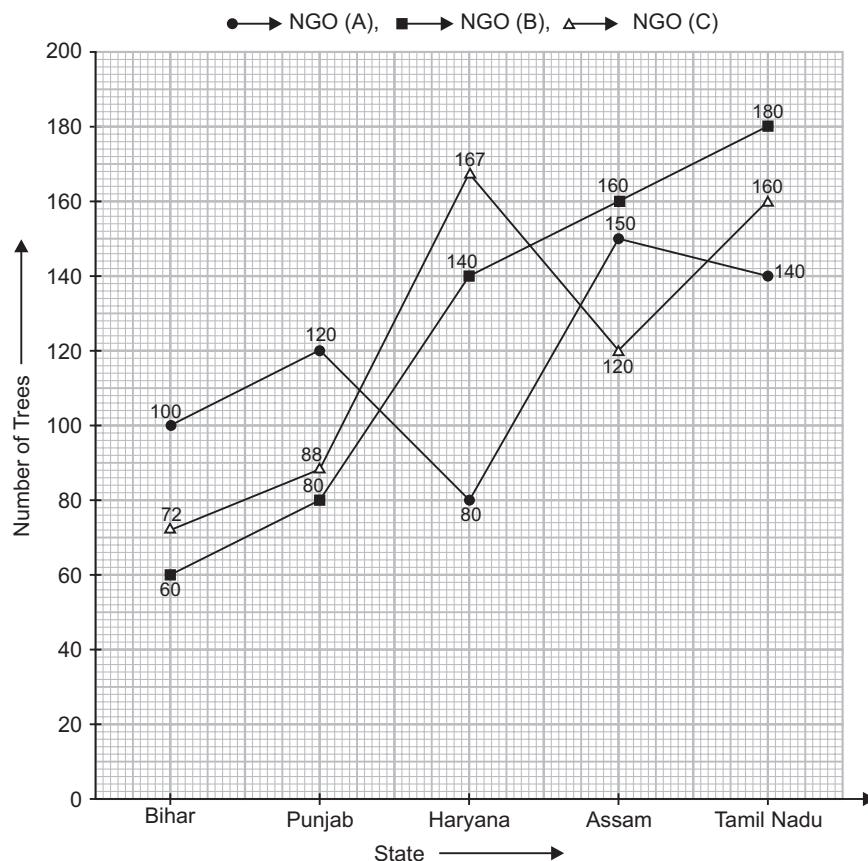
$$= \sqrt{(\sqrt{2} + \sqrt{3} + \sqrt{5})^2}$$

$$= (\sqrt{2} + \sqrt{3} + \sqrt{5})$$

Directions (Questions 6–10): Study the following graph carefully and answer the questions that follow.

(Bank P.O., 2011)

Number of Trees Planted by Three Different NGOs A, B and C in Five Different States



6. In which of the following states was the total number of trees planted by NGOs A and B together second lowest ?

(a) Bihar (b) Punjab
(c) Haryana (d) Assam
(e) Tamil Nadu

7. What was the difference between the trees planted by NGO – A in Haryana and those planted by NGO – C in Tamil Nadu ?

(a) 90 (b) 60
(c) 120 (d) 160
(e) None of these

8. What was the average number of trees planted in Haryana by all the NGOs together ?

(a) 420 (b) 140
(c) 120 (d) 390
(e) None of these

9. The total number of trees planted by NGOs A and B together in Bihar was approximately what percentage of the total number of trees planted by NGOs B and C together in Punjab ?

(a) 85 (b) 90
(c) 105 (d) 110
(e) 95

10. What was the ratio of the number of trees planted by NGO – B in Tamil Nadu, the number of trees planted by NGO – C in Assam and the number of trees planted by NGO – A in Assam ?

(a) 5 : 3 : 6 (b) 5 : 6 : 3
(c) 6 : 4 : 5 (d) 6 : 5 : 3
(e) None of these

Directions (Questions 11–15): The following line-graph gives the ratio of the amounts of imports by a company to the amount of exports from that company over the period from 2005 to 2011.

(Bank P.O., 2011)

The questions given below are based on this graph.

11. In how many of the given years were the exports more than the imports ?

(a) 1 (b) 2
(c) 3 (d) 4
(e) None of these

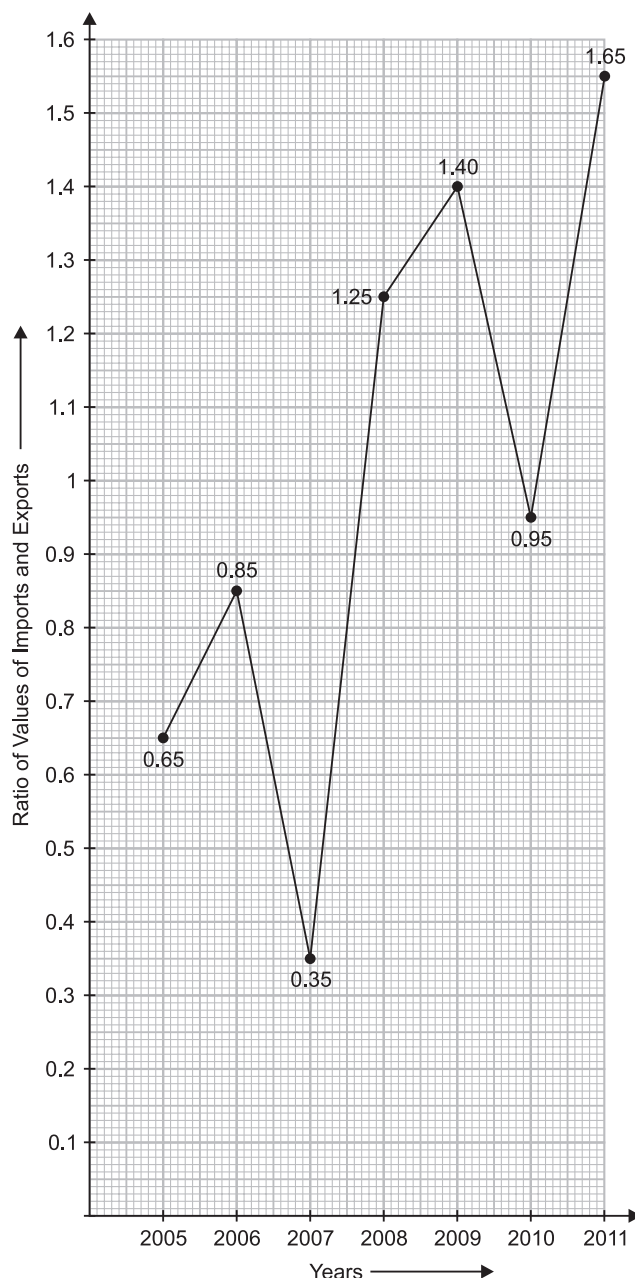
12. The imports were minimum proportionate to the exports of the company in which of the following years ?

- (a) 2005 (b) 2006
(c) 2007 (d) 2010
(e) 2011

13. If the imports of the company in 2006 be ₹ 272 crores, the exports from the company in the same year was:

- (a) ₹ 370 crores (b) ₹ 320 crores
(c) ₹ 280 crores (d) ₹ 275 crores
(e) ₹ 264 crores

Ratio of Values of Imports to Exports by a Company Over the Years



14. What was the percentage increase in imports from 2007 to 2008 ?

- (a) 72% (b) 56%
(c) 28% (d) None of these
(e) Data inadequate

15. If the imports in 2008 was ₹ 250 crores and the total exports in the years 2008 and 2009 together was ₹ 500 crores, then the imports in 2009 was

- (a) ₹ 250 crores (b) ₹ 300 crores
(c) ₹ 357 crores (d) ₹ 420 crores
(e) None of these

ANSWERS

1. (b) 2. (d) 3. (c) 4. (d) 5. (a) 6. (b) 7. (e) 8. (e) 9. (e) 10. (c)
 11. (d) 12. (c) 13. (b) 14. (e) 15. (d)

SOLUTIONS

1. Let the income of Y in 2007 be ₹ x crores. Then,

$$\frac{(x - 220)}{220} \times 100 = 35 \Rightarrow \frac{5(x - 220)}{11} = 35$$

$$\Leftrightarrow (x - 220) = \frac{35 \times 11}{5} = 77 \Rightarrow x = (77 + 220) = 297.$$

Hence, the income of Y in 2007 was ₹ 297 crores.

2. Let the income of each company in 2009 be ₹ x and let their expenditures be

E_1 and E_2 respectively. Then,

$$\frac{(x - E_1)}{E_1} \times 100 = 50 \text{ and } \frac{(x - E_2)}{E_2} \times 100 = 60$$

$$\Rightarrow \frac{x}{E_1} - 1 = \frac{50}{100} = \frac{1}{2} \text{ and } \frac{x}{E_2} - 1 = \frac{60}{100} = \frac{3}{5}$$

$$\Rightarrow \frac{x}{E_1} = \left(\frac{1}{2} + 1\right) \text{ and } \frac{x}{E_2} = \left(\frac{3}{5} + 1\right)$$

$$\Rightarrow \frac{x}{E_1} = \frac{3}{2} \text{ and } \frac{x}{E_2} = \frac{8}{5} \Rightarrow x = \frac{3E_1}{2} \text{ and } x = \frac{8E_2}{5}$$

$$\Rightarrow \frac{3E_1}{2} = \frac{8E_2}{5} \Rightarrow \frac{E_1}{E_2} = \left(\frac{8}{5} \times \frac{2}{3}\right) = \frac{16}{15}$$

$$\Rightarrow E_1 : E_2 = 16 : 15.$$

3. Let the incomes of X and Y in 2010 be ₹ $3x$ and ₹ $4x$ respectively and let their expenditures be E_1 and E_2 respectively.

$$\text{Then, } \frac{(3x - E_1)}{E_1} \times 100 = 65 \text{ and } \frac{(4x - E_2)}{E_2} \times 100 = 50$$

$$\Rightarrow \frac{3x}{E_1} - 1 = \frac{65}{100} \text{ and } \frac{4x}{E_2} - 1 = \frac{50}{100}$$

$$\Rightarrow \frac{3x}{E_1} = \left(\frac{13}{20} + 1\right) \text{ and } \frac{4x}{E_2} = \left(\frac{1}{2} + 1\right)$$

$$\Rightarrow \frac{3x}{E_1} = \frac{33}{20} \text{ and } \frac{4x}{E_2} = \frac{3}{2}$$

$$\Rightarrow E_1 = \frac{60x}{33} = \frac{20x}{11} \text{ and } E_2 = \frac{8x}{3}$$

$$\Rightarrow E_1 : E_2 = \frac{20x}{11} : \frac{8x}{3} = 60 : 88 = 15 : 22.$$

4. Let the expenditures of each of X and Y in 2006 be ₹ x crores and let the income of X in 2006 be ₹ z crores.

Then, income of Y in 2006 = ₹ $(342 - z)$ crores.

$$\therefore \frac{(z - x)}{x} \times 100 = 40 \text{ and } \frac{\{(342 - z) - x\}}{x} \times 100 = 45$$

$$\Rightarrow \frac{(z - x)}{x} = \frac{40}{100} = \frac{2}{5} \text{ and } \frac{(342 - z - x)}{x} = \frac{45}{100} = \frac{9}{20}$$

$$\Rightarrow 5z - 5x = 2x \text{ and } 6840 - 20z - 20x = 9x$$

$$\Rightarrow 7x = 5z \text{ and } 29x = 6840 - 20z$$

$$\Rightarrow x = \frac{5z}{7} \text{ and } x = \frac{6840 - 20z}{29}$$

$$\Rightarrow \frac{5z}{7} = \frac{6840 - 20z}{29} \Rightarrow 145z = 47880 - 140z$$

$$\Rightarrow 285z = 47880$$

$$\Rightarrow z = \frac{47880}{285} \Rightarrow z = 168.$$

$$\text{Putting } z = 168 \text{ in } x = \frac{5z}{7}, \text{ we get } x = \frac{5 \times 168}{7} = 120.$$

$\therefore$ Total Expenditure of X and Y in 2006

$$= ₹ 2x \text{ crores} = ₹ 240 \text{ crores.}$$

$$\text{Total Income of X and Y in 2006} = ₹ 342 \text{ crores.}$$

$\therefore$ Total Profit of X and Y in 2006

$$= ₹ (342 - 240) \text{ crores} = ₹ 102 \text{ crores.}$$

5. Let the income of X in 2008 be ₹ x crores. Then,

$$\frac{(x - 200)}{200} \times 100 = 55 \Rightarrow x - 200 = 110 \Rightarrow x = 310.$$

$\therefore$ Expenditure of X in 2011

$$= \text{Income of X in 2008} = ₹ 310 \text{ crores.}$$

Let the income of X in 2011 be ₹ z crores. Then,

$$\frac{(z - 310)}{310} \times 100 = 50 \Rightarrow \frac{(z - 310)}{310} = \frac{50}{100} = \frac{1}{2}$$

$$\Rightarrow (z - 310) = 155 \Rightarrow z = 465.$$

$\therefore$ Income of X in 2011 = ₹ 465 crores.

6. Total number of trees planted by A and B in :

$$\text{Bihar} \rightarrow (100 + 60) = 160 ;$$

$$\text{Punjab} \rightarrow (120 + 80) = 200 ;$$

$$\text{Haryana} \rightarrow (140 + 80) = 220 ;$$

$$\text{Assam} \rightarrow (150 + 160) = 310 ;$$

$$\text{Tamil Nadu} \rightarrow (140 + 180) = 320.$$

It is second lowest in Punjab.

7. (Trees planted by C in Tamil Nadu) - (Trees planted by A in Haryana) = $(160 - 80) = 80$.

8. Average number of trees planted in Haryana by 3 NGOs

$$= \frac{1}{3} (80 + 140 + 167) = \frac{387}{3} = 129.$$

9. Required% = $\left\{ \frac{(100 + 60)}{(80 + 88)} \times 100 \right\} \%$

$$= \left(\frac{160}{168} \times 100 \right) \%$$

$$= \frac{2000}{21} = 95\% \text{ (approx.)}$$

10. Required ratio = $180 : 120 : 150 = 6 : 4 : 5$.
11. Clearly, $\text{Export} > \text{Import}$ only when $\frac{\text{Import}}{\text{Export}} < 1$.

From the graph it is clear that the above ratio is less than 1 in the years 2005, 2006, 2007 and 2010.

Thus $(\text{Imports} : \text{Exports}) < 1$ in four years.

12. The imports are minimum proportionate to exports means (value of import) : (value of export) should have minimum value.

Clearly, this ratio has a minimum value of 0.35 in 2007.

13. From graph, we find that the ratio of value of import to the value of export in 2006 is 0.85.

Let the value of export in 2006 be ₹ x crores. Then,

$$\frac{272}{x} = 0.85 \Rightarrow x = \frac{272}{0.85} = \left(\frac{272 \times 100}{85} \right) = 320.$$

Hence, the value of exports in 2006 was ₹ 320 crores.

14. The graph gives only the ratio of value of imports and that of exports. In order to find the percentage increase in imports from 2007 to 2008, we require the value of import or that of export during these years.

So, we cannot find the percentage increase in imports. Hence, the data is inadequate to answer the question.

15. Let the value of export in 2008 be ₹ x crores.

Then, the value of export in 2009 = ₹ $(500 - x)$ crores.

$$\frac{250}{x} = 1.25 \Rightarrow x = \frac{250}{1.25} = \left(\frac{250 \times 100}{125} \right) = 200.$$

∴ Export in 2008 = ₹ 200 crores.

Export in 2009 = ₹ $(500 - 200)$ crores = ₹ 300 crores.

Let the value of import in 2009 be ₹ y crores.

Then,

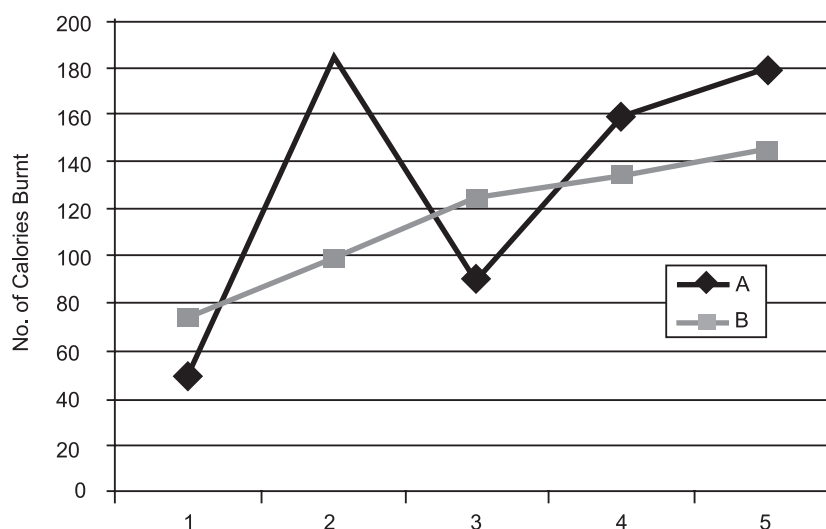
$$\frac{y}{300} = 1.40 \Rightarrow y = (1.40 \times 300) = \left(\frac{140}{100} \times 300 \right) = 420.$$

Hence, the value of import in 2009 was ₹ 420 crores.

EXERCISE – III

Directions (Questions 1-5): Refer to the graph and answer the given questions

Data Related to Number of Calories Burned by Two Individuals (A and B) on Treadmill During 5 Days



	Monday	Tuesday	Wednesday	Thursday	Friday
A	50	185	90	160	180
B	75	100	125	135	145

1. What is the respective ratio of total number of calories burned by A and B together on Wednesday and the by the same individuals together on Tuesday?

[IBPS—RRB (Off. Gr. 'B') Exam, 2015]

- (a) 45 : 59 (b) 43 : 57
(c) 41 : 57 (d) 43 : 61
(e) 47 : 61

2. If the number of calories burned by A and B increased by 10% and 20% respectively from Friday

to Saturday, what was the total number of calories burned by them together on Saturday?

[IBPS—RRB (Off. Gr. 'B') Exam, 2015]

- (a) 378 (b) 372
(c) 368 (d) 384
(e) 364

3. What is the total number of calories burned by A on Tuesday, Wednesday and Thursday together?

[IBPS—RRB (Off. Gr. 'B') Exam, 2015]

- (a) 425 (b) 440
(c) 430 (d) 445
(e) 435

4. If the average number of calories burned by B on Thursday, Friday and Saturday together is 125, what was the number of calories burned by B on Saturday?

[IBPS—RRB (Off. Gr. 'B') Exam, 2015]

- (a) 110 (b) 95
(c) 115 (d) 90
(e) 105

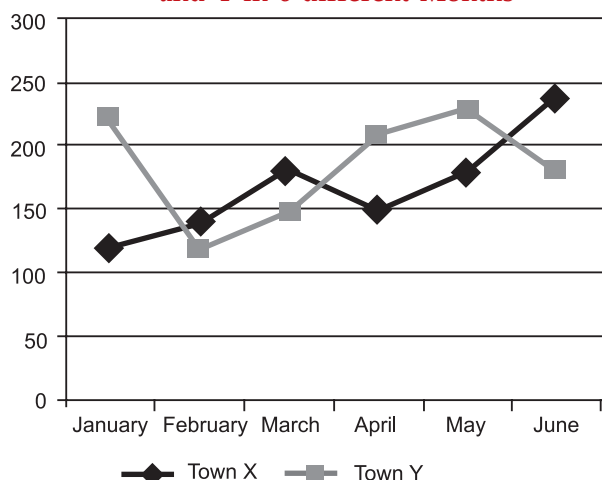
5. Number of calories burned by B increased by what percent from Monday to Thursday?

[IBPS—RRB (Off. Gr. 'B') Exam, 2015]

- (a) 80% (b) 60%
(c) 70% (d) 75%
(e) 65%

Directions (Questions 6–10): Refer to the graph and answer the given questions:

Number of watches of 'PQR' brand sold in Town X and Y in 6 different Months



Town X	120	140	180	150	180	240
Town Y	220	120	150	210	230	180

6. What is the average number of watches sold in Town X in January, February, March and June?

[IBPS—Bank Spl. Officer (IT) Exam, 2015]

- (a) 180 (b) 190
(c) 175 (d) 170
(e) 185

7. The number of watches sold in Town Y in April is what per cent more than the number of watches sold in Town X in the same month?

[IBPS—Bank Spl. Officer (IT) Exam, 2015]

- (a) 42% (b) 40%
(c) 30% (d) 50%
(e) 38%

8. The number of watches sold in Town Y increased by what per cent from February to May?

[IBPS—Bank Spl. Officer (IT) Exam, 2015]

- (a) 96% (b) $92\frac{1}{3}\%$
(c) 97% (d) $91\frac{2}{3}\%$
(e) $95\frac{2}{3}\%$

9. The number of watches sold in Town X in July was 10% more than the number of watches sold in the same town in May. What is the ratio of the number of watches sold in July to that sold in January in the same town?

[IBPS—Bank Spl. Officer (IT) Exam, 2015]

- (a) 34 : 23 (b) 32 : 25
(c) 31 : 20 (d) 33 : 23
(e) 33 : 20

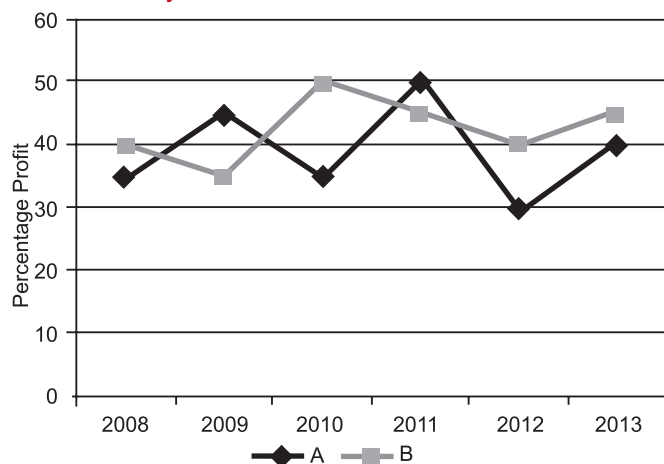
10. What is the difference between the total number of watches sold in both the towns together in June and the total number of watches sold in both the towns together in March?

[IBPS—Bank Spl. Officer (IT) Exam, 2015]

- (a) 50 (b) 90
(c) 60 (d) 70
(e) 80

Directions (Questions 11–15): Study the following graph carefully and answer the questions given below it.

Percentage profit earned by two companies A and B over the six years.



A	25	45	35	50	30	40
B	40	35	50	45	40	45

11. Expenditure of company B in 2009 and 2010 are ₹ 12 lakhs and ₹ 14.5 lakhs respectively. What was the total income of company B in 2009 and 2010 together (in lakh rupees)?

[IDBI Bank (Executive Officer's) Exam, 2015]

- (a) 39.75 (b) 37.95
(c) 38.75 (d) 38.55
(e) None of these

12. Ratio of expenditures of company A and B in 2012 was 3 : 4 respectively. What was the respective ratio of their incomes in 2012?

[IDBI Bank (Executive Officer's) Exam, 2015]

- (a) 21 : 26 (b) 13 : 14
(c) 14 : 13 (d) 26 : 21
(e) None of these

13. Total expenditure of company A in all the years together was 82.5 lakhs. What was the total income of the company A in all the years together?

[IDBI Bank (Executive Officer's) Exam, 2015]

- (a) 1.23 crores (b) 98.75 crores
(c) 99.85 crores (d) Cannot be determined
(e) None of these

14. If the expenditure of company A and B in 2013 were equal and the total incomes of the two companies was ₹ 5.7 lakhs. What was the total expenditure of the two companies in 2013?

[IDBI Bank (Executive Officer's) Exam, 2015]

- (a) 4 lakhs (b) 2 lakhs
(c) 4.2 lakhs (d) Cannot be determined
(e) None of these

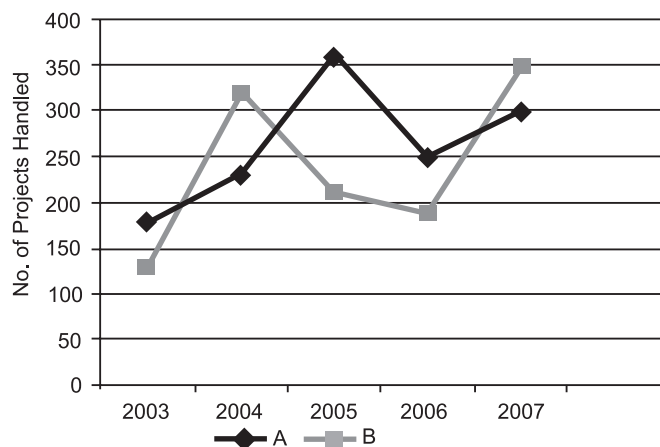
15. If the income of company B in 2010 and 2011 were in the ratio of 2 : 3 respectively, what was the respective ratio of expenditures of that company in these two years?

[IDBI Bank (Executive Officer's) Exam, 2015]

- (a) 20 : 29 (b) 9 : 10
(c) 29 : 45 (d) 10 : 29
(e) None of these

Directions (Questions 16–20): Refer to the graph and answer the given questions:

Data related to the number of projects handled by two companies A and B over 5 years.



16. Out of the total number of projects handled by Company A in 2005 and 2006 together, 20% were governmental projects. What was the total number of governmental projects handled by Company A in 2005 and 2006 together?

[IBPS—RRBs (Off. Gr. 'B') Exam, 2015]

- (a) 108 (b) 132
(c) 128 (d) 116
(e) 122

17. The projects handled by a company can be broadly classified into two types: governmental projects and non-governmental projects. If the average number of non-governmental projects handled by the same company B in 2003 and 2004 is 127. What is the total number of governmental projects handled by the same company in 2003 and 2004 together?

[IBPS—RRBs (Off. Gr. 'B') Exam, 2015]

- (a) 204 (b) 188
(c) 192 (d) 196
(e) 212

18. The number of projects handled by Company B decreased by what per cent from 2004 to 2006?

[IBPS—RRBs (Off. Gr. 'B') Exam, 2015]

- (a) $35\frac{5}{8}$ (b) $30\frac{7}{8}$
(c) $50\frac{3}{8}$ (d) $45\frac{3}{8}$
(e) $40\frac{5}{8}$

19. If the number of projects handled by company A increased by 20% from 2007 to 2008 and by 5% from 2008 to 2009, what was the number of projects handled by Company A in 2009?

[IBPS—RRBs (Off. Gr. 'B') Exam, 2015]

- (a) 378 (b) 372
(c) 384 (d) 396
(e) 368

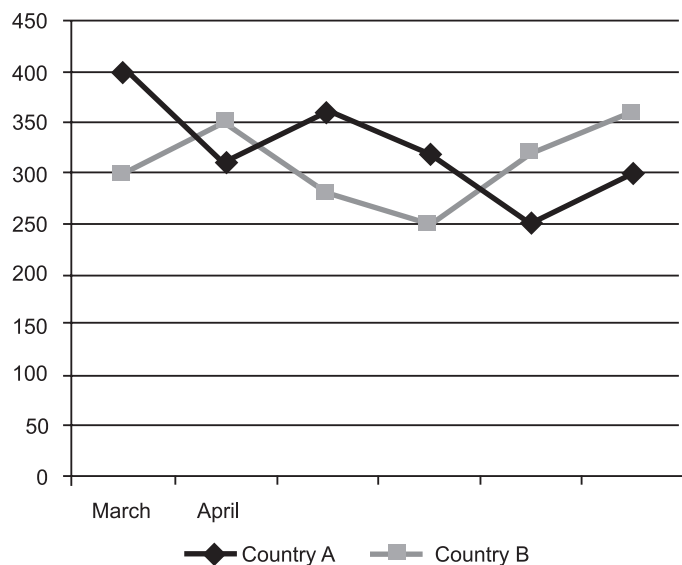
20. What is the difference between the total number of projects handled by company A in 2003 and 2004 together and the total number of projects handled by Company B in 2005 and 2007 together?

[IBPS—RRBs (Off. Gr. 'B') Exam, 2015]

- (a) 120 (b) 150
(c) 130 (d) 180
(e) 170

Directions (Questions 21–25): Read the graph and answer the given questions.

Number of visas issued by Country 'XYZ' for Country A and Country B in 6 different months.



Country A	400	310	360	320	250	300
Country B	300	350	280	250	320	360

21. What is the difference between the total number of visas issued for Country A and Country B together in April and the total number of visas issued for both the countries together in June?

[IBPS Bank PO/MT CWE-V (Pre.) Exam, 2015]

- (a) 90 (b) 70
(c) 110 (d) 100
(e) 80

22. What is the average number of visas issued for Country B in March, May, July and August?

[IBPS Bank PO/MT CWE-V (Pre.) Exam, 2015]

- (a) 315 (b) 310
(c) 320 (d) 335
(e) 325

23. The number of visas issued for Country A in March decreased by 20% from the previous month. What is the respective ratio between the number of visas issued for Country A in February and the number of visas issued for the same country in May?

[IBPS Bank PO/MT CWE-V (Pre.) Exam, 2015]

- (a) 25 : 13 (b) 25 : 18
(c) 26 : 13 (d) 24 : 13
(e) 26 : 15

24. The number of visas issued for Country A decreased by what percent from May to July?

[IBPS Bank PO/MT CWE-V (Pre.) Exam, 2015]

- (a) $35\frac{1}{3}$ (b) 33
(c) $30\frac{9}{5}$ (d) $32\frac{2}{3}$
(e) $32\frac{4}{9}$

25. The number of visas issued for Country B in March is what percent less than the number of visas issued for Country A in June?

[IBPS Bank PO/MT CWE-V (Pre.) Exam, 2015]

- (a) 8.5 (b) 7.75
(c) 4.25 (d) 6.25
(e) 5.75

ANSWERS

1. (b) 2. (b) 3. (e) 4. (b) 5. (a) 6. (d) 7. (b) 8. (d) 9. (e) 10. (b)
11. (b) 12. (e) 13. (d) 14. (a) 15. (c) 16. (e) 17. (d) 18. (e) 19. (a) 20. (b)
21. (a) 22. (a) 23. (b) 24. (c) 25. (d)

SOLUTIONS

- Total number of calories burnt by A and B together on Wednesday = $90 + 125 = 215$
Total number of calories burnt by A and B together on Tuesday = $185 + 100 = 285$
Required ratio = $215 : 285 = 43 : 57$
- Total number of calories burnt on Saturday
 $= 180 \times \frac{110}{100} + 145 \times \frac{120}{100}$
 $= 198 + 174 = 372$
- Total number of calories burnt by
 $A = 185 + 90 + 160 = 435$
- Average calories = 125 Total calories = $3 \times 125 = 375$
Number of calories burnt on Saturday = $375 - 280 = 95$

5. Required percentage = $\frac{60}{75} \times 100 = 80\%$

6. Total Number of watches sold in Town X in January, February, March and June = $120 + 140 + 180 + 240 = 680$

Required average = $\frac{680}{4} = 170$

7. The number of watches sold in Town Y in April = 210
The number of watches sold in Town X in April = 150

Required percentage = $\frac{210 - 150}{150} \times 100 = \frac{60}{150} \times 100 = 40\%$

8. Number of Watches sold in Town Y in February = 120
Number of Watches sold in Town Y in May = 230

Required percentage increase

$$= \frac{230-120}{120} \times 100 = \frac{110}{120} \times 100 = \frac{110 \times 5}{6}$$

$$= \frac{275}{3} = 91\frac{2}{3}\%$$

9. Number of watches sold in Town X in the month of May = 180

Number of watches sold in Town X in the month of July
 $= \frac{180 \times 110}{100} = 198$

Number of watches sold in Town X in the month of January = 120

$$\therefore \text{Required ratio} = \frac{198}{120} = \frac{33}{20} = 33:20$$

10. Total number of watches sold in both the towns together in the month of June = 240 + 180 = 420

Total number of watches sold in both the towns together in March = 180 + 150 = 330

$$\text{Required difference} = 420 - 330 = 90$$

11. Profit percent earned by company B in 2009 = 35%

Profit percent earned by company B in 2010 = 50%

Expenditure of company B in 2009 = 12 lakhs

Expenditure of company B in 2010 = 14.5 lakhs

$$\text{Income of company B in 2009} = 35\% = \frac{I-E}{E} \times 100$$

$$\Rightarrow 35 = \frac{(I-12 \text{ lakh})}{12 \text{ lakh}} \times 100$$

$$\Rightarrow (35 \times 12) \text{ Lakh} = 100I - 1200 \text{ lakh}$$

$$\Rightarrow 420 \text{ lakh} = 100I - 1200 \text{ lakh}$$

$$\Rightarrow I = 16.20 \text{ lakh}$$

$$\text{Income of company B in 2009} = 50\% = \frac{I-E}{E} \times 100$$

$$\Rightarrow 50 = \frac{I-14.5 \text{ lakh}}{14.5 \text{ lakh}} \times 100$$

$$\Rightarrow (50 \times 14.5) \text{ lakh} = 100I - 14.50 \text{ lakh}$$

$$\Rightarrow 2175 = (100) I \Rightarrow I = 21.75 \text{ lakhs}$$

So, total income of company B in 2009 and 2010

$$= 16.2 + 21.75$$

$$= ₹ 37.95 \text{ Lakhs}$$

12. For company A, In 2012

$$P\% = \frac{I-E}{E} \times 100$$

$$\Rightarrow 30 = \frac{I-3}{3} \times 100$$

$$\Rightarrow 90 = 100 I - 300$$

$$\Rightarrow \frac{390}{100} = I$$

$$\Rightarrow I = 3.9$$

For company B, in 2012

$$P\% = \frac{I-E}{E} \times 100$$

$$40 = v$$

$$\Rightarrow 160 = 100 I - 400$$

$$\Rightarrow \frac{560}{100} = I$$

$$\Rightarrow I = 5.6$$

$$\text{Then required ratio} = \frac{3.9}{5.6} = \frac{39}{56}$$

13. Solution cannot be determined because profit percentage of company A in all the years are not given.

14. Let the income of company A in 2013 = x

Let the income of company B in 2013 = (5.7 - x)

Expenditure of company A = Expenditure of Company

$$B = E \frac{40}{100} - \frac{x-E}{E} \times 100$$

$$\frac{x}{E} = 1.4 \dots (i) \text{ and } \frac{45}{100} = \frac{(5.7-x)-E}{E}$$

$$\Rightarrow \frac{5.7-x}{E} = 14.5 \dots (ii)$$

Divide equation (ii) by (i) we get

$$\Rightarrow \frac{5.7-x}{E} \times \frac{E}{x} = \frac{145}{140}$$

$$\Rightarrow \frac{5.7x}{x} = \frac{145}{140}$$

$$\Rightarrow 5.7 \times 140 - 140x = 145x$$

$$\Rightarrow 798 = 285x$$

$$\Rightarrow x = 2.8 \text{ lakhs}$$

$$\therefore \frac{x}{E} = 1.4$$

$$E = 2 \text{ lakhs}$$

Total expenditure of two companies

$$= (2 + 2) = 4 \text{ lakhs}$$

15. In 2010, $50 = \frac{2-E_1}{E_1} \times 100$

$$\Rightarrow 50 E_1 = 200 - 100 E_1$$

$$\Rightarrow 150 E_1 = 200$$

$$\Rightarrow E_1 = \frac{4}{3}$$

$$\text{In 2011, } 45 = \frac{(3-E_2)}{E_2} \times 100$$

$$\Rightarrow 45 E_2 = 300 - 100 E_2$$

$$\Rightarrow 145 E_2 = 300$$

$$\Rightarrow E_2 = \frac{300}{145}$$

$$\text{Then, } \frac{E_1}{E_2} = \frac{4}{3} \times \frac{145}{300}$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{29}{35}$$

16. Number of governmental projects handled by company A in 2005 and 2006 together = 360 + 250 = 610

Number of Governmental projects handled by company

$$A \text{ in 2005 and 2006 together} = 20\% \text{ of } 610 = \frac{20 \times 610}{100} = 122$$

17. Average number of non-governmental projects handled by company B = 127
 Number of Non-Governmental projects handled by Company B in 2003 and 2004 is = $127 \times 2 = 254$
 Number of Governmental projects handled by Company B in 2003 and 2004 together
 $= 130 + 320 - 254 = 40 - 254 = 196$
18. Number of projects handled by Company B in 2004 = 320
 Number of projects handled by Company B in 2006 = 190
 Required decrease percentage = $\frac{320 - 190}{320} \times 100$
 $= \frac{130}{320} \times 100 = \frac{1300}{32} \% = \frac{325}{8}$
 $= 40 \frac{5}{8} \% \text{ decrease}$
19. Number of projects handled by company A in 2008
 $= 120\% \text{ of } 300 = \frac{120 \times 300}{100} = 360$
 Number of projects handled by Company A in 2009
 $= 105\% \text{ of } 360 = \frac{105 \times 360}{100} = 378$
20. The number of projects handled by company A in 2003 and 2004 together = $180 + 230 = 410$
 The number of projects handled by company B in 2005 and 2007 together = $210 + 350 = 560$
 Required difference = $560 - 410 = 150$
21. Visas issued for country A and country B in April
 $= 310 + 350 = 660$

Visas issued for country A and country B in June
 $= 320 + 250 = 570$

Required difference
 $= 660 - 570 = 90$

22. Total number of visas issued for country B in March, May, July and August
 $= 300 + 280 + 320 + 360 = 1260$
 Required average = $\frac{300 + 280 + 320 + 360}{4}$
 $= \frac{1260}{4} = 315$
23. Number of visas for country A in February
 $= \frac{400 \times 100}{80} = 500$
 $\therefore$ Required ratio = $500 : 360$
 $= 25 : 18$
24. Number of visas issued for country A in May = 360
 Number of visas issued for country A in July = 250
 Percentage decrease = $\frac{360 - 250}{360} \times 100$
 $= \frac{1100}{36} = \frac{275}{9} = 30 \frac{5}{9} \%$
25. Number of visas issued for country B in March = 300
 Number of visas issued for country A in June = 320
 Required percent = $\frac{320 - 300}{320} \times 100$